
Foundational Mathematics in Social Sciences

A Textbook

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Preface

While developing *Foundational Mathematics*, I came to realise that a companion volume for students of the social sciences was equally important. Mathematics is often associated most strongly with the physical sciences and engineering, yet it also has a substantial role in economics, psychology, sociology, geography, political science and the wider humanities. Without acknowledging these connections, the original project would have felt incomplete.

Mathematical ideas allow social scientists to describe patterns, compare groups, measure uncertainty, evaluate evidence, and construct models of human behaviour and social change. The purpose of this book is therefore not merely to teach calculation, but also to show how mathematical reasoning can support careful interpretation and informed decision-making.

This book brings together selected material taught in Cambridge A Level, matriculation, foundation, and equivalent pre-university programmes. Rather than attempting to reproduce every topic from a particular syllabus, it emphasises the algebraic, graphical, statistical, and problem-solving skills most useful to students preparing for the social sciences. Topics more commonly associated with pure mathematics are included at an appropriate introductory level when they provide a useful foundation for ideas developed later in the book.

Some concepts from secondary-school mathematics, including index numbers and financial mathematics, are revisited where a brief review supports later learning. The intention is to connect familiar techniques with pre-university concepts without assuming that every reader has followed the same syllabus or educational pathway. Worked examples and exercises are included to encourage both procedural confidence and an understanding of what the results mean in context.

The book is intended primarily for social-science students undertaking Cambridge A Level, matriculation, foundation, or an equivalent pre-university programme. It may also be useful to undergraduates who wish to refresh the mathematics required for introductory quantitative courses.

I have no intention of making a profit from this work or of seeking publicity through it. My aim is simply to encourage readers to recognise the importance of mathematics, not only in their future studies but also in the way it shapes everyday life. Whether or not we notice it, mathematics is everywhere, and I hope this book helps readers appreciate both its usefulness and its beauty.

Future volumes may adapt the same approach for engineering, medicine, chemistry or more specialised areas of the social sciences. Until then, I hope that readers find this volume both useful and enjoyable.

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How to Use This Book

This book is written for students who use mathematics to investigate social questions. Its examples connect mathematical ideas with populations, public opinion, inequality, markets, institutions, policy, finance and human behaviour. The aim is therefore not only to obtain a numerical answer, but also to decide what that answer measures, what evidence backs it up, and what limits its interpretation.

The fifteen chapters build progressively. Numbers, functions and logic provide the language needed to describe relationships; statistics and probability support the analysis of data and uncertainty; sequences, linear models and matrices organise patterns and systems; and calculus examines change, accumulation and dynamic models. The final chapters apply these foundations to financial mathematics and index numbers. Readers following the complete course should normally work through the chapters in order, although instructors may select a shorter route after reviewing the required earlier material.

Each chapter begins with a guide to its main ideas and selected background sources. Definitions establish the notation, explanations develop the reasoning and worked examples demonstrate the methods. When you study an example, identify the observational unit or population, define the variables and their units, and note the feasible domain before calculating. Afterwards, interpret the result in the original context and consider whether it describes an exact quantity, an estimate, or a prediction from a model.

The practice sections move from core techniques to social-science applications and modelling. A complete response should show the mathematical method, retain exact values until appropriate rounding, and communicate the meaning of the result. Where data or a fitted model are involved, examine the assumptions, uncertainty, possible sources of bias and the range over which the conclusion is defensible. In particular, a mathematical association does not by itself establish causation, and an optimum produced by a model need not be the most equitable or practical policy choice.

Appendix C provides concise answer keys for checking work after an exercise has been attempted. Appendix A summarises recurring symbols and unit conventions, while Appendix B supplies the probability tables required for relevant exercises. The table of contents, lists of figures and tables, references and subject index offer additional routes through the book. These entries and internal cross-references are clickable in the digital edition.

Calculators and software may assist with arithmetic, graphs and verification, but they do not replace a transparent method or careful judgement. Report answers with appropriate units and accuracy, distinguish percentages from percentage points, and state important assumptions. In social-science work, clear interpretation is part of the mathematics rather than an optional final comment.

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Introduction

Mathematics provides the social sciences with a precise language for describing relationships, measuring variation, evaluating uncertainty, and testing whether evidence supports a conclusion. Economists use functions to study prices, income, growth, and changing markets. Psychologists and sociologists lean on probability and statistics to make sense of observations and survey responses. Geographers work with numerical models and spatial patterns, while political scientists use quantitative evidence to examine public opinion, elections, and policy outcomes. Across these disciplines, mathematics helps researchers make their assumptions visible and their reasoning open to examination.

This book develops the mathematical foundations needed for such work at A Level and pre-university depth. It assumes familiarity with basic secondary-school arithmetic and algebra, but introduces notation and new methods carefully. The emphasis is not only on obtaining an answer, but also on understanding the reasoning behind each method and interpreting the result in its proper context. Worked examples therefore connect mathematical techniques with situations involving costs, populations, measurements, surveys, risk, and decision-making.

This book is structured based on the PASUM's Social Science Mathematics lecture notes, which is why some chapters like Introduction to Statistics, Probability and Discrete Random Variables are separated instead of being merged.

Mathematical models are simplified representations of reality. They are valuable because they allow complicated relationships to be studied, but their conclusions are only as reliable as their assumptions and data. Readers should develop the habit of asking what each variable represents, whether the units are appropriate, whether a calculated value is reasonable, and whether the model is being used within a suitable range. An exact calculation may still give a misleading conclusion if the underlying model does not reflect the situation adequately.

The chapters are organised as follows:

Chapter 1: Numbers. Real and complex numbers, approximation, ratios, rates, percentages, intervals, indices, surds and logarithms are developed as the numerical and algebraic foundation for later chapters.

Chapter 2: Functions. Relations, function notation, graphs, transformations, common families, graph behaviour, rates of change, contextual models, composition and inverse functions are introduced as tools for describing how quantities depend on one another.

Chapter 3: Logic and Sets. Propositions, truth tables, set operations, counting principles, permutations, and combinations establish a language for precise reasoning and systematic counting.

Chapter 4: Introduction to Statistics. Types of data and the principal measures of centre, position, and variation are used to organise and summarise observations.

Chapter 5: Probability. Events, probability rules, conditional probability, independence, tree diagrams, and Bayes' theorem provide methods for reasoning under uncertainty.

Chapter 6: Discrete Random Variables. Probability distributions, cumulative distribution functions, expected value, variance, and linear transformations connect probability with numerical outcomes.

Chapter 7: Sequences and Series. Arithmetic and geometric patterns, finite and infinite series, sigma notation, and the binomial theorem support the study of repeated change and growth.

Chapter 8: Linear Equations and Graphs. Linear equations, inequalities, straight-line graphs, rates of change, and regression models are developed with practical applications.

Chapter 9: Matrices. Matrix operations, determinants, inverses and matrix models provide compact methods for organising data and solving simultaneous equations.

Chapter 10: Limits and Continuity. One-sided and infinite limits, asymptotes and continuity establish the conceptual foundation required for calculus.

Chapter 11: Differential Calculus. Derivatives, tangent lines, marginal quantities and rates of change are developed from the difference quotient.

Chapter 12: Integral Calculus. Antiderivatives, substitution, integration by parts, definite and numerical integration connect local rates with accumulated change, areas, average values, economic surplus and continuous flows.

Chapter 13: Differential Equations. First-order separable equations, initial conditions, equilibrium, stability, exponential and logistic growth, adjustment models and numerical approximation describe quantities whose rates change over time.

Chapter 14: Financial Mathematics. Interest, present value, annuities, loan repayments, inflation and depreciation develop practical models for personal and public finance.

Chapter 15: Index Numbers. Base-year and composite indices are extended to consumer-price indices, inflation, real-value adjustments, chain indices and alternative weighted price indices.

Readers are encouraged to work through the examples actively, reproduce the calculations independently, and attempt the exercises before consulting the answers appendix. Where an approximation is required, sufficient working should be retained before rounding. Units, domain restrictions, and the meaning of the final result should be treated as part of the solution rather than as optional details.

The aim throughout is to provide a rigorous but accessible bridge from secondary-school mathematics to quantitative study at university. The methods in this book do not replace the theories, evidence, and judgement of the social sciences; instead, they provide a framework through which these can be examined more clearly and responsibly.

Chapter 1

Numbers

Chapter guide

Begin with the real-number system and interval notation, then review quantitative conventions, complex numbers, indices, surds and logarithms before attempting the mixed applications.

Selected background sources: (Chai et al., 2024; Ministry of Education Malaysia, 2018; Wong Mee Kiong et al., 2019; Zaini bin Musa et al., 2020).

Numbers provide the language in which quantitative information is recorded, compared and interpreted. In the social sciences, a number may represent a count of people, a proportion of respondents, an income, a price index, a rate of migration or an estimate obtained from a sample. Calculation is important, but so are the meaning, units and reliability of the number being used.

Different questions require different kinds of numbers. Counts are commonly whole numbers, while measurements and averages may take any real value. Fractions and percentages express relative size; intervals describe ranges of possible or permitted values; and powers and logarithms describe change that occurs in multiplicative rather than by equal additions. Some numbers are exact, whereas others are rounded observations or model-based estimates. Recognising these distinctions prevents a mathematically correct calculation from being given an inappropriate interpretation.

This chapter develops the number systems and algebraic notation used throughout the book. It begins with real numbers, intervals and sets, then examines quantitative conventions used in applications. Complex numbers are introduced as an extension of the real-number system, while indices, surds and logarithms provide foundations for growth models, financial calculations and later work with functions and calculus.

1.1 Real numbers

Real numbers form the basis of much of elementary mathematics and include nearly all the numbers encountered in everyday calculations. They consist of both rational numbers, which can be expressed as fractions, and irrational numbers, which cannot. This section examines how the different subsets of real numbers are related, explore their fundamental properties, and learn how to represent them on a number line and using interval notation.

The set of real numbers is denoted by $\mathbb{R}$. Every real number can be represented by exactly one point on a continuous number line, and every point on that line corresponds to exactly one real number. Real numbers therefore include positive and negative numbers, zero, integers, fractions, and numbers such as $\sqrt{2}$ and π .

Several familiar number sets are contained in $\mathbb{R}$. In this book, the natural numbers are the counting numbers

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

the whole numbers are

$$\mathbb{W} = \{0, 1, 2, 3, \dots\},$$

and the integers are

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

These sets are nested: every natural number is a whole number, every whole number is an integer, and every integer is a real number. Notice that a number may belong to more than one set. For example, 5 belongs to $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$.

The real numbers are divided into two disjoint groups: rational numbers and irrational numbers. In symbols,

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \quad \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset.$$

Thus, every number that lies in the real number line is either rational or irrational, but no real number can be both.

1.1.1 Rational numbers

Definition 1.1: Rational number

A real number is *rational* if it can be written in the form

$$\frac{p}{q},$$

where $p, q \in \mathbb{Z}$ and $q \neq 0$. The set of rational numbers is denoted by $\mathbb{Q}$; hence,

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

The word *rational* is related to the word *ratio*. The integers in the fraction need not be positive, and a rational number can usually be written as a fraction in many different ways. For example,

$$-\frac{3}{4} = \frac{3}{-4} = -\frac{6}{8}.$$

The denominator cannot be zero because division by zero is undefined.

Every integer is rational because it can be written with denominator 1. For instance,

$$-7 = \frac{-7}{1}, \quad 0 = \frac{0}{1}, \quad 12 = \frac{12}{1}.$$

Consequently,

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

The decimal expansion of a rational number either *terminates* or *eventually repeats*. A

terminating decimal has only finitely many non-zero decimal places. It can be converted into a fraction by using a power of 10 as the denominator. For example,

$$0.125 = \frac{125}{1000} = \frac{1}{8}, \quad -2.35 = \frac{-235}{100} = \frac{-47}{20}.$$

A recurring decimal contains a digit or block of digits that repeats without end. A bar is placed over the repeating block; for example, $0.\overline{27} = 0.272727\dots$. To express this number as a fraction, let

$$x = 0.\overline{27}.$$

Since the repeating block contains two digits, multiplication by 100 gives

$$100x = 27.\overline{27}.$$

Subtracting the first equation from the second gives

$$99x = 27, \quad x = \frac{27}{99} = \frac{3}{11}.$$

Therefore, recurring decimals are rational as well. The same principle works when the repeating part begins after one or more non-repeating digits.

Example 1.1: Identifying rational numbers

Each of the following numbers is rational:

$$\frac{5}{9}, \quad -4, \quad 0.375, \quad 1.2\overline{36}.$$

The first is already a ratio of integers, the second can be written as $-4/1$, the third is a terminating decimal, and the fourth is an eventually repeating decimal.

1.1.2 Irrational numbers

Definition 1.2: Irrational number

An *irrational number* is a real number that cannot be written as p/q , where p and q are integers and $q \neq 0$. The set of irrational numbers is $\mathbb{R} \setminus \mathbb{Q}$.

The decimal expansion of an irrational number is non-terminating and non-repeating. In other words, its digits continue indefinitely without settling into a fixed repeating block. Common examples include

$$\sqrt{2}, \quad \sqrt{3}, \quad \pi, \quad e.$$

For example,

$$\sqrt{2} = 1.414213562\dots \quad \text{and} \quad \pi = 3.141592654\dots$$

The dots indicate that the decimal expansions continue, but no finite block of digits

repeats forever.

Not every square root is irrational. The square root of a perfect square is an integer and is therefore rational; for example, $\sqrt{49} = 7$. More generally, a prime-factor argument shows that the square root of a positive integer is rational only when that integer is a perfect square. Thus, $\sqrt{5}$ and $\sqrt{10}$ are irrational, whereas $\sqrt{1}$, $\sqrt{4}$ and $\sqrt{9}$ are rational.

The following proof illustrates the central argument. We first use the fact that if an integer square is even, then the integer itself is even: the contrapositive holds because $(2k+1)^2 = 4k(k+1) + 1$ is odd.

Proof that $\sqrt{2}$ is irrational. Assume, for a contradiction, that $\sqrt{2}$ is rational. It can then be written in lowest terms as

$$\sqrt{2} = \frac{p}{q},$$

where p and q are integers with no common factor and $q \neq 0$. Squaring both sides gives

$$2 = \frac{p^2}{q^2}, \quad \text{so} \quad p^2 = 2q^2.$$

Therefore p^2 is even, which means that p is even. Write $p = 2k$ for some integer k . Substitution gives

$$(2k)^2 = 2q^2 \implies q^2 = 2k^2.$$

Hence q is also even. This means that p and q have a common factor of 2, contradicting the assumption that p/q is in lowest terms. The original assumption is therefore false, so $\sqrt{2}$ is irrational. $\square$

Care is needed when classifying decimal expansions. A decimal that does not terminate is not automatically irrational: $0.333\ldots = 0.\bar{3} = 1/3$ is rational because it repeats. By contrast,

$$0.101001000100001\ldots$$

is irrational because the number of zeros between successive 1s keeps changing, so the decimal does not repeat in a fixed cycle.

Example 1.2: Rational or irrational?

Consider the numbers

$$\sqrt{64}, \quad \sqrt{7}, \quad 0.\overline{18}, \quad 0.110100100010\ldots$$

Here $\sqrt{64} = 8$ and $0.\overline{18}$ are rational. The number $\sqrt{7}$ is irrational because 7 is not a perfect square, while the final decimal is irrational because it neither terminates nor repeats in a fixed pattern.

1.1.3 Venn diagram representation

The main subsets of the real numbers can be represented by nested regions. The natural numbers are contained in the whole numbers, the whole numbers are contained in the

integers, and the integers are contained in the rational numbers. The irrational numbers occupy the part of the real-number set that lies outside the rational numbers.

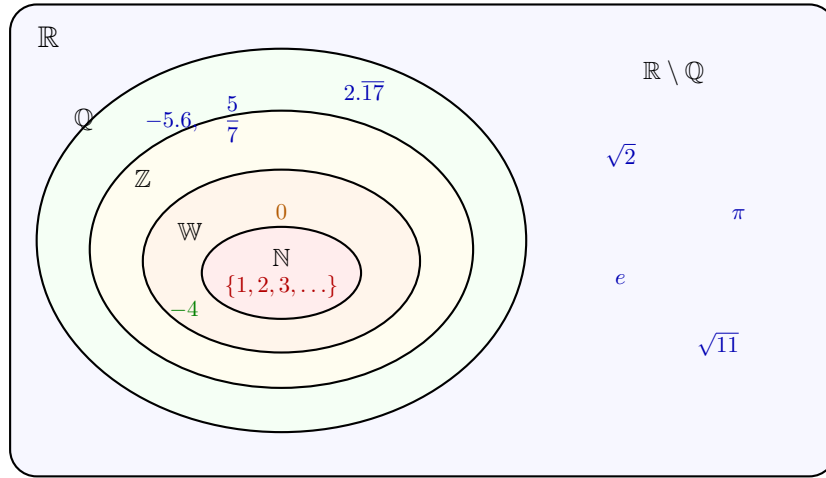


Figure 1.1: The main subsets of the real numbers.

Thus,

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R},$$

and the rational and irrational numbers form a partition of the real numbers:

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \quad \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset.$$

The nesting also shows why a number can belong to several sets at once. For example, 3 is a natural number, a whole number, an integer, a rational number, and a real number.

1.1.4 Properties of real numbers

The operations of addition and multiplication on real numbers satisfy several fundamental properties. These properties allow expressions to be rearranged and simplified without changing their values. In table 1.1, let $a, b, c \in \mathbb{R}$.

Property	Addition	Multiplication
Closure	$a + b \in \mathbb{R}$	$ab \in \mathbb{R}$
Commutative	$a + b = b + a$	$ab = ba$
Associative	$a + (b + c) = (a + b) + c$	$a(bc) = (ab)c$
Identity	$a + 0 = 0 + a = a$	$a \cdot 1 = 1 \cdot a = a$
Inverse	$a + (-a) = 0$	$a \left(\frac{1}{a} \right) = 1, a \neq 0$
Distributive	$a(b + c) = ab + ac$ and $(a + b)c = ac + bc$	

Table 1.1: Properties of addition and multiplication on the real numbers.

The *closure property* states that adding or multiplying any two real numbers produces another real number. For example,

$$\sqrt{2} + 3 \in \mathbb{R} \quad \text{and} \quad \pi\sqrt{5} \in \mathbb{R}.$$

The real numbers are also closed under subtraction and under division by a non-zero real number. Division by zero is excluded because it is undefined.

The *commutative property* permits the order of two terms or factors to be reversed, whereas the *associative property* permits their grouping to be changed. These properties must not be confused: for example,

$$4 + 7 = 7 + 4$$

uses the commutative property, while

$$4 + (7 + 2) = (4 + 7) + 2$$

uses the associative property. Subtraction and division are neither commutative nor associative in general.

The additive identity is 0, and the multiplicative identity is 1. Adding 0 or multiplying by 1 leaves a real number unchanged. The additive inverse of a is $-a$. If $a \neq 0$, its multiplicative inverse, or *reciprocal*, is $1/a$. The number 0 has no multiplicative inverse.

The *distributive property* connects multiplication and addition. For example,

$$3(x - 2) = 3x - 6.$$

Here multiplication by 3 is distributed to both terms inside the parentheses.

Multiplication by zero and the zero-product property

The *multiplicative property of zero* states that the product of zero and any real number is zero:

$$a \cdot 0 = 0 \cdot a = 0 \quad \text{for every } a \in \mathbb{R}.$$

For example,

$$0(-4) = 0, \quad m \cdot 0 = 0, \quad (27x^2y^4) \cdot 0 = 0.$$

A related result is the *zero-product property*:

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

Thus, if $(x + 3)(x - 2) = 0$, then

$$x + 3 = 0 \quad \text{or} \quad x - 2 = 0,$$

so $x = -3$ or $x = 2$. The multiplicative property of zero describes what happens when a known factor is zero; the zero-product property identifies a zero factor when a product is

known to be zero.

1.1.5 Representation on the number line

Every real number can be represented by a unique point on a horizontal *number line*. The point corresponding to 0 is called the *origin*. Positive numbers lie to the right of the origin, whereas negative numbers lie to its left. More generally, if $a < b$, then the point representing a lies to the left of the point representing b .

Both rational and irrational numbers occur on the same number line. For example, -3.5 and $0.34\bar{5}$ are rational numbers, while $\sqrt{2}$, e , and π are irrational numbers. Their approximate positions are shown in figure 1.2.

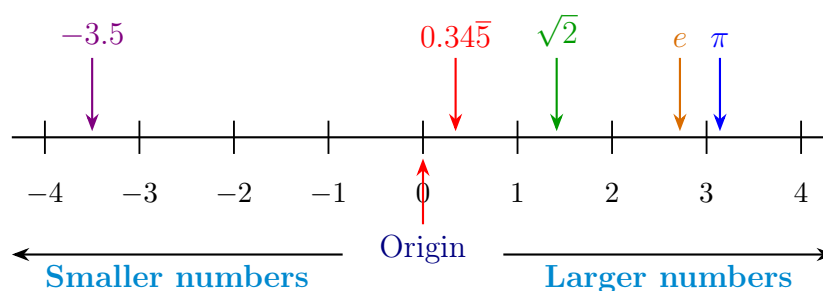


Figure 1.2: Rational and irrational numbers on the real number line.

The number line is continuous: between any two distinct real numbers there are infinitely many other real numbers. Thus, its points are not restricted to the labelled integers or to the examples displayed in the figure.

1.1.6 Intervals

Definition 1.3: Interval

An *interval* is a subset of the real numbers that contains every real number between any two of its members. If $a < b$, the numbers a and b are called the *endpoints* of a finite interval.

Square brackets indicate that a finite endpoint is included, whereas round brackets indicate that it is excluded. A filled point on a number line marks an included endpoint, and an open point marks an excluded endpoint. The symbols $-\infty$ and $+\infty$ describe directions rather than real-number endpoints, so they are always written with round brackets.

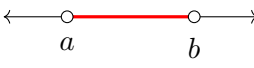
Interval type	Notation	Inequality notation	Number-line representation
Finite and closed	$[a, b]$	$a \leq x \leq b$	
Finite and open	(a, b)	$a < x < b$	
Finite and half-open	$[a, b)$	$a \leq x < b$	
	$(a, b]$	$a < x \leq b$	
Unbounded, finite endpoint included	$(-\infty, b]$	$x \leq b$	
	$[a, +\infty)$	$x \geq a$	
Unbounded, finite endpoint excluded	$(-\infty, b)$	$x < b$	
	$(a, +\infty)$	$x > a$	
All real numbers	$(-\infty, +\infty)$	$x \in \mathbb{R}$	

Table 1.2: Finite and infinite intervals of real numbers, where $a < b$.

For example, $[-2, 3)$ represents all real numbers x satisfying $-2 \leq x < 3$. Similarly, $(-\infty, 5]$ represents all real numbers satisfying $x \leq 5$.

1.1.7 Set notation for real numbers

A set is a collection of distinct objects, called its *elements*. When the elements are real numbers, the set is a subset of $\mathbb{R}$. A finite set may be written by listing its elements inside braces; for example,

$$A = \{-2, 0, 3\}.$$

For a larger or infinite set, set-builder notation is often more convenient. For instance,

$$B = \{x \in \mathbb{R} : x \geq 1\} = [1, +\infty).$$

The colon is read as “such that”, so this expression describes the set of all real numbers x such that $x \geq 1$.

table 1.3 summarises the notation used when working with sets of real numbers.

The order in which the elements of a set are listed is irrelevant, and repeated elements are written only once. Thus,

$$\{1, 2, 3\} = \{3, 2, 1\} = \{1, 1, 2, 3\}.$$

Symbol	Name	Meaning
$x \in A$	Element of	x belongs to A .
$x \notin A$	Not an element of	x does not belong to A .
$\emptyset$	Empty set	A set containing no elements.
$A \subseteq B$	Subset	Every element of A is also an element of B .
$A \not\subseteq B$	Not a subset	At least one element of A does not belong to B .
$A \subsetneq B$	Proper subset	$A \subseteq B$, but $A \neq B$.
$A = B$	Equal sets	A and B contain exactly the same elements.
$A \cup B$	Union	Elements that belong to A or B , or to both.
$A \cap B$	Intersection	Elements that belong to both A and B .

Table 1.3: Common notation for sets of real numbers.

Definition 1.4: Subset and proper subset

Let A and B be sets. The set A is a *subset* of B , written $A \subseteq B$, if every element of A is also an element of B . It is a *proper subset* of B , written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Every set is a subset of itself, so $A \subseteq A$, and the empty set is a subset of every set. However, no set is a proper subset of itself. Also, two sets are equal precisely when each is a subset of the other:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Example 1.3: Subsets and proper subsets

Let

$$C = \{1, 2, 3\}, \quad D = \{1, 3\}, \quad E = \{3, 2, 1\}.$$

Every element of D belongs to both C and E , and D is not equal to either set. Hence,

$$D \subsetneq C \quad \text{and} \quad D \subsetneq E.$$

The sets C and E contain exactly the same elements, despite their different ordering. Therefore $C = E$, so $C \subseteq E$ and $E \subseteq C$, but neither is a proper subset of the other. Finally, $C \not\subseteq D$ because $2 \in C$ but $2 \notin D$.

Definition 1.5: Union and intersection

For sets $A, B \subseteq \mathbb{R}$, their *union* is

$$A \cup B = \{x \in \mathbb{R} : x \in A \text{ or } x \in B\},$$

and their *intersection* is

$$A \cap B = \{x \in \mathbb{R} : x \in A \text{ and } x \in B\}.$$

Here “or” is inclusive: an element that belongs to both sets is included in their union.

The union contains all elements appearing in either set, whereas the intersection contains only their common elements. Sets A and B are called *disjoint* if they have no common elements; equivalently, $A \cap B = \emptyset$. These relationships are illustrated in figure 1.3.

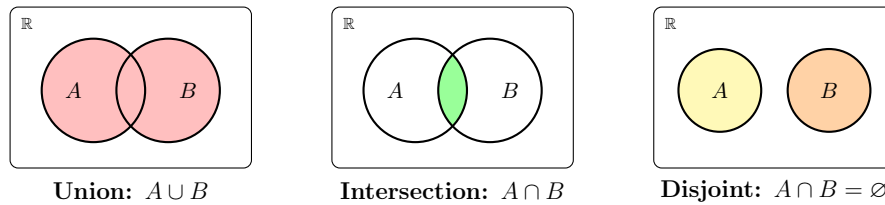


Figure 1.3: The union, intersection, and disjointness of two sets.

Set operations can be applied directly to intervals. Let

$$A = [-2, 3) \quad \text{and} \quad B = (1, 5].$$

The numbers common to both intervals are greater than 1 and less than 3, so

$$A \cap B = (1, 3).$$

Together the intervals cover every real number from -2 to 5 , including both outer endpoints. Therefore,

$$A \cup B = [-2, 5].$$

As an important example of disjoint sets, the rational and irrational numbers have no elements in common, but together they form the entire set of real numbers:

$$\mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset, \quad \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R}.$$

1.2 Numbers in quantitative applications

The numerical value of an observation cannot be separated from its meaning. For example, the number 12 might represent twelve respondents, an income of RM 12 thousand, an unemployment rate of 12%, or an index value. These quantities cannot be combined sensibly unless their definitions and units are compatible. Careful quantitative work therefore begins by identifying what each number represents and how it was obtained.

1.2.1 Discrete and continuous quantities

A *discrete quantity* takes separate, countable values. The number of households in a sample, the number of votes received and the number of hospital visits are discrete because intermediate values such as 42.6 households are not possible as actual counts. Discrete quantities are often represented by whole numbers, although their averages need not be whole. If three households own 1, 2 and 4 vehicles, the mean is $7/3$ vehicles per household.

A *continuous quantity* can, in principle, take any value in an interval. Time, distance, income measured to arbitrary precision and a proportion represented on the interval $[0, 1]$ are treated as continuous in many models. Recorded data may look discrete because instruments or surveys round the observations. A reported journey time of 24 minutes, for example, may represent a value rounded to the nearest minute rather than an exact duration.

The distinction affects the domain of a model. If n counts respondents, then $n \in \mathbb{W}$ is appropriate. If p is a population proportion, then

$$0 \leq p \leq 1,$$

or equivalently $0\% \leq 100p\% \leq 100\%$. A formula may be algebraically defined outside this range while producing values that have no sensible interpretation in the stated context.

1.2.2 Exact values, approximations and rounding

An *exact value* contains no rounding error. Fractions, surds and constants such as π can express exact values:

$$\frac{2}{3}, \quad 5\sqrt{2}, \quad \frac{3\pi}{4}.$$

The decimal 0.67, by contrast, is only an approximation to $2/3$. It is usually best to retain exact values during a calculation and round once, at the end, to the accuracy appropriate for the data.

If a value is rounded to the nearest unit u , the maximum rounding error is $u/2$. Thus, a reported sample size is normally exact, but an income reported as RM 3200 to the nearest RM 100 represents an underlying value in

$$[3150, 3250).$$

The lower endpoint is included because RM 3150 rounds to RM 3200, while RM 3250 rounds upward under the usual convention.

For an exact value x and an approximation $\tilde{x}$, the *absolute error* is

$$|\tilde{x} - x|,$$

and, when $x \neq 0$, the *relative error* is

$$\frac{|\tilde{x} - x|}{|x|}.$$

Multiplying the relative error by 100% gives the percentage error. The same absolute error is more serious when the quantity being measured is small, which is why relative error is useful when values of different sizes are compared.

Example 1.4: [

blue]Rounding a survey proportion In a survey, 317 of 500 respondents support a proposal. The exact sample proportion is

$$\hat{p} = \frac{317}{500} = 0.634.$$

Reported to the nearest percentage point, this is 63%. The rounded figure is convenient for communication, but the value 0.634 should be retained in further calculations because prematurely replacing it by 0.63 will lead to information loss.

The number of displayed decimal places should not suggest more certainty than the data provide. A model prediction of 47.382916% is not genuinely accurate to six decimal places merely because a calculator displays that many digits. Sampling variation, measurement error and model assumptions may be much more important than the final rounding error.

1.2.3 Ratios, proportions and rates

A *ratio* compares two quantities by division. If a survey includes 240 urban respondents and 160 rural respondents, their ratio is

$$240 : 160 = 3 : 2.$$

A *proportion* compares a part with its whole. The urban proportion is

$$\frac{240}{240 + 160} = 0.6 = 60\%.$$

Ratios need not compare a part with a total, so the ratio 3 : 2 should not be misread as $3/2 = 150\%$ of the entire sample.

A *rate* compares quantities with different units. Examples include ringgit per household, births per thousand people, kilometres per hour and survey responses per day. If 84 cases are processed during 6 hours, the average rate is

$$\frac{84 \text{ cases}}{6 \text{ hours}} = 14 \text{ cases per hour}.$$

Writing the units as part of the calculation helps distinguish a rate from a proportion and often reveals an incorrect operation.

Example 1.5: [

cyan]A per-capita comparison Region A provides RM 18 million for a population of 600 000, while Region B provides RM 12 million for a population of 300 000. The

total allocation is larger in Region A, but the per-capita allocations are

$$\frac{18\,000\,000}{600\,000} = \text{RM } 30 \quad \text{and} \quad \frac{12\,000\,000}{300\,000} = \text{RM } 40.$$

Region B therefore has the larger allocation per person. The appropriate comparison depends on whether the question concerns total resources or resources relative to population.

When two ratios are equal, they form a *proportion equation*. For example, if a stratified sample preserves the population ratio of 7 urban residents to 3 rural residents, then a sample of 800 should contain

$$\frac{7}{10}(800) = 560$$

urban residents and 240 rural residents.

1.2.4 Percentages and percentage points

A percentage expresses a proportion out of 100. If a quantity changes from Q_0 to Q_1 , its relative percentage change is

$$\frac{Q_1 - Q_0}{Q_0} \times 100\%, \quad Q_0 \neq 0.$$

A positive result represents an increase and a negative result a decrease. The original value belongs in the denominator because the change is measured relative to the starting point.

Percentage change is different from a change in *percentage points*. If the unemployment rate falls from 8% to 6.5%, the change is

$$6.5\% - 8\% = -1.5 \text{ percentage points},$$

whereas the relative percentage change is

$$\frac{6.5 - 8}{8} \times 100\% = -18.75\%.$$

Both statements are correct, but they answer different questions. In reports about approval ratings, participation rates or market shares, the intended measure should always be stated.

A change of $r\%$ is represented by multiplication, not addition. An increase uses the multiplier

$$1 + \frac{r}{100},$$

while a decrease uses

$$1 - \frac{r}{100}.$$

Successive percentage changes compound. An increase of 10% followed by a decrease of

10% gives the overall multiplier

$$1.10(0.90) = 0.99,$$

so the final value is 1% below the original value, not unchanged.

Example 1.6: [

orange]Nominal and inflation-adjusted change Suppose an annual income rises by 5% while the general price level rises by 3%. The income multiplier is 1.05 and the price multiplier is 1.03. The exact change in purchasing power is therefore

$$\frac{1.05}{1.03} - 1 \approx 0.01942.$$

Thus, real purchasing power rises by approximately 1.94%. Simply subtracting the rates gives 2%, which is a useful approximation for small rates but not the exact result.

These multiplier ideas are developed further in Chapters 14 and 15.

1.2.5 Units and dimensional consistency

Quantities can be added or subtracted only when they have compatible units. It is meaningful to add RM 20 to RM 35, but not to add RM 20 to 35 kilometres. Multiplication and division create derived units. If total cost C is measured in ringgit and output q in units produced, then

$$\frac{C}{q}$$

has units ringgit per unit of output.

An equation should be dimensionally consistent: every term that is added must have the same units. In the model

$$C = 2.50q + 400,$$

where C is in ringgit and q is the number of items, the coefficient 2.50 has units ringgit per item and the intercept 400 has units ringgit. Consequently, both terms on the right represent costs.

Units do not prove that a model is correct, but inconsistent units prove that something is wrong. They also guide interpretation: a slope of 0.18 ringgit per kilowatt-hour describes a rate, while an intercept of RM 24 describes a fixed predicted amount.

1.3 Complex numbers

Some equations have no solution within the real numbers. For example, the equation

$$x^2 = -1$$

cannot be satisfied by any real value of x , because the square of every real number is non-negative. Complex numbers extend the real-number system by introducing the imaginary unit i . This extension allows such equations to be solved while preserving the familiar rules of algebra. Complex numbers are also useful in fields such as economics, engineering, quantum mechanics, signal processing, and the study of oscillations.

Definition 1.6: Imaginary unit

The *imaginary unit*, denoted by i , is defined by

$$i^2 = -1.$$

Equivalently, $i = \sqrt{-1}$, provided that the square root is interpreted within the complex-number system.

Definition 1.7: Complex number

A *complex number* is a number that can be written in the form

$$z = a + bi,$$

where $a, b \in \mathbb{R}$ and $i^2 = -1$. The set of complex numbers is denoted by

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

The expression $a + bi$ is called the *Cartesian form*, or *standard form*, of a complex number.

For $z = a + bi$, the real number a is the *real part* of z , and the real number b is the *imaginary part* of z . They are denoted by

$$\operatorname{Re}(z) = a \quad \text{and} \quad \operatorname{Im}(z) = b.$$

Notice that the imaginary part is b , not bi . For example, if

$$z = -3 + 5i,$$

then $\operatorname{Re}(z) = -3$ and $\operatorname{Im}(z) = 5$.

A real number is also a complex number: if $b = 0$, then $z = a + 0i = a$. Consequently,

$$\mathbb{R} \subset \mathbb{C}.$$

If $a = 0$ and $b \neq 0$, then $z = bi$ is called a *purely imaginary number*. For instance, $-7i$ is purely imaginary, while $4 - 7i$ has both a real part and an imaginary part. The number 0 is real and complex, but it is not usually classified as purely imaginary.

Two complex numbers are equal precisely when their corresponding real and imaginary parts are equal. Thus,

$$a + bi = c + di \iff a = c \text{ and } b = d.$$

This principle allows equations involving complex numbers to be solved by comparing their real and imaginary parts.

Example 1.7: Identifying the parts of a complex number

For each complex number, identify its real and imaginary parts.

z	$\operatorname{Re}(z)$	$\operatorname{Im}(z)$
$6 - 2i$	6	-2
$-4i$	0	-4
9	9	0

In particular, the real number 9 may be written as $9 + 0i$, while the purely imaginary number $-4i$ may be written as $0 - 4i$.

1.3.1 Powers of i

The defining relation $i^2 = -1$ can be used to simplify every positive integer power of i . The first four powers are

$$i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 i = -i, \quad i^4 = i^2 i^2 = 1.$$

Multiplying by i once more begins the same pattern again:

$$i^5 = i, \quad i^6 = -1, \quad i^7 = -i, \quad i^8 = 1.$$

The powers therefore repeat in a cycle of length 4:

$$i, -1, -i, 1, i, -1, -i, 1, \dots$$

For any non-negative integer n , divide n by 4 and use the remainder. If $n = 4q + r$, where $r \in \{0, 1, 2, 3\}$, then

$$i^n = i^{4q+r} = (i^4)^q i^r = i^r.$$

Hence,

$n \bmod 4$	0	1	2	3
i^n	1	i	-1	$-i$

Here $n \bmod 4$ means the remainder when n is divided by 4. For $n = 0$, the usual convention $i^0 = 1$ applies.

Example 1.8: Simplifying a power of i

Simplify i^{47} .

Dividing 47 by 4 gives

$$47 = 4(11) + 3.$$

Therefore,

$$i^{47} = (i^4)^{11}i^3 = 1^{11}(-i) = -i.$$

The same method can simplify expressions containing several powers of i .

Example 1.9: Combining powers of i

Simplify $3i^{18} - 2i^{25} + i^{32}$.

The remainders upon division by 4 are 2, 1, and 0, respectively. Thus,

$$\begin{aligned} 3i^{18} - 2i^{25} + i^{32} &= 3i^2 - 2i^1 + i^0 \\ &= 3(-1) - 2i + 1 \\ &= -2 - 2i. \end{aligned}$$

The final answer is written in the standard form $a + bi$.

1.3.2 Venn diagram representation

The real numbers form a proper subset of the complex numbers because every real number a can be written as $a + 0i$. To distinguish the different regions of the complex-number system, let

$$\mathcal{I} = \{bi : b \in \mathbb{R}, b \neq 0\}$$

denote the set of purely imaginary numbers. With the convention that 0 is not purely imaginary, $\mathbb{R}$ and $\mathcal{I}$ are disjoint subsets of $\mathbb{C}$:

$$\mathbb{R} \subsetneq \mathbb{C}, \quad \mathcal{I} \subsetneq \mathbb{C}, \quad \mathbb{R} \cap \mathcal{I} = \emptyset.$$

These two subsets do not exhaust $\mathbb{C}$. A number $a + bi$ with both $a \neq 0$ and $b \neq 0$ is neither real nor purely imaginary; it lies in the part of $\mathbb{C}$ outside both inner regions of figure 1.4.

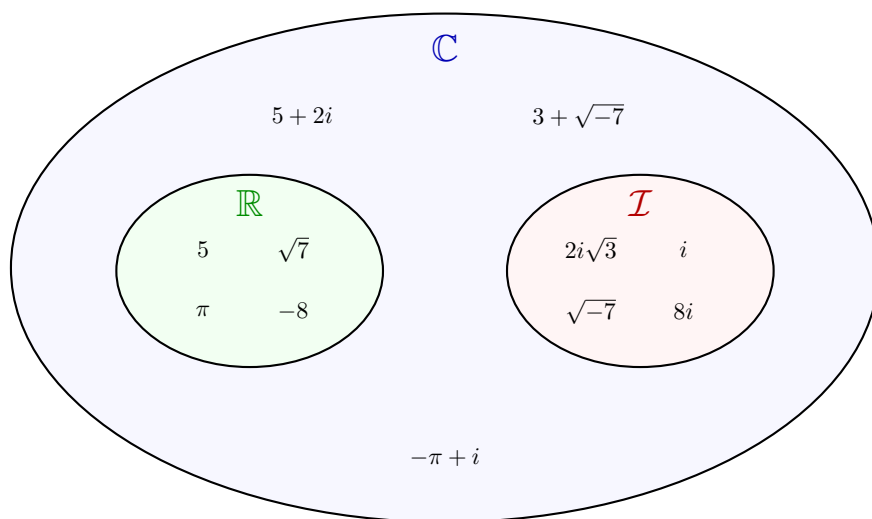


Figure 1.4: Examples of real, purely imaginary, and other complex numbers.

For example, $\sqrt{-7} = i\sqrt{7}$ is purely imaginary, whereas $3 + \sqrt{-7} = 3 + i\sqrt{7}$ has a non-zero real part and a non-zero imaginary part. The same distinction can be made directly from Cartesian form:

$$a + bi \in \begin{cases} \mathbb{R}, & b = 0, \\ \mathcal{I}, & a = 0 \text{ and } b \neq 0, \\ \mathbb{C} \setminus (\mathbb{R} \cup \mathcal{I}), & a \neq 0 \text{ and } b \neq 0. \end{cases}$$

The number sets introduced earlier remain nested inside the real-number region of $\mathbb{C}$. The rational and irrational numbers partition $\mathbb{R}$, while the natural, whole, and integer sets are nested within $\mathbb{Q}$. figure 1.5 combines these relationships in one diagram.

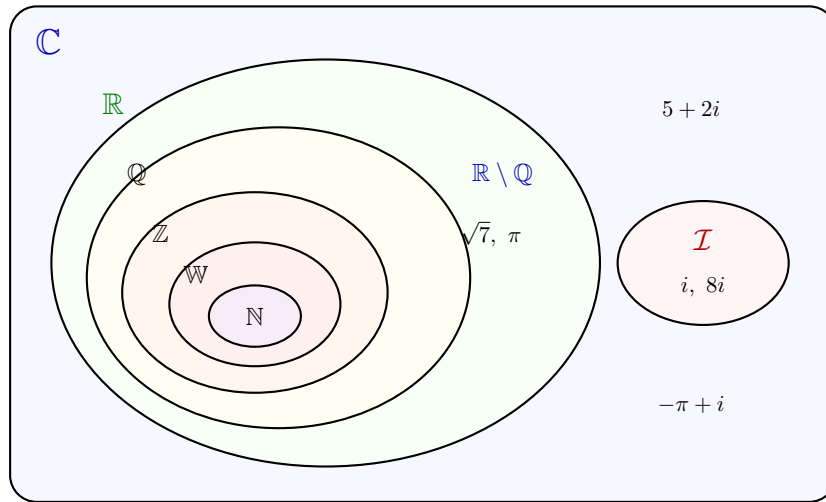


Figure 1.5: The hierarchy of real and complex number sets.

In summary,

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C},$$

and

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}), \quad \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset.$$

1.3.3 Quadratic equations with complex roots

Consider a quadratic equation with real coefficients,

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

Its solutions are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The expression

$$\Delta = b^2 - 4ac$$

is called the *discriminant* where it determines the nature of the roots of the quadratic equation:

Value of Δ	Nature of the roots
$\Delta > 0$	two distinct real roots
$\Delta = 0$	one repeated real root
$\Delta < 0$	two non-real complex roots

When $\Delta < 0$, the square root in the quadratic formula contains a negative number. Since $-\Delta = 4ac - b^2 > 0$,

$$\sqrt{\Delta} = \sqrt{-(4ac - b^2)} = i\sqrt{4ac - b^2}.$$

Hence the roots may be written as

$$x = \frac{-b}{2a} \pm \frac{\sqrt{4ac - b^2}}{2a}i.$$

They have the same real part and imaginary parts of opposite signs. Thus, if one root is

$$\alpha = u + vi,$$

then the other root is

$$\beta = u - vi.$$

The two roots are called a *complex-conjugate pair*. A quadratic equation with real coefficients cannot have only one non-real root: non-real roots always occur with their conjugates.

The discriminant also has a graphical interpretation. The real solutions of $ax^2 + bx + c = 0$ are the x -coordinates at which the graph of $y = ax^2 + bx + c$ meets the x -axis. Therefore, when $\Delta < 0$, the parabola has no x -intercepts. If $a > 0$, it lies above the x -axis; if $a < 0$, it lies below the x -axis.

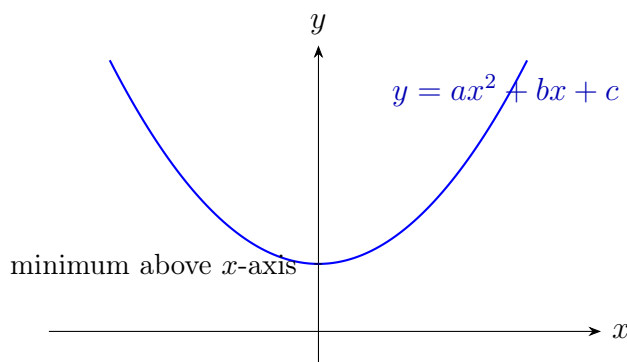


Figure 1.6: A quadratic graph with a negative discriminant and $a > 0$.

Example 1.10: Solving a quadratic equation with complex roots

Solve

$$x^2 + 54 = 0.$$

Here $a = 1$, $b = 0$, and $c = 54$, so

$$\Delta = b^2 - 4ac = 0^2 - 4(1)(54) = -216 < 0.$$

The equation may be solved directly:

$$\begin{aligned} x^2 &= -54, \\ x &= \pm\sqrt{-54}, \\ x &= \pm i\sqrt{54}, \\ x &= \pm 3\sqrt{6}i. \end{aligned}$$

Therefore, the roots are

$$\boxed{x = 3\sqrt{6}i \quad \text{or} \quad x = -3\sqrt{6}i},$$

which form a complex-conjugate pair.

1.3.4 Arithmetic operations with complex numbers

Complex numbers are added, subtracted and multiplied by collecting their real and imaginary parts. Division uses the conjugate of the denominator.

Addition of complex numbers

Add the real parts and the imaginary parts separately. Thus,

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

For example, if $z_1 = 1 + 2i$ and $z_2 = 9 + 2i$, then

$$\begin{aligned} (1 + 2i) + (9 + 2i) &= (1 + 9) + (2 + 2)i \\ &= 10 + 4i \end{aligned}$$

Subtraction of complex numbers

Subtract the real parts and the imaginary parts separately:

$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

The standard form remains $a + bi$, although the coefficient b may be negative. The conjugate of $a + bi$ is the distinct number $a - bi$.

For example, if $z_1 = 3 + 9i$ and $z_2 = 4 + 8i$, then

$$\begin{aligned}(3 + 9i) - (4 + 8i) &= (3 - 4) + (9 - 8)i \\ &= -1 + i\end{aligned}$$

Multiplication of complex numbers

Use the distributive law and replace i^2 by -1 . For ordinary algebra,

$$(a + b)^2 = a^2 + 2ab + b^2.$$

For complex numbers $a + bi$ and $c + di$, this gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

For example,

$$\begin{aligned}(9 - 5i)(7 + 7i) &= (9 \cdot 7) + (9 \cdot 7i) + (-5i \cdot 7) + (-5i \cdot 7i) \\ &= 63 + 63i - 35i - 35(i^2) \\ &= 63 + 28i + 35 \\ &= 98 + 28i\end{aligned}$$

Division of complex numbers

To divide by a non-zero complex number, multiply the numerator and denominator by the conjugate of the denominator. This is analogous to rationalising a denominator containing a surd. For example,

$$\frac{a + b\sqrt{x}}{c + d\sqrt{x}} = \frac{a + b\sqrt{x}}{c + d\sqrt{x}} \cdot \frac{c - d\sqrt{x}}{c - d\sqrt{x}}.$$

Similarly,

$$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di}.$$

Since $(c + di)(c - di) = c^2 + d^2$, the quotient can be written in standard complex-number form as

$$\frac{a + bi}{c + di} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i, \quad c^2 + d^2 \neq 0.$$

Thus, multiplying by the conjugate removes the imaginary term from the denominator.

1.3.5 Argand diagrams

An *Argand diagram* represents a complex number $z = a + bi$ as the point (a, b) in a coordinate plane. The horizontal axis shows the real part and the vertical axis shows the

imaginary part. The line from the origin to the point represents the complex number geometrically.

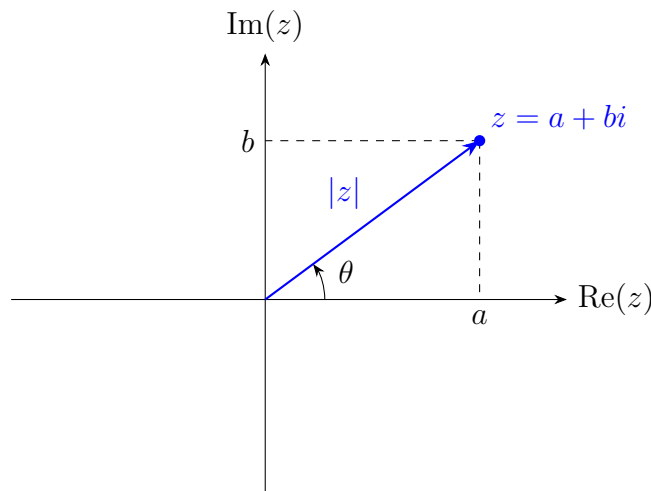


Figure 1.7: The complex number $z = a + bi$ on an Argand diagram.

The distance of the point from the origin is the *modulus*, while the angle measured anticlockwise from the positive real axis is the *argument*. Thus,

$$|z| = \sqrt{a^2 + b^2}, \quad \theta = \arg(z), \quad z = |z|(\cos \theta + i \sin \theta).$$

For most introductory social-science calculations, real numbers are sufficient. Complex numbers are included because they complete an important algebraic extension and explain how polynomial equations can retain their full set of roots. In more advanced quantitative work, they also provide an efficient language for representing cycles and oscillations. Frequency-based methods used to study seasonal economic data, repeated survey patterns and time series often perform calculations with complex numbers internally, even when the observations and final interpretations are real-valued.

The Cartesian form $a + bi$ displays two perpendicular components. Modulus and argument instead separate magnitude from direction or phase. This is similar in spirit to distinguishing the size of a periodic effect from its position within a cycle. The present chapter develops only the algebra and geometry needed as a foundation; no complex-valued social-science model is assumed in later elementary applications.

1.4 Indices, surds and logarithms

Indices, surds and logarithms provide compact ways to work with powers, roots and exponential relationships. Their laws are closely connected and are useful when simplifying expressions and solving equations.

1.4.1 Indices

An *index*, or exponent, shows how many times a base is used as a factor. For example, $a^4 = a \cdot a \cdot a \cdot a$. For non-zero bases, the main laws of indices are

$$\begin{aligned}
a^m a^n &= a^{m+n}, & \frac{a^m}{a^n} &= a^{m-n}, \\
(a^m)^n &= a^{mn}, & (ab)^n &= a^n b^n, \\
a^0 &= 1, & a^{-n} &= \frac{1}{a^n}.
\end{aligned}$$

Fractional indices represent roots:

$$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m.$$

For instance,

$$16^{3/4} = (\sqrt[4]{16})^3 = 2^3 = 8, \quad 5^{-2} = \frac{1}{25}.$$

The index laws require attention to their domains. A negative base raised to an integer power is real, but a fractional power may not be real. For example, $(-8)^3$ is real and $\sqrt[3]{-8} = -2$, whereas $(-8)^{1/2}$ is not a real number. Similarly, the expression a^{-n} requires $a \neq 0$ because it contains division by a^n .

Scientific notation and orders of magnitude

Scientific notation writes a non-zero number in the form

$$A \times 10^n, \quad 1 \leq |A| < 10,$$

where n is an integer. It is useful when values are very large or very small and makes their scale easier to compare. For example,

$$4\,750\,000 = 4.75 \times 10^6, \quad 0.000032 = 3.2 \times 10^{-5}.$$

The exponent records the order of magnitude. A value near 10^7 is roughly ten times the scale of a value near 10^6 , although the leading coefficient must be considered for an exact comparison.

The index laws make arithmetic in scientific notation systematic:

$$(3 \times 10^5)(4 \times 10^3) = 12 \times 10^8 = 1.2 \times 10^9.$$

For division,

$$\frac{7.2 \times 10^8}{3 \times 10^4} = 2.4 \times 10^4.$$

Before adding or subtracting, the powers of 10 should be made equal. Thus,

$$2.6 \times 10^6 + 7.5 \times 10^5 = 2.6 \times 10^6 + 0.75 \times 10^6 = 3.35 \times 10^6.$$

In social-science data, scientific notation may be useful for national budgets, population totals, very small probabilities or computer data sets containing millions of records. It

should not hide the units: the statements 2.5×10^6 people and RM 2.5×10^6 describe quantities of the same numerical size but entirely different meanings.

Example 1.11: [

violet]Comparing quantities on different scales A programme costs RM 3.6×10^7 and serves 1.2×10^6 people. Its average cost per person is

$$\frac{3.6 \times 10^7}{1.2 \times 10^6} = 3 \times 10^1 = \boxed{\text{RM } 30}.$$

The powers of ten show immediately that dividing a quantity of order 10^7 by one of order 10^6 should produce a value of order 10^1 .

Exponential multipliers

Repeated proportional change is represented naturally by powers. If a quantity begins at Q_0 and changes by the same proportion r in each time period, then

$$Q_n = Q_0(1 + r)^n.$$

Here r is written as a decimal: a growth rate of 4% gives $r = 0.04$ and growth factor 1.04, while a decline of 4% gives $r = -0.04$ and factor 0.96.

The exponent n counts the number of completed periods. This distinction is important when the initial observation occurs at time 0. For instance, an index that begins at 100 and grows by 3% annually has values

$$100, \quad 100(1.03), \quad 100(1.03)^2, \dots$$

at times 0, 1, 2, ... The value after five completed years is $100(1.03)^5$, not $100(1.03)^4$.

Example 1.12: [

blue]Repeated growth in participation A community programme has 2400 participants and participation is modelled as increasing by 6% each year. After n years, the model is

$$P_n = 2400(1.06)^n.$$

After four years,

$$P_4 = 2400(1.06)^4 \approx \boxed{3030}.$$

Because participants are counted, the final prediction is rounded to a whole number. The formula assumes a constant proportional rate; it does not account for capacity limits, policy changes or differences between population groups.

1.4.2 Surds

A *surd* is an irrational root that is left in exact form, such as $\sqrt{2}$ or $3\sqrt{5}$. For non-negative a and b ,

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad (b > 0).$$

More generally, if m and n are positive integers, the main properties of n -th roots are

$$\begin{aligned} \sqrt[n]{ab} &= \sqrt[n]{a} \sqrt[n]{b}, \\ \sqrt[n]{\frac{a}{b}} &= \frac{\sqrt[n]{a}}{\sqrt[n]{b}}, \quad b \neq 0, \\ \sqrt[m]{\sqrt[n]{a}} &= \sqrt[mn]{a}, \\ (\sqrt[n]{a})^n &= a, \\ \sqrt[n]{a^n} &= \begin{cases} |a|, & \text{if } n \text{ is even,} \\ a, & \text{if } n \text{ is odd.} \end{cases} \end{aligned}$$

For an even index, the radicand must be non-negative when working with real numbers; an odd-indexed root is defined for every real radicand. Roots do not distribute over addition or subtraction: in general,

$$\sqrt[n]{a+b} \neq \sqrt[n]{a} + \sqrt[n]{b}.$$

A surd is simplified by extracting perfect-square factors. Like surds may then be collected in the same way as like algebraic terms. For example,

$$\sqrt{72} - \sqrt{8} = 6\sqrt{2} - 2\sqrt{2} = 4\sqrt{2}.$$

Keeping a surd in exact form avoids cumulative rounding error. If $\sqrt{2}$ is replaced prematurely by 1.41, later products and differences may be noticeably distorted. Exact forms are especially useful in algebraic derivations; a decimal approximation can be supplied at the final stage when the context requires a measured or reported value.

Rationalising surds and rationalising factors

A *rationalising factor* of a surd is a factor that produces a rational number when multiplied by that surd. For example, $\sqrt{5}$ is a rationalising factor of $\sqrt{5}$ because

$$\sqrt{5} \cdot \sqrt{5} = 5.$$

More generally, for $0 < m < n$ and $a > 0$,

$$\sqrt[n]{a^m} \cdot \sqrt[n]{a^{n-m}} = a.$$

Thus, $\sqrt[3]{4}$ is a rationalising factor of $\sqrt[3]{2}$.

To *rationalise* a denominator containing a single square-root surd, multiply the

numerator and denominator by that surd:

$$\frac{3}{\sqrt{5}} = \frac{3}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{5}}{5}.$$

For higher roots, multiply by the rationalising factor needed to form a perfect n -th power. For example,

$$\frac{5}{\sqrt[3]{2}} = \frac{5}{\sqrt[3]{2}} \cdot \frac{\sqrt[3]{4}}{\sqrt[3]{4}} = \frac{5\sqrt[3]{4}}{2}.$$

For a denominator of the form $a + b\sqrt{c}$, its rationalising factor is the conjugate $a - b\sqrt{c}$, since

$$(a + b\sqrt{c})(a - b\sqrt{c}) = a^2 - b^2c,$$

which is rational.

Only like surds can be added or subtracted directly. Surds may also be multiplied by expanding brackets and then simplifying the resulting roots.

Example 1.13: Simplifying expressions containing surds

For example,

$$\begin{aligned} 3\sqrt{8} + 2\sqrt{18} - \sqrt{50} &= 6\sqrt{2} + 6\sqrt{2} - 5\sqrt{2} \\ &= \boxed{7\sqrt{2}}. \end{aligned}$$

Similarly,

$$\begin{aligned} (\sqrt{5} + 2)(\sqrt{5} - 1) &= 5 - \sqrt{5} + 2\sqrt{5} - 2 \\ &= \boxed{3 + \sqrt{5}}. \end{aligned}$$

The expressions $a + b\sqrt{c}$ and $a - b\sqrt{c}$ are conjugates, and their product is rational:

$$(a + b\sqrt{c})(a - b\sqrt{c}) = a^2 - b^2c.$$

For instance,

$$\begin{aligned} \frac{1}{2 + \sqrt{3}} &= \frac{1}{2 + \sqrt{3}} \cdot \frac{2 - \sqrt{3}}{2 - \sqrt{3}} \\ &= \frac{2 - \sqrt{3}}{4 - 3} = \boxed{2 - \sqrt{3}}. \end{aligned}$$

1.4.3 Logarithms

A logarithm is the inverse of an exponential expression. If $a > 0$, $a \neq 1$ and $x > 0$, then

$$a^y = x \iff \log_a x = y.$$

Thus, $\log_2 32 = 5$ because $2^5 = 32$. For positive M and N , the main laws of logarithms are

$$\begin{aligned}\log_a(MN) &= \log_a M + \log_a N, \\ \log_a\left(\frac{M}{N}\right) &= \log_a M - \log_a N, \\ \log_a(M^r) &= r \log_a M.\end{aligned}$$

Also,

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \log_a x = \frac{\log_b x}{\log_b a},$$

where the final identity is the change-of-base formula.

Since logarithmic and exponential functions are inverses,

$$\log_a(a^x) = x, \quad a^{\log_a x} = x,$$

where $x > 0$ is required in the second identity. Also, because $\log_a x$ is one-to-one,

$$\log_a M = \log_a N \iff M = N, \quad M, N > 0.$$

Two bases are used particularly often. The *common logarithm*, written $\log x$, has base 10, while the *natural logarithm*, written $\ln x$, has base e :

$$\log x = \log_{10} x, \quad \ln x = \log_e x.$$

The change-of-base formula allows any logarithm to be evaluated using either of these forms; for example,

$$\log_2 7 = \frac{\ln 7}{\ln 2} = \frac{\log 7}{\log 2}.$$

Example 1.14: Evaluating a logarithm exactly

To evaluate $\log_4 8$, let $y = \log_4 8$. Then

$$4^y = 8 \implies (2^2)^y = 2^3 \implies 2y = 3.$$

Therefore,

$$\boxed{\log_4 8 = \frac{3}{2}}.$$

Logarithms in growth models

Logarithms are used when the unknown occurs in an exponent. If

$$Q_n = Q_0 b^n,$$

where $Q_0 > 0$, $Q_n > 0$, $b > 0$ and $b \neq 1$, then

$$n = \frac{\log(Q_n/Q_0)}{\log b}.$$

The same result is obtained using natural logarithms because the change-of-base formula makes the ratio independent of the logarithm base.

Example 1.15: [

orange]Finding when a target is reached The number of users of an online public service is modelled by

$$U_n = 1200(1.06)^n.$$

To find when the model first reaches 2000 users, solve

$$2000 = 1200(1.06)^n.$$

Taking logarithms gives

$$n = \frac{\ln(2000/1200)}{\ln(1.06)} \approx 8.77.$$

The continuous solution reaches the target after about 8.77 years. If the model is evaluated only at the end of complete years, the first value at least 2000 occurs after 9 years.

For a continuous exponential model

$$Q(t) = Q_0 e^{kt},$$

the parameter k is a continuous proportional growth rate. Solving for time gives

$$t = \frac{1}{k} \ln \left(\frac{Q(t)}{Q_0} \right), \quad k \neq 0.$$

If $k > 0$, the doubling time is $\ln 2/k$; if $k < 0$, the half-life is $\ln 2/|k|$. These formulas reappear in the chapters on financial mathematics and differential equations.

Logarithmic scales and transformed data

A logarithm converts multiplication into addition and powers into coefficients:

$$\ln(AB) = \ln A + \ln B, \quad \ln(A^r) = r \ln A.$$

This makes logarithms useful when values extend across several orders of magnitude or when relative differences are more meaningful than absolute differences. On a base-10 logarithmic scale, increasing a value by one unit corresponds to multiplying the original quantity by 10.

Log transformations are also used with positive, strongly right-skewed data such as income, wealth, population size or expenditure. If two incomes are Y_1 and Y_2 , then

$$\ln Y_2 - \ln Y_1 = \ln \left(\frac{Y_2}{Y_1} \right).$$

The difference therefore describes a ratio rather than a difference in currency units. For example, if $Y_2 = 2Y_1$, then the log difference is $\ln 2$, regardless of whether the incomes are RM 1000 and RM 2000 or RM 5000 and RM 10 000.

Example 1.16: [

Interpreting a difference on a log scale Suppose two regions have average positive household expenditures E_A and E_B , and

$$\ln E_B - \ln E_A = 0.20.$$

Then

$$\frac{E_B}{E_A} = e^{0.20} \approx 1.2214.$$

Thus, Region B's average expenditure is approximately 22.14% higher than Region A's. Treating 0.20 as an absolute difference of RM 0.20 would be incorrect.

A logarithm is defined only for a positive argument. Zero or negative observations therefore require special consideration rather than automatic transformation. Moreover, a log transformation can make a pattern easier to model but does not remove poor data, omitted variables or an unsuitable research design.

1.4.4 Problem solving involving indices, surds and logarithms

The laws above are often used together with algebraic techniques. When solving an equation, it is important to preserve exact values and check any restrictions on roots, denominators and logarithms.

Example 1.17: Solving an equation involving indices

Solve

$$2^{2x} - 5(2^x) + 4 = 0.$$

Let $y = 2^x$. The equation becomes

$$y^2 - 5y + 4 = 0 \implies (y - 1)(y - 4) = 0.$$

Hence $2^x = 1$ or $2^x = 4$, giving

$$\boxed{x = 0 \quad \text{or} \quad x = 2}.$$

Example 1.18: Rationalising a surd denominator

Simplify

$$\frac{4}{3 - \sqrt{5}}.$$

Multiplying by the conjugate of the denominator gives

$$\begin{aligned}\frac{4}{3 - \sqrt{5}} &= \frac{4}{3 - \sqrt{5}} \cdot \frac{3 + \sqrt{5}}{3 + \sqrt{5}} \\ &= \frac{4(3 + \sqrt{5})}{9 - 5} = \boxed{3 + \sqrt{5}}.\end{aligned}$$

Example 1.19: Solving an equation involving a surd

Solve

$$\sqrt{x + 5} - \sqrt{x} = 1.$$

The domain requires $x \geq 0$. Isolating one radical and squaring gives

$$\begin{aligned}\sqrt{x + 5} &= \sqrt{x} + 1, \\ x + 5 &= x + 2\sqrt{x} + 1, \\ \sqrt{x} &= 2, \\ x &= 4.\end{aligned}$$

Squaring can introduce extraneous solutions, so the result must be checked:

$$\sqrt{4 + 5} - \sqrt{4} = 3 - 2 = 1.$$

Therefore,

$$\boxed{x = 4}.$$

Example 1.20: Solving a logarithmic equation

Solve

$$\log_3(x - 1) + \log_3(x + 1) = 2.$$

The logarithms require $x > 1$. Combining them gives

$$\begin{aligned}\log_3((x - 1)(x + 1)) &= 2, \\ x^2 - 1 &= 3^2, \\ x^2 &= 10.\end{aligned}$$

Only the positive solution satisfies $x > 1$, so

$$\boxed{x = \sqrt{10}}.$$

1.5 Exercises

Answers to these exercises are provided in section C.1.

The questions are arranged by topic. Work through the foundational questions first, then use the later groups for consolidation and mixed practice.

A. Real numbers, intervals and inequalities

1. State the smallest familiar number set to which each number belongs: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R} \setminus \mathbb{Q}$.

- | | |
|-----------------------|------------------------------|
| (a) -7 | (k) 32 |
| (b) 0.125 | (l) 0.542 |
| (c) $\sqrt{18}$ | (m) $\sqrt{11}$ |
| (d) $\sqrt{81}$ | (n) $\sqrt{90}$ |
| (e) $0.\overline{27}$ | (o) $0.\overline{77}$ |
| (f) 3 | (p) 0 |
| (g) π | (q) $-\sqrt{49}$ |
| (h) e^2 | (r) $\sqrt[3]{8}$ |
| (i) $\sqrt{6}$ | (s) $0.101001000100001\dots$ |
| (j) $0.\overline{23}$ | (t) -2.75 |

2. Express these numbers as a fraction in simplest form.

- | | |
|-----------------------|------------------------|
| (a) $0.\overline{36}$ | (i) $0.\overline{65}$ |
| (b) $0.\overline{33}$ | (j) $0.\overline{98}$ |
| (c) $0.\overline{55}$ | (k) $0.\overline{12}$ |
| (d) $0.\overline{76}$ | (l) $0.\overline{45}$ |
| (e) $0.\overline{27}$ | (m) $0.\overline{16}$ |
| (f) $0.\overline{99}$ | (n) $2.\overline{3}$ |
| (g) $0.\overline{11}$ | (o) $0.12\overline{3}$ |
| (h) $0.\overline{32}$ | |

3. Find the intersection and union of each pair of intervals.

- (a) $A = [-3, 4]$ and $B = (1, 6]$
(b) $C = (-\infty, 2]$ and $D = [-1, 5)$
(c) $E = (-4, 1)$ and $F = [1, 7]$

4. Solve each inequality and express the solution in interval notation.

- (a) $|2x - 5| \leq 7$

(b) $|3x + 1| < 5$

(c) $|x - 4| \geq 3$

B. Introductory complex numbers

5. Simplify.

(a) i^{37}

(b) i^{62}

(c) i^{-15}

(d) i^{2026}

(e) $1 + i + i^2 + i^3$

6. Write each result in the form $a + bi$.

(a) $(3 - 2i) + (5 + 7i)$

(b) $(2 + 3i)(1 - 4i)$

(c) $\frac{4 + 2i}{1 - i}$

(d) $(4 - 3i) - (1 + 5i)$

(e) $(2 - i)^2$

(f) $\frac{5 + i}{2 - i}$

(g) $(1 + 2i)(3 - i)$

7. Solve over the complex numbers.

(a) $x^2 - 6x + 13 = 0$

(b) $x^2 + 4x + 13 = 0$

(c) $2x^2 - 4x + 10 = 0$

C. Indices, surds, logarithms and growth

8. Simplify, giving answers with positive indices. Assume all variables are non-zero.

(a) $\frac{a^5 a^{-2}}{a}$

(b) $\frac{x^{-3} x^7}{x^2}$

(c) $\frac{(2m^3 n^{-2})^2}{4m n^{-1}}$

(d) $(p^{-2} q^3)^{-1}$

9. Simplify.

(a) $3\sqrt{8} + \sqrt{18} - \sqrt{50}$

- (b) $5\sqrt{12} - 2\sqrt{27} + \sqrt{75}$
- (c) $2\sqrt{45} + \sqrt{20} - 3\sqrt{5}$
- (d) $\sqrt{98} - \sqrt{32} + \sqrt{8}$

10. Rationalise each denominator and simplify.

- (a) $\frac{4}{\sqrt{3} + 1}$
- (b) $\frac{5}{\sqrt{6}}$
- (c) $\frac{3}{2 + \sqrt{5}}$
- (d) $\frac{2 + \sqrt{3}}{2 - \sqrt{3}}$

11. Solve, observing the domain restrictions of the logarithms.

- (a) $\log_2(x - 1) + \log_2(x + 1) = 3$
- (b) $\log_3 x + \log_3(x - 2) = 2$
- (c) $\ln(x + 4) - \ln x = \ln 3$
- (d) $\log_5(2x - 1) = 2$

12. Simplify each surd.

- (a) $\sqrt{54}$
- (b) $2\sqrt{32}$
- (c) $\frac{\sqrt{28}}{6}$
- (d) $\sqrt{\frac{12}{25}}$
- (e) $\sqrt{200}$
- (f) $\sqrt{\frac{45}{8}}$

13. Collect like surds and simplify.

- (a) $7\sqrt{2} - 3\sqrt{2}$
- (b) $9\sqrt{3} + 3\sqrt{6} - 6\sqrt{3} + 2\sqrt{6}$
- (c) $8\sqrt{5} - \sqrt{20}$
- (d) $5\sqrt{12} + 4\sqrt{27} - 3\sqrt{75}$
- (e) $4\sqrt{8} + 3\sqrt{18} - \sqrt{50}$
- (f) $6\sqrt{12} - 2\sqrt{27} + \sqrt{75}$

14. Simplify each product or quotient.

- (a) $\sqrt{6}\sqrt{7}$
- (b) $\sqrt{15} \div \sqrt{3}$
- (c) $3\sqrt{5}\sqrt{10}$
- (d) $\frac{18\sqrt{12}}{6\sqrt{2}}$
- (e) $(2\sqrt{3})(5\sqrt{6})$
- (f) $\frac{\sqrt{48}}{\sqrt{3}}$

15. Expand and simplify.

- (a) $\sqrt{2}(3\sqrt{5} + \sqrt{10})$
- (b) $2\sqrt{6}(4\sqrt{3} - 3\sqrt{6})$
- (c) $(\sqrt{7} - 3)(4 + \sqrt{7})$
- (d) $(4\sqrt{5} - 1)(2\sqrt{5} + 3)$
- (e) $(2 + \sqrt{3})(4 - \sqrt{3})$
- (f) $\sqrt{5}(\sqrt{20} - 2\sqrt{45})$

16. Expand and simplify.

- (a) $(4 + \sqrt{11})^2$
- (b) $(3\sqrt{5} - 2\sqrt{3})^2$
- (c) $(6 - \sqrt{3})(6 + \sqrt{3})$
- (d) $(3\sqrt{2} + \sqrt{5})(3\sqrt{2} - \sqrt{5})$
- (e) $(5 + 2\sqrt{6})^2$
- (f) $(2\sqrt{7} - 3\sqrt{2})(2\sqrt{7} + 3\sqrt{2})$

17. Rationalise each denominator and simplify.

- (a) $\frac{3}{\sqrt{2}}$
- (b) $\frac{2\sqrt{3}}{\sqrt{5}}$
- (c) $\frac{3\sqrt{5}}{4\sqrt{6}}$
- (d) $\frac{1 - \sqrt{2}}{\sqrt{10}}$
- (e) $\frac{7}{2\sqrt{3}}$
- (f) $\frac{3}{\sqrt{5} - \sqrt{2}}$

18. Simplify, giving answers with positive indices. Assume that all variables are non-zero.

- (a) $m^6 \div m^2$
- (b) $2s^3t^2 \times 4s^2t$
- (c) $(a^2)^5$
- (d) $(5b^3)^2$
- (e) $\frac{12x^5y^3}{3x^2y}$
- (f) $\left(\frac{2x^3}{3y}\right)^3$
- (g) $(3a^{-2}b^4)^2$
- (h) $(x^{1/2}y^{-3})^2$

19. Simplify, giving answers with positive indices.

- (a) $7m^0$
- (b) $6x^0 - (4x)^0$
- (c) x^{-3}
- (d) $\frac{8}{y^{-4}}$
- (e) $5a^2b^{-3}$
- (f) $(2x^{-2})^{-1}$
- (g) $3p^{-2}q^0$

20. Simplify, giving answers with positive indices. Assume that all variables are non-zero.

- (a) $\frac{6(mn^{-4})^3}{m^{-2}n} \left(\frac{m^2}{3n^{-3}}\right)^2$
- (b) $\left(\frac{4a^{-3}}{b^{-2}c^4}\right)^2 \div \left(\frac{2a^{-3}b}{c^2}\right)^3$
- (c) $\frac{(x^{-2}y^3)^2}{x^{-5}y}$
- (d) $(4a^2b^{-1})^{-1}(2a^{-3}b^2)$

21. Convert as directed.

- (a) Write 7.905×10^4 as an ordinary number.
- (b) Write 3.8×10^{-5} as an ordinary number.
- (c) Write 5 160 000 in scientific notation.
- (d) Write 0.00402 in scientific notation.
- (e) Write 6.02×10^{-7} as an ordinary number.
- (f) Write 0.00000091 in scientific notation.

22. Convert between surd and index notation.

- (a) Write $\sqrt{5}$ in index form.
- (b) Write $3\sqrt{3}$ in index form as a single power of 3.
- (c) Write $\sqrt[3]{9x^2}$ in index form.
- (d) Write $2\sqrt[5]{x^4}$ in index form.
- (e) Write $7^{1/4}$ in surd form.
- (f) Write $3^{2/5}$ in surd form.
- (g) Write $\sqrt[4]{16x^3}$ in index form and simplify.
- (h) Write $5^{3/2}$ in surd form and simplify.

23. Evaluate or solve.

- (a) $32^{1/5}$
- (b) $27^{-1/3}$
- (c) $2^x = 16$
- (d) $5^{2x+1} = 125^x$
- (e) $81^{3/4}$
- (f) $9^{x-1} = 27$

24. Write an exponential model for each situation, where n is the number of completed time periods.

- (a) An investment of \$400 000 earns 16% interest per year.
- (b) A tank initially contains 800 litres and loses 4% of its contents each hour.
- (c) A programme begins with 15 000 participants and grows by 2.5% each quarter.
- (d) A town has population 80 000 and is modelled as declining by 1.2% each year.

25. A house is purchased for \$600 000 and its value increases by 12% each year.

- (a) Write a formula for its value V after t years.
- (b) Find its value after 2 years and after 10 years.
- (c) After how many years will its value first reach \$1 000 000? Give the time to one decimal place.
- (d) Find the percentage increase in value after 10 years.
- (e) Find the model's doubling time, and state after how many complete years the value first exceeds twice the purchase price.

D. Consolidation and applied problems

- 26.** Express each number in the form p/q , where $p, q \in \mathbb{Z}$ and $q \neq 0$, and hence confirm that it is rational.
- (a) 0
 - (b) -12
 - (c) $7/8$
- 27.** State whether each statement is true or false, giving a brief justification.
- (a) Every natural number is a whole number.
 - (b) Every integer is a whole number.
 - (c) Every rational number is a whole number.
 - (d) Every irrational number is a real number.
 - (e) Every real number is irrational.
 - (f) Every whole number is an integer.
 - (g) $\sqrt{16}$ is irrational.
- 28.** Simplify each set and represent it on a real number line.
- (a) $[-3, 0) \cap (-3, \pi)$
 - (b) $(-1, e]$
 - (c) $(-\infty, 5] \cup [-5, \infty)$
 - (d) $[2, 4] \cup (-1, e]$
 - (e) $[2, 4] \cap (-1, e]$
 - (f) $[-3, 0) \cup (-3, \pi)$
 - (g) $(0, 3] \setminus [2, 5)$
 - (h) The complement of $[-2, 4)$ in $\mathbb{R}$
- 29.** Evaluate without a calculator.
- (a) $(0.01)^{-1/2}$
 - (b) $(0.008)^{2/3}$
 - (c) $\left(-\frac{1}{5}\right)^{-3}$
 - (d) $16^{-3/4}$
 - (e) $\left(\frac{1}{27}\right)^{-2/3}$
- 30.** Simplify, assuming the variables are positive where a square root is involved.
- (a) $\frac{(2x^3y^{-4})^2}{8xy^{-6}}$

$$(b) \frac{(9x^2y^5)^2}{6y^{-3}} \left(\frac{1}{xy^{-4}} \right)^2$$

$$(c) \sqrt{\frac{x}{y}} \sqrt{\frac{25y}{x^3}}$$

$$(d) \frac{1}{\sqrt{y}}$$

$$(e) \frac{(3a^2b^{-1})^2}{9ab^{-3}}$$

$$(f) \frac{\sqrt{18x^3}}{\sqrt{2x}}$$

31. Simplify without a calculator.

$$(a) \sqrt{\frac{3}{5}} \sqrt{\frac{75}{27}}$$

$$(b) 8\sqrt{2} - \sqrt{32} + 5\sqrt{98}$$

$$(c) \frac{3\sqrt{12}}{4\sqrt[3]{54}}$$

$$(d) (2\sqrt{3} - \sqrt{2})(1 - 2\sqrt{6})$$

$$(e) (\sqrt{5} + \sqrt{3})^2$$

$$(f) (3\sqrt{2} + 2\sqrt{3})(\sqrt{2} - \sqrt{3})$$

32. Simplify without a calculator.

$$(a) \log_3(27\sqrt{3})$$

$$(b) \log_{0.5}(0.0625)$$

$$(c) \log_3 64 \div \log_3 4$$

$$(d) 2\log 5 + \log 70 + \log \frac{45}{32} - \log \frac{45}{2}$$

$$(e) \log_2 40 - \log_2 5$$

$$(f) 2\ln 3 + \ln 4 - \ln 12$$

33. Solve, rejecting any extraneous solutions.

$$(a) (\sqrt{3})^{2x+1} - 9\sqrt{3} = 0$$

$$(b) 2\sqrt{1-x} = \sqrt{16-3x} - 1$$

$$(c) \sqrt{x} + \sqrt{3x-2} = \sqrt{6x+10}$$

$$(d) 2^{1-x} = \frac{9}{4}$$

$$(e) 4(2^x) = 3^x 5^x$$

$$(f) (\log x)^2 = \log(x^3), \text{ where } \log \text{ denotes base } 10$$

$$(g) 5 - 4\log_3 x = \log_x 3$$

$$(h) \log_2(x-1) = 3$$

(i) $\ln x + \ln(x - 3) = \ln 4$

- 34.** A forest contains 1 200 000 trees and 7% of the remaining trees are harvested each year. The number remaining after t years is $n = 1\,200\,000(0.93)^t$.

- (a) Find when half the original number remains.
(b) Estimate the number of trees remaining after 5 years.
(c) Find the percentage of the original forest remaining after 12 years.

- 35.** The cost of producing n computer chips is modelled by $C = 10n^{2/3} + 1500$.

- (a) Find the cost of producing 1000 chips.
(b) Find the greatest whole number of chips that can be produced for RM 10 000.
(c) Find the output level when $C = \text{RM } 5500$.

E. Extended complex-number practice

- 36.** Classify each number as real only, non-real complex, or both real and complex. Give a brief explanation.

- (a) 10
(b) $-3i$
(c) $4 - 2i$
(d) $\sqrt{5}$

- 37.** Perform the indicated calculations.

- (a) For $z = \frac{i - 4}{2i - 3}$, find $\text{Re}(z)$ and $\text{Im}(z)$.
(b) Write $w = \frac{3 + 2i}{1 - i}$ in the form $a + bi$.
(c) Find the modulus of the value of z in part (a).

- 38.** Let $z_1 = 2 + 3i$ and $z_2 = 2 - 3i$. Find

- | | |
|--|------------------------------|
| (a) $\text{Im}(z_1)$; | (f) $z_1 - z_2$; |
| (b) $\text{Re}(z_2)$; | (g) $\text{Im}(z_1 + z_2)$; |
| (c) $\text{Re}(z_1) - \text{Re}(z_2)$; | (h) $2z_1 + 3z_2$; |
| (d) $\text{Im}(z_1) + 2\text{Re}(z_2)$; | (i) $z_1 z_2$; |
| (e) $z_1 + z_2$; | (j) $ z_1 $. |

- 39.** Evaluate.

- (a) $\text{Im}\{(3 - 2i)^2\}$
(b) $\text{Im}\{(1 + i)^5\}$

(c) $\operatorname{Re}\{(2+i)(1-3i)\}$

40. For each value of z , find both $z + \bar{z}$ and $z\bar{z}$.

(a) $z = 6 - 3i$

(b) $z = \sqrt{5} - 3i\sqrt{5}$

(c) $z = 1 + 2i$

(d) $z = -4i$

41. Simplify.

(a) $3(2 + 4i)$

(f) $3/(1 + i)$

(b) $(2 - 7i)(3 + 4i)$

(g) $(4 + 7i)/(2 + 5i)$

(c) i^{17}

(h) $(2 - i)/(2 + i)$

(d) i^{-4}

(i) $(1 + i)^4$

(e) $1/i$

42. Given $x, y \in \mathbb{R}$, solve for x and y .

(a) $(x + iy)(2 + i) = 3 - i$

(b) $(x + iy)(1 - i) = 4 + 2i$

(c) $(x + iy)(3 + 2i) = 1 + 8i$

43. Solve over the complex numbers.

(a) $x^2 + 9 = 0$

(b) $x^2 - 4x + 7 = 0$

(c) $x^2 + 2x + 5 = 0$

(d) $x^2 - 2x + 10 = 0$

(e) $3x^2 + 6x + 15 = 0$

44. Find the monic quadratic equation with the stated roots.

(a) $4 + 2i$ and $4 - 2i$

(b) $-1 + 3i$ and $-1 - 3i$

(c) $2 + i\sqrt{5}$ and $2 - i\sqrt{5}$

45. Consider the complex numbers $2 + 2i$, $2 - 2i$, $-2 + 2i$ and $-2 - 2i$.

(a) Plot the four points on an Argand diagram.

(b) Find the modulus of each number.

(c) Describe the symmetry and the geometric figure formed by the four points.

F. Grouped skills review

46. State the smallest applicable set among $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ for each number.

(a) -2

(b) 0

(c) $\sqrt{49}$

(d) $0.\bar{2}$

(e) $\sqrt{5}$

47. Express each recurring decimal as a fraction in lowest terms.

(a) $0.\bar{1}$

(b) $0.\bar{2}$

(c) $0.\bar{3}$

(d) $0.\bar{4}$

(e) $0.\bar{5}$

48. Find each intersection.

(a) $[-3, 4] \cap (-2, 6)$

(b) $[-4, 5] \cap (-3, 7)$

(c) $[-5, 6] \cap (-4, 8)$

(d) $[-6, 7] \cap (-5, 9)$

(e) $[-7, 8] \cap (-6, 10)$

49. Simplify, giving each answer with a positive index.

(a) x^3x^4/x^2

(b) x^4x^4/x^2

(c) x^5x^4/x^2

(d) x^6x^4/x^2

(e) x^7x^4/x^2

50. Simplify each surd.

(a) $\sqrt{8}$

(b) $\sqrt{27}$

(c) $\sqrt{80}$

(d) $\sqrt{150}$

(e) $\sqrt{252}$

51. Rationalise and simplify.

- (a) $2/\sqrt{2}$
- (b) $3/\sqrt{3}$
- (c) $5/\sqrt{5}$
- (d) $6/\sqrt{6}$
- (e) $7/\sqrt{7}$

52. Evaluate.

- (a) $\log_2(2^2)$
- (b) $\log_3(3^3)$
- (c) $\log_5(5^4)$
- (d) $\log_{10}(10^5)$
- (e) $\log_4(4^6)$

53. Find the new value after the stated percentage increase.

- (a) 200 increased by 5%
- (b) 250 increased by 10%
- (c) 300 increased by 15%
- (d) 350 increased by 20%
- (e) 400 increased by 25%

54. Simplify each sum.

- (a) $(1 + 2i) + (1 - i)$
- (b) $(2 + 3i) + (3 - 2i)$
- (c) $(3 + 4i) + (5 - 3i)$
- (d) $(4 + 5i) + (7 - 4i)$
- (e) $(5 + 6i) + (9 - 5i)$

55. Evaluate i^5, i^6, i^7, i^8 and i^9 .

56. Solve each equation over the complex numbers.

- (a) $x^2 - 2x + 5 = 0$
- (b) $x^2 - 4x + 13 = 0$
- (c) $x^2 - 6x + 25 = 0$
- (d) $x^2 - 8x + 41 = 0$
- (e) $x^2 - 10x + 61 = 0$

G. Applications: quantitative reasoning and measurement

- 57.** In a survey of 600 residents, 384 support a proposed community project.
- (a) Express the supporting proportion as a decimal and percentage.
 - (b) How many surveyed residents do not support the proposal?
- 58.** A region has 1 260 000 residents and an area of 3500 km². Find its population density in residents per square kilometre.
- 59.** A community fund of RM 24 000 is divided among education, health and recreation in the ratio 5 : 3 : 2. Find the allocation to each category.
- 60.** A sample survey estimates that 43.7% of 2450 eligible voters favour a policy.
- (a) Estimate the number who favour the policy.
 - (b) Explain why the answer should not be presented as an exact count.
- 61.** A research assistant completes 126 interviews in 7 hours at a constant rate.
- (a) Find the interview rate per hour.
 - (b) At this rate, how many hours are required for 450 interviews?
- 62.** A survey team travels at an average speed of 72 km/h.
- (a) Convert this speed to metres per second.
 - (b) Find the distance travelled in 35 minutes.
- 63.** Voter turnout rises from 58% to 64.5%.
- (a) Find the increase in percentage points.
 - (b) Find the percentage increase relative to the original turnout.
- 64.** A monthly transport cost rises from RM 240 to RM 276. Find its index relative to the earlier cost and interpret the result.
- 65.** A population is 40% urban and 60% rural. Support for a programme is 72% among urban residents and 55% among rural residents. Find the overall supporting percentage.
- 66.** A local authority spends RM 18 million on a population of 240 000.
- (a) Find expenditure per resident.
 - (b) Find the new expenditure per resident if population grows by 5% while total expenditure remains unchanged.
- 67.** A population is reported as 2.4 million, correct to the nearest 0.1 million. State the interval containing the actual population.

H. Applications: percentages, rates and indices

68. A salary rises by 6% while the price level rises by 4%. Calculate the exact percentage change in the salary's purchasing power.
69. A monthly benefit of RM 2500 is reduced by 8% and is later increased by 8%.
(a) Find the final benefit.
(b) Compare it with the original benefit in ringgit and percentage terms.
70. A service has a pre-tax charge of RM 3200. A tax of 6% and a fixed administrative fee of RM 25 are added. Find the final bill.
71. After a 15% increase, a household's monthly food expenditure is RM 1840. Find its expenditure before the increase.
72. A RM 500 training fee is discounted by 20% and then by a further 10%.
(a) Find the final fee.
(b) Find the single equivalent percentage discount.
73. An unemployment rate falls from 7.5% to 6.3%. Find the fall in percentage points and the percentage decrease relative to the original rate.
74. A monthly benefit of RM 1500 was set when a cost index was 120. Find the fully indexed benefit when the index reaches 132.
75. A household has monthly income RM 4200. It spends 35% on housing and 12% on transport. Find both expenditures and the amount remaining for all other purposes.
76. A migration rate is 18 per thousand residents. Estimate the number of migrants in a population of 750 000.
77. A study sends 1200 invitations and receives 714 usable responses. Find the response rate and the number of non-responses.
78. An index falls from 125 to 110. Calculate the percentage decrease and explain why it is not 15%.

I. Applications: powers, logarithms and repeated change

79. A town with population 80 000 grows by 2.5% annually. Using an exponential model, estimate its population after four years.
80. An online service begins with 12 000 subscribers and loses 6% of the remaining subscribers each month. Estimate the number remaining after five months.
81. A social programme grows by 4% per year. Find its modelled doubling time.

82. A stock of printed questionnaires declines by 12% each month. Find the modelled time required for half the stock to remain.
83. A campaign begins with 5000 supporters and grows by 8% per period. Find when the model first reaches 7500 supporters, both as a continuous solution and in complete periods.
84. Two regions have positive average incomes A and B satisfying $\ln B - \ln A = 0.30$. Find B/A and interpret the result.
85. One data set contains values around 10^3 , while another contains values around 10^5 .
- (a) Find their approximate size ratio.
 - (b) Find the difference between their base-10 logarithms.
86. A country has 3.6×10^6 households with an average of 2.4 people per household. Estimate the population in scientific notation.
87. Two models describe programme participation after t years: $L(t) = 1000 + 120t$ and $E(t) = 1000(1.10)^t$.
- (a) Find both predictions when $t = 5$.
 - (b) Which prediction is larger, and by how much?
88. Equipment valued at RM 90 000 depreciates by 15% of its remaining value each year. Find its value after three years.
89. A participation index follows $Q(t) = 200e^{0.06t}$. Find the time when the index reaches 300.

J. Applications: modelling and interpretation

90. A public budget is reported as RM 2.8 million, correct to the nearest RM 0.1 million. Give the interval of possible actual values.
91. Two polling methods give plausible support intervals $[48, 56]$ and $[52, 60]$, measured in percentage points. Find their overlap and their union.
92. A square administrative district has area 200 km^2 . Express its side length exactly as a simplified surd and approximately to two decimal places.
93. A fieldworker's displacement is represented by $z = 6 + 8i$, where the real and imaginary parts measure kilometres east and north. Find the straight-line distance from the starting point.
94. Four stages of a seasonal cycle are represented by $1, i, -1, -i$. Starting from 1, the stage is multiplied by i each quarter. Find the stage after 14 quarters.

- 95.** A periodic component is encoded by $z = 3 + 4i$.
- (a) Find its modulus.
 - (b) Find its complex conjugate.
 - (c) State what the modulus represents in such an encoding.
- 96.** A simplified net-benefit equation is $q^2 - 6q + 13 = 0$.
- (a) Solve the equation over the complex numbers.
 - (b) Explain what the absence of real roots means for a real-valued break-even interpretation.
- 97.** A public-service cost is modelled by $C(t) = 25 + 3t$, where t is in years and C is in millions of ringgit.
- (a) State the units of the intercept and slope.
 - (b) Find and interpret $C(4)$.
- 98.** Average monthly incomes in two groups are RM 2400 and RM 3000. Find their absolute difference and ratio. Explain why the second income is 25% above the first, while the first is only 20% below the second.
- 99.** A report says that a rate “rose by 1%” when it increased from 2% to 3%. State the increase in percentage points and the percentage increase relative to the original rate.
- 100.** A city’s population is modelled by $P(t) = 50\,000(1.03)^t$.
- (a) Estimate the population after 20 years.
 - (b) State two reasons why this long-run projection should be interpreted cautiously.

Chapter 2

Functions

Chapter guide

Develop function notation and graphs first, then study transformations and common function families before composition, inverse functions and modelling applications.

Selected background sources: (Chai et al., 2024; Ministry of Education Malaysia, 2018; Schwarz, 2021).

Functions describe how one quantity depends on another. They are used to represent patterns, express mathematical models and study relationships between variables. A function may be described using words, ordered pairs, a mapping diagram, a table, a formula or a graph. Moving between these representations is an important skill: a formula supports calculation, a table records selected values and a graph reveals overall behaviour.

In applications, the variables represent measurable quantities and the parameters describe features of the relationship. A useful function model must therefore include more than an equation. Its domain, units, assumptions and limitations must also be stated, and its outputs must be interpreted in the context from which the model arose.

2.1 Relations and functions

A *relation* is any collection of ordered pairs. In an ordered pair (x, y) , the first coordinate is the input and the second coordinate is the corresponding output. For example,

$$R = \{(-2, 4), (0, 0), (3, 9)\}$$

is a relation with domain $\{-2, 0, 3\}$ and range $\{0, 4, 9\}$.

Definition 2.1: Function

Let A and B be sets. A *function* f from A to B , written $f: A \rightarrow B$, assigns every element of A to exactly one element of B .

The requirement “exactly one” is essential. Different inputs may have the same output, but a single input cannot be assigned two different outputs. Thus, a many-to-one relation can be a function, whereas a one-to-many relation cannot.

2.1.1 Domain, codomain and range

For a function $f: A \rightarrow B$:

- the *domain* is the set A of permitted inputs;

- the *codomain* is the declared set B of possible outputs;
- the *range* is the set of outputs actually produced.

The range is therefore a subset of the codomain. The mapping diagram in figure 2.1 represents the function $f(x) = x^2 + 1$ on the domain $A = \{-2, 0, 3\}$.

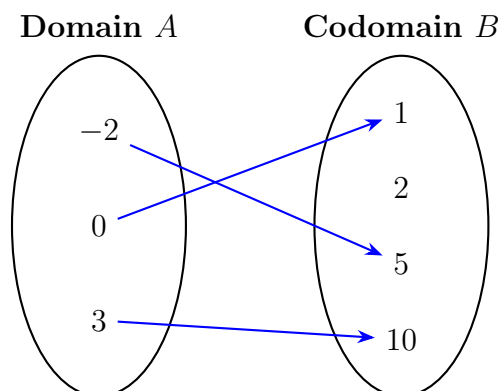


Figure 2.1: A mapping diagram for $f(x) = x^2 + 1$. The range is $\{1, 5, 10\}$.

When a function is given by a formula, its domain contains every real input for which the formula is defined, unless another domain is stated. Common restrictions include the following:

- a denominator cannot equal zero;
- the radicand of an even root must be non-negative;
- the argument of a logarithm must be positive.

For example, if

$$h(x) = \frac{\sqrt{x+2}}{x-1},$$

then $x + 2 \geq 0$ and $x \neq 1$. Hence,

$$\text{Dom}(h) = [-2, 1) \cup (1, \infty).$$

2.1.2 Natural and contextual domains

The restrictions imposed by a formula determine its *natural domain*. An application may require a smaller *contextual domain*. For example, the formula

$$C(n) = 12n + 40$$

is algebraically defined for every real number n . If n represents the number of tickets sold, however, a suitable domain is

$$n \in \{0, 1, 2, \dots\},$$

possibly with an upper limit determined by the venue's capacity. Negative or fractional inputs would be mathematically permissible in the formula but meaningless in this context.

Domain restrictions can also arise from practical conditions such as time being non-negative, a percentage lying between 0 and 100, or production being limited by available resources. A model should state these restrictions explicitly rather than relying only on the algebraic formula.

2.2 Function notation and evaluation

If f is a function, the notation $f(x)$ denotes the output associated with the input x . The variable x is often called the *independent variable*, while $y = f(x)$ is the *dependent variable*.

Example 2.1: Evaluating a function

Let

$$f(x) = 2x^2 - 3x + 1.$$

To find $f(-2)$, substitute $x = -2$ into the formula:

$$\begin{aligned} f(-2) &= 2(-2)^2 - 3(-2) + 1 \\ &= 8 + 6 + 1 \\ &= \boxed{15}. \end{aligned}$$

A function value can also be used to determine an unknown input. For example, if $g(x) = 3x - 5$ and $g(x) = 10$, then

$$3x - 5 = 10 \implies x = 5.$$

2.3 Graphs of functions

The graph of $y = f(x)$ is the set of all points $(x, f(x))$ in the coordinate plane. Each x -coordinate in the domain must correspond to only one y -coordinate. This leads to the *vertical-line test*: a graph represents a function of x if every vertical line intersects it at most once.

The domain of a graph is the set of x -coordinates covered by the graph, and its range is the set of corresponding y -coordinates. For example, the function

$$q(x) = x^2 - 4$$

has domain $\mathbb{R}$ and range $[-4, \infty)$ because its smallest value is -4 .

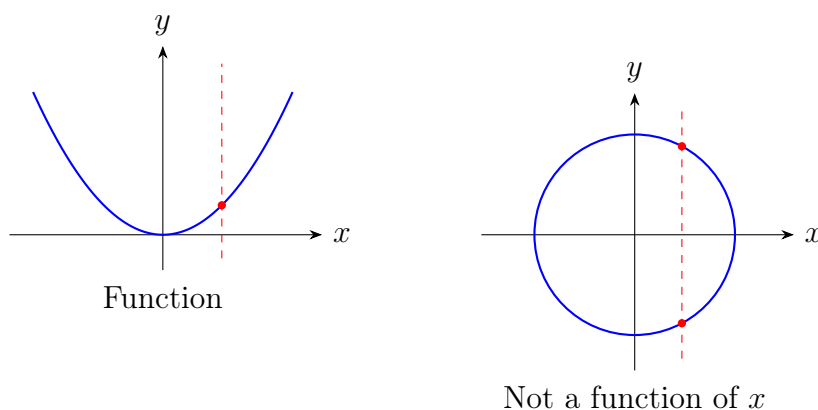


Figure 2.2: Using the vertical-line test to identify a function.

2.3.1 Intercepts, turning points and intervals of change

The x -intercepts of $y = f(x)$ occur where

$$f(x) = 0,$$

so they are also called the *zeros* or *roots* of the function. The y -intercept is $(0, f(0))$, provided 0 belongs to the domain.

A function is *increasing* on an interval when larger inputs give larger outputs throughout that interval. It is *decreasing* when larger inputs give smaller outputs. A point at which the graph changes from increasing to decreasing is a local maximum, while a change from decreasing to increasing is a local minimum. The largest or smallest value over the entire domain is called an absolute maximum or minimum.

Example 2.2: [

blue]Reading features from a quadratic Consider

$$f(x) = -(x - 2)^2 + 5.$$

The vertex $(2, 5)$ is an absolute maximum, and the axis of symmetry is $x = 2$. The function is increasing on $(-\infty, 2)$ and decreasing on $(2, \infty)$. Its y -intercept is $(0, 1)$, and its x -intercepts satisfy

$$(x - 2)^2 = 5,$$

so they are $(2 - \sqrt{5}, 0)$ and $(2 + \sqrt{5}, 0)$. The domain is $\mathbb{R}$ and the range is $(-\infty, 5]$.

2.3.2 Average rate of change

For distinct inputs x_1 and x_2 , the *average rate of change* of f over $[x_1, x_2]$ is

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Graphically, this is the slope of the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$. Its units are the units of the output divided by the units of the input.

Example 2.3: [

orange]Interpreting an average rate of change The percentage of survey respondents who recognise a public campaign rises from 42% in year 2 to 57% in year 7. The average rate of change is

$$\frac{57 - 42}{7 - 2} = \boxed{3}$$

percentage points per year. This describes the average change across the five-year interval; it does not imply that the percentage increased by exactly 3 points in every individual year.

2.3.3 Symmetry of functions

A function is *even* if

$$f(-x) = f(x)$$

for every x in a domain symmetric about zero. Its graph is symmetric about the y -axis.

A function is *odd* if

$$f(-x) = -f(x),$$

in which case its graph has rotational symmetry of 180° about the origin. For example, x^2 and $|x|$ are even, while x^3 and $1/x$ are odd. Many functions are neither even nor odd.

2.3.4 Intersections of functions

The solutions of

$$f(x) = g(x)$$

are the x -coordinates at which the graphs of $y = f(x)$ and $y = g(x)$ intersect. In an application, an intersection may represent an equilibrium, a break-even point or the time at which two competing quantities are equal.

Example 2.4: [

cyan]Finding an equilibrium by intersection Suppose demand and supply are modelled by

$$D(p) = 120 - 4p, \quad S(p) = 20 + 6p,$$

where p is price. At equilibrium, $D(p) = S(p)$, so

$$120 - 4p = 20 + 6p \implies p = 10.$$

The common output is

$$D(10) = S(10) = 80.$$

Thus, the model's equilibrium point is $(10, 80)$.

2.4 Transformations of graphs

The graph of a function can be translated, reflected or rescaled by changing its formula. table 2.1 summarises the main transformations of the graph $y = f(x)$.

New function	Transformation	Effect on a point (x, y)
$y = f(x) + k$	translate vertically by k	$(x, y) \mapsto (x, y + k)$
$y = f(x - h)$	translate horizontally by h	$(x, y) \mapsto (x + h, y)$
$y = -f(x)$	reflect in the x -axis	$(x, y) \mapsto (x, -y)$
$y = f(-x)$	reflect in the y -axis	$(x, y) \mapsto (-x, y)$
$y = af(x)$	vertical scale factor $ a $	$(x, y) \mapsto (x, ay)$
$y = f(bx)$	horizontal scale factor $1/ b $	$(x, y) \mapsto (x/b, y)$

Table 2.1: Transformations of the graph $y = f(x)$.

For translations, a positive k moves the graph upward and a negative k moves it downward. In $f(x - h)$, a positive h moves the graph to the right; the horizontal direction is therefore opposite to the sign written inside the function.

If $|a| > 1$, the graph of $y = af(x)$ is stretched vertically; if $0 < |a| < 1$, it is compressed vertically. Similarly, $|b| > 1$ compresses the graph of $y = f(bx)$ horizontally, while $0 < |b| < 1$ stretches it horizontally. A negative value of a also reflects the graph in the x -axis, and a negative value of b reflects it in the y -axis.

The combined transformation

$$y = af(b(x - h)) + k, \quad a, b \neq 0,$$

maps a point (u, v) on $y = f(x)$ to

$$\left(h + \frac{u}{b}, k + av\right).$$

This point-mapping rule is useful when several transformations occur at once.

Example 2.5: Transforming a quadratic graph

Starting with $f(x) = x^2$, consider

$$g(x) = 2(x - 3)^2 - 4.$$

The graph of g is obtained from $y = x^2$ by a vertical stretch of factor 2, followed by a translation 3 units to the right and 4 units downward. Its vertex is $(3, -4)$, its

axis of symmetry is $x = 3$, and its range is

$$[-4, \infty).$$

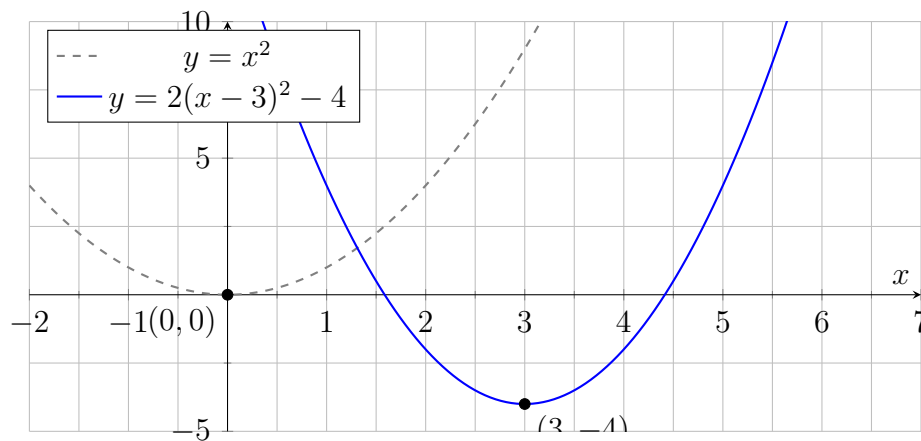
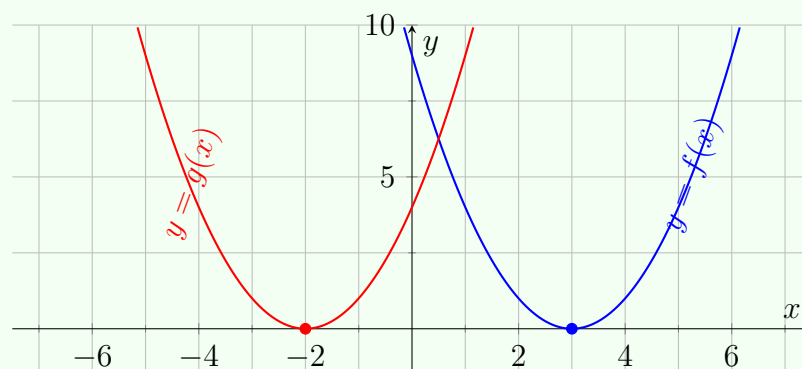


Figure 2.3: The parent parabola and its stretched and translated image.

Example 2.6: Exercise 2

The blue graph shows $y = f(x)$. Determine the equation of the red graph $y = g(x)$ in terms of $f(x)$.



2.5 Common families of functions

Many mathematical models are built from a small collection of parent functions. Their basic properties provide a starting point for sketching graphs and interpreting transformed functions.

2.5.1 Constant and linear functions

A constant function has the form $f(x) = c$ and has a horizontal graph. A linear function has the form

$$f(x) = mx + b,$$

Family	Parent form	Domain	Main feature
Linear	$f(x) = mx + b$	$\mathbb{R}$	straight line with slope m
Quadratic	$f(x) = x^2$	$\mathbb{R}$	parabola with a turning point
Polynomial	$f(x) = x^n$	$\mathbb{R}$	smooth, continuous curve
Absolute value	$f(x) = x $	$\mathbb{R}$	V-shaped graph
Reciprocal	$f(x) = 1/x$	$\mathbb{R} \setminus \{0\}$	two branches with asymptotes
Exponential	$f(x) = b^x$	$\mathbb{R}$	horizontal asymptote $y = 0$
Logarithmic	$f(x) = \log_b x$	$(0, \infty)$	vertical asymptote $x = 0$

Table 2.2: Some common families of functions, where $b > 0$ and $b \neq 1$.

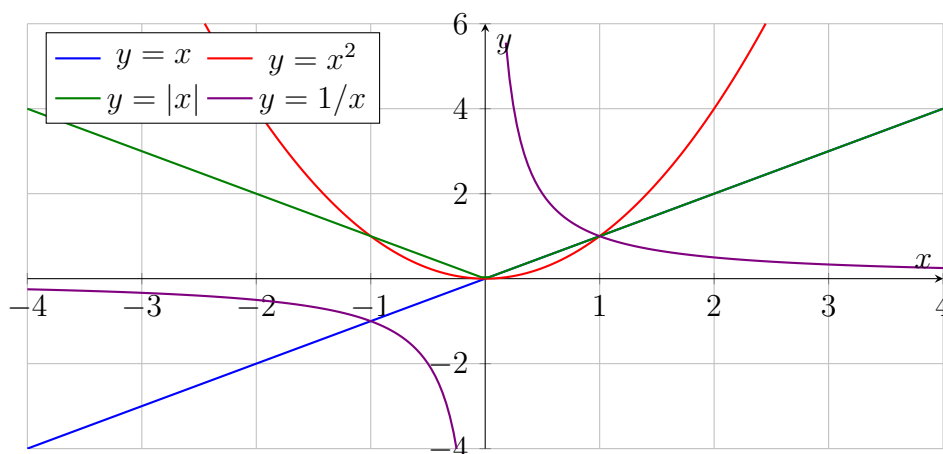


Figure 2.4: Selected algebraic parent functions.

where m is the slope and b is the y -intercept. Given two distinct points (x_1, y_1) and (x_2, y_2) , the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

The graph increases when $m > 0$, decreases when $m < 0$, and is constant when $m = 0$.

Example 2.7: Finding the equation of a line

A line has slope -3 and passes through $(2, 5)$. Using point-slope form,

$$\begin{aligned} y - 5 &= -3(x - 2), \\ y &= -3x + 11. \end{aligned}$$

Thus, the equation is $y = -3x + 11$.

2.5.2 Quadratic functions

A quadratic function has the form

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

Its graph is a parabola opening upward if $a > 0$ and downward if $a < 0$. The axis of symmetry and the x -coordinate of the vertex are

$$x = -\frac{b}{2a}.$$

Writing a quadratic in vertex form,

$$f(x) = a(x - h)^2 + k,$$

makes the vertex (h, k) and the range easy to identify.

For example,

$$x^2 - 6x + 5 = (x - 3)^2 - 4.$$

The graph therefore has vertex $(3, -4)$, axis of symmetry $x = 3$, roots $x = 1$ and $x = 5$, and range $[-4, \infty)$.

2.5.3 Polynomial and power functions

A polynomial function is a finite sum

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0.$$

Its degree is n , and the leading term $a_n x^n$ determines its end behaviour. An even-degree polynomial has both ends pointing in the same direction, whereas an odd-degree polynomial has ends pointing in opposite directions. The sign of a_n determines which end points upward.

The zeros of P are the values of x for which $P(x) = 0$ and correspond to the x -intercepts of its graph. A zero with odd multiplicity usually causes the graph to cross the x -axis, while a zero with even multiplicity causes it to touch the axis and turn.

2.5.4 Absolute-value and piecewise functions

The absolute-value function is defined by

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Its graph is V-shaped with vertex at the origin. A transformed function $a|x - h| + k$ has vertex (h, k) and opens upward when $a > 0$ and downward when $a < 0$.

More generally, a piecewise function uses different formulas on different parts of its domain. For example,

$$p(x) = \begin{cases} x + 2, & x < 0, \\ x^2, & x \geq 0. \end{cases}$$

When graphing a piecewise function, an open point indicates that an endpoint is excluded, whereas a filled point indicates that it is included.

The formula selected at a boundary is determined by its inequality. For the function above,

$$p(-2) = 0, \quad p(0) = 0, \quad p(3) = 9.$$

The value $p(0)$ is calculated using the second branch because that branch contains $x = 0$.

Piecewise functions are useful for tariffs, taxes, subsidies and other rules whose rate changes after a threshold.

Example 2.8: [

violet]A piecewise transport charge An illustrative transport charge for a journey of d kilometres is

$$C(d) = \begin{cases} 5 + 1.50d, & 0 \leq d \leq 10, \\ 20 + 1.10(d - 10), & d > 10. \end{cases}$$

For an 8-kilometre journey,

$$C(8) = 5 + 1.50(8) = \text{RM } 17.$$

For a 25-kilometre journey,

$$C(25) = 20 + 1.10(15) = \text{RM } 36.50.$$

At $d = 10$, both expressions give RM 20, so the charge has no jump at the threshold even though its rate of increase changes.

For a piecewise graph to join without a jump at a boundary, the expressions on both

sides must approach the same output and agree with the defined function value. A formal treatment of continuity is given in Chapter 10.

2.5.5 Rational functions

A rational function is a quotient of polynomials,

$$r(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0.$$

Values that make $Q(x) = 0$ are excluded from the domain. They may produce vertical asymptotes or removable discontinuities. The parent reciprocal function $r(x) = 1/x$ has vertical asymptote $x = 0$, horizontal asymptote $y = 0$ and range $\mathbb{R} \setminus \{0\}$.

An important family of rational functions is

$$f(x) = \frac{a}{x-h} + k, \quad a \neq 0.$$

It is obtained from the basic reciprocal function $y = a/x$ by translating the graph h units horizontally and k units vertically. Its domain and range are

$$\text{Dom}(f) = \mathbb{R} \setminus \{h\}, \quad \text{Ran}(f) = \mathbb{R} \setminus \{k\}.$$

The graph has vertical asymptote $x = h$ and horizontal asymptote $y = k$; the two asymptotes intersect at the centre (h, k) . If $a > 0$, the branches lie above-right and below-left of the centre. If $a < 0$, they lie above-left and below-right.

For the basic function $f(x) = a/x$, increasing $|a|$ moves the branches farther from the coordinate axes without changing the domain, range or asymptotes. It is also an odd function because

$$f(-x) = -f(x).$$

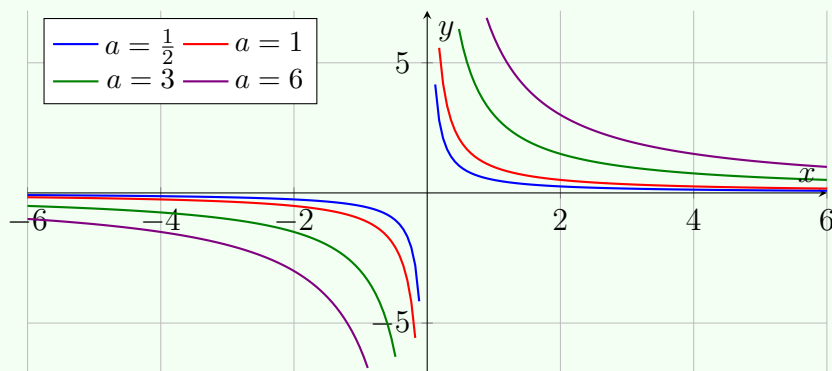
Consequently, reflecting its graph in the x -axis gives the same result as reflecting it in the y -axis.

Example 2.9: Example 1: Varying the numerator

Sketch $f(x) = a/x$ for

$$a = \frac{1}{2}, \quad 1, \quad 3, \quad 6,$$

and compare the graphs.



Every graph has domain and range

$$\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty),$$

with vertical asymptote $x = 0$ and horizontal asymptote $y = 0$.

Example 2.10: Example 2: Identifying asymptotes

Identify the vertical and horizontal asymptotes of each function.

For

$$f(x) = \frac{1}{x-1},$$

the denominator is zero at $x = 1$. Hence the vertical asymptote is $x = 1$, and the horizontal asymptote is $y = 0$.

For

$$g(x) = \frac{1}{x+3} - 2,$$

the graph is translated 3 units left and 2 units down. Hence the vertical asymptote is $x = -3$, and the horizontal asymptote is $y = -2$.

Example 2.11: Example 3: Transforming a reciprocal graph

Starting with the basic graph

$$f(x) = \frac{2}{x},$$

describe each transformation and state the domain, range and asymptotes.

For

$$g(x) = \frac{2}{x+1} - 2,$$

the graph moves 1 unit left and 2 units down. Therefore,

$$\text{Dom}(g) = \mathbb{R} \setminus \{-1\}, \quad \text{Ran}(g) = \mathbb{R} \setminus \{-2\},$$

with asymptotes $x = -1$ and $y = -2$.

For

$$h(x) = \frac{2}{x-2} + 3,$$

the graph moves 2 units right and 3 units up. Therefore,

$$\text{Dom}(h) = \mathbb{R} \setminus \{2\}, \quad \text{Ran}(h) = \mathbb{R} \setminus \{3\},$$

with asymptotes $x = 2$ and $y = 3$.

For example, the rational function

$$\frac{x+1}{x-2} = 1 + \frac{3}{x-2}.$$

This form shows that the graph is a translated reciprocal curve with vertical asymptote $x = 2$ and horizontal asymptote $y = 1$.

2.5.6 Exponential functions

An exponential function has the form

$$f(x) = b^x, \quad b > 0, \quad b \neq 1.$$

Its domain is $\mathbb{R}$, its range is $(0, \infty)$, its y -intercept is $(0, 1)$, and $y = 0$ is a horizontal asymptote. The function represents growth when $b > 1$ and decay when $0 < b < 1$.

A common growth or decay model is

$$A(t) = A_0(1+r)^t,$$

where A_0 is the initial amount and r is the rate of change per time period. A negative value of r represents decay, provided $1+r > 0$.

The natural exponential function

The *natural exponential function* is

$$f(x) = e^x,$$

where

$$e \approx 2.718281828.$$

Its domain is $\mathbb{R}$ and its range is $(0, \infty)$. It passes through $(0, 1)$ because $e^0 = 1$, is increasing for every real x , and has the x -axis, $y = 0$, as a horizontal asymptote.

A translated natural exponential function has the form

$$g(x) = e^{x-h} + k.$$

The parameter h translates the graph horizontally: it moves right when $h > 0$ and left

when $h < 0$. The parameter k translates the graph vertically: it moves upward when $k > 0$ and downward when $k < 0$. Consequently,

$$\text{Dom}(g) = \mathbb{R}, \quad \text{Ran}(g) = (k, \infty),$$

and the horizontal asymptote is $y = k$.

Reflections are obtained by changing the sign outside or inside the function:

$$y = -e^x \quad \text{is a reflection in the } x\text{-axis,}$$

whereas

$$y = e^{-x} \quad \text{is a reflection in the } y\text{-axis.}$$

The graph of $y = -e^x$ has range $(-\infty, 0)$, while $y = e^{-x}$ retains the range $(0, \infty)$ and represents exponential decay. Both have horizontal asymptote $y = 0$.

Example 2.12: Transforming the natural exponential function

Starting with $f(x) = e^x$, consider

$$g(x) = e^{x-1} + 1.$$

The graph is translated 1 unit right and 1 unit upward. Therefore,

$$\text{Dom}(g) = \mathbb{R}, \quad \text{Ran}(g) = (1, \infty),$$

and its horizontal asymptote is $y = 1$.

For

$$h(x) = -e^{x+1} + 2,$$

the graph of e^x is reflected in the x -axis, translated 1 unit left, and translated 2 units upward. Hence,

$$\text{Dom}(h) = \mathbb{R}, \quad \text{Ran}(h) = (-\infty, 2),$$

with horizontal asymptote $y = 2$.

2.5.7 Logarithmic functions

The logarithmic function

$$f(x) = \log_b x, \quad b > 0, \quad b \neq 1,$$

is the inverse of $y = b^x$. Its domain is $(0, \infty)$, its range is $\mathbb{R}$, its x -intercept is $(1, 0)$, and $x = 0$ is a vertical asymptote. It is increasing when $b > 1$ and decreasing when $0 < b < 1$.

The graphs of $y = b^x$ and $y = \log_b x$ are reflections of one another in the line $y = x$. A

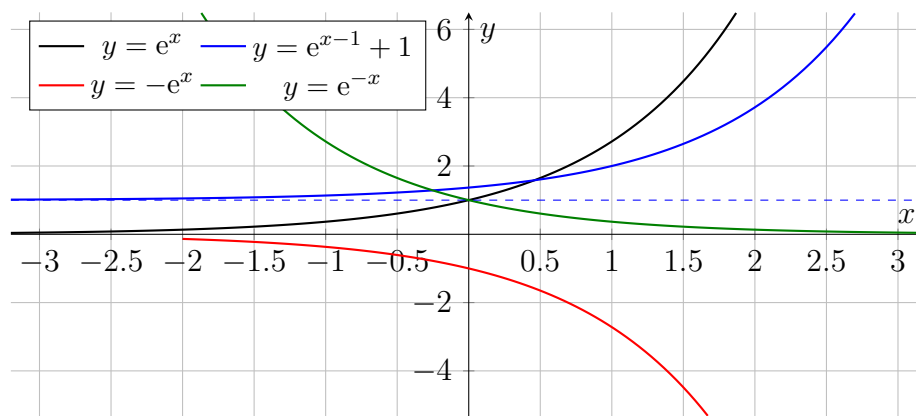


Figure 2.5: Translations and reflections of the natural exponential function.

transformation of the form

$$y = \log_b(x - h) + k$$

has vertical asymptote $x = h$ and is translated h units horizontally and k units vertically.

The natural logarithmic function

The *natural logarithmic function* is

$$f(x) = \ln x = \log_e x.$$

It is the inverse of the natural exponential function $y = e^x$. Its domain, range and vertical asymptote are

$$\text{Dom}(f) = (0, \infty), \quad \text{Ran}(f) = \mathbb{R}, \quad x = 0.$$

The graph passes through $(1, 0)$ because $\ln 1 = 0$ and is increasing on its entire domain.

A translated natural logarithmic function has the form

$$g(x) = \ln(x - h) + k.$$

The parameter h translates the graph horizontally, while k translates it vertically. Thus,

$$\text{Dom}(g) = (h, \infty), \quad \text{Ran}(g) = \mathbb{R},$$

and its vertical asymptote is $x = h$. A positive h moves the graph right and a negative h moves it left; a positive k moves it upward and a negative k moves it downward.

Reflections are formed in the usual way:

$$y = -\ln x \quad \text{is a reflection in the } x\text{-axis,}$$

and

$$y = \ln(-x) \quad \text{is a reflection in the } y\text{-axis.}$$

The first reflected function has domain $(0, \infty)$, whereas the second has domain $(-\infty, 0)$. Both have range $\mathbb{R}$ and vertical asymptote $x = 0$.

Example 2.13: Transforming the natural logarithmic function

Starting with $f(x) = \ln x$, consider

$$g(x) = \ln(x - 1) + 2.$$

The graph moves 1 unit right and 2 units upward. Therefore,

$$\text{Dom}(g) = (1, \infty), \quad \text{Ran}(g) = \mathbb{R},$$

and the vertical asymptote is $x = 1$.

For

$$h(x) = -\ln(x + 2),$$

the graph moves 2 units left and is then reflected in the x -axis. Hence,

$$\text{Dom}(h) = (-2, \infty), \quad \text{Ran}(h) = \mathbb{R},$$

with vertical asymptote $x = -2$.

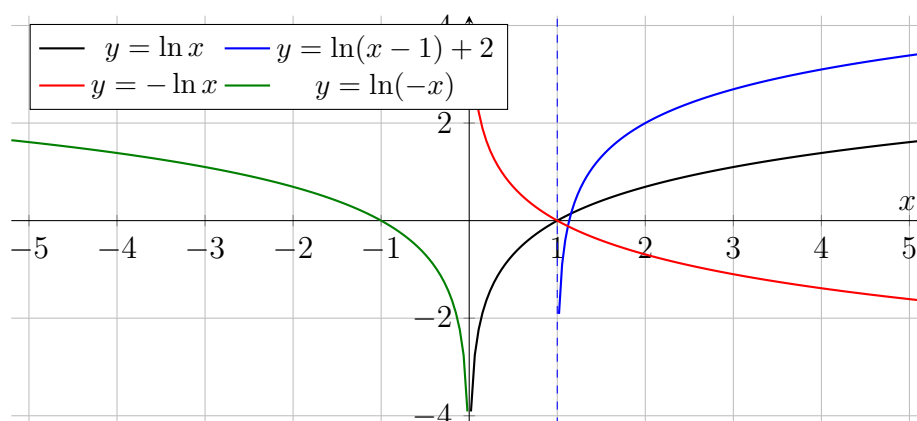


Figure 2.6: Translations and reflections of the natural logarithmic function.

2.6 Functions as mathematical models

A mathematical model uses variables and parameters to represent selected features of a real situation. If $y = f(x)$, then x is usually the explanatory or input variable and y is the response or output variable. Parameters are constants within the model that control features such as an initial value, growth rate, slope, turning point or limiting level.

When interpreting a function model, the following questions should be asked:

- What does each variable represent, and what are its units?
- Which inputs are meaningful in the context?
- What does each parameter control?

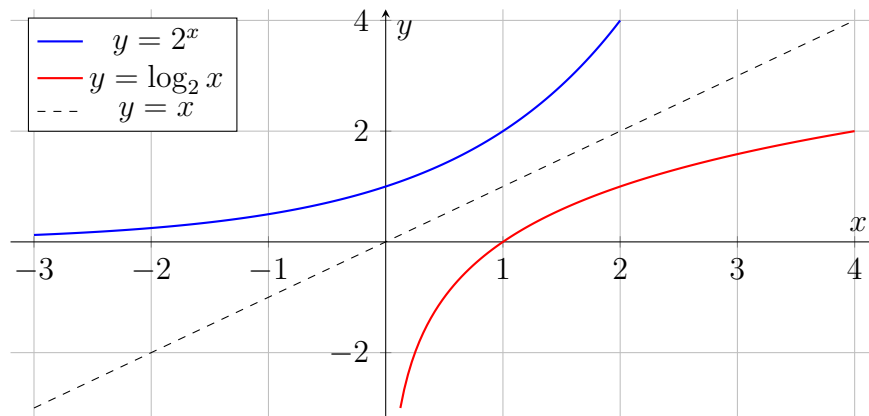


Figure 2.7: Exponential and logarithmic functions reflected in $y = x$.

- Is the model increasing, decreasing or bounded on the relevant domain?
- Does the model interpolate within observed data or extrapolate beyond them?
- Which assumptions could cause the model to fail?

Interpolation uses a model to estimate within the range of observed inputs. *Extrapolation* estimates beyond that range and is generally more uncertain because the relationship may change.

Example 2.14: [

orange] Interpreting a bounded response model The percentage of an eligible population reached by an information campaign is modelled by

$$A(t) = 75(1 - e^{-0.3t}), \quad t \geq 0,$$

where t is measured in years. The contextual domain is $[0, \infty)$ and the range is $[0, 75)$. The model begins at

$$A(0) = 0$$

and approaches, but never exceeds, 75%. After five years,

$$A(5) = 75(1 - e^{-1.5}) \approx \boxed{58.27\%}.$$

The limiting value 75 is a modelling assumption. Changes in policy, funding or public behaviour could make it inappropriate for long-term extrapolation.

2.7 One-to-one and onto functions

Functions may be classified according to how their inputs and outputs are related.

Definition 2.2: One-to-one function

A function f is *one-to-one*, or *injective*, if distinct inputs always produce distinct outputs. Equivalently,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

A graph represents a one-to-one function if every horizontal line intersects it at most once. This is called the *horizontal-line test*. For example, $f(x) = 2x - 3$ is one-to-one on $\mathbb{R}$, whereas $g(x) = x^2$ is not because $g(-x) = g(x)$.

A function is *onto*, or *surjective*, if every element of its codomain is an output of the function; that is, its range equals its codomain. A function that is both one-to-one and onto is called *bijective*. Bijective functions pair every input with a unique output and every output with a unique input.

The classification depends on the stated domain and codomain. For example, the function

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2,$$

is neither one-to-one nor onto. If its domain is restricted to $[0, \infty)$ and its codomain is $[0, \infty)$, however, it becomes bijective.

2.8 Operations on functions

Functions can be combined using the ordinary arithmetic operations. If f and g are functions, then

$$\begin{aligned} (f + g)(x) &= f(x) + g(x), \\ (f - g)(x) &= f(x) - g(x), \\ (fg)(x) &= f(x)g(x), \\ \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}, \quad g(x) \neq 0. \end{aligned}$$

The domain of each new function is the intersection of the domains of f and g , with any additional restrictions caused by division or roots.

Example 2.15: Combining two functions

Let

$$f(x) = x^2 + 1, \quad g(x) = 2x - 3.$$

Then

$$\begin{aligned} (f + g)(x) &= x^2 + 2x - 2, \\ (fg)(x) &= (x^2 + 1)(2x - 3), \\ \left(\frac{f}{g}\right)(x) &= \frac{x^2 + 1}{2x - 3}, \quad x \neq \frac{3}{2}. \end{aligned}$$

2.9 Composite functions

A function may be applied to the output of another function. The *composite function* of f with g is

$$(f \circ g)(x) = f(g(x)).$$

The function g acts first, followed by f . In general, $f \circ g \neq g \circ f$.

Example 2.16: Finding composite functions

Let

$$f(x) = 2x + 1, \quad g(x) = x^2 - 3.$$

Then

$$\begin{aligned}(f \circ g)(x) &= f(x^2 - 3) = 2(x^2 - 3) + 1 = \boxed{2x^2 - 5}, \\(g \circ f)(x) &= g(2x + 1) = (2x + 1)^2 - 3 = \boxed{4x^2 + 4x - 2}.\end{aligned}$$

The different results show that the order of composition matters.

For $(f \circ g)(x)$ to be defined, x must belong to the domain of g and $g(x)$ must belong to the domain of f .

Example 2.17: [

blue]Domains of composite functions Let

$$f(u) = \sqrt{u + 1}, \quad g(x) = 2x - 5.$$

Then

$$(f \circ g)(x) = \sqrt{2x - 4},$$

which requires $x \geq 2$. In the reverse order,

$$(g \circ f)(x) = 2\sqrt{x + 1} - 5,$$

which requires $x \geq -1$. The two composites have different formulas and different domains.

A complicated function can sometimes be understood by decomposing it into simpler functions. For example,

$$h(x) = \sqrt{3x + 1}$$

can be written as $h = f \circ g$, where

$$g(x) = 3x + 1, \quad f(u) = \sqrt{u}.$$

The inner function is applied first.

2.10 Inverse functions

An *inverse function* reverses the action of a function. If f is one-to-one, its inverse is denoted by f^{-1} and satisfies

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(x)) = x.$$

The notation $f^{-1}(x)$ does not mean $1/f(x)$. The domain of f^{-1} is the range of f , and the range of f^{-1} is the domain of f .

To find an inverse algebraically:

1. write $y = f(x)$;
2. interchange x and y ;
3. solve the resulting equation for y ;
4. write the result as $f^{-1}(x)$.

Example 2.18: Finding and checking an inverse

Let

$$f(x) = \frac{3x - 2}{5}.$$

Starting with $y = (3x - 2)/5$, interchange x and y and solve:

$$\begin{aligned} x &= \frac{3y - 2}{5}, \\ 5x &= 3y - 2, \\ y &= \frac{5x + 2}{3}. \end{aligned}$$

Therefore,

$$\boxed{f^{-1}(x) = \frac{5x + 2}{3}}.$$

Indeed,

$$f(f^{-1}(x)) = \frac{3\left(\frac{5x+2}{3}\right) - 2}{5} = x.$$

Example 2.19: [

cyan]Interpreting an inverse in context Suppose a service charge is modelled by

$$F(x) = 1.06x + 12,$$

where $x \geq 0$ is the original amount and $F(x)$ is the final amount after a 6% proportional charge and a fixed RM 12 fee. Solving

$$y = 1.06x + 12$$

for x gives

$$F^{-1}(y) = \frac{y - 12}{1.06}, \quad y \geq 12.$$

Thus, the inverse recovers the original amount from the final amount. For example, $F^{-1}(118) = 100$.

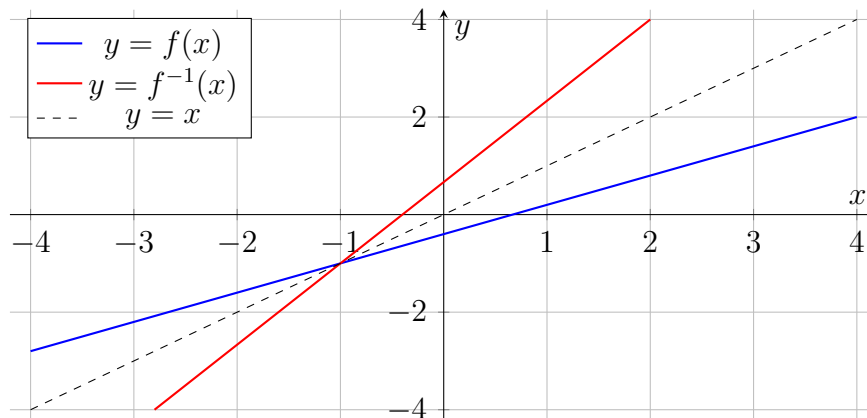


Figure 2.8: A function and its inverse reflected in the line $y = x$.

The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another in the line $y = x$. If a function is not one-to-one, its domain may sometimes be restricted so that an inverse can be defined. For example, $f(x) = x^2$ has no inverse on all of $\mathbb{R}$, but on $[0, \infty)$ its inverse is $f^{-1}(x) = \sqrt{x}$.

2.11 Exercises

Answers to these exercises are provided in section C.2.

1. Determine whether

$$R = \{(-2, 1), (0, 3), (2, 1), (2, 5)\}$$

defines a function of x . State its domain and range.

2. Find the domain of

$$h(x) = \frac{\sqrt{5-x}}{x+2}.$$

3. If $f(x) = 3x^2 + 2x - 1$, find $f(-2)$.

4. Describe the transformations that take $y = f(x)$ to

$$y = -2f(3(x-1)) + 4.$$

5. For $g(x) = -3(x+2)^2 + 5$, state the vertex, axis of symmetry, domain and range.

6. Rewrite

$$r(x) = \frac{2x+1}{x-3}$$

as a translated reciprocal function. Hence state its vertical and horizontal asymptotes, domain and range.

7. Solve $3^{2x-1} = 27$.

8. State the domain, range and vertical asymptote of $y = \ln(x-4) + 2$.

9. Let $f(x) = x+1$ and $g(x) = x^2 - 4$. Find $(f/g)(x)$ and state its domain.

10. Let $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

11. Find the inverse of

$$f(x) = \frac{4x-5}{3}.$$

12. The function $q: \mathbb{R} \rightarrow \mathbb{R}$ is given by $q(x) = x^3$. Decide whether it is one-to-one and onto, and state its inverse.

13. Describe each transformation or write the required transformed function.

- (a) In $f(x) = -2x + b$, the value of b is increased by 4. How does the graph move?
- (b) Write the function obtained when $y = |x|$ is translated 2 units to the right.
- (c) Describe the transformation from $y = x^2$ to $y = (x-2)^2 + 3$.
- (d) Describe the transformation from $y = x^2$ to $y = (x+3)^2 - 2$.
- (e) Translate $y = 2x^2 - 1$ three units to the left and write the new equation.

- (f) The graph of $f(x) = (x - 2)^2 + 4$ is transformed into $g(x) = (x - 5)^2 + 4$. Describe the transformation.
- (g) The graph of $f(x) = -3(x - 1)^2 + 2$ is transformed into $g(x) = -3(x - 1)^2 - 5$. Describe the transformation.
- 14.** Answer the following questions about stretches, compressions and reflections.
- (a) Compare the width of $y = \frac{1}{2}|x|$ with that of $y = |x|$.
- (b) What happens to the graphs of $y = x^2$ and $y = |x|$ when each output is multiplied by $\frac{1}{2}$?
- (c) Arrange $y = -5x^2$, $y = \frac{1}{2}x^2$ and $y = 3x^2$ from narrowest to widest.
- (d) For $a > 1$, compare $y = ax^2$ with $y = x^2$ and state its direction of opening.
- (e) A positive coefficient a in $y = ax^2$ is replaced by $-2a$. Describe the two effects on the graph.
- (f) If $k < -1$, describe the graph of $y = kx^2$ relative to $y = x^2$.
- (g) Describe the transformation from $y = x^2$ to $y = 2(x - 3)^2$.
- (h) Describe the transformation from $y = x^2$ to $y = 3(x - 2)^2 + 1$.
- 15.** Use function notation to describe each transformation.
- (a) Write g in terms of f when the graph of $y = f(x)$ is translated 2 units right and 1 unit up.
- (b) If $g(x) = f(x) + k$ and the graph of g lies 2 units below the graph of f , find k .
- (c) Describe the transformation from $y = f(x)$ to $y = f(x + 1) - 2$.
- 16.** Describe, in order, the transformations from $y = g(x)$ to each graph.
- (a) $y = -g(x - 2) + 3$
- (b) $y = g(-x) - 5$
- (c) $y = g(-x + 2)$
- (d) $y = -g(x - 4) + 1$
- (e) $y = -g(-x + 1) - 4$
- 17.** A piecewise-linear graph passes successively through $(-2, 2)$, $(0, 2)$, $(2, 0)$ and $(4, 4)$. Sketch the graphs of $y = f(-x)$, $y = -f(x)$ and $y = -f(-x)$.
- 18.** A curve $y = f(x)$ has a minimum point at $(2, -1)$. State the corresponding turning point for
- (a) $y = f(x + 2)$;
- (b) $y = f(x - 2)$;
- (c) $y = f(x) + 2$;

- (d) $y = f(x) - 2$;
- (e) $y = f(-x) + 2$;
- (f) $y = -f(x) + 2$.

19. Match each function to the graph description A–F.

- | | | | |
|---|-------------------------------------|---|--|
| A | vertex $(1, 2)$, narrow, opens up | B | vertex $(-1, -2)$, wide, opens up |
| C | vertex $(1, -2)$, narrow, opens up | D | vertex $(1, 2)$, narrow, opens down |
| E | vertex $(-1, 2)$, narrow, opens up | F | vertex $(-1, -2)$, narrow, opens down |

- (a) $2(x - 1)^2 - 2$
- (b) $\frac{1}{2}(x + 1)^2 - 2$
- (c) $-2(x - 1)^2 + 2$
- (d) $2(x + 1)^2 + 2$
- (e) $-2(x + 1)^2 - 2$
- (f) $2(x - 1)^2 + 2$

20. Find the transformed function $g(x)$.

- (a) $f(x) = 3x$ is reflected in either coordinate axis, translated 2 units right and 3 units down.
- (b) $f(x) = \frac{1}{2}x^2$ is reflected in the x -axis, translated 1 unit left and 2 units up.
- (c) $f(x) = 3x^2$ is translated 2 units right and 1 unit down.

21. Describe how g is obtained from f .

- (a) $f(x) = \frac{1}{2}x^2$ and $g(x) = \frac{1}{2}(x - 2)^2 - 3$
- (b) $f(x) = x^2$ and $g(x) = -(x - 2)^2 + 2$

22. For each function, state the intercepts, domain and range, and give the main features needed for a sketch.

- (a) $g(x) = -3x + 1$
- (b) $h(x) = -(x - 1)^2 + 4$
- (c) $p(x) = 2(x - 1)^2 + 1$

23. State the vertical and horizontal asymptotes, domain and range.

- (a) $f(x) = -4/x$
- (b) $f(x) = 4/(x - 1) + 1$
- (c) $f(x) = -3/(x - 1) - 1$

24. State the vertical and horizontal asymptotes, domain and range, and sketch each graph.

(a) $f(x) = 3/(x + 1) - 2$

(b) $f(x) = 3/x + 1$

(c) $f(x) = 2/(x - 3) + 1$

25. Describe how g is obtained from f .

(a) $f(x) = 3/x, g(x) = 3/(x + 3) - 2$

(b) $f(x) = e^x, g(x) = -e^{x+2} - 3$

(c) $f(x) = e^x, g(x) = 1 - e^{-x}$

(d) $f(x) = \ln x, g(x) = -\ln(1 - x) + 2$

26. Find $g(x)$ after the stated transformations.

(a) $f(x) = 2/x$: right 2, then down 1

(b) $f(x) = 3/x$: reflection in either coordinate axis, left 2, then down 2

(c) $f(x) = \ln x$: left 3, then down 2

(d) $f(x) = e^x$: left 3, then up 2

27. For each function, state all intercepts, asymptotes, domain and range, and give the main features needed for a sketch.

(a) $g(x) = -5/(x - 3) + 2$

(b) $s(x) = -2 - \ln(3 - x)$

(c) $t(x) = \ln(-2 - x) + 4$

(d) $r(x) = 2 - e^{2-x}$

28. For

$$f(x) = -2(x - 3)^2 + 8,$$

state the vertex, axis of symmetry, intercepts, domain, range and intervals on which the function is increasing or decreasing.

29. The percentage $P(t)$ of respondents supporting a proposal is 46% at $t = 2$ and 67% at $t = 8$, where t is measured in years. Find and interpret the average rate of change over this interval.

30. Determine whether each function is even, odd or neither.

(a) $f(x) = x^4 - 3x^2 + 1$

(b) $g(x) = x^3 - 2x$

(c) $h(x) = x^2 + x$

- 31.** Demand and supply are modelled by

$$D(p) = 180 - 5p, \quad S(p) = 30 + 10p.$$

Find the price at which the two functions intersect and the common output at that price.

- 32.** For the piecewise transport charge

$$C(d) = \begin{cases} 5 + 1.50d, & 0 \leq d \leq 10, \\ 20 + 1.10(d - 10), & d > 10, \end{cases}$$

find $C(8)$ and $C(25)$. Show that the two formulas agree at $d = 10$.

- 33.** For

$$A(t) = 75(1 - e^{-0.3t}), \quad t \geq 0,$$

state the contextual domain, range, initial value and horizontal asymptote. Find $A(5)$ and interpret the limiting value.

- 34.** Let $f(x) = \sqrt{x+1}$ and $g(x) = 2x - 5$. Find $f \circ g$ and $g \circ f$, stating the domain of each composite.
- 35.** A final charge is given by $F(x) = 1.06x + 12$ for $x \geq 0$. Find F^{-1} and its domain. Use it to recover the original amount when the final charge is RM 118.

- 36.** The piecewise function

$$h(x) = \begin{cases} ax + 2, & x < 1, \\ x^2 + b, & x \geq 1 \end{cases}$$

joins without a jump at $x = 1$ and satisfies $h(2) = 7$. Find a and b .

- 37.** Explain why a prediction made outside the observed range of input values should generally be treated with greater caution than one made within that range.
- 38.** A model for the number of available places in a programme is

$$V(t) = 500 - 20t,$$

where t is the number of months after registration opens. State a reasonable contextual domain and range if the model is used only until all places are filled. Find when this occurs.

- 39.** Selected values of a function are shown below.

x	0	2	5	9
$f(x)$	12	18	24	20

Find the average rate of change over $[0, 5]$ and over $[5, 9]$. Interpret the signs of the two answers.

40. Write

$$h(x) = \ln(5 - 2x)$$

as a composition $f \circ g$ of two simpler functions, and state the domain of h .

41. Let

$$f(x) = (x - 1)^2 + 4, \quad x \geq 1.$$

Explain why the domain restriction makes f one-to-one. Find f^{-1} , including its domain and range.

Additional practice: grouped skills review

42. Evaluate each function at the stated input.

- (a) $f_0(x) = 2x^2 - 3x$, at $x = 1$
- (b) $f_1(x) = 2x^2 - 3x + 1$, at $x = 2$
- (c) $f_2(x) = 2x^2 - 3x + 2$, at $x = 3$
- (d) $f_3(x) = 2x^2 - 3x + 3$, at $x = 4$
- (e) $f_4(x) = 2x^2 - 3x + 4$, at $x = 5$
- (f) $f_5(x) = 2x^2 - 3x + 5$, at $x = 6$

43. State the domain of $f_k(x) = 1/(x - k)$ for each value $k = 1, 2, 3, 4, 5, 6$.

44. State the domain of each function.

- (a) $\sqrt{x + 2}$
- (b) $\sqrt{x + 1}$
- (c) $\sqrt{x}$
- (d) $\sqrt{x - 1}$
- (e) $\sqrt{x - 2}$
- (f) $\sqrt{x - 3}$

45. Describe the transformation from $y = f(x)$ to each graph.

- (a) $y = f(x - 1) + 2$
- (b) $y = f(x - 2) + 3$
- (c) $y = f(x - 3) + 4$
- (d) $y = f(x - 4) + 5$
- (e) $y = f(x - 5) + 6$
- (f) $y = f(x - 6) + 7$

46. State the vertex, axis of symmetry and range of each quadratic.

- (a) $(x - 1)^2 - 2$

(b) $(x - 2)^2 - 1$

(c) $(x - 3)^2$

(d) $(x - 4)^2 + 1$

(e) $(x - 5)^2 + 2$

(f) $(x - 6)^2 + 3$

47. Let $f(x) = 2x + 1$ and $g_k(x) = x^2 + k$. Find $(f \circ g_k)(x)$ for $k = 0, 1, 2, 3, 4, 5$.

48. Find the inverse of each function.

(a) $f(x) = 2x + 1$

(b) $f(x) = 3x + 2$

(c) $f(x) = 4x + 3$

(d) $f(x) = 5x + 4$

(e) $f(x) = 6x + 5$

(f) $f(x) = 7x + 6$

49. Find the average rate of change of $h_k(x) = x^2 + kx$ from $x = k$ to $x = k + 2$ for $k = 0, 1, 2, 3, 4, 5$.

50. For $k = 0, 1, 2, 3, 4, 5$, a social indicator is modelled by

$$S_k(t) = (k + 2)t + (10 + 3k).$$

Find $S_k(10)$ and interpret the slope in each model.

51. For $k = 1, 2, 3, 4, 5, 6$, let

$$p_k(x) = \begin{cases} x + k, & x < 0, \\ x^2 - k, & x \geq 0. \end{cases}$$

Find $p_k(-2)$ and $p_k(3)$.

Applications: interpreting functions

52. The relation is given by

x	1	2	2	3
y	4	5	7	8

Determine whether it defines y as a function of x . State its domain and range.

53. Cumulative survey responses are modelled by $R(t) = 120 + 18t$, where t is measured in weeks.

(a) Find $R(6)$.

(b) Interpret the intercept and slope.

- (c) State a contextual domain if the model is used only until 300 responses have been collected.
- 54.** A training programme costs $C(n) = 500 + 12n$ ringgit for n participants, where at most 200 people can attend.
- (a) State an appropriate discrete domain and the corresponding range.
- (b) Find $C(150)$.
- 55.** An illustrative deprivation score is $B(x) = \sqrt{100 - x}$, where x is a percentage between 0 and 100.
- (a) State the contextual domain and range.
- (b) Find $B(64)$.
- 56.** The average administrative cost per participant is

$$A(n) = \frac{1200}{n} + 25,$$

where n is a positive whole number.

- (a) State the contextual domain.
- (b) Find $A(40)$.
- (c) Describe what happens to $A(n)$ as n becomes large.
- 57.** A grant is determined by annual income y :

$$G(y) = \begin{cases} 1200, & 0 \leq y < 20\,000, \\ 800, & 20\,000 \leq y < 30\,000, \\ 0, & y \geq 30\,000. \end{cases}$$

Find $G(18\,500)$, $G(25\,000)$ and $G(32\,000)$, and identify the income thresholds where the function jumps.

- 58.** The deviation of an index value x from a target of 50 is $D(x) = |x - 50|$.
- (a) Find $D(42)$ and $D(67)$.
- (b) State the vertex and range of the graph.
- (c) Interpret $D(x)$.

Applications: linear and piecewise models

59. A taxi fare is $F(d) = 4.50 + 1.80d$, where d is kilometres.

- (a) Find the fare for 12 kilometres.
- (b) Find the distance corresponding to a fare of RM 31.50.
- (c) Interpret the intercept and slope.

60. Weekly pay for h hours is

$$W(h) = \begin{cases} 18h, & 0 \leq h \leq 40, \\ 720 + 27(h - 40), & h > 40. \end{cases}$$

Find $W(38)$ and $W(46)$, and verify that the formulas agree at $h = 40$.

61. A project has RM 12 000 remaining and spends RM 750 per month, so $B(m) = 12\,000 - 750m$.

- (a) State the contextual domain and range until the fund is empty.
- (b) Find when the fund is exhausted.

62. Monthly rent is modelled by $R(t) = 900 + 35t$, where t is years after the start of a tenancy.

- (a) Find $R(5)$.
- (b) Find when the model predicts rent of RM 1250.
- (c) Interpret the slope.

63. A service has cost $C(q) = 2000 + 15q$ and revenue $R(q) = 25q$. Find the break-even output and the common cost and revenue there.

64. A mobile-data charge is

$$M(g) = \begin{cases} 30, & 0 \leq g \leq 5, \\ 30 + 6(g - 5), & g > 5, \end{cases}$$

where g is gigabytes. Find $M(3)$ and $M(9)$, and show that the model is continuous at $g = 5$.

65. Average monthly income is RM 2500 at $t = 0$ and RM 2900 at $t = 4$.

- (a) Construct a linear model $I(t)$.
- (b) Predict the income at $t = 7$.
- (c) State why a much longer-term prediction may be unreliable.

Applications: quadratic functions

66. Programme revenue is modelled by $R(q) = 60q - 0.2q^2$.
- (a) Find the output that maximises revenue.
 - (b) Find the maximum revenue.
 - (c) Find the zeros and state a reasonable contextual domain.
67. A satisfaction score is modelled by $S(h) = -2(h - 5)^2 + 80$, where h is weekly service time. Find the maximum score, the value of h at which it occurs, and $S(3)$.
68. A programme's operating cost is $C(x) = x^2 - 20x + 180$.
- (a) Write C in completed-square form.
 - (b) Find the minimum cost and the value of x that produces it.
69. Campaign reach is modelled by $P(t) = -5t^2 + 60t + 200$. Find the initial reach, the time of maximum reach and the maximum predicted reach.
70. The difference between two group scores is $D(x) = x^2 - 8x + 12$.
- (a) Find the values at which the scores are equal.
 - (b) Determine where $D(x) < 0$ and interpret the sign.
71. A quadratic response reaches a maximum value of 90 at $t = 4$ and has value 42 at $t = 0$. Find its equation in the form $Q(t) = a(t - 4)^2 + 90$.
72. Attendance at ticket price p is $A(p) = 500 - 20p$. Revenue is $R(p) = pA(p)$.
- (a) Write $R(p)$ as a quadratic.
 - (b) Find the price that maximises revenue.
 - (c) Find the corresponding attendance and revenue.

Applications: exponential and logarithmic functions

73. A population is modelled by $P(t) = 50\,000(1.03)^t$ for $t \geq 0$. Find $P(5)$ and interpret the factor 1.03.
74. Equipment value is $V(t) = 80\,000(0.85)^t$ for $t \geq 0$. Find $V(4)$, state the horizontal asymptote and give the contextual range.
75. An approval measure follows $A(t) = 70 - 25e^{-0.4t}$ for $t \geq 0$. Find $A(0)$, $A(5)$ and the limiting value as t becomes large.
76. An income score is $L(x) = 10 \ln(x/1000)$ for $x > 0$. State its domain and range, and find $L(1000)$ and $L(1000e)$.
77. Participation is modelled by $N(t) = 200(1.06)^t$. Find when the model first reaches 300, both as a continuous time and in complete periods.

78. For $f(x) = \ln(x - 2) + 3$:
- (a) state the domain and range;
 - (b) find $f^{-1}(x)$ and its domain.
79. Compare the models $L(t) = 100 + 12t$ and $E(t) = 100(1.10)^t$ at $t = 10$. Which is larger, and what does this illustrate about repeated percentage growth?

Applications: composition and inverse functions

80. Let $T(x) = 1.06x$ add tax and let $F(x) = x + 12$ add a fixed fee. Find $F \circ T$ and $T \circ F$. Explain why their results differ.
81. Let $g(x) = 0.90x$ represent income after a deduction and $f(x) = x + 200$ add an allowance. Find $f \circ g$ and $g \circ f$ and interpret the difference.
82. Output after t months is $q(t) = 50 + 5t$, and cost at output q is $C(q) = 20q + 500$. Find $(C \circ q)(t)$ and the cost after six months.
83. A standardised score is $S(x) = 10(x - 50)$. Find S^{-1} and use it to recover the original value associated with a standardised score of 220.
84. Demand is $q(p) = 500 - 4p$. Find the inverse demand function and the price corresponding to demand of 300 units.
85. Let $f(x) = (x - 3)^2 + 2$ for $x \geq 3$. Explain why f is one-to-one, then find its inverse, including the inverse's domain and range.
86. Let $f(x) = \sqrt{x}$ and $g(x) = 2x - 6$. Find $f \circ g$ and $g \circ f$, stating the domain of each composite.

Applications: rates, tables and fitted models

87. Selected values of an index are

t	0	2	5	8
$I(t)$	50	62	77	74

Find the average rates of change over $[0, 5]$ and $[5, 8]$ and interpret their signs.

88. Total cost is $C(q) = 1000 + 20q$ for $q > 0$. Derive the average-cost function $A(q) = C(q)/q$, then find $A(50)$ and $A(200)$.
89. A policy score is $P(t) = t^2 + 4t + 20$. Find its average rate of change from $t = 1$ to $t = 4$.
90. A fitted model is $\hat{y} = 12 + 2.5x$. For $x = 8$, find the predicted value and the residual when the observed value is 29. Interpret the sign of the residual.

91. Employment and population are modelled by $E(t) = 500 + 40t$ and $N(t) = 10\,000 + 200t$. Find the employment share $E(t)/N(t)$ at $t = 0$ and $t = 10$.

92. The table gives

x	0	1	2	3
y	5	8	11	14

Determine whether the relationship is linear, find its equation and predict y when $x = 10$.

93. A fitted line $y = 4x + 20$ was obtained using data with $10 \leq x \leq 50$. Find the predictions at $x = 30$ and $x = 80$, classify each as interpolation or extrapolation and state which is less reliable.

Applications: modelling and critical interpretation

94. Demand is modelled by $q(p) = 400 - 5p$. If price and demand must both be non-negative, state the contextual domain and range.

95. Tax is modelled by

$$T(y) = \begin{cases} 0.10y, & 0 \leq y \leq 30\,000, \\ 3000 + 0.20(y - 30\,000), & y > 30\,000. \end{cases}$$

Find $T(25\,000)$ and $T(40\,000)$ and verify that the formulas agree at the threshold.

96. A fitted crime-rate model is $C(u) = 12 - 0.4u$ for $0 \leq u \leq 20$, where u is an intervention-intensity score. Interpret the slope and intercept, find $C(20)$ and state why the slope alone does not establish causation.
97. A programme's participation doubles each period, beginning with 200 participants at $t = 0$. Choose an appropriate function family, write a model and find the prediction at $t = 5$.
98. A fitted graph rises as education spending increases. Explain why this pattern alone does not prove that increasing spending causes the response variable to rise.
99. A benefit is

$$B(y) = \begin{cases} 1000, & y < 25\,000, \\ 0, & y \geq 25\,000. \end{cases}$$

Find $B(24\,999)$ and $B(25\,000)$ and describe the practical effect of the threshold.

100. A fare has a fixed charge of RM6 plus RM2 per kilometre, but the total charge is capped at RM30.

- Write a piecewise function for distance $d \geq 0$.
- State its domain and range.
- Identify where the graph changes from increasing to constant.

Chapter 3

Logic and Sets

Chapter guide

Move from propositions and truth tables to set operations, then use the addition and multiplication principles as the foundation for permutations and combinations.

Selected background sources: (London School of Economics and Political Science, n.d.; Schwarz, 2021).

Logic provides precise rules for deciding whether an argument is valid, whereas set theory provides a language for organising mathematical objects. These topics are closely connected: logical connectives describe conditions, and sets collect the objects that satisfy those conditions. The final part of this chapter develops counting methods for determining how many selections or arrangements are possible.

3.1 Propositions and logical reasoning

A *proposition* is a declarative statement that has exactly one truth value: it is either true or false. The symbols T and F are used for these truth values.

Definition 3.1: Proposition

A proposition is a statement that is either true or false, but not both.

For example, “ $2 + 3 = 7$ ” is a proposition with truth value false, while “Shah Alam is in Selangor” is a proposition with truth value true. Questions, commands and open sentences are not propositions. Thus, “Close the window” and “Is it cold outside?” are not propositions. The sentence “ $x < 2$ ” is also not a proposition until a value of x is given or the variable is quantified.

Example 3.1: Recognising propositions

Classify each sentence.

- (a) The grass is green.
- (b) Close the window.
- (c) Is it cold outside?
- (d) x is less than 2.
- (e) $2 + 3 = 7$.

Sentences (a) and (e) are propositions because each can be assigned a truth value. Sentence (e) is false. Sentences (b) and (c) are a command and a question, respectively, while (d) is an open sentence whose truth depends on x .

Lower-case letters such as p , q and r are commonly used to represent propositions. Simple propositions can be joined by *logical connectives* to form compound propositions.

3.1.1 Negation, conjunction and disjunction

The *negation* of p , written $\neg p$, means “not p ”. It reverses the truth value of p . For example, if

p : It is raining,

then

$\neg p$: It is not raining.

The *conjunction* $p \wedge q$ means “ p and q ”. It is true only when both component propositions are true. The *disjunction* $p \vee q$ means “ p or q ”. Unless stated otherwise, mathematics uses the inclusive meaning of *or*, so $p \vee q$ is true when either proposition or both propositions are true.

p	q	$\neg p$	$p \wedge q$	$p \vee q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

Table 3.1: Truth values for negation, conjunction and disjunction.

3.1.2 Conditional statements

The *conditional* statement

$$p \rightarrow q$$

is read “if p , then q ”. The proposition p is the *hypothesis* or *antecedent*, and q is the *conclusion* or *consequent*. A conditional is false only when its hypothesis is true and its conclusion is false.

For example, suppose a parent promises:

If you enter university, then we will buy you a car.

The promise is broken only if the student enters university but the parents do not buy the car. If the student does not enter university, the conditional statement is not contradicted, regardless of whether a car is bought.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Table 3.2: Truth table for a conditional statement.

Conditional statements do not necessarily express a cause-and-effect relationship. Their truth value depends only on the truth values of the hypothesis and conclusion.

3.1.3 Constructing truth tables

A truth table lists every possible combination of truth values. With n simple propositions, the table has 2^n rows. Operations inside parentheses are evaluated first, followed by negation, conjunction, disjunction and then the conditional.

Example 3.2: A truth table for a compound proposition

Construct the truth table for

$$(p \wedge q) \rightarrow \neg p.$$

p	q	$p \wedge q$	$\neg p$	$(p \wedge q) \rightarrow \neg p$
T	T	T	F	F
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

The compound proposition is false only in the first row.

3.2 Sets and set notation

A *set* is a well-defined collection of distinct objects. The objects are called the *elements* or *members* of the set. Sets are written using braces; for example,

$$A = \{1, 2, 3, 4\}.$$

The order of the elements does not matter, and repeated elements are listed only once. Thus,

$$\{1, 2, 3, 4\} = \{3, 1, 4, 2\}.$$

The notation $a \in A$ means that a is an element of A , while $a \notin A$ means that it is not.

The *empty set*, written $\emptyset$ or $\{\}$, contains no elements.

3.2.1 Subsets and proper subsets

Set A is a *subset* of set B , written $A \subseteq B$, if every element of A is also an element of B . Every set is a subset of itself, and the empty set is a subset of every set.

Set A is a *proper subset* of B , written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$. For example, if

$$C = \{1, 2, 3\}, \quad D = \{1, 3\}, \quad E = \{3, 2, 1\},$$

then

$$D \subsetneq C, \quad C = E, \quad C \subseteq E, \quad C \not\subseteq E.$$

Care is needed to distinguish an element from a set containing that element. If $B = \{4, 9, 2, 7\}$, then

$$7 \in B \quad \text{but} \quad \{7\} \subseteq B.$$

Two sets are equal precisely when each is a subset of the other:

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

3.2.2 Cardinality and power sets

The *cardinality* of a finite set A , written $n(A)$ or $|A|$, is the number of elements in the set. The *power set* $\mathcal{P}(A)$ is the set of every subset of A , including $\emptyset$ and A itself.

For example, if $A = \{a, b, c\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

There are $8 = 2^3$ subsets. In general, if $n(A) = m$, then

$$n(\mathcal{P}(A)) = 2^m.$$

Exactly one of these subsets is not proper, namely A itself. Therefore a finite set with m elements has $2^m - 1$ proper subsets.

Example 3.3: Counting subsets

Let

$$S = \{a, b, c, d, e, f\}.$$

Since $n(S) = 6$, the number of subsets is

$$2^6 = 64,$$

and the number of proper subsets is

$$2^6 - 1 = 63.$$

Nested sets must be handled carefully. If

$$A = \{0, \{0, 1\}\},$$

then A has two elements, so

$$\mathcal{P}(A) = \{\emptyset, \{0\}, \{\{0, 1\}\}, \{0, \{0, 1\}\}\}.$$

3.3 Set operations and Venn diagrams

Let A and B be subsets of a universal set U .

- The *union* $A \cup B$ contains the elements in A , in B , or in both.
- The *intersection* $A \cap B$ contains the elements common to both sets.
- The *complement* A^c , also written A' , contains the elements of U that are not in A .

Sets are *disjoint* when they have no common elements:

$$A \cap B = \emptyset.$$

The relations

$$A \cap B \subseteq A, \quad A \cap B \subseteq B, \quad A \subseteq A \cup B, \quad B \subseteq A \cup B$$

hold for all sets A and B .

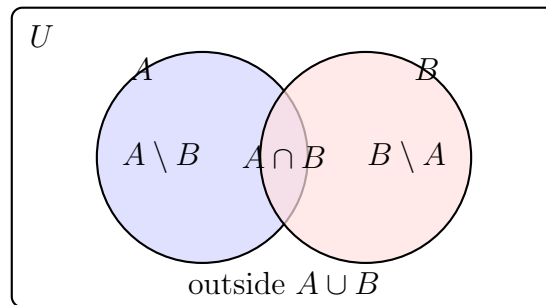


Figure 3.1: The regions determined by two sets in a universal set.

Example 3.4: Union and intersection with complements

Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\},$$

$$A = \{1, 3, 4, 7, 9\}, \quad B = \{1, 2, 7, 8\}.$$

Then

$$A \cup B = \{1, 2, 3, 4, 7, 8, 9\},$$

$$A \cap B = \{1, 7\},$$

$$A^c = \{2, 5, 6, 8\}.$$

For finite sets, the addition rule for cardinalities is

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

The intersection is subtracted because its elements were counted once in $n(A)$ and again in $n(B)$.

3.4 Basic counting principles

Counting principles determine the number of possible outcomes without requiring every outcome to be listed.

3.4.1 The addition principle

If one task can be performed in m ways and a mutually exclusive second task can be performed in n ways, then choosing one of the tasks can be done in

$$m + n$$

ways. The tasks must not be performed simultaneously.

Example 3.5: Choosing a method of travel

Karina can travel from Kuala Lumpur to Penang using one of three flights or one of two trains. The number of choices is

$$3 + 2 = \boxed{5}.$$

3.4.2 The multiplication principle

If a process consists of successive choices with m possibilities for the first choice and n possibilities for the second choice, then there are

$$mn$$

possible outcomes. More generally, if the successive stages have $n_1, n_2, \dots, n_k$ possibilities, the total is

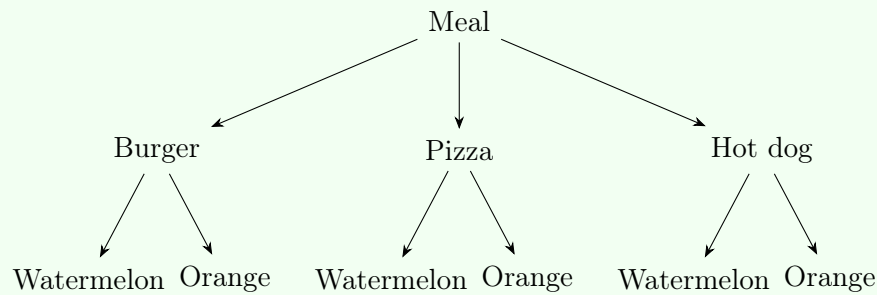
$$n_1 n_2 \cdots n_k.$$

Example 3.6: Choosing a meal

Sam chooses one main course from a burger, pizza or hot dog, and one drink from watermelon juice or orange juice. The total number of meals is

$$3 \times 2 = \boxed{6}.$$

The possibilities are represented by the following tree.

**3.4.3 Factorials**

For a positive integer n , the factorial $n!$ is defined by

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1,$$

with

$$0! = 1.$$

Factorials arise naturally when arranging distinct objects. For example, five different books can be arranged on a shelf in

$$5! = 120$$

ways.

3.5 Permutations

A *permutation* is an arrangement in which order matters. Thus, the arrangements XYZ and ZXY are different permutations.

3.5.1 Permutations with repetition

Suppose r positions are filled using any of n available objects and an object may be used more than once. Each position has n choices, so the number of arrangements is

$$n^r.$$

Example 3.7: Words and lock codes with repetition

The number of three-letter strings formed from the letters C, A and T, when repetition is allowed, is

$$3^3 = \boxed{27}.$$

A three-digit lock code using the digits 0 to 9 has

$$10^3 = \boxed{1000}$$

possible codes when digits may repeat. A code may begin with zero because it is not being interpreted as a three-digit integer.

3.5.2 Permutations without repetition

The number of ways to arrange all n distinct objects is $n!$. The number of ordered arrangements of r objects selected from n distinct objects is

$${}_nP_r = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n.$$

Equivalently,

$${}_nP_r = n(n-1)(n-2) \cdots (n-r+1).$$

Example 3.8: Basic permutation calculations

(a) Six different letters can be arranged in

$$6! = \boxed{720}$$

ways.

(b) Three people can line up for a picture in

$$3! = \boxed{6}$$

ways.

(c) The number of ordered selections of three horses from eight is

$${}_8P_3 = 8 \times 7 \times 6 = \boxed{336}.$$

(d) If six candidates are available for president, vice-president and treasurer, the positions can be filled in

$${}_6P_3 = 6 \times 5 \times 4 = \boxed{120}$$

ways.

Example 3.9: Arranging books

Ten distinct books can be arranged on a shelf in

$$10! = \boxed{3\,628\,800}$$

ways. If only four ordered spaces are available, the number of arrangements is

$${}_{10}P_4 = 10 \times 9 \times 8 \times 7 = \boxed{5040}.$$

The distinction between elements and positions explains why order matters in a permutation. Selecting A followed by B is a different outcome from selecting B followed by A.

3.5.3 Permutations containing identical objects

If n objects contain groups of $n_1, n_2, \dots, n_k$ identical objects, where

$$n_1 + n_2 + \dots + n_k = n,$$

then the number of distinguishable arrangements is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

Dividing by the factorial of each repeated group removes arrangements that look identical.

Example 3.10: Arranging the letters of STATISTICS

The word STATISTICS contains 10 letters: three S's, three T's, two I's, one A and one C. Hence the number of distinguishable arrangements is

$$\frac{10!}{3!3!2!} = \boxed{50\,400}.$$

Example 3.11: Arrangements of BANANA

The word BANANA contains three A's, two N's and one B.

(a) With no restriction, the number of arrangements is

$$\frac{6!}{3!2!} = \boxed{60}.$$

(b) If the arrangement must begin with N, arrange the remaining five letters B, A, A, A and N:

$$\frac{5!}{3!} = \boxed{20}.$$

(c) If all three A's must remain together, treat AAA as one block. The objects

are then AAA, B, N and N, giving

$$\frac{4!}{2!} = \boxed{12}.$$

3.5.4 Permutations with restrictions

Restriction problems are usually solved by arranging blocks, filling special positions first, or separating the problem into cases.

Example 3.12: Arranging boys and girls on a bench

Five distinct boys and four distinct girls are arranged on a bench.

- (a) With no restriction, there are

$$9! = \boxed{362\,880}$$

arrangements.

- (b) To alternate, the only possible gender pattern is

$$BGBGBGBGB.$$

The boys and girls can be permuted independently, giving

$$5!4! = \boxed{2880}.$$

- (c) If all girls sit together, treat them as one block. The block and five boys form six objects, while the girls can be arranged internally:

$$6!4! = \boxed{17\,280}.$$

Example 3.13: Arranging ENGLISH

The seven letters in ENGLISH are distinct. Therefore there are

$$7! = \boxed{5040}$$

unrestricted arrangements.

There are five consonants. If the first two positions must contain consonants, they can be filled in ${}_5P_2$ ways, after which the remaining five letters can be arranged freely. Thus,

$${}_5P_2 \cdot 5! = (5 \times 4)(120) = \boxed{2400}.$$

3.6 Combinations

A *combination* is a selection in which order does not matter. For example, selecting $\{A, B, C\}$ is the same as selecting $\{B, A, C\}$. The number of ways to select r objects from n distinct objects is

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n.$$

Every group of r selected objects can be arranged in $r!$ orders, so permutations and combinations are related by

$${}_nP_r = r! {}nC_r.$$

	Permutation	Combination
Order	matters	does not matter
Typical wording	arrange, rank, assign positions	choose, select, form a group
Formula	${}_nP_r = \frac{n!}{(n-r)!}$	${}_nC_r = \frac{n!}{r!(n-r)!}$

Table 3.3: The distinction between permutations and combinations.

Example 3.14: Choosing letters

The number of ways to choose three letters from A, B, C, D and E is

$${}_5C_3 = \frac{5!}{3!2!} = \boxed{10}.$$

The order of the chosen letters is irrelevant.

Example 3.15: Positions and subcommittees

From a committee of ten people:

- (a) A chairperson, vice-chairperson and secretary can be chosen in

$${}_{10}P_3 = 10 \times 9 \times 8 = \boxed{720}$$

ways because the positions are different.

- (b) A three-person subcommittee can be chosen in

$${}_{10}C_3 = \boxed{120}$$

ways because the members have no assigned order.

Example 3.16: Forming a committee under restrictions

A five-member committee is selected from six boys and four girls.

- (a) For exactly three boys and two girls,

$$\binom{6}{3} \binom{4}{2} = 20 \times 6 = \boxed{120}.$$

- (b) More boys than girls means the committee composition is (3, 2), (4, 1) or (5, 0).
Therefore the number of committees is

$$\begin{aligned} \binom{6}{3} \binom{4}{2} + \binom{6}{4} \binom{4}{1} + \binom{6}{5} \binom{4}{0} &= 120 + 60 + 6 \\ &= \boxed{186}. \end{aligned}$$

The key question in any selection problem is whether changing the order creates a new outcome. If it does, use a permutation; if it does not, use a combination.

3.7 Exercises

Answers to these exercises are provided in section C.3.

- Classify each sentence as a proposition or not a proposition. If it is a proposition, state its truth value.
 - 7 is a prime number.
 - $x + 2 = 5$.
 - Close the door.
 - $3^2 = 8$.
- Write the negation of: “Every student submitted the assignment.”
- Construct a truth table to verify that $p \rightarrow q$ is logically equivalent to $\neg p \vee q$.
- For the conditional statement “If an integer is divisible by 4, then it is even”, write its converse, inverse and contrapositive.
- Let

$$U = \{1, 2, \dots, 10\}, \quad A = \{2, 4, 6, 8, 10\}, \quad B = \{2, 3, 5, 7\}.$$
 Find $A \cap B$, $A \cup B$, A^c and $B \setminus A$.
- In a group of 60 people, 38 study economics, 27 study sociology and 12 study both. Find the number who study at least one of the subjects and the number who study neither.
- List the power set of $\{a, b, c\}$ and state its cardinality.

8. A student owns four shirts, three pairs of trousers and two pairs of shoes. How many different outfits can be formed by choosing one of each item?
9. How many distinct arrangements can be formed from all the letters of the word *MATHEMATICS*?
10. Seven different books are placed on a shelf. In how many ways can they be arranged if two particular books must remain together?
11. A committee of four is selected from six women and five men. How many committees contain exactly two women and two men?
12. A five-member committee is selected from six women and four men. How many committees contain more women than men?

Additional exercises: sets and counting

13. State whether each statement is true or false.
 - (a) $4 \in \{1, 3, 4, 5\}$
 - (b) $2 \notin \{1, 3, 4, 5\}$
 - (c) $\{2, 3, 4\} \subseteq \{3, 4, 5\}$
 - (d) $\{0, 2, 3\} \subseteq \{-1, 0, 1, 2, 3, 4\}$
 - (e) $\{1, 2, 3\}$ is a proper subset of $\{1, 2, 3\}$.
 - (f) $\{1, 2, 3\}$ is a proper subset of $\{1, 2, 3, 4\}$.
14. Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$.
 - (a) Find $n(A)$ and $n(B)$.
 - (b) State the number of subsets of A and of B .
 - (c) List $\mathcal{P}(A)$ and $\mathcal{P}(B)$.
 - (d) State the number of proper subsets of A and of B .
15. A two-set Venn diagram has
$$A \setminus B = \{4\}, \quad A \cap B = \{1, 8\}, \quad B \setminus A = \{11, 13\},$$
with $\{6\}$ outside both sets. Find
 - (a) $A \cap B$;
 - (b) $A \cup B$;
 - (c) A^c ;
 - (d) $A^c \cap B$.

16. In a three-set Venn diagram, the regions contain

region	elements
A only	$\{p\}$
B only	$\{t, u\}$
C only	$\{v\}$
$A \cap B$ only	$\{q, r\}$
$A \cap C$ only	$\emptyset$
$B \cap C$ only	$\{x\}$
$A \cap B \cap C$	$\{s\}$
outside all three sets	$\{w, y\}$

Find

- (a) $A \cap B$;
- (b) $A \cup C$;
- (c) $A^c \cap C^c$;
- (d) $A \cap B \cap C$;
- (e) $A \cap B^c \cap C$;
- (f) $(A \cup C) \cap B^c$.

17. Perform the indicated set operations.

- (a) Let $U = \{1, 2, \dots, 10\}$, let A be the set of even elements of U , and let B be the set of factors of 8 in U . Find $A \cup B$, $A \cap B^c$ and $A \cap B$.
- (b) Let A be the set of distinct letters in *FROST* and B the set of distinct letters in *BARTON*. Find $A \cap B$, $A \cup B$ and $B \setminus A$.
- (c) Let $U = \{1, 2, \dots, 10\}$, let A be the set of prime numbers in U , and let B be the set of multiples of 6 in U . Find $A \cup B$ and $A \cap B$.

18. Use the addition or multiplication principle as appropriate.

- (a) A class has 14 boys and 9 girls. In how many ways can one class monitor be selected?
- (b) A sandwich is made by choosing one of 3 breads, one of 5 fillings and one of 2 sauces. How many sandwiches are possible?
- (c) A person chooses one of 2 shirts, one of 3 pairs of trousers and one of 2 pairs of shoes. How many outfits are possible?

19. How many ordered selections of two objects can be made from three distinct objects?

20. A box contains 2 white, 3 black and 4 red balls, all distinguishable. How many selections of three balls contain at least one black ball?

21. Find the number of distinguishable arrangements of all the letters in each word.

- (a) *ALABAMA*
- (b) *MISSISSIPPI*

Tutorial exercises: logic, sets and counting

- 22.** Decide whether each sentence is a proposition. Where applicable, state its truth value.
- (a) Kuala Lumpur is the capital of Malaysia.
 - (b) Read this carefully.
 - (c) $1 + 2 = 3$.
 - (d) $x + 1 = 2$.
 - (e) What time is it?
- 23.** Let p mean “I attended the conference this month” and q mean “I presented a paper at the conference”. Express in words: (a) $\neg p$; (b) $p \vee q$; (c) $p \rightarrow q$; (d) $\neg p \rightarrow \neg q$.
- 24.** Let p mean “You get an A on the final exam”, q mean “You did every exercise in this book”, and r mean “You get an A in this class”. Translate into symbols.
- (a) You get an A in this class but did not do every exercise.
 - (b) You get an A on the final, do every exercise and get an A in the class.
 - (c) You get an A on the final but did not do every exercise; nevertheless, you get an A in the class.
 - (d) Getting an A on the final and doing every exercise is sufficient for getting an A in the class.
- 25.** In a class of 60 students, 40 like mathematics, 36 like science and 24 like both. Find the number who like mathematics only, science only, either subject, and neither.
- 26.** Erin has 5 tops, 6 skirts and 4 caps. How many outfits can she form by choosing one of each?
- 27.** A password has five characters, each of which may be an English letter or a decimal digit. Repetition is allowed. How many passwords are possible?
- 28.** Five distinct subject books are arranged on a shelf. How many arrangements are possible?
- 29.** (a) In how many ways can six people be arranged in a row? (b) How many ordered arrangements are possible if only three of the six are chosen?
- 30.** A four-speaker debate team poses for a photograph.
- (a) In how many ways can all four speakers stand in a row?
 - (b) In how many ways can a captain and vice-captain be selected?

31. How many distinguishable arrangements of the word *PARRAMATTA* are possible?
32. Six boys occupy six of eight distinct boat seats, four on each side. Find the number of arrangements
- (a) without restrictions;
 - (b) if Aaron and Blake must sit on the port side and William on the starboard side.
33. A company has seven senior and five junior officers. How many four-officer committees contain
- (a) any four officers;
 - (b) four senior officers;
 - (c) three senior and one junior officer;
 - (d) two senior and two junior officers;
 - (e) at least two senior officers?
34. A five-person committee is selected from six men and four women. How many committees are possible if
- (a) there are no restrictions;
 - (b) one specified person must be included;
 - (c) one specified woman must be excluded;
 - (d) there are three men and two women;
 - (e) all members are men;
 - (f) women form a majority?

Additional practice: grouped skills review

35. If p is true and q is false, determine the truth values of $p \wedge q$, $p \vee q$, $\neg p$, $p \Rightarrow q$, $q \Rightarrow p$ and $p \Leftrightarrow q$.
36. Negate each statement.
- (a) All 4 surveyed households own a computer.
 - (b) All 5 surveyed households own a computer.
 - (c) All 6 surveyed households own a computer.
 - (d) All 7 surveyed households own a computer.
 - (e) All 8 surveyed households own a computer.
 - (f) All 9 surveyed households own a computer.
37. Determine the truth value of $p \Rightarrow q$ in each case: $(p, q) = (F, F), (F, T), (T, F), (T, T)$. State the general conclusion when p is false.

38. Find $A \cap B$ and $A \cup B$ for each pair.

- (a) $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$
- (b) $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$
- (c) $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6\}$
- (d) $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{5, 6, 7\}$
- (e) $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{6, 7, 8\}$
- (f) $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $B = \{7, 8, 9\}$

39. Find $|A \times B|$ when $(|A|, |B|)$ is $(2, 3)$, $(3, 4)$, $(4, 5)$, $(5, 6)$, $(6, 7)$ and $(7, 8)$.

40. Find the number of subsets of a set containing 3, 4, 5, 6, 7 and 8 elements, respectively.

41. A code is formed by selecting one prefix, one middle symbol and one suffix. Find the number of codes when the numbers of choices are

$$(2, 3, 4), (3, 4, 5), (4, 5, 6), (5, 6, 7), (6, 7, 8) \quad \text{and} \quad (7, 8, 9).$$

42. Evaluate $5P_3$, $6P_3$, $7P_3$, $8P_3$, $9P_3$ and $10P_3$.

43. Evaluate $\binom{6}{2}$, $\binom{7}{2}$, $\binom{8}{2}$, $\binom{9}{2}$, $\binom{10}{2}$ and $\binom{11}{2}$.

44. Find the number of arrangements of n symbols when two are identical and the remaining symbols are distinct, for $n = 5, 6, 7, 8, 9, 10$.

45. Find the number of four-person committees containing two men and two women when the available numbers (m, w) are $(4, 3)$, $(5, 4)$, $(6, 5)$, $(7, 6)$, $(8, 7)$ and $(9, 8)$.

The legal scenarios below are simplified and hypothetical. They illustrate formal logic and counting rather than the law of any particular jurisdiction.

Applications: legal rules and logical reasoning

46. A fictional rule states that a person is eligible only when the person is a citizen and is at least 18 years old. Let p mean “the person is a citizen” and q mean “the person is at least 18”.

- (a) Express eligibility symbolically.
- (b) Negate the eligibility condition using De Morgan’s law.
- (c) Determine eligibility when p is true and q is false.

47. A permit is denied when an applicant lacks identification or has not paid the fee. Let p mean “identification is supplied” and q mean “the fee is paid”.

- (a) Express denial symbolically.
- (b) Express approval symbolically.

- (c) Explain the connection using De Morgan's law.
48. Consider the statement: "If evidence is admitted, then it has been authenticated." Write its converse, inverse and contrapositive. Which is logically equivalent to the original?
49. A filing by the deadline is necessary for an appeal to be heard. Let H mean "the appeal is heard" and D mean "the filing is on time".
- (a) Express the rule symbolically.
- (b) State its contrapositive.
- (c) Explain why filing on time need not be sufficient by itself.
50. A licence is valid if and only if its fee is paid and the required training is complete. Let these statements be V, F and T . Express the rule symbolically and determine V when F is true and T is false.
51. A hypothetical authorisation is valid when $(p \vee q) \wedge \neg r$, where p means "a warrant exists", q means "valid consent exists" and r means "the authorisation has expired". Determine validity for $(p, q, r) = (T, F, F), (F, T, F), (T, F, T)$ and (F, F, F) .
52. Negate the statement: "Every witness signed at least one statement." Write the negation in ordinary language.
53. Let M mean "the person is a minor", A mean "the person is accompanied" and E mean "the person may enter". Express the rule "an unaccompanied minor may not enter" as a conditional. Give an equivalent form describing a minor who does enter.
54. A tribunal may issue a fine or a warning, but not both. Let F and W represent the two outcomes. Write the exclusive-or condition symbolically and state whether it is true when both outcomes occur.
55. Classify each argument as valid or invalid and name the form.
- (a) If notice is served, the deadline begins. Notice is served. Therefore, the deadline begins.
- (b) If a document is signed, it is enforceable. It is not enforceable. Therefore, it was not signed.
- (c) If a case is registered, it is searchable. It is searchable. Therefore, it is registered.
56. Can the statements $p \Rightarrow q$, $q \Rightarrow r$, p and $\neg r$ all be true simultaneously? Justify the answer.

Applications: legal and administrative sets

57. A database contains 200 cases. Let C be the set of civil cases and A the set of appealed cases. If $|C| = 120$, $|A| = 70$ and $|C \cap A| = 40$, find the numbers in $C \cup A$, in neither set, in C only and in A only.
58. Among 140 applicants, 90 are eligible for legal aid, 75 have representation and 60 satisfy both descriptions. Find the number in at least one group, neither group and each group only.
59. An evidence file contains 8 documents. How many different subsets of documents can be selected? How many of these selections are non-empty?
60. A court may choose any subset of 5 possible remedies. How many remedy combinations are possible, including no remedy? How many contain exactly two remedies?
61. Four courts and six possible hearing dates are available. How many ordered court–date assignments are in the Cartesian product?
62. Let $P = \{1, 2, 3, 4, 5, 6\}$ be permit applications and $R = \{2, 4, 6\}$ the renewed applications. Find $P \setminus R$ and R^c relative to universal set P .
63. Let $A = \{a, b, c, d\}$ and $B = \{c, d, e\}$ represent two legal classifications. Find their symmetric difference.
64. Of 500 submitted forms, 420 are complete. Using complements, find the number of incomplete forms.
65. In 300 compliance files, 160 involve tax rules, 140 safety rules and 100 licensing rules. The pairwise intersections are 70, 50 and 40, and 20 files involve all three. Find the number involving at least one rule category and the number involving none.
66. Let A be the set of appellate cases and C the set of all court cases. Express “every appellate case is a court case” using set notation. If $|A| = 25$ and $|C| = 180$, explain why the inclusion is proper.
67. Determine whether the sets

$$A = \{\text{fraud, theft, bribery}\}, \quad B = \{\text{bribery, fraud, theft}\}$$

are equal. Explain why order does not affect the answer.

Applications: counting in legal procedures

68. A hearing assignment uses one of 3 court codes, one of 4 case categories and one of 3 daily sessions. How many assignments are possible?
69. Five judges are seated in a row for a ceremonial session. How many seating orders are possible?
70. How many three-member review panels can be selected from eight available members?
71. How many six-person juries can be selected from ten eligible citizens if order is irrelevant?
72. A panel must contain two lawyers chosen from five and three community representatives chosen from seven. How many panels are possible?
73. Six agenda items are to be ordered, with two specified items required to appear consecutively. How many orders are possible?
74. For the same six agenda items, how many orders keep the two specified items apart?
75. How many distinguishable arrangements can be formed from the letters of the word APPEAL?
76. Four distinct cases are assigned to four judges, one case per judge. How many assignments are possible?
77. A chair and secretary are selected from nine committee members. How many choices are possible when one person cannot hold both positions?
78. A filing code uses two distinct letters chosen from five followed by three digits, with digit repetition allowed. How many codes are possible?

Applications: surveys and public administration

79. In a survey of 500 people, 320 support a policy, 280 have heard of it and 210 both support it and have heard of it. Find the numbers in at least one set, neither set, support only and awareness only.
80. Respondents may select any non-empty subset of three policy priorities. How many response patterns are possible?
81. Four policy proposals are to be ranked from first to fourth. How many complete rankings are possible?
82. Eight proposals are available and a committee selects an ordered top three. How many outcomes are possible?
83. How many four-person advisory committees can be selected from twelve people?

84. A committee of five is selected from eight adults and four youths. How many committees contain at least one youth?
85. Three distinct grants are awarded to ten organisations, with no organisation receiving more than one grant. How many award patterns are possible?
86. A questionnaire contains six yes–no items. How many complete response patterns are possible?
87. An administrative record contains four binary indicators and one of three regional labels. How many distinct records can be formed?
88. Of 1000 residents, 620 use an education service, 480 use a health service and 250 use both. Find the number using at least one service and the number using neither.
89. Seven interview questions are arranged in order, but the opening and closing questions are fixed. How many arrangements remain for the other questions?

Applications: mixed reasoning and interpretation

90. A rule says that an applicant must submit a passport or a birth certificate, and submitting both is allowed. Is the word “or” inclusive or exclusive? Express the requirement symbolically.
91. Consider: “If a person is licensed, then the person’s name is on the register.” Explain why the converse need not follow from the original statement.
92. A conviction requires admissible evidence. Let C mean “conviction occurs” and E mean “admissible evidence exists”. Express the rule symbolically and explain why observing E does not logically prove C .
93. A benefit requires both residency and low income. If R and L are the relevant sets, express the eligible set. Among 200 applicants, 150 are residents, 90 have low income and 70 are in both sets. Find the number eligible and the number in neither set.
94. An appeal route is determined by one of 3 initial tribunals, one of 2 appellate panels and one of 2 hearing modes. How many routes are possible?
95. Find the number of distinguishable arrangements of the words COURT and LETTER. Explain why the longer word can have more arrangements despite repeated letters.
96. A four-person committee is selected from six lawyers and four researchers. How many committees contain at least one person from each profession?
97. Given $p \Rightarrow q$ and $q \Rightarrow r$, what follows when p is true? What can be concluded about p when r is false?
98. A report says that set A is a subset of set B , but also reports $|A| = 90$ and $|B| = 80$. Explain why the claims are inconsistent.

- 99.** In a population of 250, sets A and B have sizes 140 and 110, and 40 people are in neither set. Find $|A \cap B|$.
- 100.** An automated screening rule flags a record when condition A or condition B is present. Write the logical rule and give two reasons why a logically correct implementation might still produce poor or unfair practical decisions.

Chapter 4

Introduction to Statistics

Chapter guide

Identify populations, samples and variable types before choosing measures of centre, position and variation; finish by interpreting grouped data and statistical displays.

Selected background sources: (Madsen, 2016; Moore et al., 2021).

Statistics provides methods for learning from data. It helps researchers describe observed information, compare groups, identify patterns and use a sample to draw carefully qualified conclusions about a larger population.

4.1 The nature of statistics

Definition 4.1: Statistics

Statistics is the discipline concerned with collecting, organising, summarising, analysing and interpreting data.

Statistical methods are divided into two broad areas.

- *Descriptive methods* organise and summarise observed data using tables, graphs and numerical measures such as the mean, median, range and standard deviation.
- *Inferential statistics* uses data from a sample to estimate, predict or test claims about a population. Confidence intervals, hypothesis tests and regression models are inferential methods.

For example, reporting the average income of the people who answered a survey is descriptive. Using that sample average to estimate the average income of all residents in a city is inferential.

4.1.1 Population and sample

The *population* is the complete collection of individuals, objects or measurements of interest. A *sample* is a subset selected from that population and studied in order to learn about the population.

Example 4.1: Identifying populations and samples

- (a) A paediatrician surveys ten parents of patients about diapers. The population is all parents of the paediatrician's relevant patients, and the sample is the ten parents surveyed.
- (b) A school surveys 100 seniors about cafeteria lunches. The population is all seniors at the school, and the sample is the 100 seniors surveyed.
- (c) An inspector tests seven randomly selected classrooms in an elementary school. The population is all classrooms in that school, and the sample is the seven selected classrooms.

A sample should represent the population as closely as possible. Random selection helps reduce selection bias, although sampling variability remains: different random samples will generally produce different results.

4.1.2 Parameters and statistics

A numerical measure describing a population is a *parameter*. A numerical measure calculated from a sample is a *statistic*. Parameters are usually fixed but unknown, whereas sample statistics vary from sample to sample and are used to estimate parameters.

Measure	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Size	N	n

Table 4.1: Common notation for population parameters and sample statistics.

The average weight of every cat in Penang is a parameter. The average height of 500 selected Malaysian women is a statistic. Similarly, the average test score of all students in a programme is a parameter, whereas the average IQ of 100 students selected from a class of 500 is a statistic.

4.2 Variables and types of data

A *variable* is a characteristic that can take different values among the individuals or objects being studied.

- A *qualitative*, or categorical, variable records a category. A *nominal* variable has no natural ordering, such as blood type or country. An *ordinal* variable has ordered categories, such as low, medium and high satisfaction.

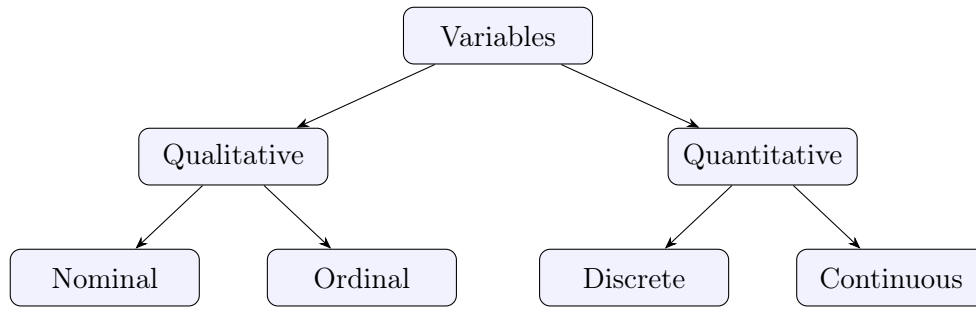


Figure 4.1: A classification of variables.

- A *quantitative* variable records a numerical amount. A *discrete* variable has countable values, such as the number of children in a household. A *continuous* variable may take any value in an interval, such as height, time or mass.

The type of variable determines which summaries and graphs are appropriate. For example, a mean is meaningful for quantitative data but usually not for nominal categories.

4.3 Measures of central tendency

A measure of central tendency describes a typical or central value of a data set. The three most common measures are the mean, median and mode.

4.3.1 Mean for ungrouped data

The arithmetic mean is the sum of the observations divided by their number. For a population,

$$\mu = \frac{\sum_{i=1}^N x_i}{N},$$

and for a sample,

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

The mean uses every observation, but it can be strongly affected by extreme values.

Example 4.2: Mean age of employees

The ages of all eight employees in a small company are

53, 32, 61, 27, 39, 44, 49, 57.

Their population mean age is

$$\mu = \frac{53 + 32 + 61 + 27 + 39 + 44 + 49 + 57}{8} = \frac{362}{8} = \boxed{45.25 \text{ years}}.$$

4.3.2 Median for ungrouped data

The median is the middle observation after the data have been arranged in ascending order. If n is odd, the median is the observation in position

$$\frac{n+1}{2}.$$

If n is even, it is the average of the observations in positions $n/2$ and $n/2+1$. The median is resistant to extreme values.

Example 4.3: Finding medians

For

38, 43, 50, 55, 31, 47, 52,

the ordered data are

31, 38, 43, 47, 50, 52, 55,

so the median is 47.

For

9, 5, 7, 5, 11, 15, 18, 12,

the ordered data are

5, 5, 7, 9, 11, 12, 15, 18.

Thus,

$$\text{Median} = \frac{9+11}{2} = \text{10}.$$

4.3.3 Mode

The mode is the value or category with the greatest frequency. A data set may have one mode, several modes or no mode. Unlike the mean and median, the mode can be used for nominal data.

Example 4.4: Finding modes

(a) The mode of

77, 69, 74, 81, 71, 68, 74, 73

is 74.

(b) In

21, 19, 27, 22, 29, 19, 25, 21, 22, 30,

the values 19, 21 and 22 each occur twice. The data are trimodal.

(c) The values

26, 65, 34, 47, 13, 45, 78

are all distinct, so there is no mode.

Example 4.5: The effect of an extreme value

Consider the house prices

\$2 000 000, \$500 000, \$300 000, \$100 000, \$100 000.

The mean, median and mode are

\$600 000, \$300 000, \$100 000,

respectively. The unusually expensive house pulls the mean upward, while the median and mode are less affected.

4.3.4 Mean for grouped data

When observations are reported in class intervals, the exact values are not known. Each class is represented by its midpoint x_i . If f_i is the class frequency, the estimated grouped mean is

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}.$$

For a complete population distribution, the same calculation estimates μ .

Example 4.6: Mean of a grouped frequency distribution

Class	f_i	x_i	$f_i x_i$
6–10	1	8	8
11–15	2	13	26
16–20	3	18	54
21–25	5	23	115
26–30	4	28	112
31–35	3	33	99
36–40	2	38	76
Total	20		490

Therefore,

$$\bar{x} = \frac{490}{20} = \boxed{24.5}.$$

4.3.5 Median and mode for grouped data

For a grouped distribution, the median is estimated by interpolation within the median class:

$$\tilde{x} = L + \left(\frac{N/2 - F}{f_m} \right) c,$$

where L is the lower class boundary of the median class, N is the total frequency, F is the cumulative frequency before the median class, f_m is the frequency of the median class and c is the class width.

The modal class has the greatest frequency. An interpolated estimate of the mode is

$$\hat{x} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) c,$$

where Δ_1 is the modal-class frequency minus the previous frequency, and Δ_2 is the modal-class frequency minus the following frequency.

Example 4.7: Grouped median and mode

Consider the distribution

Class	Boundaries	f_i	Cumulative f
148–152	147.5–152.5	6	6
153–157	152.5–157.5	11	17
158–162	157.5–162.5	14	31
163–167	162.5–167.5	9	40
168–172	167.5–172.5	3	43
173–177	172.5–177.5	2	45

Since $N/2 = 22.5$, the median class is 158–162. Hence,

$$\tilde{x} = 157.5 + \left(\frac{22.5 - 17}{14} \right) 5 \approx \boxed{159.46}.$$

The same class is modal. Here $\Delta_1 = 14 - 11 = 3$ and $\Delta_2 = 14 - 9 = 5$, so

$$\hat{x} = 157.5 + \left(\frac{3}{3 + 5} \right) 5 = \boxed{159.375}.$$

4.3.6 Comparing the mean, median and mode

For a roughly symmetric unimodal distribution, the mean, median and mode are close together. In a positively skewed distribution, the long right tail usually gives

$$\text{mode} < \text{median} < \text{mean}.$$

In a negatively skewed distribution, the long left tail usually gives

$$\text{mean} < \text{median} < \text{mode}.$$

The mean is appropriate when all values should influence the centre and there are no severe outliers. The median is often preferable for skewed data, while the mode is useful

for identifying the most common value or category.

4.4 Measures of position

Measures of position describe where an observation lies within an ordered data set. Quartiles divide the data into four parts:

- Q_1 has approximately 25% of the observations below it;
- Q_2 is the median, with approximately 50% below it;
- Q_3 has approximately 75% below it.

4.4.1 Quartiles for ungrouped data

After arranging n observations in ascending order, this chapter uses the positions

$$\text{pos}(Q_1) = \frac{n+1}{4}, \quad \text{pos}(Q_2) = \frac{n+1}{2}, \quad \text{pos}(Q_3) = \frac{3(n+1)}{4}.$$

If a position is not an integer, interpolate between the two neighbouring observations. Other statistical conventions may produce slightly different quartile values, so the chosen convention should always be stated.

Example 4.8: Quartiles for ungrouped data

For

4, 8, 3, 10, 1, 17, 7,

the ordered data are 1, 3, 4, 7, 8, 10, 17. The quartile positions are 2, 4 and 6, giving

$$Q_1 = 3, \quad Q_2 = 7, \quad Q_3 = 10.$$

For

3, 2, 3, 8, 10, 7, 18, 14,

the ordered data are 2, 3, 3, 7, 8, 10, 14, 18. The positions are 2.25, 4.5 and 6.75. Linear interpolation gives

$$Q_1 = 3, \quad Q_2 = 7.5, \quad Q_3 = 13.$$

For

5, 2, 11, 8, 10, 7, 19, 15,

the ordered data are 2, 5, 7, 8, 10, 11, 15, 19, so

$$Q_1 = 5.5, \quad Q_2 = 9, \quad Q_3 = 14.$$

4.4.2 Quartiles for grouped data

The k th quartile of grouped data is estimated by

$$Q_k = L + \left(\frac{kN/4 - F}{f_Q} \right) c, \quad k = 1, 2, 3,$$

where L is the lower boundary of the quartile class, F is the cumulative frequency before that class, f_Q is its frequency, c is the class width and N is the total frequency.

Example 4.9: Quartiles for grouped data

Using the grouped distribution from the previous section, $N = 45$.

The position of Q_1 is $45/4 = 11.25$, which lies in the second class:

$$Q_1 = 152.5 + \left(\frac{11.25 - 6}{11} \right) 5 \approx \boxed{154.89}.$$

The position of Q_2 is 22.5, giving

$$Q_2 = 157.5 + \left(\frac{22.5 - 17}{14} \right) 5 \approx \boxed{159.46}.$$

The position of Q_3 is 33.75, which lies in the fourth class:

$$Q_3 = 162.5 + \left(\frac{33.75 - 31}{9} \right) 5 \approx \boxed{164.03}.$$

4.4.3 Five-number summaries and box plots

The *five-number summary* consists of

$$x_{\min}, \quad Q_1, \quad Q_2, \quad Q_3, \quad x_{\max}.$$

A box plot displays these values using a box from Q_1 to Q_3 , a line at the median and whiskers extending towards the smallest and largest non-outlying observations.

Example 4.10: A five-number summary

The lengths, in minutes, of Yoshi's phone calls are

$$5, 8, 12, 15, 16, 16, 18, 20, 21, 25, 30, 35, 36, 40, 50.$$

There are 15 observations, so

$$\boxed{x_{\min} = 5, \quad Q_1 = 15, \quad Q_2 = 20, \quad Q_3 = 35, \quad x_{\max} = 50}.$$

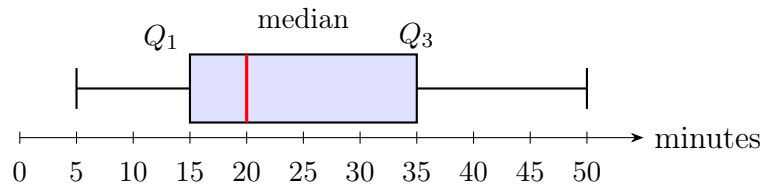


Figure 4.2: A box plot for Yoshi's phone-call lengths.

4.5 Measures of variation

A measure of centre alone may hide important differences between data sets. For example, both

50 000, 60 000, 70 000, 80 000, 90 000

and

10 000, 20 000, 70 000, 120 000, 130 000

have mean and median 70 000, but the second set is much more dispersed. Measures of variation describe this spread.

4.5.1 Range

For ungrouped data,

$$\text{range} = x_{\max} - x_{\min}.$$

For grouped data, it may be estimated by subtracting the lower boundary of the lowest class from the upper boundary of the highest class.

The range is simple but depends only on two observations and is highly sensitive to outliers. For example, replacing the largest value 5 in a data set whose minimum is 1 by 120 changes the range from 4 to 119, even if every other value remains unchanged.

Example 4.11: Range of grouped scores

Suppose test scores are grouped from 21–25 through 46–50. The lowest and highest class boundaries are 20.5 and 50.5, so

$$\text{range} = 50.5 - 20.5 = \boxed{30}.$$

4.5.2 Interquartile range and outliers

The *interquartile range* measures the spread of the middle 50% of the observations:

$$\text{IQR} = Q_3 - Q_1.$$

Because it ignores the most extreme quarters of the data, the IQR is a resistant measure.

The 1.5 IQR rule defines the lower and upper fences as

$$Q_1 - 1.5 \text{ IQR} \quad \text{and} \quad Q_3 + 1.5 \text{ IQR}.$$

An observation outside these fences is treated as a potential outlier.

For Yoshi's phone calls,

$$\text{IQR} = 35 - 15 = 20.$$

The fences are

$$15 - 1.5(20) = -15, \quad 35 + 1.5(20) = 65.$$

Every observation lies between -15 and 65 , so there are no outliers under this rule.

4.5.3 Variance and standard deviation

Variance measures the average squared distance from the mean. Population variance and standard deviation are

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}, \quad \sigma = \sqrt{\sigma^2}.$$

For a sample,

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}, \quad s = \sqrt{s^2}.$$

The denominator $n - 1$ corrects the tendency of a sample to underestimate population variation.

Useful computational forms are

$$\sigma^2 = \frac{\sum x_i^2}{N} - \mu^2$$

and

$$s^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n - 1}.$$

Notice that $\sum x_i^2$ means square each value and then add; it is generally not equal to $(\sum x_i)^2$.

The variance is measured in squared units. The standard deviation has the same units as the original data and is therefore easier to interpret. A small standard deviation indicates observations concentrated near the mean, while a large standard deviation indicates greater spread.

Example 4.12: Population variance and standard deviation

The annual earnings, in thousands of dollars, of all six employees in a small company are

$$88.50, 108.40, 65.50, 52.50, 79.80, 54.60.$$

Here

$$\sum x_i = 449.30, \quad \sum x_i^2 = 35\,978.51, \quad \mu = 74.8833.$$

Since the data include the entire population of employees,

$$\sigma^2 = \frac{35\,978.51}{6} - (74.8833)^2 \approx \boxed{388.90}.$$

Thus,

$$\sigma = \sqrt{388.90} \approx \boxed{19.72}$$

thousand dollars.

4.5.4 Variance and standard deviation for grouped data

For grouped data, use each class midpoint x_i to approximate the values in that class. The population formulas are

$$\mu = \frac{\sum f_i x_i}{N}, \quad \sigma^2 = \frac{\sum f_i x_i^2}{N} - \mu^2.$$

For a sample,

$$s^2 = \frac{\sum f_i x_i^2 - (\sum f_i x_i)^2/n}{n-1}.$$

Example 4.13: Variation in grouped data

Consider the distribution

Class	Frequency	Midpoint
4–6	2	5
7–9	3	8
10–12	7	11
13–15	17	14
16–18	11	17
19–21	8	20

The totals are

$$N = 48, \quad \sum f_i x_i = 696, \quad \sum f_i x_i^2 = 10\,800.$$

Therefore,

$$\mu = \frac{696}{48} = 14.5,$$

$$\sigma^2 = \frac{10\,800}{48} - (14.5)^2 = \boxed{14.75},$$

and

$$\sigma = \sqrt{14.75} \approx \boxed{3.84}.$$

Because midpoints replace the unknown individual observations, these grouped

results are estimates.

Together, measures of centre, position and variation provide a concise description of a distribution. Sound statistical interpretation should also consider the data-collection method, possible bias, unusual observations and the context in which the data were obtained.

4.6 Exercises

Answers to these exercises are provided in section C.4.

1. A university surveys 250 of its 8 000 students and finds that 62% of those surveyed use public transport. Identify the population, sample, parameter and statistic.
2. Classify each variable as categorical or quantitative. For a quantitative variable, also state whether it is discrete or continuous.

- (a) blood group
- (b) number of children in a household
- (c) time taken to complete a test
- (d) satisfaction rating: low, medium or high

3. Find the mean, median and mode of

4, 7, 7, 9, 13.

4. The values 1, 2, 3, 4 have frequencies 2, 3, 4, 1, respectively. Find the mean, median and mode.
5. Estimate the mean of the following grouped distribution.

Class	0–4	5–9	10–14
Frequency	3	5	2

6. Using the quartile-position convention in this chapter, find the five-number summary and interquartile range of

2, 4, 5, 7, 8, 9, 15.

Determine whether any observation is an outlier under the 1.5 IQR rule.

7. Treating 2, 4, 4, 6 as a population, find the range, variance and standard deviation.
8. Treating the same values 2, 4, 4, 6 as a sample, find the sample variance and sample standard deviation.
9. For the data 10, 11, 12, 13, 54, find the mean and median. Which measure better represents a typical value, and why?
10. Classify each variable as discrete or continuous: (a) commuting time; (b) number of credit cards; (c) number of employees; (d) amount of a drug injected; (e) number of cars stolen in a week.
11. Classify each variable as quantitative or qualitative: (a) body temperature; (b) insurance type; (c) blood type; (d) number of previous hospital visits; (e) ethnicity.

12. A national study tests 400 safety gloves obtained from six manufacturers for resistance to cuts, punctures and abrasion. Identify the population, sample and variables of interest.
13. Treat the following data as a population and calculate the mean, median, mode, variance and standard deviation:

9, 4, 8, 12, 5, 1, 8, 18, 9, 13, 18, 1, 5, 10, 15, 8, 15, 8, 8, 5.

14. The weekly hours spent playing video games are grouped below.

Hours	0–4	5–9	10–14	15–19	20–24
Frequency	5	12	20	8	3

Estimate the mean, median and mode.

15. The test scores are

75, 80, 68, 53, 99, 58, 76, 73, 85, 88, 91, 79.

Find Q_1 , Q_2 , Q_3 , the interquartile range and the five-number summary. Describe the skewness indicated by the corresponding box plot.

16. For

23, 17, 32, 60, 22, 52, 29, 38, 42, 92, 27, 46,

find the five-number summary, describe the skewness and identify any outliers using the 1.5 IQR rule.

17. Complete the grouped-data calculations for the distribution

Class	4–6	7–9	10–12	13–15	16–18	19–21
Frequency	2	3	7	17	11	8

by finding the midpoints, $f_i x_i$ and $f_i x_i^2$. Hence estimate the sample variance and standard deviation.

18. For the grouped distribution

Class	25–30	31–36	37–42	43–48	49–54	55–60
Frequency	7	14	35	28	11	5

estimate the range, Q_1 , Q_3 and the interquartile range.

19. A distribution has a long right-hand tail, with the upper whisker of its box plot much longer than the lower whisker. Describe its shape and justify the answer.

Additional practice: interpretation and applications

- 20.** A municipality wishes to estimate the proportion of adult residents who are satisfied with local public services. An interviewer surveys the first 120 adults entering the central library on a Saturday morning.

- (a) Identify the target population and the sample.
- (b) Explain one likely source of bias.
- (c) Suggest a more representative sampling method.

- 21.** A student's module scores and credit weights are shown below.

Score	68	74	81	70
Credit weight	3	4	2	1

Calculate the credit-weighted mean score and explain why the four scores should not simply be given equal weight.

- 22.** The annual household incomes, in thousands of ringgit, are

24, 30, 30, 36, 60.

- (a) Find the mean, median, mode and range.
 - (b) Each household receives an additional RM 5000. State the new mean, median, mode and range.
 - (c) Each original income is instead increased by 10%. State the new mean, median, mode and range, and describe what happens to the standard deviation.
- 23.** Group A contains 40 observations with mean 68, while Group B contains 60 observations with mean 74.

- (a) Find the mean of all 100 observations.
- (b) Explain why taking the simple average of 68 and 74 is not appropriate here.

- 24.** The daily time spent on a social-media platform is grouped below.

Time (hours)	0–< 2	2–< 4	4–< 6	6–< 8
Frequency	8	15	12	5

Estimate the mean, median and mode.

- 25.** Consider the two populations

$A : 48, 49, 50, 51, 52,$ $B : 30, 40, 50, 60, 70.$

- (a) Find the mean, median and range of each population.

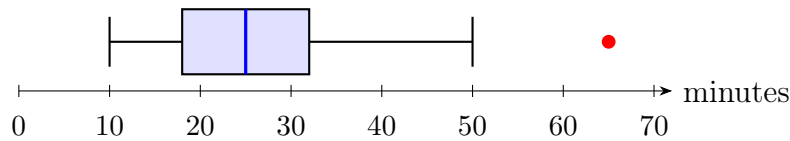
- (b) Find the population standard deviation of each.
 (c) Which population is more consistent? Explain.

26. A sample of eight observations has

$$\sum x_i = 96, \quad \sum x_i^2 = 1240.$$

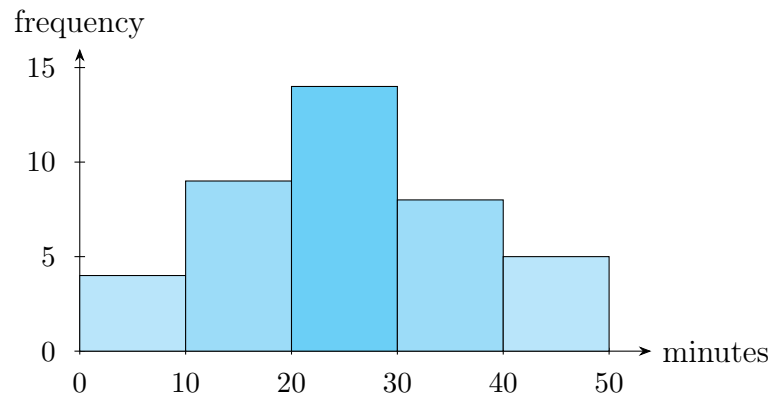
Find the sample mean, sample variance and sample standard deviation.

27. The following box plot summarises commuting times, in minutes. The isolated red point is shown as a potential outlier.



- (a) State the minimum non-outlier, Q_1 , median, Q_3 and maximum non-outlier.
 (b) Find the interquartile range and the two outlier fences.
 (c) Confirm whether 65 minutes is an outlier.
 (d) Describe the skewness suggested by the plot.

28. The histogram shows commuting times for 40 workers.



- (a) State the modal class.
 (b) Estimate the mean commuting time using class midpoints.
 (c) Estimate the median by interpolation.
 (d) Briefly describe the shape of the distribution.

29. Using the quartile convention in this chapter, analyse

12, 15, 16, 18, 20, 22, 24, 27, 31, 60.

Find the five-number summary, interquartile range and outlier fences, and identify any outlier.

30. The values 1, 2, 3, 4, 5 have respective frequencies 2, 5, 6, 5, 2. Treat the distribution as a population and find its mean, variance and standard deviation.
31. Two regions have the following household incomes, in thousands of ringgit:

$$A : 28, 30, 31, 33, 38, \quad B : 20, 22, 24, 26, 68.$$

Compare their means, medians and ranges. Explain why the mean alone gives an incomplete comparison.

32. For each variable, recommend an appropriate measure of centre and graphical display. Give a brief reason.
- (a) political-party preference
 - (b) strongly right-skewed annual household income
 - (c) number of children per household
 - (d) satisfaction recorded as low, medium or high
33. A poll of 1000 randomly selected voters reports that 54% support a policy.
- (a) Identify the sample statistic and the corresponding population parameter.
 - (b) Explain why another random sample need not produce exactly 54%.
 - (c) State one reason why the result could still be biased despite the large sample size.

Additional practice: grouped skills review

34. Classify each variable as categorical nominal, categorical ordinal, quantitative discrete or quantitative continuous.
- (a) municipality of residence
 - (b) satisfaction recorded as low, medium or high
 - (c) number of dependants in a household
 - (d) annual household income
 - (e) employment status
 - (f) commuting time
 - (g) agreement recorded on a five-level ordered scale
 - (h) number of court cases filed in a month
35. For each study, identify the population, sample, parameter and statistic.
- (a) From 120 000 adult residents, 600 are surveyed and 54% support a transport proposal.
 - (b) A random sample of 80 case files from all applications processed during a year has mean processing time 17.2 days.

- 36.** Find the mean, median and mode of each data set. State which measure of centre is most resistant to the unusually large value in part (d).

- (a) 3, 5, 5, 8, 9
- (b) 12, 12, 14, 18, 24
- (c) 2, 4, 6, 8
- (d) 20, 22, 23, 25, 70

- 37.** Find the mean, median and mode for each frequency distribution.

Value	0	1	2	3	4	Value	10	20	30
Frequency in A	2	5	6	4	3	Frequency in B	4	3	3

- 38.** For each grouped distribution, estimate the mean, median and mode.

Class	0– < 10	10– < 20	20– < 30	30– < 40
Frequency in A	4	9	5	2

Class	20– < 30	30– < 40	40– < 50	50– < 60
Frequency in B	3	7	8	2

- 39.** Using the quartile-position convention in this chapter, find the five-number summary, IQR and outlier fences for each data set. Identify any outlier.

- (a) 2, 4, 5, 7, 8, 11, 16
- (b) 4, 5, 6, 7, 8, 9, 10, 30

- 40.** A distribution has mean 42, median 40, range 18 and standard deviation 6. State these four summaries after every observation is transformed as follows.

- (a) $y = x + 5$
- (b) $z = 1.2x$
- (c) $w = 1.2x + 5$

- 41.** For the data 2, 4, 6, 8:

- (a) calculate the population variance and standard deviation;
- (b) calculate the sample variance and standard deviation;
- (c) explain why the two variances differ.

- 42.** Calculate each weighted mean.

- (a) A group of 30 students has mean 65 and a group of 70 students has mean 72. Find the combined mean.
- (b) Household expenditure indices 110, 125, 140 have weights 0.40, 0.35, 0.25, respectively.

43. Compare the two populations

$$A : 40, 42, 44, 46, 48, \quad B : 20, 40, 44, 48, 68.$$

Find the mean, median, range and population standard deviation of each, then explain what is concealed by comparing centre alone.

Applications: surveys, sampling and data quality**44.** A university posts a transport survey on the student union's social-media page and treats the first 300 replies as representative of all students.

- (a) Identify the target population and sample.
- (b) Explain two possible sources of bias.
- (c) Propose a more representative design.

45. A health survey covers three districts containing 60%, 30% and 10% of the region's population. A proportionally stratified sample of 500 residents is planned.

- (a) Allocate the sample among the districts.
- (b) Explain why stratification may improve representation.

46. A council has an ordered list of 4000 households. After choosing a random starting point among the first 20 entries, it selects every twentieth household.

- (a) Find the sample size.
- (b) Name the sampling method.
- (c) State one circumstance in which the method could be biased.

47. Of 1000 people invited to an online policy survey, 420 respond and 70% of respondents support the policy.

- (a) Identify the reported statistic.
- (b) How many respondents support the policy?
- (c) Explain why the response rate matters to interpretation.

48. An administrative register reports 18 400 people as unemployed. Explain why this count need not equal the total number of unemployed residents. Distinguish measurement error from incomplete coverage in this context.**49.** A questionnaire records employment sector, age, household size and trust in government on a five-level ordered scale. Classify the four variables and identify the observational unit if each row represents one respondent.**50.** Postcodes are stored as five-digit numbers. Explain why calculating their arithmetic mean is not substantively meaningful, even though software permits the calculation. Suggest a suitable summary.

51. A school data set contains one row per school and columns for district, enrolment, teacher count and pass rate.
- Identify the observational unit.
 - Classify each variable.
 - Explain why the mean of school pass rates need not equal the overall pupil pass rate.
52. A public report contains a table cell describing only two people in a small community. Give two reasons for aggregating or suppressing the cell, and state one disadvantage of doing so.

Applications: measures of centre and frequency distributions

53. Household incomes, in thousands of ringgit, are

28, 30, 31, 33, 128.

Find the mean, median, mode and range. Which measure of centre gives the more defensible description of a typical household, and why?

54. Monthly earnings in two districts, in hundreds of ringgit, are

$A : 30, 32, 34, 36, 38,$ $B : 20, 22, 34, 46, 48.$

Compare their means, medians, ranges and population standard deviations. Which district is more economically homogeneous under this measure?

55. Monthly transport spending has the distribution

Spending (RM)	100	150	200	300
Households	12	18	14	6

Find the mean, median and mode.

56. A rural sample of 120 households has mean internet use 42 GB per month, while an urban sample of 80 households has mean 57 GB. Find the combined mean and explain why the unweighted average of the two means is incorrect.
57. A service records waiting times 5, 6, 6, 7, 8 minutes. Find the mean and median before and after a waiting time of 28 minutes is added. Which measure changes more?
58. Five ordered observations are 12, 15, x , 22, 30. Their mean is 20. Find x and hence the median.
59. A population has mean income RM 3200, median income RM 3000 and standard deviation RM 400. State the new summaries under each separate policy.
- Every income increases by RM 250.

- (b) Every income increases by 8%.
60. Annual income, measured in thousands of ringgit, has mean 48 and standard deviation 12. It is converted to monthly income in the same units. Find the new mean, variance and standard deviation.
61. Satisfaction responses are distributed as follows.

Response	very dissatisfied	dissatisfied	neutral	satisfied	very satisfied
Frequency	5	8	12	20	15

Find the median category and mode. Explain why a numerical mean depends on an additional coding assumption.

62. The values 1, 2, 3, 4 have frequencies 4, 6, x , 2. If the mean is 2.2, find x , the total frequency, the median and the mode.
63. Eight observations have reported mean 14.5. One value was entered as 19 instead of 91.
- (a) Find the corrected mean.
- (b) Explain why the corrected median cannot be recovered from the information given.

Applications: quartiles, outliers and box plots

64. For

6, 8, 9, 10, 12, 14, 18, 22, 40,

find the five-number summary, IQR and outlier fences. Identify any outlier and state the largest non-outlying observation.

65. The comparative box plots summarise service times for two offices. Read the five-number summaries from the diagram, compare their medians and IQRs, and identify which office is less consistent.

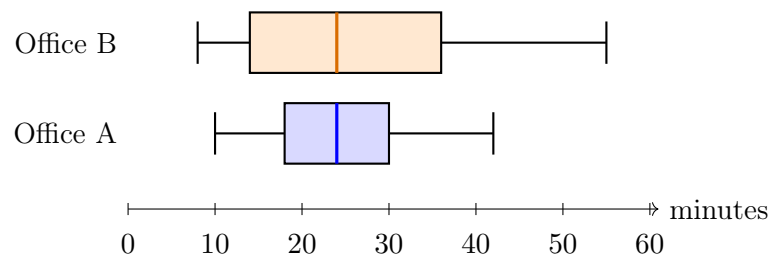


Figure 4.3: Comparative service-time box plots for Exercise 65.

66. A data set has $Q_1 = 20$, median 28 and $Q_3 = 36$.
- (a) Find the outlier fences.

- (b) Classify an observation of 75.
- (c) Explain why statistical classification as an outlier does not by itself justify deleting the observation.
- 67.** Explain how two distributions can share the same five-number summary but differ in clustering, gaps and frequency within the four quartile intervals. What information should accompany a box plot when these features matter?
- 68.** The distribution below represents travel time in minutes. Estimate Q_1 , Q_3 and the IQR by interpolation.

Time	0– < 10	10– < 20	20– < 30	30– < 40	40– < 50
Frequency	5	9	12	8	6

- 69.** For a household-income distribution, $Q_1 = 30$ and $Q_3 = 50$ thousand ringgit.
- (a) Find the fences.
- (b) Classify an income of RM 110 000.
- (c) Explain why this value could be genuine and substantively important even though it is statistically unusual.
- 70.** Emergency-department waiting times are

12, 14, 15, 16, 18, 19, 22, 24, 25, 75.

Find the median, quartiles, IQR and fences. Recommend two summaries for reporting a typical wait and its spread.

- 71.** Neighbourhood rent summaries, in ringgit, are

	min	Q_1	median	Q_3	max
A	900	1050	1200	1350	1500
B	800	1000	1200	1600	2500

Compare centre and spread, and describe the feature of Neighbourhood B that the common median conceals.

- 72.** A response-time distribution has $Q_1 = 9$, median 11 and $Q_3 = 15$ minutes. The smallest observation is 6 and the largest is 28.
- (a) Find the IQR and fences.
- (b) Determine whether 28 is an outlier.
- (c) Describe the likely direction of skewness.
- 73.** Construct a box plot for

2, 3, 4, 5, 5, 6, 7, 8, 20.

Give the quartiles, fences, whisker endpoints and any plotted outlier.

74. A survey has 19 ordered scores. Under the chapter's convention, state the positions of Q_1 , the median and Q_3 . If the fifth, tenth and fifteenth scores are 32, 48 and 71, state the three quartiles and interpret Q_1 .

Applications: variation and comparison

75. Compare the populations

$$A : 18, 20, 22, 24, 26, \quad B : 10, 16, 22, 28, 34.$$

Find each mean, population variance and standard deviation. Which is more consistent?

76. For the data 4, 7, 9, 10, calculate both the population and sample variances and standard deviations. Explain the role of the denominator in each calculation.
77. A sample of 12 observations has $\sum x = 240$ and $\sum x^2 = 5020$. Find the sample mean, variance and standard deviation.
78. A population has mean 40, variance 25 and standard deviation 5. For $y = 3x - 4$, find the mean, variance and standard deviation of y .
79. A rate recorded in units is accidentally reported in hundredths of a unit, so every observation is multiplied by 100. If the original mean is 50 and variance is 16, find the reported mean, variance and standard deviation.
80. Treat the frequency distribution as a population and find its mean, variance and standard deviation.

x	1	2	3	4
f	1	3	4	2

81. Two small groups have values

$$A : 10, 12, 14, \quad B : 20, 22, 24.$$

Find the population standard deviation within each group and for the six observations combined. Explain why the combined spread is much larger.

82. Compare 10, 11, 12, 13, 14 with the data set obtained by replacing 14 with 40. For both data sets find the range, IQR and population standard deviation. Which measures are most affected by the extreme value?
83. Office A has mean service time 18 minutes and standard deviation 2.5 minutes. Office B has mean 16 minutes and standard deviation 6 minutes. Compare typical speed and consistency, and state one further feature of the distributions that should be checked.
84. Waiting time is measured in hours. State the units of its variance and standard deviation. If the data are converted to minutes, state how each measure changes.

85. Sample A contains 100 observations and Sample B contains 10; both have standard deviation 4. Explain what the equal standard deviations do and do not establish about the two samples and their population estimates.

Applications: grouped data and visual interpretation

86. The histogram represents the travel times of 40 workers.

- (a) State the modal class.
- (b) Estimate the mean, median and mode.
- (c) Describe the distribution's shape.

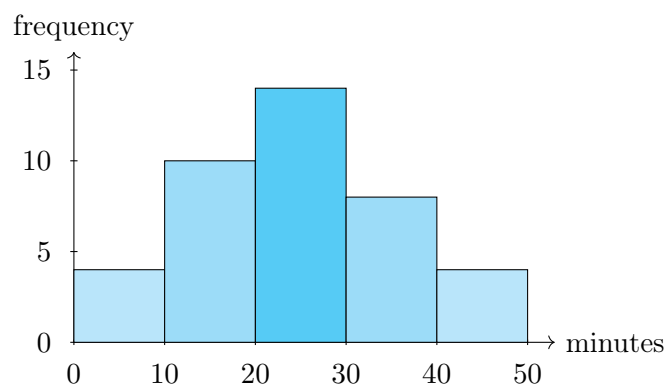


Figure 4.4: Grouped travel-time data for Exercise 86.

87. A report compares policy support of 48% in Region A with 52% in Region B using the truncated-axis chart below.

- (a) Find the difference in percentage points.
- (b) Find the relative percentage difference using Region A as the reference.
- (c) Explain how the vertical scale may exaggerate the visual difference and suggest a clearer presentation.



Figure 4.5: A truncated vertical scale makes a modest difference appear visually dominant.

88. Recommend a suitable display and measure of centre for each case, with a brief justification.
- (a) political-party preference
 - (b) strongly right-skewed household income
 - (c) number of children per household
 - (d) ordered satisfaction categories
89. A grouped sample has the following distribution.

Class	$0- < 10$	$10- < 20$	$20- < 30$	$30- < 40$
Frequency	3	7	8	2

Using class midpoints, estimate the sample mean, variance and standard deviation.

90. A distribution has classes below 20, $20- < 40$, $40- < 60$ and 60 or above, with frequencies 5, 12, 20, 13.
- (a) Estimate the grouped median using boundaries 39.5 and 59.5 for the median class.
 - (b) Explain why the mean cannot be estimated reliably without an additional assumption about the open-ended classes.
91. Give three limitations of grouped-data estimates based on class midpoints. Include one reason why two different raw data sets can produce the same grouped table.
92. The classes $0- < 10$, $10- < 20$ and $20- < 30$ have frequencies 4, x , 6. If the estimated mean is 16, find x and the total frequency.
93. The cumulative frequencies for classes $0- < 5$, $5- < 10$, $10- < 15$, $15- < 20$ and $20- < 25$ are 4, 11, 25, 36, 40.
- (a) Recover the class frequencies.
 - (b) Identify the modal class.
 - (c) Estimate the median.

Applications: responsible statistical interpretation

94. A programme's mean participant income rises after enrolment. Explain why this does not prove that every participant became better off or that the programme caused the increase.
95. Explain why a very large convenience sample can remain more biased than a smaller probability sample. Distinguish sample size from representativeness.
96. Give three checks that should be made before deleting a statistical outlier from a social-science data set.

- 97.** A researcher reports the mean of responses coded 1 = disagree, 2 = neutral and 3 = agree. State the assumption required to interpret the mean numerically and give an alternative summary that does not require equal spacing.
- 98.** Explain why the median and IQR are often reported for income, wealth or waiting-time distributions. State one useful feature that these two summaries still omit.
- 99.** Give four principles for an honest statistical graphic, including one concerning axes and one concerning missing or excluded data.
- 100.** Give a six-stage checklist for collecting, summarising, checking and communicating a descriptive statistical result in a social-science study.

Chapter 5

Probability

Chapter guide

Start with outcomes, events and probability rules, then build conditional and independent-event calculations using tables, trees and Bayes' theorem.

Selected background sources: (Madsen, 2016; Moore et al., 2021).

Probability measures uncertainty. It provides a mathematical language for describing chance, combining uncertain events and updating beliefs when new information becomes available.

5.1 Experiments, outcomes and events

A *probability experiment* is a repeatable process whose result is not known in advance, such as tossing a coin or rolling a die. An *outcome* is one possible result of a single trial. The *sample space*, denoted by S , is the set of all possible outcomes. An *event* is any subset of the sample space.

For a single coin toss,

$$S = \{H, T\}.$$

For a roll of a six-sided die,

$$S = \{1, 2, 3, 4, 5, 6\},$$

and the event of obtaining an even number is

$$E = \{2, 4, 6\}.$$

Definition 5.1: Probability

The probability $P(A)$ of an event A is a number between 0 and 1 that describes how likely the event is to occur.

An impossible event has probability 0, while a certain event has probability 1:

$$0 \leq P(A) \leq 1, \quad P(\emptyset) = 0, \quad P(S) = 1.$$

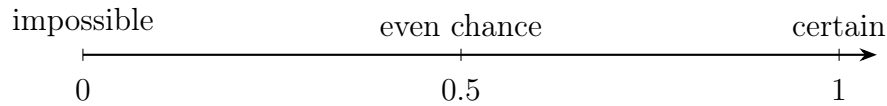


Figure 5.1: The probability scale from 0 to 1.

5.1.1 Equally likely outcomes

When a finite sample space contains equally likely outcomes,

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Example 5.1: [

blue]A coin and a die For a fair coin,

$$P(H) = \frac{1}{2}.$$

For a fair six-sided die, the event of rolling an even number is $E = \{2, 4, 6\}$, so

$$P(E) = \frac{3}{6} = \boxed{\frac{1}{2}}.$$

Example 5.2: [

cyan]Two four-sector spinners Two fair spinners are numbered 1 to 4. Their 16 ordered outcomes are equally likely.

	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Exactly four outcomes have sum 5, so

$$P(\text{sum} = 5) = \frac{4}{16} = \boxed{\frac{1}{4}}.$$

Six outcomes have sum greater than 5, so

$$P(\text{sum} > 5) = \frac{6}{16} = \boxed{\frac{3}{8}}.$$

5.2 Probability rules and event notation

Events are combined using the set operations introduced in Chapter 3.

- $A \cap B$ means that both A and B occur.
- $A \cup B$ means that A or B or both occur.
- A^c means that A does not occur.

The *complement rule* is

$$P(A^c) = 1 - P(A).$$

The general *addition rule* is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The intersection is subtracted because it was included once in $P(A)$ and again in $P(B)$.

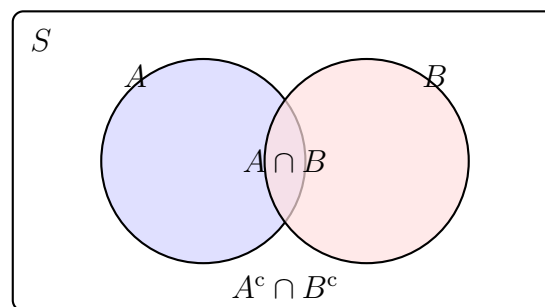


Figure 5.2: Events represented as regions of a sample space.

5.2.1 A three-event Venn problem

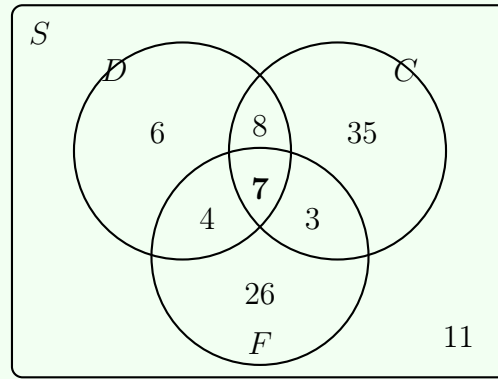
When pairwise intersection totals include observations in all three sets, the central region must be entered first. The other regions are then found by subtraction.

Example 5.3: [

orange]Ownership of household pets A veterinarian surveys 100 clients. Let D , C and F represent owning a dog, cat and tropical fish. The survey gives

$$\begin{aligned} n(D) &= 25, & n(C) &= 53, & n(F) &= 40, \\ n(D \cap C) &= 15, & n(D \cap F) &= 11, & n(C \cap F) &= 10, \\ n(D \cap C \cap F) &= 7. \end{aligned}$$

The three pair-only regions contain 8, 4 and 3, respectively. Therefore the dog-only, cat-only and fish-only regions contain 6, 35 and 26. The union contains 89 clients, leaving 11 outside all three sets.



Hence,

$$\begin{aligned}
 P(\text{dog only}) &= \boxed{0.06}, \\
 P(F^c) &= 1 - \frac{40}{100} = \boxed{0.60}, \\
 P((D \cup C \cup F)^c) &= \boxed{0.11}.
 \end{aligned}$$

5.2.2 Two-way tables and complements

A two-way table organises counts for two categorical variables. Probabilities are obtained by dividing the relevant cell or marginal total by the appropriate overall or conditional total.

Example 5.4: [

violet]Driving-test results

	Male	Female	Total
Pass	57	62	119
Fail	12	19	31
Total	69	81	150

For a randomly selected test-taker,

$$P(\text{fail}) = \frac{31}{150}, \quad P(\text{female and pass}) = \frac{62}{150} = \frac{31}{75}.$$

If a male test-taker is selected, the relevant total is 69, so

$$P(\text{fail} \mid \text{male}) = \frac{12}{69} = \boxed{\frac{4}{23}}.$$

Example 5.5: [

teal]Using the complement rule Of 1000 employees, 366 are aged from 25 to 34. Therefore,

$$P(\text{not aged 25-34}) = 1 - \frac{366}{1000} = \boxed{0.634}.$$

5.3 Conditional probability

The conditional probability of B given that A has occurred is

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0.$$

Conditioning reduces the sample space to the outcomes in A .

Example 5.6: [

blue]IQ and the presence of a gene

	Gene present	Gene absent	Total
High IQ	33	19	52
Normal IQ	39	11	50
Total	72	30	102

Given that a child has the gene, the conditional sample space contains 72 children. Thus,

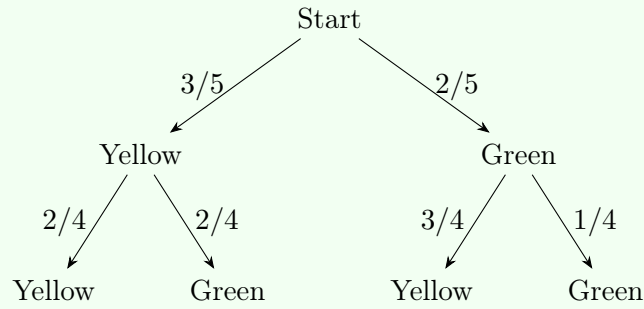
$$P(\text{high IQ} | \text{gene}) = \frac{33}{72} = \boxed{\frac{11}{24}}.$$

5.3.1 Tree diagrams

A tree diagram displays sequential events. Branch probabilities leaving each node must sum to 1, and the probability of a complete path is found by multiplying its branch probabilities.

Example 5.7: [

orange]Selecting counters without replacement A bag contains three yellow counters and two green counters. Two counters are selected without replacement.



The probability of selecting counters of the same colour is

$$\frac{3}{5} \cdot \frac{2}{4} + \frac{2}{5} \cdot \frac{1}{4} = \frac{3}{10} + \frac{1}{10} = \boxed{\frac{2}{5}}.$$

The probability of different colours is

$$\frac{3}{5} \cdot \frac{2}{4} + \frac{2}{5} \cdot \frac{3}{4} = \boxed{\frac{3}{5}}.$$

5.4 Independent and dependent events

The general multiplication rule is

$$P(A \cap B) = P(A)P(B | A).$$

Events A and B are *independent* when the occurrence of either event does not change the probability of the other. Equivalent tests for independence are

$$P(A \cap B) = P(A)P(B),$$

$$P(B | A) = P(B) \quad \text{or} \quad P(A | B) = P(A),$$

provided the conditional probabilities are defined.

If the first event changes the probabilities for the second event, the events are *dependent*. Sampling without replacement usually produces dependent events.

5.4.1 Independent-event examples

Example 5.8: [

blue]A coin and a die A fair coin toss and a fair die roll are independent. Therefore,

$$P(H \text{ and } 3) = \frac{1}{2} \cdot \frac{1}{6} = \boxed{\frac{1}{12}}.$$

Example 5.9: [

cyan]Marbles selected with replacement A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles, for a total of 16. A marble is replaced before the second selection. The probability of first selecting green and then yellow is

$$\frac{5}{16} \cdot \frac{6}{16} = \boxed{\frac{15}{128}}.$$

If the order is not specified, either green–yellow or yellow–green is allowed, so

$$2 \left(\frac{5}{16} \cdot \frac{6}{16} \right) = \boxed{\frac{15}{64}}.$$

Example 5.10: [

teal]Repeated independent selections If the probability that a randomly selected student likes pizza is $9/10$, then three selections with replacement give

$$P(\text{all three like pizza}) = \left(\frac{9}{10} \right)^3 = \boxed{0.729}.$$

5.4.2 Dependent-event examples**Example 5.11:** [

orange]Marbles selected without replacement Using the same jar, the probability of first selecting green and then yellow without replacement is

$$\frac{5}{16} \cdot \frac{6}{15} = \boxed{\frac{1}{8}}.$$

If either order is allowed,

$$\frac{5}{16} \cdot \frac{6}{15} + \frac{6}{16} \cdot \frac{5}{15} = \boxed{\frac{1}{4}}.$$

Example 5.12: [

red]Selecting two male goldfish An aquarium contains six male and four female goldfish. Two fish are selected without replacement. Thus,

$$P(\text{both male}) = \frac{6}{10} \cdot \frac{5}{9} = \boxed{\frac{1}{3}}.$$

5.4.3 Testing for independence

Example 5.13: [

violet]Mathematics and science classes Suppose

$$P(M) = 0.6, \quad P(S) = 0.5, \quad P(M \cap S) = 0.3.$$

Since

$$P(M)P(S) = 0.6(0.5) = 0.3 = P(M \cap S),$$

the events are independent.

Example 5.14: [

purple]Gender and hair length Suppose

$$P(F) = 0.60, \quad P(L) = 0.50, \quad P(F \cap L) = 0.45.$$

Here

$$P(F)P(L) = 0.30 \neq 0.45,$$

so the events are not independent. Equivalently,

$$P(L | F) = \frac{0.45}{0.60} = 0.75 \neq P(L).$$

5.5 Mutually exclusive events and addition rules

Events A and B are *mutually exclusive*, or disjoint, when they cannot occur in the same trial:

$$A \cap B = \emptyset \quad \text{and} \quad P(A \cap B) = 0.$$

For mutually exclusive events, the addition rule simplifies to

$$P(A \cup B) = P(A) + P(B).$$

For one roll of a die, the events “roll a 3” and “roll a 5” are mutually exclusive. The events “roll an even number” and “roll a prime number” are not mutually exclusive because the outcome 2 belongs to both.

Independence and mutual exclusivity are different ideas. Two mutually exclusive events with positive probabilities cannot be independent: if one occurs, the other is impossible.

Example 5.15: [

red]Blood types and Rh factors The following table summarises 409 blood donors.

	O	A	B	AB	Total
Rh-positive	156	139	37	12	344
Rh-negative	28	25	8	4	65
Total	184	164	45	16	409

Blood types O and A are mutually exclusive, so

$$P(O \cup A) = \frac{184 + 164}{409} = \boxed{\frac{348}{409}}.$$

The events “type B” and “Rh-negative” overlap in 8 donors. Therefore,

$$P(B \cup \text{Rh}^-) = \frac{45 + 65 - 8}{409} = \boxed{\frac{102}{409}}.$$

5.6 Bayes' theorem

Conditional probability gives

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

and

$$P(B | A) = \frac{P(A \cap B)}{P(A)}.$$

Since

$$P(A \cap B) = P(B | A)P(A),$$

substitution gives *Bayes' theorem*:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}, \quad P(B) > 0.$$

The probability $P(A)$ is the *prior probability*; $P(A | B)$ is the *posterior probability* after observing evidence B .

5.6.1 The law of total probability

Let $A_1, A_2, \dots, A_n$ form a partition of the sample space: the events are mutually exclusive and their union is S . For any event E ,

$$P(E) = \sum_{j=1}^n P(A_j)P(E | A_j).$$

Bayes' theorem then becomes

$$P(A_i | E) = \frac{P(A_i)P(E | A_i)}{\sum_{j=1}^n P(A_j)P(E | A_j)}.$$

Example 5.16: [

violet]Identifying the supplier of a defective part A manufacturer obtains 65% of its parts from Supplier 1 and 35% from Supplier 2. The defective rates are 2% and 5%, respectively. Let D denote a defective part.

The total probability of a defect is

$$P(D) = 0.65(0.02) + 0.35(0.05) = 0.0305.$$

Hence,

$$P(S_1 | D) = \frac{0.65(0.02)}{0.0305} \approx \boxed{0.4262},$$

and

$$P(S_2 | D) = \frac{0.35(0.05)}{0.0305} \approx \boxed{0.5738}.$$

Although Supplier 2 provides fewer parts, its higher defective rate makes it the more likely source of a randomly observed defective part.

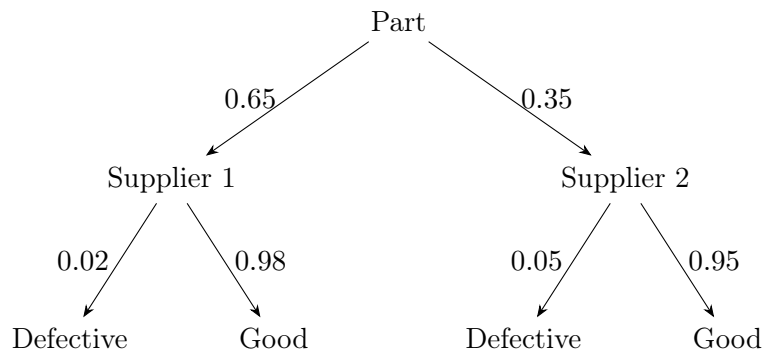


Figure 5.3: A probability tree for the supplier example.

Example 5.17: [

teal]Updating the probability of a successful bid A consulting firm initially believes that a bid has a 50% chance of success. An agency requests additional information for 75% of successful bids and 40% of unsuccessful bids. Let S mean a successful bid and R mean a request for information.

The prior probability is

$$P(S) = 0.50,$$

and

$$P(R | S) = 0.75.$$

The total probability of a request is

$$P(R) = 0.50(0.75) + 0.50(0.40) = 0.575.$$

Therefore the updated probability of success is

$$P(S | R) = \frac{0.50(0.75)}{0.575} = \frac{15}{23} \approx \boxed{0.6522}.$$

The request for additional information raises the estimated chance of success from 50% to about 65.2%.

Probability problems become more manageable when the sample space and events are defined before selecting a rule. Venn diagrams are useful for unions and intersections, tables for categorical data, tree diagrams for sequential events and Bayes' theorem for reversing a conditional probability.

5.7 Exercises

Answers to these exercises are provided in section C.5.

1. A fair six-sided die is rolled. Find the probability of obtaining a prime number.
2. Suppose $P(A) = 0.55$, $P(B) = 0.40$ and $P(A \cap B) = 0.20$. Find $P(A \cup B)$ and the probability that neither event occurs.
3. A fair die is rolled four times. Find the probability of obtaining at least one six.
4. The following table records course outcomes for 120 students.

	Pass	Fail	Total
Online	42	18	60
Campus	54	6	60
Total	96	24	120

Find $P(\text{pass})$, $P(\text{online and fail})$ and $P(\text{fail} | \text{online})$.

5. A bag contains five red and three blue counters. Two counters are selected without replacement. Find the probability that
 - (a) both are red;
 - (b) one is red and one is blue.

6. Given $P(M) = 0.6$, $P(S) = 0.5$ and $P(M \cap S) = 0.3$, determine whether M and S are independent.
7. Events A and B are mutually exclusive, with $P(A) = 0.30$ and $P(B) = 0.40$. Find $P(A \cup B)$ and explain why the events are not independent.
8. A factory receives 70% of its components from Supplier A and 30% from Supplier B. Their defective rates are 1% and 4%, respectively. Find the probability that a randomly selected component is defective and, given that it is defective, the probability that it came from Supplier B.
9. In a survey of 200 people,

$$\begin{aligned} n(A) &= 90, & n(B) &= 80, & n(C) &= 70, \\ n(A \cap B) &= 30, & n(A \cap C) &= 25, & n(B \cap C) &= 20, \\ n(A \cap B \cap C) &= 10. \end{aligned}$$

Find the number in at least one set and the number in none of the sets.

Tutorial exercises: probability

10. A fair coin is tossed and an eight-sector fair spinner numbered 1 to 8 is spun. Find the probability of a tail and an odd number.
11. Two fair five-sided spinners are numbered 1, 2, 3, 4, 5 and 2, 3, 5, 7, 11. The positive difference between the scores is recorded.
 - (a) Construct the 5×5 sample-space table.
 - (b) Find the probability of a difference of zero.
 - (c) Find the probability of a difference greater than four.
12. In a village, 70% of families own a dog, 45% own both a dog and a cat, and 18% own neither.
 - (a) Complete a two-set Venn diagram.
 - (b) Find the percentage that do not own a cat.
 - (c) Find the probability of owning a dog or a cat, but not both.
13. A knee operation succeeds with probability 0.85. For three independent operations, find the probabilities that (a) all succeed; (b) none succeeds; (c) at least one succeeds.
14. A survey produced the following voting table.

Age	Yes	No	Total
18–25	50	12	62
26–40	104	65	169
41–65	60	76	136
Over 65	35	61	96
Total	249	214	463

Find the probability of selecting (a) a voter aged 26–40; (b) a voter over 65 or a No voter.

15. A class of 30 students is represented by the disjoint Venn regions

$$\begin{aligned} A \text{ only} &= 3, & A \cap B &= 4, & B \text{ only} &= 5, \\ B \cap C &= 10, & C \text{ only} &= 7, & \text{none} &= 1. \end{aligned}$$

Find (a) $P(B \cup C)$; (b) whether A and B are independent.

16. Five of 100 machine parts are defective. Two are selected without replacement. Find the probability that both are defective.
17. The table summarises preferred study-abroad destinations.

	UK	US	Australia	Total
Junior	90	80	70	240
Senior	80	95	95	270
Total	170	175	165	510

Find the probability that a selected student

- (a) is not a junior;
- (b) did not select the US;
- (c) is a junior and selected Australia;
- (d) is a junior and did not select Australia;
- (e) selected the UK, given that the student is a senior;
- (f) is a junior, given that the student selected the US.

18. In a three-event diagram the disjoint regions are

$$\begin{aligned} A \text{ only} &= p, & A \cap B &= 0.10, & B \text{ only} &= 0.25, \\ B \cap C &= 0.15, & C \text{ only} &= q, & \text{none} &= r. \end{aligned}$$

There is no overlap between A and C .

- (a) Given that A and B are independent, find p .

- (b) Given also that $P(B | C) = 0.75$, find q and r .
19. Suppose $P(A \cup B) = 3/5$ and $P(A) = 2/5$. Find $P(B)$ when
- (a) A and B are mutually exclusive;
 - (b) A and B are independent;
 - (c) $P(B | A) = 1/5$.
20. An urn contains eight blue and five yellow balls. Six are selected without replacement. Find the probability that
- (a) four blue and two yellow are selected;
 - (b) all six are blue;
 - (c) more than three yellow are selected.
21. A shelf contains eight mystery, six science-fiction and five fantasy books. Four books are selected without replacement. Find the probability that
- (a) exactly three are science fiction;
 - (b) at least three are mysteries;
 - (c) no fantasy book is selected.
22. A box is equally likely to contain a cat or a dog. A cat is quiet with probability 0.8 and a dog with probability 0.3. Find (a) the probability that the pet is quiet; (b) $P(\text{cat} | \text{quiet})$.
23. During a journey the probabilities of dry weather, rain and snow are $1/2$, $1/3$ and $1/6$. The corresponding probabilities of arriving on time are $4/5$, $2/5$ and $1/10$.
- (a) Draw a probability tree.
 - (b) Find the probability of arriving late.
 - (c) Given a late arrival, find the probability that it rained.

Additional practice: grouped skills review

24. Use the complement rule to find the missing probability.
- (a) $P(A) = 0.18$; find $P(A^c)$.
 - (b) $P(B) = 0.64$; find $P(B^c)$.
 - (c) $P(C) = 7/20$; find $P(C^c)$.
 - (d) $P(D^c) = 0.27$; find $P(D)$.
25. Use the addition rule in each case.
- (a) $P(A) = 0.45$, $P(B) = 0.38$, $P(A \cap B) = 0.20$; find $P(A \cup B)$.

- (b) $P(A) = 0.62$, $P(B) = 0.50$, $P(A \cap B) = 0.30$; find $P(A \cup B)$.
 - (c) $P(A \cup B) = 0.70$, $P(A) = 0.40$, $P(B) = 0.45$; find $P(A \cap B)$.
 - (d) Mutually exclusive events have probabilities 0.28 and 0.35; find their union probability.
- 26.** Find the requested conditional or joint probability.
- (a) $n(A) = 120$ and $n(A \cap B) = 42$; find $P(B | A)$.
 - (b) $P(A \cap B) = 0.18$ and $P(B) = 0.45$; find $P(A | B)$.
 - (c) $P(A | B) = 0.60$ and $P(B) = 0.25$; find $P(A \cap B)$.
 - (d) Of 80 policy supporters, 30 are aged under 30; find $P(\text{under 30} | \text{support})$.
- 27.** Determine whether the events are independent, giving a numerical test.
- (a) $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$
 - (b) $P(A) = 0.3$, $P(B) = 0.6$, $P(A \cap B) = 0.15$
 - (c) $P(A) = 0.25$ and $P(A | B) = 0.25$, where $P(B) > 0$
 - (d) $P(A) = 0.7$, $P(B) = 0.2$, $P(A \cup B) = 0.76$
- 28.** Answer each question about mutually exclusive events.
- (a) If $P(A) = 0.25$ and $P(B) = 0.30$, find the intersection and union when the events are mutually exclusive.
 - (b) If $P(B) > 0$, find $P(A | B)$ for mutually exclusive events.
 - (c) When can two events be both mutually exclusive and independent?
 - (d) Explain why an event and its complement are exhaustive as well as mutually exclusive.
- 29.** Use the law of total probability.
- (a) A population is 60% Group A and 40% Group B. Their success probabilities are 0.75 and 0.35. Find the overall success probability.
 - (b) Three regions contain 50%, 30% and 20% of a population. Their response probabilities are 0.8, 0.6 and 0.4. Find the overall response probability.
- 30.** Use Bayes' theorem with the data in Exercise 29.
- (a) Given success, find the probability of belonging to Group A.
 - (b) Given non-response, find the probability of belonging to the third region.
- 31.** A container holds six red and four blue counters.
- (a) Two are selected without replacement. Find the probability that both are red.
 - (b) Find the probability of selecting one of each colour without replacement.

- (c) Among 12 records, three contain errors. Two are selected without replacement. Find the probability that neither contains an error.
 - (d) Repeat part (c) if the first record is replaced before the second selection.
- 32.** Independent trials have the stated success probability. Find the requested probability.
- (a) At least one success in three trials when $p = 0.2$.
 - (b) At least one failure in two trials when $p = 0.65$.
 - (c) Exactly one success in two trials when $p = 0.3$.
 - (d) No successes in four trials when $p = 0.1$.
- 33.** Translate A^c , $A \cup B$, $A \cap B^c$ and $P(A | B)$ into words. Then explain why a large value of $P(A | B)$ does not by itself show that B causes A .

Applications: events, surveys and probability rules

- 34.** In a survey of 500 commuters, 280 use public transport, 190 are students and 130 are students who use public transport. Find the probabilities of public-transport use, student status, both events, at least one event and being a student given public-transport use.
- 35.** Of 400 households, 230 receive rent assistance, 180 receive energy assistance and 90 receive both.
- (a) Find the numbers receiving only each form of assistance.
 - (b) Find the numbers receiving at least one, neither and exactly one form of assistance.
- 36.** Among 1000 residents, 520 use Service A, 410 use Service B and 300 use Service C. The pairwise overlaps are 180, 140 and 120, and 60 use all three services.
- (a) Find the number using at least one service and the number using none.
 - (b) Find the numbers using exactly one and at least two services.
- 37.** An administrative file contains an error with probability 0.07. Find the probability that a file is error-free. In a collection of 500 files, interpret the number $500(0.07)$ without claiming that exactly this many errors must occur.
- 38.** For each proposed probability statement, decide whether it can be valid and justify the decision.
- (a) $P(A) = 1.10$
 - (b) $P(A \cap B) > P(A)$
 - (c) $P(A \cup B) < P(B)$
 - (d) $P(A) = 0.35$ and $P(A^c) = 0.65$

39. The diagram gives probabilities in four disjoint regions. Find $P(A)$, $P(B)$, $P(A \cup B)$, $P((A \cup B)^c)$, the probability of exactly one event and $P(A | B)$.

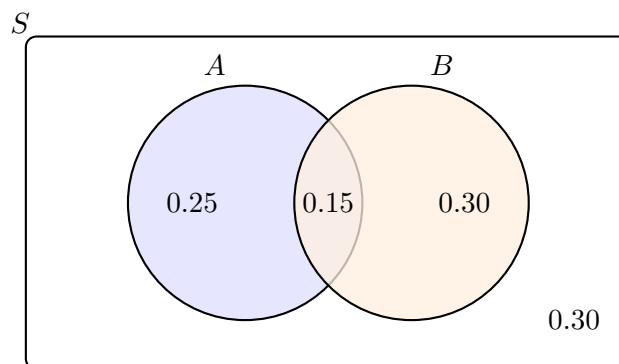


Figure 5.4: Event probabilities for Exercise 39.

40. In a city survey of 1000 adults, 620 use home internet, 480 use a smartphone for government services and 400 do both. Find the numbers using exactly one method, neither method and at least one method.
41. Let R denote registration for a youth programme and A actual attendance. Suppose $P(R) = 0.65$, $P(A) = 0.50$ and $P(R \cap A) = 0.42$. Find $P(R \cup A)$, $P(A | R)$, $P(R | A)$ and the probability of neither event.
42. Suppose $P(\text{rain}) = 0.30$, $P(\text{cancellation}) = 0.20$ and $P(\text{rain and cancellation}) = 0.12$. Find the probabilities of rain or cancellation, rain only, cancellation only, neither, and cancellation given rain.
43. Events satisfy $P(A) = 0.65$ and $P(B) = 0.55$.
- State the smallest and largest possible values of $P(A \cap B)$.
 - If $P(A \cup B) = 0.85$, find $P(A \cap B)$.
44. A study finds a high value of $P(\text{employed} | \text{degree})$. Give three reasons why this conditional probability alone does not establish the causal effect of obtaining a degree.

Applications: conditional probability and two-way tables

45. Employment outcomes are recorded below.

	Employed	Not employed	Total
Trained	144	36	180
Not trained	126	54	180
Total	270	90	360

Find $P(E)$, $P(T)$, $P(E | T)$, $P(E | T^c)$ and $P(T | E)$. Are employment and training independent?

46. The table compares two programmes within low- and high-risk groups.

	Programme A		Programme B	
	Success	Total	Success	Total
Low risk	9	10	80	100
High risk	30	100	2	10

- Compare success probabilities within each risk group.
- Compare the overall programme success probabilities.
- Explain the reversal using the distribution of risk groups.

47. A policy survey gives

	Yes	No	Total
Women	120	80	200
Men	90	110	200
Total	210	190	400

Find $P(\text{Yes})$, $P(\text{Yes} \mid \text{woman})$ and $P(\text{woman} \mid \text{Yes})$. Test independence.

48. Suppose $P(A) = 0.40$, $P(B) = 0.50$ and $P(A \mid B) = 0.40$. Determine whether the events are independent, then find $P(A \cap B)$ and $P(A \cup B)$.

49. In a population,

$$P(\text{unemployed} \mid \text{young}) = 0.12, \quad P(\text{young}) = 0.25, \quad P(\text{unemployed}) = 0.06.$$

Find $P(\text{young} \mid \text{unemployed})$ and explain why the two conditional probabilities differ.

50. In a sample of 200, the disjoint regions are A only = 40, $A \cap B = 30$, B only = 50 and neither = 80. Find $P(A \mid B)$, $P(B \mid A)$ and $P(A^c \mid B)$.

51. Ten case files include four appeals. Two files are selected without replacement. Find the probability that the second is an appeal given that the first is (a) an appeal; (b) not an appeal. Explain what the comparison shows.

52. Assume births are independent and each child is equally likely to be a boy or girl. For a two-child household, find the probability of at least one girl and the probability of two girls given at least one girl.

53. An error-detection system produces the table

	Flagged	Not flagged	Total
Error	45	5	50
No error	95	855	950
Total	140	860	1000

Find $P(\text{error})$, $P(\text{flagged})$, $P(\text{error} \mid \text{flagged})$ and $P(\text{flagged} \mid \text{error})$.

54. Suppose $P(A) = 0.60$, $P(B | A) = 0.40$ and $P(B | A^c) = 0.20$. Find $P(B)$, $P(A \cap B)$ and $P(A | B)$.
55. Explain the error in treating $P(E | I)$ as if it were $P(I | E)$, where E is observed evidence and I is innocence. State what additional probabilities are needed to reverse the conditioning correctly.

Applications: independence, dependence and repeated trials

56. Two independent appointments each have non-attendance probability 0.10. Find the probabilities that both are missed, at least one is missed and exactly one is missed.
57. Two independent information systems have failure probabilities 0.05 and 0.08. Find the probabilities that both fail, at least one works and at least one fails.
58. Four independent survey contacts each receive a response with probability 0.30. Find the probabilities of no response, at least one response, exactly one response and exactly two responses.
59. Eight records are complete and two are incomplete. Two are selected without replacement. Find the probabilities that both are incomplete, exactly one is incomplete, and the second is incomplete given that the first is incomplete.
60. A table has the following counts.

	B	B^c	Total
A	30	20	50
A^c	30	20	50
Total	60	40	100

Test whether A and B are independent using both a product test and a conditional-probability test.

61. Two policy choices are mutually exclusive, with probabilities 0.40 and 0.35. Find $P(A | B)$ and demonstrate numerically why the choices are not independent.
62. Independent events have $P(A) = 0.30$ and $P(B) = 0.40$. Find $P(A^c \cap B^c)$ and $P(A \cup B)$, and explain why the complements are also independent.
63. Three independent projects each succeed with probability 0.80. Find the probabilities that exactly two succeed, all three succeed and at least two succeed.
64. A container holds five approved and three rejected applications. Two are selected. Compare the probability that both are approved when sampling is (a) without replacement; (b) with replacement.
65. Passing an interview and a written test have probabilities 0.70 and 0.60.

- (a) Assuming independence, find the probability of passing both and the probability of the written-test pass given an interview pass.
- (b) If instead $P(\text{written pass} \mid \text{interview pass}) = 0.75$, find the probability of passing both.
66. Give three reasons why financial hardship events for two people in the same household should not automatically be modelled as independent.

Applications: total probability, trees and Bayes' theorem

67. Applications arrive online, by phone and in person with probabilities 0.50, 0.30 and 0.20. Their completion probabilities are 0.80, 0.70 and 0.90. Find the overall completion probability.
68. Using Exercise 67, find the probabilities that a completed application arrived (a) in person; (b) online.
69. Eight per cent of applications are ineligible. The screening system flags an ineligible application with probability 0.85 and flags an eligible application with probability 0.10.
- (a) Use the tree to find the probability of a flag.
- (b) Given a flag, find the probability that the application is ineligible.
- (c) Interpret why this posterior probability is well below 0.85.

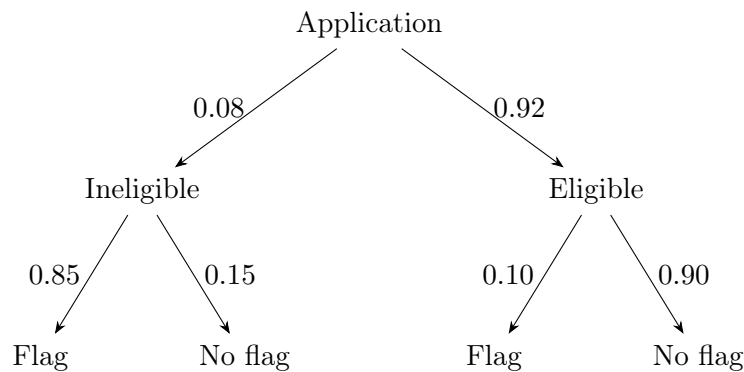


Figure 5.5: Screening outcomes for Exercise 69.

70. Thirty-five per cent of jobseekers participate in a training programme. Employment probabilities are 0.72 for participants and 0.55 for non-participants. Find the overall employment probability and $P(\text{participant} \mid \text{employed})$.
71. Sixty per cent of survey invitations are sent by web and 40% by phone. Response probabilities are 0.50 and 0.75. Find the overall response probability and the probability that a response came from the phone group.
72. Weather is dry, rainy or stormy with probabilities 0.50, 0.30 and 0.20. Late-arrival probabilities are 0.10, 0.40 and 0.80. Find the probability of lateness and, given lateness, the probability of stormy weather.

73. Regions A, B and C contain 50%, 30% and 20% of voters. Policy-support probabilities are 0.40, 0.55 and 0.70. Find the overall support probability and $P(C \mid \text{support})$.
74. Complaints originate from Departments A, B and C in proportions 0.50, 0.30 and 0.20. Their complaint rates are 0.02, 0.04 and 0.08. Find the overall complaint probability and the probability that a complaint originated in Department C.
75. A test has sensitivity 0.95 and specificity 0.90.
- (a) Find the positive predictive probability when prevalence is 0.01.
 - (b) Repeat when prevalence is 0.20.
 - (c) Explain the effect of the changed base rate.
76. A flagging rule has $P(F \mid I) = 0.80$ and $P(F \mid I^c) = 0.10$. If $P(I \mid F) = 0.50$, find the prior probability $P(I)$.
77. A population is 40% Group A and 60% Group B. The success probability for Group A is 0.80 and the overall success probability is 0.62. Find the success probability for Group B and $P(B \mid \text{success})$.

Applications: finite sampling without replacement

78. A committee of three is selected from seven women and five men. Find the probabilities of selecting (a) exactly two women; (b) three women.
79. Among 15 audited files, four contain errors. Three are selected without replacement. Find the probabilities of selecting no error, exactly one error and at least one error.
80. Twenty case files include six appeals. Four files are selected. Find the probability that exactly two are appeals.
81. A finite population contains 30 policy supporters and 20 non-supporters. Two people are selected without replacement. Find the probabilities that both support the policy and that exactly one supports it.
82. A sample of three households is selected from ten urban and eight rural households. Find the probability of selecting exactly one urban and two rural households.
83. Of 20 households, three are classified as highly vulnerable. Four are sampled without replacement. Find the probability that at least one highly vulnerable household is selected.
84. A five-person committee is selected from six women and three men. Find the probability that at least two men are selected.
85. A chair and secretary are selected in order from eight people, of whom three are senior staff. Find the probabilities that both officers are senior and that exactly one is senior.

86. Of 20 insurance claims, four are marked as suspect. Five are audited without replacement. Find the probability that none of the selected claims is suspect.
87. One record is selected from each of two separate groups. Group A contains two problematic records among ten, and Group B contains three among fifteen. Assuming the selections are independent, find the probabilities that at least one and both records are problematic.
88. A box contains seven Type A and three Type B forms. Three forms are selected. Compare the probability that all are Type A when sampling is without replacement and with replacement. Explain the direction of the difference.

Applications: responsible probability interpretation

89. A data set shows $P(\text{high income} \mid \text{private education})$ is large. Explain why this is evidence of association rather than proof of a causal effect.
90. Explain the prosecutor's fallacy: confusing a small probability of evidence given innocence with a small probability of innocence given the evidence.
91. Explain base-rate neglect using a rare-condition screening example. Why can a highly sensitive test still have a modest positive predictive probability?
92. Give three checks that should precede an assumption that two social events are independent.
93. Distinguish mutually exclusive events from exhaustive events. Give one example of each distinction using public-transport choices.
94. A report says a policy "doubles risk" because an outcome rises from 1% to 2%. State the relative increase and the absolute percentage-point increase, and explain why both should be reported.
95. A high value of $P(\text{approved} \mid \text{complete file})$ is observed. Give three possible selection or confounding mechanisms that could affect this conditional probability.
96. Explain how missing outcomes can bias an estimated conditional probability even when the arithmetic on the observed cases is correct.
97. Interpret a probability forecast of 0.70 as a long-run statement. Explain why one event occurring or not occurring does not by itself validate or refute the forecast.
98. A risk model gives different false-positive probabilities for two demographic groups. State four quantities that should be reported before the model is used for a high-stakes decision.
99. Explain why replacing dependent household outcomes by independent trials may underestimate the probability of several adverse events occurring together.

-
- 100.** Give a six-stage checklist for defining events, selecting a probability rule, calculating, checking and communicating a probability result in a social-science application.

Chapter 6

Discrete Random Variables

Chapter guide

Construct and read probability mass and cumulative distribution functions before calculating expectation, variance and transformed or decision-based quantities.

Selected background sources: (Madsen, 2016; Moore et al., 2021).

Random variables convert the outcomes of a probability experiment into numerical values. This makes it possible to describe an uncertain quantity by its distribution and to calculate measures such as its expected value and variance. This chapter considers random variables whose possible values can be listed.

6.1 Discrete random variables

Definition 6.1: Random variable

A *random variable* is a function that assigns a real number to each outcome in a sample space. A random variable is *discrete* if its possible values form a finite or countably infinite set.

Random variables are usually denoted by capital letters such as X , while a particular possible value is denoted by the corresponding lower-case letter x . For example, when a fair die is rolled, the random variable

$$X = \text{the number shown on the die}$$

has possible values 1, 2, 3, 4, 5, 6.

A discrete variable has separated possible values. Counts such as the number of customers arriving, defective items found or goals scored are discrete. By contrast, measurements such as time, mass and temperature are usually modelled using continuous random variables.

6.1.1 Probability mass functions

Definition 6.2: Probability mass function

The *probability mass function* (PMF) of a discrete random variable X is

$$p_X(x) = P(X = x).$$

It assigns a probability to every possible value of X .

Some elementary texts use the terms *probability function* or *discrete probability distribution*. This book uses *probability mass function* and the abbreviation PMF so that it is not confused with the probability density function of a continuous random variable.

For a function to be a valid PMF, it must satisfy

$$0 \leq p_X(x) \leq 1 \quad \text{for every } x, \quad \sum_x p_X(x) = 1.$$

Example 6.1: [

blue]The score on a fair die For a fair six-sided die,

$$p_X(x) = \begin{cases} \frac{1}{6}, & x = 1, 2, 3, 4, 5, 6, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, the distribution can be displayed as a table:

x	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

The probabilities are non-negative and sum to 1, so this is a valid probability distribution.

Example 6.2: [

cyan]Checking proposed probability functions Determine whether each proposed function is a PMF on the stated support.

For

$$p(x) = \frac{x^2}{55}, \quad x = 1, 2, 3, 4, 5,$$

all probabilities are non-negative and

$$\sum_{x=1}^5 \frac{x^2}{55} = \frac{1 + 4 + 9 + 16 + 25}{55} = 1.$$

Therefore it is a valid PMF.

The function $p(x) = x/5$ for $x = -1, 0, 1, 2, 3$ is invalid because $p(-1) = -1/5 < 0$.

The function

$$p(x) = \frac{2x-3}{10}, \quad x = 2, 4, 6,$$

is also invalid because its probabilities sum to

$$\frac{1 + 5 + 9}{10} = \frac{3}{2} \neq 1.$$

6.1.2 Calculating probabilities from a PMF

The probability of an event involving X is obtained by adding the masses of all values that satisfy the event. Thus,

$$P(X \in A) = \sum_{x \in A} p_X(x).$$

Example 6.3: [

violet]Probabilities from a formula Suppose

$$p_X(x) = \frac{x^2}{14}, \quad x = 0, 1, 2, 3.$$

The distribution is

x	0	1	2	3
$P(X = x)$	0	1/14	4/14	9/14

Hence,

$$P(X = 2) = \boxed{\frac{2}{7}},$$

$$P(X > 1) = \frac{4}{14} + \frac{9}{14} = \boxed{\frac{13}{14}},$$

$$P(X \leq 2) = \frac{0}{14} + \frac{1}{14} + \frac{4}{14} = \boxed{\frac{5}{14}}.$$

Example 6.4: [

orange]Finding an unknown probability A biased five-sector spinner has the distribution

x	1	2	3	4	5
$P(X = x)$	0.15	0.24	k	0.25	0.19

Since the probabilities sum to 1,

$$k = 1 - (0.15 + 0.24 + 0.25 + 0.19) = \boxed{0.17}.$$

Therefore,

$$P(X \geq 4) = 0.25 + 0.19 = \boxed{0.44},$$

$$P(X < 5) = 1 - 0.19 = \boxed{0.81},$$

$$P(2 < X \leq 4) = 0.17 + 0.25 = \boxed{0.42}.$$

Example 6.5: [

teal]A piecewise probability function Let

$$p_X(x) = \begin{cases} kx^2, & x = 0, 1, 2, 3, \\ k, & x = 4, 5, \\ 0, & \text{otherwise.} \end{cases}$$

Normalisation gives

$$k(0^2 + 1^2 + 2^2 + 3^2) + 2k = 16k = 1, \quad k = \boxed{\frac{1}{16}}.$$

It follows that

$$P(0 < X \leq 3) = \frac{1 + 4 + 9}{16} = \boxed{\frac{7}{8}},$$

$$P(|X - 2| < 1) = P(X = 2) = \boxed{\frac{1}{4}},$$

$$P(X \text{ is even}) = P(X = 0) + P(X = 2) + P(X = 4) = \boxed{\frac{5}{16}}.$$

6.1.3 Probability histograms

A probability histogram represents each possible value by a bar whose area equals its probability. When the bars have unit width, the height of the bar at x is simply $P(X = x)$.

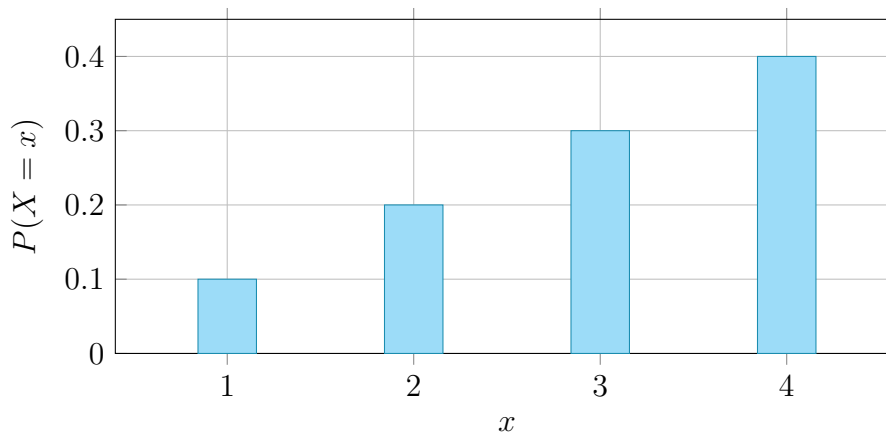


Figure 6.1: The PMF $p_X(x) = x/10$ for $x = 1, 2, 3, 4$.

6.1.4 An application using combinations

Example 6.6: [

purple]Selecting peppers without replacement A basket contains 3 red, 4 green and 5 yellow peppers. Three peppers are selected without replacement. There are

$$\binom{12}{3} = 220$$

equally likely selections.

The probability of selecting one pepper of each colour is

$$\frac{\binom{3}{1}\binom{4}{1}\binom{5}{1}}{\binom{12}{3}} = \frac{60}{220} = \boxed{\frac{3}{11}}.$$

Let X be the number of green peppers selected. Since there are 4 green and 8 non-green peppers,

$$P(X = x) = \frac{\binom{4}{x}\binom{8}{3-x}}{\binom{12}{3}}, \quad x = 0, 1, 2, 3.$$

Thus,

x	0	1	2	3
$P(X = x)$	14/55	28/55	12/55	1/55

In particular, the probability of exactly two green peppers is 12/55.

6.2 Cumulative distribution functions

Definition 6.3: Cumulative distribution function

The *cumulative distribution function* (CDF) of a random variable X is

$$F_X(x) = P(X \leq x).$$

For a discrete random variable,

$$F_X(x) = \sum_{t \leq x} p_X(t).$$

A CDF is non-decreasing, takes values between 0 and 1, and approaches 0 to the far left and 1 to the far right. The CDF of a discrete random variable is a step function. Each jump at x has size $P(X = x)$.

For any real numbers $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

The notation $F_X(a^-)$ denotes the value immediately before a , so

$$P(X = a) = F_X(a) - F_X(a^-).$$

Example 6.7: [

blue]The cumulative distribution of a fair die For a fair die,

x	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6
$F_X(x)$	1/6	2/6	3/6	4/6	5/6	1

For example,

$$P(X \leq 4) = F_X(4) = \frac{4}{6} = \frac{2}{3}.$$

Example 6.8: [

orange]The spinner CDF For the biased spinner considered earlier, the cumulative probabilities are

x	1	2	3	4	5
$P(X = x)$	0.15	0.24	0.17	0.25	0.19
$F_X(x)$	0.15	0.39	0.56	0.81	1.00

Hence,

$$\begin{aligned} P(X < 3) &= P(X \leq 2) = \boxed{0.39}, \\ P(2 < X \leq 5) &= F_X(5) - F_X(2) = \boxed{0.61}, \\ P(2 \leq X < 5) &= 0.24 + 0.17 + 0.25 = \boxed{0.66}. \end{aligned}$$

Example 6.9: [

violet]Recovering a PMF from a CDF Suppose X takes the values 0, 1, 2, 3 and

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \leq x < 1, \\ 0.6, & 1 \leq x < 2, \\ 0.9, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

The probabilities are the jump sizes, giving

x	0	1	2	3
$P(X = x)$	0.2	0.4	0.3	0.1

For example,

$$\begin{aligned} P(X \leq 1) &= 0.6, & P(X > 1) &= 1 - 0.6 = 0.4, \\ P(X = 1) &= 0.6 - 0.2 = 0.4, & P(0 < X \leq 2) &= 0.9 - 0.2 = 0.7. \end{aligned}$$

Also, $P(1 < X < 3) = P(X = 2) = 0.3$.

Example 6.10: [

teal]Daily car sales A car dealer models the number of cars sold in one day as follows:

x	0	1	2	3	4	5
$P(X = x)$	0.1	0.1	0.2	0.3	0.2	0.1
$F_X(x)$	0.1	0.2	0.4	0.7	0.9	1.0

Therefore,

$$P(X \leq 2) = \boxed{0.4}.$$

The complete CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.1, & 0 \leq x < 1, \\ 0.2, & 1 \leq x < 2, \\ 0.4, & 2 \leq x < 3, \\ 0.7, & 3 \leq x < 4, \\ 0.9, & 4 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$

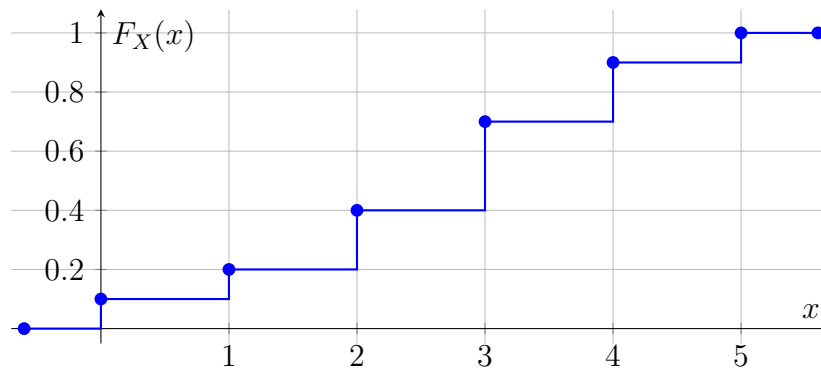


Figure 6.2: The cumulative distribution function for daily car sales.

6.3 Expected value

Definition 6.4: Expected value

If X is a discrete random variable with PMF $p_X(x)$, its *expected value*, or mean, is

$$\mu = E(X) = \sum_x x p_X(x),$$

provided the sum exists.

The expected value is the long-run average obtained over many repetitions of the experiment. It need not itself be a possible value of X .

Example 6.11: [

blue]Expected score on a fair die For a fair die,

$$E(X) = \sum_{x=1}^6 x \left(\frac{1}{6}\right) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \boxed{3.5}.$$

Although a single roll cannot equal 3.5, the average of many rolls tends towards 3.5.

Example 6.12: [

cyan]Expectation from a formula Let

$$p_X(x) = \frac{x^2}{30}, \quad x = 0, 1, 2, 3, 4.$$

Since $0^2 + 1^2 + 2^2 + 3^2 + 4^2 = 30$, this is a valid PMF. Its expectation is

$$E(X) = \sum_{x=0}^4 x \frac{x^2}{30} = \frac{0^3 + 1^3 + 2^3 + 3^3 + 4^3}{30} = \boxed{\frac{10}{3}}.$$

6.3.1 Expected monetary value

Expected value is often used to assess games, insurance risks and financial decisions. Gains are represented by positive values and losses by negative values.

Example 6.13: [

orange]Expected gain from a dice game A fair die is rolled. A player wins RM 6 if a 6 is rolled and loses RM 1 otherwise. If X is the player's net gain, then

$$P(X = 6) = \frac{1}{6}, \quad P(X = -1) = \frac{5}{6}.$$

Therefore,

$$E(X) = 6 \left(\frac{1}{6} \right) - 1 \left(\frac{5}{6} \right) = \boxed{\frac{1}{6}}.$$

The game gives an expected gain of about RM 0.17 per play, although the actual outcome of any one play is either RM 6 or RM -1 .

6.3.2 Expectation of a function

If $g(X)$ is a function of X , then

$$E(g(X)) = \sum_x g(x)p_X(x).$$

In particular,

$$E(a) = a, \quad E(aX) = aE(X), \quad E(aX + b) = aE(X) + b,$$

where a and b are constants.

Example 6.14: [

violet]Comparing two expectations Suppose

x	1	2	3
$P(X = x)$	0.1	0.4	0.5

Then

$$E(X) = 1(0.1) + 2(0.4) + 3(0.5) = 2.4,$$

so

$$E(2X) = 2E(X) = \boxed{4.8}.$$

However,

$$E(X^2) = 1^2(0.1) + 2^2(0.4) + 3^2(0.5) = \boxed{6.2}.$$

Thus, in general, $E(X^2) \neq [E(X)]^2$.

Example 6.15: [

teal]Expected weekly profit A bicycle shop models weekly sales by

x	0	1	2	3	4
$P(X = x)$	0.04	0.24	0.33	0.36	0.03

The expected number sold is

$$E(X) = 0(0.04) + 1(0.24) + 2(0.33) + 3(0.36) + 4(0.03) = 2.10.$$

If the profit per bicycle is RM 160 and the weekly fixed cost is RM 50, then the weekly profit is

$$g(X) = 160X - 50.$$

Hence,

$$E(g(X)) = 160E(X) - 50 = 160(2.10) - 50 = \boxed{\text{RM } 286}.$$

6.4 Variance and standard deviation

Expected value describes the centre of a distribution. Variance describes how widely the possible values are dispersed about that centre.

Definition 6.5: Variance and standard deviation

For a discrete random variable with mean $\mu = E(X)$,

$$\text{Var}(X) = E((X - \mu)^2) = \sum_x (x - \mu)^2 p_X(x).$$

An equivalent computational formula is

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

The standard deviation is

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Variance is measured in squared units, while standard deviation has the same units as X .

Example 6.16: [

blue]Calculating a variance Let

x	1	2	3
$P(X = x)$	0.1	0.5	0.4

First,

$$E(X) = 1(0.1) + 2(0.5) + 3(0.4) = 2.3,$$

and

$$E(X^2) = 1^2(0.1) + 2^2(0.5) + 3^2(0.4) = 5.7.$$

Therefore,

$$\text{Var}(X) = 5.7 - (2.3)^2 = \boxed{0.41}, \quad \sigma_X = \sqrt{0.41} \approx \boxed{0.640}.$$

6.4.1 Variance under a linear transformation

For constants a and b ,

$$\text{Var}(a) = 0, \quad \text{Var}(aX) = a^2 \text{Var}(X),$$

and

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Adding a constant shifts every value equally and therefore does not change the spread. Multiplying by a multiplies the standard deviation by $|a|$ and the variance by a^2 :

$$\sigma_{aX+b} = |a|\sigma_X.$$

Example 6.17: [

orange]Transforming a random variable Suppose

x	1	2	3
$P(X = x)$	0.3	0.4	0.3

Then

$$E(X) = 2, \quad E(X^2) = 4.6, \quad \text{Var}(X) = 4.6 - 2^2 = 0.6.$$

If $Y = 10X + 5$, then

$$\begin{aligned} E(Y) &= 10E(X) + 5 = \boxed{25}, \\ \text{Var}(Y) &= 10^2 \text{Var}(X) = \boxed{60}, \\ \sigma_X &= \sqrt{0.6} \approx 0.775, \\ \sigma_Y &= 10\sqrt{0.6} \approx \boxed{7.746}. \end{aligned}$$

6.5 Mode and median

The *mode* of a discrete random variable is a value having the greatest probability. A distribution may have more than one mode.

A *median* is a value m satisfying

$$P(X \leq m) \geq \frac{1}{2} \quad \text{and} \quad P(X \geq m) \geq \frac{1}{2}.$$

For an ordered finite support, it is common to select the smallest possible value m for which $F_X(m) \geq 1/2$.

Example 6.18: [

purple]A distribution with unequal gaps Let

$$p_X(x) = \begin{cases} \frac{x}{8}, & x = 1, 2, 5, \\ 0, & \text{otherwise.} \end{cases}$$

Its CDF is

$$F_X(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{8}, & 1 \leq x < 2, \\ \frac{3}{8}, & 2 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$

The first value at which the CDF reaches at least $1/2$ is 5, so the median is 5. The greatest probability is also attached to 5, so the mode is 5.

Furthermore,

$$E(X) = \frac{1^2 + 2^2 + 5^2}{8} = \frac{15}{4}, \quad E(X^2) = \frac{1^3 + 2^3 + 5^3}{8} = \frac{67}{4},$$

and hence

$$\text{Var}(X) = \frac{67}{4} - \left(\frac{15}{4}\right)^2 = \frac{43}{16}.$$

If $Y = 3X + 5$, then

$$E(Y) = \boxed{\frac{65}{4}}, \quad \text{Var}(Y) = \boxed{\frac{387}{16}}.$$

Example 6.19: [

teal]A symmetric distribution Consider

x	0	1	2	3	4
$P(X = x)$	0.1	0.2	0.4	0.2	0.1

The cumulative probabilities are 0.1, 0.3, 0.7, 0.9, 1.0. Therefore the median is 2,

and the mode is also 2. By symmetry,

$$E(X) = 2.$$

If $Y = X - 3$, then

$$E(Y) = E(X) - 3 = \boxed{-1}.$$

Finally,

$$P(X > E(X)) = P(X > 2) = 0.2 + 0.1 = \boxed{0.3}.$$

6.6 Exercises

Answers to these exercises are provided in section C.6.

1. A random variable Y has

$$P(Y = y) = \frac{ay}{8}, \quad y = 1, 2, 3, 4.$$

Find a , $E(Y)$ and $\text{Var}(Y)$.

2. The distribution of X is

x	0	1	2	3	4
$P(X = x)$	0.26	k	$3k$	0.05	0.09

Find k , $E(X)$ and $\text{Var}(X)$. If $Y = 2X - 7$, also find $E(Y)$ and $\text{Var}(Y)$.

3. A discrete random variable has PMF

$$p_X(x) = \begin{cases} \frac{1}{6}, & x = 1, 5, 9, \\ \frac{1}{4}, & x = 3, 7, \\ 0, & \text{otherwise.} \end{cases}$$

Find $E(X)$ and $\text{Var}(X)$. If $Y = 3X - 2$, find $E(Y)$ and $\text{Var}(Y)$.

4. The number of enquiries received by an office in one hour has distribution

x	0	1	2	3	4	5
$P(X = x)$	0.08	0.17	0.30	0.25	0.15	0.05

Construct the CDF and calculate $P(1 < X \leq 4)$, the mean and the standard deviation.

Tutorial exercises: discrete random variables

5. Determine whether each formula defines a valid probability function, and justify the answer.

(a) $P(X = x) = (2x - 1)/16$, $x = 1, 2, 3, 4$

(b) $P(X = x) = x/3$, $x = -1, 0, 1, 2$

(c) $P(X = x) = 15/[23(1 + 2x)]$, $x = 0, 1, 2$

6. A random variable has probabilities 0.1 at $x = 2, 4, 6, 8$, probabilities 0.2 at $x = 1, 3, 5$, and probability zero otherwise. Verify that it is a PMF and find (a) $P(X = 4)$; (b) $P(X < 4)$; (c) $P(3 \leq X < 6)$.

7. A random variable has probabilities 0.1 at $x = -1, 0, 5$, probabilities a at $x = 1, 3$, and probability 0.3 at $x = 4$. Construct its distribution table, find a , calculate $P(X \geq 3)$ and obtain its CDF.

8. The distribution of X is

x	1	2	3	4	5
$P(X = x)$	0.20	0.25	0.40	k	0.05

Find k , $P(1 \leq X \leq 3)$, $P(X \geq 3)$, $P(2 < X < 5)$ and the CDF.

9. The CDF of X has successive levels

$$0, \quad 0.05, \quad 0.30, \quad 0.60, \quad 0.75, \quad 1$$

on the intervals cut by 0.1, 0.2, 0.3, 0.4, 0.5. Find $P(X \leq 0.4)$, $P(X > 0.3)$, $P(X = 0.3)$ and the PMF.

10. For the same CDF as in Exercise 9, find the PMF, $E(X)$, $\text{Var}(X)$ and, for $Y = 2X - 7$, find $E(Y)$ and $\text{Var}(Y)$.

11. A random variable has PMF

$$P(X = x) = \begin{cases} \frac{kx}{x^2 + 1}, & x = 2, 3, \\ \frac{2kx}{x^2 - 1}, & x = 4, 5. \end{cases}$$

Show that $k = 20/33$. Then find $P(X < 3 \text{ or } X > 4)$, $E(X)$, $\text{Var}(X - 2)$ and the CDF.

12. The number of spelling mistakes has distribution

x	0	1	2	3	4
$P(X = x)$	0.23	0.31	0.27	0.14	0.05

Find the mean μ , standard deviation σ , $P(X > \mu + \sigma)$, mode and median.

13. For $x = 1, 2, 3, 4$, a CDF is defined by $F(x) = 1 - (1 - x/4)^x$, with the usual stepwise extension between these values. Estimate the median, and find $E(X)$ and $\text{Var}(3X)$.

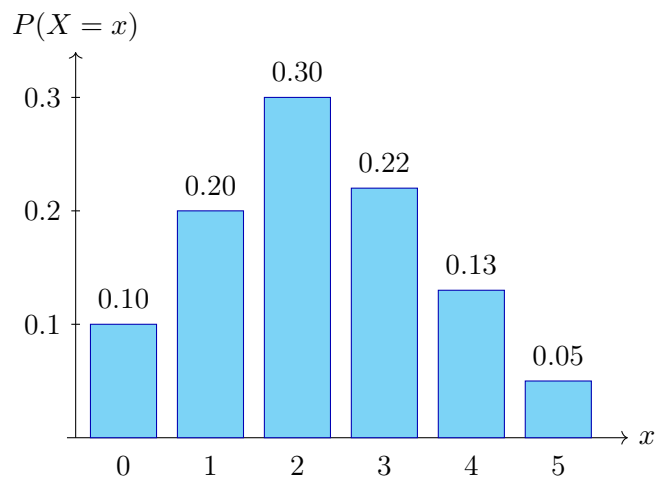
Additional practice: modelling and interpretation

14. A random variable X has probability mass function

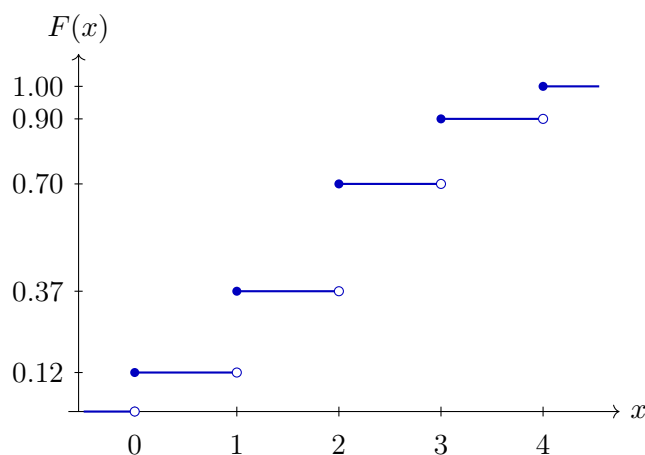
$$P(X = x) = k(x + 1), \quad x = 0, 1, 2, 3.$$

Find k , $P(X \geq 2)$, $E(X)$, $\text{Var}(X)$, the mode and the median.

15. The probability histogram shows the number X of complaints received by a service centre during a randomly selected hour.



- Write the probability distribution in tabular form.
 - Find $P(1 \leq X \leq 3)$ and $P(X \geq 4)$.
 - Construct the CDF.
 - Find the mean, standard deviation, mode and median.
16. The graph below is the CDF of a discrete random variable X .



- (a) Recover the PMF of X .
- (b) Find $P(1 < X \leq 3)$.
- (c) Find $E(X)$ and $\text{Var}(X)$.
- (d) State the mode and median.

17. A municipal pilot project has the following possible net financial outcomes after one year.

Net outcome (RM)	-2000	0	3000	6000
Probability	0.15	0.25	0.40	0.20

Find the expected net outcome, the probability of a positive outcome and the expected combined outcome from 20 independent projects with this same distribution. Explain why the last answer is not a guaranteed amount.

18. The number X of support sessions used by a student has distribution

x	1	2	3
$P(X = x)$	0.2	0.5	0.3

and the programme cost for that student is

$$C = 100 + 40X + 10X^2.$$

Find $E(X)$, $E(X^2)$ and $E(C)$. Calculate $C(E(X))$ and explain why it does not equal $E(C)$.

19. A random variable X has $E(X) = 12$ and $\text{Var}(X) = 9$. Without reconstructing its distribution, find the mean, variance and standard deviation of

- (a) $Y = 50 + 4X$;
- (b) $Z = (X - 12)/3$.

20. A random variable X takes the values 0, 1, 2 and 3. Their respective probabilities are 0.2, a , b and 0.1. Given that $E(X) = 1.4$, find a and b . Then construct the CDF and find $\text{Var}(X)$.

21. Two discrete distributions are given below.

x	0	1	2
$P(A = x)$	0.25	0.50	0.25
$P(B = x)$	0.45	0.10	0.45

Find the mean and variance of each distribution. Which variable is more predictable, and what feature of the distributions explains the difference?

- 22.** A participant pays RM5 to enter a fundraising game. The net return G , after allowing for the entry payment, is

g	-5	3	20
$P(G = g)$	0.6	0.3	0.1

Find the expected net return per play, the probability of making a profit and the expected combined net return over 100 plays. Interpret the sign of the expected value.

- 23.** Of ten surveyed voters, six support a proposal and four do not. Three voters are selected at random without replacement. Let X be the number among the three who support the proposal.
- Construct the probability distribution of X using combinations.
 - Find the probability that a majority of those selected support the proposal.
 - Find $E(X)$ and $\text{Var}(X)$.

- 24.** The number X of claims made on a microinsurance policy in one year has distribution

x	0	1	2	3
$P(X = x)$	0.70	0.20	0.08	0.02

The insurer pays RM1200 per claim and charges an annual premium of RM600. Find the expected annual payout, the standard deviation of the payout and the insurer's expected net income per policy before other expenses.

- 25.** The number V of visits made to a community advice centre in one month has distribution

v	0	1	2	3	4
$P(V = v)$	0.10	0.25	0.30	0.25	0.10

Construct the CDF and find $P(1 \leq V \leq 3)$, $E(V)$, $\text{Var}(V)$, the mode and the median.

- 26.** A random variable takes the values 1, 2, 4 with probabilities a , $2/5$, $3/5 - a$, respectively. Given that $E(X) = 14/5$, find a , construct the CDF and calculate $\text{Var}(X)$.
- 27.** In a survey of 200 students, the observed number X of news sources consulted during one week was recorded as follows.

x	0	1	2	3	4
Frequency	18	42	65	51	24

Use relative frequencies to estimate the PMF. Hence estimate $P(X \geq 3)$, the mean, variance and standard deviation. State one reason why these values should be interpreted as sample estimates rather than exact population quantities.

Additional practice: grouped skills review

- 28.** For a Bernoulli random variable with success probability $p = 0.15, 0.40$ and 0.65 , respectively:
- state the PMF;
 - find the mean and variance.
- 29.** Determine the missing constant or explain why the proposed PMF is invalid.
- $P(X = x) = kx$, for $x = 1, 2, 3, 4$
 - probabilities $0.10, 0.20, k, 0.25$ at $x = 0, 1, 2, 3$
 - $P(X = x) = (x - 1)/6$, for $x = 1, 2, 3$
 - $P(X = x) = k2^x$, for $x = 0, 1, 2$
- 30.** A random variable has probabilities $0.10, 0.25, 0.40, 0.25$ at $x = -1, 0, 1, 2$. Find $P(X \geq 1)$, $P(-1 < X < 2)$, $P(X \neq 0)$ and $P(|X| = 1)$.
- 31.** The PMF at $x = 0, 1, 2, 3$ is $0.20, 0.35, 0.30, 0.15$.
- Construct the CDF.
 - Find $P(X \leq 1)$, $P(1 < X \leq 3)$ and $P(X > 2)$.
- 32.** A CDF has values $0.10, 0.40, 0.75, 1$ at $x = 0, 1, 2, 3$, respectively, with the usual stepwise extension. Recover the PMF and find $P(X \geq 2)$.
- 33.** Find the expected value in each case.
- $X = 0, 2$ with probabilities $0.30, 0.70$
 - $X = -10, 5, 20$ with probabilities $0.20, 0.50, 0.30$
 - a Bernoulli variable with success probability 0.60
 - $X = 1, 2, 3$ with probabilities $0.20, 0.50, 0.30$
- 34.** For probabilities $0.10, 0.20, 0.40, 0.30$ at $x = 0, 1, 2, 3$, find $E(X)$, $E(X^2)$, $\text{Var}(X)$ and the standard deviation.
- 35.** Suppose $E(X) = 5$ and $\text{Var}(X) = 4$. Find the mean, variance and standard deviation of $Y = 3X - 2$, $Z = 10 - X$ and $W = X/2$.
- 36.** Find the mode and median of each distribution.

x	0	1	2
$P(A = x)$	0.20	0.50	0.30
$P(B = x)$	0.45	0.10	0.45

Also find the mode and median when $X = 1, 2, 3$ has probabilities $0.50, 0.25, 0.25$.

- 37.** Decision A yields RM 100 with probability 0.40 and loses RM 20 otherwise. Decision B yields RM 50 with probability 0.50 and RM 0 otherwise. Find both expected values, identify the larger, and explain why the expected value need not be a possible individual outcome.

Applications: PMFs, CDFs and recorded counts

- 38.** The number of enquiries received by a help desk in ten minutes has distribution

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

Find $P(X \geq 3)$, the mean, variance, standard deviation, mode and median.

- 39.** The number of unresolved cases has probabilities 0.20, k , $2k$, 0.20 at $x = 0, 1, 2, 3$. Find k , construct the CDF, and calculate the mean and variance.

- 40.** A sample of 100 service periods records the following frequencies.

x	0	1	2	3	4
f	15	25	35	20	5

Form the empirical PMF and estimate $P(X \geq 3)$, the mean, variance and standard deviation.

- 41.** The histogram represents the number of requests received during a randomly selected interval.

- (a) Write the PMF in a table.
 (b) Find $P(1 \leq X \leq 3)$ and $P(X \geq 4)$.
 (c) Find the mean, variance, standard deviation, mode and median.

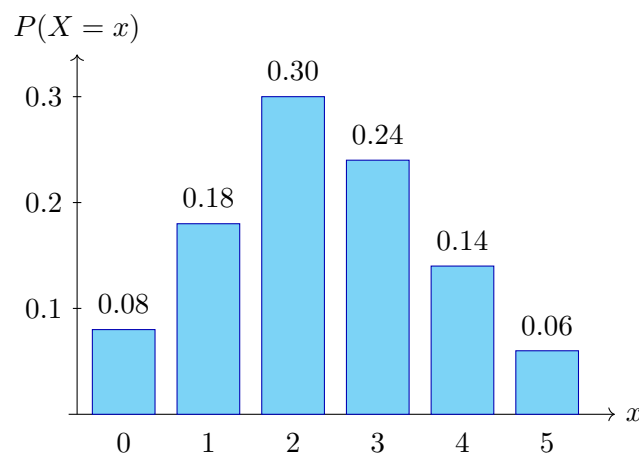


Figure 6.3: Probability histogram for Exercise 41.

- 42.** The graph is the CDF of a discrete random variable.

- (a) Recover the PMF.
- (b) Find $P(1 < X \leq 3)$.
- (c) Find the mean, variance, mode and median.

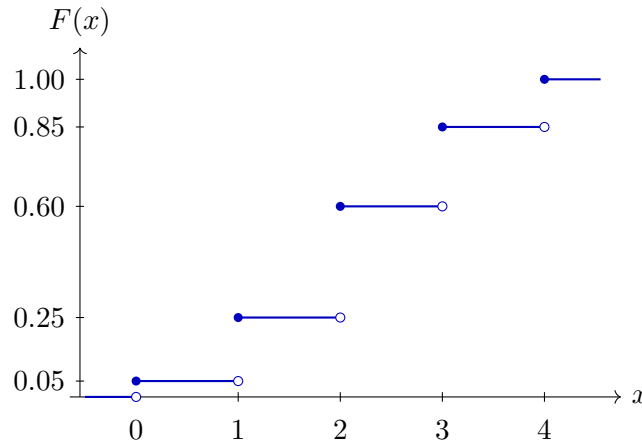


Figure 6.4: Step CDF for Exercise 42.

- 43. A random variable takes 0, 1, 2, 3 with probabilities 0.10, a , b , 0.20. Given $E(X) = 1.7$, find a and b , construct the CDF and calculate the variance.
- 44. Household size X has probabilities 0.10, 0.20, 0.35, 0.25, 0.10 at $x = 0, 1, 2, 3, 4$. Construct the CDF and find $P(X \leq 2)$, $P(1 < X < 4)$, the median and $P(X > \text{median})$.
- 45. A CDF satisfies $F(0) = 0.20$, $F(1) = 0.45$, $F(2) = 0.80$ and $F(3) = 1$. Find the PMF, mode, median and the smallest value whose cumulative probability is at least 0.75.
- 46. Explain why income band, education level and satisfaction category may be discrete recorded variables without being numerical counts. State one danger of treating arbitrary category codes as measured magnitudes.
- 47. The numbers of government applications used by 150 respondents are

x	0	1	2	3	4
f	12	33	48	39	18

Estimate the PMF, $P(X \geq 3)$, the mean, variance and standard deviation. Explain why these are sample estimates.

- 48. A random variable has

$$P(X = x) = k(4 - x), \quad x = 0, 1, 2, 3.$$

Find k , construct the CDF, and calculate the mean and variance.

Applications: expected value and decisions

49. The number of support sessions used has probabilities 0.10, 0.40, 0.35, 0.15 at $x = 0, 1, 2, 3$. Find the mean, mode and median, and interpret the expected value.
50. Let $X = 0, 1, 2$ with probabilities 0.20, 0.50, 0.30. A programme's cost is $C = 50 + 20X + 5X^2$. Find $E(X)$, $E(X^2)$, $E(C)$ and $C(E(X))$. Explain the difference between the last two values.
51. A fundraising game has net outcomes RM -10 , RM 5 and RM 40 with probabilities 0.60, 0.30 and 0.10. Find the expected net return, the probability of a profit and the expected combined net return over 200 plays.
52. A public project has possible net outcomes, in ringgit, $-2000, 0, 5000, 9000$ with probabilities 0.20, 0.30, 0.35, 0.15. Find the expected outcome, probability of a positive outcome and expected combined outcome from ten comparable projects.
53. Demand from one service user is $X = 0, 1, 2$ units with probabilities 0.30, 0.50, 0.20. Find $E(X)$ and the expected total demand from 500 comparable users. Explain why independence is not required for this expectation calculation.
54. Annual claim count X has probabilities 0.65, 0.25, 0.08, 0.02 at $x = 0, 1, 2, 3$. Each claim costs RM 1000 and the premium is RM 650. Find the expected payout, standard deviation of payout and expected net income before other expenses.
55. Decision A pays RM 0 or RM 100 with equal probabilities. Decision B pays RM 40 or RM 60 with equal probabilities. Compare their means and variances. Which is more predictable?
56. Claim count $X = 0, 1, 2$ has probabilities 0.80, 0.15, 0.05 and each claim costs RM 2000. Find the break-even premium based only on expected claim payments.
57. Let I_i indicate whether applicant i succeeds, with $P(I_i = 1) = 0.40$. For 300 applicants, find the expected number of successes. Explain why the result does not claim that exactly this many applicants will succeed.
58. For $X = 1, 2, 3$ with probabilities 0.20, 0.50, 0.30, find $E[(X - 2)^2]$.
59. A benefit is RM 0 with probability 0.90 and RM 1000 with probability 0.10. Find the expected value, median and mode. Explain which summary better describes the most common individual outcome.

Applications: variance, transformations and predictability

60. For $X = 0, 1, 2, 3$ with probabilities 0.20, 0.30, 0.30, 0.20, find the mean, $E(X^2)$, variance and standard deviation.
61. Given $E(X) = 4$ and $E(X^2) = 19$, find the variance and standard deviation. State the identity used.
62. Suppose $E(X) = 8$ and $\text{Var}(X) = 16$. Find the mean, variance and standard deviation of $Y = 3X + 5$ and $Z = (X - 8)/4$.
63. A variable has mean 20 and standard deviation 5. For $Z = (X - 20)/5$, find $E(Z)$ and $\text{Var}(Z)$ and explain the purpose of this standardisation.
64. Distributions A and B place probability on $x = 0, 2, 4$ as follows:

x	0	2	4
$P(A = x)$	0.25	0.50	0.25
$P(B = x)$	0.40	0.20	0.40

Find both means and variances. Which is more concentrated around its mean?

65. A variable has mean 30, variance 36 and standard deviation 6. Find these summaries for $X + 10$, $0.5X$ and $2X - 5$.
66. The probability histograms show two distributions with the same support. Calculate their means and variances and identify the more predictable distribution.

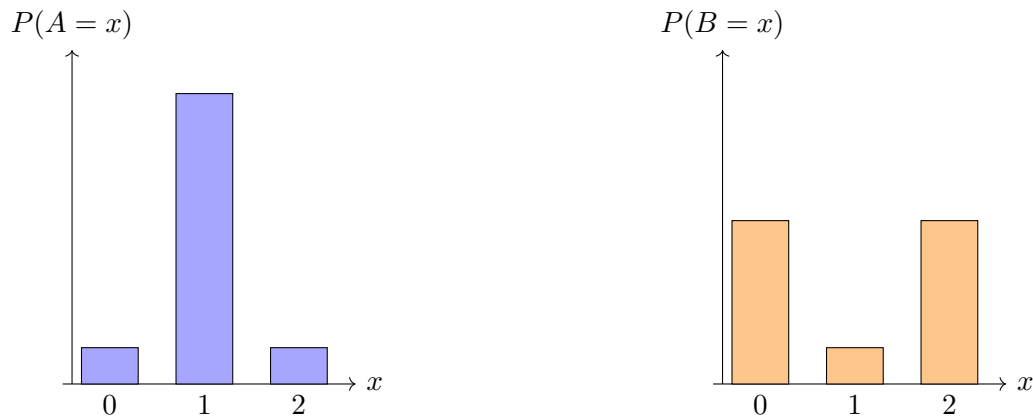


Figure 6.5: Equal means can be accompanied by very different spreads.

67. Weekly service time has mean 6 hours and variance 4 hours squared. Convert the mean, variance and standard deviation to minutes.
68. For Bernoulli variables with $p = 0.10, 0.50, 0.90$, calculate the variance and explain why it is greatest at $p = 0.50$.
69. A report claims that $E(X) = 3$ and $E(X^2) = 8$. Show why these summaries cannot belong to any random variable with finite second moment.

70. An empirical distribution has values 0, 1, 2, 3 with frequencies 10, 20, 15, 5. Treating relative frequencies as probabilities, find the mean, variance and standard deviation.

Applications: distributions derived from finite samples

71. Seven of twelve voters support a proposal. Four voters are selected without replacement, and X is the number of supporters selected. Construct the PMF, then find $P(X \geq 3)$, $E(X)$ and $\text{Var}(X)$.
72. A committee of three is selected from five women and three men. Let X be the number of women selected. Construct the PMF and find $E(X)$.
73. Four of fifteen records contain errors. Two are selected without replacement. Let X be the number of error-containing records. Construct the PMF and find $E(X)$.
74. Using the distribution in Exercise 73, construct the CDF and find $P(X \geq 1)$ and the median.
75. A population contains twelve policy supporters and eight non-supporters. Three people are selected without replacement. Construct the distribution of the number X of supporters, calculate $P(X \geq 2)$ and find $E(X)$.
76. Two independent appointments are each attended with probability 0.70. Construct the distribution of the number attended and find its mean and variance.
77. A web contact succeeds with probability 0.60 and a phone contact succeeds with probability 0.80, independently. Construct the PMF of the number of successful contacts and find its mean and variance.
78. Sixty per cent of residents are in Region A and 40% in Region B. Conditional visit-count distributions are

x	0	1	2
$P(X = x A)$	0.50	0.40	0.10
$P(X = x B)$	0.20	0.50	0.30

Find the unconditional PMF and its mean.

79. Two samples give empirical distributions

x	0	1	2	3
f_A	20	50	25	5
f_B	10	40	40	10

Find the empirical mean and variance for each sample.

80. A distribution has probabilities $a, 0.40, b$ at $x = 0, 1, 2$ and mean 1.1. Find a, b , the CDF and the variance.
81. If $P(X = x) = k(x + 2)$ for $x = 0, 1, 2, 3$, find k , $P(X \geq 2)$ and $E(X)$.

Applications: risk, programmes and social decisions

- 82.** Claim count $X = 0, 1, 2, 3$ has probabilities 0.75, 0.18, 0.06, 0.01. Each claim costs RM 1500 and the premium is RM 500. Find expected payout, payout standard deviation and expected net income before other expenses.

- 83.** A benefit-processing error has the following cost distribution:

Cost (RM)	0	200	500
Probability	0.70	0.20	0.10

Find the expected cost, standard deviation and expected total cost over 100 comparable cases.

- 84.** A project has net outcome, in millions of ringgit, $-1, 0.5$ or 3 with probabilities $0.25, 0.50, 0.25$. Find its expected value and standard deviation. State why the mean alone is insufficient for a public decision.

- 85.** A compensation payment has the following distribution:

Payment (RM)	100	500	2000
Probability	0.60	0.30	0.10

Find its mean, median and mode. Explain why the mean is not the payment received by a typical claimant.

- 86.** Arrivals $X = 0, 1, 2, 3, 4$ have probabilities $0.05, 0.20, 0.35, 0.25, 0.15$. Find $E(X)$ and $P(X \geq 3)$. If three service slots are available, let $U = \max(3 - X, 0)$ be unused slots; find $E(U)$.
- 87.** A loan produces loss RM 0, RM 5000 or RM 20 000 with probabilities $0.94, 0.05, 0.01$. Find expected loss and standard deviation. Explain what the large standard deviation indicates.
- 88.** A fundraising game has net returns RM -2 , RM 3 and RM 18 with probabilities $0.70, 0.25, 0.05$. Find expected return, probability of profit and expected total return over 100 plays.
- 89.** Three households independently adopt a service with probability 0.40. Construct the distribution of the number adopting and find its mean and variance.
- 90.** Policy A yields 0 or 100 benefit units with equal probability; Policy B yields 40 or 60 with equal probability. Compare expected benefit and standard deviation. State one reason a decision-maker might prefer each policy.
- 91.** Explain why an expected outcome calculated from a fitted programme distribution is predictive rather than proof of the programme's causal effect.

Applications: responsible interpretation

92. Explain why expected value need not describe a typical outcome, using a distribution with a rare large value as an example.
93. Explain why two policies with equal expected benefit can represent substantially different risk and distributional consequences.
94. An empirical PMF is estimated from a survey. Give three reasons why it may differ from the population PMF.
95. Published probabilities rounded to two decimals sum to 0.99. Give two defensible ways to handle this and state what should be disclosed.
96. Give three limitations of reducing social outcomes to expected monetary value.
97. Explain how one overall PMF can conceal different distributions across regions or demographic groups.
98. An allocation rule maximises expected total benefit. Give four fairness or governance considerations that the calculation omits.
99. Explain why a PMF fitted before a major policy or economic change may give unreliable later predictions.
100. Give a six-stage checklist for constructing, calculating, checking and communicating a discrete-random-variable model in a social-science application.

For a discrete distribution, the PMF describes the probability at each possible value and the CDF accumulates those probabilities. Expected value summarises the long-run centre, while variance and standard deviation summarise spread. Linear-transformation rules allow these measures to be updated without reconstructing the entire distribution.

Chapter 7

Sequences and Series

Chapter guide

Compare arithmetic and geometric patterns, develop their finite and infinite sums, then use sigma notation and the binomial theorem in algebraic and applied settings.

Selected background sources: (Chai et al., 2024; Ministry of Education Malaysia, 2018).

Sequences describe ordered numerical patterns, while series describe the sums of their terms. These ideas are used in financial growth, depreciation, population models, instalment plans and many other situations in which a quantity changes repeatedly according to a fixed rule.

7.1 Sequences and series

Definition 7.1: Sequence

A *sequence* is an ordered list of numbers. Each number is called a *term*. The first, second and n -th terms are commonly denoted by a_1 , a_2 and a_n , respectively.

The order of a sequence is significant. Thus, the sequences

$$1, 4, 7 \quad \text{and} \quad 7, 4, 1$$

are different, even though they contain the same numbers. This contrasts with sets, for which order is irrelevant.

A sequence is *finite* if it has a final term and *infinite* if it continues indefinitely:

$$1, 2, 3, \dots, n \quad \text{and} \quad 1, 2, 3, \dots$$

A formula for a_n in terms of the position n is called an *explicit formula*, or general term.

Example 7.1: [

blue]Using a general term Let

$$a_n = 4(-2)^n, \quad n = 1, 2, 3, \dots$$

Then

$$\begin{aligned}a_1 &= 4(-2) = -8, \\a_3 &= 4(-2)^3 = -32, \\a_6 &= 4(-2)^6 = 256, \\a_{10} &= 4(-2)^{10} = 4096.\end{aligned}$$

The alternating signs arise because the base -2 is negative.

Definition 7.2: Series

A *series* is the sum of the terms of a sequence. A finite sequence $a_1, a_2, \dots, a_n$ produces the finite series

$$a_1 + a_2 + \cdots + a_n,$$

while an infinite sequence produces an infinite series

$$a_1 + a_2 + a_3 + \cdots.$$

7.1.1 Sigma notation

The Greek capital letter sigma, Σ , provides a compact notation for a sum:

$$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \cdots + a_n.$$

The expression below Σ gives the first index, the expression above it gives the final index, and the expression to the right gives the term to be added.

For example,

$$1 + 2 + 3 + 4 + 5 + 6 = \sum_{i=1}^6 i.$$

Example 7.2: [

cyan]Evaluating finite sums Expanding the terms gives

$$\begin{aligned}\sum_{i=1}^4 (4i + 1) &= 5 + 9 + 13 + 17 = \boxed{44}, \\ \sum_{k=2}^5 (2 + k^3) &= 10 + 29 + 66 + 127 = \boxed{232}, \\ \sum_{n=1}^4 (2n + 4) &= 6 + 8 + 10 + 12 = \boxed{36}.\end{aligned}$$

The index is a temporary variable; changing its letter does not change the value of the sum.

Useful summation rules include

$$\begin{aligned}\sum_{i=1}^n (cu_i + dv_i) &= c \sum_{i=1}^n u_i + d \sum_{i=1}^n v_i, \\ \sum_{i=1}^n c &= nc, \\ \sum_{i=1}^n i &= \frac{n(n+1)}{2},\end{aligned}$$

where c and d are constants.

7.2 Arithmetic sequences

Definition 7.3: Arithmetic sequence

An *arithmetic sequence* is a sequence in which the difference between successive terms is constant. This constant is the *common difference* d .

If the first term is a , an arithmetic sequence has the form

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \dots$$

Its explicit formula is

$$\boxed{a_n = a + (n - 1)d}.$$

It may also be defined recursively by

$$a_1 = a, \quad a_n = a_{n-1} + d \quad (n \geq 2).$$

Example 7.3: [

teal]Using the recursive rule Suppose $a_1 = 3$ and $d = 5$. Then

$$a_2 = 3 + 5 = 8, \quad a_3 = 8 + 5 = 13, \quad a_4 = 13 + 5 = 18.$$

The first four terms are therefore

$$\boxed{3, 8, 13, 18}.$$

Example 7.4: [

orange]Finding an explicit rule The sequence

$$3, 7, 11, 15, 19, \dots$$

has $a = 3$ and $d = 4$. Therefore,

$$a_n = 3 + (n - 1)4 = \boxed{4n - 1}.$$

For example, $a_5 = 4(5) - 1 = 19$.

Example 7.5: [

violet]Finding a sequence from two terms Suppose $a_8 = 25$ and $a_{12} = 41$. Since

$$a_{12} - a_8 = 4d,$$

it follows that

$$41 - 25 = 4d, \quad d = 4.$$

Using $a_8 = a + 7d$,

$$25 = a + 7(4), \quad a = -3.$$

Hence,

$$a_n = -3 + 4(n - 1) = \boxed{4n - 7}.$$

7.3 Arithmetic series

The sum of the first n terms of an arithmetic sequence is denoted by S_n . If the first term is a and the last term is ℓ , write the sum in forward and reverse order:

$$S_n = a + (a + d) + (a + 2d) + \cdots + \ell,$$

$$S_n = \ell + (\ell - d) + (\ell - 2d) + \cdots + a.$$

Adding corresponding terms gives n copies of $a + \ell$, so

$$2S_n = n(a + \ell).$$

Therefore,

$$\boxed{S_n = \frac{n}{2}(a + \ell)}.$$

Since $\ell = a + (n - 1)d$, an equivalent formula is

$$\boxed{S_n = \frac{n}{2}[2a + (n - 1)d]}.$$

Example 7.6: [

blue]Adding the first fifty integers The series

$$1 + 2 + 3 + \cdots + 49 + 50$$

is arithmetic with $n = 50$, $a = 1$ and $\ell = 50$. Thus,

$$S_{50} = \frac{50}{2}(1 + 50) = 25(51) = \boxed{1275}.$$

Example 7.7: [

cyan]A sum specified by its number of terms Find the sum of the first 30 terms of

$$3 + 7 + 11 + 15 + \cdots .$$

Here $a = 3$, $d = 4$ and $n = 30$. Hence,

$$S_{30} = \frac{30}{2} [2(3) + 29(4)] = 15(122) = \boxed{1830}.$$

Example 7.8: [

purple]A business-sales model A small business records sales of RM 20 000 in its first year and aims to increase annual sales by RM 15 000 each year. If the target is achieved, the annual sales form an arithmetic sequence with

$$a = 20\,000, \quad d = 15\,000.$$

The twentieth-year sales are

$$a_{20} = 20\,000 + 19(15\,000) = 305\,000.$$

Therefore, total sales over the first 20 years are

$$S_{20} = \frac{20}{2}(20\,000 + 305\,000) = \boxed{\text{RM } 3\,250\,000}.$$

7.4 Geometric sequences

Definition 7.4: Geometric sequence

A *geometric sequence* is a sequence in which the ratio of each term to the preceding term is constant. This constant is the *common ratio* r .

If the first term is a , a geometric sequence has the form

$$a, \quad ar, \quad ar^2, \quad ar^3, \dots$$

Its explicit formula is

$$a_n = ar^{n-1},$$

and its recursive form is

$$a_1 = a, \quad a_n = ra_{n-1} \quad (n \geq 2).$$

Example 7.9: [

blue]Finding a distant geometric term A geometric sequence has first term 20 and common ratio 1.05. Its fifteenth term is

$$a_{15} = 20(1.05)^{14} \approx \boxed{39.60}.$$

The common ratio 1.05 corresponds to an increase of 5% from one term to the next.

Example 7.10: [

teal]Writing a recursive rule For

$$96, 48, 24, 12, \dots,$$

the common ratio is

$$r = \frac{48}{96} = \frac{1}{2}.$$

Thus,

$$a_1 = 96, \quad a_n = \frac{1}{2}a_{n-1}.$$

The next two terms are

$$a_5 = 6, \quad a_6 = 3.$$

Example 7.11: [

orange]Finding a geometric rule from two terms Suppose

$$a_2 = -18, \quad a_5 = \frac{2}{3}.$$

Since $a_2 = ar$ and $a_5 = ar^4$,

$$r^3 = \frac{a_5}{a_2} = \frac{2/3}{-18} = -\frac{1}{27}.$$

Therefore $r = -1/3$, and

$$a = \frac{-18}{-1/3} = 54.$$

The explicit rule is

$$a_n = 54 \left(-\frac{1}{3} \right)^{n-1}.$$

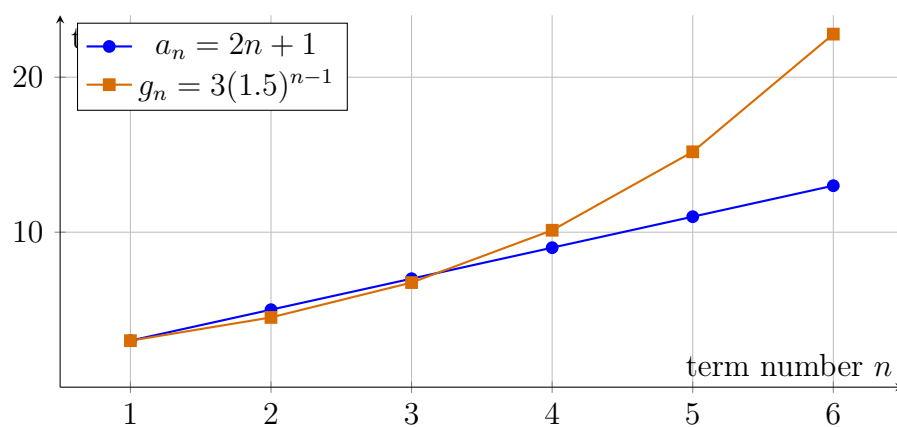


Figure 7.1: Arithmetic growth is linear, whereas geometric growth is multiplicative.

7.5 Geometric series

A finite geometric series has the form

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1}.$$

Multiplying by r gives

$$rS_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n.$$

Subtracting the first equation from the second,

$$(r - 1)S_n = a(r^n - 1).$$

For $r \neq 1$,

$$\boxed{S_n = \frac{a(r^n - 1)}{r - 1}} \quad \text{or} \quad \boxed{S_n = \frac{a(1 - r^n)}{1 - r}}.$$

If $r = 1$, every term is a and $S_n = na$.

Example 7.12: [

violet]A rapidly growing geometric series For

$$9 + 36 + 144 + 576 + \cdots,$$

the first term is $a = 9$ and the common ratio is $r = 4$. The sum of the first ten terms is

$$S_{10} = \frac{9(4^{10} - 1)}{4 - 1} = 3(1\,048\,575) = \boxed{3\,145\,725}.$$

7.5.1 Infinite geometric series

For a geometric series,

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

If

$$|r| < 1,$$

then $r^n \rightarrow 0$ as $n \rightarrow \infty$. The series converges to

$$\boxed{S_\infty = \frac{a}{1 - r}}.$$

If $|r| \geq 1$ and $a \neq 0$, the terms do not approach zero and the infinite series does not have a finite sum.

Example 7.13: [

cyan]Finding a common ratio from the sum to infinity A geometric series has first term 20 and sum to infinity 15. Then

$$15 = \frac{20}{1 - r}.$$

Hence,

$$15(1 - r) = 20, \quad r = \boxed{-\frac{1}{3}}.$$

The value satisfies $|r| < 1$, so the infinite sum exists.

Example 7.14: [

teal]An infinite sum in sigma notation Consider

$$\sum_{n=1}^{\infty} \frac{5}{10^n} = \frac{5}{10} + \frac{5}{10^2} + \frac{5}{10^3} + \cdots.$$

This is geometric with $a = 1/2$ and $r = 1/10$. Therefore,

$$S_{\infty} = \frac{1/2}{1 - 1/10} = \boxed{\frac{5}{9}}.$$

7.5.2 Recurring decimals

A recurring decimal can be written as an infinite geometric series.

Example 7.15: [

orange]Converting a recurring decimal to a fraction The decimal

$$0.717171 \dots$$

can be written as

$$\frac{71}{100} + \frac{71}{100^2} + \frac{71}{100^3} + \cdots.$$

This is an infinite geometric series with $a = 71/100$ and $r = 1/100$. Hence,

$$0.717171 \dots = \frac{71/100}{1 - 1/100} = \boxed{\frac{71}{99}}.$$

7.6 Arithmetic and geometric means

For numbers $x_1, x_2, \dots, x_n$, the *arithmetic mean* is

$$\text{AM} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

For positive numbers, the *geometric mean* is

$$\text{GM} = \sqrt[n]{x_1 x_2 \cdots x_n}.$$

The arithmetic mean is used for additive data, while the geometric mean is particularly useful for multiplicative changes such as compound growth rates. For positive numbers,

$$\text{AM} \geq \text{GM},$$

with equality precisely when all the numbers are equal.

Example 7.16: [

blue]Calculating arithmetic and geometric means For the sequence 3, 7, 11, 15, 19,

$$\text{AM} = \frac{3 + 7 + 11 + 15 + 19}{5} = \boxed{11}.$$

For the positive numbers 2, 8, 32,

$$\text{GM} = \sqrt[3]{2 \cdot 8 \cdot 32} = \sqrt[3]{512} = \boxed{8}.$$

7.6.1 Inserting arithmetic means

To insert k arithmetic means between two values, treat the endpoints and the inserted values as an arithmetic sequence containing $k + 2$ terms.

Example 7.17: [

purple]Five arithmetic means Insert five arithmetic means between 18 and 54. There are seven terms in total, so

$$54 = 18 + 6d, \quad d = 6.$$

The completed sequence is

$$18, \boxed{24, 30, 36, 42, 48}, 54.$$

7.6.2 Inserting geometric means

To insert k geometric means between two positive values, form a geometric sequence of $k + 2$ terms and solve for its common ratio.

Example 7.18: [

violet]Four geometric means Insert four geometric means between 4 and 972. There are six terms, so

$$972 = 4r^5, \quad r^5 = 243 = 3^5.$$

Thus $r = 3$, and the completed sequence is

$$4, \boxed{12, 36, 108, 324}, 972.$$

7.7 Applications of arithmetic and geometric models

The explicit-term and sum formulas become especially useful when a repeated change is described in words. A constant absolute change produces an arithmetic model, while a constant percentage or proportional change produces a geometric model.

Example 7.19: [

blue]A decreasing stamp collection A collector begins with 500 stamps and gives away 5 stamps each week. After n weeks, the number remaining is

$$a_n = 500 - 5n.$$

At the end of the tenth week,

$$a_{10} = 500 - 5(10) = \boxed{450}.$$

This is an arithmetic model because the collection changes by a fixed number, rather than a fixed percentage.

Example 7.20: [

teal]Accumulating monthly savings A worker saves RM 20 in the first month and increases the monthly saving by RM 5 thereafter. The total saved after n months is

$$S_n = \frac{n}{2}[2(20) + (n-1)5].$$

Now,

$$S_{11} = \frac{11}{2}(40 + 50) = 495,$$

whereas

$$S_{12} = \frac{12}{2}(40 + 55) = 570.$$

Therefore the worker first reaches at least RM 500 during the

$$\boxed{\text{twelfth month}}.$$

Example 7.21: [

orange]The height of a bouncing ball A ball is dropped from a height of 100 cm and rebounds to 90% of its previous height. The height after the n -th impact is

$$h_n = 100(0.9)^n.$$

After the tenth impact, it rises to

$$h_{10} = 100(0.9)^{10} \approx \boxed{34.87 \text{ cm}}.$$

The heights form a decreasing geometric sequence.

Example 7.22: [

violet]The economic multiplier Suppose a tax rebate of \$1200 is received and 80% of every receipt is spent again. The successive amounts spent are

$$960, \quad 768, \quad 614.4, \dots$$

This is an infinite geometric series with $a = 960$ and $r = 0.8$. Hence,

$$S_{\infty} = \frac{960}{1 - 0.8} = \boxed{\$4800}.$$

The model assumes that the same spending proportion continues indefinitely.

7.8 Standard finite sums

Several frequently occurring finite sums have closed formulas:

$$\boxed{\sum_{r=1}^n r = \frac{n(n+1)}{2}},$$

$$\boxed{\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}},$$

and

$$\boxed{\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2}.$$

Together with the linearity rules for summation, these identities allow any polynomial in the index of degree at most three to be summed directly.

Example 7.23: [

cyan]Applying the standard formulas The formulas give

$$\begin{aligned} 1 + 2 + \cdots + 15 &= \frac{15(16)}{2} = \boxed{120}, \\ 1^2 + 2^2 + \cdots + 8^2 &= \frac{8(9)(17)}{6} = \boxed{204}, \\ 1^3 + 2^3 + \cdots + 6^3 &= \left[\frac{6(7)}{2} \right]^2 = \boxed{441}. \end{aligned}$$

Also,

$$\begin{aligned} \sum_{r=1}^n r(r+1) &= \sum_{r=1}^n (r^2 + r) \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\ &= \boxed{\frac{n(n+1)(n+2)}{3}}. \end{aligned}$$

Example 7.24: [

purple]Rewriting less familiar sums Consider

$$3(3) + 4(5) + 5(7) + \cdots + 17(31).$$

Its general term is $r(2r-3)$ for $r = 3, \dots, 17$. Thus,

$$\begin{aligned} \sum_{r=3}^{17} r(2r-3) &= 2 \sum_{r=3}^{17} r^2 - 3 \sum_{r=3}^{17} r \\ &= 2(1785 - 5) - 3(153 - 3) \\ &= \boxed{3110}. \end{aligned}$$

Similarly,

$$\sum_{r=1}^{10} (r+7)(r-1) = \sum_{r=1}^{10} (r^2 + 6r - 7) = 385 + 330 - 70 = \boxed{645},$$

and

$$3 \sum_{r=1}^{18} (r+1)^3 = 3 \left(\sum_{k=1}^{19} k^3 - 1 \right) = 3(36100 - 1) = \boxed{108297}.$$

7.9 Factorials and binomial coefficients

7.9.1 Factorial notation

For a positive integer n ,

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1,$$

and, by definition,

$$0! = 1.$$

For example, $10! = 3\,628\,800$. Factorials frequently simplify by cancellation:

$$\frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

Example 7.25: [

blue]Evaluating factorial expressions First,

$$6! = \boxed{720},$$

$$\frac{8!}{3!} = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = \boxed{6720},$$

and

$$\frac{n!}{(n-3)!} = \boxed{n(n-1)(n-2)}.$$

7.9.2 Binomial coefficients

The binomial coefficient is

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n.$$

It also counts the number of ways to select r objects from n objects, without regard to order. Important properties include

$$\binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{r} = \binom{n}{n-r}.$$

Adjacent entries satisfy *Pascal's identity*:

$$\boxed{\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}}.$$

Indeed,

$$\begin{aligned} \binom{n}{r-1} + \binom{n}{r} &= \frac{n!}{(r-1)!(n-r+1)!} + \frac{n!}{r!(n-r)!} \\ &= \frac{rn! + (n-r+1)n!}{r!(n-r+1)!} \\ &= \frac{(n+1)!}{r!(n+1-r)!} = \binom{n+1}{r}. \end{aligned}$$

7.9.3 Pascal's triangle

Pascal's triangle lists the binomial coefficients. Every interior entry is the sum of the two entries directly above it:

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & & 1 & \\ & & & 1 & & 2 & \\ & & 1 & & 3 & & 1 \\ & 1 & & 4 & & 6 & \\ 1 & & 5 & & 10 & & 10 & \\ & & & & & 5 & & 1 \end{array}$$

The entries in row n are

$$\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}.$$

7.10 The binomial theorem

Definition 7.5: Binomial theorem

For a non-negative integer n ,

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r.$$

The general term is

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r, \quad r = 0, 1, \dots, n.$$

Proof sketch. The result can be proved by induction on n . For $n = 0$, both sides equal 1. Assume that

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r.$$

Multiplying by $a+b$ produces two sums. After shifting the index of the second sum, the

coefficient of $a^{n+1-r}b^r$ is

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$$

by Pascal's identity, with the two endpoint coefficients equal to 1. Therefore the formula holds for $n+1$, completing the induction. $\square$

The expansion has $n+1$ terms. In successive terms, the power of a decreases from n to 0, while the power of b increases from 0 to n . The total degree of each term is n , and the coefficients are symmetric.

For the important special case $a = 1$,

$$(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \cdots + x^n.$$

7.10.1 Expanding binomial expressions

Example 7.26: [

teal]Expansions using Pascal's triangle The fourth row of Pascal's triangle contains 1, 4, 6, 4, 1, so

$$(x-y)^4 = x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4.$$

Using the fifth row,

$$(x-2y)^5 = x^5 - 10x^4y + 40x^3y^2 - 80x^2y^3 + 80xy^4 - 32y^5.$$

Example 7.27: [

orange]Expansions using the theorem First,

$$(1+x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5.$$

Also,

$$(2-y)^5 = 32 - 80y + 80y^2 - 40y^3 + 10y^4 - y^5.$$

Finally,

$$(2+3x)^4 = 16 + 96x + 216x^2 + 216x^3 + 81x^4.$$

7.10.2 Finding a specified coefficient

The general term can be used without writing the complete expansion.

Example 7.28: [

violet]The coefficient of a selected power In the expansion of $(2 + 3x)^{10}$, the term containing x^2 occurs when $r = 2$:

$$T_3 = \binom{10}{2} 2^8 (3x)^2.$$

Therefore, the coefficient of x^2 is

$$\binom{10}{2} 2^8 3^2 = 45(256)(9) = \boxed{103680}.$$

Example 7.29: [

purple]Analysing a general binomial term For $(y - 2x^3)^6$, the general term is

$$T_{r+1} = \binom{6}{r} y^{6-r} (-2x^3)^r = \binom{6}{r} (-2)^r y^{6-r} x^{3r}.$$

The term containing x^6 has $3r = 6$, so $r = 2$. It is

$$\binom{6}{2} (-2)^2 y^4 x^6 = 60y^4 x^6,$$

and the coefficient of x^6 is $\boxed{60y^4}$.

The first two terms are

$$y^6 - 12y^5 x^3,$$

the term independent of x is y^6 , and the last two terms are

$$-192yx^{15} + 64x^{18}.$$

7.10.3 Numerical approximation

Binomial expansion can evaluate powers of numbers close to a convenient base.

Example 7.30: [

cyan]Approximating a decimal power Write $2.02 = 2 + 0.02$. Then

$$\begin{aligned} (2.02)^6 &= 2^6 + 6(2^5)(0.02) + 15(2^4)(0.02)^2 \\ &\quad + 20(2^3)(0.02)^3 + 15(2^2)(0.02)^4 \\ &\quad + 6(2)(0.02)^5 + (0.02)^6. \end{aligned}$$

Evaluating the terms gives

$$(2.02)^6 = 67.937289638464 \dots \approx \boxed{67.9373}$$

to four decimal places.

7.11 Exercises

Answers to these exercises are provided in section C.7.

1. The first three terms of an arithmetic sequence are

$$12, \quad k^2, \quad 5k.$$

Find the two possible values of k .

2. Write a recursive formula for each sequence.

(a) $9, 14, 19, 24, \dots$

(b) $2, -3, -8, -13, \dots$

3. Find:

(a) the sum of the first 100 positive integers;

(b) the sum of the first 50 positive integers.

4. Find the sum

$$151 + 147 + 143 + 139 + \dots + (-5).$$

5. In an arithmetic sequence,

$$a_{41} + a_9 = -232, \quad a_{27} = -130.$$

Find the sum of the first 21 terms.

6. The third term of a geometric sequence is 36 and the sixth term is 972. Find its first term and common ratio.

7. Give the first four terms and state whether each sequence is arithmetic or geometric.

(a) $a_1 = 3, a_n = 4a_{n-1}$

(b) $a_1 = 7, a_n = a_{n-1} + 5$

8. A business has first-year sales of RM 20 000 and increases sales by RM 15 000 each year. Find its total sales during the first 20 years.

9. The second term of a geometric series is 40 and the fifth term is 20.48.

(a) Find the first term and common ratio.

(b) Find the sum of the first ten terms.

(c) Find the sum to infinity.

10. Express each recurring decimal as an infinite geometric series and hence as a fraction in simplest form.

(a) $0.312312312\dots$

(b) $4.086086086\dots$

11. Evaluate

(a) $\sum_{r=1}^{10} (r+7)(r-1);$

(b) $3 \sum_{r=1}^{18} (r+1)^3.$

12. Simplify

(a) $6!;$

(b) $8!/3!;$

(c) $n!/(n-3)!.$

13. Expand

(a) $(x-y)^4;$

(b) $(x-2y)^5.$

14. Find the coefficient of x^2 in $(2+3x)^{10}.$

15. For the expansion of $(y-2x^3)^6$, find the general term, the coefficient of x^6 , the term independent of x , and the last two terms.

16. Use the binomial theorem to evaluate $(2.02)^6$ correct to four decimal places.

Tutorial exercises: sequences and series

17. Consider $a_n = 2n + 1$ and $S_n = \sum_{k=1}^n (2k + 1).$

(a) Identify the sequence and the series.

(b) Explain the difference between a_n and S_n .

(c) Find the first four terms of a_n .

(d) Find S_4 .

18. In an arithmetic progression, the sum of the third and fourth terms is 18, and the seventh term exceeds the fifth by 14. Find the first term and common difference.

19. A series has $S_n = 5n + n^2$. Find its n th term and show that the terms form an arithmetic progression.

20. Insert five arithmetic means between (a) 3 and 33; (b) $1/3$ and $10/3$.

21. Find the sum of all multiples of 3 between 1 and 400.

22. Find the eighth term and the n th term of each geometric progression.

- (a) $\frac{1}{4}, \frac{1}{6}, \frac{1}{9}, \dots$
 - (b) $4, 1.2, 0.36, 0.108, \dots$
 - (c) $1, -\sqrt{3}, 3, -3\sqrt{3}, \dots$
- 23.** The fourth term of a geometric progression is 15 and the seventh is -960 . Find the first term and common ratio.
- 24.** The first three positive terms of a geometric progression are $x - 2, x, 2x - 3$. Find x and the fifth term.
- 25.** In a geometric progression, the second term exceeds the first by 20, and the fourth exceeds the second by 15. Find the possible first terms.
- 26.** Find the geometric mean of 3, 5, 2 and 4.
- 27.** For $12 + 18 + 27 + \dots$, find the smallest n for which the sum of the first n terms exceeds 1000.
- 28.** A geometric progression has $S_{10} = 4$, and the sum of terms 11 through 30 is 48. Find the sum of terms 31 through 60.
- 29.** Express each recurring decimal as an infinite geometric series and hence as a fraction:
(a) $0.05555\dots$; (b) $3.22222\dots$
- 30.** A shop sold 150 computers in 2000 and 10 more in every subsequent year.
(a) Show that it sold 220 in 2007.
(b) Find the total sold from 2000 through 2013.
- 31.** A company makes 200 phones in week 1 and increases weekly production by 20 until it reaches 600 in week N .
(a) Find N .
(b) It then produces 600 each week. Find the total production during the first 52 weeks.
- 32.** A player scores 140, 160, 180, $\dots$ points for successive captures. Find (a) the score for the twentieth capture; (b) the total for the first twenty captures.
- 33.** A company's predicted profit is RM 120 000 in 2023 and grows by 5% per year.
(a) Verify the predicted profit for 2026.
(b) Find the first year in which profit exceeds RM 200 000.
(c) Find the total predicted profit from 2023 through 2033.
- 34.** Evaluate or simplify.
(a) $4^2 + 5^2 + \dots + 12^2$

- (b) $2(5) + 3(7) + 4(9) + \cdots$ for n terms
 (c) $1(1) + 2(4) + 3(7) + 4(10) + \cdots$ for n terms

35. Evaluate.

- (a) $\sum_{r=1}^n (2r - 1)$
 (b) $\sum_{r=1}^n (n + 2r + 3)$
 (c) $\sum_{r=1}^{10} (2r + r^2)$
 (d) $\sum_{i=1}^{80} (i^3 + i^2)$

36. Expand in ascending powers of x .

- (a) $(2 - 3x)^4$
 (b) $(1 - \frac{3}{2}x)^4$
 (c) $(x - 3y)^3$
 (d) $(3x - 4)^5$

37. Find the first four terms of $(1 + x/4)^8$ in ascending powers of x . Hence estimate $(1.025)^8$ to four decimal places.

38. For $(x^2 + x^{-3})^{10}$, find

- (a) the first two terms;
 (b) the last two terms;
 (c) the general term;
 (d) the term independent of x ;
 (e) the seventh term;
 (f) the coefficient of x^{-15} .

Additional practice: grouped skills review

39. For each arithmetic sequence, find the stated term.

Case	A	B	C	D
a	4	15	-5	120
d	3	-2	7	15
n	12	9	8	6

40. For each case in Exercise 39, find the sum of the first n terms.

41. For each geometric sequence, find the stated term.

Case	A	B	C	D
a	3	160	5	81
r	2	$1/2$	-2	$1/3$
n	7	6	5	5

42. For each case in Exercise 41, find the sum of the first n terms.

43. Find the sum to infinity where it exists, and identify any divergent series.

(a) $12 + 3 + 0.75 + \cdots$

(b) $-8 - 4.8 - 2.88 - \cdots$

(c) $5 - 2 + 0.8 - \cdots$

(d) $7 + 7.35 + 7.7175 + \cdots$

44. Evaluate.

(a) $\sum_{r=1}^{20} (3r - 2)$

(b) $\sum_{r=1}^{12} r^2$

(c) $\sum_{r=1}^8 (2r^3 + r)$

(d) $\sum_{r=0}^6 3(2^r)$

45. Insert the stated means.

(a) Four arithmetic means between 10 and 40.

(b) Three arithmetic means between -8 and 12 .

(c) Two positive geometric means between 4 and 108.

(d) Three positive geometric means between 2 and 162.

46. Evaluate or solve.

(a) $8!/5!$

(b) $\binom{10}{3}$

(c) $\binom{12}{10}$

(d) $\binom{n}{2} = 45$, where n is a positive integer.

47. Expand.

(a) $(1 + 2x)^4$

- (b) $(2 - x)^5$
- (c) $(a - 3b)^4$
- (d) The first four terms of $(1 - x/2)^6$ in ascending powers of x .

48. Find:

- (a) the coefficient of x^3 in $(2 + x)^8$;
- (b) the coefficient of x^4 in $(1 - 3x)^6$;
- (c) the term independent of x in $(x^2 + 2/x)^6$;
- (d) the general term in $(3x^2 - 1/x)^7$.

Applications: arithmetic change and public planning

- 49. A legal-aid office opens 18 new cases in week 1, and the number opened rises by 3 each week. Find the number opened in week 12 and the total opened during the first 12 weeks.
- 50. A housing programme completes 240 homes in its first year and 18 more homes in each succeeding year. Find the number completed in year 8 and the total completed during the first eight years.
- 51. A scholarship budget is RM 1.20 million in year 1 and rises by RM 0.08 million each year. Find the year-10 budget and total funding over the first ten years.
- 52. A backlog model is

$$B_n = 1200 - 75n,$$

where B_n is the number of unresolved cases after n months.

- (a) Find B_8 .
 - (b) Find the first month for which $B_n \leq 300$.
 - (c) Explain why the formula should not be extrapolated indefinitely.
- 53. A training programme begins with 40 participants in cohort 1 and adds 5 participants to each new cohort. Find the first cohort containing 100 participants and the total number trained through that cohort.
- 54. An annual household subsidy is RM 600 in its first year and increases by RM 45 per year. Find the subsidy in year 7 and the total paid to one continuously eligible household over seven years.
- 55. An auditorium has 18 rows. The first row contains 24 seats and each successive row contains two more seats than the preceding row. Find the number of seats in the final row and the total capacity.

56. A starting annual salary is RM 36 000 and increases by RM 2400 at the beginning of each subsequent year. Find the salary in year 5 and total salary received over the first five years. State one limitation of the constant-increment model.
57. A sequence of service targets is arithmetic. The fifth target is 140 cases and the twelfth is 210 cases. Find the first target, the common difference and the sum of the first twelve targets.
58. Plan A supplies 100 units in period 1 and increases by 20 units per period. Plan B supplies 160 units initially and increases by 10 units per period.
- Find the period in which their outputs are equal.
 - Compare their cumulative outputs through that period.
 - State which plan has the greater output in the following period.
59. Recorded waiting-time indices are 42, 39, 36, 33 for four successive periods. If an arithmetic pattern is fitted, find the tenth value and the sum of the first ten values. Give one reason the extrapolated values may be unreliable.

Applications: geometric growth, decay and accumulation

60. A district population is 80 000 in year 1 and is modelled to grow by 2.5% per year. Find the population in year 10 and the sum of the ten annual population figures.
61. An unresolved caseload is modelled by $N_n = 15\,000(0.92)^n$ after n months. Find N_6 and the first whole month for which fewer than half the initial cases remain.
62. Two service-demand models, measured from the same starting period, are

$$A(t) = 100 + 15t, \quad G(t) = 100(1.12)^t.$$

Compare the models at $t = 5$, and find the first non-negative whole value of t for which $G(t) > A(t)$. Use figure 7.2 to explain the long-run difference between equal additions and compounding.

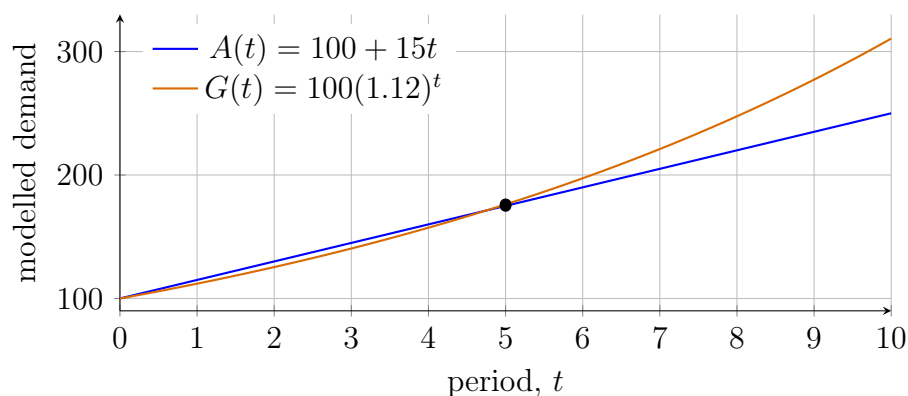


Figure 7.2: Arithmetic and geometric change in Exercise 62. The models begin together, but proportional growth eventually produces increasingly large additions.

63. Equipment worth RM 120 000 loses 15% of its value each year. Find its value and total loss after five years, and the first whole year in which its value is below half the original amount.
64. A campaign reaches 500 new people in its first wave and each later wave reaches 60% as many new people as the preceding wave. Find the total reached in the first six waves and the limiting total predicted by the model.
65. A cohort has 2000 participants in year 1, and 75% of each year's participants remain in the next year. Find the year-5 number and the sum of the first five annual participant counts.
66. A public grant is RM 2 million in year 1 and grows by 4% annually. Find the grant in year 6 and the total awarded over the first six years.
67. Express $0.272727\dots$ as an infinite geometric series and hence as a fraction. State the common ratio and why the series converges.
68. Subscribers number 12 000 in year 1 and 7680 in year 3. Assuming a positive constant geometric ratio, find the ratio, the year-5 number and the sum of the first five annual counts.
69. A programme produces benefit 1000 units in its first period and 70% as much in each later period.
- Find the limiting accumulated benefit.
 - Find the first number of periods for which the accumulated benefit is at least 95% of the limit.
 - Explain the approach to the limit using figure 7.3.

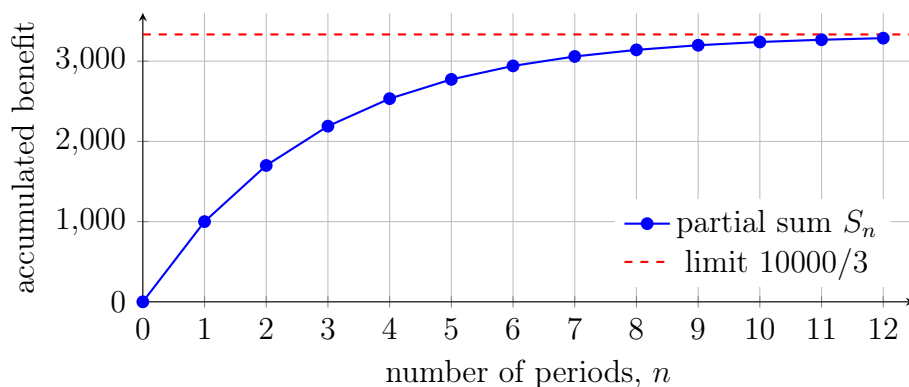


Figure 7.3: Partial sums for Exercise 69. Each additional benefit is positive but smaller than the preceding one, so the total approaches a finite limit.

70. A stock begins at 1000 units. An arithmetic model removes 80 units per year, while a geometric model retains 92% per year. Compare the two predictions after five and ten years. Explain why the arithmetic model eventually becomes unsuitable.

Applications: summation and aggregated quantities

71. Evaluate using standard finite-sum formulas.

(a) $\sum_{r=1}^{50} r$

(b) $\sum_{r=1}^{20} r^2$

(c) $\sum_{r=1}^{10} r^3$

72. Monthly workload is $W_r = 40 + 3r$ cases in month r . Find the total workload over months 1 to 12.

73. A recorded monthly cost is $C_r = 100 + 5r + 0.2r^2$ ringgit-equivalent units. Find $\sum_{r=1}^{10} C_r$.

74. Derive a formula for

$$\sum_{r=1}^n r(2r+1),$$

then evaluate it for $n = 20$.

75. Write $25 + 30 + 35 + \cdots + 70$ in sigma notation and evaluate it.

76. A cumulative-output model is $S_n = 4n^2 + 10n$. Find the output in period n , show that the period outputs form an arithmetic sequence, and find total output through period 5.

77. A rollout incurs cost $2000 + 300r$ ringgit in stage r . Find the total cost of stages 1 through 8.

78. A spatial survey samples 1^2 blocks in zone 1, 2^2 blocks in zone 2, and so on through zone 15. Find the total number of sampled blocks.

79. The sum of the first 30 positive integers is reported after one term was accidentally omitted. If the reported total is 438, identify the omitted term.

80. Show that

$$\sum_{r=1}^n (3r+2) = \sum_{j=0}^{n-1} (3j+5),$$

and find a closed form for the common value. Explain why changing the index does not change the terms being added.

Applications: binomial coefficients and approximation

81. Use the first three non-zero terms of the binomial expansion to estimate $(1.03)^8$. Compare with the calculator value and state the approximation error.
82. Find the first three terms of $(100 + x)^4$ in ascending powers of x . Explain why these terms may provide a useful approximation when $|x|$ is small relative to 100.
83. Find the coefficient of x^3 in $(2 + 3x)^8$.
84. For $(x^2 - 2/x)^7$:
- (a) write the general term;
 - (b) find the coefficient of x^5 ;
 - (c) determine whether a term independent of x exists.
85. In $(1 + x)^n$, the coefficient of x^2 is 45. Find the positive integer n .
86. Write row 8 of Pascal's triangle and explain its symmetry using $\binom{n}{r} = \binom{n}{n-r}$.
87. Estimate $(1.01)^{12}$ using first a linear and then a quadratic binomial approximation. Compare both estimates with the calculator value.
88. Five independent policy components may each be excluded or included. Explain how the coefficients in $(1 + x)^5$ count selections by the number of included components, and state the number of selections containing exactly two components.
89. Find the term independent of x in $(2x - x^{-2})^9$.
90. Find the coefficient of x^3 in $(1 + x/2)^{12}$.

Applications: model evaluation and responsible interpretation

91. Distinguish a sequence from a series using annual programme costs as an example.
92. Observed values are 100, 120, 144, 172.8. Identify a suitable model, state its parameters and predict the next value. Explain what feature rules out an arithmetic model.
93. Give three reasons why an arithmetic or geometric pattern fitted to a short historical series may fail when extrapolated far ahead.
94. State the convergence condition for an infinite geometric series and explain why a formally convergent series can still be a poor applied model.
95. A report adds projected benefits over many years. Give three checks needed before treating the resulting series as a valid total social benefit.
96. Explain why a constant percentage increase and an equal annual addition represent different assumptions, even when they give similar short-run predictions.

97. Explain why a truncated binomial expansion should include an error assessment when used in a financial or policy calculation.
98. A sigma expression is applied to administrative data. Give three data-quality or unit checks that should be performed before summing.
99. A rapidly growing geometric forecast is used to allocate public resources. Give four ethical, distributional or governance issues that should accompany the calculation.
100. Give a six-stage checklist for constructing, calculating, checking and communicating a sequence-or-series model in a social-science application.

Arithmetic sequences change by repeated addition, so their terms grow linearly. Geometric sequences change by repeated multiplication, so their behaviour may show growth, decay or alternating signs. Standard summation formulas handle polynomial terms efficiently, while the binomial theorem expands integer powers and isolates particular coefficients without requiring the complete expansion.

Chapter 8

Linear Equations and Graphs

Chapter guide

Solve linear equations and inequalities before linking slope, intercepts and line equations to applied models, regression, residuals and break-even decisions.

Selected background sources: (Chai et al., 2024; London School of Economics and Political Science, n.d.).

Linear equations describe relationships in which a variable has power one. They arise naturally in budgeting, pricing, measurement, forecasting and many other applications. This chapter develops methods for solving linear equations and inequalities, representing their solution sets, drawing and interpreting straight-line graphs, and using linear models to describe data.

8.1 Linear equations in one variable

A first-degree equation in one variable can be written in the form

$$ax + b = 0, \quad a \neq 0.$$

The value of x that makes the equation true is called its *solution* or *root*. Solving an equation means replacing it by simpler equivalent equations until the variable is isolated.

The same operation must be performed on both sides of an equation. In particular, one may add or subtract the same number, or multiply or divide by the same non-zero number, without changing the solution set.

Example 8.1: [

blue]Solving and checking a linear equation Solve

$$8x - 3(x - 4) = 3(x - 4) + 6.$$

Expanding and collecting like terms gives

$$8x - 3x + 12 = 3x - 12 + 6,$$

$$5x + 12 = 3x - 6,$$

$$2x = -18,$$

$$x = \boxed{-9}.$$

To check, substitute $x = -9$ into the original equation:

$$8(-9) - 3(-9 - 4) = -33 \quad \text{and} \quad 3(-9 - 4) + 6 = -33.$$

Both sides have the same value, so the solution is correct.

When an equation contains fractions, it is often helpful to multiply every term by the lowest common denominator.

Example 8.2: [

violet]A linear equation containing fractions Solve

$$\frac{x+2}{2} - \frac{x}{3} = 5.$$

The lowest common denominator is 6. Multiplying every term by 6 gives

$$\begin{aligned} 3(x+2) - 2x &= 30, \\ 3x + 6 - 2x &= 30, \\ x &= \boxed{24}. \end{aligned}$$

Indeed, $(24+2)/2 - 24/3 = 13 - 8 = 5$.

An equation may have one solution, no solution or infinitely many solutions. For example,

$$2x + 1 = 2x + 5$$

reduces to the false statement $1 = 5$, so it has no solution. By contrast,

$$3(x+2) = 3x + 6$$

reduces to an identity and is true for every real x .

8.1.1 Applications of linear equations

The essential modelling step is to define the unknown quantity before translating the information into an equation.

Example 8.3: [

orange]Purchase price before tax and fees A car costs \$28 400 after an 8.5% sales tax and a fixed licence fee of \$190 have been added. Let p be the purchase price before tax. Then

$$p + 0.085p + 190 = 28\,400.$$

Therefore,

$$\begin{aligned} 1.085p &= 28\,210, \\ p &= \boxed{26\,000}. \end{aligned}$$

Example 8.4: [

cyan]Retail price and percentage mark-up Suppose a shop sets the retail price R at 40% above the wholesale price W . Then

$$R = W + 0.40W = 1.40W.$$

If $W = \$300$, then $R = 1.40(300) = \boxed{\$420}$. If the retail price is \$77, then

$$W = \frac{77}{1.40} = \boxed{\$55}.$$

8.2 Linear inequalities

A *linear inequality* is obtained when the equality sign in a linear equation is replaced by $<$, $>$, $\leq$ or $\geq$. Unlike an equation, an inequality usually has an interval of solutions rather than a single solution.

For real numbers x , y and z :

$$\begin{aligned} x < y &\implies x + z < y + z, \\ x < y, z > 0 &\implies xz < yz, \\ x < y, z < 0 &\implies xz > yz. \end{aligned}$$

Thus, adding or subtracting the same value preserves the inequality. When multiplying or dividing by a negative value, however, the direction of the inequality must be reversed.

8.2.1 Interval notation

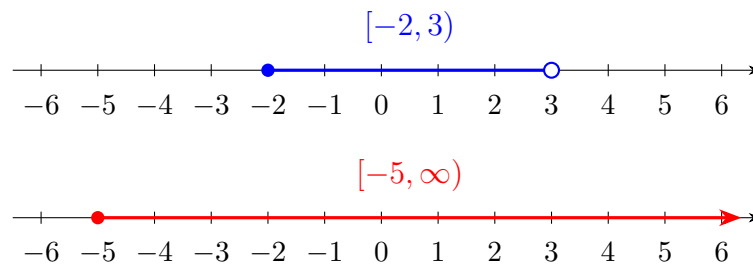
Intervals provide a compact way to describe sets of real numbers. A square bracket includes an endpoint, whereas a round bracket excludes it. Infinity is never an endpoint that can be reached, so it is always written with a round bracket.

8.2.2 Solving linear inequalities

The algebraic procedure resembles the method for solving equations, with the important additional rule concerning multiplication or division by a negative number.

Inequality	Interval	Description
$a < x < b$	(a, b)	neither endpoint included
$a \leq x \leq b$	$[a, b]$	both endpoints included
$a \leq x < b$	$[a, b)$	a included, b excluded
$x > a$	(a, ∞)	all real numbers greater than a
$x \leq b$	$(-\infty, b]$	all real numbers at most b

Table 8.1: Inequalities and their interval notation.

Figure 8.1: Number-line representations of $-2 \leq x < 3$ and $x \geq -5$.**Example 8.5:** [

orange]Solving a linear inequality Solve

$$3(x - 2) > 4(2x + 1).$$

Then

$$3x - 6 > 8x + 4,$$

$$-10 > 5x,$$

$$x < \boxed{-2}.$$

The last line may also be written $(-\infty, -2)$.**Example 8.6:** [

blue]An inequality containing fractions Solve

$$\frac{x}{6} \leq \frac{x - 3}{5}.$$

Multiplying by the positive number 30 does not reverse the inequality:

$$\begin{aligned} 5x &\leq 6(x - 3), \\ 5x &\leq 6x - 18, \\ x &\geq \boxed{18}. \end{aligned}$$

Thus, the solution set is $[18, \infty)$.

Example 8.7: [

violet]A compound inequality Solve

$$-3 < 2x + 3 \leq 9.$$

Subtracting 3 throughout and then dividing by 2 gives

$$-6 < 2x \leq 6 \implies \boxed{-3 < x \leq 3}.$$

In interval notation, the solution is $(-3, 3]$.

For a compound inequality, every operation must be applied to all three parts. For example,

$$\begin{aligned} -6 &\leq 4 - 7x < 3, \\ -10 &\leq -7x < -1, \\ \frac{10}{7} &\geq x > \frac{1}{7}. \end{aligned}$$

Therefore,

$$\boxed{\frac{1}{7} < x \leq \frac{10}{7}}.$$

The inequality signs reverse when all three parts are divided by -7 .

8.3 Graphs and straight lines

A linear equation in two variables has the standard form

$$Ax + By = C,$$

where A and B are not both zero. Its graph is a straight line. If $B \neq 0$, the equation can be rearranged as

$$y = -\frac{A}{B}x + \frac{C}{B}.$$

If $B = 0$, the result is a vertical line $x = C/A$.

8.3.1 Slope and intercepts

The *slope*, or gradient, measures the rate at which y changes with respect to x . For two distinct points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{vertical change}}{\text{horizontal change}}.$$

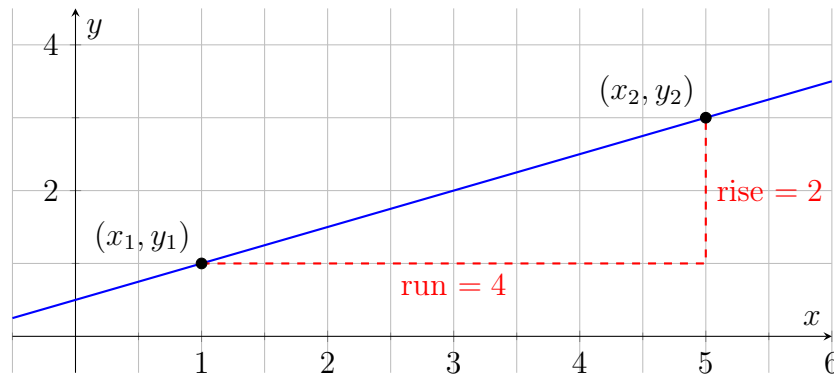


Figure 8.2: The slope is the ratio of rise to run; here $m = 2/4 = 1/2$.

The sign of the slope describes the direction of the line as x increases:

- $m > 0$: the line rises from left to right;
- $m < 0$: the line falls from left to right;
- $m = 0$: the line is horizontal;
- a vertical line has undefined slope because its horizontal change is zero.

The x -intercept is found by setting $y = 0$, and the y -intercept is found by setting $x = 0$.

8.3.2 Forms of the equation of a line

Form	Equation	Useful information
Standard	$Ax + By = C$	convenient for intercepts
Slope-intercept	$y = mx + b$	slope m , y -intercept b
Point-slope	$y - y_1 = m(x - x_1)$	slope and one point
Horizontal	$y = b$	slope 0
Vertical	$x = a$	slope undefined

Table 8.2: Common forms of a straight-line equation.

In slope-intercept form,

$$y = mx + b,$$

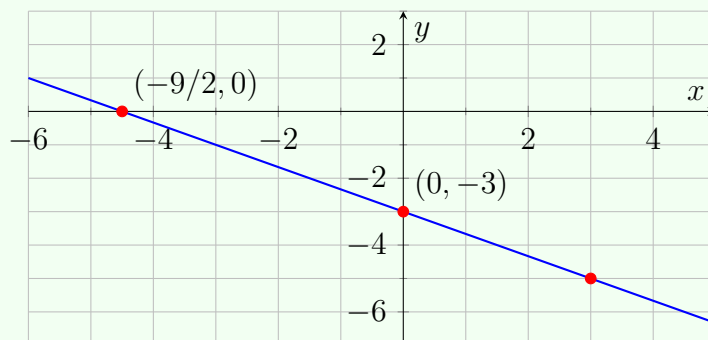
the line crosses the y -axis at $(0, b)$. A second point can be found by using the slope as a rise-to-run ratio.

Example 8.8: [

blue]Drawing a line from slope–intercept form For

$$y = -\frac{2}{3}x - 3,$$

the slope is $-2/3$ and the y -intercept is $(0, -3)$. From $(0, -3)$, moving 3 units to the right and 2 units downward gives another point $(3, -5)$. Equivalently, the x -intercept is $(-9/2, 0)$.



Example 8.9: [

orange]Writing a line from its slope and intercept A line has slope $2/3$ and y -intercept -2 . Substitution into $y = mx + b$ gives

$$y = \frac{2}{3}x - 2.$$

The point–slope form is especially useful when the slope and a point are known.

Example 8.10: [

violet]A line through a given point Find the equation of the line with slope $1/2$ passing through $(-4, 3)$. Using point–slope form,

$$\begin{aligned} y - 3 &= \frac{1}{2}(x + 4), \\ y &= \frac{1}{2}x + 5. \end{aligned}$$

Thus, one form of the answer is

$$x - 2y = -10.$$

If two points are known, calculate the slope first and then use either point.

Example 8.11: [

cyan]A line through two points Find the equation of the line through $(-3, 2)$ and $(-4, 5)$. Its slope is

$$m = \frac{5 - 2}{-4 - (-3)} = \frac{3}{-1} = -3.$$

Using $(-3, 2)$,

$$\begin{aligned} y - 2 &= -3(x + 3), \\ y &= -3x - 7. \end{aligned}$$

Therefore, the equation is $y = -3x - 7$.

8.4 Linear models and rate of change

In a model of the form

$$y = mx + b,$$

the intercept b is the predicted value of y when $x = 0$, while the slope m is the predicted change in y for each one-unit increase in x . The units of a slope are therefore

$$\frac{\text{units of } y}{\text{units of } x}.$$

Example 8.12: [

orange]A manufacturing cost model A skateboard manufacturer has a fixed monthly cost of \$300. The total cost is \$4300 when 100 skateboards are produced. Taking the points $(0, 300)$ and $(100, 4300)$, the slope is

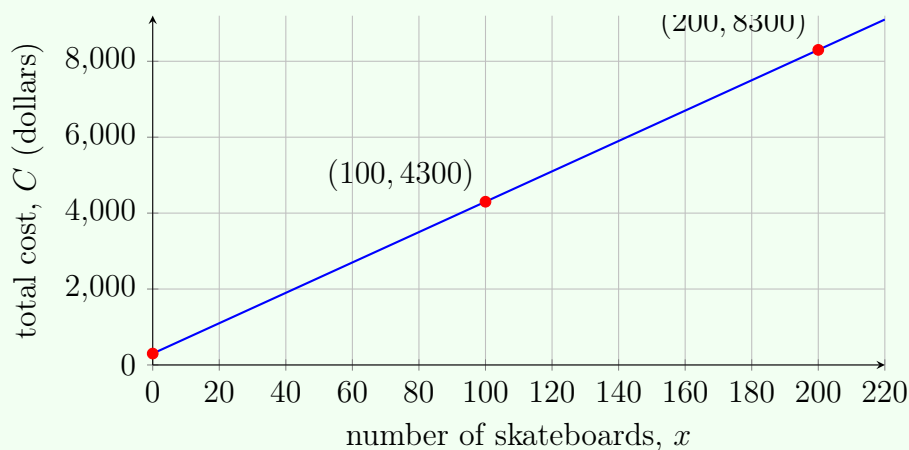
$$m = \frac{4300 - 300}{100 - 0} = 40.$$

If x is the number produced and C is the total cost in dollars, then

$$C = 40x + 300.$$

The fixed cost is \$300, and the variable cost is \$40 per skateboard. For 200 skateboards,

$$C(200) = 40(200) + 300 = \$8300.$$

**Example 8.13:** [

blue]Interpreting a rate of change The body-surface area a , in square inches, of a dog of weight w pounds is modelled by

$$a = 16.12w + 375.6.$$

The slope is 16.12 square inches per pound. According to the model, each additional pound of weight is associated with an increase of approximately 16.12 square inches in body-surface area. The intercept 375.6 is the model's predicted value at $w = 0$; it is an algebraic intercept and may not have a meaningful physical interpretation.

8.5 Linear regression

Measurements rarely lie exactly on a straight line. A *linear regression model* uses a line

$$\hat{y} = a + bx$$

to summarise the general relationship between an explanatory variable x and a response variable y . The symbol $\hat{y}$ denotes a predicted value. For an observed value y , the difference

$$e = y - \hat{y}$$

is called a *residual*. The least-squares regression line is chosen to minimise the sum of the squared residuals.

The slope describes the predicted change in the response for a one-unit increase in the explanatory variable. Predictions made far outside the range of the original data are called *extrapolations* and should be treated with caution.

Example 8.14: [

violet]A regression model for diamond prices Suppose the price p , in dollars, of a diamond of weight c carats is modelled by

$$p = 6140c - 480.$$

The slope indicates a predicted price increase of \$6 140 for each additional carat. For a 0.85-carat diamond,

$$p = 6140(0.85) - 480 = \boxed{\$4\,739}.$$

For a 1.20-carat diamond,

$$p = 6140(1.20) - 480 = \boxed{\$6\,888}.$$

If the predicted price is \$4 000, then

$$4000 = 6140c - 480 \implies c = \frac{4480}{6140} \approx \boxed{0.73 \text{ carats}}.$$

Example 8.15: [

cyan]Atmospheric carbon-dioxide concentration Let t be the number of years since 1995. A linear model for atmospheric carbon-dioxide concentration C , in parts per million, is

$$C = 360 + 2.04t.$$

The slope predicts an average increase of 2.04 parts per million per year. For 2025,

$t = 30$, so

$$C = 360 + 2.04(30) = 421.2 \approx \boxed{421 \text{ ppm}}.$$

For 1990, $t = -5$, giving

$$C = 360 + 2.04(-5) = \boxed{349.8 \text{ ppm}}.$$

The numerical results are predictions from the stated linear model, not a claim that the relationship remains exactly linear for all years.

Example 8.16: [

orange]A linear model for body weight A simple model for the weight w , in kilograms, of a woman who is more than five feet tall is

$$w = 49 + 1.7h,$$

where h is the number of inches above five feet. The slope means that the model adds 1.7 kilograms for each additional inch of height.

For a height of five feet four inches, $h = 4$, so

$$w = 49 + 1.7(4) = \boxed{55.8 \text{ kg}}.$$

If $w = 60$ kilograms, then

$$60 = 49 + 1.7h \implies h = \frac{11}{1.7} \approx 6.47.$$

The corresponding height is approximately five feet 6.47 inches.

8.6 Exercises

Answers to these exercises are provided in section C.8.

1. Solve $5(2x - 1) - 3 = 4x + 16$ and check the result.

2. Solve

$$\frac{2x - 1}{3} + \frac{x + 2}{4} = 5.$$

3. Solve $2 + (x - 3)/4 \geq (2x + 1)/5$ and write the answer in interval notation.

4. Express each set in interval notation:

(a) $-4 < x \leq 6$;

(b) $x < 3$;

(c) $x \geq -2$.

5. Find the slope and both intercepts of $3x + 2y = 12$.

6. Find the equation of the line with slope -4 through $(2, 5)$.

7. Find the equation of the line through $(-1, 4)$ and $(3, -2)$.

8. A taxi fare consists of a fixed charge of \$4.50 and a distance charge of \$1.80 per kilometre.

(a) Write a linear model for the fare F after d kilometres.

(b) Find the fare for a 12-kilometre journey.

(c) Interpret the slope and intercept.

9. A regression model for monthly household electricity cost E , in dollars, and electricity use u , in kilowatt-hours, is $E = 0.18u + 24$.

(a) Interpret the slope and intercept.

(b) Predict the cost when $u = 650$.

(c) Find the predicted usage when $E = 150$.

10. Solve and check.

(a) $\frac{1}{2}(x + 5) - 4 = \frac{1}{3}(2x - 1)$

(b) $\frac{3}{x - 2} = \frac{1}{x - 1} + \frac{7}{(x - 1)(x - 2)}$

(c) $(2y + 1)(y - 1) = (y + 5)(2y - 5)$

(d) $\frac{3x}{x - 1} + 2 = \frac{3}{x - 1}$

11. A total of \$18 000 is divided between stocks and bonds. The amount in bonds is half the amount in stocks. Find both amounts.

12. An employee earns \$435 for 52 hours, with time-and-a-half paid for hours beyond 40. Find the regular hourly wage.

13. Solve and express each answer in interval notation.

(a) $-2x \geq 8$

(b) $3(2x + \frac{1}{5}) \geq 3x - \frac{2}{5}$

(c) $\frac{y}{3} - \frac{y-2}{2} \leq \frac{y}{4} - 4$

(d) $\frac{2}{3}(y+4) - \frac{y}{4} > \frac{1}{2}(3-y) + \frac{y}{6}$

14. Solve the compound inequalities.

(a) $-6 < 3y - 2 \leq 4$

(b) $24 \leq \frac{2}{3}(y-5) < 36$

(c) $-2 \leq \frac{7-3x}{4} < 10$

15. Find the equation of each line.

(a) Slope $2/3$ through $(6, -2)$, in the form $Ax + By = C$ with $A > 0$.

(b) Through $(2, -3)$ and $(4, 3)$, in slope-intercept form.

16. A can manufacturer has fixed daily costs of \$250 and total costs of \$3450 at an output of 80 cans. Assuming a linear model, find the slope, obtain $C(x)$, and describe its graph for $0 \leq x \leq 160$.

17. Diamond weights and prices are

Weight (carats)	0.5	0.6	0.7	0.8	0.9	1.0
Price (dollars)	1677	2353	2718	3218	3982	4510

A model is $p = 5600c - 1100$.

- (a) Draw a scatter plot and the model line.
- (b) Interpret the slope.
- (c) Estimate the prices of 0.85- and 1.20-carat diamonds.
- (d) Estimate the weight of a \$4000 diamond.
18. A balsam fir's height is modelled by $h = 3.8d + 18.73$, where d is trunk diameter in inches and h is height in feet. Interpret the slope, estimate h for $d = 8$, and estimate d for $h = 30$, using the requested units and suitable rounding.
19. Solve or classify each equation. State whether it has one solution, no solution or infinitely many solutions.

(a) $3(2x - 5) - 4 = 2(x + 7)$

(b) $5(2x - 1) = 10x + 3$

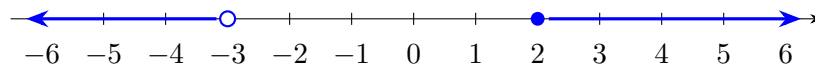
(c) $4(3x + 2) = 12x + 8$

20. Solve each compound inequality and give the answer in interval notation.

(a) $-7 \leq 5 - 2x < 11$

(b) $3x - 4 \leq 2(x + 5) < 5x + 8$

21. The number line below represents a solution set.

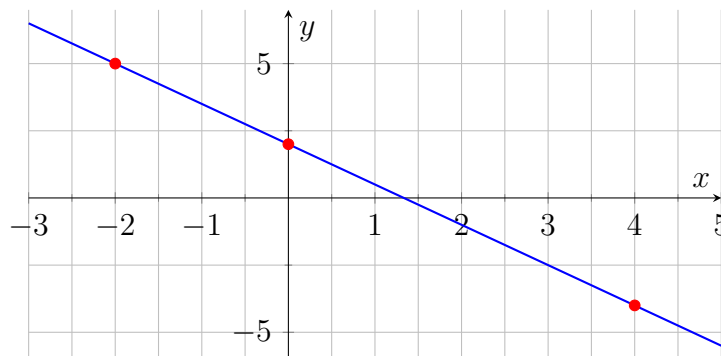


(a) Write the set using inequalities and interval notation.

(b) State whether -3 and 2 belong to the set.

(c) Write the complement of the set in interval notation.

22. The graph of a straight line is shown below.



Find its slope, equation and both intercepts. Hence find the value of y when $x = 3$.

23. Two mobile-data plans have monthly costs

$$A = 20 + 2.5n, \quad B = 50 + 1.5n,$$

where n is the number of gigabytes used.

(a) Interpret the slope and intercept of each model.

(b) Find the usage and cost at which the plans are equal.

(c) Determine which plan is cheaper below and above this usage.

(d) Sketch both lines for $0 \leq n \leq 50$, marking their intersection.

24. A public programme begins with a fund of RM 240 000. Its remaining balance after m months is modelled by

$$B = 240\,000 - 12\,000m.$$

Interpret the slope and intercept, find the balance after eight months, and determine when the model predicts that the fund is exhausted. State a sensible domain for the model.

25. Temperature in degrees Fahrenheit is related to temperature in degrees Celsius by

$$F = 1.8C + 32.$$

Find F when $C = 25$, find C when $F = 95$, and rearrange the model to make C the subject. Interpret the slope and intercept of the original equation.

26. The following table is believed to follow a linear relationship.

x	1	3	5	7
y	14	22	30	38

Verify that the rate of change is constant and find the equation of the line. Use it to find y when $x = 10$ and x when $y = 70$. If x represents hours of training and y a performance score, interpret the slope and intercept.

27. A regression model for monthly income Y , in thousands of ringgit, from years of experience x is

$$\hat{Y} = 42 + 3.2x.$$

The following observations are recorded.

x	2	4	6	8	10
Observed Y	47	56	60	70	73

Calculate the predicted value and residual for each observation. Identify the largest absolute residual and explain the meanings of positive and negative residuals in this context.

28. A transport study based on observations from 2020 to 2026 uses the model

$$R = 3.8 + 0.22t,$$

where R is annual ridership in millions and t is the number of years after 2020. Predict ridership in 2025 and 2035. Classify each prediction as interpolation or extrapolation and explain why one should be treated more cautiously.

29. A small producer has total cost and revenue models

$$C = 4500 + 18q, \quad R = 30q,$$

where q is the number of units sold.

- (a) Find the break-even output and the corresponding revenue.
- (b) Determine the outputs for which the model predicts a profit.
- (c) Find the predicted profit when $q = 600$.
- (d) Interpret the slopes and the cost intercept.

30. A simplified monthly savings model is

$$S = 0.18I - 240,$$

where income I and savings S are measured in ringgit. Find the income at which predicted savings are zero, calculate predicted savings when $I = 3000$, and solve the inequality $S > 0$. Discuss whether the intercept has a reasonable contextual interpretation.

31. A regression line passes through the predicted points $(10, 65)$ and $(30, 105)$.

- (a) Find its equation in slope–intercept form.
- (b) Find the predicted response when $x = 20$.
- (c) If the observed response at $x = 20$ is 82, calculate and interpret the residual.

32. Weekly leisure time L , in hours, is modelled from weekly paid working time h by

$$L = 18 - 0.35h.$$

The model was fitted using observations with $20 \leq h \leq 50$. Interpret its slope and intercept, predict L when $h = 40$, and explain why using the model at $h = 70$ produces an unsuitable conclusion.

Additional practice: grouped skills review

33. Solve.

- (a) $3x + 7 = 25$
- (b) $5 - 2x = 17$
- (c) $\frac{x-3}{4} + \frac{x+1}{2} = 5$
- (d) $4(2x-1) - 3(x+2) = 11$

34. Solve or classify each equation as having one solution, no solution or infinitely many solutions.

- (a) $3(x + 2) = 3x + 6$
- (b) $5(2x - 1) = 10x + 4$
- (c) $4x + 7 = 2x - 1$

35. Solve and give interval notation.

- (a) $3x - 5 \leq 10$
- (b) $-2x + 7 > 15$
- (c) $\frac{x - 2}{3} \geq \frac{x + 4}{5}$
- (d) $2 \leq 3x - 1 < 11$

36. Convert between inequalities and interval notation.

- (a) $-3 < x \leq 5$
- (b) $x \geq 2$
- (c) $x < -4$ or $x \geq 1$
- (d) $[-2, 6)$

37. Find the slope through each pair of points, stating when the slope is zero or undefined.

Case	P	Q
A	$(1, 3)$	$(5, 11)$
B	$(-2, 4)$	$(4, -5)$
C	$(3, -1)$	$(3, 7)$
D	$(-4, 6)$	$(2, 6)$

38. Find the equation of each line.

- (a) Slope 3 through $(2, -1)$.
- (b) Through $(-1, 5)$ and $(3, -3)$.
- (c) The line $4x + 2y = 10$ in slope-intercept form.
- (d) Parallel to $y = -\frac{1}{2}x + 4$ through $(2, 5)$.

39. The line $3x - 2y = 8$ is given.

- (a) State the slopes of lines parallel and perpendicular to it.
- (b) Find the parallel line through $(0, -4)$.
- (c) Find the perpendicular line through $(3, 1)$.

40. Find all existing coordinate-axis intercepts.

- (a) $2x + 5y = 20$
- (b) $y = -3x + 12$

(c) $x = -6$

41. For each cost model, calculate the stated cost and interpret the slope and intercept.

Case	F	v	q	$C(q) = F + vq.$
A	500	12	40	
B	1200	8.5	100	
C	0	25	36	

42. For each regression model, calculate the prediction and residual $e = y - \hat{y}$.

Case	Model	x	y
A	$\hat{y} = 10 + 2.5x$	8	32
B	$\hat{y} = 50 - 1.2x$	15	29
C	$\hat{y} = -4 + 0.8x$	20	13

Applications: linear equations and quantitative constraints

43. A RM 720 000 programme budget is divided among education, health and administration. Health receives twice the education allocation, while administration receives RM 120 000 more than education. Find all three allocations.
44. An employee works 48 hours and earns RM 780. Hours above 40 are paid at one-and-a-half times the regular hourly rate. Find the regular hourly rate.
45. A survey contains 640 respondents. The number from urban areas is 40 more than twice the number from rural areas. Find both counts.
46. A programme has fixed cost RM 18 000 and variable cost RM 45 per participant. Each participant contributes RM 60. Find the break-even number of participants and state when participant contributions exceed the modelled cost.
47. A standardised score is converted to a reporting scale by $S = 50 + 10z$. Find z when $S = 68$, and find S when $z = 2.5$.
48. Thirty concession and full-price tickets produce RM 345 revenue. A full-price ticket costs RM 15 and a concession ticket RM 8. Find the number of each type sold.
49. An employee receives RM 3500 after an 8% contribution and a fixed RM 180 deduction. If gross income is I , solve
- $$0.92I - 180 = 3500.$$
50. A response rate is $r = 100x/N$, where x people respond from a sample of size N .
- (a) Find N when $x = 378$ and $r = 63$.
- (b) Find x when $N = 800$ and $r = 72.5$.

51. A city begins with population 500 000. During the year it records 8000 births, 5000 deaths and net migration M . If the final population is 506 500, find M .
52. Monthly rent is modelled by $R = 1200 + 75t$, where t is years after the starting date. Find when the model reaches RM 1800 and interpret both coefficients.
53. Explain the contextual difference between the equations

$$4x + 12 = 4(x + 3) \quad \text{and} \quad 5x + 10 = 5x + 7.$$

Classify the solution set of each.

Applications: straight lines, rates and comparisons

54. A performance index has values 48 at $x = 2$ and 78 at $x = 8$. Assuming a linear relationship, find the model, predict the index at $x = 12$, and interpret the slope.
55. Total cost is RM 900 at output 20 and RM 1500 at output 50. Find the linear cost model and interpret its slope and intercept.
56. A public fund has balance $B = 250\,000 - 15\,000t$ after t months. Find the balance after ten months, the predicted exhaustion time and a sensible contextual domain.
57. The data

x	0	2	5	9
y	12	18	27	39

follow a linear relationship. Verify the constant rate, find the equation and calculate y when $x = 12$.

58. A main route follows $y = \frac{3}{2}x + 2$. Find the equation of a parallel route through $(4, -1)$ and a perpendicular access route through $(3, 4)$.
59. An approval measure is 48 percentage points in 2022 and 57 percentage points in 2025. Fit a linear model using t years after 2022, predict the 2028 value, and interpret the slope carefully.
60. Two service plans have costs

$$A = 30 + 4x, \quad B = 60 + 2x.$$

Find their intersection and identify the cheaper plan below and above it. Use figure 8.3 to support the comparison.

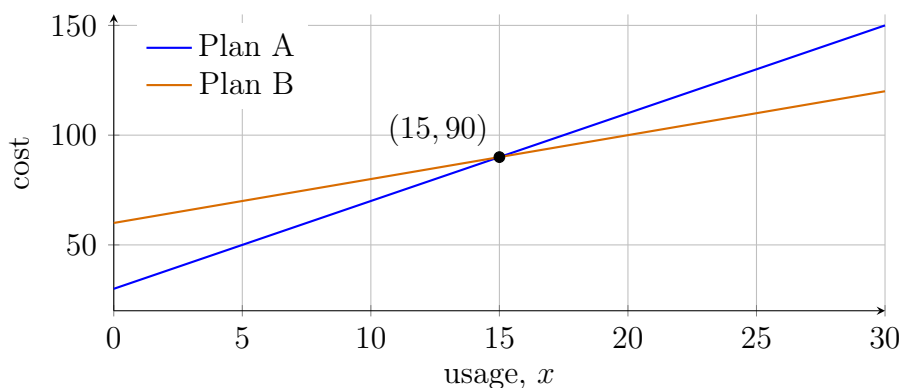


Figure 8.3: The service plans in Exercise 60. The intersection separates the usage ranges in which each plan is cheaper.

61. Demand and supply are $D = 200 - 5p$ and $S = 20 + 4p$. Find the price at which they are equal and the corresponding quantity. Interpret the signs of the two slopes.
62. A regional indicator is 2.4 million in 2018 and 3.3 million in 2024. Construct a linear interpolation model, estimate the 2021 value and extrapolate to 2030. State why the latter estimate is less secure.
63. The budget boundary $3x + 2y = 24$ represents non-negative quantities of two services. Find the intercepts, slope and feasible line segment in the first quadrant.
64. A monthly trend has slope 0.8 units per month. Express the slope in units per year. Explain what happens to the intercept if the time origin is moved to a different date.

Applications: inequalities and feasible ranges

65. An eligibility index is $0.25I + 300$, where monthly income I is in ringgit. Eligibility requires the index not to exceed 1800. Find the feasible income interval, assuming $I \geq 0$.
66. A project costs RM 12 000 plus RM 80 per participant and has a budget of RM 50 000. Find the greatest possible whole-number participation level.
67. A programme is open to people aged at least 20 but under 65. Write the range as inequalities and interval notation, and explain the endpoint conventions using figure 8.4.

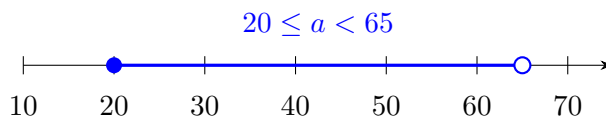


Figure 8.4: The eligibility interval in Exercise 67. A filled endpoint is included, while an open endpoint is excluded.

68. Solve $3(2x - 1) > 5x + 7$ and express the result in interval notation.

69. A service level must satisfy $x \geq 12$ and $4x + 100 \leq 220$. Find the combined feasible interval.
70. A reserve is modelled by $R = 500 - 18t$. Find the time interval for which $R \geq 200$, assuming $t \geq 0$, and state the greatest whole number of periods allowed.
71. An event has fixed cost RM 500 and cost RM 35 per attendee. With a RM 2500 budget, find the greatest possible whole-number attendance.
72. A percentage model is $P = 80 - 4t$. Find the domain on which the prediction remains non-negative, assuming $t \geq 0$.
73. A benefit is available when income is greater than RM 2500 and at most RM 5000. Give inequality and interval notation, and explain the different endpoint choices.
74. Plans cost $A = 40 + 3x$ and $B = 70 + 1.5x$. Solve $A < B$, identify the equality point and interpret the result for non-negative usage.
75. Solve $-5 \leq 2x + 1 < 9$, then intersect the solution with the contextual restriction $x \geq 0$.

Applications: regression, prediction and residuals

76. A fitted model is $\hat{y} = 25 + 1.8x$. Find the prediction at $x = 12$ and the residual when the observed value is 44.
77. For $\hat{y} = 60 - 2x$, calculate all predictions and residuals.

x	0	5	10	15
y	61	47	42	27

Identify the largest absolute residual.

78. A regression line passes through predicted points (4, 18) and (14, 43). Find its equation and the prediction at $x = 10$.
79. Monthly income Y , in thousands of ringgit, is modelled by $\hat{Y} = 2.4 + 0.18e$, where e is years of education. Interpret the slope and intercept, and explain why neither interpretation is automatically causal.
80. Observations and the fitted line $\hat{y} = 9 + 3x$ are shown in figure 8.5.

x	1	2	3	4	5
y	12	15	19	20	24

Calculate every residual and identify the observations above and below the line.

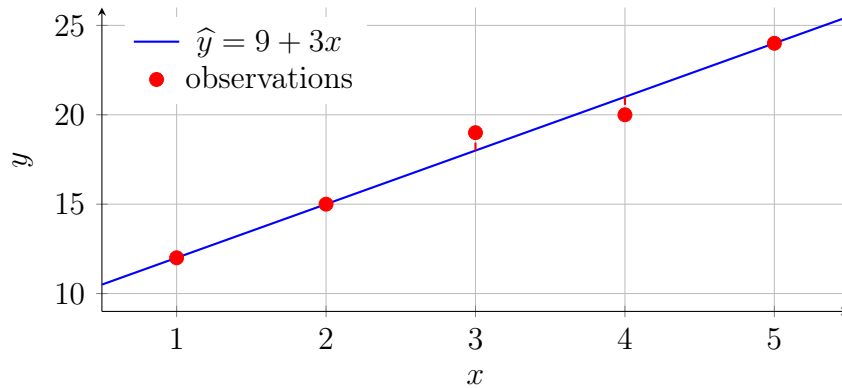


Figure 8.5: Observed values and fitted predictions for Exercise 80. Vertical differences are residuals; their signs indicate whether an observation lies above or below the line.

81. A model fitted using data from 2019–2025 is $R = 1.2 + 0.08t$, where t is years after 2019. Predict R in 2023 and 2035, classifying each prediction as interpolation or extrapolation.
82. Residuals from a fitted model are $-2, 1, 0, 3, -2$. Find their mean. Explain why a zero mean residual does not prove that a linear model is adequate.
83. A model is $y = 5 + 0.002x$, where x is measured in ringgit. Rewrite it using $X = x/1000$, where X is measured in thousands of ringgit, and explain the change in the numerical slope.
84. Explain how one high-leverage or extreme observation can alter a fitted regression line and its predictions.
85. Explain why a non-zero regression slope establishes association but not necessarily causation.
86. Group A has model $\hat{y} = 20 + 2x$, while Group B has model $\hat{y} = 40 + 0.5x$. Compare their predictions at $x = 10$ and explain what an overall pooled line might conceal.

Applications: intersections, break-even points and decisions

87. Cost and revenue are $C = 5000 + 12q$ and $R = 20q$. Find the break-even output and revenue, the outputs predicting profit, and the profit at $q = 800$.
88. Demand and supply are $D = 300 - 4p$ and $S = 60 + 2p$. Find their intersection and interpret it as an equilibrium in the model.
89. A department allocates 120 staff-hours between outreach O and training T , with outreach receiving twice as many hours as training. Find O and T .
90. Transport plans cost $A = 25 + 2.2n$ and $B = 55 + 1.2n$. Find their break-even usage and common cost, and identify the cheaper plan on either side.
91. Classify the intersections of each pair and explain their geometric meaning.

(a) $y = 2x + 1$ and $y = 2x - 3$.

(b) $y = 2x + 1$ and $2y = 4x + 2$.

92. The lines $y = ax + 5$ and $y = -x + 17$ meet when $x = 4$. Find a and the intersection point.
93. A service has cost $C = 8000 + 25q$ and benefit $B = 40q$. Find the smallest whole-number scale q for which benefit is at least cost, and calculate the surplus at that scale.
94. A policy produces benefit $B = 90x$ and cost $C = 2000 + 50x$. Find the threshold at which net benefit becomes positive and the net benefit at $x = 80$.

Applications: responsible use of linear models

95. Explain why a positive slope in an observational model does not by itself show that increasing the explanatory variable causes the response to rise.
96. Explain why the intercept of a fitted line may have little contextual meaning when zero lies outside the observed domain.
97. Give four reasons why extrapolating a linear trend far beyond its data range may be unreliable.
98. Explain why small residuals within the fitted sample do not guarantee accurate future predictions.
99. A linear model is used to allocate services across demographic groups. Give four fairness, data-quality or governance checks that should accompany the calculation.
100. Give a six-stage checklist for constructing, calculating, checking and communicating a linear model in a social-science application.

Chapter 9

Matrices

Chapter guide

Establish matrix notation and operations, then use determinants and inverses to solve systems before studying data and transition-matrix applications.

Selected background sources: (Chai et al., 2024; Hassan & Mazlin, 2025–2026).

Matrices organise numerical information in rectangular arrays and provide a compact method for representing transformations, networks, economic tables and systems of simultaneous equations.

9.1 Matrix notation and types

A matrix with m rows and n columns has order $m \times n$. Its entry in row i and column j is denoted by a_{ij} :

$$A = [a_{ij}]_{m \times n}.$$

A row matrix has one row, a column matrix has one column, and a square matrix has the same number of rows and columns. A zero matrix contains only zeros. The identity matrix I_n has ones on its main diagonal and zeros elsewhere.

A square matrix is diagonal if every off-diagonal entry is zero. It is upper triangular if every entry below the main diagonal is zero, and lower triangular if every entry above the main diagonal is zero.

9.2 Matrix operations

Matrices of the same order are added entry by entry. Scalar multiplication multiplies every entry by the scalar. The transpose A^T is obtained by interchanging rows and columns.

If A has order $m \times n$ and B has order $n \times p$, then AB is defined and has order $m \times p$. Its (i, j) entry is the dot product of row i of A with column j of B :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Matrix multiplication is generally not commutative: AB and BA may be different, or one product may be undefined.

Example 9.1: [

blue]Multiplying two matrices Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 1(4) + 2(5) \\ 3(4) - 1(5) \end{pmatrix} = \begin{pmatrix} 14 \\ 7 \end{pmatrix}.$$

9.3 Determinants, minors and cofactors

For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For a square matrix A , the minor M_{ij} is the determinant obtained by deleting row i and column j . The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Expanding along any row or column gives

$$\det(A) = \sum_j a_{ij} C_{ij}.$$

The cofactor matrix is $C = [C_{ij}]$, and the adjugate is $\text{adj}(A) = C^T$.

9.4 Inverse matrices and linear systems

A square matrix is invertible when its determinant is non-zero. In that case,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A), \quad AA^{-1} = A^{-1}A = I.$$

A system $A\mathbf{x} = \mathbf{b}$ with invertible coefficient matrix has the unique solution

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

For a 2×2 matrix, the inverse formula follows directly rather than being merely a rule to memorise. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

then multiplication gives

$$AB = BA = (ad - bc)I_2.$$

When $ad - bc \neq 0$, division by this non-zero scalar gives

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, the same calculation cannot be divided through, and the rows or columns are linearly dependent; consequently, no inverse exists.

Example 9.2: [

violet]Solving a two-variable system The system

$$2x + y = 7, \quad x - y = 2$$

can be written

$$\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}.$$

Since the coefficient matrix is invertible, multiplication by its inverse gives $x = 3$ and $y = 1$.

9.5 Matrices as data models

Matrices are useful when the same variables are recorded for several groups. Rows and columns must be labelled carefully because an entry has meaning only in relation to both labels. Multiplication can then combine group sizes, prices, rates or weights without repeating many separate calculations.

For example, suppose the row vector

$$H = (80 \quad 120 \quad 50)$$

gives the numbers of low-, middle- and high-income households in a sample, while

$$E = \begin{pmatrix} 400 & 180 & 600 \\ 520 & 250 & 850 \\ 350 & 140 & 500 \end{pmatrix}$$

gives average monthly expenditure on food, transport and housing for the three groups. Then

$$HE = (111\,900 \quad 51\,400 \quad 175\,000)$$

gives the total expenditure in each category for all sampled households. The units and ordering of the rows and columns are part of the model.

9.6 Transition matrices

A *state vector* records how a population is distributed among a fixed set of categories. A transition matrix describes movement between those categories from one period to the next. Using column vectors, the entry t_{ij} is the proportion moving from category j to category i , so each column of a transition matrix sums to 1.

If $\mathbf{p}_n$ is the state vector after period n , then

$$\mathbf{p}_{n+1} = T\mathbf{p}_n, \quad \mathbf{p}_{n+k} = T^k\mathbf{p}_n.$$

Such models can describe changes in voter preference, employment status, residential location or consumer choice. They assume that the transition rates remain stable over the periods being modelled.

Example 9.3: [

orange]A voter-preference transition model Let the categories be Party A, Party B and undecided. Suppose

$$T = \begin{pmatrix} 0.80 & 0.10 & 0.25 \\ 0.10 & 0.75 & 0.25 \\ 0.10 & 0.15 & 0.50 \end{pmatrix}, \quad \mathbf{p}_0 = \begin{pmatrix} 0.50 \\ 0.30 \\ 0.20 \end{pmatrix}.$$

The projected distribution after one period is

$$\mathbf{p}_1 = T\mathbf{p}_0 = \begin{pmatrix} 0.480 \\ 0.325 \\ 0.195 \end{pmatrix}.$$

Thus the model predicts 48% support for Party A, 32.5% for Party B and 19.5% undecided. These figures are conditional on the assumed transition rates and should not be treated as certain forecasts.

9.7 Exercises

Answers are provided in section C.9.

1. Given

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \\ 4 & -4 & 4 \end{pmatrix},$$

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 1 & 0 \\ -1 & 5 & 2 & 8 \end{pmatrix}, \quad E = \begin{pmatrix} 2 \\ -1 \\ 0 \\ 3 \end{pmatrix}, \quad F = \begin{pmatrix} 2 & 4 & 0 \\ 0 & 1 & 6 \\ 0 & 0 & 5 \end{pmatrix},$$

identify all row, column, identity, diagonal, square, zero, upper triangular and lower triangular matrices.

2. Let

$$P = \begin{pmatrix} 4 & 0 & -3 \\ -1 & -2 & 3 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 4 \end{pmatrix}, \quad R = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix},$$

and

$$S = \begin{pmatrix} 2 & -3 & 0 & 1 \\ 5 & -1 & -4 & 2 \\ -1 & 0 & 0 & 3 \end{pmatrix}.$$

Find, or state that the product is undefined:

$$P + Q, \quad 2Q - P, \quad QP, \quad PS, \quad QR, \quad RR, \\ SR, \quad P^T, \quad Q^T S, \quad R^T Q^T.$$

3. For $A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & -1 & 4 \\ 5 & 6 & -2 \end{pmatrix}$, find M_{11}, M_{23}, M_{32} and the corresponding cofactors.

4. Find the cofactor matrix of

(a) $\begin{pmatrix} 4 & 1 & 3 \\ 2 & -2 & 0 \\ 1 & 5 & 6 \end{pmatrix};$

(b) $\begin{pmatrix} -2 & 4 & 1 \\ 3 & 1 & -1 \\ 0 & 5 & 8 \end{pmatrix}.$

5. Find the determinants of $\begin{pmatrix} 3 & 1 \\ 2 & -4 \end{pmatrix}$ and $\begin{pmatrix} 7 & 6 & 5 \\ 1 & 2 & 1 \\ 3 & -3 & 1 \end{pmatrix}.$

6. Find the inverse of each matrix using the adjugate formula.

(a) $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} -1 & 0 & 0 \\ 4 & -1 & -2 \\ -2 & 1 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$

7. Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ and suppose $A^2 + pA + qI = 0$. Find p and q .

8. Labour hours per shelving unit and hourly wages are

$$M = \begin{pmatrix} 0.7 & 0.8 & 0.3 \\ 1.1 & 1.3 & 0.4 \\ 1.6 & 1.7 & 0.5 \end{pmatrix}, \quad N = \begin{pmatrix} 12.45 & 9.95 \\ 9.55 & 7.35 \\ 7.75 & 5.65 \end{pmatrix}.$$

Rows of M represent small, medium and large units; its columns represent cutting, assembly and packaging. Columns of N represent the Italian and Polish plants.

- Find the labour cost of a medium unit made in Italy.
- Find the labour cost of a small unit made in Poland.
- Explain why MN is meaningful and NM is not.
- Calculate and label MN .

9. Solve by the inverse-matrix method:

$$\begin{pmatrix} 2 & 1 & -3 \\ 3 & -1 & -4 \\ 5 & 2 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 5 \end{pmatrix}.$$

10. Write the following system as a matrix equation and solve it:

$$5x + 2y + 3z = 48, \quad 7x + 3y + 4z = 67, \quad x + 2y + 3z = 32.$$

11. Foods X, Y, Z supply nutrients per gram according to

$$\begin{pmatrix} 10 & 0 & 20 \\ 10 & 20 & 10 \\ 5 & 10 & 30 \end{pmatrix},$$

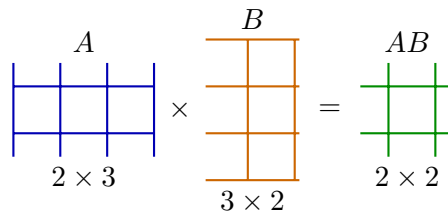
where the rows are protein, carbohydrate and vitamins. Find the grams of each food required to provide $(340, 280, 440)^T$ milligrams.

12. A survey contains 120, 80 and 50 respondents from three age groups. Within these groups, the proportions supporting policy X, supporting policy Y and undecided are the rows of

$$A = \begin{pmatrix} 0.60 & 0.30 & 0.10 \\ 0.40 & 0.40 & 0.20 \\ 0.20 & 0.50 & 0.30 \end{pmatrix}.$$

Use matrix multiplication to estimate the numbers in the three response categories.

13. For the voter-preference model above, calculate the projected state vector after two periods and interpret its entries.
14. The diagram illustrates the dimension rule for matrix multiplication.



Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & -2 \\ 5 & 3 \end{pmatrix}.$$

- State the orders of AB and BA before calculating them.
 - Calculate both products.
 - Explain why the results demonstrate that matrix multiplication is not commutative.
15. Given

$$2X - A = B, \quad A = \begin{pmatrix} 3 & -1 \\ 5 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 7 \\ -3 & 4 \end{pmatrix},$$

find X . Verify the result by substitution, then find X^T and $\det(X)$.

16. Let

$$M(k) = \begin{pmatrix} k & 2 \\ 3 & k-1 \end{pmatrix}.$$

- Find $\det M(k)$ and factorise it.
- Find the values of k for which $M(k)$ is singular.
- Find $M(4)^{-1}$.

17. For

$$C = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{pmatrix},$$

find the three cofactors in the first row. Hence calculate $\det(C)$ by expansion along that row.

18. A theatre sold 180 adult and child tickets. Adult tickets cost RM 12 and child tickets cost RM 7, producing total revenue of RM 1710.

- Write the information as a 2×2 matrix equation.
- Find the inverse of the coefficient matrix.
- Use the inverse to find the numbers of adult and child tickets sold.

19. Survey counts from two regions are arranged as

$$R = \begin{pmatrix} 120 & 80 & 50 \\ 90 & 110 & 70 \end{pmatrix},$$

where the columns represent support, opposition and undecided responses.

- Use multiplication by a suitable column vector to find each region's total sample size.
- Use multiplication by a suitable row vector to find the total in each response category.
- Find the overall response proportions.

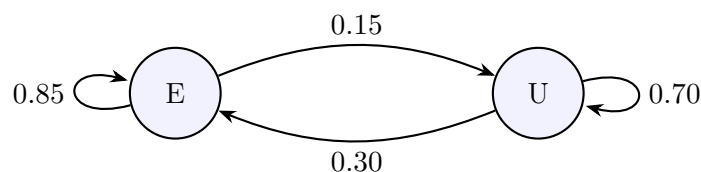
20. Three districts receive scores on access, affordability and reliability:

$$S = \begin{pmatrix} 72 & 65 & 80 \\ 60 & 75 & 70 \\ 85 & 55 & 90 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 0.40 \\ 0.35 \\ 0.25 \end{pmatrix}.$$

Rows correspond to Districts A, B and C, while $\mathbf{w}$ contains the indicator weights.

- Explain why $S\mathbf{w}$ is defined and state its order.
- Calculate the weighted score for each district.
- Rank the districts and explain how changing the weights could alter the ranking.

21. The following diagram models movement between employed and unemployed states during one period.



Using column state vectors:

- (a) construct the transition matrix T ;
 - (b) verify that its columns sum to 1;
 - (c) starting from 800 employed and 200 unemployed people, find the projected state vectors after one and two periods;
 - (d) state one important assumption behind these projections.
- 22.** Determine whether each matrix is a valid transition matrix under the column-vector convention. Justify each answer.

$$A = \begin{pmatrix} 0.75 & 0.20 \\ 0.25 & 0.80 \end{pmatrix}, \quad B = \begin{pmatrix} 0.70 & 0.40 \\ 0.20 & 0.60 \end{pmatrix}, \quad C = \begin{pmatrix} 1.05 & 0.10 \\ -0.05 & 0.90 \end{pmatrix}.$$

- 23.** A two-state opinion model uses

$$T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}.$$

After one period, the state vector is $\mathbf{p}_1 = (0.62, 0.38)^\top$.

- (a) Show that T is invertible and find T^{-1} .
 - (b) Recover the preceding state vector $\mathbf{p}_0 = T^{-1}\mathbf{p}_1$.
 - (c) Verify the result by calculating $T\mathbf{p}_0$.
- 24.** Resource requirements per service are

$$R = \begin{pmatrix} 4 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 100 \\ 80 \\ 60 \end{pmatrix}.$$

The first row of R gives staff hours and the second gives material units for three service types. Staff hours cost RM 25 and material units cost RM 8.

- (a) Calculate $R\mathbf{v}$ and interpret its entries.
 - (b) Write a row vector of unit costs and use a single matrix product to calculate the total resource cost.
 - (c) Explain why reversing the order of these matrices is not meaningful.
- 25.** Solve the matrix equation $AX = B$, where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix}.$$

Find A^{-1} , calculate $X = A^{-1}B$, and verify that $AX = B$.

26. Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

- (a) Explain without expansion why $\det(A) = 0$.
 - (b) State the consequence for A^{-1} .
 - (c) Solve $A\mathbf{x} = (6, 12, 2)^T$, expressing the infinitely many solutions using a parameter.
 - (d) Explain why $A\mathbf{x} = (6, 13, 2)^T$ has no solution.
27. A transition model predicts that a programme's participation share will rise from 45% to 58% over several periods.
- (a) Give three assumptions that should be checked before treating the projection as reliable.
 - (b) Explain how observed data could be used to update the transition matrix.
 - (c) State why a matrix calculation can be correct even when its practical prediction is poor.

Additional practice: grouped skills review

28. State the order of each matrix and identify all applicable types among row, column, square, identity, diagonal and zero.

$$A = \begin{pmatrix} 1 & 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

29. Let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 5 \\ -2 & 0 \end{pmatrix}.$$

Find $A + B$, $2A - B$ and $3B$.

30. Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & -2 \\ 5 & 3 \end{pmatrix}.$$

- (a) State the orders of AB and BA before calculation.
- (b) Calculate both products.
- (c) Explain why the results show that matrix multiplication is not commutative.

31. For the matrices in Exercise 30, find A^T , $(AB)^T$ and $B^T A^T$, and verify the transpose rule for a product.

32. Find the determinants.

(a) $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 5 & -2 \\ 3 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$

(d) $\begin{pmatrix} 2 & 1 & 4 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{pmatrix}$

33. Find the inverse where it exists.

(a) $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$

34. Solve by a matrix method.

(a) $2x + y = 7, x - y = 2.$

(b) $3x + 2y = 14, x + y = 5.$

35. For

$$C = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{pmatrix},$$

find the three cofactors in its first row and hence calculate $\det(C)$ by expansion along that row.

36. Solve the matrix equations.

(a) $2X - A = B$, where $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 0 \\ -1 & 2 \end{pmatrix}.$

(b) $A\mathbf{x} = \mathbf{b}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}.$

37. Let

$$T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}, \quad \mathbf{p}_0 = \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix}.$$

Verify that T is a valid transition matrix under the column-vector convention, then calculate and interpret $\mathbf{p}_1 = T\mathbf{p}_0$.

Applications: matrices as social-science data tables

38. Survey counts from three regions are

$$R = \begin{pmatrix} 120 & 80 & 50 \\ 90 & 110 & 70 \\ 60 & 75 & 65 \end{pmatrix},$$

where columns are support, opposition and undecided.

- (a) Use multiplication by a suitable column vector to find the regional sample sizes.
- (b) Use multiplication by a suitable row vector to find the totals in the three response categories.
- (c) Find the overall response proportions.

39. Three age groups contain 200, 150 and 100 respondents. Their response proportions are the rows of

$$P = \begin{pmatrix} 0.50 & 0.30 & 0.20 \\ 0.40 & 0.40 & 0.20 \\ 0.25 & 0.45 & 0.30 \end{pmatrix}.$$

Use a matrix product to estimate the combined counts in the three response categories.

40. Four districts receive scores on three indicators:

$$S = \begin{pmatrix} 70 & 80 & 65 \\ 85 & 60 & 75 \\ 60 & 75 & 90 \\ 78 & 72 & 74 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 0.40 \\ 0.35 \\ 0.25 \end{pmatrix}.$$

Calculate $S\mathbf{w}$, rank the districts and state why the result depends on the chosen weights.

41. Quantities of three goods purchased in two regions are

$$Q = \begin{pmatrix} 100 & 80 & 60 \\ 120 & 70 & 90 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} 5 \\ 8 \\ 4 \end{pmatrix}.$$

Find $Q\mathbf{p}$ and interpret its entries as regional expenditure.

42. Counts in two quarters are

$$A = \begin{pmatrix} 120 & 80 & 50 \\ 90 & 110 & 70 \end{pmatrix}, \quad B = \begin{pmatrix} 130 & 75 & 55 \\ 95 & 105 & 80 \end{pmatrix}.$$

Calculate $A + B$ and $B - A$, and interpret the difference matrix.

43. A matrix of costs is

$$X = \begin{pmatrix} 100 & 120 \\ 80 & 150 \end{pmatrix}.$$

Find the matrix after every cost rises by 5%, then after an additional fixed charge of RM 10 is added to every entry.

44. A 2×3 matrix records two regions in rows and three categories in columns. Explain the interpretation of its transpose and why transposing changes the presentation but not the underlying six observations.
45. Let R be a 2×3 resource matrix, $\mathbf{v}$ a 3×1 service vector and $\mathbf{c}$ a 1×2 cost vector. State the orders and meanings of $R\mathbf{v}$, $\mathbf{c}R$ and $\mathbf{c}R\mathbf{v}$. Explain why $\mathbf{v}R$ is undefined.
46. Group counts are $\mathbf{n} = (100, 80, 120)$ and group-specific shares are

$$P = \begin{pmatrix} 0.60 & 0.40 \\ 0.50 & 0.50 \\ 0.25 & 0.75 \end{pmatrix}.$$

Find the category counts $\mathbf{n}P$. If category unit costs are $\mathbf{c} = (10, 20)^\top$, calculate $\mathbf{n}P\mathbf{c}$.

47. For

$$Q = \begin{pmatrix} 20 & 30 & 40 & 10 \\ 15 & 25 & 35 & 25 \\ 30 & 20 & 10 & 40 \end{pmatrix},$$

use vectors of ones to find all row totals and all column totals.

48. A data table is

$$M = \begin{pmatrix} 20 & 30 & x \\ 15 & 25 & 40 \end{pmatrix}.$$

The first row totals 90 and the third column totals 80. Find x and verify that both statements agree.

Applications: determinants, inverses and linear systems

49. Let

$$M(k) = \begin{pmatrix} k & 2 \\ 3 & k-1 \end{pmatrix}.$$

Find and factorise the determinant, identify the singular values of k , and find $M(4)^{-1}$.

50. Recorded totals satisfy

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 90 \\ 50 \end{pmatrix}.$$

Use the inverse matrix to recover x and y and interpret them as the underlying category counts.

51. A theatre sells 180 adult and child tickets. Adult tickets cost RM 12, child tickets RM 7, and total revenue is RM 1710. Write and solve the 2×2 matrix equation.

52. Three programme allocations satisfy

$$x + y + z = 120, \quad 2x + y = 140, \quad y + 3z = 120.$$

Write the matrix equation and solve for the three allocations.

53. Two foods provide nutrients according to

$$\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11 \\ 7 \end{pmatrix}.$$

Find the required quantities x and y .

54. For

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 14 \\ 12 \end{pmatrix},$$

find A^{-1} and solve $A\mathbf{x} = \mathbf{b}$.

55. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

- (a) Solve $A\mathbf{x} = (5, 10)^T$, using a parameter.
- (b) Explain why $A\mathbf{x} = (5, 11)^T$ has no solution.
- (c) Relate both results to $\det(A)$.

56. Calculate the determinant of

$$C = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{pmatrix}$$

by cofactor expansion, and state whether C^{-1} exists.

57. The observation vector satisfies

$$\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 11 \\ 8 \end{pmatrix}.$$

Recover $\mathbf{x}$ using the inverse matrix.

58. Solve $AX = B$, where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix},$$

and verify the result by multiplication.

59. Compare

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Find both determinants. For matrix A , compare the solutions for right-hand sides $(2, 2.01)^\top$ and $(2, 2.02)^\top$. Explain the sensitivity revealed by the small determinant.

Applications: resource, expenditure and allocation models

60. Resource requirements per service and planned service quantities are

$$R = \begin{pmatrix} 4 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 100 \\ 80 \\ 60 \end{pmatrix}.$$

Calculate $R\mathbf{v}$. If resource unit costs are $\mathbf{c} = (25, 8)$, find $\mathbf{c}R\mathbf{v}$.

61. Household counts are $(120, 80, 50)$ and average expenditures are $(500, 650, 900)^\top$ ringgit. Use a matrix product to find total expenditure.

62. A three-resource, three-service model is

$$R = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 100 \\ 80 \\ 60 \end{pmatrix}, \quad \mathbf{c} = (30 \quad 12 \quad 5).$$

Find the resource totals and total cost. Use figure 9.1 to explain how the dimensions and labels determine the multiplication order.

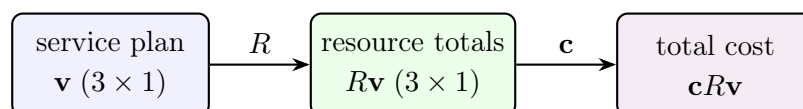


Figure 9.1: The resource-cost chain in Exercise 62. Multiplication is meaningful only when adjacent dimensions and the represented categories agree.

63. Two alternative service plans are the columns of

$$Q = \begin{pmatrix} 100 & 120 \\ 80 & 60 \\ 60 & 90 \end{pmatrix}.$$

Using R and $\mathbf{c}$ from Exercise 62, calculate RQ and $\mathbf{c}RQ$. Which plan has the lower total resource cost?

64. Programme funds of RM 2, RM 3 and RM 1 million are distributed among three districts using

$$D = \begin{pmatrix} 0.50 & 0.20 & 0.30 \\ 0.30 & 0.50 & 0.40 \\ 0.20 & 0.30 & 0.30 \end{pmatrix}.$$

Find each district's total allocation and verify that all funding is distributed.

- 65.** Sales of three products in two regions are

$$S = \begin{pmatrix} 100 & 120 \\ 80 & 60 \\ 50 & 90 \end{pmatrix}.$$

With price row vector $\mathbf{p} = (5, 8, 4)$, find regional revenue $\mathbf{p}S$.

- 66.** District scores are

$$S = \begin{pmatrix} 72 & 65 & 80 \\ 60 & 75 & 70 \\ 85 & 55 & 90 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 0.40 \\ 0.35 \\ 0.25 \end{pmatrix}.$$

Find the weighted scores and rank the districts.

- 67.** Group sizes are $(300, 200, 100)$ and average service-use vectors are the rows of

$$U = \begin{pmatrix} 2 & 1 & 0.5 \\ 1 & 2 & 1 \\ 0.5 & 1 & 3 \end{pmatrix}.$$

Find the total use of each service using one matrix product.

- 68.** A resource matrix is $R = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$. The original plan is $\mathbf{v} = (100, 60)^\top$ and the revised plan changes quantities by $\Delta\mathbf{v} = (20, -10)^\top$. Find the change in resources $R\Delta\mathbf{v}$ and verify it by comparing the two full plans.
- 69.** Explain why $\mathbf{c}R\mathbf{v}$ and $(\mathbf{c}R)\mathbf{v} = \mathbf{c}(R\mathbf{v})$ give the same scalar, while reversing the factor order generally fails.
- 70.** A weighted allocation is $S\mathbf{w}$. Explain how a sensitivity analysis can be performed by replacing $\mathbf{w}$ with alternative defensible weight vectors and comparing the resulting rankings.

Applications: transition and state-change models

- 71.** Employment movement is modelled by

$$T = \begin{pmatrix} 0.85 & 0.30 \\ 0.15 & 0.70 \end{pmatrix}.$$

Starting from $(800, 200)^\top$, find the projected employed and unemployed counts after one and two periods.

72. Determine whether each matrix is a valid transition matrix under the column-vector convention.

$$A = \begin{pmatrix} 0.75 & 0.20 \\ 0.25 & 0.80 \end{pmatrix}, \quad B = \begin{pmatrix} 0.70 & 0.40 \\ 0.20 & 0.60 \end{pmatrix}, \quad C = \begin{pmatrix} 1.05 & 0.10 \\ -0.05 & 0.90 \end{pmatrix}.$$

73. An opinion model uses $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$ and $\mathbf{p}_0 = (0.64, 0.36)^\top$. Find $\mathbf{p}_1$ and $\mathbf{p}_2$.
74. figure 9.2 shows movement among support (S), opposition (O) and undecided (U). Construct the transition matrix under the column-vector convention and apply it to $\mathbf{p}_0 = (0.50, 0.30, 0.20)^\top$.

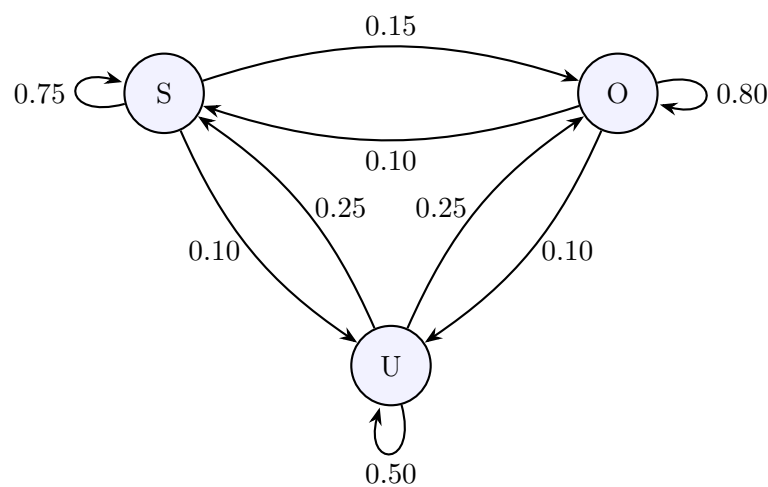


Figure 9.2: Three-state transition model for Exercise 74. Arrows leaving a state provide one column of the transition matrix.

75. Find the stationary probability vector for $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$, subject to the entries summing to 1.
76. After one period, $\mathbf{p}_1 = (0.62, 0.38)^\top$ under $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$. Find T^{-1} and recover $\mathbf{p}_0$.
77. Observed transitions are: from state A, 80 remain in A and 20 move to B; from state B, 30 move to A and 70 remain in B. Estimate the transition matrix.
78. Starting from $(0.5, 0.5)^\top$, compare the one-period predictions of

$$T_1 = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}.$$

79. A programme has active and completed states with

$$T = \begin{pmatrix} 0.7 & 0 \\ 0.3 & 1 \end{pmatrix}.$$

Explain why completion is absorbing. Starting from $(1000, 0)^T$, find the projected counts after two periods.

80. For $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$ and $\mathbf{p}_0 = (0.2, 0.8)^T$, calculate $T^5 \mathbf{p}_0$ and compare it with the stationary vector.
81. Explain why $T^n \mathbf{p}_0$ represents n repeated transitions, and give two reasons why a real system may not follow the same matrix for every period.

Applications: matrix sensitivity and decision analysis

82. Solve $3X - A = B$, where

$$A = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 5 \\ -1 & 6 \end{pmatrix}.$$

83. Find the value of k for which $A(k) = \begin{pmatrix} 1 & k \\ 2 & 4 \end{pmatrix}$ is singular. Explain what singularity means for a decision system $A(k)\mathbf{x} = \mathbf{b}$.

84. For

$$S = \begin{pmatrix} 72 & 65 & 80 \\ 60 & 75 & 70 \\ 85 & 55 & 90 \end{pmatrix},$$

compare the rankings produced by

$$\mathbf{w}_1 = \begin{pmatrix} 0.40 \\ 0.35 \\ 0.25 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 0.10 \\ 0.80 \\ 0.10 \end{pmatrix}.$$

85. Two service quantities satisfy resource capacities

$$\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 260 \\ 220 \end{pmatrix}.$$

Solve for x and y .

86. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. Calculate AB and BA , and explain why applying two transformations in different orders can produce different outcomes.
87. A reporting table has regions in rows and indicators in columns. Explain how transposition changes the reporting perspective and identify one labelling error that could make the transposed table misleading.

88. The transition matrix changes from

$$T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix} \quad \text{to} \quad T' = \begin{pmatrix} 0.78 & 0.35 \\ 0.22 & 0.65 \end{pmatrix}.$$

For $\mathbf{p} = (0.6, 0.4)^\top$, compare the one-period predictions.

89. Verify that $I_3\mathbf{v} = \mathbf{v}$ for $\mathbf{v} = (4, -2, 7)^\top$, and explain the role of the identity matrix in matrix algebra and transition modelling.

90. Scenario matrices are

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 100 \\ 50 \end{pmatrix}.$$

Calculate $A\mathbf{v}$, $B\mathbf{v}$ and $(A - B)\mathbf{v}$.

91. Recorded categories satisfy

$$\mathbf{y} = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix} \mathbf{x}, \quad \mathbf{y} = \begin{pmatrix} 520 \\ 480 \end{pmatrix}.$$

Use the inverse to estimate the underlying category counts $\mathbf{x}$, and state one caution about interpreting the corrected values.

Applications: responsible use of matrix models

92. Explain why row and column labels, units and ordering are essential when multiplying matrices from different data sources.
93. Give four assumptions that should be checked before applying a transition matrix repeatedly to forecast social behaviour.
94. A weighted-score matrix ranks districts for funding. Give four fairness or governance questions that should accompany the ranking.
95. Explain how aggregate matrix products can conceal subgroup differences even when the arithmetic is correct.
96. Explain the practical meaning of a singular coefficient matrix and distinguish systems with no solution from those with infinitely many solutions.
97. Explain why a matrix product describing association or movement does not by itself establish a causal mechanism.
98. Published transition probabilities are rounded and their column sums differ slightly from 1. Give two defensible responses and state what should be disclosed.
99. A matrix contains small-cell administrative counts. Give four privacy or disclosure safeguards that should be considered before publication.

- 100.** Give a six-stage checklist for constructing, calculating, checking and communicating a matrix-based social-science model.

Chapter 10

Limits and Continuity

Chapter guide

Interpret one-sided and two-sided limits numerically and graphically, develop algebraic techniques, then connect continuity, asymptotes and the Intermediate Value Theorem to calculus.

Selected background sources: (Strang & Herman, 2016a).

Limits describe the value approached by a function when its input moves arbitrarily close to a specified point. The input need not actually reach the point, and the value of the function at that point may be different from the limit or may not exist at all. This distinction allows mathematics to study local behaviour even when a formula contains a hole, a jump or an unbounded value.

In social-science applications, limits are used to describe long-run equilibria, saturation levels, threshold effects and marginal change. A population share may approach a stable level, an average cost may settle as output grows, or a policy rule may change abruptly when an eligibility boundary is crossed. In each case, the limit describes nearby or long-run behaviour rather than merely substituting a number into a formula.

This chapter develops limits from numerical, graphical and algebraic perspectives. It introduces one-sided limits, limit laws, removable and non-removable discontinuities, end behaviour, asymptotes and continuity on intervals. The Intermediate Value Theorem is then used to establish the existence of solutions and to motivate a simple numerical method for locating them. These ideas provide the foundation for differential and integral calculus in the following chapters.

10.1 One-sided and two-sided limits

The notation

$$\lim_{x \rightarrow a} f(x) = L$$

means that $f(x)$ approaches L as x approaches a . The one-sided limits $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ describe approach from the left and right. A finite two-sided limit exists precisely when both one-sided limits exist and are equal.

The value $f(a)$ need not equal the limit, and it may even be undefined. Infinite limits describe unbounded behaviour near a vertical asymptote.

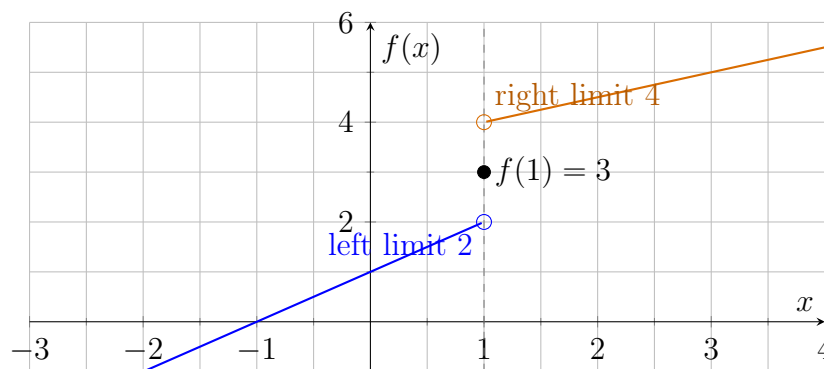


Figure 10.1: Unequal one-sided limits at $x = 1$. The two-sided limit does not exist, regardless of the separately assigned value $f(1) = 3$.

10.2 Estimating limits numerically and graphically

A limit can often be estimated before it is evaluated algebraically. A table of values should use inputs on both sides of the point and increasingly close to it. The purpose is to identify a trend, not to substitute the point itself. Values that are far from the point may conceal local behaviour, while rounded calculator output may suggest equality before it has actually been proved.

Example 10.1: [

Estimating a limit from a table For $x \neq 1$, let

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

The following values approach $x = 1$ from both sides.

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	1.9	1.99	1.999	2.001	2.01	2.1

The values suggest that

$$\lim_{x \rightarrow 1} f(x) = 2.$$

The expression is undefined at $x = 1$, but that does not prevent the limit from existing. Algebraically, $f(x) = x + 1$ whenever $x \neq 1$, which confirms the estimate.

When reading a graph, the direction of approach matters. Trace the curve towards the specified input from the left and from the right, without being distracted by a separately plotted filled point. A graph or table can provide strong evidence for a limit, but an exact algebraic argument is preferable when one is available.

10.3 Algebraic evaluation of limits

10.3.1 Limit laws and direct substitution

Suppose

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} g(x) = M,$$

and let c be a constant. The principal limit laws are

$$\begin{aligned} \lim_{x \rightarrow a} (f(x) + g(x)) &= L + M, \\ \lim_{x \rightarrow a} cf(x) &= cL, \\ \lim_{x \rightarrow a} f(x)g(x) &= LM, \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \frac{L}{M}, \quad M \neq 0. \end{aligned}$$

The corresponding difference and integer-power laws follow from these. For a root, the limiting value must lie in the domain of the root being used.

Polynomials are continuous everywhere, so their limits may be found by direct substitution. Rational functions are continuous wherever their denominators are non-zero. Exponential functions are continuous for every real input, while logarithmic and root functions are continuous on their respective domains.

Example 10.2: [

blue]Using the limit laws Suppose $f(x) \rightarrow 4$ and $g(x) \rightarrow -2$ as $x \rightarrow a$. Then

$$\lim_{x \rightarrow a} \frac{3f(x) - g(x)}{f(x) + 5} = \frac{3(4) - (-2)}{4 + 5} = \boxed{\frac{14}{9}}.$$

The denominator approaches 9, so the quotient law is applicable.

10.3.2 Indeterminate forms and algebraic simplification

When direct substitution gives $0/0$, the result is an *indeterminate form*, not an answer. It indicates that further simplification is needed. Common techniques include factorisation and cancellation, rationalisation, and combining fractions over a common denominator.

For example,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4.$$

The factor $x - 2$ may be cancelled while taking the limit because the nearby inputs satisfy $x \neq 2$. The cancellation does not define the original function at $x = 2$.

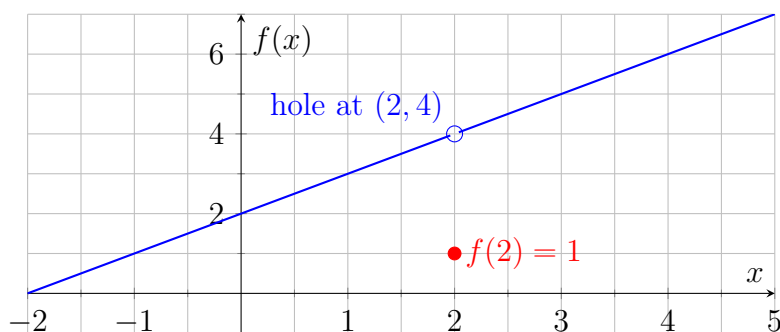


Figure 10.2: A removable discontinuity: nearby values approach 4, even though the illustrated function assigns a different value at $x = 2$.

Example 10.3: [

orange]Rationalising an indeterminate form Evaluate

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}.$$

Multiplying by the conjugate gives

$$\begin{aligned} \frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} &= \frac{x - 4}{(x - 4)(\sqrt{x} + 2)} \\ &= \frac{1}{\sqrt{x} + 2}, \quad x \neq 4. \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \frac{1}{2 + 2} = \boxed{\frac{1}{4}}.$$

Example 10.4: [

violet]Combining fractions Evaluate

$$\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{1+x} - 1 \right).$$

The expression in brackets has the common-denominator form

$$\frac{1 - (1+x)}{1+x} = -\frac{x}{1+x}.$$

Hence, for $x \neq 0$,

$$\frac{1}{x} \left(-\frac{x}{1+x} \right) = -\frac{1}{1+x},$$

and the required limit is $\boxed{-1}$.

An algebraic transformation used in a limit must be valid throughout a punctured

neighbourhood of the point: it may exclude the point itself, but it must agree with the original expression for all sufficiently close permitted inputs.

10.3.3 Limits at infinity and end behaviour

For limits at infinity, the leading terms determine the end behaviour. In a rational function, divide numerator and denominator by the highest power of x in the denominator. If P and Q are polynomials, the behaviour of $P(x)/Q(x)$ depends on their degrees.

Comparison of degrees	Limit as $x \rightarrow \pm\infty$
$\deg P < \deg Q$	0
$\deg P = \deg Q$	ratio of the leading coefficients
$\deg P > \deg Q$	generally unbounded or governed by polynomial division; no finite horizontal limit

Example 10.5: [

Long-run value of a rational model Suppose an average administrative cost is modelled by

$$C(n) = \frac{1200 + 18n}{n}, \quad n > 0,$$

where n is the number of cases processed. Rewriting gives

$$C(n) = \frac{1200}{n} + 18.$$

Therefore,

$$\lim_{n \rightarrow \infty} C(n) = \boxed{18}.$$

The fixed cost per case tends to zero, so the modelled average cost approaches RM 18 per case. The limit does not mean that the total cost approaches RM 18.

If the limit is $+\infty$ or $-\infty$, the function is unbounded rather than approaching a finite number. The sign should be determined carefully from the leading term and the direction of approach.

10.4 Continuity

A function is continuous at $x = a$ when

$$f(a) \text{ is defined, } \lim_{x \rightarrow a} f(x) \text{ exists, } \lim_{x \rightarrow a} f(x) = f(a).$$

For a piecewise function, continuity at a boundary is checked by comparing the left-hand limit, right-hand limit and stated function value.

Continuity is a local property. A graph is continuous at a point when no break is encountered there, but the formal three-part test is more reliable than the informal instruction to “draw without lifting the pen”. It also applies when the graph is difficult to sketch.

Example 10.6: [

blue]Choosing a parameter for continuity Let

$$f(x) = \begin{cases} kx + 1, & x < 2, \\ x^2 - 1, & x \geq 2. \end{cases}$$

For continuity at $x = 2$, the two branches must approach the same value:

$$2k + 1 = 2^2 - 1 = 3.$$

Thus $k = \boxed{1}$. With this value, the left-hand limit, right-hand limit and function value are all 3.

10.4.1 Continuity on intervals

A function is continuous on an open interval when it is continuous at every point in that interval. On a closed interval $[a, b]$, continuity means continuity at every interior point together with right-continuity at a and left-continuity at b .

The standard elementary functions are continuous wherever they are defined:

- polynomials are continuous on $\mathbb{R}$;
- rational functions are continuous wherever the denominator is non-zero;
- exponential functions are continuous on $\mathbb{R}$;
- logarithmic functions are continuous for positive arguments;
- square-root functions are continuous where their arguments are non-negative.

Sums, differences, products and compositions of continuous functions remain continuous wherever the resulting expressions are defined. Quotients remain continuous wherever the denominator is non-zero.

For example,

$$h(x) = \frac{\ln(x+4)}{x-1}$$

requires $x > -4$ and $x \neq 1$. It is therefore continuous on

$$(-4, 1) \cup (1, \infty).$$

10.4.2 Types of discontinuity

Discontinuities can be classified by comparing one-sided limits and the function value.

Type	Defining behaviour
Removable	The two-sided limit is finite, but the function is undefined or has a different value at the point.
Jump	The left- and right-hand limits are finite but unequal.
Infinite	At least one one-sided limit is $+\infty$ or $-\infty$; a vertical asymptote is present.
Oscillatory	Nearby values do not settle towards a single value because the function oscillates increasingly rapidly.

The first three types are illustrated in figures 10.1 to 10.3. A removable discontinuity can be repaired by redefining one function value. Jump and infinite discontinuities cannot be removed in this way.

In applications, a jump need not represent an error. Tax brackets, eligibility rules and administrative fees may deliberately create threshold effects. The modeller must decide whether the discontinuity reflects the actual rule or is an unintended consequence of joining incompatible formulas.

10.5 Asymptotes

The line $x = a$ is a vertical asymptote when at least one one-sided limit is infinite. The line $y = L$ is a horizontal asymptote when $f(x) \rightarrow L$ as $x \rightarrow \infty$ or $x \rightarrow -\infty$. A cancelled factor produces a hole rather than a vertical asymptote.

The two ends of a graph may have different horizontal asymptotes. A graph may also cross a horizontal asymptote; the asymptote describes end behaviour, not a boundary that the graph is forbidden to cross. By contrast, the function is undefined on a vertical asymptote.

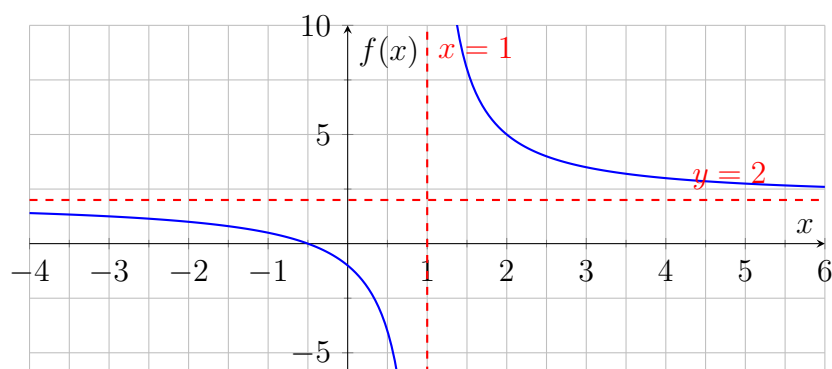


Figure 10.3: For $f(x) = 2 + 3/(x - 1)$, $x = 1$ is a vertical asymptote and $y = 2$ is a horizontal asymptote.

When the numerator of a rational function has degree exactly one greater than the denominator, polynomial division may reveal an *oblique* or *slant asymptote*. For example,

$$\frac{x^2 + 1}{x - 1} = x + 1 + \frac{2}{x - 1}.$$

Since $2/(x - 1) \rightarrow 0$ as $x \rightarrow \pm\infty$, the line $y = x + 1$ is an oblique asymptote. There is also a vertical asymptote at $x = 1$.

10.6 Limits in social-science models

In an applied model, a limit often describes long-run behaviour rather than a value reached at a particular finite time. For example, if an approval index is modelled by

$$A(t) = 70 - 20e^{-0.3t},$$

then

$$\lim_{t \rightarrow \infty} A(t) = 70.$$

The model predicts that the index approaches 70, although it does not claim that the index becomes exactly 70 after a specified number of periods. Horizontal asymptotes can similarly represent market saturation, a limiting population share or a long-run equilibrium price.

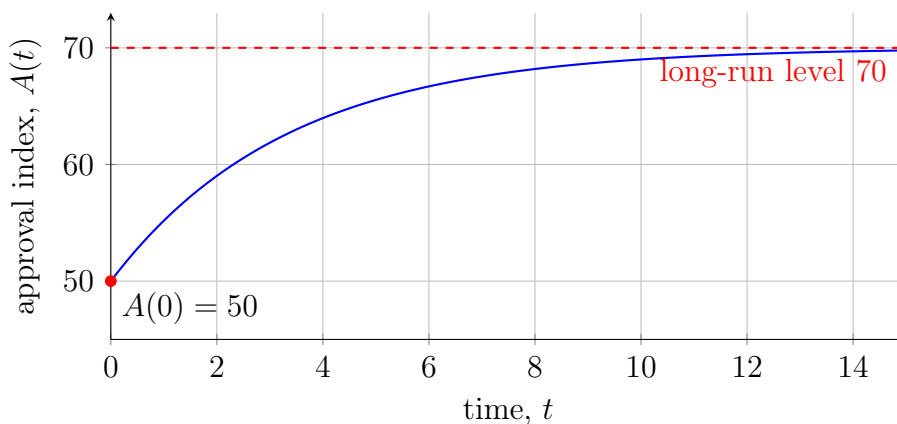


Figure 10.4: The model $A(t) = 70 - 20e^{-0.3t}$ approaches its long-run level without reaching it at a finite time.

A discontinuity may represent a genuine policy threshold, such as a tax rate that changes abruptly when income passes a stated boundary. It may instead signal that a formula has been used outside the range for which it was designed. Mathematical continuity should therefore be interpreted together with the assumptions of the model.

10.7 The Intermediate Value Theorem

If f is continuous on $[a, b]$ and a number N lies between $f(a)$ and $f(b)$, then there is at least one $c \in [a, b]$ such that $f(c) = N$. In particular, if $f(a)$ and $f(b)$ have opposite signs, a root lies between a and b . This theorem establishes existence; it does not by itself give the exact value of the root.

The continuity assumption is essential. A discontinuous function can change from a negative value to a positive value by jumping over zero. The theorem also guarantees only that at least one required value occurs. It does not establish uniqueness, provide a formula for the solution or state how rapidly a numerical procedure will find it.

Example 10.7: [

violet]Locating a break-even point Suppose the net benefit of a programme is

$$B(x) = x^3 - 4x - 1.$$

Because B is continuous, $B(2) = -1$ and $B(3) = 14$ show that at least one break-even value satisfying $B(x) = 0$ lies in the interval $(2, 3)$.

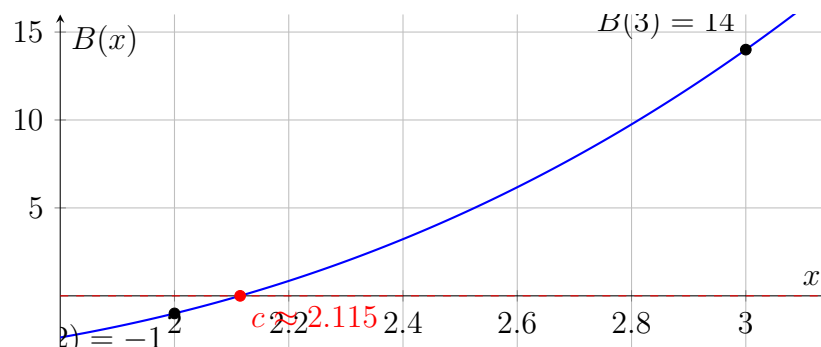


Figure 10.5: A continuous curve changing sign between $x = 2$ and $x = 3$ must cross the axis somewhere in that interval.

10.7.1 Refining a root by bisection

Once a sign change has been found for a continuous function, the interval can be narrowed by the *bisection method*. Evaluate the function at the midpoint and retain the half-interval whose endpoints have opposite signs. Repeating the process halves the interval width at every step.

For the break-even model $B(x) = x^3 - 4x - 1$, begin with $[2, 3]$.

Current interval	Midpoint	Value at midpoint	Retained interval
$[2, 3]$	2.5	4.625	$[2, 2.5]$
$[2, 2.5]$	2.25	1.390625	$[2, 2.25]$
$[2, 2.25]$	2.125	0.095703	$[2, 2.125]$

After three bisections, the root is known to lie in an interval of width 0.125. Further iterations give $x \approx 2.115$. Bisection is dependable when continuity and a sign change are known, although faster numerical methods are often used in more advanced work.

10.8 Limits and instantaneous rates of change

Suppose a quantity $Q(x)$ changes as the explanatory variable moves from a to $a + h$. Its average rate of change over that interval is

$$\frac{Q(a + h) - Q(a)}{h}.$$

If this expression approaches a finite value as $h \rightarrow 0$, the limit describes the instantaneous rate of change at $x = a$:

$$\lim_{h \rightarrow 0} \frac{Q(a + h) - Q(a)}{h}.$$

This limit is the derivative introduced in Chapter 11.

Example 10.8: [

orange]A marginal cost from a limit Let the total cost of producing q units be

$$C(q) = 500 + 20q + 0.02q^2.$$

At $q = 100$, the difference quotient is

$$\begin{aligned} \frac{C(100 + h) - C(100)}{h} &= \frac{24h + 0.02h^2}{h} \\ &= 24 + 0.02h, \quad h \neq 0. \end{aligned}$$

Therefore,

$$\lim_{h \rightarrow 0} \frac{C(100 + h) - C(100)}{h} = \boxed{24}.$$

Near an output of 100 units, the modelled cost of one additional unit is approximately RM 24. This is a local interpretation; it need not equal the average cost per unit.

10.9 Exercises

Answers are provided in section C.10.

1. Use the four schematic graphs in figure 10.6 to find the indicated limits.

- For graph A, find the limits as $x \rightarrow 0^+$, $x \rightarrow 0^-$, $x \rightarrow 0$, $x \rightarrow \infty$ and $x \rightarrow -\infty$.
- For graph B, find the limits as $x \rightarrow 1^+$, $x \rightarrow 1^-$, $x \rightarrow 1$, $x \rightarrow \infty$ and $x \rightarrow -\infty$.
- For graph C, find the limits as $x \rightarrow -3^+$, $x \rightarrow -3^-$, $x \rightarrow -3$, $x \rightarrow \infty$ and $x \rightarrow -\infty$.
- For graph D, find the limits as $x \rightarrow 0.5^+$, $x \rightarrow 0.5^-$, $x \rightarrow 0.5$, $x \rightarrow \infty$ and $x \rightarrow -\infty$.

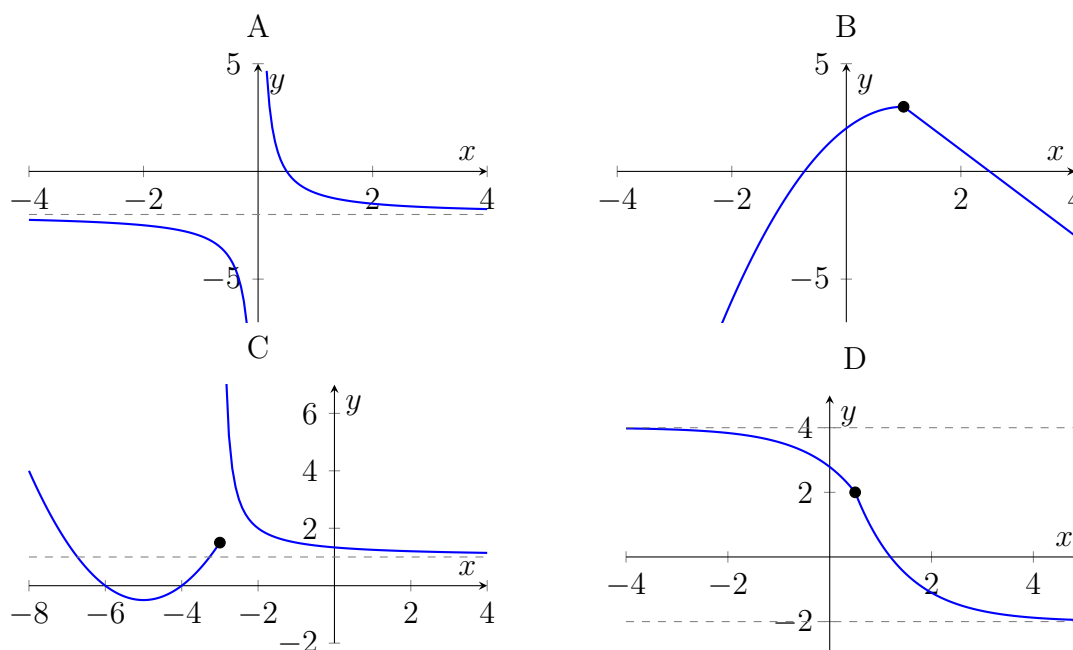


Figure 10.6: Schematic graphs for Exercise 1.

2. Find the limit at the stated boundary.

- $\lim_{x \rightarrow 2} f(x)$, where

$$f(x) = \begin{cases} 2x^2 - x + 1, & x \geq 2, \\ -1 + 4x, & x < 2; \end{cases}$$

- $\lim_{x \rightarrow -3} f(x)$, where

$$f(x) = \begin{cases} \frac{3}{x-4}, & x > -3, \\ \sqrt{5-x}, & x \leq -3. \end{cases}$$

3. Find the constants that make the indicated limits exist.

$$\begin{aligned} \text{(a)} \quad f(x) &= \begin{cases} \frac{1}{2}x - 3, & x \geq 1, \\ ax^2, & x < 1, \end{cases} \text{ at } x = 1; \\ \text{(b)} \quad f(x) &= \begin{cases} 7, & x \geq 4, \\ ax + b, & 1 \leq x < 4, \\ -2, & x < 1, \end{cases} \text{ at both } x = 1 \text{ and } x = 4. \end{aligned}$$

4. Find each limit.

$$\begin{aligned} \text{(a)} \quad & \lim_{x \rightarrow -\infty} 2x^2(x + 1) \\ \text{(b)} \quad & \lim_{x \rightarrow -\infty} 3(2x + 1)^2 \\ \text{(c)} \quad & \lim_{x \rightarrow -\infty} \frac{x^3 - 2x^2 + 5x - 1}{2x^2 - 3x + 1} \\ \text{(d)} \quad & \lim_{x \rightarrow -\infty} \left(\frac{2x}{x - 1} + \frac{3x}{2x + 1} \right) \end{aligned}$$

5. Find all vertical and horizontal asymptotes, distinguishing any removable holes.

$$\begin{aligned} \text{(a)} \quad & (x^2 - 4)/(x - 2) \\ \text{(b)} \quad & 2/x^2 \\ \text{(c)} \quad & (x^2 - 2)/(x^2 - x - 2) \\ \text{(d)} \quad & (x^2 - 6x - 7)/(x + 1) \end{aligned}$$

6. Let

$$f(x) = \begin{cases} 1, & x \leq -1, \\ 1 - ax, & -1 < x < 1, \\ x - 1, & x \geq 1. \end{cases}$$

Given that f is continuous at $x = 1$, find a and determine whether f is continuous at $x = -1$.

7. A participation rate is modelled by $P(t) = 80 - 30/(t + 1)$ for $t \geq 0$. Find its long-run limit and interpret the result. Does the model ever give $P(t) = 80$ for a finite value of t ?
8. The net social benefit of a policy is $N(x) = x^3 - 6x - 4$. Use the Intermediate Value Theorem to show that a break-even point lies between 2 and 3. State clearly what the theorem does and does not establish.
9. Evaluate each limit algebraically.

$$\begin{aligned} \text{(a)} \quad & \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \\ \text{(b)} \quad & \lim_{x \rightarrow 0} \frac{\sqrt{9 + x} - 3}{x} \end{aligned}$$

$$(c) \lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$$

$$(d) \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

10. Suppose that $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} g(x) = -2$. Use the limit laws to find

$$(a) \lim_{x \rightarrow a} (3f(x) - 2g(x));$$

$$(b) \lim_{x \rightarrow a} f(x)g(x);$$

$$(c) \lim_{x \rightarrow a} \frac{f(x)}{g(x)};$$

$$(d) \lim_{x \rightarrow a} \sqrt{f(x) + 5}.$$

11. Classify the discontinuity at the stated point as removable, jump or infinite. Give a brief reason.

$$(a) f(x) = (x^2 - 1)/(x - 1) \text{ for } x \neq 1, \text{ with } f(1) = 5;$$

$$(b) f(x) = -1 \text{ for } x < 0 \text{ and } f(x) = 1 \text{ for } x \geq 0, \text{ at } x = 0;$$

$$(c) f(x) = 1/(x - 2), \text{ at } x = 2;$$

$$(d) f(x) = (x + 3)/(x^2 - 9), \text{ at } x = -3.$$

12. State the interval or union of intervals on which each function is continuous.

$$(a) f(x) = 1/(x^2 - 9)$$

$$(b) g(x) = \sqrt{5 - 2x}$$

$$(c) h(x) = \ln(x + 4)/(x - 1)$$

$$(d) p(x) = x^4 - 3x + 7$$

13. Find the constants required for continuity.

$$(a) f(x) = \begin{cases} kx + 1, & x < 2, \\ x^2 - 1, & x \geq 2, \end{cases} \text{ at } x = 2.$$

$$(b) g(x) = \begin{cases} ax + 3, & x < 1, \\ 5, & x = 1, \\ x^2 + b, & x > 1, \end{cases} \text{ at } x = 1.$$

14. Find all vertical, horizontal and oblique asymptotes, as applicable. Identify any removable holes.

$$(a) \frac{3x^2 + x - 2}{x^2 - 4}$$

$$(b) \frac{2x + 1}{x^2 + 1}$$

$$(c) \frac{2x^2 + 3}{x - 1}$$

(d) $\frac{x^2 - 4}{x^2 - x - 2}$

15. Find each limit, including its sign when the function is unbounded.

(a) $\lim_{x \rightarrow \infty} \frac{5x^3 - x}{2x^3 + 7}$

(b) $\lim_{x \rightarrow -\infty} \frac{3x^2 + 1}{x^3 - 2}$

(c) $\lim_{x \rightarrow \infty} \frac{2x^4}{1 - x^2}$

(d) $\lim_{x \rightarrow -\infty} \frac{-x^3 + 4x}{2x^2 + 1}$

16. Let $Q(x) = x^3 + x - 12$.

- (a) Use the Intermediate Value Theorem to show that a root lies in $(2, 3)$.
- (b) Perform three bisection steps, beginning with $[2, 3]$, and state the resulting interval containing the root.
- (c) Use the midpoint of the final interval as an approximation and give a bound for its absolute error.

17. A satisfaction index is modelled by $S(t) = 60 + 25e^{-0.2t}$ for $t \geq 0$.

- (a) Find $S(0)$.
- (b) Find $\lim_{t \rightarrow \infty} S(t)$ and interpret it.
- (c) Does the index approach its limiting value from above or from below?
- (d) Find when the model first gives $S(t) = 65$.

18. A processing charge, in RM, is modelled by

$$C(x) = \begin{cases} 0.05x, & 0 \leq x < 2000, \\ 100 + 0.08(x - 2000), & x \geq 2000, \end{cases}$$

where x is the transaction value in RM.

- (a) Find the left- and right-hand limits at $x = 2000$.
- (b) Determine whether C is continuous at $x = 2000$.
- (c) Find $C(2500)$.
- (d) Interpret the change that occurs at the threshold even though the charge itself is continuous.

19. Revenue from selling q units is modelled by $R(q) = 60q - 0.1q^2$.

- (a) Simplify $(R(100 + h) - R(100))/h$ for $h \neq 0$.
- (b) Find its limit as $h \rightarrow 0$.

(c) Interpret the result in context.

20. For the approval model $A(t) = 70 - 20e^{-0.3t}$, a researcher claims: “The index reaches 70 after a sufficiently long but finite time, and the continuity of the formula proves that real approval data must change continuously.” Explain both errors in this statement.

Additional practice: grouped skills review

21. Evaluate

$$\lim_{x \rightarrow k} (x^2 + 2x + 3)$$

for $k = 1, 2, 3, 4, 5, 6, 7$ and 8.

22. Evaluate

$$\lim_{x \rightarrow k} \frac{2x + 1}{x + 2}$$

for $k = 1, 2, 3, 4, 5, 6, 7$ and 8.

23. Evaluate

$$\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a}$$

for $a = 2, 3, 4, 5, 6, 7, 8$ and 9.

24. For $m = 2, 3, 4, 5, 6, 7, 8$ and 9, evaluate

$$\lim_{x \rightarrow m-1} \frac{\sqrt{x + m^2 - m + 1} - m}{x - (m - 1)}.$$

25. In each case, the left-hand limit is L and the right-hand limit is $L + 2$. Determine whether the two-sided limit exists for $L = 2, 3, 4, 5, 6, 7, 8$ and 9.

26. Evaluate

$$\lim_{x \rightarrow \infty} \frac{mx^2 + 1}{(m + 1)x^2 - 4}$$

for $m = 2, 3, 4, 5, 6, 7, 8$ and 9.

27. For $m = 2, 3, 4, 5, 6, 7, 8$ and 9, state the vertical and horizontal asymptotes of

$$f_m(x) = m + \frac{m + 1}{x - (m - 1)}.$$

28. For $m = 2, 3, 4, 5, 6, 7, 8$ and 9, find c so that

$$f(x) = \begin{cases} mx + m - 2, & x < m - 1, \\ c, & x \geq m - 1 \end{cases}$$

is continuous at $x = m - 1$.

29. For each $m = 2, 3, 4, 5, 6, 7, 8$ and 9, use the Intermediate Value Theorem to show that

$$f_m(x) = x^2 - (m^2 + 1)$$

has a root in $(m, m + 1)$.

30. Evaluate

$$\lim_{h \rightarrow 0} \frac{(a + h)^2 - a^2}{h}$$

for $a = 1, 2, 3, 4, 5, 6, 7$ and 8.

Applications: numerical and local behaviour

31. An approval index has the recorded values

t	4.9	4.99	5.01	5.1
$A(t)$	69.7	69.97	70.03	70.3

and $A(5) = 75$. Estimate the left-hand, right-hand and two-sided limits at $t = 5$, then compare the limit with the recorded value.

32. A waiting-time index is

$$W(x) = \begin{cases} 2 + 0.1x, & x < 10, \\ 5, & x = 10, \\ 4 - 0.1x, & x > 10. \end{cases}$$

Find both one-sided limits at $x = 10$, the two-sided limit and $W(10)$. Classify the discontinuity.

33. A household receives RM1000 if income $y < 30\,000$ and receives no payment if $y \geq 30\,000$. Find the one-sided limits of the payment as $y \rightarrow 30\,000$, and explain why the two-sided limit does not exist.

34. A congestion model is

$$T(u) = \frac{10}{100 - u}, \quad 0 \leq u < 100,$$

where u is capacity utilisation in percentage points. Find $\lim_{u \rightarrow 100^-} T(u)$ and interpret the result.

35. A highly unstable polling model is

$$V(t) = 50 + \sin\left(\frac{1}{t-2}\right), \quad t \neq 2.$$

Explain why $\lim_{t \rightarrow 2} V(t)$ does not exist and why assigning a value to $V(2)$ cannot repair the discontinuity.

36. A processing fee is

$$F(x) = \begin{cases} 0.02x, & x < 1000, \\ 20, & x = 1000, \\ 25 + 0.01(x - 1000), & x > 1000. \end{cases}$$

Find the one-sided limits at $x = 1000$, determine whether the two-sided limit exists, and compare the limits with $F(1000)$.

37. Suppose $f(x) \rightarrow 12$ and $g(x) \rightarrow 3$ as $x \rightarrow a$. Use the limit laws to find the limits of

- (a) $2f(x) - g(x)$;
- (b) $f(x)g(x)$;
- (c) $f(x)/g(x)$;
- (d) $\sqrt{f(x) + 4}$.

38. Average administrative cost is modelled by $M(n) = 40 + 600/n$. Calculate $M(100)$, $M(1000)$ and $M(10\,000)$, then infer and interpret the long-run limit.

39. A direction indicator is $D(x) = x/|x|$ for $x \neq 0$. Find the left-hand and right-hand limits at 0 and determine whether the two-sided limit exists.

40. Explain why a table of increasingly close values may provide good evidence for a limit without proving its exact value. Include the possible effect of rounding.

Applications: algebraic limit techniques

41. Evaluate $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$ by factorisation.

42. Evaluate $\lim_{x \rightarrow 4} (\sqrt{x} - 2)/(x - 4)$ by rationalisation.

43. Evaluate

$$\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{1+x} - 1 \right).$$

44. Evaluate $\lim_{x \rightarrow 2} (x^3 - 8)/(x - 2)$.

45. Evaluate

$$\lim_{x \rightarrow 0} \frac{\frac{1}{2+x} - \frac{1}{2}}{x}.$$

46. Evaluate $\lim_{x \rightarrow 3} (x^4 - 81)/(x - 3)$.

47. Total cost is $C(q) = 1000 + 20q + 0.05q^2$. Simplify

$$\frac{C(200 + h) - C(200)}{h}, \quad h \neq 0,$$

and find its limit as $h \rightarrow 0$.

48. Programme participation is $P(n) = n(100 - n)$. Simplify

$$\frac{P(20 + h) - P(20)}{h}, \quad h \neq 0,$$

and find its limit as $h \rightarrow 0$.

49. Demand is $D(p) = 500 - 5p$. Evaluate

$$\lim_{h \rightarrow 0} \frac{D(p + h) - D(p)}{h}$$

and interpret its sign.

50. A student notes that $(x^2 + x)/x = x + 1$ and substitutes $x = 0$. Explain why the original expression is undefined at 0 but why its limit as $x \rightarrow 0$ is still 1.

Applications: continuity and policy thresholds

51. A grant rule is

$$G(x) = \begin{cases} kx + 200, & x < 1000, \\ 1200, & x \geq 1000. \end{cases}$$

Find k so that the rule is continuous at $x = 1000$.

52. A service fee is

$$F(x) = \begin{cases} 0.02x, & x < 5000, \\ d + 0.03(x - 5000), & x \geq 5000. \end{cases}$$

Find d so that the total fee is continuous at $x = 5000$.

53. A response index is $R(t) = (t^2 - 9)/(t - 3)$ for $t \neq 3$, with $R(3) = 10$. Find the limit at 3 and classify the discontinuity.

54. A transit fare is RM 2.50 for journeys of at most 10 km and RM 4.00 for longer journeys. Find the one-sided limits at 10 km and classify the discontinuity.

55. A monthly benefit is

$$B(y) = \begin{cases} 1200, & y < 3000, \\ 0, & y \geq 3000, \end{cases}$$

where y is monthly income in RM. Find the one-sided limits and $B(3000)$, classify the discontinuity and interpret the policy effect shown in figure 10.7.

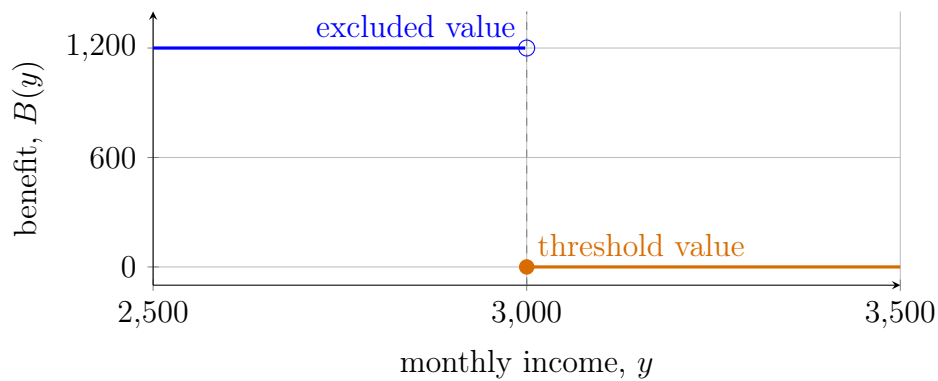


Figure 10.7: The benefit cliff in Exercise 55. Crossing the threshold produces a discrete loss rather than a small marginal change.

56. A tax rule is

$$T(y) = \begin{cases} 0.05y, & y \leq 20\,000, \\ 1000 + 0.10(y - 20\,000), & y > 20\,000. \end{cases}$$

Show that the total tax is continuous at RM 20 000, and describe what nevertheless changes at the threshold.

57. Average delay is $D(u) = 30/(1 - u)$ for $0 \leq u < 1$. Classify the discontinuity at $u = 1$ and interpret it in the congestion model.

58. State the intervals on which

$$f(x) = \frac{\ln(x + 5)}{x - 2}$$

is continuous.

59. Define $f(4)$ so that

$$f(x) = \frac{x^2 - 16}{x - 4}, \quad x \neq 4,$$

becomes continuous at $x = 4$.

60. Give two reasons why a discontinuity in a fitted policy model may represent either a genuine institutional rule or a modelling error. State what evidence would help distinguish the two.

Applications: asymptotes and long-run behaviour

61. Average administrative cost is $C(n) = 5000/n + 25$. Find its long-run limit and interpret both terms of the model.

62. Programme participation is

$$P(t) = \frac{80}{1 + 3e^{-0.4t}}, \quad t \geq 0.$$

Find $P(0)$ and the long-run limit. Does the model reach its limit at a finite time?

63. An unemployment index is $U(t) = 4 + 6e^{-0.2t}$. Find its long-run limit, state the direction of approach and determine when it is within 0.5 index point of the limit.
64. Expenditure per recipient is $E(x) = (120 + 5x)/(10 + x)$. Find its limit as $x \rightarrow \infty$ and state whether the model approaches the limit from above or below for $x \geq 0$.
65. A funding model is $F(x) = 50 + 1000/(x - 20)$. Find the vertical and horizontal asymptotes and the one-sided behaviour as $x \rightarrow 20$.
66. Two adoption models are

$$A(t) = 80 - 60e^{-0.2t}, \quad B(t) = 80 - 60e^{-0.5t}.$$

Find their initial and long-run values, compare them at $t = 10$ and use figure 10.8 to explain the role of the rate parameter.

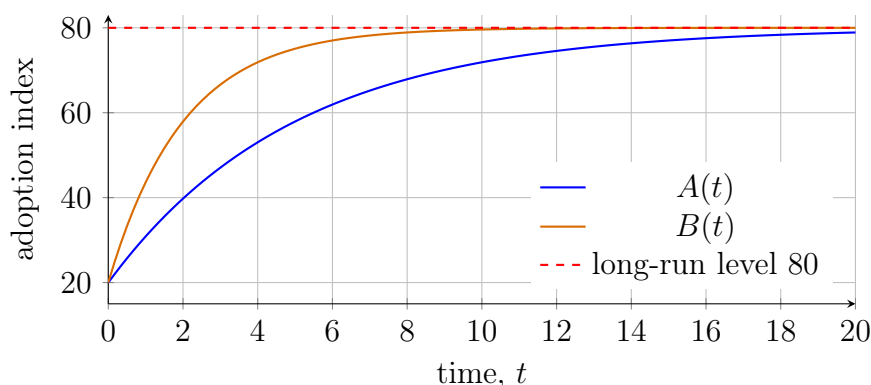


Figure 10.8: The models in Exercise 66 share an initial value and a long-run limit, but the larger rate parameter produces faster convergence.

67. A social indicator is $I(x) = (3x^2 + 2)/(x^2 + 4)$. Find its horizontal asymptote and determine whether the graph approaches it from above or below.
68. Infrastructure cost is modelled algebraically by $C(x) = (x^2 + 100)/(x + 10)$. Find its vertical and oblique asymptotes. If the contextual domain is $x \geq 0$, explain which asymptote is directly relevant to long-run interpretation.
69. A fitted probability is $H(x) = 1/(1 + e^{-x})$. Find its limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$, and identify both horizontal asymptotes.
70. Show that $A(t) = 70 + e^{-t} \sin t$ has horizontal asymptote $y = 70$. Explain why this example shows that a horizontal asymptote need not be an upper or lower bound.

Applications: existence and bisection

71. For $N(x) = x^3 - 7x - 1$, use the Intermediate Value Theorem to show that a break-even point lies in $(2, 3)$.
72. Demand and supply are $D(p) = 100 - 2p$ and $S(p) = 20 + p$. Apply the Intermediate Value Theorem to the excess-demand function to show that an equilibrium price lies in $(26, 27)$, then find the exact equilibrium price.
73. A budget balance is $B(t) = t^3 - 5t - 3$. Show that a balanced point lies in $(2, 3)$.
74. A satisfaction model is $S(t) = 60 + 20(1 - e^{-0.3t})$. Use the Intermediate Value Theorem to show that the target $S = 75$ is met between $t = 4$ and $t = 5$, then find the exact modelled time.
75. Perform three bisection steps for the root of $N(x) = x^3 - 7x - 1$ beginning with $[2, 3]$. State the final interval, its midpoint and an upper bound for the midpoint error.
76. Housing demand is $H(p) = 500 - 10p$. Use the Intermediate Value Theorem to show that demand equals 275 for some $p \in (20, 25)$, then find the exact price.
77. A continuous function has positive values at both endpoints of an interval. Explain why this does not prove that the function has no root inside. Give a counterexample.
78. Let

$$J(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Although $J(-1) < 0 < J(1)$, explain why J has no root and why this does not contradict the Intermediate Value Theorem. Use figure 10.9.

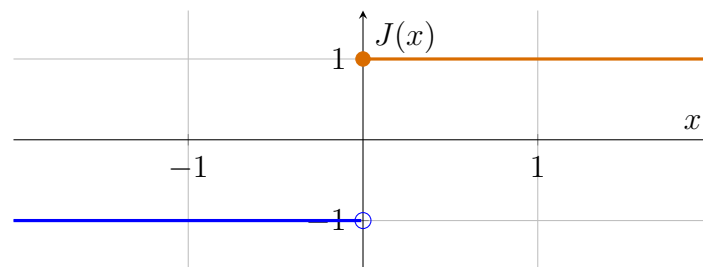


Figure 10.9: The discontinuous sign change in Exercise 78 jumps over zero. The continuity hypothesis of the Intermediate Value Theorem is therefore essential.

79. A continuous function has the values

x	1	2	3	4
$f(x)$	-6	-1	2	10

Identify an interval on which the Intermediate Value Theorem guarantees a root and state the evidence used.

80. Starting with an interval of width 4, how many bisection steps are required to guarantee that the midpoint approximation has absolute error less than 0.001?

Applications: instantaneous rates from limits

81. Total cost is $C(q) = 2000 + 30q + 0.02q^2$. Simplify the difference quotient at $q = 100$ and find its limit as $h \rightarrow 0$.
82. Revenue is $R(q) = 100q - 0.1q^2$. Simplify the difference quotient at $q = 200$ and find its limit as $h \rightarrow 0$.
83. Net programme benefit is $B(x) = 50x - x^2$. Simplify the difference quotient at $x = 10$ and find its limit as $h \rightarrow 0$.
84. Household income is $I(t) = 3000 + 120t - 3t^2$. Simplify the difference quotient at $t = 10$ and find its limit as $h \rightarrow 0$.
85. Survey responses are $S(n) = 200 + 15n + 0.5n^2$. Simplify the difference quotient at $n = 20$ and find its limit as $h \rightarrow 0$.
86. For the cost model in Exercise 81, calculate the forward quotient for $h = 1, 0.1$ and 0.01 . Explain how the results support the limiting value.
87. If $P(t)$ is measured in people and t in years, state the units and interpretation of

$$\lim_{h \rightarrow 0} \frac{P(t+h) - P(t)}{h}.$$

88. Evaluate the symmetric difference quotient

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - (3-h)^2}{2h}.$$

89. For $f(x) = |x|$, compare the left- and right-hand limits of $[f(h) - f(0)]/h$ as $h \rightarrow 0$. Explain what the result says about an instantaneous rate at the kink.
90. A continuous but kinked cost rule is

$$C(q) = \begin{cases} 10q, & q \leq 100, \\ 1000 + 15(q - 100), & q > 100. \end{cases}$$

Find the left- and right-hand difference-quotient limits at $q = 100$ and explain why no single instantaneous marginal cost exists there.

Applications: model evaluation and responsible interpretation

91. Explain why $\lim_{t \rightarrow \infty} A(t) = L$ does not necessarily mean that $A(t) = L$ at any finite time.
92. Explain why continuity of a fitted formula does not prove that the underlying observed social process changes continuously.
93. Give four reasons why a long-run limit estimated from a short data series may be unreliable.
94. Distinguish a horizontal asymptote from a legal cap, physical bound or guaranteed policy target.
95. A fitted model has a jump at a policy date. Give three checks for deciding whether the jump reflects the actual rule, a data problem or an unsuitable functional form.
96. Explain why rounded table values that appear to converge to L are insufficient by themselves to establish that the exact limit is L .
97. A congestion model tends to infinity as utilisation approaches capacity. Give three reasons why its numerical predictions may become unreliable very close to the asymptote.
98. State four conclusions that the Intermediate Value Theorem does not provide even when all its hypotheses are satisfied.
99. A bisection calculation uses estimated values subject to sampling error. Explain how uncertainty in signs or continuity affects the claimed root interval.
100. Give a six-stage checklist for calculating, checking and communicating a limit-based social-science result.

Chapter 11

Differential Calculus

Chapter guide

Build the derivative from rates of change, practise the differentiation rules, and then apply derivatives to marginal analysis, approximation, optimisation and elasticity.

Selected background sources: (Strang & Herman, 2016a).

Differentiation measures instantaneous change. It converts a function that describes the level of a quantity into a new function that describes how rapidly that quantity is changing. The derivative may therefore represent a slope, growth rate, marginal effect or local sensitivity, depending on the context and units of the variables.

In the social sciences, derivatives are used to study marginal cost and revenue, the response of demand to price, the changing rate of programme adoption, population growth and the optimisation of welfare or profit. A derivative does not by itself establish causation: it describes the local change predicted by the stated function or model. Its practical meaning depends on the variables, units, feasible domain and assumptions used.

This chapter begins with the derivative as a limit of average rates of change. It then develops the main differentiation rules, higher derivatives, linear approximation, marginal analysis, optimisation and elasticity. The emphasis throughout is on interpreting a derivative as well as calculating it.

11.1 Average and instantaneous rates of change

The average rate of change of f from $x = a$ to $x = b$ is

$$\frac{f(b) - f(a)}{b - a}.$$

The derivative is the limiting value of this quotient over a shrinking interval:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}.$$

Geometrically, $f'(a)$ is the slope of the tangent to $y = f(x)$ at $(a, f(a))$.

If x is measured in years and $f(x)$ in percentage points, then $f'(x)$ is measured in percentage points per year. If total cost is measured in RM and output in units, then the derivative has units RM per unit. A derivative without its units and point of evaluation is often incomplete as an applied answer.

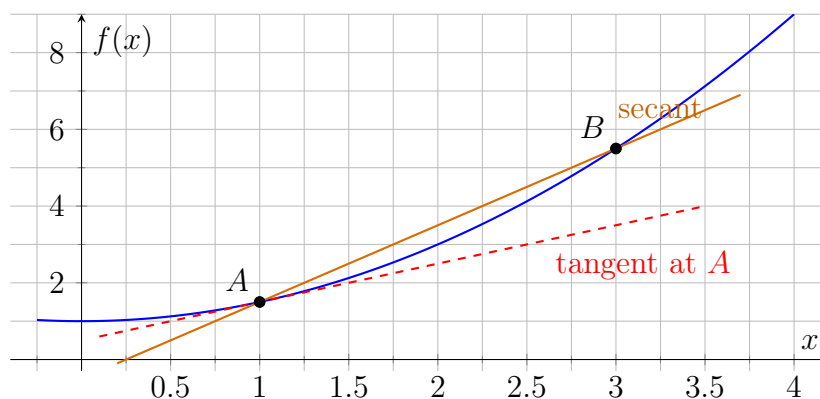


Figure 11.1: The secant slope gives an average rate over an interval. As the second point moves towards A , the secant approaches the tangent and its slope approaches the derivative.

Example 11.1: [

Deriving a rule from the limit definition Let $f(x) = x^2 + 3x$. Then

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h} \\
 &= \lim_{h \rightarrow 0} (2x + h + 3) \\
 &= \boxed{2x + 3}.
 \end{aligned}$$

The cancellation is valid for $h \neq 0$ during the limiting process. The limit is taken only after the quotient has been simplified.

11.2 Basic differentiation rules

For constants c and real exponents n ,

$$\frac{d}{dx}(c) = 0, \quad \frac{d}{dx}(x^n) = nx^{n-1}.$$

Derivatives distribute over sums and constant multiples. Negative and fractional powers should first be written using index notation. The tangent at $x = a$ has equation

$$y - f(a) = f'(a)(x - a).$$

A horizontal tangent occurs wherever $f'(x) = 0$.

If $f'(a) \neq 0$, the normal line at $x = a$ is perpendicular to the tangent and has slope $-1/f'(a)$. Thus its equation is

$$y - f(a) = -\frac{1}{f'(a)}(x - a).$$

The exponential and logarithmic rules used in many growth models are

$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}, \quad x > 0.$$

Combined with the chain rule, these give

$$\frac{d}{dx}e^{u(x)} = u'(x)e^{u(x)}, \quad \frac{d}{dx}\ln u(x) = \frac{u'(x)}{u(x)}$$

where the expressions are defined.

Three further rules are needed for many applied models:

$$\begin{aligned} \frac{d}{dx}[u(x)v(x)] &= u'(x)v(x) + u(x)v'(x), \\ \frac{d}{dx}\left[\frac{u(x)}{v(x)}\right] &= \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}, \\ \frac{d}{dx}f(g(x)) &= f'(g(x))g'(x). \end{aligned}$$

These are the product, quotient and chain rules, respectively. The chain rule is especially important when a response depends on an intermediate quantity, such as consumption depending on income that itself changes with time.

Proof sketch. For the product rule, add and subtract $u(x+h)v(x)$ in the difference quotient:

$$\begin{aligned} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} &= u(x+h)\frac{v(x+h) - v(x)}{h} \\ &\quad + v(x)\frac{u(x+h) - u(x)}{h}. \end{aligned}$$

Differentiability implies continuity, so $u(x+h) \rightarrow u(x)$. Taking the limit as $h \rightarrow 0$ gives $u(x)v'(x) + v(x)u'(x)$. The quotient rule follows by applying the product rule to $u(x)[v(x)]^{-1}$, where $v(x) \neq 0$. $\square$

Structure	Derivative
$cu(x) + v(x)$	$cu'(x) + v'(x)$
$u(x)v(x)$	$u'(x)v(x) + u(x)v'(x)$
$u(x)/v(x)$	$[u'(x)v(x) - u(x)v'(x)]/[v(x)]^2$
$[u(x)]^n$	$n[u(x)]^{n-1}u'(x)$
$e^{u(x)}$	$u'(x)e^{u(x)}$
$\ln u(x)$	$u'(x)/u(x)$

Example 11.2: [

cyan]Product and chain rules in a growth model Suppose the number of monthly interactions with a public information service is modelled by

$$I(t) = (100 + 5t)e^{0.02t}.$$

Using the product rule,

$$\begin{aligned} I'(t) &= 5e^{0.02t} + (100 + 5t)(0.02)e^{0.02t} \\ &= (7 + 0.1t)e^{0.02t}. \end{aligned}$$

At $t = 10$,

$$I'(10) = 8e^{0.2} \approx \boxed{9.77}$$

interactions per month per additional month. This is the modelled local rate, not the number of interactions itself.

Common differentiation errors include applying the power rule to a product without using the product rule, omitting the inner derivative in the chain rule, and reversing the numerator terms in the quotient rule. Substitution should normally be performed after differentiating so that the derivative function remains visible and can be checked.

11.3 Marginal quantities

Example 11.3: [

blue]Marginal interpretation If $C(x)$ is total cost, then $C'(x)$ is marginal cost: the approximate change in cost caused by producing one additional unit near output x . For small h ,

$$C(x + h) \approx C(x) + hC'(x).$$

If $R(q)$ is total revenue and $C(q)$ is total cost, then

$$\text{MR}(q) = R'(q), \quad \text{MC}(q) = C'(q),$$

and profit satisfies

$$\Pi(q) = R(q) - C(q), \quad \Pi'(q) = \text{MR}(q) - \text{MC}(q).$$

Thus an interior stationary profit level occurs where marginal revenue equals marginal cost. This equality identifies a candidate; feasibility and the maximum test must still be checked.

When output is restricted to whole units, a derivative is an approximation to the

change caused by one additional unit. The exact change is

$$C(q+1) - C(q),$$

which is generally close to $C'(q)$ when the cost function varies smoothly.

Example 11.4: [

orange]Marginal cost and an exact one-unit change Let

$$C(q) = 800 + 18q + 0.04q^2.$$

Then

$$C'(q) = 18 + 0.08q, \quad C'(100) = \boxed{26}.$$

The exact increase from 100 to 101 units is

$$C(101) - C(100) = \boxed{26.04}.$$

The derivative gives a close local estimate and has units RM per unit.

11.4 Higher derivatives and concavity

The derivative $f'(x)$ is itself a function and may be differentiated again. The second derivative

$$f''(x) = \frac{d}{dx}f'(x)$$

measures how the first derivative changes. In a time model, it describes the change in the growth rate; in motion, it is acceleration.

A graph is *concave upward* where $f''(x) > 0$ and *concave downward* where $f''(x) < 0$. A point of inflexion is a point where the concavity changes. The equation $f''(x) = 0$ provides possible inflexion points, but a sign change must be checked.

Example 11.5: [

violet]Growth that continues but slows Suppose programme adoption is modelled by

$$A(t) = 80 - 60e^{-0.4t}, \quad t \geq 0.$$

Then

$$A'(t) = 24e^{-0.4t} > 0, \quad A''(t) = -9.6e^{-0.4t} < 0.$$

Adoption is increasing, but it is increasing at a decreasing rate. At $t = 5$,

$$A'(5) = 24e^{-2} \approx \boxed{3.25}$$

percentage points per year. The concave-down shape is consistent with an approach towards a saturation level.

11.5 Linear approximation

Near $x = a$, a differentiable function is approximated by its tangent line:

$$f(x) \approx f(a) + f'(a)(x - a).$$

The right-hand side is called the *linearisation* of f at a . The approximation is usually more accurate for inputs closer to a and for functions with less curvature in the relevant interval.

Example 11.6: [

cyan]Estimating a survey-response index Suppose a response index is modelled by

$$S(t) = 40 + 15 \ln(t + 1).$$

At $t = 3$,

$$S(3) = 40 + 15 \ln 4, \quad S'(3) = \frac{15}{4} = 3.75.$$

The linearisation at $t = 3$ is

$$L(t) = 40 + 15 \ln 4 + 3.75(t - 3).$$

Hence,

$$S(3.2) \approx L(3.2) = 40 + 15 \ln 4 + 0.75 \approx \boxed{61.54}.$$

The estimate uses the local trend at $t = 3$ rather than recalculating the full non-linear model.

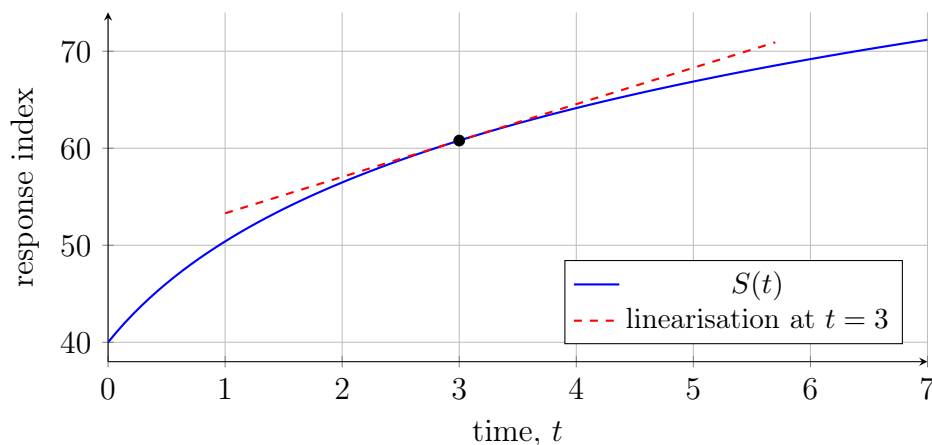


Figure 11.2: A tangent-line approximation is accurate locally but gradually separates from the curved model farther from the point of contact.

11.6 Motion

If $s(t)$ is position, velocity is $v(t) = s'(t)$. Average velocity over an interval uses the ordinary difference quotient, while instantaneous velocity uses the derivative. Acceleration is

$$a(t) = v'(t) = s''(t).$$

Positive velocity means movement in the positive direction and negative velocity means movement in the opposite direction. Speed is $|v(t)|$, so an object can have increasing speed even when its velocity is negative.

The same notation applies more broadly. If $P(t)$ is a population, then $P'(t)$ is its instantaneous net growth in people per unit time, while $P''(t)$ describes whether that growth rate is rising or falling.

11.7 Increasing, decreasing and optimisation

A differentiable function is increasing where $f'(x) > 0$ and decreasing where $f'(x) < 0$. A stationary point occurs where $f'(x) = 0$. The sign of f' on either side can classify the point, while the second derivative gives the useful test

$$f''(a) > 0 \Rightarrow \text{local minimum}, \quad f''(a) < 0 \Rightarrow \text{local maximum}.$$

In an economic model, maximum profit occurs where marginal profit is zero, equivalently where marginal revenue equals marginal cost, provided the point is feasible and actually gives a maximum.

For optimisation on a closed interval, compare the value of the objective function at every interior stationary point and at both endpoints. A stationary point outside the feasible domain must be rejected. The first derivative sign test remains valid even when the second derivative is zero or inconclusive:

- a change from $f' > 0$ to $f' < 0$ gives a local maximum;
- a change from $f' < 0$ to $f' > 0$ gives a local minimum;
- no sign change means the stationary point is neither a local maximum nor a local minimum.

Example 11.7: [

orange]Maximising profit Suppose demand is $p = 80 - 0.2q$ and total cost is $C(q) = 20q + 500$. Revenue and profit are

$$R(q) = 80q - 0.2q^2, \quad \Pi(q) = 60q - 0.2q^2 - 500.$$

Since $\Pi'(q) = 60 - 0.4q$, the stationary output is $q = 150$. Moreover, $\Pi''(q) = -0.4 < 0$, so this output maximises profit. The corresponding price is 50 and the maximum modelled profit is 4000 monetary units.

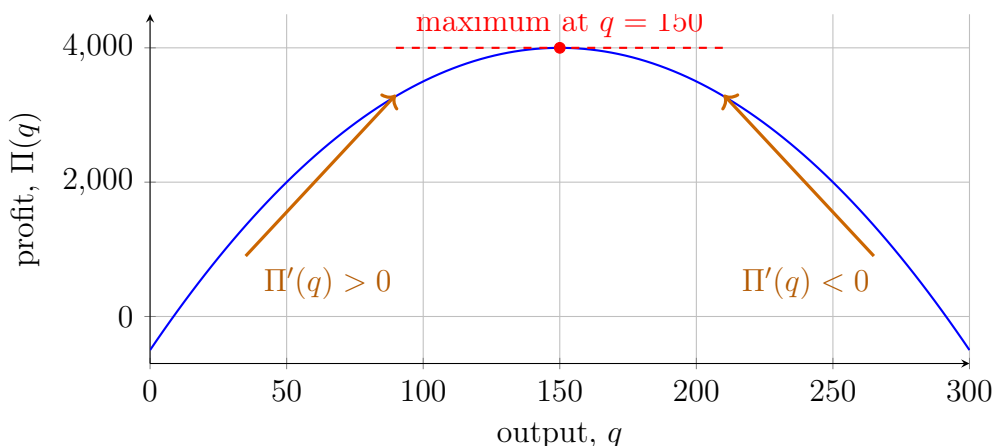


Figure 11.3: Profit rises while marginal profit is positive and falls after marginal profit becomes negative.

11.8 Elasticity of demand

If quantity demanded q is a differentiable function of price p , the point price elasticity of demand is

$$E(p) = -\frac{p}{q(p)}q'(p).$$

Demand is called elastic when $E > 1$, unit elastic when $E = 1$ and inelastic when $0 < E < 1$. Elasticity is dimensionless and estimates the percentage change in quantity associated with a one-percent change in price, locally.

For an ordinary downward-sloping demand function, $q'(p) < 0$, so the minus sign makes elasticity positive. Elasticity compares proportional rather than absolute change. This allows responsiveness to be compared across products whose prices and quantities use different units or scales.

For $q(p) = 500 - 5p$, at $p = 60$ the quantity is 200 and

$$E(60) = -\frac{60}{200}(-5) = 1.5.$$

Thus demand is elastic at that price.

Elasticity also connects price changes with revenue. Since

$$R(p) = pq(p),$$

the product rule gives

$$R'(p) = q(p) + pq'(p) = q(p)(1 - E(p)).$$

Provided $q(p) > 0$:

- if $E < 1$, increasing price raises revenue locally;
- if $E > 1$, increasing price lowers revenue locally;

- if $E = 1$, revenue is stationary with respect to price.

Example 11.8: [

violet]Elasticity under exponential demand Let

$$q(p) = 1000e^{-0.02p}.$$

Then $q'(p) = -20e^{-0.02p}$, and

$$E(p) = -\frac{p}{1000e^{-0.02p}}(-20e^{-0.02p}) = 0.02p.$$

Demand is unit elastic at $p = 50$. The revenue function

$$R(p) = 1000pe^{-0.02p}$$

has derivative

$$R'(p) = 1000e^{-0.02p}(1 - 0.02p),$$

so revenue is maximised at $p = 50$ within the positive-price domain.

11.9 Interpreting derivative-based models

A derivative is a property of a mathematical function. When that function is used as a social-science model, the derivative inherits all of the model's assumptions and limitations. A careful interpretation should address the following questions:

- What do the independent and dependent variables represent?
- What are the units of the derivative?
- At which input is the derivative evaluated?
- Is the input inside the feasible and observed domain?
- Is the derivative an exact continuous rate or an approximation for a discrete change?
- Which other influences are being held constant by the model?

A positive derivative indicates association within the stated model, not necessarily a causal effect in observed data. Similarly, an optimising value is conditional on the objective function and constraints: a profit maximum, for example, need not maximise employment, access, fairness or social welfare.

11.10 Exercises

Answers are provided in section C.11.

1. Use the limit definition to find $f'(x)$.

(a) $f(x) = 3 - 5x$

(b) $f(x) = \sqrt{3x}$

(c) $f(x) = 3/x$

(d) $f(x) = 2x^2 - 5x + 2$

2. Differentiate with respect to x .

(a) $y = \frac{1}{4x^5} + 3x^2 - \sqrt{2x}$

(b) $y = \frac{7x^2 - 5x}{3x^5}$

(c) $y = \frac{x^3 + 3}{\sqrt{x}}$

(d) $y = \frac{5}{x} - x^{5/3} + \frac{5}{3}x^2$

3. Differentiate with respect to x .

(a) $f(x) = x^{-2} + 7$

(b) $f(x) = 4x^3 - \frac{1}{3\sqrt{x}} + 2$

(c) $f(x) = 2x^{3/4} + \frac{3}{x^4}$

(d) $f(x) = \frac{8x - 2x}{4x^2 - 5x^3}$

4. Revenue from selling x infant car seats is $R(x) = 60x - 0.025x^2$, $0 \leq x \leq 2400$.

(a) Find the change and average rate of change in revenue when output rises from 1000 to 1050.

(b) Find $R'(x)$ from the limit definition.

(c) Find $R(1000)$ and $R'(1000)$ and interpret both values.

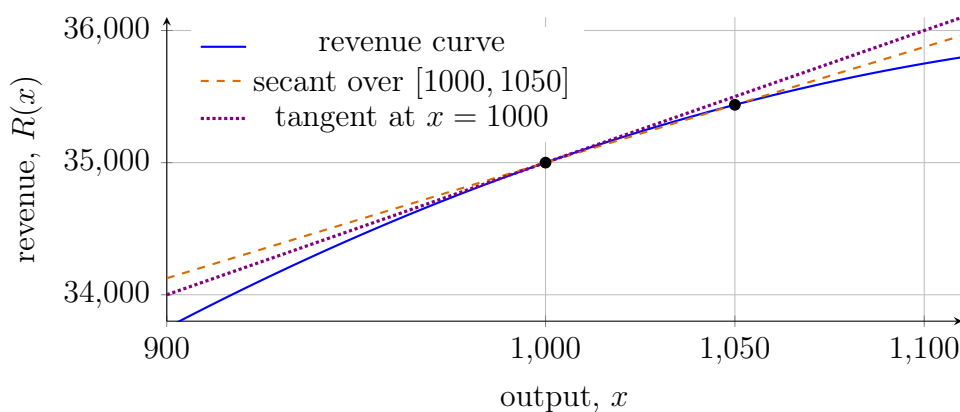


Figure 11.4: The secant and tangent interpretations used in Exercise 4. Their slopes represent average and instantaneous rates of change in revenue.

5. Total sales, in millions of dollars, are $S(t) = \sqrt{t} + 8$, where t is measured in months.
 - (a) Find $S'(t)$ from the limit definition.
 - (b) Find and interpret $S(9)$ and $S'(9)$.
 - (c) Use linear approximation at $t = 9$ to estimate sales at $t = 10$ and $t = 11$.
6. An object's position is $s(t) = t^2 + t$ metres.
 - (a) Find its average velocity from $t = 1$ to $t = 3$.
 - (b) Find its average velocity from $t = 1$ to $t = 1 + h$.
 - (c) Find its instantaneous velocity at $t = 1$.
7. An object's position is $s(t) = t^3 - 15t^2 + 72t$ feet.
 - (a) Find the velocity function.
 - (b) Find the velocity at $t = 2$ and $t = 5$.
 - (c) Find when the velocity is zero.
8. Let $k(x) = x^4 - 8x^3 + 7$.
 - (a) Find $k'(x)$.
 - (b) Find the tangent equation at $x = 1$.
 - (c) Find all values of x where the tangent is horizontal.
9. Differentiate.
 - (a) $y = (x^2 + 1)e^x$
 - (b) $y = (3x - 1)/(x^2 + 1)$
 - (c) $y = (2x + 5)^4$
10. A firm faces demand $p = 80 - 0.2q$ and has total cost $C(q) = 20q + 500$. Find the output and price that maximise profit, and calculate the maximum profit.

11. For the demand function $q(p) = 500 - 5p$, calculate the price elasticity of demand at $p = 60$. Classify demand and interpret the result in percentage terms.
12. Differentiate each function, simplifying where appropriate.
- (a) $y = (3x^2 - 1)e^{2x}$
 - (b) $y = \ln(2x^2 + 5)$
 - (c) $y = e^{\sqrt{x}}, x > 0$
 - (d) $y = (x^2 + 1)/(x - 2)$
 - (e) $y = (5 - 3x)^{-2}$
13. Let $f(x) = x^3 - 2x + 1$.
- (a) Find the point on the curve where $x = 1$.
 - (b) Find the equation of the tangent at that point.
 - (c) Find the equation of the normal at that point.
14. Programme adoption is modelled by $A(t) = 80 - 60e^{-0.4t}$ for $t \geq 0$.
- (a) Find $A'(t)$ and $A''(t)$.
 - (b) Find $A'(0)$ and $A'(5)$.
 - (c) State where the model is increasing and describe its concavity.
 - (d) Explain how the signs of the derivatives are consistent with movement towards a saturation level.

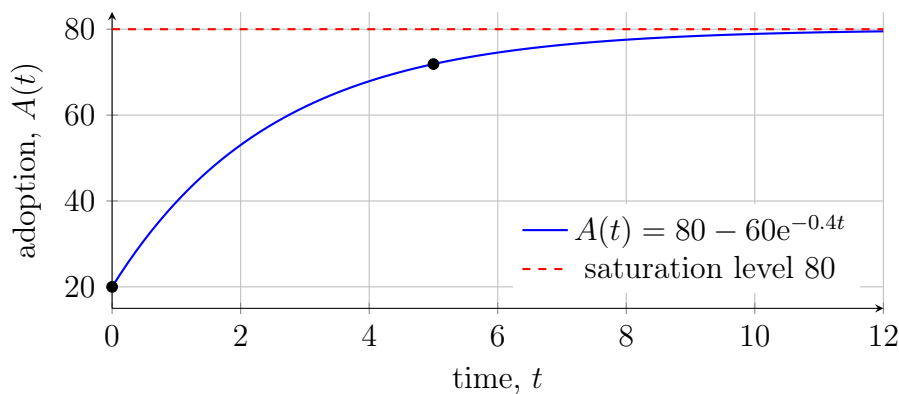


Figure 11.5: The adoption model in Exercise 14 increases towards a saturation level while its slope gradually decreases.

15. For $f(x) = x^3 - 6x^2 + 9x + 4$:
- (a) find the stationary points and classify them;
 - (b) state the intervals on which f is increasing and decreasing;
 - (c) find the point of inflexion and state the concavity intervals.

16. A survey-response index is $G(t) = 50 + 20 \ln(t + 1)$.

- Find the linearisation of G at $t = 4$.
- Use it to estimate $G(4.1)$.
- Explain why the approximation should not automatically be trusted for a value of t far from 4.

17. A firm's total cost and revenue are

$$C(q) = 1200 + 25q + 0.05q^2, \quad R(q) = 90q - 0.1q^2.$$

- Find marginal cost and marginal revenue.
- Obtain the profit function and marginal profit.
- Find the continuous output level that maximises profit and verify that it is a maximum.
- Calculate the maximum modelled profit.

18. Total cost is $C(q) = 500 + 12q + 0.02q^2$ for $q > 0$.

- Find the average-cost function $\bar{C}(q) = C(q)/q$.
- Find the output that minimises average cost.
- Show that marginal cost equals average cost at this output.

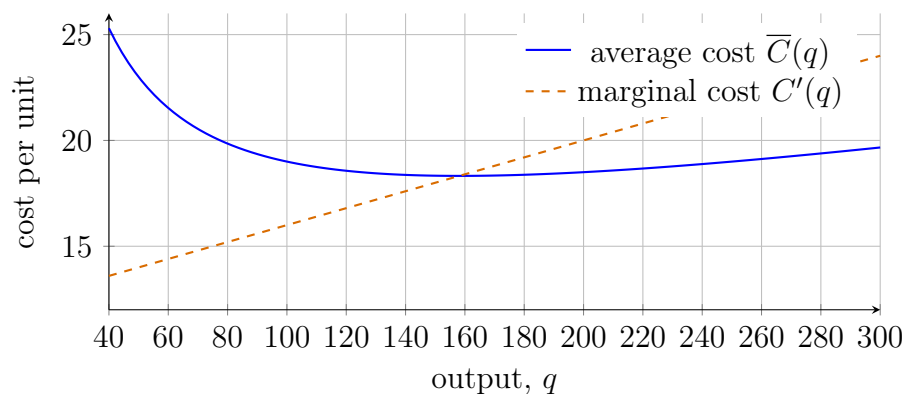


Figure 11.6: Average and marginal cost for Exercise 18. The minimum of the average-cost curve occurs where marginal cost crosses it.

19. The gross benefit and implementation cost of a programme are

$$B(x) = 120x - 2x^2, \quad C(x) = 20x + 200,$$

where $0 \leq x \leq 50$ is the programme scale.

- Form the net-benefit function.
- Find the scale that maximises net benefit.

- (c) Calculate the maximum net benefit and compare it with the endpoint values.
- 20.** A policy-performance score is modelled by $W(x) = -x^2 + 12x + 5$ on $0 \leq x \leq 10$. Find the absolute maximum and minimum values on the stated domain, including the inputs at which they occur.
- 21.** Demand is $q(p) = 1000e^{-0.02p}$.
- Find $q'(p)$ and the elasticity function $E(p)$.
 - Classify demand at $p = 30$ and at $p = 80$.
 - Find the unit-elastic price.
 - Use differentiation to show that revenue is maximised at this price.

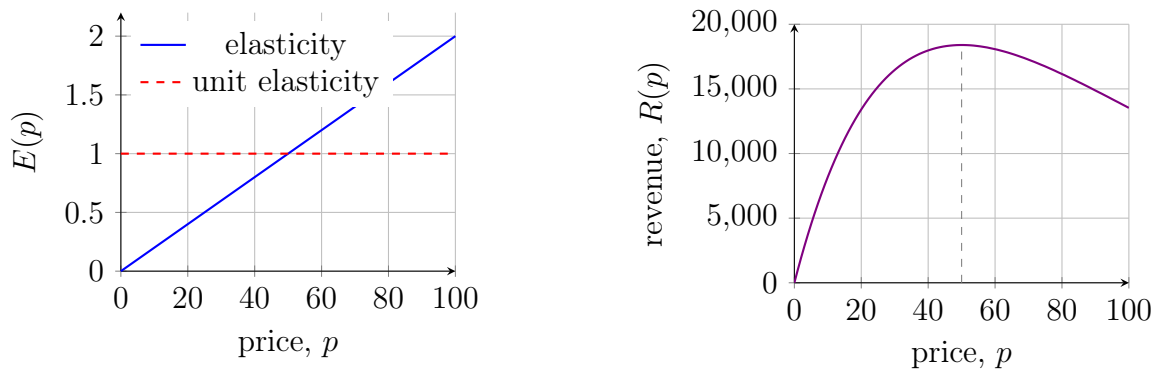


Figure 11.7: Elasticity and revenue for Exercise 21. The unit-elastic price corresponds to the turning point of the revenue curve.

- 22.** For the linear demand function $q(p) = 600 - 3p$:
- state the economically meaningful price domain;
 - find the elasticity function;
 - find the unit-elastic price;
 - find the price, quantity and revenue at which revenue is maximised.
- 23.** Income is modelled by $I(t) = 2000 + 150t - 3t^2$, and consumption is $C = 100 + 0.8I$.
- Find dI/dt .
 - Use the chain rule to find dC/dt .
 - Evaluate both rates at $t = 10$ and interpret the consumption rate.
- 24.** A differentiable function satisfies $f'(x) = (x + 1)(x - 2)$.
- Construct a sign chart for f' .
 - State the intervals on which f is increasing and decreasing.
 - Classify the stationary points at $x = -1$ and $x = 2$.

25. A survey records the following approval-index values near $t = 5$ years:

t	4.9	5.0	5.1
$A(t)$	62.80	63.10	63.38

- (a) Estimate $A'(5)$ using the centred difference based on $t = 4.9$ and $t = 5.1$.
 (b) Calculate the corresponding backward and forward estimates.
 (c) Interpret the centred estimate and explain why the three estimates need not agree exactly.
- 26.** A fitted model has $M'(x) > 0$ throughout the observed data range. A researcher concludes that increasing x causes M to increase and that x should be made as large as possible, even beyond the observed range. Give three reasons why these conclusions do not follow from the derivative alone.

Additional practice: grouped skills review

27. Differentiate $y = x^n$ for $n = 2, 3, 4, 5, 6, 7, 8$ and 9 .

28. Differentiate $y = ax^2 + bx - 4$ for each case.

Case	A	B	C	D	E	F	G	H
a	2	3	4	5	6	7	8	9
b	1	4	7	10	13	16	19	22

29. Differentiate $y = x^n e^x$ for $n = 1, 2, 3, 4, 5, 6, 7$ and 8 .

30. For $k = 2, 3, 4, 5, 6, 7, 8$ and 9 , differentiate

$$y = \frac{kx + k - 1}{x + k + 1}.$$

31. For $k = 2, 3, 4, 5, 6, 7, 8$ and 9 , differentiate $y = (kx + k - 1)^{k+1}$.

32. For $k = 2, 3, 4, 5, 6, 7, 8$ and 9 , differentiate $y = \ln(kx + k - 1)$.

33. For $k = 2, 3, 4, 5, 6, 7, 8$ and 9 , find the tangent to

$$y = x^2 + (k - 2)x$$

at $x = k - 1$.

34. For each total-cost model $C(q) = F + bq + cq^2$, find and interpret $C'(10)$.

Case	A	B	C	D	E	F	G	H
F	500	600	700	800	900	1000	1100	1200
b	20	21	22	23	24	25	26	27
c	0.10	0.12	0.14	0.16	0.18	0.20	0.22	0.24

35. Find the output that maximises each profit function and calculate the maximum profit.

Case	$\Pi(q)$	Case	$\Pi(q)$
A	$-q^2 + 20q - 50$	E	$-q^2 + 28q - 90$
B	$-q^2 + 22q - 60$	F	$-q^2 + 30q - 100$
C	$-q^2 + 24q - 70$	G	$-q^2 + 32q - 110$
D	$-q^2 + 26q - 80$	H	$-q^2 + 34q - 120$

36. Demand is $q = 500 - 5p$. Find and classify point elasticity at the prices

$$p = 40, 45, 50, 55, 60, 65, 70 \quad \text{and} \quad 75.$$

Applications: rates, tangents and local change

37. An approval index is $A(t) = 40 + 12\sqrt{t}$ for $t > 0$. Find $A'(t)$ and interpret $A'(9)$, including its units.
38. Cumulative public spending is $S(t) = 2t^3 - 15t^2 + 36t + 100$ million ringgit. Find the marginal spending rates at $t = 2$ and $t = 5$ and interpret their signs.
39. Revenue is $R(q) = 80q - 0.05q^2$. Find the change and average rate of change as output rises from 500 to 520, then compare the average rate with $R'(500)$.
40. A public-confidence index is $G(x) = 50 + 10\ln(x + 1)$. Find the tangent at $x = 4$ and use it to estimate $G(4.2)$.
41. Find the tangent and normal to $f(x) = x^2 - 4x + 7$ at $x = 3$.
42. Income is $Y(t) = 3000 + 100t - 2t^2$ and consumption is $C = 200 + 0.75Y$. Find dY/dt and dC/dt , then evaluate both at $t = 10$.
43. A population model is $P(t) = 100\,000e^{0.03t}$. Find $P'(t)$ and interpret $P'(5)$.
44. Survey non-response follows $N(t) = 500e^{-0.2t}$. Find $N'(t)$ and interpret $N'(4)$.
45. Programme benefit is $B(x) = 60x/(x + 10)$. Find $B'(x)$ and interpret $B'(20)$ as a marginal effect.
46. For $C(q) = 1000 + 15q + 0.03q^2$, compare $C'(100)$ with the exact one-unit change $C(101) - C(100)$.

Applications: marginal economics and social programmes

47. Total cost is $C(q) = 2000 + 30q + 0.05q^2$. Find and interpret marginal cost at $q = 100$.
48. Total revenue is $R(q) = 120q - 0.2q^2$. Find and interpret marginal revenue at $q = 200$.
49. A firm has $R(q) = 100q - 0.1q^2$ and $C(q) = 1000 + 20q + 0.05q^2$. Find the continuous output that maximises profit and calculate the maximum profit.
50. Tax revenue is modelled by $T(r) = 500r - 20r^2$, where r is a tax rate measured in percentage points. Find the rate that maximises revenue and the maximum modelled revenue.
51. Gross programme benefit is $B(x) = 150x - 3x^2$ and cost is $C(x) = 30x + 300$. Find the scale that maximises net benefit and the maximum net benefit.
52. For $C(q) = 500 + 12q + 0.02q^2$, compare marginal cost at $q = 200$ with the exact increase from 200 to 201 units.
53. Total cost is $C(q) = 1000 + 10q + 0.04q^2$. Find the output that minimises average cost, the minimum average cost, and verify that marginal cost equals average cost there.
54. Output from labour is $Q(L) = 50L - 0.5L^2$. Find the marginal product at $L = 20$ and the labour input that maximises total output.
55. Consumption is $C(Y) = 500 + 0.8Y - 0.00001Y^2$. Find the marginal propensity to consume and evaluate it at $Y = 20\,000$.
56. Cumulative learning benefit is $A(n) = 80(1 - e^{-0.1n})$. Find the marginal benefit $A'(n)$ and evaluate it at $n = 10$. Explain the diminishing-return pattern.

Applications: optimisation and shape

57. A policy-welfare score is $W(x) = -2x^2 + 40x + 100$ on $0 \leq x \leq 15$. Find its absolute maximum and minimum.
58. Programme benefit and cost are $B(x) = 100x - x^2$ and $C(x) = 20x + 500$ on $0 \leq x \leq 60$. Find the scale that maximises net benefit and compare the maximum with both endpoint values.
59. Administrative cost is $C(s) = s^2 - 40s + 900$ for $10 \leq s \leq 35$ staff. Find its absolute minimum and maximum.
60. For $f(x) = x^3 - 6x^2 + 9x + 2$, find and classify the stationary points, state the increasing and decreasing intervals, and find the point of inflexion. Use figure 11.8 to check that the classifications agree with the shape of the graph.

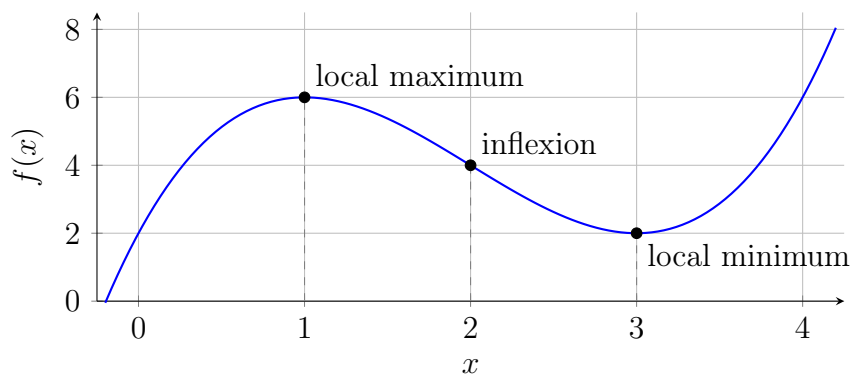


Figure 11.8: The cubic model in Exercise 60. Stationary points concern changes in monotonicity, while the inflexion point concerns a change in concavity.

61. Profit is $P(q) = -q^2 + 30q - 80$ on $0 \leq q \leq 20$. Find the absolute maximum and minimum values.
62. A programme's adoption rate is $A'(t) = 12e^{-0.3t}$. Find when the rate first falls below 2 percentage points per year.
63. Demand is $q = 800 - 4p$. Find the price and quantity that maximise revenue, and calculate maximum revenue.
64. A firm faces $p = 100 - 0.25q$ and cost $C(q) = 20q + 1000$. Find the profit-maximising output, price and profit.
65. Social loss is $L(x) = (x - 6)^2 + 40$ on $0 \leq x \leq 10$. Find its absolute minimum and maximum.
66. Output must be a whole number and profit is $\Pi(q) = -2q^2 + 50q - 100$. Find the continuous optimum and determine the best integer output or outputs.

Applications: elasticity and response

67. For $q(p) = 600 - 3p$, find the elasticity function, classify demand at $p = 50$ and $p = 150$, and find the unit-elastic price.
68. For $q(p) = 2000e^{-0.01p}$, find the elasticity function, classify demand at $p = 60$ and $p = 140$, and find the unit-elastic price.
69. Demand is $q(p) = 500p^{-1.5}$. Find its price elasticity and explain why the classification is the same at every price.
70. For $q(p) = 1000e^{-0.02p}$, find the price, quantity and revenue at which revenue is maximised.
71. For $q(p) = 500 - 5p$, find the revenue-maximising price, quantity and revenue using elasticity or differentiation.

72. Demand is $q(p) = 1000/(p + 10)$. Find the elasticity function and classify demand at $p = 40$. Does the model reach unit elasticity at a finite non-negative price?
73. Enrolment at fee f is $N(f) = 2000e^{-0.005f}$. Find $N'(f)/N(f)$ and interpret the result as an approximate percentage response to a one-ringgit fee increase.
74. Sales response to advertising is $S(a) = 100 + 40 \ln(a + 1)$. Find the elasticity $E(a) = aS'(a)/S(a)$ and evaluate it at $a = 9$.
75. Revenue from a tax rate is $T(r) = r(100 - r)$. Find the signed elasticity $\varepsilon_T = rT'(r)/T(r)$ at $r = 20$ and $r = 60$ and interpret the change of sign using figure 11.9.

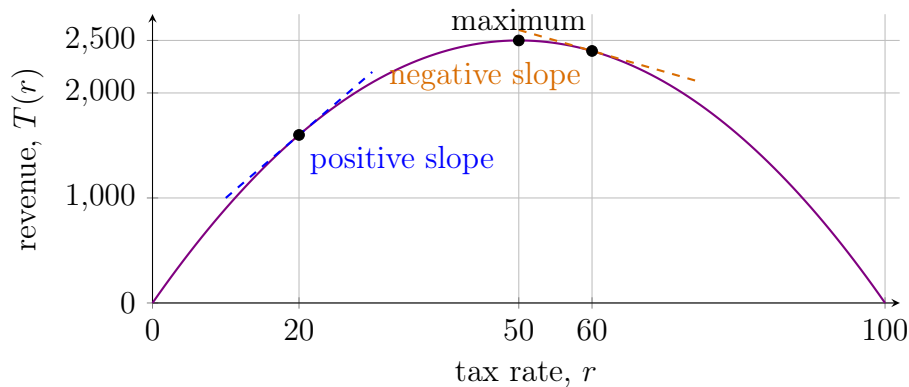


Figure 11.9: Tax revenue in Exercise 75. Before the maximum, a higher rate raises revenue locally; after the maximum, it lowers revenue, producing a negative signed elasticity.

76. Compare $q_1(p) = 100 - 2p$ and $q_2(p) = 1000 - 20p$ at $p = 20$. Calculate both slopes and elasticities, then explain why different slopes can correspond to the same elasticity.

Applications: approximation and derivative estimates

77. For $G(t) = 50 + 20 \ln(t + 1)$, find the linearisation at $t = 4$ and use it to estimate $G(4.1)$. Compare with the model value.
78. Use linear approximation at $q = 100$ for $C(q) = 1000 + 20q + 0.05q^2$ to estimate $C(102)$. Compare with the exact value.
79. Approval data near $t = 3$ are

t	2.9	3.0	3.1
$A(t)$	54.2	55.0	55.7

Calculate backward, forward and centred estimates of $A'(3)$.

80. Recorded costs are

q	99	100	101
$C(q)$	3470.05	3500	3530.05

Find the backward, forward and centred marginal-cost estimates at $q = 100$.

81. Unemployment is modelled by $U(t) = 8e^{-0.1t}$. Use linearisation at $t = 5$ to estimate $U(5.2)$ and compare with the model value.
82. Explain whether the tangent approximation to $G(t) = 50 + 20\ln(t + 1)$ at $t = 4$ overestimates or underestimates nearby values, using concavity.
83. Equally spaced observations are 100, 104, 103 at times 0, 1, 2. Estimate the derivative at time 1 using backward, forward and centred differences. Explain the sensitivity to measurement noise with reference to figure 11.10.

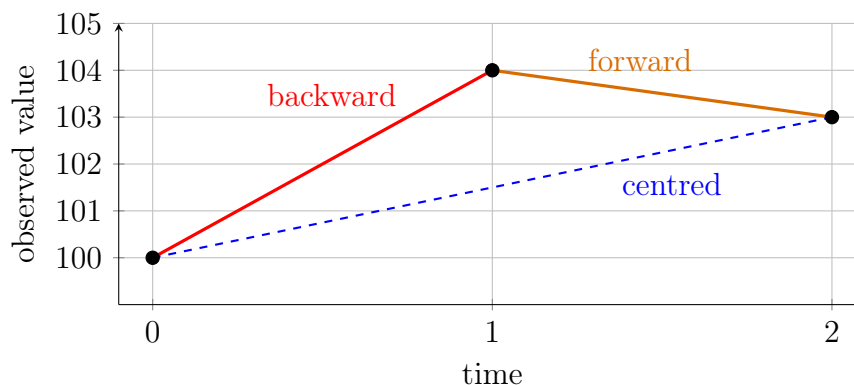


Figure 11.10: The three finite-difference estimates in Exercise 83 use different pairs of observations. A small change in one recorded value can therefore alter the estimated local slope substantially.

84. A fitted approval model is $A(t) = 40 + 5t - 0.2t^2$. Find and interpret $A'(10)$.
85. A participation percentage is $P(t) = 30 + 2t - 0.1t^2$. At $t = 5$, find the change in percentage points per year and the corresponding relative percentage rate.
86. A fitted slope is 3 index points per year, while a centred difference from observed data gives 2.6. Find the discrepancy and give two possible explanations.

Applications: model evaluation and responsible interpretation

87. Explain why $M'(x) > 0$ in a fitted model establishes neither causation nor the desirability of increasing x .
88. Explain why $f'(a) = 0$ does not by itself prove that f has a local maximum or minimum at a .
89. A policy is chosen by maximising total net benefit. Give three reasons why the resulting optimum need not be socially preferred.
90. Distinguish a local maximum from an absolute maximum and state why endpoints matter on a closed feasible interval.
91. Explain why a stationary point outside the feasible domain must be rejected even when the algebra is correct.

-
92. Give two examples showing why the units of a derivative are essential to interpretation.
 93. Explain when a derivative is a useful approximation to a one-unit change and when the exact discrete difference should be preferred.
 94. Give three reasons why an estimated price elasticity may change across periods or population groups.
 95. A reform changes the parameters of a fitted model at a known date. Explain why a piecewise model and separate derivatives may be more appropriate than one equation for the whole period.
 96. A benefit rule has a jump at an eligibility threshold. Explain why the derivative is undefined there and how one-sided changes can still be described.
 97. Explain why a negative second derivative need not mean that the modelled quantity is decreasing.
 98. If income is converted from ringgit to thousands of ringgit, explain how the numerical derivative changes while the substantive relationship remains the same.
 99. Explain how omitted variables can make a partial derivative or fitted marginal effect misleading in a social-science model.
 100. Give a six-stage checklist for calculating, checking and communicating a derivative-based social-science result.

Chapter 12

Integral Calculus

Chapter guide

Develop antiderivatives and integration techniques before interpreting definite integrals as accumulation, area, average value, present value and economic surplus.

Selected background sources: (Strang & Herman, 2016a, 2016b).

Integration reverses differentiation and measures the accumulation of continuously varying quantities. It is used to recover totals from marginal functions, calculate areas, estimate quantities from tabulated data and find average values over intervals. In social-science applications, an integral may represent total cost, cumulative participation, person-time, economic surplus, discounted income or an area used to summarise inequality.

The value of an integral relies not only on an algebraic technique but also on the meaning and units of the integrand. Integrating a rate measured in people per year over time produces a number of people; integrating a price with respect to quantity produces a monetary amount. These interpretations are essential when a result is used outside pure mathematics.

A useful distinction is between a *stock*, measured at a particular time, and a *flow*, measured per unit time. Population, account balance and the number of active programme participants are stocks. Migration per year, spending per month and registrations per day are flows. Integrating a net flow over an interval gives the change in the corresponding stock. The initial stock must still be known if its final level is required.

12.1 Antiderivatives and indefinite integrals

If $F'(x) = f(x)$, then F is an *antiderivative* of f . The indefinite integral represents the family of all antiderivatives:

$$\int f(x) \, dx = F(x) + C.$$

For $n \neq -1$,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C,$$

while

$$\int \frac{1}{x} \, dx = \ln |x| + C, \quad \int e^x \, dx = e^x + C.$$

Integration is linear. For constants a and b ,

$$\int [af(x) + bg(x)] \, dx = a \int f(x) \, dx + b \int g(x) \, dx.$$

Some frequently used rules are summarised below.

Integrand	Antiderivative
1	$x + C$
$x^n, n \neq -1$	$x^{n+1}/(n+1) + C$
$1/x$	$\ln x + C$
$e^{kx}, k \neq 0$	$e^{kx}/k + C$
$g'(x)/g(x)$	$\ln g(x) + C$

The constant C is necessary because every member of the family has the same derivative. An initial value determines a particular antiderivative. An indefinite integral should be checked by differentiating the result. The check is especially valuable after substitution or integration by parts, where a missing constant factor is a common error.

Example 12.1: [

blue]Recovering total cost from marginal cost Suppose marginal cost is

$$C'(q) = 30 + 0.04q$$

and fixed cost is $C(0) = 500$. Integrating gives

$$C(q) = 30q + 0.02q^2 + K.$$

The condition $C(0) = 500$ gives $K = 500$, so

$$C(q) = 500 + 30q + 0.02q^2.$$

The integration constant represents the cost already present when production is zero.

If an antiderivative is required on more than one interval separated by a point outside the domain, the constants need not agree across those intervals. For example, $\int 1/x \, dx = \ln |x| + C$ is considered separately on $(-\infty, 0)$ and $(0, \infty)$.

12.2 Substitution

Substitution reverses the chain rule. If $u = g(x)$ and $du = g'(x) \, dx$, then

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du.$$

Algebraic rearrangement may be needed to express all remaining factors in terms of u .

A useful substitution usually identifies an inner expression whose derivative also appears in the integrand. The main steps are:

1. choose $u = g(x)$;
2. calculate $du = g'(x) \, dx$;
3. rewrite the entire integral in terms of u ;
4. integrate and substitute back, unless the limits were changed.

The derivative of the chosen inner expression need only appear up to a non-zero constant multiple. If $du = 6x \, dx$ but the integral contains $2x \, dx$, then $2x \, dx = du/3$. Every factor involving the original variable must be replaced before integrating in u .

Example 12.2: [

orange]Substitution in an indefinite integral Evaluate

$$\int 2x(x^2 + 3)^4 \, dx.$$

Let $u = x^2 + 3$, so $du = 2x \, dx$. Then

$$\int 2x(x^2 + 3)^4 \, dx = \int u^4 \, du = \frac{u^5}{5} + C.$$

Therefore,

$$\boxed{\int 2x(x^2 + 3)^4 \, dx = \frac{(x^2 + 3)^5}{5} + C}.$$

For a definite integral, the limits may be converted to u -values. If this is done, there is no need to substitute back to x .

Example 12.3: [

violet]Changing the limits under substitution Evaluate

$$\int_0^2 xe^{x^2} \, dx.$$

With $u = x^2$ and $du = 2x \, dx$, the bounds become $u = 0$ and $u = 4$. Hence,

$$\int_0^2 xe^{x^2} \, dx = \frac{1}{2} \int_0^4 e^u \, du = \boxed{\frac{e^4 - 1}{2}}.$$

12.3 Integration by parts

Integration by parts reverses the product rule. From

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx},$$

one obtains

$$\int u \, dv = uv - \int v \, du.$$

It is particularly useful when an algebraic factor is multiplied by an exponential or logarithmic function. The choice of u should usually become simpler when differentiated.

When one factor is logarithmic, it is normally chosen as u . A lone logarithm can be treated as a product with 1.

Example 12.4: [

cyan]Integrating an algebraic–exponential product To evaluate

$$\int x e^{0.2x} \, dx,$$

choose $u = x$ and $dv = e^{0.2x} \, dx$. Then $du = dx$ and $v = 5e^{0.2x}$. Therefore,

$$\begin{aligned} \int x e^{0.2x} \, dx &= 5x e^{0.2x} - \int 5e^{0.2x} \, dx \\ &= \boxed{5x e^{0.2x} - 25e^{0.2x} + C}. \end{aligned}$$

Differentiating the answer provides a direct check.

Example 12.5: [

violet]Integrating a logarithm To evaluate $\int \ln x \, dx$ for $x > 0$, choose

$$u = \ln x, \quad dv = dx.$$

Then $du = dx/x$ and $v = x$, so

$$\int \ln x \, dx = x \ln x - \int 1 \, dx = \boxed{x \ln x - x + C}.$$

This result is useful when averaging logarithmic response or utility models.

12.4 Definite integrals and area

The Fundamental Theorem of Calculus states

$$\int_a^b f(x) \, dx = F(b) - F(a),$$

where $F' = f$. Useful properties include

$$\int_a^a f(x) \, dx = 0, \quad \int_b^a f(x) \, dx = -\int_a^b f(x) \, dx,$$

and, for any c between a and b ,

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

The Fundamental Theorem also connects accumulation and instantaneous rate. If

$$A(x) = \int_a^x f(t) \, dt,$$

then

$$A'(x) = f(x).$$

Thus, accumulating a rate and then differentiating the accumulated quantity returns the original rate.

Proof sketch. Let $A(x) = \int_a^x f(t) \, dt$. For $h \neq 0$,

$$\frac{A(x+h) - A(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

The right-hand side is the average value of f on the short interval between x and $x+h$. If f is continuous at x , these nearby values approach $f(x)$ as $h \rightarrow 0$. Hence $A'(x) = f(x)$. Applying this result to an antiderivative F gives $\int_a^b f(x) \, dx = F(b) - F(a)$. $\square$

12.4.1 Stock–flow relationships

If $S(t)$ is a stock and $r(t)$ is its net flow rate, then

$$S'(t) = r(t) \quad \implies \quad S(b) = S(a) + \int_a^b r(t) \, dt.$$

A positive integral increases the stock, while a negative integral decreases it. When inflow and outflow are given separately,

$$S'(t) = \text{inflow rate} - \text{outflow rate}.$$

The integral of the net rate gives net change, not the total volume of activity in both directions.

Example 12.6: [

cyan]Population change from a net migration rate Suppose a region begins with 50 000 residents and its net migration rate is modelled by

$$m(t) = 1200 - 80t$$

people per year for $0 \leq t \leq 10$. The population change is

$$\int_0^{10} (1200 - 80t) \, dt = 12\,000 - 4000 = 8000.$$

The model therefore gives

$$P(10) = 50\,000 + 8000 = \boxed{58\,000}.$$

This calculation treats $m(t)$ as a net rate; it does not reveal the separate numbers entering and leaving the region.

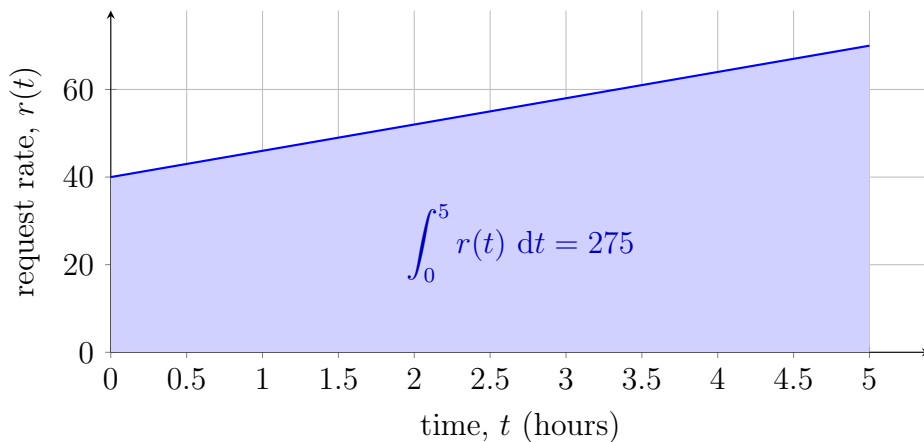


Figure 12.1: The area under a non-negative flow-rate graph represents the accumulated number of requests over the time interval.

12.4.2 Signed area and total geometric area

A definite integral is a signed area. Where $f(x) > 0$, the integral contributes positive area; where $f(x) < 0$, it contributes negative area. The total geometric area between a curve and the x -axis is

$$\int_a^b |f(x)| \, dx,$$

which may require the interval to be divided at the zeros of f .

Example 12.7: [

blue]Signed area compared with total area For $f(x) = x^2 - 4$ on $[-3, 3]$, symmetry gives the signed integral

$$\int_{-3}^3 (x^2 - 4) \, dx = -6.$$

The curve crosses the axis at $x = -2$ and $x = 2$. Its total geometric area is

$$2 \left[\int_0^2 (4 - x^2) \, dx + \int_2^3 (x^2 - 4) \, dx \right] = \boxed{\frac{46}{3}}.$$

The signed integral and total area answer different questions.

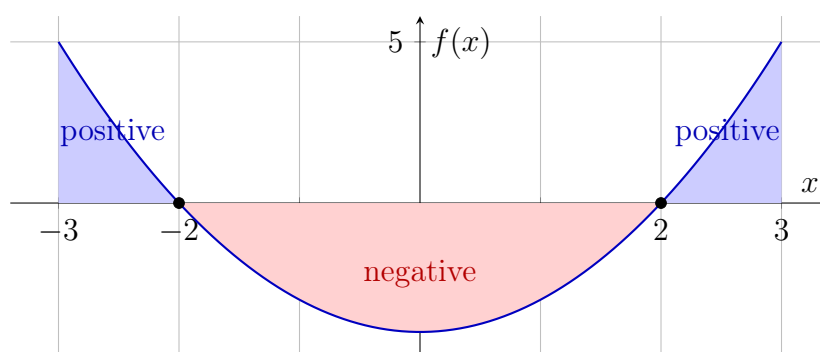


Figure 12.2: A signed integral subtracts the red region from the blue regions, whereas total geometric area adds all three magnitudes.

12.4.3 Area between two curves

If $f(x) \geq g(x)$ on $[a, b]$, then the enclosed area is

$$\int_a^b [f(x) - g(x)] \, dx.$$

The intersection points are found first by solving $f(x) = g(x)$. If the upper curve changes, the integral must be split.

Example 12.8: [

orange]Area between a line and a parabola The curves $y = 2x + 3$ and $y = x^2$ intersect when

$$x^2 = 2x + 3,$$

giving $x = -1$ and $x = 3$. The line lies above the parabola between these points, so the enclosed area is

$$\int_{-1}^3 (2x + 3 - x^2) \, dx = \boxed{\frac{32}{3}}.$$

12.5 Average value and accumulated change

The average value of f on $[a, b]$ is

$$f_{\text{av}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

Its units are the same as the units of the function. The factor $1/(b-a)$ converts the accumulated total into a representative level over the interval.

If f is continuous, the Mean Value Theorem for integrals guarantees at least one $c \in [a, b]$ for which

$$f(c) = f_{\text{av}}.$$

This is an existence statement; it does not say that the average occurs at the midpoint or that it is attained at an observed data point.

Example 12.9: [

violet]Average participation over time Suppose the number of participants, in hundreds, is modelled by

$$H(t) = 20 + 4t - t^2, \quad 0 \leq t \leq 4.$$

The average number over the four-year interval is

$$\frac{1}{4} \int_0^4 (20 + 4t - t^2) \, dt = \frac{68}{3}.$$

Thus, the modelled average is approximately 22.67 hundreds, or about 2267 participants.

If $C'(x)$ is marginal cost, the increase in total cost from output a to b is

$$\int_a^b C'(x) \, dx.$$

More generally, if $r(t)$ is a flow rate, then

$$\int_a^b r(t) \, dt$$

is the net accumulated change over the interval. For example, if service requests arrive at the rate $r(t) = 40 + 6t$ requests per hour, then the number arriving during the first five hours is

$$\int_0^5 (40 + 6t) \, dt = \boxed{275}.$$

If a rate changes sign, its integral gives the *net* accumulated change. The total amount of movement or activity may instead require integrating $|r(t)|$ or treating inflow and

outflow separately.

12.6 Numerical integration and the trapezoidal rule

Observed data are often available only at selected input values, so an exact antiderivative may not be known. The trapezoidal rule approximates the area under the graph by straight-sided trapezia. For equally spaced values $x_0, x_1, \dots, x_n$ with step width h ,

$$\int_{x_0}^{x_n} f(x) \, dx \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})],$$

where $y_i = f(x_i)$. The approximation improves when the intervals are made smaller and the function is reasonably smooth.

Example 12.10: [

cyan]Estimating visits from tabulated rates The observed arrival rates at a centre are

t (hours)	0	1	2	3	4
$r(t)$ (visits per hour)	120	150	180	160	130.

Using $h = 1$, the estimated number of visits is

$$\frac{1}{2}[120 + 130 + 2(150 + 180 + 160)] = \boxed{615}.$$

The result is an estimate because the rates between observation times are unknown.

For unequal spacing, each adjacent pair is treated separately:

$$\int_a^b f(x) \, dx \approx \sum_i \frac{x_{i+1} - x_i}{2} [f(x_i) + f(x_{i+1})].$$

For a smooth convex function, trapezia tend to lie above the curve and the trapezoidal rule tends to overestimate. For a smooth concave function, it tends to underestimate. This observation provides a directional check, but the size of the error still depends on the interval widths and curvature.

12.6.1 Simpson's rule

When there is an even number n of equal subintervals, Simpson's rule fits quadratic arcs through successive groups of data. With step width h ,

$$\int_{x_0}^{x_n} f(x) \, dx \approx \frac{h}{3} [y_0 + y_n + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})].$$

The endpoint coefficients are 1, the odd-positioned interior coefficients are 4, and the remaining interior coefficients are 2.

Example 12.11: [

violet]Comparing two numerical estimates For the visit-rate data

$$120, 150, 180, 160, 130$$

at hourly intervals, Simpson's rule gives

$$\frac{1}{3}[120 + 130 + 4(150 + 160) + 2(180)] = \boxed{\frac{1850}{3} \approx 616.67}.$$

The trapezoidal estimate was 615. Neither value is automatically exact, because the rates between the recorded times are unknown.

12.7 Continuous flows and present value

If $R(t)$ is a monetary flow measured in currency per unit time, then

$$\int_0^T R(t) \, dt$$

is the undiscounted amount received over $[0, T]$. If money is discounted continuously at rate $r > 0$, the present value of the flow is

$$\boxed{\text{PV} = \int_0^T R(t)e^{-rt} \, dt}.$$

The factor e^{-rt} converts a payment at time t into its value at time 0. This continuous model differs from the discrete annuities considered in Chapter 14.

Example 12.12: [

orange]Present value of a continuous income stream An income stream is received at rate

$$R(t) = 10\,000e^{0.02t}$$

ringgit per year for five years. If the continuous discount rate is 5%, then

$$\begin{aligned} \text{PV} &= \int_0^5 10\,000e^{0.02t}e^{-0.05t} \, dt \\ &= 10\,000 \int_0^5 e^{-0.03t} \, dt \\ &= \frac{10\,000}{0.03}(1 - e^{-0.15}) \\ &\approx \boxed{\text{RM } 46\,430.67}. \end{aligned}$$

12.8 Consumer and producer surplus

Let $p = D(q)$ be an inverse demand curve and $p = S(q)$ an inverse supply curve. If equilibrium occurs at (q_e, p_e) , then

$$\text{consumer surplus} = \int_0^{q_e} [D(q) - p_e] \, dq,$$

$$\text{producer surplus} = \int_0^{q_e} [p_e - S(q)] \, dq.$$

These areas measure benefits relative to the common market price within the model. They depend on the assumed demand and supply curves and do not capture every social cost or benefit.

Example 12.13: [

cyan]Surplus in a linear market For $D(q) = 100 - 2q$ and $S(q) = 20 + q$, equilibrium satisfies $100 - 2q = 20 + q$. Hence $q_e = 80/3$ and $p_e = 140/3$. The surpluses are

$$\text{CS} = \frac{6400}{9}, \quad \text{PS} = \frac{3200}{9}.$$

For linear curves these integrals agree with the areas of the corresponding triangles.

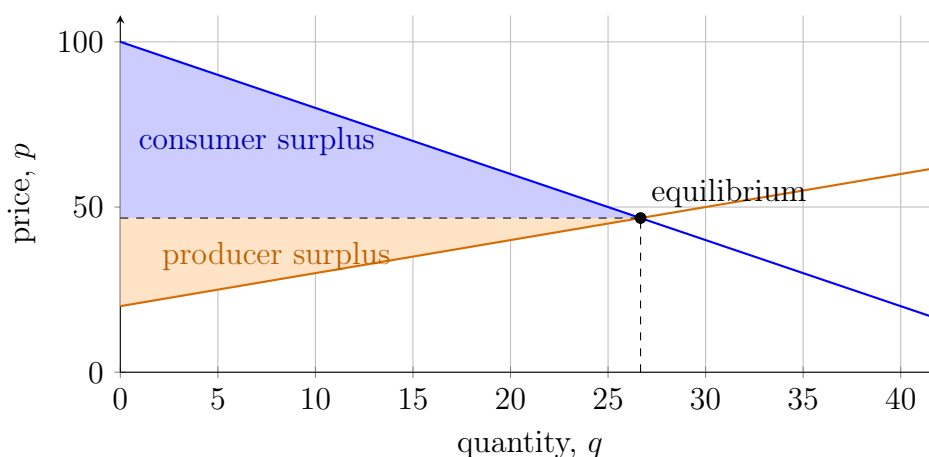


Figure 12.3: Consumer and producer surplus are the areas above and below the equilibrium price within the demand–supply model.

Surplus calculations are model-based welfare measures. Taxes, externalities, distributional concerns, market power and non-price effects may require a broader analysis than the two shaded areas alone.

12.9 Lorenz curves and the Gini coefficient

A Lorenz curve $L(x)$ gives the cumulative proportion of income received by the lowest proportion x of a population. For a standard distribution of non-negative income, it

satisfies $L(0) = 0$ and $L(1) = 1$, is non-decreasing and convex, and lies at or below the line $L(x) = x$ of perfect equality. The Gini coefficient is

$$G = 1 - 2 \int_0^1 L(x) \, dx.$$

Values closer to 0 indicate greater equality, while values closer to 1 indicate greater inequality. For the illustrative Lorenz curve $L(x) = x^2$,

$$G = 1 - 2 \int_0^1 x^2 \, dx = \frac{1}{3}.$$

Equivalently, the Gini coefficient is twice the area between the line of perfect equality and the Lorenz curve:

$$G = 2 \int_0^1 [x - L(x)] \, dx.$$

The coefficient summarises a distribution but does not explain why the distribution arose or how living standards compare across populations. Two populations can have the same Gini coefficient while having different income levels or differently shaped distributions.

A valid Lorenz curve is constructed after arranging the population from lowest to highest income. Its slope therefore cannot decrease as progressively higher incomes are included, which gives the standard curve its convex shape. Multiplying every income by the same positive constant leaves the Lorenz curve and Gini coefficient unchanged, which is why the coefficient measures relative inequality rather than the overall standard of living.

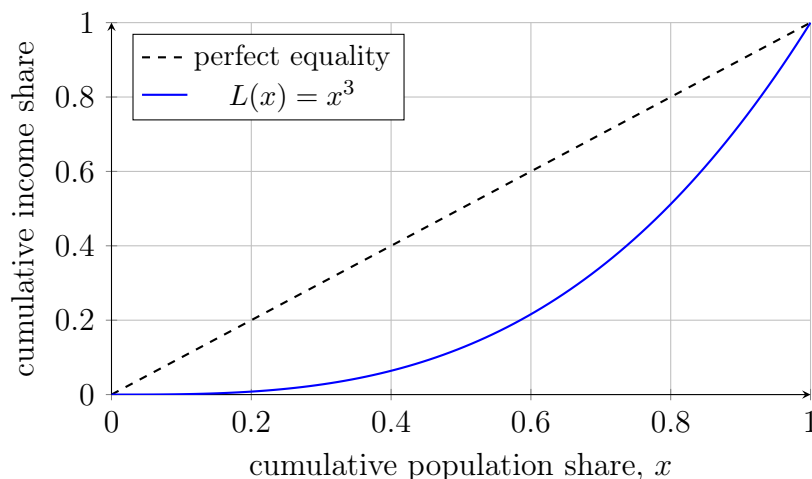


Figure 12.4: The area between the equality line and the Lorenz curve determines the Gini coefficient. A more bowed curve indicates greater inequality.

12.10 Exercises

Answers are provided in section C.12.

1. Find each indefinite integral.

(a) $\int \frac{4}{4x-5} \, dx$

(b) $\int 4a^2 e^x \, dx$

(c) $\int \left(4 + \frac{5}{2x}\right) \, dx$

(d) $\int 4e^{3x/4} \, dx$

(e) $\int \frac{3x^3 - 5x}{x^7} \, dx$

(f) $\int (x^2 - 3x^4)^2 \, dx$

(g) $\int \frac{e^x}{2b^2} \, dx$

(h) $\int (6x^6 + 4x^{2/3} + 8) \, dx$

2. Use substitution to find (a) $\int 3x^2(x^3 - 2)^3 \, dx$; (b) $\int (2 + \ln \beta)^3 \, d\beta/\beta$.

3. Use the indicated substitution.

(a) $\int x^5 \sqrt{x^3 + 1} \, dx, u = x^3 + 1$

(b) $\int 2x^3 \sqrt{x^2 + 1} \, dx, u = x^2 + 1$

(c) $\int \frac{2x}{\sqrt[3]{x^2 - 5}} \, dx, u = x^2 - 5$

(d) $\int \frac{t^2 + 2t}{\sqrt[7]{t^3 + 3t^2 + 10}} \, dt, u = t^3 + 3t^2 + 10$

4. Given $\int_1^3 f(x) \, dx = 5$ and $\int_3^8 f(x) \, dx = 10$, find

(a) $\int_3^3 f(x) \, dx$;

(b) $\int_3^1 f(x) \, dx$;

(c) $\int_1^3 4f(x) \, dx$;

(d) $\int_1^8 f(x) \, dx$;

(e) $\int_3^8 [f(x) + 5] \, dx$.

5. Evaluate.

(a) $\int_1^2 (x - 1/x)(x + 1/x) \, dx$

(b) $\int_0^3 (1 - x)\sqrt{x} \, dx$

(c) $\int_1^2 (2 - 3x)^2 \, dx$

(d) $\int_1^e (x - 1/x) \, dx$

(e) $\int_2^5 (3a - 10)^4 \, da$

(f) $\int_1^e \frac{t+1}{t^2} \, dt$

(g) $\int_1^3 \frac{x}{2x^2 - 1} \, dx$

(h) $\int_1^2 \frac{e^{-x} + e^x}{e^x} \, dx$

6. Find $\int 2x/\sqrt{2x-7} \, dx$ by substitution. Hence evaluate $\int_4^8 2x/\sqrt{2x-7} \, dx$.

7. Find the area bounded by $y = -x^2 + 8x - 12$, the line $x = 7$ and the x -axis.

8. Find the enclosed area for each curve–line pair.

(a) $y = x(x - 2)$ and $y = x$

(b) $y = 4 - 3x - x^2$ and $2x + y + 2 = 0$

9. Find the area enclosed by each pair of curves.

(a) $y = x(2 - x)$ and $y = x(x - 1)$

(b) $y = 3x^2 - 6x$ and $y = x^2 - 5x$

(c) $x^2 = 4y$ and $y^2 = 4x$

10. A bicycle manufacturer has marginal cost $C'(x) = 500 - x/3$ for $0 \leq x \leq 900$. Find the increase in total cost when production rises from 300 to 900 bicycles.

11. A machine has marginal accumulated cost and revenue rates, in thousands of RM per year,

$$C'(t) = t/11, \quad R'(t) = 5te^{-t^2}.$$

Define its economically useful life as the time when marginal revenue first equals marginal cost after $t = 0$.

(a) Find this time to the nearest year.

(b) Find the accumulated profit over this useful life, assuming zero initial accumulated cost and revenue.

12. The supply equation is $S(x) = 10e^{0.05x}$. Find the average price over the supply interval $[20, 30]$.
13. The total cost of printing x dictionaries is $C(x) = 20\,000 + 10x$.
- (a) Find the average cost per dictionary when 1000 are printed.
 - (b) Find the average value of C on $[0, 1000]$.
 - (c) Explain why the two answers represent different quantities.
14. A market has inverse demand $D(q) = 120 - 3q$ and inverse supply $S(q) = 20 + 2q$.
- (a) Find the equilibrium quantity and price.
 - (b) Calculate consumer surplus and producer surplus.
 - (c) Find the total surplus represented by the two areas.
15. A simplified income distribution has Lorenz curve $L(x) = x^3$. Verify its endpoint values and calculate the corresponding Gini coefficient.

Further exercises: methods and applications

16. Marginal cost is $C'(q) = 24 + 0.06q$ and fixed cost is $C(0) = 900$. Find $C(q)$ and calculate $C(200)$.
17. Evaluate.
- (a) $\int 3x^2 e^{x^3} dx$
 - (b) $\int \frac{x}{x^2 + 4} dx$
18. Use substitution, changing the limits, to evaluate
- $$\int_0^1 2x(1 + x^2)^3 dx.$$
19. Use integration by parts to evaluate.
- (a) $\int x e^{2x} dx$
 - (b) $\int_1^e \ln x dx$
20. Service requests arrive at rate $r(t) = 60 + 8t$ requests per hour for $0 \leq t \leq 6$.
- (a) Find the total number of requests over the interval.
 - (b) Find the average arrival rate.
21. For $f(x) = x - 2$ on $[0, 5]$, calculate
- (a) the signed integral $\int_0^5 f(x) dx$;

(b) the total geometric area between the graph and the x -axis.

22. Find the area enclosed by $y = 2x + 3$ and $y = x^2$.

23. The rate of service requests is recorded below.

t (hours)	0	2	4	6
$r(t)$ (requests per hour)	30	45	39	24

Use the trapezoidal rule to estimate the total number of requests during the six hours.

24. A continuous income stream pays RM 12 000 per year for three years. Find its present value under a continuous discount rate of 4% per year.

25. A Lorenz curve is given by $L(x) = x^4$.

(a) Verify the endpoint conditions.

(b) Find the Gini coefficient.

(c) Compare it with the coefficient for $L(x) = x^2$.

Extended practice: accumulation and social-science models

26. The number of registered participants satisfies $N'(t) = 300e^{-0.1t}$ participants per year, with $N(0) = 1200$.

(a) Find $N(t)$.

(b) Estimate $N(5)$.

(c) Find the long-run limit of the model and interpret it.

27. Find each indefinite integral.

(a) $\int \frac{6x}{3x^2 + 4} dx$

(b) $\int te^{-t^2/2} dt$

(c) $\int \ln x dx, x > 0$

(d) $\int x^2 e^x dx$

28. Evaluate by substitution, changing the bounds where convenient.

(a) $\int_0^1 4x(1 + 2x^2)^3 dx$

(b) $\int_1^e \frac{\ln x}{x} dx$

(c) $\int_0^2 \frac{3t^2}{t^3 + 1} dt$

29. Define

$$A(x) = \int_1^x (3t^2 - 2t) \, dt.$$

- (a) Find $A'(x)$ without first evaluating the integral.
- (b) Find $A(1)$.
- (c) Obtain an explicit formula for $A(x)$.
- (d) Find $A(2)$ and interpret it as accumulated change.

30. For $f(x) = x^2 - 1$ on $[-2, 2]$:

- (a) calculate the signed integral;
- (b) calculate the total geometric area between the graph and the x -axis;
- (c) explain why the two answers differ.

31. A participation level is modelled by $M(t) = 100 + 20t - 2t^2$ on $0 \leq t \leq 5$.

- (a) Find its average value over the interval.
- (b) Find a value $c \in [0, 5]$ at which $M(c)$ equals this average.

32. A service centre records the following arrival rates at unequally spaced times:

t (hours)	0	1	2.5	4	6
$r(t)$ (arrivals per hour)	50	70	80	65	40

Use separate trapezia to estimate the total number of arrivals during the six hours. State the units of the answer.

33. The following equally spaced rates are recorded over six hours:

t	0	1	2	3	4	5	6
$r(t)$	20	28	35	39	36	30	22

- (a) Estimate the accumulated total using the trapezoidal rule.
- (b) Estimate it using Simpson's rule.
- (c) Explain why the estimates may differ.

34. A city's population stock is initially 10 000. Its inflow and outflow rates are

$$I(t) = 500 + 20t, \quad O(t) = 300 + 10t$$

people per year for $0 \leq t \leq 8$.

- (a) Find the total inflow and total outflow.
- (b) Find the net population change.
- (c) Find the population after eight years.

35. An income stream is received at rate $R(t) = 15\,000e^{0.03t}$ ringgit per year for four years.

- (a) Find the undiscounted total income.
- (b) Find its present value using a continuous discount rate of 6% per year.
- (c) Explain why the present value is smaller.

36. A market has inverse demand and supply curves

$$D(q) = 100 - q^2, \quad S(q) = 20 + q^2.$$

- (a) Find the positive equilibrium quantity and price.
- (b) Calculate consumer and producer surplus.
- (c) Find total surplus and state one limitation of interpreting it as a complete measure of social welfare.

37. Consider the family of Lorenz curves $L(x) = x^\alpha$ with $\alpha > 1$.

- (a) Derive the Gini coefficient in terms of α .
- (b) Find the coefficient when $\alpha = 5/2$.
- (c) Compare it with the coefficient for $L(x) = x^2$.

38. A researcher integrates a fitted expenditure-rate model well beyond the observed time interval and reports the result as an exact causal effect of a policy. Give three reasons why the integration may be mathematically correct but the conclusion still unreliable.

Additional practice: grouped skills review

39. Find $\int x^n dx$ for $n = 2, 3, 4, 5, 6, 7$ and 8.

40. Find $\int (ax^2 + b) dx$ for each case.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
<i>a</i>	2	3	4	5	6	7	8
<i>b</i>	1	2	3	4	5	6	7

41. Find $\int a/(ax + b) dx$ for the seven (a, b) pairs in Exercise 40.

42. Find

$$\int 2x(x^2 + c)^c dx$$

for $c = 1, 2, 3, 4, 5, 6$ and 7.

43. Use integration by parts to find $\int xe^{ax} dx$ for $a = 1, 2, 3, 4, 5, 6$ and 7.

44. Evaluate

$$\int_0^n (2x + n - 2) dx$$

for $n = 2, 3, 4, 5, 6$ and 7.

45. Find the area under $y = (n - 1)x + n$ from $x = 0$ to $x = n$ for $n = 2, 3, 4, 5, 6$ and 7 .
46. Find the average value of $f(x) = (m - 3)x + (m + 6)$ on $[0, m]$ for $m = 4, 5, 6, 7, 8$ and 9 .
47. Use the trapezoidal rule with width 1 for each row of three consecutive observations.

Case	A	B	C	D	E	F
y_0	10	11	12	13	14	15
y_1	14	16	18	20	22	24
y_2	18	19	20	21	22	23

48. For each linear inverse-demand model, find consumer surplus at the stated quantity q_e .

Case	A	B	C	D	E	F
$p(q)$	$100 - 2q$	$105 - 3q$	$110 - 4q$	$115 - 5q$	$120 - 6q$	$125 - 7q$
q_e	10	11	12	13	14	15

Applications: accumulated social quantities

49. Calls arrive at a public-information hotline at rate $r(t) = 120 + 15t$ calls per hour for $0 \leq t \leq 8$.
- (a) Find the total number of calls received.
- (b) Find the average arrival rate over the eight hours.
50. A city has inflow and outflow rates

$$I(t) = 900 + 40t, \quad O(t) = 600 + 10t$$

people per year for $0 \leq t \leq 10$. Find the total inflow, total outflow and net population change. If the initial population is 85 000, find the population after ten years.

51. A vaccination programme has delivery rate $v(t) = 5000e^{-0.2t}$ doses per month. If 20 000 doses had already been delivered at $t = 0$, find the accumulated total $N(t)$, calculate $N(6)$ and interpret the long-run limit.
52. A survey team completes interviews at rate $q(t) = 40 + 12t - 2t^2$ interviews per hour for $0 \leq t \leq 6$. Find the total completed, the average hourly rate and the time at which the rate is greatest.
53. A complaints stock changes at net rate $C'(t) = 30 - 5t$ complaints per month, with $C(0) = 400$ and $0 \leq t \leq 10$. Find $C(t)$, its maximum value and $C(10)$.
54. Participation is modelled by $M(t) = 60 + 12t - t^2$ on $0 \leq t \leq 8$. Find its average value and a value $c \in [0, 8]$ at which $M(c)$ equals that average.

55. A programme's net-benefit rate is $b(t) = t - 3$ million ringgit per year on $0 \leq t \leq 6$. Find the signed accumulated benefit and the total geometric area between the rate graph and the time axis. Explain the difference, using figure 12.5.

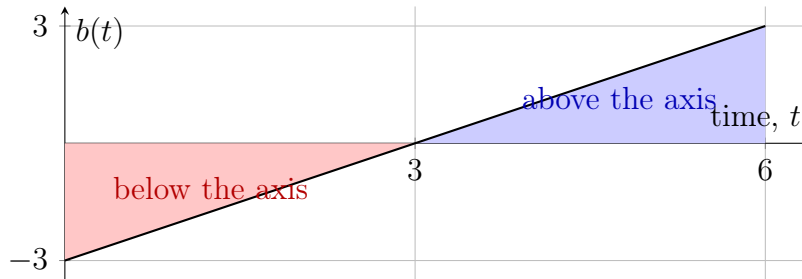


Figure 12.5: Positive and negative regions for Exercise 55. A signed integral retains the signs, whereas total geometric area counts the magnitude of both regions.

56. Marginal social benefit and marginal social cost are

$$B(q) = 20 + 8q - q^2, \quad C(q) = 8 + 2q.$$

Find the accumulated net benefit from $q = 0$ to $q = 6$, and find the positive output at which the two marginal quantities are equal.

57. Housing completions occur at rate $h(t) = 2000 + 300\sqrt{t}$ homes per year for $0 \leq t \leq 4$. Estimate the total number completed and the average annual rate.
58. A food bank receives net donations at rate $F'(t) = 800/(t + 1)$ kilograms per month, with $F(0) = 5000$. Find $F(t)$ and $F(4)$, stating the units of the result.

Applications: economics and public finance

59. A producer has marginal cost $C'(q) = 30 + 0.04q$ and fixed cost $C(0) = 1200$. Find $C(q)$ and the cost of producing 500 units.
60. Marginal revenue and marginal cost are

$$R'(q) = 100 - 0.2q, \quad C'(q) = 20 + 0.1q,$$

with $R(0) = 0$ and $C(0) = 1000$. Find $R(q)$, $C(q)$ and the output that maximises profit. Calculate the maximum profit.

61. A public project produces benefits continuously at rate $2e^{0.02t}$ million ringgit per year for five years.
- Find the undiscounted total benefit.
 - Find its present value using a continuous discount rate of 5% per year.
62. A delayed programme benefit of RM 1.5 million per year is received continuously from $t = 3$ to $t = 7$. Find its present value at a continuous discount rate of 4% per year.

- 63.** A market has inverse demand $D(q) = 120 - 2q$ and inverse supply $S(q) = 20 + q$. Find equilibrium quantity and price, consumer surplus, producer surplus and total surplus.
- 64.** A market initially has $D(q) = 100 - q$ and $S(q) = 20 + q$. A tax of RM 20 per unit is imposed on sellers.
- Find the before-tax equilibrium and total surplus.
 - Find the after-tax quantity, buyer price and seller price.
 - Calculate consumer surplus, producer surplus, tax revenue and deadweight loss after the tax.

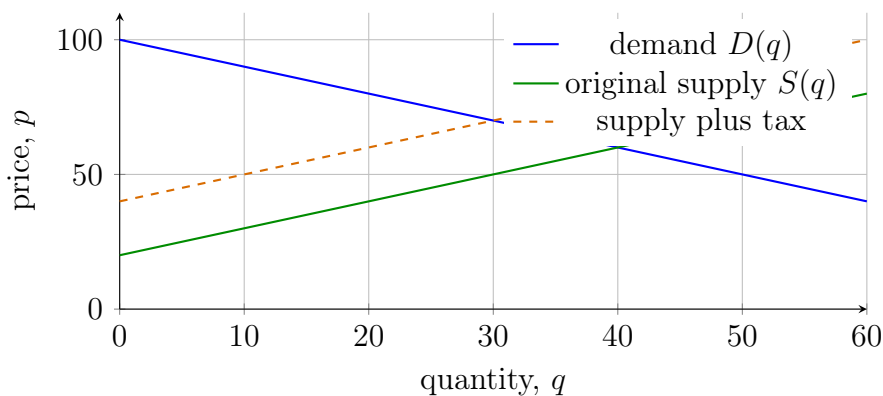


Figure 12.6: The per-unit tax in Exercise 64 shifts the buyer-facing supply curve vertically. The two intersections determine the before-tax and after-tax market outcomes.

- 65.** A social programme generates benefit and cost flows, in millions of ringgit per year, $B(t) = 4 + 0.8t$ and $C(t) = 3 + 0.5t$ for $0 \leq t \leq 6$. Find total benefit, total cost and accumulated net benefit.
- 66.** An infrastructure project incurs cost continuously at RM 5 million per year from $t = 0$ to $t = 2$, then produces benefits at RM 3 million per year from $t = 2$ to $t = 8$. At a continuous discount rate of 4%, find the present value of costs and benefits, the NPV and the benefit–cost ratio.
- 67.** Total cost is $C(q) = 10\,000 + 25q + 0.02q^2$.
- Find average cost per unit at $q = 500$.
 - Find the average value of the total-cost function on $[0, 500]$.
 - Explain why the two averages differ.
- 68.** A municipal reserve is initially RM 5 million. Revenue and spending rates are $r(t) = 3 + 0.1t$ and $s(t) = 2 + 0.4t$ million ringgit per month for $0 \leq t \leq 10$. Find the reserve function, the time at which it returns to its initial level, and its value at $t = 10$.

Applications: inequality and distribution

69. For the Lorenz curve $L(x) = x^{5/2}$, verify the endpoint conditions and calculate the Gini coefficient.
70. Populations A and B have Lorenz curves $L_A(x) = x^2$ and $L_B(x) = x^4$. Calculate both Gini coefficients and identify the distribution represented as more unequal. Use figure 12.7 to support the interpretation.

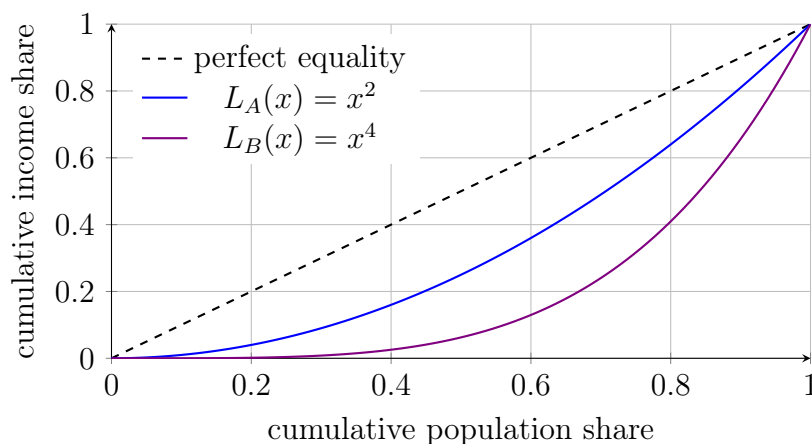


Figure 12.7: The Lorenz curves in Exercise 70. Greater bowing away from the equality line corresponds to a larger area and a larger Gini coefficient.

71. A report proposes $L(x) = x(2 - x)$ as a Lorenz curve on $[0, 1]$. Check its endpoints and its position relative to the equality line. Explain why it is not a valid Lorenz curve for an ordered income distribution.
72. Before a transfer policy, $L_0(x) = x^3$. Afterwards, $L_1(x) = \frac{1}{2}x^2 + \frac{1}{2}x^3$. Calculate both Gini coefficients and the change in the coefficient.
73. Under $L(x) = x^2$, find the income shares received by the bottom 40%, the middle 20% and the top 40% of the population.
74. For $L(x) = x^3$, find the area between the Lorenz curve and the line of equality, then recover the Gini coefficient from that area.
75. Consider the piecewise Lorenz curve

$$L(x) = \begin{cases} 0.5x, & 0 \leq x \leq 0.8, \\ 3x - 2, & 0.8 < x \leq 1. \end{cases}$$

Verify continuity and the endpoint conditions, then calculate its Gini coefficient.

76. A fitted Lorenz family is $L(x) = x^\alpha$, $\alpha > 1$. If the Gini coefficient is 0.45, find α .
77. A cumulative tax-payment curve is $T(x) = x^{3/2}$, where households are ordered from lowest to highest income. Find the shares of tax paid by the bottom and top halves, and interpret the result.

78. Explain why multiplying every income by the same positive constant leaves the Lorenz curve and Gini coefficient unchanged, whereas adding the same amount to every income need not do so.

Applications: numerical integration and recorded data

79. A service records hourly arrival rates

t	0	1	2	3	4
$r(t)$	50	70	90	80	60

Use the trapezoidal rule to estimate total arrivals. The geometric construction is shown in figure 12.8.

80. Use Simpson's rule on the data in Exercise 79. Compare the estimate with the trapezoidal estimate.

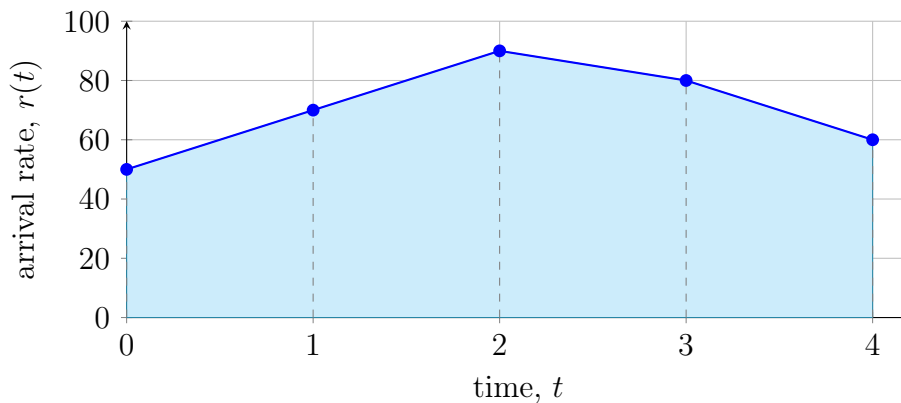


Figure 12.8: The trapezoidal construction for Exercises 79–80. Straight segments between observations form the upper edges of the trapezia.

81. Estimate the accumulated total from the unequally spaced data

t	0	1.5	3	5
$r(t)$	20	32	28	16

by applying a separate trapezium to each interval.

82. For $r(t) = 10 + 3t^2$ on $[0, 4]$:
- (a) find the exact integral;
 - (b) use the trapezoidal rule with width 2;
 - (c) use Simpson's rule with width 2 and compare the errors.
83. A benefit-delivery rate is recorded as

t	0	1	2	3	4	5
$b(t)$	100	140	180	160	120	80

Use successive trapezia to estimate the total by $t = 5$. At which recorded time does the cumulative estimate first exceed 500?

84. A net-flow rate is recorded as

t	0	2	4	6
$r(t)$	-10	20	30	-5

Use the trapezoidal rule to find the signed accumulated change, and explain how negative rates enter the calculation.

85. Annual net-migration rates for four consecutive one-year intervals are represented at times 0, 1, 2, 3, 4 by 800, 1000, 900, 700, 600 people per year. Estimate the four-year net migration and state the units.
86. Three equally spaced rate observations are 10, x , 30. If the trapezoidal estimate over the two-unit interval is 50, find x .
87. State whether the trapezoidal rule overestimates or underestimates the integral of a convex function. Explain why Simpson's rule is exact for any polynomial of degree at most three when its conditions are satisfied.
88. A statistical agency revises one observation in a published rate series. Explain how the accumulated estimate should be updated and give three checks that should accompany the revision.

Applications: model evaluation and responsible interpretation

89. Distinguish a stock from a flow using unemployment as an example. Explain why integrating a net flow changes the stock.
90. Explain the role of the constant of integration when a marginal or rate function is used to recover a total function. State what additional information identifies the constant.
91. A rate is measured in households per month. State the units of its integral over time and of its average value over a time interval.
92. A researcher integrates a fitted programme-participation rate and calls the result the programme's causal effect. Explain why the calculation alone does not justify that conclusion.
93. Give four reasons why integrating a fitted expenditure-rate model far beyond its observed domain may produce an unreliable forecast.
94. A continuous benefit flow of RM 10 000 per year lasts ten years. Find its present value at continuous discount rates of 3% and 7%. Explain the direction of the difference.
95. Give three reasons why consumer and producer surplus may be incomplete measures of the welfare effect of a public policy.
96. Explain why two societies can have the same Gini coefficient while differing substantially in median income, poverty and the location of inequality within the distribution.
97. Explain when a signed integral and a total geometric area answer different substantive questions in a social-science application.

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98. Give three considerations for choosing among exact integration, the trapezoidal rule and Simpson's rule for observed data.
 99. An accumulated-benefit estimate is used to allocate public funds. Give four ethical or governance safeguards that should accompany the calculation.
 100. Give a six-stage checklist for constructing, calculating, checking and communicating an integration-based social-science model.

Chapter 13

Differential Equations

Chapter guide

Translate rate assumptions into differential equations, solve separable models, analyse equilibria and logistic growth, and finish with parameter estimation and Euler's method.

Selected background sources: (Hassan & Mazlin, 2025–2026; Strang & Herman, 2016b).

A differential equation relates an unknown function to one or more of its derivatives. Unlike an ordinary algebraic equation, whose solution is usually a number, a differential equation is solved by finding a function that has a specified rate of change. Such equations are useful whenever the present state of a system influences how it changes, as in models of population, employment, prices, public-service adoption, migration and the spread of information.

This chapter focuses on first-order ordinary differential equations that can be solved by separating their variables. The algebraic solution is only one part of the modelling process. The variables and parameters must also be interpreted, their units checked and the assumptions of the model compared with the situation being studied.

In social-science applications, it is especially important to distinguish a *level* from a *rate*. The number of people receiving a service is a level, while the number entering or leaving the service per month is a rate. A differential equation connects the two. It describes how a level is expected to evolve under stated assumptions; it does not, by itself, establish that one variable causes another.

13.1 Basic terminology

An *ordinary differential equation* contains derivatives with respect to one independent variable. A first-order equation can often be written as

$$\frac{dy}{dx} = F(x, y).$$

Its *order* is the order of the highest derivative present. For example,

$$\frac{dy}{dx} = x - y$$

is first order, whereas

$$\frac{d^2y}{dx^2} + 4y = 0$$

is second order.

A solution may be written explicitly as $y = f(x)$ or implicitly as a relation $H(x, y) = C$. A *general solution* contains an arbitrary constant and represents a family of solution curves. A condition such as

$$y(x_0) = y_0$$

is called an *initial condition*. It selects a *particular solution* from the family, and the differential equation together with the condition is called an *initial-value problem*.

13.1.1 Levels, rates and units

If $P(t)$ is a population measured in people and t is measured in years, then dP/dt is measured in people per year. If $A(t)$ is an adoption percentage, then dA/dt is measured in percentage points per unit of time, not as a percentage of the existing value unless the equation explicitly uses a proportional rate.

For example, the statement

$$\frac{dP}{dt} = 800 - 0.06P$$

contains a constant inflow of 800 people per year and an outflow equal to 6% of the current population per year. Both terms on the right therefore have the same units as dP/dt . The units would be inconsistent if 800 were a number of people rather than a number of people per year.

The derivative is an instantaneous rate. Over a short interval of length Δt ,

$$\Delta y \approx \frac{dy}{dt} \Delta t.$$

This approximation links differential equations with the marginal and rate-of-change ideas developed in Chapter 11.

13.1.2 Checking a proposed solution

A proposed solution is checked by differentiating it and substituting the result into the original equation. It must also satisfy any stated initial condition.

Example 13.1: [

blue]Verifying a solution Consider

$$\frac{dy}{dt} = 2y, \quad y(0) = 5.$$

For $y = 5e^{2t}$,

$$\frac{dy}{dt} = 10e^{2t} = 2(5e^{2t}) = 2y,$$

and $y(0) = 5e^0 = 5$. Therefore, $y = 5e^{2t}$ satisfies both the differential equation and the initial condition.

The differential equation determines a family of possible trajectories. The initial condition identifies which member of that family represents the case being studied.

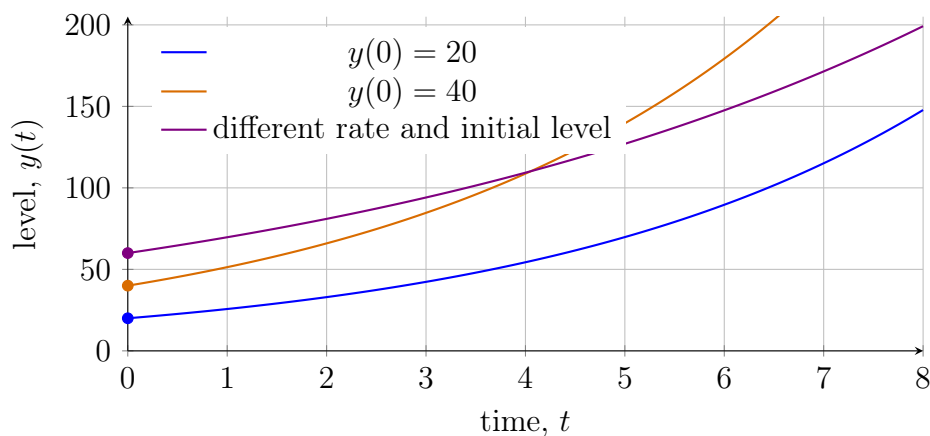


Figure 13.1: A differential-equation model needs both a rate rule and an initial condition. Changing either one changes the trajectory.

13.2 Separable differential equations

A first-order equation is *separable* if it can be rearranged so that all expressions involving y occur with dy and all expressions involving x occur with dx . For example, if

$$\frac{dy}{dx} = f(x)h(y),$$

then, where $h(y) \neq 0$,

$$\frac{1}{h(y)} dy = f(x) dx.$$

Integrating both sides gives

$$\int \frac{1}{h(y)} dy = \int f(x) dx + C.$$

The result may remain implicit if solving explicitly for y is difficult or unnecessary.

A reliable procedure is:

1. rearrange the equation to separate the variables;
2. record any values excluded by division;
3. integrate both sides, using one arbitrary constant;
4. use the initial condition, if given, to determine the constant;

5. check the result and state an appropriate domain.

Division during separation may remove constant solutions. If a factor such as $h(y)$ is divided out, every value satisfying $h(y) = 0$ should be checked separately in the original equation.

Example 13.2: [

orange]A separable growth equation Solve

$$\frac{dy}{dx} = 3xy.$$

For $y \neq 0$,

$$\frac{dy}{y} = 3x \, dx \implies \ln |y| = \frac{3}{2}x^2 + C.$$

Exponentiating and absorbing the sign into the arbitrary constant gives

$$y = Ae^{3x^2/2}.$$

The constant solution $y = 0$, which was temporarily excluded by division, is included by allowing $A = 0$.

Example 13.3: [

violet]An initial-value problem Solve

$$\frac{dy}{dx} = x(1 + y^2), \quad y(0) = 1.$$

Separating and integrating gives

$$\frac{dy}{1 + y^2} = x \, dx \implies \tan^{-1} y = \frac{x^2}{2} + C.$$

The initial condition gives $C = \pi/4$, so

$$y = \tan \left(\frac{x^2}{2} + \frac{\pi}{4} \right).$$

The solution containing $x = 0$ is defined for

$$-\sqrt{\frac{\pi}{2}} < x < \sqrt{\frac{\pi}{2}}.$$

This example shows that a solution of a well-defined differential equation may still become unbounded at a finite value of the independent variable.

Example 13.4: [

A changing proportional rate Suppose a survey's daily response rate $R(t)$ declines according to

$$\frac{dR}{dt} = -0.08tR, \quad R(0) = 70,$$

where R is measured in responses per day and t in weeks. The proportional decline is not constant: its magnitude increases with time. Separating gives

$$\frac{dR}{R} = -0.08t \, dt \implies \ln R = -0.04t^2 + C.$$

The initial condition gives $C = \ln 70$, so

$$R(t) = 70e^{-0.04t^2}.$$

After four weeks, the model gives $R(4) \approx 36.91$ responses per day. The equation describes an increasingly strong fatigue effect, an assumption that should be checked against observed response data.

When a logarithm appears during integration, absolute-value signs should be retained until the sign of the solution is known. In contextual models, a non-negative initial value and the structure of the differential equation often restrict the relevant branch, but that restriction should be stated rather than assumed silently.

13.3 Autonomous equations and equilibrium

An equation of the form

$$\frac{dy}{dt} = F(y)$$

is called *autonomous* because its rate depends on the current state y , but not explicitly on time. A value $y = y^*$ satisfying

$$F(y^*) = 0$$

is an *equilibrium solution*. If the system begins at an equilibrium, the model predicts no change.

The sign of $F(y)$ describes the direction of movement:

- if $F(y) > 0$, then y increases;
- if $F(y) < 0$, then y decreases;
- if nearby solutions move towards y^* , the equilibrium is stable;
- if nearby solutions move away from y^* , it is unstable.

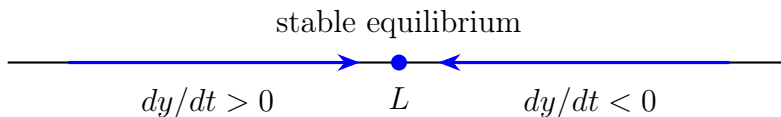


Figure 13.2: A phase line for $dy/dt = k(L - y)$: solutions move towards L .

For the adjustment equation $dy/dt = k(L - y)$ with $k > 0$, values below L increase and values above L decrease. Thus, L is a stable equilibrium.

This qualitative analysis is useful even before the equation is solved. It can reveal whether a proposed formula behaves consistently with the model. For models with a meaningful state space, such as $0 \leq P \leq 100$ for a percentage, the arrows should also be checked at the boundaries. If the equation directs solutions outside the meaningful interval, the model may be unsuitable even if its algebra is correct.

13.4 Proportional growth and decay

If the rate of change of a quantity is proportional to its current value, then

$$\frac{dy}{dt} = ky.$$

For $y \neq 0$,

$$\frac{dy}{y} = k \, dt \implies \ln |y| = kt + C.$$

If $y(0) = y_0$, the particular solution is

$$\boxed{y(t) = y_0 e^{kt}}.$$

Positive k produces exponential growth and negative k produces exponential decay. The parameter k has units of inverse time and represents the continuous proportional rate of change:

$$\frac{1}{y} \frac{dy}{dt} = k.$$

For growth, the doubling time is

$$T_2 = \frac{\ln 2}{k}.$$

For decay, the half-life is

$$T_{1/2} = \frac{\ln 2}{|k|}.$$

If a quantity changes by a discrete proportion g once per period, its multiplier is $1 + g$. The equivalent continuous rate is

$$k = \ln(1 + g),$$

while a continuous rate k corresponds to the one-period discrete change $e^k - 1$. These rates are close when the changes are small, but they are not identical.

Example 13.5: [

cyan]Estimating a continuous growth rate The number of registered users of a community information service rises from 1200 to 1680 over four years. Under the model $N' = kN$,

$$1680 = 1200e^{4k},$$

so

$$k = \frac{\ln(1.4)}{4} \approx 0.08412.$$

The fitted model is

$$N(t) = 1200e^{0.08412t}.$$

The value of k is approximately 0.08412 per year. It is not exactly the same as an annual discrete growth rate of 8.412% because continuous and period-by-period compounding use different conventions.

Proportional growth can approximate population, participation or the spread of information over a limited interval. It assumes that the relative growth rate remains constant and that no capacity constraint becomes important. These assumptions usually become less realistic over a long period.

Example 13.6: [

orange]Estimating a decay model from two observations Interest in a news topic, measured by a search index, falls from 80 to 52 over five days. Under the model $I' = kI$,

$$52 = 80e^{5k} \implies k = \frac{1}{5} \ln\left(\frac{52}{80}\right) \approx -0.08616.$$

Thus,

$$I(t) = 80e^{-0.08616t}.$$

The model's half-life is approximately

$$\frac{\ln 2}{0.08616} \approx 8.04 \text{ days}.$$

The calculation summarises the observed decline; it does not identify why public attention changed.

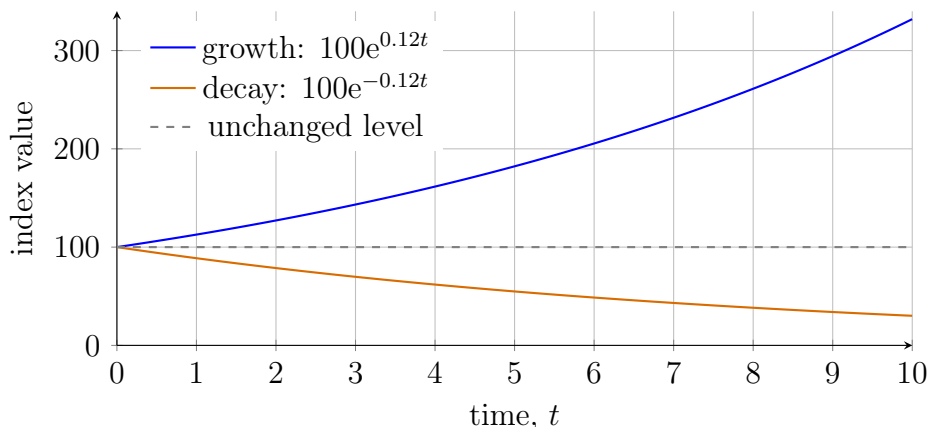


Figure 13.3: Proportional growth and decay have constant relative rates, although their absolute changes become larger or smaller with the level.

13.5 Adjustment towards an equilibrium

A quantity that moves towards a long-run level L can be modelled by

$$\frac{dy}{dt} = k(L - y), \quad k > 0.$$

Separating variables gives

$$\frac{dy}{L - y} = k \, dt,$$

and hence

$$-\ln |L - y| = kt + C.$$

Using $y(0) = y_0$ produces

$$y(t) = L + (y_0 - L)e^{-kt}.$$

The gap from equilibrium satisfies

$$y(t) - L = (y_0 - L)e^{-kt},$$

so a constant proportion of the remaining gap is removed per unit of time. The idealised model approaches L without reaching it at a finite time. The time required for the absolute gap to halve is

$$T_{\text{gap}} = \frac{\ln 2}{k}.$$

This is sometimes easier to interpret than k : it shows how rapidly the system closes the distance to its long-run level.

More generally, an equation with a constant inflow and proportional outflow,

$$\frac{dy}{dt} = a - by, \quad b > 0,$$

has equilibrium $L = a/b$ and solution

$$y(t) = \frac{a}{b} + \left(y_0 - \frac{a}{b}\right)e^{-bt}.$$

Example 13.7: [

blue]Adoption of a public service Suppose the adoption percentage $A(t)$ satisfies

$$\frac{dA}{dt} = 0.4(100 - A), \quad A(0) = 20.$$

Then

$$A(t) = 100 - 80e^{-0.4t}.$$

After five years, the model predicts

$$A(5) \approx \boxed{89.17\%}.$$

The ceiling of 100% is an assumption of the model, not a guarantee about actual public behaviour.

Example 13.8: [

orange]Constant entry and proportional exit A public agency receives 900 active cases per year, while resolving cases at a continuous rate equal to 12% of the current caseload per year. If $C(0) = 5000$, then

$$\frac{dC}{dt} = 900 - 0.12C.$$

The equilibrium caseload is $900/0.12 = 7500$, and

$$C(t) = 7500 - 2500e^{-0.12t}.$$

Since the initial caseload is below equilibrium, the model predicts that it will rise towards 7500. The equilibrium is a balance of flows, not a claim that entries or resolved cases cease.

13.6 Logistic growth

Pure exponential growth has no upper limit. When growth is initially rapid but slows as a finite capacity is approached, a *logistic model* may be more appropriate:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right), \quad r > 0, \quad K > 0.$$

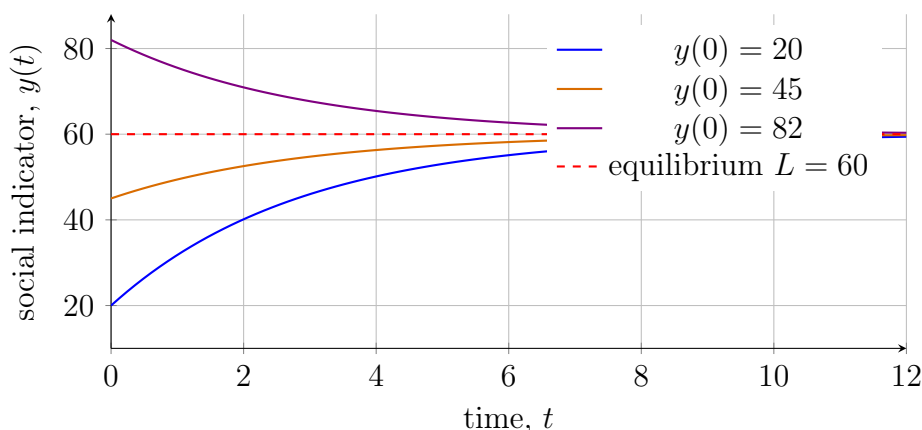


Figure 13.4: For $y' = 0.35(60 - y)$, trajectories starting on either side move towards the same stable equilibrium.

Here K is the *carrying capacity* or saturation level. The equilibria are $P = 0$ and $P = K$. For $0 < P < K$, the quantity grows towards K .

For a non-equilibrium solution,

$$\frac{K \, dP}{P(K - P)} = r \, dt.$$

Using

$$\frac{K}{P(K - P)} = \frac{1}{P} + \frac{1}{K - P}$$

and integrating gives

$$\ln \left| \frac{P}{K - P} \right| = rt + C.$$

If $P(0) = P_0$ with $0 < P_0 < K$, then

$$P(t) = \frac{K}{1 + Ae^{-rt}}, \quad A = \frac{K - P_0}{P_0}.$$

The growth rate $rP(1 - P/K)$ is greatest when $P = K/2$. Before this level, the absolute rate accelerates; afterwards, it slows as saturation approaches.

Example 13.9: [

violet]A bounded adoption model The percentage $P(t)$ of an eligible population using a programme is modelled by

$$\frac{dP}{dt} = 0.5P \left(1 - \frac{P}{80} \right), \quad P(0) = 10.$$

Since $K = 80$ and $A = (80 - 10)/10 = 7$,

$$P(t) = \frac{80}{1 + 7e^{-0.5t}}.$$

After six years,

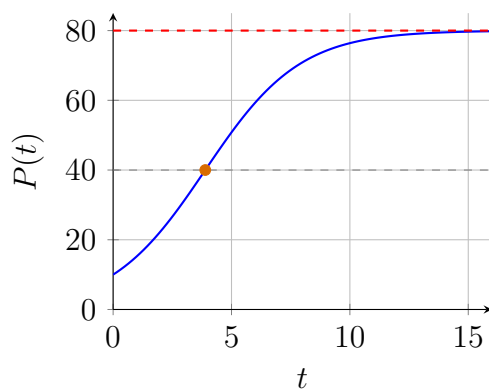
$$P(6) \approx 59.32\%.$$

The model reaches 60% when

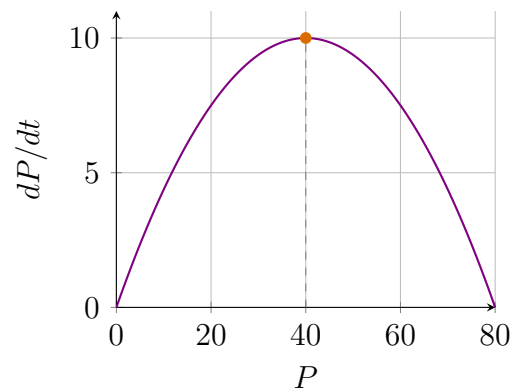
$$60 = \frac{80}{1 + 7e^{-0.5t}},$$

which gives $t = 2 \ln 21 \approx 6.09$ years.

The logistic model is still a simplification. It assumes fixed parameters, a stable saturation level and a smooth response. Campaigns, policy changes, social networks and differences between population groups can alter the observed pattern.



(a) The level follows an S-shaped path towards $K = 80$.



(b) Growth is fastest at the midpoint $P = K/2$.

Figure 13.5: The logistic level and its corresponding rate of change for $P' = 0.5P(1 - P/80)$.

13.7 Estimating parameters and comparing models

Differential equations often contain parameters that are not known in advance. They may be estimated from observations. The resulting model should then be checked against data that were not used to estimate the parameters. Agreement with two observations is not strong evidence that the model will remain accurate over a long period: many different curves can pass through the same two points.

For proportional growth, observations $y(0) = y_0$ and $y(T) = y_T$ give

$$k = \frac{1}{T} \ln \left(\frac{y_T}{y_0} \right).$$

For adjustment towards a known level L ,

$$y(t) - L = (y_0 - L)e^{-kt},$$

so one later observation can be used to estimate k . For a logistic model, the saturation level K must either be estimated or justified separately; otherwise the data may not identify a unique trajectory.

Example 13.10: [

cyan]Estimating an adjustment speed An index is assumed to adjust towards the long-run level 80 according to

$$\frac{dI}{dt} = k(80 - I).$$

Suppose $I(0) = 40$ and $I(2) = 60$. Since

$$I(t) = 80 - 40e^{-kt},$$

the second observation gives

$$60 = 80 - 40e^{-2k} \implies e^{-2k} = \frac{1}{2}.$$

Thus $k = \ln 2/2 \approx 0.3466$ per year. The gap halves every two years, and the model predicts $I(6) = 75$. This conclusion depends on treating 80 as a fixed long-run level.

An exponential model may be reasonable during an early growth phase, while a logistic model may be preferable when participation is bounded. An adjustment model is appropriate when the rate depends on a gap from a target or equilibrium. The choice should reflect the proposed mechanism, not merely which formula produces an attractive curve.

Useful checks include:

- comparing predicted and observed changes over several periods;
- examining whether parameter values and units are plausible;
- checking whether the model respects natural bounds;
- testing sensitivity to alternative initial values or parameters;
- reconsidering the model after a policy or structural change.

13.8 Translating assumptions into equations

Many modelling problems begin with a verbal statement about a rate. The following translations are useful.

Modelling statement	Differential equation
rate proportional to the current amount	$dy/dt = ky$
rate proportional to the remaining gap from L	$dy/dt = k(L - y)$
constant inflow a and proportional outflow by	$dy/dt = a - by$
rate proportional to both the current amount and remaining capacity	$dy/dt = ry(1 - y/K)$
fixed intervention c combined with proportional growth or decay	$dy/dt = c + ky$
time-varying external input $u(t)$ and proportional adjustment	$dy/dt = u(t) - by$

Table 13.1: Common verbal assumptions and their rate equations.

Units provide an important check. If y is measured in people and t in years, then dy/dt has units people per year. In $dy/dt = ky$, k must have units per year. In $dy/dt = a - by$, a has units people per year and b has units per year.

After solving a model, the result should be checked for plausible values, appropriate domains and consistency with the assumptions. A mathematically correct solution may still be unsuitable if it predicts negative quantities, uses fixed parameters during a period of structural change or extrapolates far beyond the observed data.

The sign of a parameter matters. In $y' = k(L - y)$, the usual adjustment model requires $k > 0$; if $k < 0$, trajectories move away from L . In $y' = a - by$, a positive b represents proportional removal, whereas a negative b would produce self-reinforcing growth. Parameter restrictions are therefore part of the substantive model, not merely algebraic decoration.

13.9 Euler's method

Some differential equations cannot be solved conveniently in closed form. If

$$\frac{dy}{dt} = F(t, y), \quad y(t_0) = y_0,$$

Euler's method uses the current slope to approximate the next value. With step size h ,

$$t_{n+1} = t_n + h, \quad y_{n+1} = y_n + hF(t_n, y_n).$$

It replaces the solution curve locally by a tangent-line segment. Smaller steps usually improve accuracy but require more calculations.

Example 13.11: [

orange]Approximating an adjustment model For

$$\frac{dU}{dt} = 0.5(4 - U), \quad U(0) = 10,$$

use Euler's method with $h = 1$. Then

$$U_1 = 10 + 0.5(4 - 10) = 7,$$

$$U_2 = 7 + 0.5(4 - 7) = 5.5,$$

$$U_3 = 5.5 + 0.5(4 - 5.5) = 4.75.$$

Thus, $U(3) \approx 4.75$. The exact solution is $U(t) = 4 + 6e^{-0.5t}$, which gives $U(3) \approx 5.34$. The difference illustrates the numerical error produced by a relatively large step size.

The calculation can be organised in a table. At each row, the slope is evaluated using the current approximate value; it is not recalculated halfway through the step.

n	t_n	U_n	$0.5(4 - U_n)$
0	0	10.000	-3.000
1	1	7.000	-1.500
2	2	5.500	-0.750
3	3	4.750	-0.375

Euler's method has accumulated, or *global*, error because the result of one approximate step becomes the starting value for the next. Halving the step size does not guarantee an exact answer, but it commonly reduces the error for a smooth solution over a fixed interval.

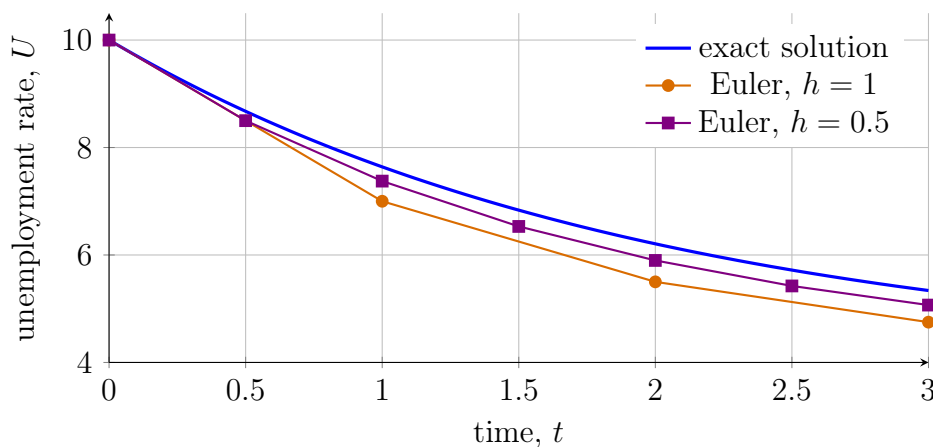


Figure 13.6: A smaller Euler step follows the curvature of the exact adjustment path more closely in this example.

13.10 A modelling workflow

A differential-equation calculation is most useful when it is embedded in a clear modelling process.

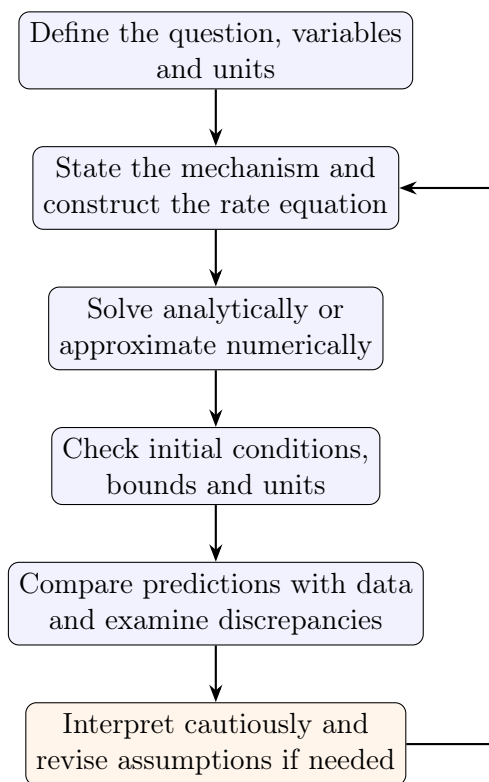


Figure 13.7: A modelling cycle for differential equations in applied social research. Empirical checking may lead to a revised rate equation.

In a written solution, the final interpretation should identify the quantity being predicted, its units, the relevant time period and at least one major assumption. Statements such as “the model predicts” are generally more appropriate than claims that the calculated trajectory must occur.

13.11 Exercises

Answers are provided in section C.13.

1. Express each equation in the form $g(y) \, dy = f(x) \, dx$.

(a) $(x^2 + 1) \frac{dy}{dx} = x\sqrt{y+1}$

(b) $(5 - x) \, dy + (y + 4) \, dx = 0$

2. Obtain the general solution of each equation in Exercise 1.

3. Find the particular solution.

(a) $\frac{dy}{dx} = 1 - \frac{2}{y}$, with $y = 3$ when $x = 3$

(b) $3y^2 \frac{dy}{dx} = 2x + 1$, with $y = 2$ when $x = 1$

4. Show that

$$(4xy + 12y) \, dx + (xy - 3x) \, dy = 0$$

is separable, and find its general solution.

5. A population $P(t)$ satisfies $dP/dt = 0.03P$ and $P(0) = 200\,000$.

(a) Find $P(t)$.

(b) Estimate the population after 10 years.

(c) Find the doubling time.

6. An adoption percentage satisfies $dA/dt = 0.4(100 - A)$ with $A(0) = 20$.

(a) Derive the particular solution.

(b) Find $A(5)$.

(c) Find when adoption first reaches 80%.

7. An unemployment-rate model is $dU/dt = 0.5(4 - U)$ with $U(0) = 10$, where rates are percentages. Find $U(t)$ and the time at which the model first reaches 5%.

8. Verify that

$$y(t) = \frac{4}{1 + 3e^{-2t}}$$

satisfies $dy/dt = 2y(1 - y/4)$ and $y(0) = 1$.

9. Solve

$$\frac{dy}{dx} = x(1 + y^2), \quad y(0) = 0,$$

and state the largest interval containing 0 on which the solution is defined.

10. For $dy/dt = y(6 - y)$, find the equilibrium solutions and classify their stability for non-negative values of y .

11. A participation percentage $S(t)$ satisfies

$$\frac{dS}{dt} = 0.4S \left(1 - \frac{S}{75} \right), \quad S(0) = 15.$$

- (a) Find $S(t)$.
 - (b) Find $S(5)$.
 - (c) Find when S first reaches 60%.
12. The number of registered users of a community service grows from 1200 to 1680 in four years. Assuming $N' = kN$:
- (a) estimate k ;
 - (b) predict the number of users after ten years;
 - (c) find the model's doubling time.

13. Use Euler's method with step size $h = 1$ to approximate $U(1)$, $U(2)$ and $U(3)$ for

$$\frac{dU}{dt} = 0.5(4 - U), \quad U(0) = 10.$$

Compare the approximation of $U(3)$ with the exact value.

14. Write a differential equation for each model and identify any equilibrium value.
- (a) A programme gains 50 participants per month, while 10% of its current participants leave each month.
 - (b) The rate at which a survey platform gains users is proportional to both its current number of users and the difference between that number and its capacity K .

Further exercises: methods and social-science models

15. For each equation, state its order, whether it is autonomous and whether it is a first-order separable equation. For an autonomous first-order equation, also find its equilibrium solutions.

- (a) $\frac{dy}{dt} = 0.6(12 - y)$
- (b) $\frac{dy}{dt} = t(1 - y^2)$
- (c) $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + y = 0$
- (d) $\frac{dy}{dt} = y(5 - y)$

16. A policy-support index is proposed to follow

$$P(t) = 80 - 50e^{-0.2t}.$$

- (a) Verify that P satisfies $P' = 0.2(80 - P)$ and $P(0) = 30$.
- (b) Find $P(5)$.
- (c) Find when the model first gives $P = 70$.
- (d) Interpret the equilibrium and the parameter 0.2.

17. Solve each initial-value problem.

- (a) $\frac{dy}{dx} = 2x(y + 3)$, $y(0) = -2$.
- (b) $\frac{dy}{dt} = \frac{t+1}{y}$, $y(0) = 2$, taking the positive branch.
- (c) $\frac{dy}{dx} = \frac{e^x}{1+y}$, $y(0) = 1$.

18. Online readership of a local newspaper falls from 40 000 to 29 000 over three months. Assume $R' = kR$.

- (a) Estimate k and state its units.
- (b) Write the fitted model.
- (c) Predict the readership after 12 months.
- (d) Find the model's half-life.
- (e) State one reason why the fitted model may be unreliable over a much longer period.

19. An index begins at 100 and changes continuously at a rate of -0.12 per year.

- (a) Write and solve its differential equation.
- (b) Find its value after five years.
- (c) Find the equivalent one-year discrete percentage change.
- (d) Explain why the answer to part (c) is not exactly -12% .

20. An approval index adjusts towards a long-run level of 68:

$$\frac{dA}{dt} = 0.25(68 - A), \quad A(0) = 44.$$

- (a) Find $A(t)$ and $A(4)$.
- (b) Find when the index first lies within two points of its equilibrium.
- (c) Interpret the sign of A' when $A < 68$ and when $A > 68$.

21. A public-service caseload receives 120 new cases per month, while 15% of the current caseload is resolved per month. Let $C(0) = 500$.

- (a) Construct the differential equation and find its equilibrium.
- (b) Solve the initial-value problem.
- (c) Estimate the caseload after 12 months.
- (d) Find when the caseload first reaches 700.
- (e) Explain why the equilibrium does not mean that no cases enter or leave the system.

22. Consider the autonomous equation

$$\frac{dy}{dt} = (y - 2)(6 - y).$$

- (a) Find and classify the equilibria.
- (b) Draw a phase line and state the direction of movement in each interval.
- (c) Describe the long-run behaviour when $y(0) = 4$.
- (d) If y represents a non-negative index, explain one concern about using the model when $0 < y(0) < 2$.

23. Adoption of a digital public service is modelled by

$$\frac{dM}{dt} = 0.3M \left(1 - \frac{M}{90} \right), \quad M(0) = 18,$$

where M is a percentage.

- (a) Find $M(t)$.
 - (b) Estimate $M(8)$.
 - (c) Find when adoption reaches 45%.
 - (d) Find the maximum rate of increase and state when it occurs.
- 24.** A logistic participation model has saturation level 100 and satisfies $P(0) = 20$ and $P(4) = 50$.
- (a) Find the logistic parameter r .
 - (b) Predict $P(8)$.
 - (c) Find when participation reaches 75.
 - (d) State why the assumed saturation level is important when estimating r .

25. Two models begin with 10 users:

$$E' = 0.4E, \quad L' = 0.4L \left(1 - \frac{L}{100} \right).$$

- (a) Find both initial rates of change.
- (b) Find $E(5)$ and $L(5)$.

- (c) Find when the exponential model first reaches 100 users.
 - (d) Does the logistic model reach 100 at a finite time?
 - (e) Explain why the models diverge even though their parameters both contain 0.4.
- 26.** An indicator follows $Y' = k(75 - Y)$. Observations give $Y(0) = 35$ and $Y(3) = 55$.
- (a) Estimate k .
 - (b) Interpret the corresponding gap half-life.
 - (c) Predict $Y(6)$.
 - (d) Identify one assumption required for this prediction.
- 27.** Translate each statement into a differential equation and identify any equilibrium values. State the units of each rate parameter.
- (a) A membership scheme gains 200 members per month, while 5% of its current members leave per month.
 - (b) A price index adjusts each year at 30% of the gap between its current value and a target of 125.
 - (c) Awareness spreads at a rate proportional to both the number aware and the number not yet aware in a population of size N .
 - (d) A trust index receives a constant campaign contribution of three points per year but otherwise decays proportionally to its current level.
- 28.** For $U' = 0.5(4 - U)$ with $U(0) = 10$, use Euler's method with $h = 0.5$ to approximate $U(0.5)$, $U(1)$, $U(1.5)$ and $U(2)$. Compare the result for $U(2)$ with both the exact value and the approximation obtained using $h = 1$.
- 29.** A participation percentage satisfies

$$P' = 0.4P \left(1 - \frac{P}{100} \right), \quad P(0) = 20.$$

Use Euler's method with $h = 1$ to estimate $P(1)$, $P(2)$ and $P(3)$. Compare $P(3)$ with the exact logistic value.

- 30.** A researcher fits an exponential model to the first two years of programme adoption and uses it to forecast the next twenty years.
- (a) Give three assumptions or limitations that should be reported.
 - (b) Describe two patterns in later data that would suggest a logistic or adjustment model instead.
 - (c) Explain why a close fit to the first two observations does not demonstrate a causal relationship.
 - (d) Suggest how the model should be reconsidered after a major policy change.

Additional practice: grouped skills review

31. For each (P_0, r) pair, solve $P' = rP$, $P(0) = P_0$, and find $P(5)$:

Case	A	B	C	D	E	F	G
P_0	1000	1200	1400	1600	1800	2000	2200
r	0.02	0.04	0.06	0.08	0.10	0.12	0.14

32. Find the half-life of $N' = -kN$ for $k = 0.10, 0.11, 0.12, 0.13, 0.14, 0.15$ and 0.16 .

33. Solve $y' = k(L - y)$, $y(0) = y_0$, and find $y(3)$ for each case.

Case	A	B	C	D	E	F	G
k	0.20	0.22	0.24	0.26	0.28	0.30	0.32
L	60	65	70	75	80	85	90
y_0	20	22	24	26	28	30	32

34. For $P' = rP(1 - P/100)$ with $P(0) = 20$, find $P(t)$ and $P(5)$ for

$$r = 0.20, 0.22, 0.24, 0.26, 0.28, 0.30, 0.32.$$

35. Find and classify the non-negative equilibria of $y' = y(K - y)$ for $K = 5, 6, 7, 8, 9, 10$ and 11 .

36. Solve $dy/dx = nxy$, $y(0) = 1$, for $n = 1, 2, 3, 4, 5, 6$ and 7 .

37. Verify the form $y = L + (y_0 - L)e^{-0.3t}$ for the pairs

Case	A	B	C	D	E	F	G
L	50	55	60	65	70	75	80
y_0	20	21	22	23	24	25	26

Show that each satisfies $y' = 0.3(L - y)$ and its stated initial condition.

38. Under $N' = rN$, find r when the doubling time is $2, 3, 4, 5, 6, 7$ and 8 years.

39. Use one Euler step of size 0.5 for $y' = k(80 - y)$ in each case.

Case	A	B	C	D	E	F	G
k	0.20	0.22	0.24	0.26	0.28	0.30	0.32
$y(0)$	20	23	26	29	32	35	38

40. Give four reasons why a constant-parameter differential-equation model may become unreliable after a major policy, economic or institutional change.

Applications: population and demographic change

41. An urban population satisfies $P' = 0.025P$ with $P(0) = 50\,000$. Find $P(t)$, estimate $P(10)$ and find the doubling time.
42. A district gains 1200 residents per year while 4% of its current population leaves each year. If $P(0) = 20\,000$:
- (a) construct the differential equation and find its equilibrium;
 - (b) solve the initial-value problem;
 - (c) estimate the population after five years.
43. A registered-voter roll declines according to $N' = -0.06N$, with $N(0) = 80\,000$. Estimate the roll after eight years and find its modelled half-life.
44. Region A begins with 100 000 people and grows continuously at 2% per year. Region B begins with 130 000 people and declines continuously at 1% per year. Find when the models predict equal populations.
45. A population index satisfies $I' = 0.05I$, $I(0) = 100$. Find the one-year index and the equivalent discrete annual percentage growth rate.
46. A community grows from 10 000 to 12 500 people in four years. Assuming $N' = kN$, estimate k , predict $N(10)$ and find the doubling time.
47. A town gains 500 residents per year while 1% of its current population leaves annually. Given $P(0) = 20\,000$, find the equilibrium, solve for $P(t)$ and estimate $P(10)$.
48. A cohort of 12 000 programme participants leaves continuously at rate 8% per year. Estimate the number remaining after five years and find when the model first falls below 8000.
49. The urbanisation percentage S adjusts towards 65 according to $S' = 0.2(65 - S)$, with $S(0) = 45$. Find $S(5)$ and the time at which S reaches 60.
50. A city of 50 000 people initially gains 1000 people per year. Compare the ten-year predictions of the linear model $L(t) = 50\,000 + 1000t$ and the exponential model $E' = 0.02E$, $E(0) = 50\,000$. Explain why the predictions diverge.

Applications: public services and administrative systems

51. A case backlog receives 200 new cases per month while 10% of the current backlog is resolved monthly. If $B(0) = 1000$, find the equilibrium, $B(t)$ and $B(6)$.
52. A service has capacity index 500 and adjusts according to $A' = 0.3(500 - A)$ with $A(0) = 200$. Find $A(4)$ and interpret the long-run level.
53. An unemployment-rate model is $U' = 0.25(5 - U)$ with $U(0) = 9$. Find $U(t)$ and the time at which the model reaches 6%.

54. Public approval satisfies $A' = 0.18(70 - A)$, $A(0) = 40$. Find $A(6)$ and the time at which approval first lies within five points of its equilibrium.
55. A complaints system receives 90 new complaints per month and resolves 15% of its current stock each month. If $C(0) = 800$, find the equilibrium, $C(12)$ and the direction of movement.
56. Staff vacancies satisfy $V' = 40 - 0.2V$, $V(0) = 300$. Find the equilibrium, solve the model and estimate $V(5)$.
57. A trust index satisfies $T' = 3 - 0.1T$, $T(0) = 20$. Find its equilibrium and value after ten years. Interpret the two terms in the rate equation.
58. A hospital waiting list satisfies $W' = 150 - 0.05W$ with $W(0) = 2000$. Find the equilibrium and the time at which the list reaches 2500.
59. A budget index adjusts towards 120 according to $B' = 0.4(120 - B)$, $B(0) = 80$. Find $B(3)$ and the half-life of the remaining gap from the target.
60. Two service indices begin at 40 and adjust towards 100. Model A has rate parameter 0.2 and Model B has rate parameter 0.5. Find both values after four periods and compare their speeds of adjustment.

Applications: political and economic adjustment

61. An inflation rate satisfies $I' = 0.3(2 - I)$ with $I(0) = 8$. Find $I(t)$ and the time at which the rate reaches 3%.
62. A rent index adjusts towards 100 according to $R' = 0.15(100 - R)$, $R(0) = 70$. Find $R(5)$ and the time at which it reaches 90.
63. Public debt, in millions of ringgit, satisfies $D' = 0.04D + 2$ with $D(0) = 50$. Solve the model, estimate $D(10)$ and explain why there is no non-negative equilibrium.
64. A consumption index satisfies $C' = 0.25(60 - C)$, $C(0) = 45$. Find $C(4)$ and interpret the sign of C' above and below 60.
65. Consider $y' = 0.01(y - 20)(80 - y)$.
- find and classify its equilibria;
 - describe the phase-line directions;
 - state one concern if y is required to remain non-negative and $0 < y(0) < 20$.
66. An inequality index adjusts towards 40 according to $G' = 0.1(40 - G)$, $G(0) = 55$. Find $G(5)$ and interpret the modelled direction of change.
67. Before a policy change, an index satisfies $Y' = 0.2(100 - Y)$ with $Y(0) = 60$. At $t = 3$, the target changes instantly to 80 while the rate parameter remains 0.2. Find $Y(3)$ and $Y(6)$.

68. A tax-compliance percentage satisfies $C' = 0.4(95 - C)$ with $C(0) = 65$. Find when the model first reaches 90%.
69. For $y' = 0.02(y - 40)(70 - y)$, find and classify the equilibria and describe the long-run behaviour when $y(0) = 50$.
70. For each proposed model, identify the equilibrium and state whether the equation directs nearby non-negative values towards or away from it:

$$H' = 50 - 0.1H, \quad S' = 0.2S(1 - S/100), \quad R' = -0.05R, \quad T' = 2 + 0.01T.$$

Applications: diffusion and capacity constraints

71. Adoption follows $A' = 0.5A(1 - A/100)$ with $A(0) = 10$. Find $A(t)$, $A(5)$ and the time at which adoption reaches 50.
72. Participation follows $P' = 0.3P(1 - P/80)$ with $P(0) = 20$. Find $P(6)$ and the maximum growth rate.
73. The proportion aware of a campaign satisfies $p' = 0.8p(1 - p)$, $p(0) = 0.1$. Find $p(4)$ and interpret the result as a percentage.
74. A logistic model has capacity 100, $P(0) = 20$ and $P(3) = 50$. Estimate its rate parameter and predict $P(6)$.
75. Exponential and logistic models both begin at 10 with rate parameter 0.3; the logistic capacity is 100. Find both predictions at $t = 10$ and explain their difference.
76. Under $P' = 0.4P(1 - P/100)$, find the time required for participation to rise from 20 to 80.
77. A logistic membership model has capacity 2000 and rate parameter 0.6. Find the membership level and rate of change at the point of maximum growth.
78. Explain how an incorrectly specified capacity changes logistic estimates of the rate parameter, inflexion point and long-run prediction.
79. Market share follows $S' = 0.25S(1 - S/70)$ with $S(0) = 14$. Find $S(8)$ and interpret the saturation level.
80. Verify that

$$P(t) = \frac{K}{1 + Ae^{-rt}}$$

satisfies $P' = rP(1 - P/K)$. Express A in terms of K and $P(0)$.

Applications: numerical approximation and data checking

81. Use Euler's method with $h = 1$ for $y' = 0.3(50 - y)$, $y(0) = 10$. Estimate the values at $t = 1, 2, 3$, then compare the final estimate with the exact value.
82. Repeat Exercise 81 using $h = 0.5$ to estimate $y(3)$. Compare the error with the $h = 1$ result.
83. Use Euler's method with $h = 1$ for $p' = 0.5p(1 - p/100)$, $p(0) = 10$, to estimate $p(1), p(2), p(3)$. Compare $p(3)$ with the exact logistic value.
84. Explain why a smaller Euler step usually improves accuracy but increases computational work. State why a smaller step does not correct a poorly specified differential equation.
85. A caseload model is $C' = 100 - 0.05C$ with $C(0) = 500$. Use one Euler step of length one to predict $C(1)$. If the observed value is 560, calculate the residual and suggest one possible interpretation.
86. User counts are $N(0) = 1000$ and $N(2) = 1210$. Fit $N' = kN$ and predict $N(4)$. If the observed value is 1400, find the residual.
87. An adjustment model has target 100, with $Y(0) = 40$ and $Y(2) = 70$. Estimate k in $Y' = k(100 - Y)$ and interpret the gap half-life.
88. Euler's method with a large step can overshoot an equilibrium. For $y' = 2(1 - y)$, $y(0) = 0$, compare one step with $h = 1$ and one step with $h = 0.25$.
89. Distinguish local truncation error, accumulated numerical error and model error. Explain which of these is reduced by decreasing the Euler step size.
90. Write Euler recursions for exponential, adjustment and logistic models. State one units or bounds check that should accompany each numerical calculation.

Applications: model evaluation and responsible interpretation

91. A fitted model shows programme participation increasing over time. Explain why this does not establish that the programme caused the increase, and identify three possible confounding influences.
92. Before a reform, approval satisfies $A' = 0.2(70 - A)$ with $A(0) = 40$. At $t = 4$, the target changes to 85 and the rate parameter to 0.3. Construct the piecewise model and find $A(4)$ and $A(7)$.
93. Explain why an equilibrium represents balanced flows rather than an absence of activity. Illustrate using $C' = 120 - 0.15C$.
94. An exponential population model starts at 100 000 with rate 2% per year. Find its fifty-year prediction and give three reasons why such a long extrapolation may be unreliable.

95. Logistic models with the same rate 0.3 and initial value 20 use capacities 100 and 120. Compare their predictions at $t = 5$ and explain the sensitivity to the assumed capacity.
96. Explain why two observations are generally insufficient to choose confidently among exponential, logistic and adjustment models.
97. Give four empirical or substantive criteria for choosing between exponential, adjustment, logistic and piecewise models.
98. For $P' = 0.2P(1 - P/100)$ with P measured as a percentage, check the units, equilibria and direction of change. State the meaningful state space.
99. A differential-equation forecast is used to allocate public funds. Give four ethical or governance safeguards that should accompany the calculation.
100. Give a six-stage checklist for constructing, solving, checking and communicating a social-science differential-equation model.

Chapter 14

Financial Mathematics

Chapter guide

Follow cash flows through interest accumulation and discounting, then study annuities, loans, investment appraisal, inflation and depreciation with explicit timelines and units.

Selected background sources: (Cambridge University Press & Assessment, n.d.; London School of Economics and Political Science, n.d.).

Financial mathematics studies how money changes value over time. Interest, inflation, regular saving, borrowing and depreciation all involve sequences of cash flows whose timing matters. Two amounts can be compared directly only after they have been valued at the same date using a suitable interest or discount rate.

The methods in this chapter support decisions involving savings goals, consumer credit, mortgages, pensions and public or business projects. They are mathematical models rather than descriptions of every contract. Fees, taxes, changing rates, missed payments, credit risk and non-monetary effects may alter an actual outcome. Unless otherwise stated, rates are assumed to remain fixed over the given period and payments are made exactly when due.

Careful work begins by matching the rate to the payment period, drawing a cash-flow timeline and choosing a focal date. Monetary answers should be rounded only at the end of a calculation so that repeated rounding does not distort long sequences of payments.

14.1 The time value of money and cash flows

An amount invested for one period at rate i grows by the *accumulation factor* $1 + i$. Conversely, the present value of one unit due one period later is found using the *discount factor*

$$v = \frac{1}{1+i} = (1+i)^{-1}.$$

After n periods, the corresponding factors are $(1+i)^n$ and v^n .

A cash flow received is commonly shown as positive and a payment made as negative, although another sign convention may be used if it is applied consistently. A *focal date* is the date at which all cash flows are compared. Moving money forward multiplies by an accumulation factor; moving it backward multiplies by a discount factor.

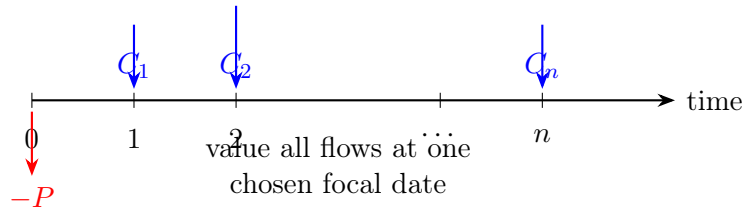


Figure 14.1: A cash-flow timeline records both the amount and timing of each payment.

14.2 Simple and compound interest

Under simple interest, interest is calculated only on the original principal P . If the rate per period is i for n periods, then

$$I = Pin, \quad A = P(1 + in),$$

where I is the interest earned and A is the accumulated amount.

Under compound interest, each period's interest is added to the balance and earns interest in later periods:

$$A = P(1 + i)^n.$$

Compound growth is therefore geometric rather than linear. When comparing financial products, the interest period and any fees must be considered as well as the quoted annual rate.

If a nominal annual rate j is compounded m times per year for t years, then

$$A = P \left(1 + \frac{j}{m} \right)^{mt}.$$

Under continuous compounding at annual rate r ,

$$A = Pe^{rt}.$$

If the rates vary from period to period, the accumulated amount is

$$A = P(1 + i_1)(1 + i_2) \cdots (1 + i_n).$$

An average of the rates cannot generally replace this product because compound changes act multiplicatively.

Example 14.1: [

blue]Comparing simple and compound interest An investment of RM 10 000 earns 5% per year for four years. With simple interest,

$$A = 10\,000[1 + 0.05(4)] = \text{RM } 12\,000.$$

With annual compounding,

$$A = 10\,000(1.05)^4 \approx \text{RM } 12\,155.06.$$

The difference arises because compound interest earns interest on earlier interest.

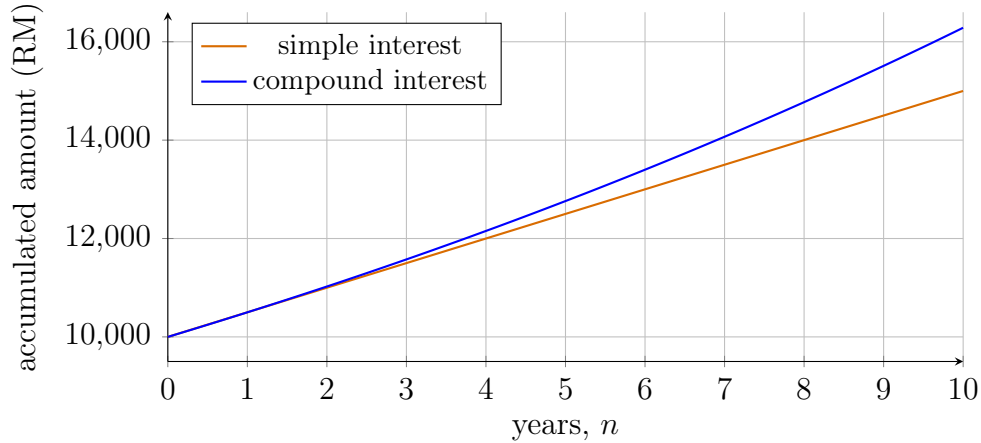


Figure 14.2: At the same positive annual rate, compound interest increasingly exceeds simple interest as the investment period grows.

14.3 Present value and effective rates

The present value of a future amount F due after n periods is

$$P = F(1 + i)^{-n} = \frac{F}{(1 + i)^n}.$$

Present value allows cash flows occurring at different dates to be compared at a common point in time.

If a nominal annual rate j is compounded m times per year, the rate per conversion period is $i = j/m$. The effective annual rate is

$$i_{\text{eff}} = \left(1 + \frac{j}{m}\right)^m - 1.$$

The effective rate is the actual proportional increase over one year after allowing for compounding.

Two rates are *equivalent* when they produce the same accumulation over the same time interval. If i_{eff} is an effective annual rate, the equivalent nominal annual rate convertible m times per year is

$$j = m \left[(1 + i_{\text{eff}})^{1/m} - 1 \right].$$

For a continuously compounded annual rate r , the effective annual rate is

$$e^r - 1.$$

Quoted rates should not be compared until their compounding conventions are made consistent. An account quoting 4.8% nominal compounded monthly has effective annual rate

$$(1.004)^{12} - 1 \approx 4.907\%,$$

which is slightly greater than an account quoting 4.9% effective annually, before considering fees or other conditions.

14.4 Equations of value

An *equation of value* states that the values of two sets of cash flows are equal at a chosen focal date. If the focal date is time 0, a sequence of future payments $C_1, \dots, C_n$ has present value

$$V_0 = \sum_{k=1}^n C_k v^k.$$

If the focal date is time n , earlier cash flows are accumulated forward:

$$V_n = \sum_{k=0}^n C_k (1+i)^{n-k}.$$

The same focal date must be used for every term.

Example 14.2: [

cyan]Replacing two debts by one payment A payment of RM 5000 due at the end of year 2 and RM 7000 due at the end of year 5 are replaced by a single payment X at the end of year 3. At an effective annual rate of 5%, value all three amounts at year 3:

$$X = 5000(1.05) + 7000(1.05)^{-2}.$$

Therefore,

$$X \approx \boxed{\text{RM } 11\,599.21}.$$

The earlier debt is accumulated one year, while the later debt is discounted two years.

14.5 Annuities and regular saving

An *ordinary annuity* consists of equal payments R made at the end of each period. Its accumulated value immediately after the n th payment is

$$S_n = R \frac{(1+i)^n - 1}{i} = Rs_{\overline{n}|i}.$$

Its present value one period before the first payment is

$$V_0 = R \frac{1 - (1 + i)^{-n}}{i} = Ra_{\overline{n}|i}.$$

Here $s_{\overline{n}|i}$ is the accumulated-value annuity factor and $a_{\overline{n}|i}$ is the present-value annuity factor.

For an annuity due, payments occur at the beginning of each period, so the corresponding ordinary-annuity value is multiplied by $1 + i$.

If an ordinary annuity is deferred for d periods, first find its value one period before its first payment and then discount that value through the deferment period. In the usual convention,

$$V_0 = v^d Ra_{\overline{n}|i}.$$

A level perpetuity pays R at the end of every period without a final payment date. If $i > 0$, its present value is

$$V_0 = \frac{R}{i}.$$

This formula assumes that the first payment is one period from now and that the rate and payments continue indefinitely.

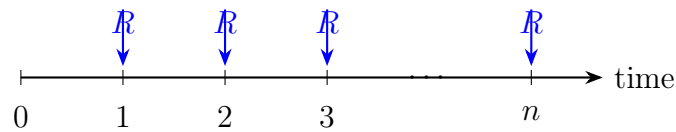


Figure 14.3: End-of-period payments in an ordinary annuity.

Example 14.3: [

violet]A regular savings plan RM 300 is deposited at the end of each month for three years into an account paying a nominal annual rate of 4.8% compounded monthly. Here $i = 0.048/12 = 0.004$ and $n = 36$, so

$$S_{36} = 300 \frac{(1.004)^{36} - 1}{0.004} \approx \text{RM } 11\,591.43.$$

The total deposits are RM 10 800; the remainder is interest.

To accumulate a target amount F using end-of-period deposits, rearrange the ordinary annuity formula:

$$R = F \frac{i}{(1 + i)^n - 1}.$$

This is often called a sinking-fund calculation.

Example 14.4: [

blue]Monthly deposits for a savings target A student wants to accumulate RM 50 000 in five years using end-of-month deposits. The account pays a nominal annual rate of 3.6% compounded monthly, so $i = 0.003$ and $n = 60$. The required deposit is

$$R = 50\,000 \frac{0.003}{(1.003)^{60} - 1} \approx \boxed{\text{RM } 761.83}.$$

If the deposits were made at the beginning of each month, the required amount would be divided by 1.003, giving approximately RM 759.55.

14.6 Loans and amortisation

If a loan of present value P is repaid by n equal end-of-period payments, then the payment is

$$R = P \frac{i}{1 - (1 + i)^{-n}}.$$

After k payments, the outstanding balance can be found from

$$B_k = P(1 + i)^k - R \frac{(1 + i)^k - 1}{i}.$$

This *retrospective method* accumulates the original loan and subtracts the accumulated value of the payments already made. The equivalent *prospective method* values the remaining $n - k$ payments at time k :

$$B_k = R \frac{1 - (1 + i)^{-(n-k)}}{i}.$$

Both methods should give the same result when the same rate and payment are used.

An amortisation payment contains both interest and principal. Early payments usually contain a larger interest component because the outstanding balance is larger.

If B_{k-1} is the balance immediately after payment $k - 1$, then the interest and principal portions of payment k are

$$I_k = iB_{k-1}, \quad Q_k = R - I_k.$$

After the payment, the new balance is $B_k = B_{k-1} - Q_k$.

Example 14.5: [

orange]Monthly repayment on a loan A loan of RM 120 000 is repaid monthly over ten years at a nominal annual rate of 6% compounded monthly. With $i = 0.005$ and $n = 120$,

$$R = 120\,000 \frac{0.005}{1 - (1.005)^{-120}} \approx \text{RM } 1\,332.25.$$

The formula excludes additional fees, insurance and changes in interest rates that may occur in an actual loan contract.

Example 14.6: [

violet]Outstanding balance and payment components For the RM 120 000 loan above, the balance immediately after 36 monthly payments is

$$B_{36} = R \frac{1 - (1.005)^{-84}}{0.005} \approx \boxed{\text{RM } 91\,196.29}.$$

The interest charged before payment 37 is approximately

$$0.005B_{36} = \text{RM } 455.98.$$

Thus approximately RM 876.26 of payment 37 reduces the principal. The principal component grows as the balance and interest component fall.

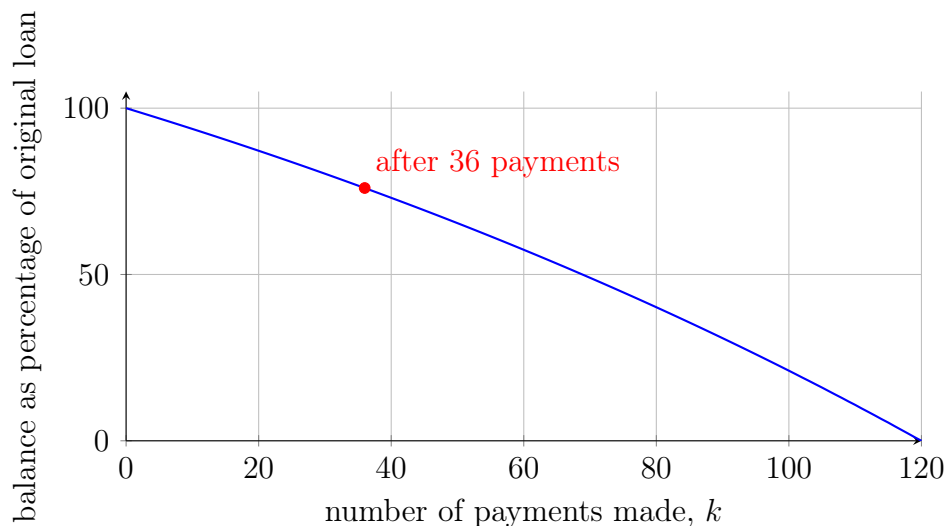


Figure 14.4: A level-payment loan balance often falls slowly at first because early payments contain a larger interest component.

14.7 Investment appraisal

The *net present value* (NPV) of a project is the present value of its future net cash flows minus its initial cost. If C_0 is the time-0 cash flow and C_t is the net cash flow at time t , then

$$\text{NPV} = C_0 + \sum_{t=1}^n \frac{C_t}{(1+i)^t}.$$

An NPV above zero means that the discounted inflows exceed the discounted outflows at the chosen rate. The conclusion is conditional on the cash-flow estimates, discount rate

and project horizon.

The *internal rate of return* (IRR) is a rate that makes the NPV equal to zero. It is normally found numerically. Some non-standard cash-flow patterns may have more than one IRR or no useful IRR, so NPV is often the more direct decision measure.

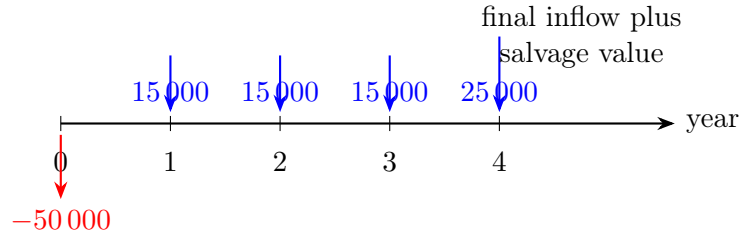


Figure 14.5: A project cash-flow timeline. Each future amount must be discounted according to its payment date.

Example 14.7: [

blue]Net present value of a project A project costs RM 50 000 initially, produces RM 15 000 at the end of each of four years and has an additional RM 10 000 salvage value at the end of year 4. At a discount rate of 8%,

$$\begin{aligned} \text{NPV} &= -50\,000 + 15\,000 \frac{1 - (1.08)^{-4}}{0.08} + 10\,000(1.08)^{-4} \\ &\approx \boxed{\text{RM } 7032.20}. \end{aligned}$$

The project has positive NPV under the stated assumptions. This financial calculation does not measure externalities, distributional effects or other non-monetary consequences.

14.8 Inflation, real change and depreciation

If a monetary quantity grows at nominal rate g while prices rise at inflation rate r , its exact real growth rate is

$$g_{\text{real}} = \frac{1 + g}{1 + r} - 1.$$

The approximation $g - r$ is convenient when both rates are small, but the fractional formula is more accurate.

Reducing-balance depreciation at rate d gives

$$V_n = V_0(1 - d)^n.$$

This is an exponential model. It assumes the same proportional loss in every period and may not represent unusual market conditions or maintenance costs.

Under straight-line depreciation, the same monetary amount is removed each period. If an asset costs V_0 , has residual value S after N periods and annual depreciation amount

D , then

$$D = \frac{V_0 - S}{N}, \quad V_n = V_0 - nD.$$

Straight-line and reducing-balance methods allocate the loss differently and need not approximate market value equally well.

Example 14.8: [

cyan]Nominal and real wage growth If wages rise by 5% while prices rise by 3%, then

$$g_{\text{real}} = \frac{1.05}{1.03} - 1 \approx 0.0194.$$

Real purchasing power rises by approximately 1.94%, slightly less than the simple difference of two percentage points.

Example 14.9: [

orange]Comparing depreciation methods An asset costs RM 80 000 and is expected to have residual value RM 20 000 after five years. Straight-line depreciation is

$$D = \frac{80\,000 - 20\,000}{5} = \text{RM } 12\,000$$

per year, giving book value RM 44 000 after three years.

The reducing-balance rate that reaches the same five-year residual value satisfies

$$80\,000(1 - d)^5 = 20\,000,$$

so

$$d = 1 - (0.25)^{1/5} \approx \boxed{24.21\%}.$$

Under this method, the three-year value is approximately RM 34 822.02.

14.9 Interpreting financial models

Financial formulas can be evaluated exactly while still giving a misleading comparison if important features are omitted. Before drawing a conclusion, check:

- whether quoted rates are nominal, effective or continuous;
- whether the rate period matches the payment period;
- whether payments occur at the beginning or end of each period;
- whether fees, taxes, penalties or variable rates are excluded;
- whether inflation and purchasing power matter;
- whether a project has social costs or benefits outside its cash flows.

A smaller periodic loan payment does not necessarily imply a cheaper loan, because a longer term can produce substantially more total interest. Likewise, the project with the highest financial NPV need not be the project with the greatest social value.

14.10 Exercises

Answers are provided in section C.14.

1. RM 5000 is invested for three years at 4% simple interest. Find the interest earned and the final amount.
2. RM 8000 is invested for six years at 5% per year compounded annually. Find the accumulated amount.
3. Find the present value of RM 20 000 due in four years if money is valued at 6% per year compounded annually.
4. A bank quotes a nominal annual rate of 4.8% compounded monthly. Find the effective annual rate.
5. RM 300 is deposited at the end of every month for three years at a nominal annual rate of 4.8% compounded monthly. Find the accumulated value and the interest earned.
6. A loan of RM 120 000 is amortised by equal monthly payments over ten years at a nominal annual rate of 6% compounded monthly. Find the monthly payment and the total of all payments.
7. A vehicle valued at RM 90 000 depreciates by 15% each year on a reducing-balance basis. Find its modelled value after five years and the total loss in value.
8. An annual salary of RM 48 000 rises by 6% while the price level rises by 4%.
 - (a) Find the exact real percentage increase.
 - (b) Express the new salary in the purchasing power of the original year's money.

Further exercises: cash flows and financial decisions

9. RM 12 000 is invested for six years at 4.5% per year.
 - (a) Find the amount under simple interest.
 - (b) Find the amount under annual compound interest.
 - (c) Find the amount under continuous compounding.
 - (d) Explain why the three values differ.
10. A lender quotes a nominal annual rate of 5.4% compounded monthly.
 - (a) Find the effective annual rate.

- (b) Find the equivalent nominal annual rate compounded quarterly.
11. RM 20 000 earns successive annual returns of 4%, 5.5%, -2% and 6%.
- (a) Find the value after four years.
- (b) Find the overall percentage return.
- (c) Find the equivalent constant annual effective rate.
12. A payment of RM 5000 due at the end of year 2 and RM 7000 due at the end of year 5 are replaced by one payment at the end of year 3. Find the replacement payment at an effective annual rate of 5%.
13. An account will pay RM 4000 at the end of each of years 1, 2 and 3, followed by RM 8000 at the end of year 4. Find the present value at an effective annual rate of 6%.
14. A saver wants RM 50 000 after five years. The account pays a nominal annual rate of 3.6% compounded monthly.
- (a) Find the required end-of-month deposit.
- (b) Find the required beginning-of-month deposit.
- (c) Explain why the second deposit is smaller.
15. A scholarship pays RM 6000 at the end of years 5, 6, 7 and 8. Find its present value at time 0 using an effective annual rate of 5%.
16. A fund promises RM 2400 at the end of every year indefinitely, with the first payment one year from now. Find its present value at an effective annual rate of 4.8%.
17. A loan of RM 180 000 is repaid monthly over 20 years at a nominal annual rate of 5.4% compounded monthly.
- (a) Find the monthly payment.
- (b) Find the balance immediately after 60 payments.
- (c) Find the interest and principal portions of payment 61.
18. A RM 100 000 loan is offered at a nominal annual rate of 6% compounded monthly. Compare repayment terms of five years and ten years by finding, for each term,
- (a) the monthly payment;
- (b) the total of all payments;
- (c) the total interest paid.
19. For the RM 120 000, ten-year loan in the worked example, find the balance after 24 payments using both the retrospective and prospective methods. Verify that the results agree.

20. A liability of RM 100 000 is due in eight years. Equal deposits are made at the end of each year into a fund earning 5% per year. Find the required annual deposit.
21. A public-service project costs RM 80 000 initially and produces net cash inflows of RM 25 000 at the end of each of four years. It also has a RM 5000 salvage value at the end of year 4.
- (a) Calculate its NPV at a discount rate of 7%.
 - (b) State the financial decision suggested by the NPV.
 - (c) Give one non-financial consideration that the calculation does not capture.
22. Two community projects each cost RM 40 000 initially. Project A returns RM 15 000 at the end of each of three years. Project B returns RM 8000, RM 14 000 and RM 25 000 at the ends of years 1, 2 and 3, respectively.
- (a) Find the NPV of each project at 6%.
 - (b) Which project has the greater financial NPV?
 - (c) Explain why this comparison need not determine the socially preferable project.
23. A project costs RM 10 000 and returns RM 6000 at the end of each of the next two years. Find its positive internal rate of return.
24. A salary of RM 60 000 rises by 7% while the price level rises by 4.5%.
- (a) Find the exact real percentage increase.
 - (b) Express the new salary in the purchasing power of the original year.
25. An asset costs RM 80 000 and is expected to have residual value RM 20 000 after five years.
- (a) Find the annual straight-line depreciation and the book value after three years.
 - (b) Find the reducing-balance rate that gives the same five-year residual value.
 - (c) Find the value after three years under reducing-balance depreciation.
26. Find the present value of RM 25 000 due in seven years under a continuously compounded discount rate of 4% per year.
27. A loan advertisement emphasises a low monthly repayment obtained by extending the repayment term. Give three quantities or conditions that should be examined before concluding that the loan is cheaper.

Additional practice: grouped skills review

28. Find the simple interest earned over three years in each case.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
<i>P</i> (RM)	1000	1500	2000	2500	3000	3500	4000	4500
<i>r</i>	4%	4.5%	5%	5.5%	6%	6.5%	7%	7.5%

29. Find each accumulated value after five years with annual compounding.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
<i>P</i> (RM)	2000	2500	3000	3500	4000	4500	5000	5500
<i>r</i>	4%	4.5%	5%	5.5%	6%	6.5%	7%	7.5%

30. At an annual effective rate of 5%, find the present value of amounts RM 5000, RM 5500, RM 6000, RM 6500, RM 7000, RM 7500, RM 8000 and RM 8500, each due in four years.

31. Find the effective annual rate equivalent to each nominal annual rate compounded monthly: 3.6%, 4.2%, 4.8%, 5.4%, 6.0%, 6.6% and 7.2%.

32. Find the accumulated value of twelve end-of-month deposits at 0.5% per month when each deposit is RM 100, RM 120, RM 140, RM 160, RM 180, RM 200 and RM 220.

33. Loans of RM 10 000, RM 11 000, RM 12 000, RM 13 000, RM 14 000, RM 15 000 and RM 16 000 are repaid by 24 equal end-of-month payments at 0.6% per month. Find each monthly payment.

34. For each loan in Exercise 33, find the outstanding balance immediately after payment 12.

35. Assets initially have values

RM 20 000, RM 22 000, RM 24 000, RM 26 000, RM 28 000, RM 30 000, RM 32 000.

Each asset depreciates by 10% per year. Find the values after five years.

36. Prices rise by 3%. Find the exact real-income growth rate when nominal income rises by 5%, 5.5%, 6%, 6.5%, 7%, 7.5% and 8%, respectively.

37. Each project returns the stated amount at the end of each of three years. Find its NPV at 6%.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
Initial cost (RM)	10000	11000	12000	13000	14000	15000	16000
Annual return (RM)	4000	4500	5000	5500	6000	6500	7000

Applications: household saving and borrowing

- 38.** A household places RM 5000 in an emergency fund for three years at 4% per year.
- (a) Find the amount under simple interest.
 - (b) Find the amount with annual compounding.
 - (c) Find the additional amount produced by compounding.
- 39.** A worker deposits RM 300 at the end of every month for 24 months into an account earning 0.4% per month.
- (a) Find the accumulated value.
 - (b) Find total deposits and interest earned.
 - (c) State one reason the modelled value may differ from an actual account balance.
- 40.** Repeat Exercise 39 when deposits are made at the beginning of each month. Compare the two accumulated values and explain the difference.
- 41.** A household wants RM 20 000 after 36 months. The account earns 0.45% per month.
- (a) Find the required end-of-month deposit.
 - (b) Find the required beginning-of-month deposit.
 - (c) Explain why the second amount is smaller.
- 42.** A RM 30 000 personal loan is repaid monthly over five years at 0.5% per month.
- (a) Find the monthly payment.
 - (b) Find total repayments and total interest.
 - (c) State one cost that may not be included in this calculation.
- 43.** For the loan in Exercise 42:
- (a) find the balance immediately after payment 24;
 - (b) split payment 25 into interest and principal;
 - (c) explain why the interest component declines over time.
- 44.** A computer has cash price RM 5000. An instalment plan requires RM 500 immediately and 24 monthly payments of RM 220.
- (a) Find the total paid and the finance charge above the cash price.
 - (b) Explain why the finance charge divided by RM 5000 is not a valid annual interest rate.
- 45.** Payments of RM 3000 due in one year and RM 5000 due in three years are replaced by one payment at the end of year 2. Find the replacement payment using an annual effective rate of 6%.

46. A savings account earns 5%, then loses 3%, then earns 8% over three successive years. Find the final value of RM 10 000 and the overall percentage return. Explain why the rates should not simply be added.
47. Compare RM 10 000 invested for five years at 4% compounded annually with 4% compounded continuously. Find both values and their difference.
48. University tuition costs RM 50 000 today and is expected to rise by 4% annually for eight years.
- (a) Estimate the future tuition cost.
 - (b) Find the lump sum required today if savings earn 6% annually.
 - (c) State one uncertainty ignored by the calculation.

Applications: housing, education and retirement

49. A RM 250 000 mortgage is repaid monthly over 20 years at 0.45% per month. Find the monthly payment, total repayments and total interest.
50. Compare 15-year and 25-year repayment periods for the same RM 250 000 mortgage at 0.45% per month. Find each payment and total interest, then describe the trade-off.
51. For the 20-year mortgage in Exercise 49, find the balance after 60 payments and split payment 61 into interest and principal.
52. A house costs RM 300 000. A buyer pays a 10% deposit and RM 5000 in initial fees, borrowing the remainder for 20 years at 0.45% per month.
- (a) Find the initial cash required and the loan principal.
 - (b) Find the monthly mortgage payment.
 - (c) Explain why affordability should not be judged from the payment alone.
53. A RM 40 000 education loan is repaid monthly over eight years at 0.35% per month. Find the payment and total interest.
54. An institution needs RM 60 000 in five years to replace computer equipment. Find the required end-of-quarter deposit if the fund earns 1% per quarter.
55. A worker deposits RM 6000 at the end of each year for 25 years at 5% annually.
- (a) Find the retirement fund.
 - (b) Recalculate for beginning-of-year deposits.
 - (c) Find the advantage of depositing at the beginning of each year.
56. Find the fund required at retirement to provide RM 30 000 at the end of each year for 15 years when the fund earns 4%.

57. A scholarship endowment must pay RM 3000 at the end of every year indefinitely. Find the required fund at an annual rate of 5%.
58. An endowment pays RM 5000 immediately and then RM 5000 at the beginning of every following year indefinitely. Find its value at an annual rate of 5%.
59. An investment earns 7% while inflation is 4%.
- (a) Find the exact real return.
 - (b) Find the purchasing-power value after five years of an initial RM 100 000.
 - (c) Explain why subtracting the rates is only approximate.

Applications: public finance and social programmes

60. A social programme promises RM 2 million at the end of each year for ten years. Find the present value at 4%.
61. A local authority deposits RM 500 000 at each year-end for six years into a maintenance reserve earning 3%. Find the value of the reserve after the sixth deposit.
62. An infrastructure project costs RM 10 million now and produces benefits valued at RM 2 million at each year-end for seven years. Find its NPV at 5% and state the financial decision indicated.
63. At a discount rate of 6%, compare:
- Project A: cost RM 8 million, annual benefit RM 1.9 million for six years;
 - Project B: cost RM 10 million, annual benefit RM 2.5 million for six years.

Find both NPVs and state which has the larger financial NPV.

64. A government payment of RM 20 million is due in five years. Find its present value at 4% and interpret the result.
65. A grant pays RM 1 million annually for five years, with the first payment at the end of year 4. Find its present value at 5%.
66. A monthly benefit is RM 1800. It is fully adjusted for successive annual inflation rates of 3%, 4% and 5%. Find the benefit after all three adjustments and the cumulative percentage increase.
67. A public-sector wage rises by 5% while prices rise by 6.5%. Find the exact real-wage change and interpret it.
68. A programme costs RM 2 million today and its cost grows by 3% annually. Estimate its cost after four years and state one reason actual costs may differ.
69. A subsidised housing loan of RM 200 000 charges 0.25% per month for 20 years. A market loan charges 0.45% per month. Find each payment and total interest, and quantify the subsidy to the borrower under these assumptions.

70. A programme costs RM 12 000 now and returns RM 7000 at the end of each of the next two years. Find its internal rate of return and explain what the rate represents.

Applications: organisations and project appraisal

71. Equipment costs RM 100 000, has residual value RM 20 000 and useful life eight years. Find annual straight-line depreciation and book value after three years.
72. An asset worth RM 100 000 depreciates by 20% yearly. Find its value and accumulated depreciation after three years.
73. An asset is reduced from RM 80 000 to RM 25 000 in five years. Find the annual reducing-balance depreciation rate by solving

$$80\,000(1 - d)^5 = 25\,000.$$

74. An asset costs RM 80 000, has residual value RM 16 000 and useful life eight years. Compare its book value after four years under straight-line depreciation and under the constant reducing-balance rate that reaches the residual value after eight years.
75. A project costs RM 50 000, returns RM 15 000 at each year-end for four years and RM 20 000 at the end of year 5. Find its NPV at 8%.
76. At 7%, compare Project A, costing RM 40 000 and returning RM 12 000 annually for five years, with Project B, costing RM 50 000 and returning RM 16 000 annually for five years.
77. A public project costs RM 5 million now, requires RM 0.2 million at each year-end for five years and produces benefits of RM 1.4 million at each year-end for five years. At 4%, find the NPV and benefit–cost ratio.
78. A project costs RM 20 000 and returns RM 8000 at each year-end for three years.
- (a) Find the undiscounted payback period.
 - (b) Find NPV at 10%.
 - (c) Explain why the two criteria may give different impressions.
79. Equipment may be bought for RM 30 000, with RM 2000 maintenance at each year-end for four years and resale value RM 8000 at year 4. Alternatively, it may be leased for RM 9000 at each year-end for four years. Compare present costs at 6%.
80. An endowment contains RM 2 million and earns 4.5% annually. Find the perpetual annual grant supported without reducing capital. Discuss the effect of paying RM 100 000 annually instead.

Applications: inflation, depreciation and rate sensitivity

81. A salary rises by 6% while prices rise by 4%. Find exact real salary growth and compare it with the subtraction approximation.
82. An investment earns 8% while inflation is 5%. Find its exact real return.
83. A price rises by 12% and then falls by 10%. Find the overall percentage change and explain the asymmetry.
84. A household has RM 50 000 when the price index is 125 relative to the relevant base. Express the amount in base-period purchasing power.
85. An asset worth RM 150 000 depreciates by 12% yearly. Find its value and total loss after four years.
86. Equipment costs RM 60 000, has residual value RM 12 000 and useful life six years.
 - (a) Find annual straight-line depreciation.
 - (b) Find the equivalent constant reducing-balance rate.
 - (c) Find the reducing-balance value after three years.
87. Inflation acts continuously at 3% per year. Find the nominal amount required after ten years to equal the purchasing power of RM 10 000 today.
88. A nominal discount rate is 8% and expected inflation is 3%. Find the exact real discount rate.
89. A project costs RM 50 000 and returns RM 14 000 at each year-end for five years. Calculate its NPV at 5% and 8% and explain the effect of the discount rate.
90. Compare RM 100 000 loans repaid monthly over ten years at nominal annual rates of 4% and 6%, compounded monthly. Find each payment and total interest.

Applications: evaluating financial claims and decisions

91. An advertisement quotes “12% per year, compounded monthly”. Find the effective annual rate and explain why it exceeds 12%.
92. A three-year RM 10 000 loan charges flat interest of 5% of the original principal each year.
 - (a) Find total interest, total repayment and the monthly payment.
 - (b) Explain why the quoted flat rate understates the cost relative to a reducing-balance loan rate.
93. An investment earns 4% during a year, after which a management fee equal to 1% of the resulting balance is deducted. Find the net annual return and compare it with the rough subtraction $4\% - 1\%$.

-
94. A borrower receives RM 200 and repays RM 230 after two weeks. Find the two-week rate, the simple annualised rate using 26 two-week periods and the effective annual rate if the transaction were repeatedly renewed. Comment on the comparison.
 95. Explain why a positive NPV need not be sufficient for accepting a public or social project. Give four relevant non-financial or distributional considerations.
 96. A social project costs RM 800 000 and produces benefits valued at RM 100 000 at each year-end for ten years. Find NPV at 3% and 8%, then explain why the choice of discount rate matters.
 97. A fixed pension is RM 3000 per month. If prices rise by 5% annually for four years, express the pension in today's purchasing power after four years and discuss who bears inflation risk.
 98. Loan A charges a nominal annual rate of 5% compounded monthly plus a RM 500 fee. Loan B charges 5.8% compounded monthly with no fee. Both lend RM 20 000 for 36 months. Compare monthly payments and total stated cost.
 99. A mortgage projection assumes one constant monthly rate for 20 years. State four circumstances that could make the projected payment path or total cost unreliable.
 100. Give a six-point checklist for comparing savings, loans or projects fairly. Include timing, effective rates, fees, inflation, risk and non-financial consequences.

Chapter 15

Index Numbers

Chapter guide

Begin with simple and chained comparisons, then construct weighted indices and apply them to inflation, real values and responsible social-science interpretation.

Selected background sources: (Cambridge University Press & Assessment, n.d.; Hassan & Mazlin, 2025–2026).

Index numbers provide a compact way to measure how a quantity, or a group of related quantities, changes relative to a chosen reference period. By assigning the reference period an index of 100, values recorded in different years, regions or groups can be compared on a common scale even when their original units or magnitudes differ. Index numbers are therefore widely used to summarise movements in prices, wages, production, trade, employment, population indicators and many other forms of social and economic data.

A single index may describe the change in one item, while a composite index combines several components according to their relative importance. Such summaries make complicated data sets easier to communicate, but they must be interpreted carefully: the selected base period, components and weights can all affect the result. This chapter develops simple and composite indices, then applies them to inflation, purchasing power and comparisons of nominal and real values. It also introduces rebasing, chain indices and weighted aggregative methods, together with the assumptions and limitations that should be considered when index numbers are used in practical decisions.

Index numbers describe *relative* position, not absolute size. An index of 140 may refer to a price of RM 7, a wage of RM 3500 or an output of 2 million units, depending on the base value. They also describe change rather than its cause. A price index can show that a representative basket became more expensive, but further evidence is needed to explain why.

15.1 Simple index numbers

Let Q_0 be the value of a quantity in the *base period* and let Q_1 be its value in the *current period*. The index number for period 1 relative to period 0 is

$$I_{1/0} = \frac{Q_1}{Q_0} \times 100.$$

The base-period index is always 100. An index above 100 represents an increase from the base period, while an index below 100 represents a decrease.

Index value	Interpretation relative to the base period
$I = 100$	no change
$I > 100$	increase of $(I - 100)\%$
$I < 100$	decrease of $(100 - I)\%$

The index is dimensionless: the units in the numerator and denominator cancel. It does not report the original amount, so the base value must be known if an actual price, wage or quantity is required.

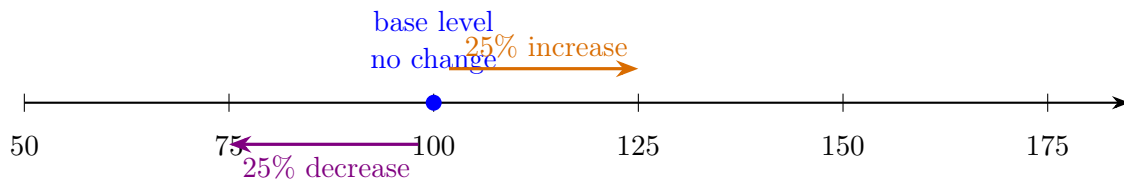


Figure 15.1: The index scale is centred on the base value 100. An index of 125 means 25% above the base, while 75 means 25% below it.

Example 15.1: [

blue]Constructing and interpreting a price index The price of a kilogram of rice rises from RM 4.80 in the base year to RM 6.00 in the current year. Its price index is

$$I = \frac{6.00}{4.80} \times 100 = \boxed{125}.$$

The current price is 125% of the base-year price, which means that it has increased by 25%. It does not mean that the price increased by 125%.

The index formula may be rearranged to recover either value:

$$Q_1 = \frac{I_{1/0}}{100} Q_0, \quad Q_0 = \frac{100 Q_1}{I_{1/0}}.$$

Example 15.2: [

orange]Recovering a value from an index A monthly transport cost has index 118 relative to a base-year cost of RM 250. The current cost is

$$Q_1 = \frac{118}{100} (250) = \boxed{\text{RM } 295}.$$

Conversely, if RM 295 and the index 118 were known, the base-year cost would be $100(295)/118 = \text{RM } 250$.

15.1.1 Index points, percentages and reverse comparisons

An index-point change is not generally the same as a percentage change. If an index rises from 120 to 132, it gains 12 index points, but the percentage increase is

$$\left(\frac{132}{120} - 1\right) 100\% = 10\%.$$

Reversing the direction of comparison also changes the index. If $I_{1/0} = 125$, then

$$I_{0/1} = \frac{10000}{125} = 80.$$

Thus, period 1 is 25% above period 0, while period 0 is 20% below period 1. Percentage increases and decreases are not symmetric: a 25% increase followed by a 20% decrease returns to the original value.

15.2 Comparing periods and changing the base

Index numbers can be linked across several periods. If the index for period 1 relative to period 0 is $I_{1/0}$ and the index for period 2 relative to period 1 is $I_{2/1}$, then

$$I_{2/0} = \frac{I_{1/0}I_{2/1}}{100}.$$

The division by 100 is necessary because each index has already been scaled by 100.

Example 15.3: [

violet]Linking two index numbers An average rental index is 112 in 2024 with 2022 as the base year. Between 2024 and 2026, rents increase by 7%, giving an index of 107 for 2026 relative to 2024. Hence,

$$I_{2026/2022} = \frac{112(107)}{100} = \boxed{119.84}.$$

Thus, the modelled average rent in 2026 is 19.84% above its 2022 level.

Suppose a published series I_t uses one base year, but a new base year b is required. The rebased series is

$$I_t^{\text{new}} = \frac{I_t^{\text{old}}}{I_b^{\text{old}}} \times 100.$$

Rebasing changes the reference point but not the proportional relationship between any two periods.

A useful base period should be clearly defined, supported by reliable data and reasonably typical of the phenomenon being studied. An abnormal year may make later comparisons appear unusually large or small. The base should also be updated when the original basket, technology or economic structure is no longer representative.

Example 15.4: [

orange]Rebasing an employment-cost series Suppose an employment-cost series with 2020 as base year has values

$$100, \quad 108, \quad 117, \quad 126$$

for 2020–2023. To make 2022 the new base year, divide every value by 117 and multiply by 100. The rebased values are approximately

$$85.47, \quad 92.31, \quad 100, \quad 107.69.$$

The 2023-to-2021 comparison is unchanged:

$$\frac{126}{108} = \frac{107.69}{92.31} \approx 1.1667.$$

Only the reference point has changed.

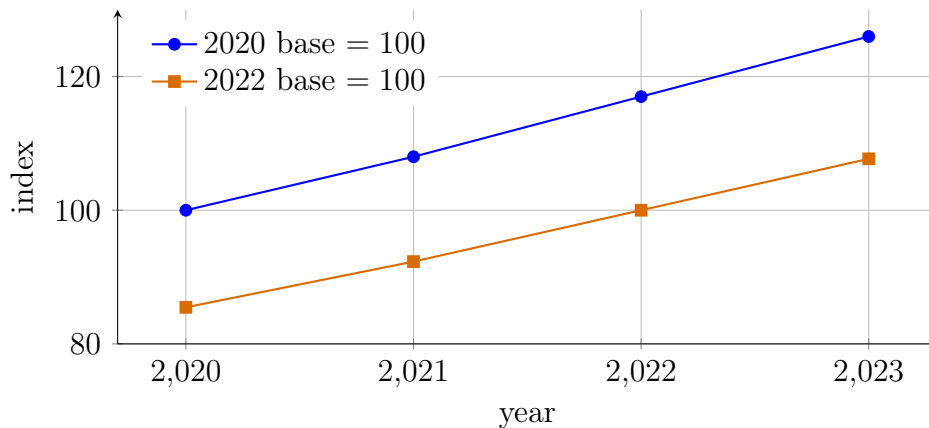


Figure 15.2: Rebasing changes the numerical scale but preserves every proportional comparison between periods.

15.3 Composite index numbers

A simple index describes one item or quantity. A *composite index* combines several component indices into a single summary measure. If all components are treated equally, the unweighted composite index is

$$I_{\text{unweighted}} = \frac{I_1 + I_2 + \cdots + I_n}{n}.$$

Equal weighting is rarely appropriate when the components differ greatly in importance. If component i has weight w_i , the weighted composite index is

$$I_{\text{composite}} = \frac{\sum_{i=1}^n w_i I_i}{\sum_{i=1}^n w_i}.$$

Weights may represent quantities purchased, expenditure shares, relative importance or another meaningful measure. Multiplying every weight by the same constant leaves the composite index unchanged.

Example 15.5: [

cyan]A weighted household-cost index Suppose the component indices for food, housing and transport are 112, 125 and 108. A household assigns them weights 4, 3 and 3, respectively. The composite index is

$$\frac{4(112) + 3(125) + 3(108)}{4 + 3 + 3} = \frac{1147}{10} = \boxed{114.7}.$$

On the assumptions represented by these weights, the combined cost is 14.7% higher than in the base period.

The contribution of component i to the overall percentage change is

$$c_i = \frac{w_i}{\sum_j w_j} (I_i - 100).$$

For the preceding example, food contributes 4.8 index points, housing contributes 7.5 and transport contributes 2.4. Their sum is the overall increase of 14.7 points. A component can therefore have a high individual index but a modest effect on the composite when its weight is small.

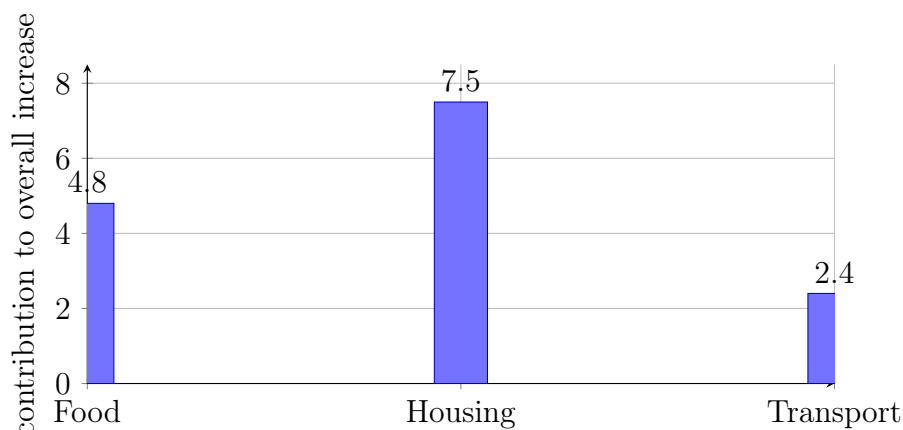


Figure 15.3: Component contributions depend jointly on price movement and weight. Here housing contributes most to the composite increase.

An unknown component index or weight can be found by substituting the known values into the composite-index formula and solving the resulting linear equation.

15.4 Consumer-price indices and inflation

A *consumer-price index* (CPI) measures how the cost of a representative basket of consumer goods and services changes over time. The basket is weighted so that components on which consumers spend more generally have a larger influence on the index. The detailed construction varies between statistical agencies, but the central idea is a weighted comparison of prices with a reference period.

Constructing a CPI requires decisions about which households, products, outlets and regions are represented. Expenditure surveys are commonly used to estimate weights. Prices must then be collected for comparable products over time, with procedures for missing items, quality change and the arrival of new goods. Consequently, a CPI is a carefully designed statistical estimate rather than a direct observation of one universal cost of living.

The CPI level and the inflation rate are related but are not the same. If C_{t-1} and C_t are CPI values in consecutive periods, then

$$\text{inflation rate} = \left(\frac{C_t}{C_{t-1}} - 1 \right) \times 100\%.$$

The inflation rate measures the percentage change in the index, not the number of index points gained.

Example 15.6: [

blue]Inflation from two CPI values Suppose the CPI rises from 124 to 130. The index increases by 6 points, but the inflation rate is

$$\left(\frac{130}{124} - 1 \right) 100\% \approx \boxed{4.84\%}.$$

Reporting the change as 6% would confuse an index-point change with a percentage change.

For several periods, cumulative inflation is found from the ratio of the final CPI to the initial CPI:

$$\text{cumulative inflation} = \left(\frac{C_T}{C_0} - 1 \right) 100\%.$$

Annual inflation rates compound. Adding them gives only an approximation and can be misleading when rates are large or the period is long.

15.4.1 Index-linked adjustments

Wages, pensions, maintenance payments and budget allocations may be linked to a published index. If an amount V_a is stated when the index is C_a , then full indexation to a later index C_b gives

$$V_b = V_a \frac{C_b}{C_a}.$$

Example 15.7: [

violet]Updating a social benefit A monthly benefit is RM 1800 when the CPI is 125. If it is fully adjusted when the CPI reaches 132.5, the new amount is

$$1800 \left(\frac{132.5}{125} \right) = \boxed{\text{RM } 1908}.$$

The adjustment preserves purchasing power relative to the chosen CPI, but the recipient's personal spending pattern may differ from the national basket.

15.4.2 Nominal and real values

A *nominal value* is measured in the money of the period in which it is observed. A *real value* adjusts for changes in the price level. If the base-year CPI is 100, then

$$\text{real value in base-year money} = \frac{\text{nominal value}}{C_t/100}.$$

This process is called *deflation*. It allows monetary observations from different periods to be compared in approximately constant purchasing-power terms.

Example 15.8: [

orange]Deflating a nominal income A monthly income is RM 3600 when the CPI is 130, with the base-year CPI equal to 100. Its value in base-year purchasing power is

$$\frac{3600}{1.30} \approx \boxed{\text{RM } 2769.23}.$$

The calculation does not claim that the worker receives RM 2769.23; it expresses RM 3600 in the purchasing power of the base year's money.

For two consecutive periods, a nominal growth rate g and inflation rate r give the exact real growth rate

$$g_{\text{real}} = \frac{1+g}{1+r} - 1,$$

as introduced in Chapter 14.

The same comparison can be made using indices. If a nominal income index is W_t and the price index is C_t , with both using the same base period, the real-income index is

$$R_t = \frac{W_t}{C_t} \times 100.$$

Similarly, a simple index of the purchasing power of money is

$$M_t = \frac{10000}{C_t}.$$

When the CPI is 125, money has an index of 80: one unit of money has approximately

80% of its base-period purchasing power.

Example 15.9: [

Nominal wage growth and real wage growth Suppose a wage rises from RM 3000 to RM 3450, while the CPI rises from 100 to 112. The nominal wage index is 115, but the real-wage index is

$$\frac{115}{112}(100) \approx \boxed{102.68}.$$

The wage has therefore increased by about 2.68% in purchasing-power terms, not by the nominal increase of 15%.

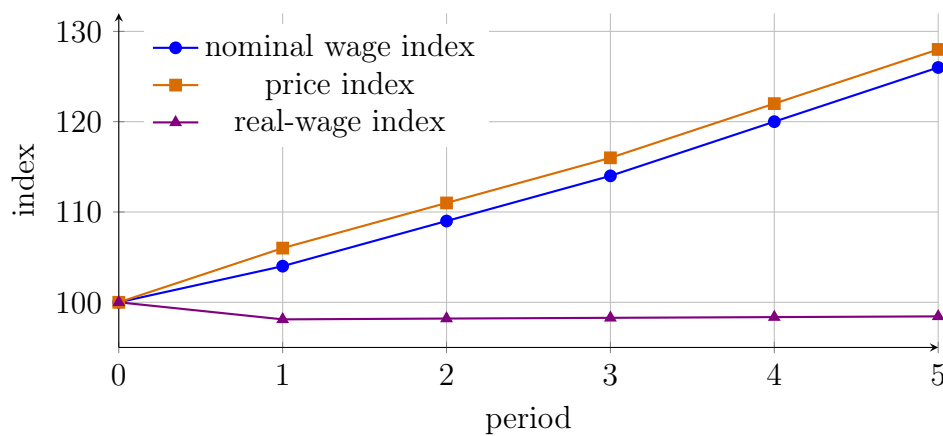


Figure 15.4: Nominal wages can rise while real purchasing power falls when the price index increases faster.

15.5 Fixed-base and chain indices

In a *fixed-base index*, every period is compared directly with the same base period. This makes comparisons easy, although the original basket or weights may become less representative over time.

In a *chain index*, each period is first compared with the immediately preceding period. The resulting *link relatives* are then multiplied. If L_j is the index for period j relative to period $j - 1$, then

$$I_{t/0} = 100 \prod_{j=1}^t \frac{L_j}{100}.$$

Equivalently,

$$I_t = I_{t-1} \frac{L_t}{100}, \quad L_t = 100 \frac{I_t}{I_{t-1}}.$$

These forms allow a fixed-base series to be constructed recursively or its link relatives to be recovered.

Example 15.10: [

violet]Constructing a chain index Annual link relatives are 104, 106 and 103. Beginning from 100, the fixed-base index after three years is

$$100(1.04)(1.06)(1.03) = \boxed{113.5472}.$$

The accumulated increase over the three years is therefore approximately 13.55%, not $4 + 6 + 3 = 13\%$. Successive percentage changes compound.

Chain indices can accommodate regularly updated baskets and weights, but revisions and rounding may make a long series less transparent. The choice between fixed and chain methods depends on the purpose and the available data.

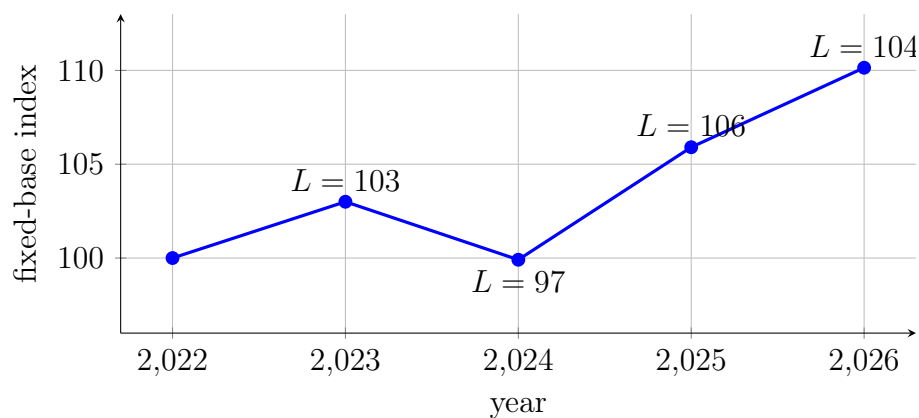


Figure 15.5: A chain series can fall in one period and still show a higher level over the full interval because all link relatives are compounded.

15.6 Weighted aggregative price indices

At A-Level and introductory university level, price indices may be constructed directly from price and quantity data. Let p_0, p_1 denote base and current prices, and let q_0, q_1 denote base and current quantities.

The *Laspeyres price index* uses the base-period basket:

$$L = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100.$$

The *Paasche price index* uses the current-period basket:

$$P = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100.$$

The *Fisher ideal index* is the geometric mean of the two:

$$F = \sqrt{LP}.$$

Laspeyres asks what the old basket would cost at current prices, relative to its old cost. Paasche asks how the current basket's cost changes when the same current quantities are valued at current rather than base prices.

When consumers substitute away from goods whose prices rise most, the Laspeyres index may exceed the Paasche index because it retains the older, higher-weight quantities. This is a common pattern rather than a universal rule. Fisher's geometric mean provides a compromise between the two weighting conventions.

Example 15.11: [

cyan]Laspeyres, Paasche and Fisher indices Consider the following prices and quantities.

Item	p_0	p_1	q_0	q_1
A	4.00	5.00	100	90
B	2.00	2.50	200	220
C	10.00	12.00	50	55

The required expenditure totals are

$$\begin{aligned}\sum p_0 q_0 &= 1300, & \sum p_1 q_0 &= 1600, \\ \sum p_0 q_1 &= 1350, & \sum p_1 q_1 &= 1660.\end{aligned}$$

Therefore,

$$\begin{aligned}L &= \frac{1600}{1300}(100) \approx 123.08, \\ P &= \frac{1660}{1350}(100) \approx 122.96, \\ F &= \sqrt{(123.08)(122.96)} \approx \boxed{123.02}.\end{aligned}$$

All three measures indicate a price increase of about 23%, although their weighting conventions differ.

The *value index*

$$V = \frac{\sum p_1 q_1}{\sum p_0 q_0} \times 100$$

compares total current expenditure with total base-period expenditure. It is not a pure price index because both prices and quantities may have changed. For the worked data,

$$V = \frac{1660}{1300}(100) \approx 127.69.$$

Thus expenditure rose by about 27.69%, while the price indices rose by about 23%. The difference reflects changes in quantities as well as the choice of price-index formula.

15.7 Using indices in social-science comparisons

Indices can compare regions, demographic groups or institutions as well as time periods. For example, a regional cost index can be used to express nominal incomes in the purchasing power of a common reference region:

$$\text{cost-adjusted income} = \frac{\text{nominal income}}{\text{regional cost index}/100}.$$

Example 15.12: [

blue]Comparing regional purchasing power Region A has an average monthly income of RM 5000 and a cost index of 125. Region B has an average income of RM 4500 and a cost index of 110. Relative to the common base,

$$\text{A: } \frac{5000}{1.25} = \text{RM } 4000, \quad \text{B: } \frac{4500}{1.10} \approx \text{RM } 4090.91.$$

Although Region A has the higher nominal income, Region B has the slightly higher cost-adjusted income under this index. The conclusion depends on the representativeness of the regional cost basket.

Index-based comparisons are also used to adjust poverty thresholds, public budgets, school funding, maintenance payments and historical monetary values. Before applying an index, the analyst should check that its population, geographical coverage, basket and time period match the question being asked.

15.8 Interpretation and limitations

An index is a summary measure rather than a complete description of every individual experience. Its interpretation depends on the chosen base, components, weights and data-collection methods. Important limitations include the following:

- **Representativeness.** A national basket may not match the spending pattern of a particular household, region or income group.
- **Changing behaviour.** Consumers may substitute away from items whose prices rise rapidly, whereas a fixed basket initially holds quantities constant.
- **Quality change.** A higher price may partly reflect an improvement in quality rather than pure inflation.
- **New and disappearing products.** A basket must be updated as consumption patterns and technologies change.
- **Choice of weights and base period.** Old weights can become unrepresentative, while an unusual base year can distort comparisons.

- **Sampling and measurement.** Prices are collected from a sample of products and outlets. Missing observations, discounts and data revisions may affect the published result.
- **Group differences.** An average index can conceal different changes across income groups, household types or regions.
- **Formula choice.** Laspeyres, Paasche, Fisher and chain methods use different weighting conventions and need not give identical answers.

When an index is used in policy, wage negotiations or comparisons of living standards, both the numerical result and the method used to construct it should be examined.

15.8.1 Communicating an index responsibly

A clear report should state:

- what the index measures and which population or basket it covers;
- the base period and the periods being compared;
- whether the reported change is in index points or percentage terms;
- the weighting and linking method, where relevant;
- whether the figures are provisional, revised or seasonally adjusted;
- important limitations for the decision or group under discussion.

For example, saying that “the CPI is 132” is incomplete without its base period. Saying that it “rose by four” is ambiguous unless the writer states whether this means four index points or a 4% increase. Transparent wording prevents a correct calculation from being given a misleading interpretation.

15.9 Exercises

Answers are provided in section C.15.

1. The price of a notebook rises from RM 2.40 in the base year to RM 3.00 in the current year. Find and interpret its price index.
2. A salary has index 118 when its current value is RM 2950. Find the salary in the base period.
3. An output index is 115 in 2022 relative to 2020. The index for 2024 relative to 2022 is 108. Find the index for 2024 relative to 2020.
4. Four component indices are 110, 125, 95 and 120.
 - (a) Find their unweighted composite index.
 - (b) Find their weighted composite index using weights 5, 2, 2 and 1.
5. Three component indices are 120, 110 and x , with weights 3, 2 and 5. If their composite index is 116, find x .
6. The CPI rises from 126.4 to 132.1. Calculate the inflation rate to two decimal places.
7. A household's nominal annual income is RM 54 000 when the CPI is 135, with base-year CPI 100. Express the income in base-year purchasing power.
8. The annual link relatives for three successive years are 104, 106 and 103. Construct the fixed-base index at the end of the third year, beginning from 100.
9. Using the worked example's price and quantity data, calculate the Laspeyres, Paasche and Fisher indices. State which quantities are used by each of the first two methods.
10. A household budget has the following component indices and percentage weights: food 125 with weight 30; housing 118 with weight 35; transport 140 with weight 20; and other items 110 with weight 15. Find and interpret the composite index.
11. Give two reasons why the inflation experienced by a particular household may differ from the published national inflation rate.

Further exercises: interpretation and applications

12. A price index is 125 relative to its base period.
 - (a) State the percentage increase from the base period.
 - (b) Find the reverse index for the base period relative to the current period.
 - (c) Explain why the two percentage changes have different magnitudes.
 - (d) Show that a 25% increase followed by a 20% decrease returns a value to its original level.

- 13.** An employment index uses 250 000 workers in 2022 as its base value. The indices for 2021–2024 are 96, 100, 104.8 and 109.2, respectively.

- (a) Estimate employment in each year.
- (b) Calculate the percentage change from 2023 to 2024.
- (c) Explain why the 4.4-point increase in the index is not a 4.4% increase in employment.

- 14.** A series with 2020 as base year has the following values:

Year	2019	2020	2021	2022	2023
Index	92	100	108	117	126

- (a) Rebase the series so that 2022 equals 100.
 - (b) Verify using both series that the proportional increase from 2021 to 2023 is unchanged.
 - (c) State one reason for choosing a new base year.
- 15.** A rental index is 110 in 2022 relative to 2020. Its link relatives are 104 for 2023 relative to 2022 and 98 for 2024 relative to 2023.

- (a) Find the 2024 index relative to 2020.
- (b) Find the overall percentage change from 2020 to 2024.
- (c) Find the reverse index for 2020 relative to 2024.
- (d) If the average rent in 2024 is RM 2400, estimate the 2020 rent.

- 16.** A household-cost index has the following components:

Component	Food	Housing	Transport	Other
Weight	35	30	20	15
Index	108	115	125	102

- (a) Find the composite index.
 - (b) Calculate each component's contribution to the change from 100.
 - (c) Identify the component making the largest contribution and explain why this need not be the component with the largest weight.
- 17.** Component indices for food, housing, transport and leisure are 120, 110, 140 and 105.
- (a) Find Household A's index using weights 45, 35, 15, 5.
 - (b) Find Household B's index using weights 20, 30, 20, 30.
 - (c) Which household experiences the larger modelled increase?
 - (d) Explain how the result illustrates household-specific inflation.
- 18.** The CPI values for 2024, 2025 and 2026 are 118.0, 123.9 and 129.5.

- (a) Find the inflation rate during 2025.
 - (b) Find the inflation rate during 2026.
 - (c) Find cumulative inflation from 2024 to 2026.
 - (d) Explain why adding the two annual rates does not give the exact cumulative rate.
- 19.** A monthly social benefit is RM 1800 when the CPI is 125.
- (a) Find the fully indexed benefit when the CPI reaches 132.5.
 - (b) Find the percentage adjustment.
 - (c) If the benefit is increased by only 5%, find its value in the purchasing power of the earlier period.
 - (d) State one limitation of using the national CPI for this adjustment.
- 20.** A monthly wage rises from RM 3000 to RM 3450, while the CPI rises from 100 to 112.
- (a) Find the nominal wage index.
 - (b) Find the real-wage index.
 - (c) Express the current wage in base-period purchasing power.
 - (d) Interpret the difference between nominal and real growth.
- 21.** The CPI is 125 in one period and 120 in a later period.
- (a) Find the purchasing-power-of-money index in each period.
 - (b) Calculate the percentage change in purchasing power.
 - (c) Explain why a falling CPI raises this index.
- 22.** Beginning with a fixed-base index of 100 in 2022, the link relatives for 2023–2026 are 103, 97, 106 and 104.
- (a) Construct the fixed-base series for 2022–2026.
 - (b) Find the total percentage change over the period.
 - (c) Identify the year with the largest annual percentage increase.
 - (d) Explain how the 2024 fixed-base index can be below 100 even though the 2023 index was above 100.
- 23.** A chain index begins at 100. Its first two link relatives are 102 and 105, and the fixed-base index after the third link is 117.81.
- (a) Find the fixed-base indices after the first and second links.
 - (b) Find the missing third link relative.
 - (c) Interpret the missing link relative.

24. Consider the data

Item	p_0	p_1	q_0	q_1
A	40	46	100	90
B	80	84	60	70
C	25	30	80	60

- (a) Calculate $\sum p_0 q_0$, $\sum p_1 q_0$, $\sum p_0 q_1$ and $\sum p_1 q_1$.
 - (b) Find the Laspeyres, Paasche and Fisher price indices.
 - (c) Explain why the Laspeyres index is slightly larger here.
 - (d) State which index you would report if a compromise between the two baskets were required.
25. Using the data in Exercise 24:
- (a) calculate the value index;
 - (b) compare the change in total expenditure with the Fisher price index;
 - (c) explain why a value index should not be interpreted as a pure measure of inflation.
26. Region A has a monthly income of RM 4800 and a cost index of 120. Region B has a monthly income of RM 4400 and a cost index of 105.
- (a) Find each cost-adjusted income.
 - (b) Which region has the higher nominal income?
 - (c) Which has the higher cost-adjusted income, and by how much?
 - (d) Give one caution about drawing a welfare conclusion from these figures alone.
27. A composite index contains food, housing and other items with indices 110, 130 and 100. Housing has weight 40, food has weight w , and the remaining weight is assigned to other items. The weights total 100 and the composite index is 117.
- (a) Find w and the weight of other items.
 - (b) Find the contribution of each component to the overall 17-point increase.
 - (c) Explain why a component with index 100 can still be important to the interpretation of the composite.
28. A national nominal-wage index is 108. The national CPI is 106, while an index designed for a lower-income household group is 111.
- (a) Calculate real wage growth using the national CPI.
 - (b) Calculate real wage growth for the lower-income group.
 - (c) Explain why the statement “real wages increased” may be incomplete or misleading.
29. A report states: “Living costs rose by 50% because the index increased from 150 to 200.”

- (a) Calculate the actual percentage increase between the two periods.
- (b) Explain the error in the report.
- (c) State four pieces of contextual information that should accompany the index comparison.

Additional practice: grouped skills review

30. For each case, calculate the simple index $100Q_1/Q_0$.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Q_0	50	60	70	80	90	100	110	120
Q_1	52.5	64.2	76.3	88.8	101.7	115	128.7	142.8

31. Recover the current value Q_1 in each case.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
Q_0 (RM)	200	250	300	350	400	450	500
Index	110	111	112	113	114	115	116

32. Find the reverse index corresponding to each index: 110, 112, 114, 116, 118, 120 and 122.

33. Rebase each three-value series so that its final value is 100.

- (a) 90, 100, 110
- (b) 91, 100, 112
- (c) 92, 100, 114
- (d) 93, 100, 116
- (e) 94, 100, 118
- (f) 95, 100, 120
- (g) 96, 100, 122

34. Find each weighted composite index, using weight 40 for the first component and weight 60 for the second.

Case	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
I_1	105	106	107	108	109	110	111
I_2	120	119	118	117	116	115	114

35. Calculate the percentage inflation rate for each consecutive CPI pair:

(120, 125), (121, 127), (122, 129), (123, 131), (124, 133), (125, 135), (126, 137).

36. Find the real-income index for each (nominal-income index, CPI) pair:

(110, 105), (112, 106), (114, 107), (116, 108), (118, 109), (120, 110), (122, 111).

37. Beginning from 100, find the fixed-base index after both links for link-relative pairs (102, 98), (103, 99), (104, 100), (105, 101), (106, 102), (107, 103) and (108, 104).

38. Two items have the following price and base-quantity data:

Item	p_0	p_1	q_0
A	2.00	2.50	100
B	4.00	4.40	50

In every case $q_{1B} = 60$, while q_{1A} takes the values 80, 85, 90, 95, 100, 105 and 110. For each value of q_{1A} , calculate the Laspeyres, Paasche and Fisher price indices.

39. Give four reasons why a national price index may differ from the inflation experienced by a particular household. For each reason, state the aspect of the household or index that differs.
40. An index rises from 145 to 150.
- If the original base is 100, how far is the new level above the base?
 - Find the percentage increase from 145 to 150.
 - Find the reverse index for the base period relative to the period whose index is 150.
 - Explain why a rise of 5 index points is not a 5% rise.

Applications: public budgets and social protection

41. A monthly child benefit is RM 1200 when the CPI is 125. The CPI later reaches 132.5.
- Find the fully indexed benefit.
 - Find the required percentage adjustment.
 - If the benefit is set at RM 1248, express it in the purchasing power of the earlier period.
 - Find the real percentage change in the benefit.
42. A pension is RM 2400 per month in a base period with CPI 100. The CPI later becomes 118, but the pension is raised by only 12%.
- Find the pension required for full indexation.
 - Find the actual pension and the nominal shortfall.
 - Express the actual pension in base-period purchasing power.

- 43.** A public programme has a base-year budget of RM 8 million. Its service-cost index rises to 115 while its eligible-population index rises to 108.
- Estimate the budget needed to preserve the same real service provision per eligible person.
 - If the new budget is RM 9.5 million, find the shortfall.
 - State one modelling assumption behind the calculation.

- 44.** A school-meal cost index uses food, labour and transport weights 70, 20 and 10. Their component indices are 112, 108 and 125.
- Find the composite index.
 - Estimate the current cost of a meal that cost RM 4 in the base period.
 - Find each component's contribution to the overall increase.

- 45.** For 2022–2025, a welfare programme has the index series

Year	2022	2023	2024	2025
Caseload	100	106	111	109
Average cost per case	100	104	108	113

- Construct an expenditure index for each year.
 - Find the total expenditure change from 2022 to 2025.
 - Explain how expenditure can rise from 2024 to 2025 even though the caseload index falls.
- 46.** A social allowance has index 109 while the CPI is 112, with both indices using the same base.
- Find the real-allowance index.
 - Interpret the result.
 - If the base allowance was RM 1500, find the current nominal allowance and its base-period purchasing power.
- 47.** A social-policy budget combines health, education and housing indices 120, 110 and 130 with weights 45, 35 and 20.
- Find the composite index.
 - Find each component's contribution to the increase above 100.
 - Identify the largest contribution.
- 48.** A national poverty threshold is RM 1800 in the reference region. Regional cost indices are 120 for Region A and 105 for Region B.
- Find the cost-adjusted threshold in each region.

- (b) Compare incomes of RM 2100 in A and RM 1950 in B with the appropriate thresholds.
 - (c) State one limitation of this classification.
49. Between two reviews, the CPI rises from 140 to 147, while a benefit index rises from 140 to 145.
- (a) Find the inflation rate.
 - (b) Find the nominal benefit growth rate.
 - (c) Find the exact real change in the benefit.
50. A report states: “The benefit rose by 15% because its index is now 115.” Explain when this statement is correct. If the index rose from 110 to 115, calculate the actual percentage increase.

Applications: wages and living standards

51. A monthly wage rises from RM 3200 to RM 3520, while the CPI rises from 125 to 132.5.
- (a) Find nominal wage growth and inflation.
 - (b) Find the exact real-wage growth rate.
 - (c) Express the current wage in the earlier period’s purchasing power.
52. Food, housing, transport and other-cost indices are 118, 112, 130 and 105. Household A uses weights 40, 30, 20, 10; Household B uses 25, 40, 10, 25.
- (a) Find each household-specific cost index.
 - (b) Which household experiences the greater modelled increase?
 - (c) Explain the source of the difference.
53. A nominal-wage index is 108. The national CPI is 105, while a lower-income household CPI is 111.
- (a) Find the real-wage index using each CPI.
 - (b) Interpret the two results.
 - (c) Explain why a single national statement about real wages may conceal distributional differences.
54. Salary indices for 2020–2023 are 100, 104, 109, 116; the corresponding CPI values are 100, 102, 108, 114.
- (a) Construct the real-salary index series.
 - (b) Identify the year with the highest real salary relative to 2020.
 - (c) Explain why the nominal and real rankings differ.

- 55.** Base-period average monthly wages are RM 4000 for Group M and RM 3200 for Group W. Their current wage indices are 112 and 118, respectively.
- (a) Find both current average wages.
 - (b) Calculate the wage gap as a percentage of Group M's wage in each period.
 - (c) Explain why the larger index for Group W does not imply a higher current wage level.
- 56.** A labour-productivity index is 115, a nominal-wage index is 108 and an output-price index is 104.
- (a) Find the real-wage index.
 - (b) Find an index of nominal unit labour cost using wage divided by productivity.
 - (c) Interpret both results cautiously.
- 57.** A benefit programme's recipient-count index is 110, its total spending index is 121 and the CPI is 105.
- (a) Find the nominal average-benefit index.
 - (b) Find the real average-benefit index.
 - (c) State why neither index describes how benefits are distributed among recipients.
- 58.** A wage agreement grants 60% of the measured inflation rate. The CPI rises from 130 to 139.1.
- (a) Find inflation and the nominal wage adjustment.
 - (b) Find the exact real-wage change after the adjustment.
 - (c) Explain the purchasing-power consequence.
- 59.** Region A has nominal monthly income RM 4800 and cost index 120. Region B has income RM 4400 and cost index 105.
- (a) Find each cost-adjusted income.
 - (b) Compare the nominal and adjusted rankings.
 - (c) State one reason the adjusted figures are not complete welfare measures.
- 60.** A median nominal-income index is 125 and the CPI is 118.
- (a) Find the real median-income index.
 - (b) Interpret the change from the base period.
 - (c) Explain why this result does not establish that every household became better off.

Applications: housing, regions and transport

- 61.** A rent index is 112 in 2023 relative to 2022. Its next two link relatives are 105 and 98.
- (a) Find the 2025 rent index relative to 2022.
 - (b) Find the modelled 2025 rent when the 2022 rent was RM 1500.
 - (c) Explain the effect of the final link relative.
- 62.** A housing-cost index combines rent, utilities and maintenance with weights 70, 20, 10 and indices 125, 115, 130.
- (a) Find the composite index.
 - (b) Update a base-period monthly housing cost of RM 1800.
 - (c) State why a renter and an owner may require different weights.
- 63.** Region A has a house-price index of 140 and a base price of RM 300 000. Region B has index 125 and base price RM 400 000.
- (a) Find both current prices.
 - (b) Which region has the larger proportional increase?
 - (c) Which has the higher current price?
 - (d) Explain why an index alone cannot answer the final question.
- 64.** A regional rent series is 90, 100, 108, 120. Rebase it so that the third value equals 100, and verify that the proportional change from the second to the fourth period is preserved.
- 65.** A bus-fare cost index uses fuel, wages and maintenance weights 40, 35, 25 with component indices 135, 115, 125.
- (a) Find the composite index.
 - (b) Estimate the fare corresponding to a base fare of RM 2.50.
 - (c) Find the contribution of fuel to the overall increase.
- 66.** A public-transport ridership index is 92 while the population index is 108.
- (a) Find the per-capita ridership index.
 - (b) Interpret the result.
 - (c) Give one possible explanation that the index calculation alone cannot confirm.
- 67.** A road-maintenance budget has index 110, while the construction- cost index is 118.
- (a) Find the real purchasing-capacity index of the budget.
 - (b) If the base budget was RM 50 million, compare the current budget with the amount needed for full cost adjustment.

- (c) State one non-price reason why equal real spending may not maintain equal service quality.
- 68.** A common poverty line is RM 1600 in the base region. Cost indices are 125 in Region A and 95 in Region B. Compare incomes of RM 1900 in A and RM 1600 in B with their regional thresholds.
- 69.** A rental index rises from 150 to 165.
- (a) Find the percentage rise.
 - (b) Find the reverse index for the earlier period relative to the later period.
 - (c) Explain why the 15-point rise is not a 15% rise.
- 70.** City cost indices are 110, 125 and 105 for Cities A, B and C. Their population weights are 40%, 35% and 25%.
- (a) Find the population-weighted regional index.
 - (b) Explain why expenditure weights might produce a different result.
 - (c) State one danger of treating the result as every resident's personal cost increase.

Applications: education, health and public services

- 71.** A tuition-fee index is 130 while a scholarship index is 118.
- (a) Find the scholarship-coverage index.
 - (b) If the scholarship covered 60% of tuition in the base period, estimate the current coverage percentage.
 - (c) Interpret the result.
- 72.** A household health-cost index uses medicine, consultation and transport weights 50, 30, 20 with indices 140, 115, 110.
- (a) Find the composite index.
 - (b) Update a base monthly health budget of RM 240.
 - (c) Identify the largest contribution to the increase.
- 73.** A hospital-output index is 108 and its staff index is 105.
- (a) Find the output-per-staff index.
 - (b) Interpret the result.
 - (c) Give one reason this is not, by itself, a complete measure of healthcare productivity.
- 74.** A school system has budget index 119, enrolment index 112 and education-cost index 107.
- (a) Find the nominal budget-per-student index.

- (b) Find the real budget-per-student index.
 - (c) Interpret whether real provision per student has been maintained.
- 75.** A literacy-rate index is 104.4 relative to a base literacy rate of 75%.
- (a) Estimate the current literacy rate.
 - (b) Find the change in percentage points.
 - (c) Distinguish the percentage-point change from the relative percentage change.
- 76.** A healthcare-utilisation index is 120 and its service-price index is 115.
- (a) Estimate the total-expenditure index.
 - (b) Update base expenditure of RM 2 million.
 - (c) Explain why the expenditure increase should not be described entirely as inflation.
- 77.** A wellbeing index combines health, education and income indices 108, 115, 105 with weights 0.40, 0.35, 0.25.
- (a) Find the composite index.
 - (b) Find each component's contribution to the increase above 100.
 - (c) State one limitation of combining unlike indicators.
- 78.** Component indices are 120, 110, 95.
- (a) Calculate the composite using old weights 50, 30, 20.
 - (b) Recalculate it using new weights 30, 40, 30.
 - (c) Explain why reweighting changes the result although the component indices are unchanged.
- 79.** A vaccination programme has a cost-per-dose index of 112 and a doses-delivered index of 125.
- (a) Find the total-cost index.
 - (b) Update a base programme cost of RM 10 million.
 - (c) Explain why the larger budget may represent expanded service rather than only higher prices.
- 80.** A waiting-time index is 85, where lower is better, while a patient-satisfaction index is 110, where higher is better. Calculate their simple average and explain why it is a misleading performance index without first aligning their directions and scales.

Applications: production, trade and the environment

81. An emissions index is 90 while an output index is 120.

- (a) Find the emissions-intensity index.
- (b) Interpret the result.
- (c) Explain why lower intensity does not necessarily mean that every environmental impact has fallen.

82. An energy-price index is 130 and an energy-use index is 92.

- (a) Find the energy-expenditure index.
- (b) Interpret the separate effects of price and use.
- (c) State why the product is only an approximation if the indices cover different baskets or populations.

83. Agricultural data are

Product	p_0	p_1	q_0	q_1
Rice	3.00	3.60	100	90
Vegetables	4.00	4.40	50	60

- (a) Find the Laspeyres, Paasche and Fisher price indices.
 - (b) Find the value index.
 - (c) Explain why the value index differs from the price indices.
- 84.** Tourism-arrival link relatives for three years are 95, 110 and 104. Construct the fixed-base index from 100, find the overall change and explain why adding the three annual percentage changes is not exact.
- 85.** Sector-output indices are 115 for manufacturing, 108 for services and 102 for agriculture, with weights 50, 40, 10.
- (a) Find the composite output index.
 - (b) Find each sector's contribution to the increase.
 - (c) State one reason the weights may need regular revision.
- 86.** An export-value index is 125 and an export-price index is 110. Estimate the export-volume index and interpret the result.
- 87.** An electricity tariff has index 118. Total household consumption has index 105, and the household-count index is 110.
- (a) Find the total household-expenditure index.
 - (b) Find the expenditure-per-household index.
 - (c) Explain why the average may conceal unequal changes across households.

88. A carbon charge was RM 50 when the CPI was 100. The CPI is now 128, but the charge is RM 60.
- (a) Find the fully indexed charge.
 - (b) Find the real-charge index of the actual RM 60 charge.
 - (c) Interpret the degree of under-indexation.
89. A chain index rises by 10% in one year and falls by 10% in the next. Starting from 100, find the final index and explain why the opposite percentage changes do not cancel.
90. Rebase the series 100, 120, 90, 135 so that the third period equals 100. Show that rebasing preserves the ranking and every proportional comparison between periods.

Applications: evaluating and communicating indices

91. A news report says, “The CPI is 132, so prices have risen by 132%.” Correct the statement and identify the information that must accompany the index value.
92. A provisional index is revised from 125.4 to 124.8.
- (a) Find the revision in index points.
 - (b) Find the revision as a percentage of the provisional value.
 - (c) Explain why reports should identify provisional data.
93. A product normally cost RM 95, but a temporary discount reduced its base-period price to RM 80. Its current price is RM 100.
- (a) Find the index using the discounted base price.
 - (b) Find the index using RM 95 as a more typical comparison.
 - (c) Explain how an abnormal base period affects interpretation.
94. A composite uses indices 110, 125, 95. The first component has weight 30, the second has weight w , and the third receives the remaining weight out of 100. If the composite is 111.5, find both unknown weights.
95. Components have weights 40, 35, 25 and indices 120, x , 108. If their composite index is 114.9, find x and its contribution to the overall increase.
96. A price study reports Laspeyres index 124 and Paasche index 118.
- (a) Find the Fisher index.
 - (b) Find the spread between Laspeyres and Paasche.
 - (c) Give one possible behavioural explanation for the difference.
97. Healthcare, housing, food and other-cost indices are 140, 120, 115, 105. An older-household index uses weights 30, 40, 20, 10, while the national index uses 15, 30, 25, 30.

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- (a) Calculate both indices.
 - (b) Interpret the difference.
 - (c) Explain why publishing subgroup indices can improve policy analysis.
- 98.** Series A, with 2020 as base, is 100, 110, 121. Rebase it to 2021. Series B for the same periods, already based on 2021, is 91, 100, 111. Compare the growth patterns and give one reason the series may still differ slightly.
- 99.** A laptop price rises from RM 1000 to RM 1100, while a quality measure rises by 20%.
- (a) Find the unadjusted price index.
 - (b) Calculate a simple price-per-unit-of-quality index.
 - (c) Explain why statistical agencies attempt quality adjustment.
- 100.** A policy claim relies on an index comparison. List six questions an analyst should ask before accepting the claim, including the base, coverage, weights, method, uncertainty and interpretation. Explain why an index movement alone does not establish its cause.

Appendix A

Notation Glossary

This glossary records the conventions used throughout the book. A symbol may acquire a more specialised meaning within a particular model; when it does, the local definition takes priority. Variables, units, populations and time scales should always be stated when a result is interpreted.

General mathematical notation

Symbol	Meaning	Convention in this book
$\mathbb{N}$	natural numbers	$\{1, 2, 3, \dots\}$; zero is excluded.
$\mathbb{W}$	whole numbers	$\{0, 1, 2, 3, \dots\}$.
$\mathbb{Z}$	integers	Positive and negative whole numbers, including zero.
$\mathbb{Q}$	rational numbers	Numbers expressible as p/q with integers p, q and $q \neq 0$.
$\mathbb{R}$	real numbers	Rational and irrational numbers.
$\mathbb{C}$	complex numbers	Numbers of the form $a + bi$, where $i^2 = -1$.
$[a, b], (a, b)$	closed and open intervals	A square bracket includes an endpoint; a round bracket excludes it.
A^c	complement of A	The universal set must be clear from context.
$ x ; A $	absolute value; set size	The intended meaning is determined by whether the argument is a number or a set.
$\approx$	approximately equal	Used for rounded or model-based numerical values; $=$ is reserved for exact equality.
e	base of natural logarithms	Printed upright to distinguish the constant from a variable.
$\ln x$	natural logarithm	Logarithm to base e ; other bases are shown explicitly.

Functions, vectors and matrices

Symbol	Meaning	Convention in this book
$f: A \rightarrow B$	function from A to B	A is the domain and B is the codomain.
f^{-1}	inverse function	Used only when the relevant restriction makes f one-to-one. It does not mean $1/f$.
$\mathbf{x}$	vector	Vectors are normally bold lower-case letters.
A	matrix	Matrices are normally upper-case letters.
$A^{\top}$	transpose	Rows and columns are interchanged.
A^{-1}	inverse matrix	Defined only for an invertible square matrix.
I_n	identity matrix	The $n \times n$ matrix with ones on the main diagonal.
$\det(A)$	determinant	A scalar associated with a square matrix.

Probability and statistics

Symbol	Meaning	Convention in this book
$P(A)$	probability of event A	Probabilities lie between 0 and 1.
$P(A \mid B)$	conditional probability	Probability of A given that B has occurred, with $P(B) > 0$.
$X; x$	random variable; realised value	Random variables use capital letters and their possible values use lower-case letters.
$p_X(x)$	probability mass function	$p_X(x) = P(X = x)$ for a discrete random variable.
$F_X(x)$	cumulative distribution function	$F_X(x) = P(X \leq x)$.
$E(X)$	expected value	The probability-weighted long-run mean.
$\text{Var}(X)$	variance	$E[(X - E(X))^2]$; measured in squared units.
μ, σ	population mean and standard deviation	Parameters of a population or probability model.
$\bar{x}, s$	sample mean and standard deviation	Statistics calculated from sample data.
Q_1, Q_2, Q_3	quartiles	Q_2 is the median.

Sequences, calculus and models

Symbol	Meaning	Convention in this book
a_n	n th term of a sequence	Indexing normally begins at $n = 1$ unless stated otherwise.
S_n	partial sum	Sum of the first n terms.
$\sum$	summation	The lower and upper limits specify the included terms.
$\lim_{x \rightarrow a} f(x)$	limit	Describes nearby behaviour and need not equal $f(a)$.
$f'(x); \frac{dy}{dx}$	first derivative	An instantaneous rate of change with output units per input unit.
$f''(x)$	second derivative	The rate of change of the first derivative.
$\int f(x) \, dx$	indefinite integral	A family of antiderivatives, including a constant of integration.
$\int_a^b f(x) \, dx$	definite integral	Signed accumulation from a to b .
P_0, Q_0	base-period values	Subscript 0 denotes the reference period; subscript 1 denotes a comparison period.
$I_{1/0}$	index for period 1 relative to period 0	The base-period index is 100.

Units and numerical reporting

- Currency amounts in Malaysian ringgit are written with the prefix RM, for example RM 2 500. Other currencies retain their stated symbols.
- A percentage describes a relative change. A percentage-point change is the arithmetic difference between two percentages; the terms are not interchangeable.
- A rate includes both numerator and denominator units, such as people per year or ringgit per unit of output.
- Time units in a rate, model and input must agree. Convert annual, monthly and daily quantities before combining them.
- Exact forms are retained where they clarify structure. Decimal answers use $\approx$ and should be rounded only at the final step unless an application requires otherwise.

Appendix B

Statistical Tables

The tables in this appendix follow the conventions used by Murdoch and Barnes (1998). Probabilities are rounded to four decimal places, so displayed boundary values should be interpreted using the notes accompanying each table.

Table 1: Cumulative Binomial Probabilities

If p is the probability of success in a single trial, n is the number of trials, and r is the greatest permitted number of successes, the table gives

$$P(X \leq r) = \sum_{x=0}^r \binom{n}{x} p^x (1-p)^{n-x}.$$

An empty entry represents a probability below 0.00005; a displayed value of 1.0000 represents a probability above 0.99995. $n = 2$

r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9801	.9604	.9409	.9216	.9025	.8836	.8649	.8464	.8281
1	1.0000	.9999	.9996	.9991	.9984	.9975	.9964	.9951	.9936	.9919
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 5$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9510	.9039	.8587	.8154	.7738	.7339	.6957	.6591	.6240
1	1.0000	.9990	.9962	.9915	.9852	.9774	.9681	.9575	.9456	.9326
2	1.0000	1.0000	.9999	.9997	.9994	.9988	.9980	.9969	.9955	.9937
3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9998	.9997
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 10$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.9044	.8171	.7374	.6648	.5987	.5386	.4840	.4344	.3894
1	1.0000	.9957	.9838	.9655	.9418	.9139	.8824	.8483	.8121	.7746
2	1.0000	.9999	.9991	.9972	.9938	.9885	.9812	.9717	.9599	.9460
3	1.0000	1.0000	1.0000	.9999	.9996	.9990	.9980	.9964	.9942	.9912
4	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9997	.9994	.9990
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 20$										
r	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	.8179	.6676	.5438	.4420	.3585	.2901	.2342	.1887	.1516
1	1.0000	.9831	.9401	.8802	.8103	.7358	.6605	.5869	.5169	.4516
2	1.0000	.9990	.9929	.9790	.9561	.9245	.8850	.8390	.7879	.7334
3	1.0000	1.0000	.9994	.9973	.9926	.9841	.9710	.9529	.9294	.9007
4	1.0000	1.0000	1.0000	.9997	.9990	.9974	.9944	.9893	.9817	.9710
5	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9991	.9981	.9962	.9932
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9994	.9987
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 50$										

[illegible]

$n = 100$									
r	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	.3660	.1326	.0476	.0169	.0059	.0021	.0007	.0002	.0001
1	.7358	.4033	.1946	.0872	.0371	.0152	.0060	.0023	.0009
2	.9206	.6767	.4198	.2321	.1183	.0566	.0258	.0113	.0048
3	.9816	.8590	.6472	.4295	.2578	.1430	.0744	.0367	.0173
4	.9966	.9492	.8179	.6289	.4360	.2768	.1632	.0903	.0474
5	.9995	.9845	.9192	.7884	.6160	.4407	.2914	.1799	.1045
6	.9999	.9959	.9688	.8936	.7660	.6064	.4443	.3032	.1940
7	1.0000	.9991	.9894	.9525	.8720	.7483	.5988	.4471	.3128
8	1.0000	.9998	.9968	.9810	.9369	.8537	.7340	.5926	.4494
9	1.0000	1.0000	.9991	.9932	.9718	.9225	.8380	.7220	.5875
10	1.0000	1.0000	.9998	.9978	.9885	.9624	.9092	.8243	.7171
11	1.0000	1.0000	1.0000	.9993	.9957	.9832	.9531	.8972	.8124
12	1.0000	1.0000	1.0000	.9998	.9985	.9931	.9776	.9441	.8862
13	1.0000	1.0000	1.0000	1.0000	.9995	.9974	.9901	.9718	.9355
14	1.0000	1.0000	1.0000	1.0000	.9999	.9991	.9959	.9867	.9659
15	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9984	.9942	.9831
16	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9994	.9976	.9922
17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9991	.9966
18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9986
19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995
20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 2$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.8100	.7225	.6400	.5625	.4900	.4225	.3600	.3025	.2500
1	.9900	.9775	.9600	.9375	.9100	.8775	.8400	.7975	.7500
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 5$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.5905	.4437	.3277	.2373	.1681	.1160	.0778	.0503	.0312
1	.9185	.8352	.7373	.6328	.5282	.4284	.3370	.2562	.1875
2	.9914	.9734	.9421	.8965	.8369	.7648	.6826	.5931	.5000
3	.9995	.9978	.9933	.9844	.9692	.9460	.9130	.8688	.8125
4	1.0000	.9999	.9997	.9990	.9976	.9947	.9898	.9815	.9688
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$n = 10$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.3487	.1969	.1074	.0563	.0282	.0135	.0060	.0025	.0010
1	.7361	.5443	.3758	.2440	.1493	.0860	.0464	.0233	.0107
2	.9298	.8202	.6778	.5256	.3828	.2616	.1673	.0996	.0547
3	.9872	.9500	.8791	.7759	.6496	.5138	.3823	.2660	.1719
4	.9984	.9901	.9672	.9219	.8497	.7515	.6331	.5044	.3770
5	.9999	.9986	.9936	.9803	.9527	.9051	.8338	.7384	.62

Table 1: Cumulative Binomial Probabilities—continued

$n = 50$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0	.0052	.0003							
1	.0338	.0029	.0002						
2	.1117	.0142	.0013	.0001					
3	.2503	.0460	.0057	.0005					
4	.4312	.1121	.0185	.0021	.0002				
5	.6161	.2194	.0480	.0070	.0007	.0001			
6	.7702	.3613	.1034	.0194	.0025	.0002			
7	.8779	.5188	.1904	.0453	.0073	.0008	.0001		
8	.9421	.6681	.3073	.0916	.0183	.0025	.0002		
9	.9755	.7911	.4437	.1637	.0402	.0067	.0008	.0001	
10	.9906	.8801	.5836	.2622	.0789	.0160	.0022	.0002	
11	.9968	.9372	.7107	.3816	.1390	.0342	.0057	.0006	
12	.9990	.9699	.8139	.5110	.2229	.0661	.0133	.0018	.0002
13	.9997	.9868	.8894	.6370	.3279	.1163	.0280	.0045	.0005
14	.9999	.9947	.9393	.7481	.4468	.1878	.0540	.0104	.0013
15	1.0000	.9981	.9692	.8369	.5692	.2801	.0955	.0220	.0033
16	1.0000	.9993	.9856	.9017	.6839	.3889	.1561	.0427	.0077
17	1.0000	.9998	.9937	.9449	.7822	.5060	.2369	.0765	.0164
18	1.0000	.9999	.9975	.9713	.8594	.6216	.3356	.1273	.0325
19	1.0000	1.0000	.9991	.9861	.9152	.7264	.4465	.1974	.0595
20	1.0000	1.0000	.9997	.9937	.9522	.8139	.5610	.2862	.1013
21	1.0000	1.0000	.9999	.9974	.9749	.8813	.6701	.3900	.1611
22	1.0000	1.0000	1.0000	.9990	.9877	.9290	.7660	.5019	.2399
23	1.0000	1.0000	1.0000	.9996	.9944	.9604	.8438	.6134	.3359
24	1.0000	1.0000	1.0000	.9999	.9976	.9793	.9022	.7160	.4439
25	1.0000	1.0000	1.0000	1.0000	.9991	.9900	.9427	.8034	.5561
26	1.0000	1.0000	1.0000	1.0000	.9997	.9955	.9686	.8721	.6641
27	1.0000	1.0000	1.0000	1.0000	.9999	.9981	.9840	.9220	.7601
28	1.0000	1.0000	1.0000	1.0000	1.0000	.9993	.9924	.9556	.8389
29	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9966	.9765	.8987
30	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9986	.9884	.9405
31	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995	.9947	.9675
32	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9978	.9836
33	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9991	.9923
34	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9997	.9967
35	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9987
36	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995
37	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
38	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1 gives binomial probabilities for a limited range of values of n and p . Use exact binomial probabilities when they are available. For values outside the table, an approximation may be suitable when its conditions are checked.

As a reasonable working rule:

- (i) use the Poisson approximation when n is large and p is small, putting $m = np$;
- (ii) use the normal approximation when both $np \geq 5$ and $n(1 - p) \geq 5$, putting $\mu = np$ and $\sigma = \sqrt{np(1 - p)}$, and applying a continuity correction;
- (iii) when p is close to 1, work in terms of failures and consider a Poisson approximation with mean $m = n(1 - p)$.

For $p > 0.5$, work in terms of failures, whose probability is $q = 1 - p$.

Example. If at least 40 of 50 seeds are to germinate when the germination probability is 70%, then at most 10 seeds may fail. With failure probability 0.30, Table 1 gives $P(X \leq 10) = 0.0789$.

$n = 100$									
r	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
0									
1	.0003								
2	.0019								
3	.0078	.0001							
4	.0237	.0004							
5	.0576	.0016							
6	.1172	.0047	.0001						
7	.2061	.0122	.0003						
8	.3209	.0275	.0009						
9	.4513	.0551	.0023						
10	.5832	.0994	.0057	.0001					
11	.7030	.1635	.0126	.0004					
12	.8018	.2473	.0253	.0010					
13	.8761	.3474	.0469	.0025	.0001				
14	.9274	.4572	.0804	.0054	.0002				
15	.9601	.5683	.1285	.0111	.0004				
16	.9794	.6725	.1923	.0211	.0010				
17	.9900	.7633	.2712	.0376	.0022	.0001			
18	.9954	.8372	.3621	.0630	.0045	.0001			
19	.9980	.8935	.4602	.0995	.0089	.0003			
20	.9992	.9337	.5595	.1488	.0165	.0008			
21	.9997	.9607	.6540	.2114	.0288	.0017			
22	.9999	.9779	.7389	.2864	.0479	.0034	.0001		
23	1.0000	.9881	.8109	.3711	.0755	.0066	.0003		
24	1.0000	.9939	.8686	.4617	.1136	.0121	.0006		
25	1.0000	.9970	.9125	.5535	.1631	.0211	.0012		
26	1.0000	.9986	.9442	.6417	.2244	.0351	.0024	.0001	
27	1.0000	.9994	.9658	.7224	.2964	.0558	.0046	.0002	
28	1.0000	.9997	.9800	.7925	.3768	.0848	.0084	.0004	
29	1.0000	.9999	.9888	.8505	.4623	.1236	.0148	.0008	
30	1.0000	1.0000	.9939	.8962	.5491	.1730	.0248	.0015	
31	1.0000	1.0000	.9969	.9307	.6331	.2331	.0398	.0030	.0001
32	1.0000	1.0000	.9984	.9554	.7107	.3029	.0615	.0055	.0002
33	1.0000	1.0000	.9993	.9724	.7793	.3803	.0913	.0098	.0004
34	1.0000	1.0000	.9997	.9836	.8371	.4624	.1303	.0166	.0009
35	1.0000	1.0000	.9999	.9906	.8839	.5458	.1795	.0272	.0018
36	1.0000	1.0000	.9999	.9948	.9201	.6269	.2386	.0429	.0033
37	1.0000	1.0000	1.0000	.9973	.9470	.7024	.3068	.0651	.0060
38	1.0000	1.0000	1.0000	.9986	.9660	.7699	.3822	.0951	.0105
39	1.0000	1.0000	1.0000	.9993	.9790	.8276	.4621	.1343	.0176
40	1.0000	1.0000	1.0000	.9997	.9875	.8750	.5433	.1831	.0284
41	1.0000	1.0000	1.0000	.9999	.9928	.9123	.6225	.2415	.0443
42	1.0000	1.0000	1.0000	.9999	.9960	.9406	.6967	.3087	.0666
43	1.0000	1.0000	1.0000	1.0000	.9979	.9611	.7635	.3828	.0967
44	1.0000	1.0000	1.0000	1.0000	.9989	.9754	.8211	.4613	.1367
45	1.0000	1.0000	1.0000	1.0000	.9995	.9850	.8689	.5413	.1841
46	1.0000	1.0000	1.0000	1.0000	.9997	.9912	.9070	.6196	.2421
47	1.0000	1.0000	1.0000	1.0000	.9999	.9950	.9362	.6931	.3086
48	1.0000	1.0000	1.0000	1.0000	.9999	.9973	.9577	.7596	.3822
49	1.0000	1.0000	1.0000	1.0000	1.0000	.9985	.9729	.8173	.4602
50	1.0000	1.0000	1.0000	1.0000	1.0000	.9993	.9832	.8654	.5398
51	1.0000	1.0000	1.0000	1.0000	1.0000	.9996	.9900	.9040	.6178
52	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9942	.9338	.6914
53	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9968	.9559	.7579
54	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9983	.9716	.8159
55	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991	.9824	.8644
56	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9996	.9894	.9033
57	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9939	.9334
58	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9966	.9577
59	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9982	.9816
60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991	.9824
61	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9995	.9895
62	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9940
63	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9967
64	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9982
65	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9991
66	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9996
67	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
68	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
69	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities

The table gives the probability that r or fewer random events occur in an interval when the average number of events is m :

$$P(X \leq r) = \sum_{x=0}^r e^{-m} \frac{m^x}{x!}.$$

An empty entry represents a probability below 0.00005; a displayed value of 1.0000 represents a probability above 0.99995.

[illegible]

Table 2: Cumulative Poisson Probabilities—continued

r	7.2	7.4	7.6	7.8	8	8.2	8.4	8.6	8.8	9
0	.0007	.0006	.0005	.0004	.0003	.0003	.0002	.0002	.0002	.0001
1	.0061	.0051	.0043	.0036	.0030	.0025	.0021	.0018	.0015	.0012
2	.0255	.0219	.0188	.0161	.0138	.0118	.0100	.0086	.0073	.0062
3	.0719	.0632	.0554	.0485	.0424	.0370	.0323	.0281	.0244	.0212
4	.1555	.1395	.1249	.1117	.0996	.0887	.0789	.0701	.0621	.0550
5	.2759	.2526	.2307	.2103	.1912	.1736	.1573	.1422	.1284	.1157
6	.4204	.3920	.3646	.3384	.3134	.2896	.2670	.2457	.2256	.2068
7	.5689	.5393	.5100	.4812	.4530	.4254	.3987	.3728	.3478	.3239
8	.7027	.6757	.6482	.6204	.5925	.5647	.5369	.5094	.4823	.4557
9	.8096	.7877	.7649	.7411	.7166	.6915	.6659	.6400	.6137	.5874
10	.8867	.8707	.8535	.8352	.8159	.7955	.7743	.7522	.7294	.7060
11	.9371	.9265	.9148	.9020	.8881	.8731	.8571	.8400	.8220	.8030
12	.9673	.9609	.9536	.9454	.9362	.9261	.9150	.9029	.8898	.8758
13	.9841	.9805	.9762	.9714	.9658	.9595	.9524	.9445	.9358	.9261
14	.9927	.9908	.9886	.9859	.9827	.9791	.9749	.9701	.9647	.9585
15	.9969	.9959	.9948	.9934	.9918	.9898	.9875	.9848	.9816	.9780
16	.9987	.9983	.9978	.9971	.9963	.9953	.9941	.9926	.9909	.9889
17	.9995	.9993	.9991	.9988	.9984	.9979	.9973	.9966	.9957	.9947
18	.9998	.9997	.9996	.9995	.9993	.9991	.9989	.9985	.9981	.9976
19	.9999	.9999	.9999	.9998	.9997	.9997	.9995	.9994	.9992	.9989
20	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998	.9998	.9997	.9996
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9999	.9998
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

r	9.2	9.4	9.6	9.8	10	11	12	13	14	15
0	.0001	.0001	.0001	.0001						
1	.0010	.0009	.0007	.0006	.0005	.0002	.0001			
2	.0053	.0045	.0038	.0033	.0028	.0012	.0005	.0002	.0001	.0002
3	.0184	.0160	.0138	.0120	.0103	.0049	.0023	.0011	.0005	.0004
4	.0486	.0429	.0378	.0333	.0293	.0151	.0076	.0037	.0018	.0009
5	.1041	.0935	.0838	.0750	.0671	.0375	.0203	.0107	.0055	.0028
6	.1892	.1727	.1574	.1433	.1301	.0786	.0485	.0259	.0142	.0076
7	.3010	.2792	.2584	.2388	.2202	.1432	.0895	.0540	.0316	.0180
8	.4296	.4042	.3796	.3558	.3328	.2320	.1550	.0998	.0621	.0374
9	.5611	.5349	.5089	.4832	.4579	.3405	.2424	.1658	.1094	.0699
10	.6820	.6576	.6329	.6080	.5830	.4599	.3472	.2517	.1757	.1185
11	.7832	.7626	.7412	.7193	.6968	.5793	.4616	.3532	.2600	.1848
12	.8607	.8448	.8279	.8101	.7916	.6887	.5760	.4631	.3585	.2676
13	.9156	.9042	.8919	.8786	.8645	.7813	.6815	.5730	.4644	.3632
14	.9517	.9441	.9357	.9265	.9165	.8540	.7720	.6751	.5704	.4657
15	.9738	.9691	.9638	.9579	.9513	.9074	.8444	.7636	.6694	.5681
16	.9865	.9838	.9806	.9770	.9730	.9441	.8987	.8355	.7559	.6641
17	.9934	.9919	.9902	.9881	.9857	.9678	.9370	.8905	.8272	.7489
18	.9969	.9962	.9952	.9941	.9928	.9823	.9626	.9302	.8826	.8195
19	.9986	.9983	.9978	.9972	.9965	.9907	.9787	.9573	.9235	.8752
20	.9994	.9992	.9990	.9987	.9984	.9953	.9884	.9750	.9521	.9170
21	.9998	.9997	.9996	.9995	.9993	.9977	.9939	.9859	.9712	.9469
22	.9999	.9999	.9998	.9998	.9997	.9990	.9970	.9924	.9833	.9673
23	1.0000	1.0000	.9999	.9999	.9999	.9995	.9985	.9960	.9907	.9805
24	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9993	.9980	.9950	.9888
25	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9990	.9974	.9938
26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9987	.9967
27	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9994	.9983
28	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9991
29	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9996
30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
31	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
32	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

r	16	17	18	19	20	21	22	23	24	25
3	.0001									
4	.0004	.0002	.0001							
5	.0014	.0007	.0003	.0002	.0001					
6	.0040	.0021	.0010	.0005	.0003	.0001	.0001			
7	.0100	.0054	.0029	.0015	.0008	.0004	.0002	.0001		
8	.0220	.0126	.0071	.0039	.0021	.0011	.0006	.0003	.0002	.0001
9	.0433	.0261	.0154	.0089	.0050	.0028	.0015	.0008	.0004	.0002
10	.0774	.0491	.0304	.0183	.0108	.0063	.0035	.0020	.0011	.0006
11	.1270	.0847	.0549	.0347	.0214	.0129	.0076	.0044	.0025	.0014
12	.1931	.1350	.0917	.0606	.0390	.0245	.0151	.0091	.0054	.0031
13	.2745	.2009	.1426	.0984	.0661	.0434	.0278	.0174	.0107	.0065
14	.3675	.2808	.2081	.1497	.1049	.0716	.0477	.0311	.0198	.0124
15	.4667	.3715	.2867	.2148	.1565	.1111	.0769	.0520	.0344	.0223
16	.5660	.4677	.3751	.2920	.2211	.1629	.1170	.0821	.0563	.0377
17	.6593	.5640	.4686	.3784	.2970	.2270	.1690	.1228	.0871	.0605
18	.7423	.6550	.5622	.4695	.3814	.3017	.2325	.1748	.1283	.0920
19	.8122	.7363	.6509	.5606	.4703	.3843	.3060	.2377	.1803	.1336
20	.8682	.8055	.7307	.6472	.5591	.4710	.3869	.3101	.2426	.1855
21	.9108	.8615	.7991	.7255	.6437	.5577	.4716	.3894	.3139	.2473
22	.9418	.9047	.8551	.7931	.7206	.6405	.5564	.4723	.3917	.3175
23	.9633	.9367	.8989	.8490	.7875	.7160	.6374	.5551	.4728	.3939
24	.9777	.9594	.9317	.8933	.8432	.7822	.7117	.6346	.5540	.4734
25	.9869	.9748	.9554	.9269	.8878	.8377	.7771	.7077	.6319	.5529
26	.9925	.9848	.9718	.9514	.9221	.8826	.8324	.7723	.7038	.6294
27	.9959	.9912	.9827	.9687	.9475	.9175	.8775	.8274	.7677	.7002
28	.9978	.9950	.9887	.9805	.9657	.9436	.9129	.8726	.8225	.7634
29	.9989	.9973	.9941	.9882	.9782	.9626	.9398	.9085	.8679	.8179
30	.9994	.9986	.9967	.9930	.9865	.9758	.9595	.9360	.9042	.8633
31	.9997	.9993	.9982	.9960	.9919	.9848	.9735	.9564	.9322	.8999
32	.9999	.9996	.9990	.9978	.9953	.9907	.9831	.9711	.9533	.9285
33	.9999	.9998	.9995	.9988	.9973	.9945	.9895	.9813	.9686	.9502
34	1.0000	.9999	.9998	.9994	.9985	.9968	.9936	.9882	.9794	.9662
35	1.0000	1.0000	.9999	.9997	.9992	.9982	.9962	.9927	.9868	.9775
36	1.0000	1.0000	.9999	.9998	.9996	.9990	.9978	.9956	.9918	.9854
37	1.0000	1.0000	1.0000	.9999	.9998	.9995	.9988	.9974	.9950	.9908
38	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9993	.9985	.9970	.9943
39	1.0000	1.0000	1.0000	1.0000	.9999	.9999	.9996	.9992	.9983	.9966
40	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9996	.9990	.9980
41	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9995	.9988
42	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9993
43	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9996
44	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998
45	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
46	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
47	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 2: Cumulative Poisson Probabilities—continued

<i>r</i>	26	27	28	29	30	32	34	36	38	40
9	.0001	.0001								
10	.0003	.0002	.0001							
11	.0008	.0004	.0002	.0001	.0001					
12	.0018	.0010	.0006	.0003	.0002					
13	.0038	.0022	.0013	.0007	.0004	.0001				
14	.0076	.0046	.0027	.0016	.0009	.0003	.0001			
15	.0142	.0088	.0054	.0033	.0019	.0007	.0002	.0001		
16	.0248	.0160	.0101	.0063	.0039	.0014	.0005	.0002		
17	.0411	.0274	.0179	.0115	.0073	.0028	.0010	.0003	.0001	
18	.0646	.0445	.0300	.0199	.0129	.0052	.0020	.0007	.0002	.0001
19	.0968	.0687	.0478	.0326	.0219	.0093	.0037	.0014	.0005	.0002
20	.1387	.1015	.0727	.0511	.0353	.0159	.0068	.0027	.0010	.0004
21	.1905	.1436	.1060	.0767	.0544	.0260	.0116	.0049	.0019	.0007
22	.2517	.1952	.1483	.1104	.0806	.0406	.0191	.0085	.0035	.0014
23	.3209	.2559	.1998	.1529	.1146	.0610	.0302	.0141	.0062	.0026
24	.3959	.3242	.2599	.2042	.1572	.0881	.0460	.0224	.0103	.0045
25	.4739	.3979	.3272	.2637	.2084	.1228	.0674	.0345	.0166	.0076
26	.5519	.4744	.3997	.3301	.2673	.1656	.0953	.0513	.0259	.0123
27	.6270	.5509	.4749	.4014	.3329	.2162	.1306	.0736	.0389	.0193
28	.6967	.6247	.5500	.4753	.4031	.2741	.1733	.1023	.0565	.0294
29	.7593	.6935	.6226	.5492	.4757	.3380	.2235	.1379	.0796	.0432
30	.8134	.7553	.6903	.6206	.5484	.4061	.2804	.1806	.1089	.0617
31	.8589	.8092	.7515	.6874	.6186	.4765	.3427	.2303	.1448	.0855
32	.8958	.8546	.8051	.7479	.6845	.5468	.4089	.2861	.1875	.1153
33	.9249	.8918	.8505	.8011	.7444	.6150	.4772	.3470	.2365	.1514
34	.9472	.9213	.8879	.8465	.7973	.6792	.5454	.4115	.2914	.1939
35	.9637	.9441	.9178	.8841	.8426	.7379	.6117	.4778	.3510	.2424
36	.9756	.9612	.9411	.9144	.8804	.7901	.6744	.5442	.4138	.2963
37	.9840	.9737	.9587	.9381	.9110	.8352	.7319	.6087	.4784	.3547
38	.9897	.9825	.9717	.9562	.9352	.8732	.7834	.6699	.5430	.4160
39	.9936	.9887	.9810	.9697	.9537	.9044	.8283	.7263	.6059	.4790
40	.9961	.9928	.9875	.9795	.9677	.9293	.8664	.7771	.6657	.5419
41	.9976	.9955	.9920	.9864	.9779	.9488	.8981	.8217	.7211	.6033
42	.9986	.9973	.9950	.9911	.9852	.9636	.9237	.8599	.7712	.6618
43	.9992	.9984	.9969	.9944	.9903	.9747	.9439	.8919	.8155	.7162
44	.9996	.9991	.9981	.9965	.9937	.9827	.9596	.9181	.8538	.7657
45	.9998	.9995	.9989	.9978	.9960	.9884	.9714	.9391	.8861	.8097
46	.9999	.9997	.9994	.9987	.9975	.9924	.9801	.9555	.9128	.8479
47	.9999	.9998	.9996	.9992	.9985	.9951	.9864	.9680	.9343	.8804
48	1.0000	.9999	.9998	.9996	.9991	.9969	.9909	.9775	.9514	.9075
49	1.0000	1.0000	.9999	.9998	.9995	.9981	.9940	.9844	.9647	.9297
50	1.0000	1.0000	.9999	.9999	.9997	.9988	.9961	.9894	.9747	.9474
51	1.0000	1.0000	1.0000	.9999	.9998	.9993	.9976	.9929	.9822	.9613
52	1.0000	1.0000	1.0000	1.0000	.9999	.9996	.9985	.9953	.9877	.9719
53	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9991	.9970	.9916	.9800
54	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9994	.9981	.9944	.9860
55	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9988	.9963	.9903
56	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9993	.9976	.9934
57	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995	.9985	.9956
58	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997	.9990	.9971
59	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998	.9994	.9981
60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9996	.9988
61	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9998	.9992
62	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9995
63	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999	.9997
64	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9998
65	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
66	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	.9999
67	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

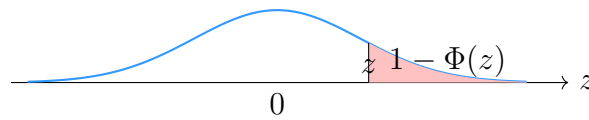
For values of m greater than 30, the normal table may be used to approximate Poisson probabilities by taking $\mu = m$ and $\sigma = \sqrt{m}$. Apply a continuity correction before standardising a discrete boundary.

Table 3: Upper-tail areas for the normal distribution

The function tabulated is $1 - \Phi(z)$, where $\Phi(z)$ is the cumulative distribution function of a standard normal variable:

$$1 - \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-t^2/2} dt.$$

Thus $1 - \Phi(z)$ is the probability that a standard normal variable exceeds z . For an observation x from a normal distribution with mean μ and standard deviation σ , the corresponding standardised value is $z = (x - \mu)/\sigma$.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.00990	.00964	.00939	.00914	.00889	.00866	.00842
2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
3.7	.00011	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00007	.00007
3.8	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005	.00005	.00005
3.9	.00004	.00004	.00004	.00004	.00004	.00003	.00003	.00003	.00003	.00003
4.0	.00003	.00003	.00003	.00002	.00002	.00002	.00002	.00002	.00002	.00002

$$1 - \Phi(5.0) = 0.0000002867.$$

$$1 - \Phi(5.5) = 0.0000000190.$$

$$1 - \Phi(6.0) = 0.0000000010.$$

Appendix C

Answers to Practice Exercises

This appendix brings together the answer keys supplied for the practice sections throughout all chapters. Equivalent mathematical forms are also acceptable unless a particular form was requested.

C.1 Chapter 1: Numbers

Return to the corresponding questions in Chapter 1.

1. (a) $\mathbb{Z}$; (b) $\mathbb{Q}$; (c) $\mathbb{R} \setminus \mathbb{Q}$; (d) $\mathbb{N}$; (e) $\mathbb{Q}$; (f) $\mathbb{N}$; (g) $\mathbb{R} \setminus \mathbb{Q}$; (h) $\mathbb{R} \setminus \mathbb{Q}$; (i) $\mathbb{R} \setminus \mathbb{Q}$; (j) $\mathbb{Q}$; (k) $\mathbb{N}$; (l) $\mathbb{Q}$; (m) $\mathbb{R} \setminus \mathbb{Q}$; (n) $\mathbb{R} \setminus \mathbb{Q}$; (o) $\mathbb{Q}$; (p) $\mathbb{W}$; (q) $\mathbb{Z}$; (r) $\mathbb{N}$; (s) $\mathbb{R} \setminus \mathbb{Q}$; (t) $\mathbb{Q}$.
2. (a) $4/11$; (b) $1/3$; (c) $5/9$; (d) $76/99$; (e) $3/11$; (f) 1 ; (g) $1/9$; (h) $32/99$; (i) $65/99$; (j) $98/99$; (k) $4/33$; (l) $5/11$; (m) $1/6$; (n) $7/3$; (o) $37/300$.
3. (a) $A \cap B = (1, 4)$ and $A \cup B = [-3, 6]$; (b) $C \cap D = [-1, 2]$ and $C \cup D = (-\infty, 5)$; (c) $E \cap F = \emptyset$ and $E \cup F = (-4, 7]$.
4. (a) $[-1, 6]$; (b) $(-2, 4/3)$; (c) $(-\infty, 1] \cup [7, \infty)$.
5. (a) i ; (b) -1 ; (c) i ; (d) -1 ; (e) 0 .
6. (a) $8 + 5i$; (b) $14 - 5i$; (c) $1 + 3i$; (d) $3 - 8i$; (e) $3 - 4i$; (f) $9/5 + (7/5)i$; (g) $5 + 5i$.
7. (a) $x = 3 \pm 2i$; (b) $x = -2 \pm 3i$; (c) $x = 1 \pm 2i$.
8. (a) a^2 ; (b) x^2 ; (c) m^5/n^3 ; (d) p^2/q^3 .
9. (a) $4\sqrt{2}$; (b) $9\sqrt{3}$; (c) $5\sqrt{5}$; (d) $5\sqrt{2}$.
10. (a) $2\sqrt{3} - 2$; (b) $5\sqrt{6}/6$; (c) $3\sqrt{5} - 6$; (d) $7 + 4\sqrt{3}$.
11. (a) $x = 3$; (b) $x = 1 + \sqrt{10}$; (c) $x = 2$; (d) $x = 13$.
12. (a) $3\sqrt{6}$; (b) $8\sqrt{2}$; (c) $\frac{\sqrt{7}}{3}$; (d) $\frac{2\sqrt{3}}{5}$; (e) $10\sqrt{2}$; (f) $3\sqrt{10}/4$.
13. (a) $4\sqrt{2}$; (b) $3\sqrt{3} + 5\sqrt{6}$; (c) $6\sqrt{5}$; (d) $7\sqrt{3}$; (e) $12\sqrt{2}$; (f) $11\sqrt{3}$.
14. (a) $\sqrt{42}$; (b) $\sqrt{5}$; (c) $15\sqrt{2}$; (d) $3\sqrt{6}$; (e) $30\sqrt{2}$; (f) 4 .
15. (a) $3\sqrt{10} + 2\sqrt{5}$; (b) $24\sqrt{2} - 36$; (c) $\sqrt{7} - 5$; (d) $37 + 10\sqrt{5}$; (e) $5 + 2\sqrt{3}$; (f) -20 .

16. (a) $27 + 8\sqrt{11}$; (b) $57 - 12\sqrt{15}$; (c) 33; (d) 13; (e) $49 + 20\sqrt{6}$; (f) 10.
17. (a) $\frac{3\sqrt{2}}{2}$; (b) $\frac{2\sqrt{15}}{5}$; (c) $\frac{\sqrt{30}}{8}$; (d) $\frac{\sqrt{10} - 2\sqrt{5}}{10}$; (e) $7\sqrt{3}/6$; (f) $\sqrt{5} + \sqrt{2}$.
18. (a) m^4 ; (b) $8s^5t^3$; (c) a^{10} ; (d) $25b^6$; (e) $4x^3y^2$; (f) $\frac{8x^9}{27y^3}$; (g) $9b^8/a^4$; (h) x/y^6 .
19. (a) 7; (b) 5; (c) $\frac{1}{x^3}$; (d) $8y^4$; (e) $\frac{5a^2}{b^3}$; (f) $x^2/2$; (g) $3/p^2$.
20. (a) $\frac{2m^9}{3n^7}$; (b) $\frac{2a^3b}{c^2}$; (c) xy^5 ; (d) $b^3/(2a^5)$.
21. (a) 79 050; (b) 0.000038; (c) 5.16×10^6 ; (d) 4.02×10^{-3} ; (e) 0.000000602; (f) 9.1×10^{-7} .
22. (a) $5^{1/2}$; (b) $3^{3/2}$; (c) $9^{1/3}x^{2/3}$; (d) $2x^{4/5}$; (e) $\sqrt[4]{7}$; (f) $\sqrt[5]{9}$; (g) $2x^{3/4}$; (h) $5\sqrt{5}$.
23. (a) 2; (b) $\frac{1}{3}$; (c) $x = 4$; (d) $x = 1$; (e) 27; (f) $x = 5/2$.
24. (a) $A = 400\,000(1.16)^n$; (b) $A = 800(0.96)^n$; (c) $P = 15\,000(1.025)^n$; (d) $P = 80\,000(0.988)^n$.
25. (a) $V = 600\,000(1.12)^t$; (b) $V(2) = \$752\,640$ and $V(10) \approx \$1\,863\,509$; (c) approximately 4.5 years, or after 5 complete annual growth periods; (d) approximately 210.58%; (e) approximately 6.12 years, or after 7 complete years.
26. (a) $0 = 0/1$; (b) $-12 = -12/1$; (c) $7/8$ is already in the required form.
27. (a) True; (b) false; (c) false; (d) true; (e) false. (f) True; (g) false because $\sqrt{16} = 4$ is rational.
28. (a) $(-3, 0)$; (b) $(-1, e]$; (c) $\mathbb{R}$; (d) $(-1, 4]$; (e) $[2, e]$; (f) $[-3, \pi)$. (g) $(0, 2)$; (h) $(-\infty, -2) \cup [4, \infty)$.
29. (a) 10; (b) $1/25$; (c) -125 ; (d) $1/8$; (e) 9.
30. (a) $x^5/(2y^2)$; (b) $27x^2y^{21}/2$; (c) $5/x$; (d) $\sqrt{y}/y$; (e) a^3b ; (f) $3x$.
31. (a) $\sqrt{15}/3$; (b) $39\sqrt{2}$; (c) $\sqrt{3}\sqrt[3]{4}/4$; (d) $6\sqrt{3} - 13\sqrt{2}$; (e) $8 + 2\sqrt{15}$; (f) $-\sqrt{6}$.
32. (a) $7/2$; (b) 4; (c) 3; (d) $\log(875/8)$; (e) 3; (f) $\ln 3$.
33. (a) $x = 2$; (b) $x = -3$; (c) $x = 9$; (d) $x = \log_2(8/9)$; (e) $x = \log(4)/\log(15/2)$; (f) $x = 1, 1000$; (g) $x = 3, \sqrt[4]{3}$; (h) $x = 9$; (i) $x = 4$.
34. (a) $t = \log(0.5)/\log(0.93) \approx 9.55$ years, so half remains after about 10 complete annual harvests; (b) approximately 834 826 trees; (c) approximately 41.86%.
35. (a) RM 2500; (b) $n \leq 850^{3/2} \approx 24\,781.55$, so the greatest whole number is 24 781 chips; (c) 8000 chips.

36. (a) Both real and complex; (b) non-real complex; (c) non-real complex; (d) both real and complex.
37. (a) $z = 14/13 + (5/13)i$, so $\operatorname{Re}(z) = 14/13$ and $\operatorname{Im}(z) = 5/13$; (b) $w = 1/2 + (5/2)i$; (c) $|z| = \sqrt{221}/13$.
38. (a) 3; (b) 2; (c) 0; (d) 7; (e) 4; (f) $6i$; (g) 0; (h) $10 - 3i$; (i) 13; (j) $\sqrt{13}$.
39. (a) -12 ; (b) -4 ; (c) 5.
40. (a) $z + \bar{z} = 12$, $z\bar{z} = 45$; (b) $z + \bar{z} = 2\sqrt{5}$, $z\bar{z} = 50$; (c) $z + \bar{z} = 2$, $z\bar{z} = 5$; (d) $z + \bar{z} = 0$, $z\bar{z} = 16$.
41. (a) $6 + 12i$; (b) $34 - 13i$; (c) i ; (d) 1; (e) $-i$; (f) $3/2 - (3/2)i$; (g) $43/29 - (6/29)i$; (h) $3/5 - (4/5)i$; (i) -4 .
42. (a) $x = 1$, $y = -1$; (b) $x = 1$, $y = 3$; (c) $x = 19/13$, $y = 22/13$.
43. (a) $x = \pm 3i$; (b) $x = 2 \pm i\sqrt{3}$; (c) $x = -1 \pm 2i$; (d) $x = 1 \pm 3i$; (e) $x = -1 \pm 2i$.
44. (a) $x^2 - 8x + 20 = 0$; (b) $x^2 + 2x + 10 = 0$; (c) $x^2 - 4x + 9 = 0$.
45. (a) The points are $(2, 2)$, $(2, -2)$, $(-2, 2)$ and $(-2, -2)$; (b) each modulus is $2\sqrt{2}$; (c) the points form a square centred at the origin and are symmetric about both coordinate axes.

Exercises 46–56: grouped skills review.

46. (a) $\mathbb{Z}$; (b) $\mathbb{W}$; (c) $\mathbb{N}$; (d) $\mathbb{Q}$; (e) $\mathbb{R} \setminus \mathbb{Q}$.
47. (a) $1/9$; (b) $2/9$; (c) $1/3$; (d) $4/9$; (e) $5/9$.
48. (a) $(-2, 4]$; (b) $(-3, 5]$; (c) $(-4, 6]$; (d) $(-5, 7]$; (e) $(-6, 8]$.
49. (a) x^5 ; (b) x^6 ; (c) x^7 ; (d) x^8 ; (e) x^9 .
50. (a) $2\sqrt{2}$; (b) $3\sqrt{3}$; (c) $4\sqrt{5}$; (d) $5\sqrt{6}$; (e) $6\sqrt{7}$.
51. (a) $\sqrt{2}$; (b) $\sqrt{3}$; (c) $\sqrt{5}$; (d) $\sqrt{6}$; (e) $\sqrt{7}$.
52. (a) 2; (b) 3; (c) 4; (d) 5; (e) 6.
53. (a) 210; (b) 275; (c) 345; (d) 420; (e) 500.
54. (a) $2 + i$; (b) $5 + i$; (c) $8 + i$; (d) $11 + i$; (e) $14 + i$.
55. $i, -1, -i, 1, i$, respectively.
56. (a) $x = 1 \pm 2i$; (b) $x = 2 \pm 3i$; (c) $x = 3 \pm 4i$; (d) $x = 4 \pm 5i$; (e) $x = 5 \pm 6i$.

Exercises 57–67: quantitative reasoning and measurement.

57. (a) $384/600 = 0.64 = 64\%$; (b) 216 residents.
58. $1\,260\,000/3500 = 360$ residents per square kilometre.
59. Education RM 12 000; health RM 7200; recreation RM 4800.
60. (a) $0.437(2450) = 1070.65$, so approximately 1071 voters; (b) it is a sample-based estimate and the calculated value is not a whole-number observation.
61. (a) 18 interviews per hour; (b) 25 hours.
62. (a) 20 m/s; (b) 42 km.
63. (a) 6.5 percentage points; (b) $(64.5/58 - 1)100\% \approx 11.21\%$.
64. The index is $(276/240)100 = 115$; the cost is 15% above its earlier value.
65. $0.40(72\%) + 0.60(55\%) = 61.8\%$.
66. (a) RM 75 per resident; (b) approximately RM 71.43 per resident.
67. [2.35, 2.45) million.

Exercises 68–78: percentages, rates and indices.

68. $(1.06/1.04 - 1)100\% \approx 1.92\%$ increase.
69. (a) $2500(0.92)(1.08) = \text{RM } 2484$; (b) RM 16, or 0.64%, below the original benefit.
70. $3200(1.06) + 25 = \text{RM } 3417$.
71. $1840/1.15 = \text{RM } 1600$.
72. (a) $500(0.80)(0.90) = \text{RM } 360$; (b) 28%.
73. A fall of 1.2 percentage points and a relative decrease of $(1.2/7.5)100\% = 16\%$.
74. $1500(132/120) = \text{RM } 1650$.
75. Housing RM 1470; transport RM 504; remaining RM 2226.
76. $(18/1000)(750\,000) = 13\,500$ migrants.
77. Response rate $= 714/1200 = 59.5\%$; non-responses $= 486$.
78. $(1 - 110/125)100\% = 12\%$. The 15-point fall is measured relative to the starting index 125, not to 100.

Exercises 79–89: powers, logarithms and repeated change.

79. $80\,000(1.025)^4 \approx 88\,305$ residents.
80. $12\,000(0.94)^5 \approx 8807$ subscribers.
81. $\ln 2 / \ln 1.04 \approx 17.67$ years.
82. $\ln 0.5 / \ln 0.88 \approx 5.42$ months.
83. $n = \ln(1.5) / \ln(1.08) \approx 5.27$ periods; the first complete period meeting the target is period 6.
84. $B/A = e^{0.30} \approx 1.350$; Region B's average income is about 35.0% higher.
85. (a) $10^5/10^3 = 100$; (b) $5 - 3 = 2$.
86. $(3.6 \times 10^6)(2.4) = 8.64 \times 10^6$ people.
87. (a) $L(5) = 1600$ and $E(5) = 1610.51$; (b) the exponential prediction is larger by 10.51 participants.
88. $90\,000(0.85)^3 = \text{RM } 55\,271.25$.
89. $t = \ln(300/200)/0.06 \approx 6.76$ time units.

Exercises 90–100: modelling and interpretation.

90. $[2.75, 2.85)$ million ringgit.
91. Overlap = $[52, 56]$; union = $[48, 60]$.
92. $\sqrt{200} = 10\sqrt{2}$ km ≈ 14.14 km.
93. $|z| = \sqrt{6^2 + 8^2} = 10$ km.
94. $i^{14} = -1$, the third stage in the stated cycle.
95. (a) $|z| = 5$; (b) $\bar{z} = 3 - 4i$; (c) the modulus represents the component's magnitude or amplitude.
96. (a) $q = 3 \pm 2i$; (b) the model has no real break-even value of q .
97. (a) The intercept is in millions of ringgit and the slope is in millions of ringgit per year; (b) $C(4) = 37$, predicting a cost of RM 37 million after four years.
98. Difference = RM 600 and ratio = $3000/2400 = 1.25$. The percentage comparisons use different starting incomes: $600/2400 = 25\%$ but $600/3000 = 20\%$.
99. An increase of 1 percentage point and a relative increase of $(3/2 - 1)100\% = 50\%$.
100. (a) $50\,000(1.03)^{20} \approx 90\,306$ residents; (b) any two suitable cautions, such as the assumption of a constant growth rate, migration or policy changes, economic shocks, resource constraints, or uncertainty in the initial data.

C.2 Chapter 2: Functions

Return to the corresponding questions in Chapter 2.

1. It is not a function because the input 2 has two outputs. Domain = $\{-2, 0, 2\}$; range = $\{1, 3, 5\}$.
2. $(-\infty, -2) \cup (-2, 5]$.
3. $f(-2) = 7$.
4. Horizontal compression by factor $1/3$, translation one unit right, vertical stretch by factor 2 with reflection in the x -axis, and translation four units upward.
5. Vertex $(-2, 5)$; axis $x = -2$; domain $\mathbb{R}$; range $(-\infty, 5]$.
6. $r(x) = 2 + \frac{7}{x-3}$; vertical asymptote $x = 3$; horizontal asymptote $y = 2$; domain $\mathbb{R} \setminus \{3\}$; range $\mathbb{R} \setminus \{2\}$.
7. $x = 2$.
8. Domain $(4, \infty)$; range $\mathbb{R}$; vertical asymptote $x = 4$.
9. $\left(\frac{f}{g}\right)(x) = \frac{x+1}{x^2-4}$, with $x \neq -2, 2$.
10. $(f \circ g)(x) = 2x^2 - 1$ and $(g \circ f)(x) = 4x^2 - 12x + 10$.
11. $f^{-1}(x) = \frac{3x+5}{4}$.
12. The function is both one-to-one and onto, so it is bijective; $q^{-1}(x) = \sqrt[3]{x}$.
13. (a) Up 4 units; (b) $y = |x - 2|$; (c) right 2 and up 3; (d) left 3 and down 2; (e) $y = 2(x+3)^2 - 1$; (f) right 3 units; (g) down 7 units.
14. (a) It is wider; equivalently, it is vertically compressed by factor $1/2$. (b) Both graphs are vertically compressed by factor $1/2$ and therefore appear wider. (c) $y = -5x^2$, $y = 3x^2$, $y = \frac{1}{2}x^2$. (d) It is narrower and opens upward. (e) Reflection in the x -axis and a further vertical stretch by factor 2. (f) It opens downward and is narrower, with vertical scale factor $|k|$. (g) Vertical stretch by factor 2, then translation 3 units right. (h) Vertical stretch by factor 3, translation 2 units right and 1 unit up.
15. (a) $g(x) = f(x-2) + 1$; (b) $k = -2$; (c) translation 1 unit left and 2 units down.
16. (a) Right 2, reflect in the x -axis, up 3; (b) reflect in the y -axis, down 5; (c) reflect in the y -axis, right 2; (d) right 4, reflect in the x -axis, up 1; (e) reflect in the y -axis, right 1, reflect in the x -axis, down 4.

17. The successive vertices are

$$\begin{aligned} f(-x) &: (-4, 4), (-2, 0), (0, 2), (2, 2), \\ -f(x) &: (-2, -2), (0, -2), (2, 0), (4, -4), \\ -f(-x) &: (-4, -4), (-2, 0), (0, -2), (2, -2). \end{aligned}$$

18. (a) $(0, -1)$; (b) $(4, -1)$; (c) $(2, 1)$; (d) $(2, -3)$; (e) $(-2, 1)$; (f) $(2, 3)$, now a maximum.

19. (a) C; (b) B; (c) D; (d) E; (e) F; (f) A.

20. (a) $g(x) = -3(x-2) - 3 = -3x + 3$; (b) $g(x) = -\frac{1}{2}(x+1)^2 + 2$; (c) $g(x) = 3(x-2)^2 - 1$.

21. (a) Right 2 and down 3; (b) right 2, reflect in the x -axis and up 2.

22. (a) x -intercept $(1/3, 0)$, y -intercept $(0, 1)$, domain and range $\mathbb{R}$. (b) vertex $(1, 4)$, x -intercepts $(-1, 0), (3, 0)$, y -intercept $(0, 3)$, domain $\mathbb{R}$, range $(-\infty, 4]$. (c) vertex $(1, 1)$, no real x -intercept, y -intercept $(0, 3)$, domain $\mathbb{R}$, range $[1, \infty)$.

23. (a) asymptotes $x = 0, y = 0$, domain and range $\mathbb{R} \setminus \{0\}$; (b) asymptotes $x = 1, y = 1$, domain $\mathbb{R} \setminus \{1\}$, range $\mathbb{R} \setminus \{1\}$; (c) asymptotes $x = 1, y = -1$, domain $\mathbb{R} \setminus \{1\}$, range $\mathbb{R} \setminus \{-1\}$.

24. (a) $x = -1, y = -2$; domain $\mathbb{R} \setminus \{-1\}$, range $\mathbb{R} \setminus \{-2\}$. (b) $x = 0, y = 1$; domain $\mathbb{R} \setminus \{0\}$, range $\mathbb{R} \setminus \{1\}$. (c) $x = 3, y = 1$; domain $\mathbb{R} \setminus \{3\}$, range $\mathbb{R} \setminus \{1\}$.

25. (a) Left 3, down 2; (b) left 2, reflect in the x -axis, down 3; (c) reflect in the y -axis, reflect in the x -axis, up 1; (d) reflect in the y -axis, right 1, reflect in the x -axis, up 2.

26. (a) $g(x) = 2/(x-2) - 1$; (b) $g(x) = -3/(x+2) - 2$; (c) $g(x) = \ln(x+3) - 2$; (d) $g(x) = e^{x+3} + 2$.

27. (a) asymptotes $x = 3, y = 2$; intercepts $(11/2, 0)$ and $(0, 11/3)$; domain $\mathbb{R} \setminus \{3\}$, range $\mathbb{R} \setminus \{2\}$. (b) vertical asymptote $x = 3$; intercepts $(3 - e^{-2}, 0)$ and $(0, -2 - \ln 3)$; domain $(-\infty, 3)$, range $\mathbb{R}$. (c) vertical asymptote $x = -2$; x -intercept $(-2 - e^{-4}, 0)$, no y -intercept; domain $(-\infty, -2)$, range $\mathbb{R}$. (d) horizontal asymptote $y = 2$; intercepts $(2 - \ln 2, 0)$ and $(0, 2 - e^2)$; domain $\mathbb{R}$, range $(-\infty, 2)$.

28. Vertex $(3, 8)$; axis $x = 3$; x -intercepts $(1, 0)$ and $(5, 0)$; y -intercept $(0, -10)$; domain $\mathbb{R}$; range $(-\infty, 8]$; increasing on $(-\infty, 3)$ and decreasing on $(3, \infty)$.

29. $(67 - 46)/(8 - 2) = 3.5$ percentage points per year. On average, support rises by 3.5 percentage points for each year across the stated interval.

30. (a) Even; (b) odd; (c) neither.

31. $180 - 5p = 30 + 10p$ gives $p = 10$ and the common output is 130.

32. $C(8) = \text{RM } 17$ and $C(25) = \text{RM } 36.50$. At $d = 10$, both branches give RM 20.

33. Domain $[0, \infty)$; range $[0, 75)$; $A(0) = 0$; horizontal asymptote $y = 75$; $A(5) \approx 58.27\%$. The model approaches a limiting reach of 75%.
34. $(f \circ g)(x) = \sqrt{2x - 4}$ with domain $[2, \infty)$; $(g \circ f)(x) = 2\sqrt{x + 1} - 5$ with domain $[-1, \infty)$.
35. $F^{-1}(x) = (x - 12)/1.06$, with domain $[12, \infty)$; $F^{-1}(118) = 100$, so the original amount is RM 100.
36. Since $h(2) = 4 + b = 7$, $b = 3$. Joining at $x = 1$ requires $a + 2 = 1 + b = 4$, so $a = 2$.
37. Extrapolation assumes that the observed relationship continues beyond the available data. Structural changes, new constraints or different behaviour outside the observed range may make that assumption unreliable.
38. A suitable domain is $0 \leq t \leq 25$, with range $0 \leq V \leq 500$. All places are filled when $500 - 20t = 0$, giving $t = 25$ months.
39. Over $[0, 5]$, the average rate is $(24 - 12)/(5 - 0) = 12/5 = 2.4$ output units per input unit. Over $[5, 9]$, it is $(20 - 24)/(9 - 5) = -1$. The positive sign represents an average increase, while the negative sign represents an average decrease.
40. One choice is $g(x) = 5 - 2x$ and $f(u) = \ln u$, so $h = f \circ g$. The condition $5 - 2x > 0$ gives domain $(-\infty, 5/2)$.
41. On $[1, \infty)$, the function is increasing and each output has only one permitted input. From $y = (x - 1)^2 + 4$ with $x \geq 1$,

$$f^{-1}(x) = 1 + \sqrt{x - 4}.$$

Its domain is $[4, \infty)$ and its range is $[1, \infty)$.

Exercises 42–51: grouped skills review.

42. (a) -1 ; (b) 3 ; (c) 11 ; (d) 23 ; (e) 39 ; (f) 59 .
43. For each k , the domain is $\mathbb{R} \setminus \{k\}$.
44. (a) $[-2, \infty)$; (b) $[-1, \infty)$; (c) $[0, \infty)$; (d) $[1, \infty)$; (e) $[2, \infty)$; (f) $[3, \infty)$.
45. Respectively: right 1 and up 2; right 2 and up 3; right 3 and up 4; right 4 and up 5; right 5 and up 6; right 6 and up 7.
46. The vertices are $(1, -2), (2, -1), (3, 0), (4, 1), (5, 2), (6, 3)$; the corresponding axes are $x = 1, 2, 3, 4, 5, 6$ and the ranges are $[-2, \infty), [-1, \infty), [0, \infty), [1, \infty), [2, \infty)$ and $[3, \infty)$.
47. $2x^2 + 1, 2x^2 + 3, 2x^2 + 5, 2x^2 + 7, 2x^2 + 9, 2x^2 + 11$.
48. (a) $(x - 1)/2$; (b) $(x - 2)/3$; (c) $(x - 3)/4$; (d) $(x - 4)/5$; (e) $(x - 5)/6$; (f) $(x - 6)/7$.
49. $2, 5, 8, 11, 14, 17$, respectively.

50. $S_k(10) = 30, 43, 56, 69, 82, 95$, respectively. The slopes are 2, 3, 4, 5, 6, 7 indicator units per time period.
51. The pairs $(p_k(-2), p_k(3))$ are $(-1, 8), (0, 7), (1, 6), (2, 5), (3, 4), (4, 3)$.

Exercises 52–58: interpreting functions.

52. It is not a function because input 2 has outputs 5 and 7. Domain = $\{1, 2, 3\}$; range = $\{4, 5, 7, 8\}$.
53. (a) $R(6) = 228$; (b) 120 is the initial response count and 18 is the increase per week; (c) $0 \leq t \leq 10$, or integer weeks 0 through 10 if only complete weeks are recorded.
54. (a) Domain = $\{0, 1, \dots, 200\}$ and range = $\{500 + 12n : n = 0, 1, \dots, 200\}$; (b) $C(150) = \text{RM } 2300$.
55. (a) Domain $[0, 100]$ and range $[0, 10]$; (b) $B(64) = 6$.
56. (a) $n \in \mathbb{N}$; (b) $A(40) = 55$; (c) $A(n)$ approaches RM 25 from above.
57. $G(18\,500) = 1200$, $G(25\,000) = 800$ and $G(32\,000) = 0$. The jumps occur at incomes RM 20 000 and RM 30 000.
58. (a) $D(42) = 8$ and $D(67) = 17$; (b) vertex $(50, 0)$ and range $[0, \infty)$; (c) $D(x)$ is the absolute number of index points between x and the target 50.

Exercises 59–65: linear and piecewise models.

59. (a) $F(12) = \text{RM } 26.10$; (b) $d = 15$ km; (c) RM 4.50 is the fixed charge and RM 1.80 is charged per kilometre.
60. $W(38) = \text{RM } 684$ and $W(46) = \text{RM } 882$. Both formulas give RM 720 at $h = 40$.
61. (a) Domain $[0, 16]$ months and range $[0, 12\,000]$ ringgit; (b) the fund is exhausted after 16 months.
62. (a) $R(5) = \text{RM } 1075$; (b) $t = 10$ years; (c) rent increases by RM 35 per year in the model.
63. $2000 + 15q = 25q$ gives $q = 200$ and a common value of RM 5000.
64. $M(3) = \text{RM } 30$ and $M(9) = \text{RM } 54$. Both formulas give RM 30 at $g = 5$.
65. (a) $I(t) = 2500 + 100t$; (b) $I(7) = \text{RM } 3200$; (c) the linear rate may not remain constant far beyond the observed period.

Exercises 66–72: quadratic functions.

66. (a) $q = 150$; (b) $R(150) = 4500$; (c) zeros 0 and 300, giving contextual domain $[0, 300]$.
67. Maximum score 80 at $h = 5$; $S(3) = 72$.
68. (a) $C(x) = (x - 10)^2 + 80$; (b) minimum 80 at $x = 10$.
69. $P(0) = 200$; maximum occurs at $t = 6$ and equals 380.
70. (a) $x = 2, 6$; (b) $D(x) < 0$ for $2 < x < 6$, meaning the first group score is lower than the second under the stated difference convention.
71. $42 = 16a + 90$ gives $a = -3$, so $Q(t) = -3(t - 4)^2 + 90$.
72. (a) $R(p) = 500p - 20p^2$; (b) $p = 12.5$; (c) attendance 250 and maximum revenue 3125.

Exercises 73–79: exponential and logarithmic functions.

73. $P(5) \approx 57\,964$. The factor 1.03 represents annual growth of 3%.
74. $V(4) = \text{RM } 41\,760.50$; horizontal asymptote $V = 0$; contextual range $(0, 80\,000]$.
75. $A(0) = 45$, $A(5) \approx 66.62$ and the limiting value is 70.
76. Domain $(0, \infty)$ and range $\mathbb{R}$; $L(1000) = 0$ and $L(1000e) = 10$.
77. $t = \ln(1.5)/\ln(1.06) \approx 6.96$ periods, so the first complete period meeting the target is period 7.
78. (a) Domain $(2, \infty)$ and range $\mathbb{R}$; (b) $f^{-1}(x) = e^{x-3} + 2$, with domain $\mathbb{R}$.
79. $L(10) = 220$ and $E(10) \approx 259.37$; the exponential model is larger, illustrating compounding rather than equal additions.

Exercises 80–86: composition and inverse functions.

80. $(F \circ T)(x) = 1.06x + 12$ and $(T \circ F)(x) = 1.06x + 12.72$. In the second order, tax is also applied to the fee.
81. $(f \circ g)(x) = 0.90x + 200$ and $(g \circ f)(x) = 0.90x + 180$. In the second order, the allowance is reduced by the deduction as well.
82. $(C \circ q)(t) = 1500 + 100t$ and $(C \circ q)(6) = 2100$.
83. $S^{-1}(x) = x/10 + 50$ and $S^{-1}(220) = 72$.
84. $q^{-1}(x) = 125 - x/4$; demand 300 corresponds to price 50.
85. The restriction $x \geq 3$ selects the increasing branch. $f^{-1}(x) = 3 + \sqrt{x - 2}$, with domain $[2, \infty)$ and range $[3, \infty)$.
86. $(f \circ g)(x) = \sqrt{2x - 6}$ with domain $[3, \infty)$; $(g \circ f)(x) = 2\sqrt{x} - 6$ with domain $[0, \infty)$.

Exercises 87–93: rates, tables and fitted models.

87. Over $[0, 5]$ the rate is $(77 - 50)/5 = 5.4$ units per time period; over $[5, 8]$ it is $(74 - 77)/3 = -1$. The index first rises on average and then falls.
88. $A(q) = 1000/q + 20$; $A(50) = 40$ and $A(200) = 25$.
89. $[P(4) - P(1)]/(4 - 1) = (52 - 25)/3 = 9$.
90. $\hat{y} = 32$ and residual $= 29 - 32 = -3$; the observed value is 3 units below the prediction.
91. At $t = 0$, $E/N = 500/10\,000 = 5\%$; at $t = 10$, $E/N = 900/12\,000 = 7.5\%$.
92. The first differences are constantly 3, so $y = 3x + 5$ and $y(10) = 35$.
93. The predictions are 140 and 340. The first is interpolation and the second is extrapolation; the extrapolation is less reliable.

Exercises 94–100: modelling and critical interpretation.

94. Contextual domain $[0, 80]$ and range $[0, 400]$.
95. $T(25\,000) = 2500$ and $T(40\,000) = 5000$. Both formulas give 3000 at $y = 30\,000$.
96. The slope predicts a decrease of 0.4 rate units for each additional intervention unit; the intercept predicts rate 12 at zero intervention. $C(20) = 4$. A fitted association may reflect confounding variables and does not by itself establish causation.
97. An exponential model is appropriate: $P(t) = 200(2)^t$, giving $P(5) = 6400$.
98. An increasing fitted relationship shows association within the data and model. Other variables, selection effects or reverse causation may explain some or all of the pattern.
99. $B(24\,999) = 1000$ and $B(25\,000) = 0$. Crossing the threshold by one ringgit causes an abrupt RM 1000 loss under this rule.
100. (a) $C(d) = \begin{cases} 6 + 2d, & 0 \leq d \leq 12, \\ 30, & d > 12; \end{cases}$ (b) domain $[0, \infty)$ and range $[6, 30]$; (c) the graph becomes constant at $d = 12$ kilometres.

C.3 Chapter 3: Logic and Sets

Return to the corresponding questions in Chapter 3.

- (a) Proposition, true; (b) not a proposition; (c) not a proposition; (d) proposition, false.
- At least one student did not submit the assignment.
- The final two columns agree:

p	q	$p \rightarrow q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

4. Converse: if an integer is even, then it is divisible by 4. Inverse: if an integer is not divisible by 4, then it is not even. Contrapositive: if an integer is not even, then it is not divisible by 4.

5. $A \cap B = \{2\}$; $A \cup B = \{2, 3, 4, 5, 6, 7, 8, 10\}$; $A^c = \{1, 3, 5, 7, 9\}$; $B \setminus A = \{3, 5, 7\}$.

6. 53 study at least one subject and 7 study neither.

7.

$$\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\},$$

with cardinality 8.

8. 24 outfits.

9. $\frac{11!}{2!2!2!} = 4\,989\,600$.

10. $2(6!) = 1\,440$.

11. $\binom{6}{2}\binom{5}{2} = 150$.

12. $\binom{6}{3}\binom{4}{2} + \binom{6}{4}\binom{4}{1} + \binom{6}{5} = 186$.

13. (a) True; (b) true; (c) false; (d) true; (e) false; (f) true.

14. (a) $n(A) = 2$ and $n(B) = 3$. (b) A has 4 subsets and B has 8 subsets. (c) $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ and

$$\mathcal{P}(B) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

(d) A has 3 proper subsets and B has 7.

15. (a) $\{1, 8\}$; (b) $\{1, 4, 8, 11, 13\}$; (c) $\{6, 11, 13\}$; (d) $\{11, 13\}$.

16. (a) $\{q, r, s\}$; (b) $\{p, q, r, s, v, x\}$; (c) $\{t, u, w, y\}$; (d) $\{s\}$; (e) $\emptyset$; (f) $\{p, v\}$.

17. (a) $A \cup B = \{1, 2, 4, 6, 8, 10\}$, $A \cap B^c = \{6, 10\}$ and $A \cap B = \{2, 4, 8\}$. (b) $A \cap B = \{o, r, t\}$, $A \cup B = \{a, b, f, n, o, r, s, t\}$ and $B \setminus A = \{a, b, n\}$. (c) $A \cup B = \{2, 3, 5, 6, 7\}$ and $A \cap B = \emptyset$.

18. (a) 23; (b) 30; (c) 12.

19. ${}_3P_2 = 6$.

20. $\binom{9}{3} - \binom{6}{3} = 84 - 20 = 64$.
21. (a) $\frac{7!}{4!} = 210$; (b) $\frac{11!}{4!4!2!} = 34\,650$.
22. (a) Proposition, true; (b) not a proposition; (c) proposition, true; (d) not a proposition until x is fixed; (e) not a proposition.
23. (a) I did not attend the conference this month. (b) I attended the conference or presented a paper, or both. (c) If I attended, then I presented a paper. (d) If I did not attend, then I did not present a paper.
24. (a) $r \wedge \neg q$; (b) $p \wedge q \wedge r$; (c) $p \wedge \neg q \wedge r$; (d) $(p \wedge q) \rightarrow r$.
25. Mathematics only = 16; science only = 12; either subject = 52; neither = 8.
26. $5(6)(4) = 120$.
27. $36^5 = 60\,466\,176$.
28. $5! = 120$.
29. (a) $6! = 720$; (b) ${}_6P_3 = 120$.
30. (a) $4! = 24$; (b) ${}_4P_2 = 12$.
31. $10!/(4!2!2!) = 37\,800$.
32. (a) ${}_8P_6 = 20\,160$; (b) ${}_4P_2(4){}_5P_3 = 2\,880$.
33. (a) $\binom{12}{4} = 495$; (b) $\binom{7}{4} = 35$; (c) $\binom{7}{3}\binom{5}{1} = 175$; (d) $\binom{7}{2}\binom{5}{2} = 210$; (e) $210 + 175 + 35 = 420$.
34. (a) $\binom{10}{5} = 252$; (b) $\binom{9}{4} = 126$; (c) $\binom{9}{5} = 126$; (d) $\binom{6}{3}\binom{4}{2} = 120$; (e) $\binom{6}{5} = 6$; (f) $\binom{4}{3}\binom{6}{2} + \binom{4}{4}\binom{6}{1} = 66$.

Exercises 35–45: grouped skills review.

35. $p \wedge q$ is false; $p \vee q$ is true; $\neg p$ is false; $p \Rightarrow q$ is false; $q \Rightarrow p$ is true; and $p \Leftrightarrow q$ is false.
36. In each part: at least one of the stated number of surveyed households does not own a computer.
37. The values are true, true, false and true, respectively. A conditional with a false antecedent is true.
38. (a) $A \cap B = \{2, 3\}$ and $A \cup B = \{1, 2, 3, 4\}$;
(b) $A \cap B = \{3, 4\}$ and $A \cup B = \{1, 2, 3, 4, 5\}$;
(c) $A \cap B = \{4, 5\}$ and $A \cup B = \{1, 2, 3, 4, 5, 6\}$;
(d) $A \cap B = \{5, 6\}$ and $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$;
(e) $A \cap B = \{6, 7\}$ and $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$;

(f) $A \cap B = \{7, 8\}$ and $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

39. 6, 12, 20, 30, 42, 56.

40. 8, 16, 32, 64, 128, 256.

41. 24, 60, 120, 210, 336, 504.

42. 60, 120, 210, 336, 504, 720.

43. 15, 21, 28, 36, 45, 55.

44. 60, 360, 2520, 20 160, 181 440, 1 814 400.

45. 18, 60, 150, 315, 588, 1008.

Exercises 46–56: legal rules and logical reasoning.

46. (a) $p \wedge q$; (b) $\neg p \vee \neg q$; (c) not eligible because the conjunction is false.

47. (a) $\neg p \vee \neg q$; (b) $p \wedge q$; (c) the approval condition is the negation of denial by De Morgan's law.

48. Converse: if authenticated, then admitted. Inverse: if not admitted, then not authenticated. Contrapositive: if not authenticated, then not admitted. Only the contrapositive is logically equivalent to the original.

49. (a) $H \Rightarrow D$; (b) $\neg D \Rightarrow \neg H$; (c) other requirements may also be necessary, so D alone need not imply H .

50. $V \Leftrightarrow (F \wedge T)$. When F is true and T is false, V is false.

51. The values are true, true, false and false, respectively.

52. At least one witness signed no statement.

53. $M \wedge \neg A \Rightarrow \neg E$; equivalently, $M \wedge E \Rightarrow A$.

54. $(F \vee W) \wedge \neg(F \wedge W)$. It is false when both outcomes occur.

55. (a) Valid, modus ponens; (b) valid, modus tollens; (c) invalid, affirming the consequent.

56. No. From p and $p \Rightarrow q$, obtain q ; then $q \Rightarrow r$ gives r , contradicting $\neg r$.

Exercises 57–67: legal and administrative sets.

57. $|C \cup A| = 150$; neither = 50; civil only = 80; appealed only = 30.
58. At least one = 105; neither = 35; eligible only = 30; represented only = 15.
59. $2^8 = 256$ subsets, of which 255 are non-empty.
60. $2^5 = 32$ combinations including none; $\binom{5}{2} = 10$ contain exactly two remedies.
61. $4(6) = 24$ assignments.
62. $P \setminus R = \{1, 3, 5\}$ and $R^c = \{1, 3, 5\}$ relative to P .
63. $A \triangle B = \{a, b, e\}$.
64. $500 - 420 = 80$ incomplete forms.
65. At least one = $160 + 140 + 100 - 70 - 50 - 40 + 20 = 260$; none = 40.
66. $A \subset C$; it is proper because $|A| < |C|$, so the sets cannot be equal.
67. Yes. The sets contain exactly the same elements; order is irrelevant in a set.

Exercises 68–78: counting in legal procedures.

68. $3(4)(3) = 36$.
69. $5! = 120$.
70. $\binom{8}{3} = 56$.
71. $\binom{10}{6} = 210$.
72. $\binom{5}{2} \binom{7}{3} = 10(35) = 350$.
73. Treating the two specified items as a block gives $2(5!) = 240$.
74. $6! - 240 = 480$.
75. $6!/(2!2!) = 180$.
76. $4! = 24$.
77. $9P_2 = 9(8) = 72$.
78. $5P_2(10^3) = 20\,000$.

Exercises 79–89: surveys and public administration.

79. At least one = 390; neither = 110; support only = 110; awareness only = 70.
80. $2^3 - 1 = 7$.
81. $4! = 24$.
82. $8P_3 = 336$.
83. $\binom{12}{4} = 495$.
84. $\binom{12}{5} - \binom{8}{5} = 792 - 56 = 736$.
85. $10P_3 = 720$.
86. $2^6 = 64$.
87. $2^4(3) = 48$.
88. At least one = $620 + 480 - 250 = 850$; neither = 150.
89. $5! = 120$.

Exercises 90–100: mixed reasoning and interpretation.

90. It is inclusive or: $p \vee q$.
91. The converse would say that being on the register implies being licensed. The original conditional does not rule out other reasons for appearing on the register.
92. $C \Rightarrow E$. Observing E and concluding C would affirm the consequent; other requirements for conviction may fail.
93. Eligible set = $R \cap L$; eligible = 70. Since $|R \cup L| = 150 + 90 - 70 = 170$, neither = 30.
94. $3(2)(2) = 12$ routes.
95. COURT: $5! = 120$; LETTER: $6!/(2!2!) = 180$. The extra position more than offsets the repeated letters.
96. $\binom{10}{4} - \binom{6}{4} - \binom{4}{4} = 210 - 15 - 1 = 194$.
97. If p is true, then q and hence r are true. If r is false, the contrapositive chain gives p false.
98. A subset cannot have greater cardinality than the finite set that contains it, so the two reported claims cannot both be correct.
99. $|A \cup B| = 250 - 40 = 210$, so $|A \cap B| = 140 + 110 - 210 = 40$.
100. The rule is $A \vee B$. Suitable cautions include biased or inaccurate input data, poorly chosen definitions of A and B , unequal error costs, missing context, and discriminatory effects even when the Boolean calculation is implemented correctly.

C.4 Chapter 4: Introduction to Statistics

Return to the corresponding questions in Chapter 4.

1. Population: all 8 000 students; sample: the 250 surveyed students; parameter: the true proportion of all students who use public transport; statistic: 0.62.
2. (a) Categorical; (b) quantitative discrete; (c) quantitative continuous; (d) categorical ordinal.
3. Mean = 8, median = 7, mode = 7.
4. Mean = 2.4, median = 2.5, mode = 3.
5. Estimated mean = 6.5.
6. Five-number summary (2, 4, 7, 9, 15); IQR = 5; fences -3.5 and 16.5 , so there are no outliers.
7. Range = 4; $\sigma^2 = 2$; $\sigma = \sqrt{2} \approx 1.414$.
8. $s^2 = \frac{8}{3} \approx 2.667$; $s = \sqrt{8/3} \approx 1.633$.
9. Mean = 20 and median = 12. The median is more representative because the unusually large value 54 pulls the mean upward.
10. (a) Continuous; (b) discrete; (c) discrete; (d) continuous; (e) discrete.
11. (a) Quantitative; (b) qualitative; (c) qualitative; (d) quantitative; (e) qualitative.
12. Population: all safety gloves produced for use on Malaysian construction sites by the manufacturers of interest. Sample: the 400 collected gloves. Variables: measured resistance to cuts, punctures and abrasion, with manufacturer as a possible categorical variable.
13. Mean = 9; median = 8; mode = 8; population variance = 23.3; population standard deviation ≈ 4.827 .
14. Estimated mean ≈ 11.167 hours; median ≈ 11.25 hours; mode ≈ 11.5 hours.
15. $Q_1 = 70.5$, $Q_2 = 77.5$, $Q_3 = 86.5$, and IQR = 16. The five-number summary is

(53, 70.5, 77.5, 86.5, 99).

The box plot is roughly symmetric with a slightly longer lower tail.

16. Five-number summary (17, 25, 35, 49, 92); IQR = 24; fences -11 and 85 , so 92 is an outlier. The distribution is positively skewed; the upper non-outlier whisker ends at 60.

17. The working values are

x_i	5	8	11	14	17	20
$f_i x_i$	10	24	77	238	187	160
$f_i x_i^2$	50	192	847	3332	3179	3200

Estimated sample variance ≈ 15.064 and sample standard deviation ≈ 3.881 .

18. Range ≈ 35 ; $Q_1 \approx 37.19$; $Q_3 \approx 46.57$; IQR ≈ 9.39 .
19. Positively skewed: the right tail and upper whisker are longer, and the median lies left of the centre of the box.
20. (a) Population: all adult residents of the municipality; sample: the first 120 adult library visitors surveyed. (b) Library visitors on a Saturday morning may differ systematically from residents who do not visit the library, so the sample may suffer from undercoverage and selection bias. (c) One improvement is a simple random or stratified random sample drawn from residents across neighbourhoods, with responses collected at varied times and by more than one method.
21. The weighted mean is

$$\frac{3(68) + 4(74) + 2(81) + 70}{3 + 4 + 2 + 1} = \boxed{73.2}.$$

Equal weighting would ignore the different credit values.

22. (a) Mean = 36, median = 30, mode = 30, range = 36 thousand ringgit. (b) Mean = 41, median = 35, mode = 35, range = 36. (c) Mean = 39.6, median = 33, mode = 33, range = 39.6; the standard deviation is multiplied by 1.10.
23. (a) Combined mean = $[40(68) + 60(74)]/100 = \boxed{71.6}$. (b) The groups have different sizes, so their means require weights 40 and 60 rather than equal weights.
24. Using midpoints 1, 3, 5, 7, the estimated mean is $148/40 = \boxed{3.7}$ hours. The estimated median is $2 + [(20-8)/15](2) = \boxed{3.6}$ hours, and the estimated mode is $2 + [7/(7+3)](2) = \boxed{3.4}$ hours.
25. (a) Both populations have mean and median 50; their ranges are 4 for A and 40 for B . (b) $\sigma_A = \sqrt{2} \approx 1.414$ and $\sigma_B = 10\sqrt{2} \approx 14.142$. (c) Population A is more consistent because its observations have much smaller spread around the common mean.
26. $\bar{x} = 96/8 = 12$,

$$s^2 = \frac{1240 - 96^2/8}{7} = \frac{88}{7} \approx 12.571, \quad s \approx 3.545.$$

27. (a) Minimum non-outlier = 10, $Q_1 = 18$, median = 25, $Q_3 = 32$, maximum non-outlier = 50. (b) IQR = 14; fences = -3 and 53. (c) Since $65 > 53$, it is an outlier. (d) The longer upper whisker and high outlier suggest positive skewness.

28. (a) Modal class = 20–< 30 minutes. (b) Estimated mean = $[4(5)+9(15)+14(25)+8(35)+5(45)]/40 = \boxed{25.25}$ minutes. (c) Estimated median = $20 + [(20 - 13)/14](10) = \boxed{25}$ minutes. (d) The distribution is unimodal and roughly symmetric around the 20–< 30 minute class.
29. Five-number summary (12, 15.75, 21, 28, 60); IQR = 12.25; fences –2.625 and 46.375. Therefore, 60 is an outlier.
30. $\mu = 3$, $\sigma^2 = 1.3$, and $\sigma = \sqrt{1.3} \approx 1.140$.
31. Region A: mean 32, median 31, range 10. Region B: mean 32, median 24, range 48. Region B's high income of 68 raises its mean and produces much greater spread, so the equal means conceal important differences.
32. (a) Mode with a bar chart, because the variable is nominal. (b) Median and IQR with a box plot or histogram, because the data are quantitative and strongly skewed. (c) Mean or median with a frequency bar chart or dot plot, because the variable is quantitative discrete. (d) Median or mode with an ordered bar chart, because the variable is ordinal. Other well-justified choices are acceptable.
33. (a) Statistic $\hat{p} = 0.54$; parameter: the unknown true proportion p of all voters who support the policy. (b) Random samples contain different voters and therefore exhibit sampling variability. (c) Possible sources include non-response, incomplete population coverage or wording that influences answers; a large sample does not remove systematic bias.

Exercises 34–43: grouped skills review.

34. (a) Categorical nominal; (b) categorical ordinal; (c) quantitative discrete; (d) quantitative continuous; (e) categorical nominal; (f) quantitative continuous; (g) categorical ordinal; (h) quantitative discrete.
35. (a) Population: all 120 000 adult residents; sample: the 600 surveyed adults; parameter: the true population proportion supporting the proposal; statistic: 0.54.
(b) Population: all applications processed during the year; sample: the 80 selected case files; parameter: the population mean processing time; statistic: 17.2 days.
36. (a) Mean 6, median 5, mode 5.
(b) Mean 16, median 14, mode 12.
(c) Mean 5, median 5, no mode.
(d) Mean 32, median 23, no mode. The median is more resistant to the unusually large value 70.
37. Distribution A has mean $41/20 = 2.05$, median 2 and mode 2. Distribution B has mean 19, median 20 and mode 10.

38. For A, the estimated mean is 17.5, median is

$$10 + \frac{10-4}{9}(10) \approx 16.67,$$

and mode is $10 + [5/(5+4)](10) \approx 15.56$. For B, the estimates are mean 39.5, median 40 and mode $40 + [1/(1+6)](10) \approx 41.43$.

39. (a) Five-number summary (2, 4, 7, 11, 16); IQR = 7; fences -6.5 and 21.5, so there is no outlier. (b) Five-number summary (4, 5.25, 7.5, 9.75, 30); IQR = 4.5; fences -1.5 and 16.5, so 30 is an outlier.

	Transformation	Mean	Median	Range	Standard deviation
40.	$x + 5$	47	45	18	6
	$1.2x$	50.4	48	21.6	7.2
	$1.2x + 5$	55.4	53	21.6	7.2

41. The mean is 5 and the squared deviations sum to 20. Thus

$$\sigma^2 = 5, \quad \sigma = \sqrt{5} \approx 2.236, \quad s^2 = \frac{20}{3} \approx 6.667, \quad s \approx 2.582.$$

The population calculation divides by $N = 4$; the sample calculation uses $n - 1 = 3$ to estimate population variation from a sample.

42. (a) The combined mean is

$$\frac{30(65) + 70(72)}{100} = \boxed{69.9}.$$

(b) The weighted mean is

$$0.40(110) + 0.35(125) + 0.25(140) = \boxed{122.75}.$$

43. Both populations have mean and median 44. Population A has range 8 and standard deviation $\sqrt{8} \approx 2.828$; Population B has range 48 and standard deviation $\sqrt{236.8} \approx 15.388$. Equal centres therefore conceal the much greater dispersion in B.

Exercises 44–52: surveys, sampling and data quality.

44. (a) Target population: all university students; sample: the first 300 respondents. (b) The design is vulnerable to voluntary-response bias and undercoverage of students who do not follow or use the social-media page. (c) A simple random or proportionally stratified sample drawn from the student register would be more representative.
45. (a) The allocations are 300, 150 and 50 residents. (b) Stratification ensures that each district is represented in proportion to its population and may reduce imbalance caused by a purely accidental sample composition.

46. (a) $4000/20 = 200$ households. (b) Systematic sampling with a random start. (c) Bias can arise if the ordering contains a pattern with period 20, or if the list omits some types of household.
47. (a) The statistic is 0.70, or 70% support among respondents. (b) $0.70(420) = 294$ respondents. (c) The 580 non-respondents may have systematically different views, producing non-response bias.
48. People who are unemployed but do not register, are ineligible, or are recorded incorrectly are absent from the count. Measurement error concerns incorrect recorded status for included people; incomplete coverage concerns unemployed residents missing from the register entirely.
49. Employment sector is categorical nominal; age is quantitative continuous; household size is quantitative discrete; trust on the ordered scale is categorical ordinal. The observational unit is one respondent.
50. A postcode is an identifier rather than a measured magnitude, so arithmetic distances and the mean have no geographical meaning. Frequencies, proportions by postcode, or the mode are suitable.
51. (a) The observational unit is one school. (b) District is categorical nominal; enrolment and teacher count are quantitative discrete; pass rate is quantitative continuous. (c) The mean of school rates weights every school equally, whereas the overall pupil rate weights schools by the number of pupils assessed.
52. Very small cells can permit re-identification and disclosure of sensitive attributes, undermining confidentiality. Aggregation or suppression reduces those risks, but also removes detail and may conceal meaningful differences affecting small groups.

Exercises 53–63: measures of centre and frequency distributions.

53. Mean = 50, median = 31, no mode and range = 100 thousand ringgit. The median is the more defensible typical value because the income 128 strongly raises the mean.
54. Both districts have mean and median 34. District A has range 8 and population standard deviation $\sqrt{8} \approx 2.828$; District B has range 28 and standard deviation $\sqrt{136} \approx 11.662$. District A is more homogeneous.
55. There are 50 households and total spending RM 8500. Hence the mean is RM 170, while the median and mode are both RM 150.
56. The combined mean is

$$\frac{120(42) + 80(57)}{200} = \boxed{48} \text{ GB.}$$

The unweighted mean 49.5 GB incorrectly gives equal influence to groups of different sizes.

57. Initially, mean = 6.4 and median = 6. After adding 28, mean = 10 and median = 6.5. The mean rises by 3.6, compared with a median increase of only 0.5.
58. The total must be $5(20) = 100$. Since the known values total 79, $x = 21$ and the median is 21.
59. (a) Mean RM 3450, median RM 3250, standard deviation RM 400. (b) Mean RM 3456, median RM 3240, standard deviation RM 432.
60. Dividing every annual value by 12 gives monthly mean 4, variance 1 and standard deviation 1, in the corresponding thousand-ringgit units.
61. There are 60 responses. The thirtieth and thirty-first responses are both *satisfied*, which is also the modal category. A mean would require numerical scores and the assumption that successive categories are equally spaced.
62. Solving
- $$\frac{24 + 3x}{12 + x} = 2.2$$
- gives $x = 3$. The total frequency is 15, and both the median and mode are 2.
63. The reported total is $8(14.5) = 116$. The corrected total is $116 - 19 + 91 = 188$, giving corrected mean 23.5. The median depends on the ordering of all eight observations, which is not supplied.

Exercises 64–74: quartiles, outliers and box plots.

64. Five-number summary (6, 8.5, 12, 20, 40); IQR = 11.5; fences -8.75 and 37.25 . Thus 40 is an outlier and 22 is the largest non-outlying observation.
65. The summaries are
- $$\begin{aligned}\text{Office A: } & (10, 18, 24, 30, 42), \quad \text{IQR} = 12, \\ \text{Office B: } & (8, 14, 24, 36, 55), \quad \text{IQR} = 22.\end{aligned}$$
- Their medians are equal, but Office B is less consistent because its middle half and overall range are wider.
66. (a) IQR = 16, giving fences -4 and 60 . (b) Since $75 > 60$, it is an upper outlier. (c) It may be a valid extreme case with substantive importance; its source, measurement and context should be checked before any decision to retain, correct or exclude it.
67. A five-number summary fixes only endpoints and quartile boundaries; it does not record how observations are arranged within the four intervals. A dot plot, histogram, frequency table or the raw sample size should accompany the box plot when gaps and clustering matter.

68. With $N = 40$, the first quartile is in the $10 - < 20$ class and the third is in the $30 - < 40$ class:

$$Q_1 = 10 + \frac{10 - 5}{9}(10) \approx 15.56, \quad Q_3 = 30 + \frac{30 - 26}{8}(10) = 35.$$

Thus $\text{IQR} \approx 19.44$ minutes.

69. (a) $\text{IQR} = 20$, so the fences are 0 and 80 thousand ringgit. (b) RM 110 000 is an upper outlier. (c) A genuinely high income may describe real inequality rather than an error and can strongly affect totals, means and policy interpretation.
70. $Q_1 = 14.75$, median = 18.5, $Q_3 = 24.25$ and $\text{IQR} = 9.5$ minutes. The fences are 0.5 and 38.5, so 75 is an outlier. Median and IQR are suitable resistant summaries.
71. Both neighbourhoods have median RM 1200. A has IQR RM 300 and range RM 600; B has IQR RM 600 and range RM 1700. B therefore has much greater spread and a long upper tail concealed by the common median.
72. $\text{IQR} = 6$ and the fences are 0 and 24. Hence 28 is an upper outlier. The high extreme and longer possible upper tail suggest positive skewness.
73. $Q_1 = 3.5$, median = 5, $Q_3 = 7.5$, so $\text{IQR} = 4$ and the fences are -2.5 and 13.5 . The whiskers end at 2 and 8, with 20 plotted separately as an outlier.
74. The positions are $(19 + 1)/4 = 5$, $(19 + 1)/2 = 10$ and $3(19 + 1)/4 = 15$. Thus $Q_1 = 32$, median = 48 and $Q_3 = 71$. Approximately one quarter of the ordered scores are at or below 32 under this convention.

Exercises 75–85: variation and comparison.

75. Both populations have mean 22. For A, $\sigma^2 = 8$ and $\sigma \approx 2.828$; for B, $\sigma^2 = 72$ and $\sigma \approx 8.485$. Population A is more consistent.
76. The mean is 7.5 and the squared deviations total 21. Population variance = $21/4 = 5.25$ and standard deviation ≈ 2.291 ; sample variance = $21/3 = 7$ and standard deviation ≈ 2.646 . The population formula averages over all four population values, while the sample formula uses $n - 1$.
77. $\bar{x} = 240/12 = 20$ and

$$s^2 = \frac{5020 - 240^2/12}{11} = 20, \quad s = \sqrt{20} \approx 4.472.$$

78. The transformed mean is $3(40) - 4 = 116$. Variance is $3^2(25) = 225$ and standard deviation is $3(5) = 15$.
79. The reported mean is 5000, variance is $100^2(16) = 160\,000$, and standard deviation is 400.

80. $N = 10$, $\sum fx = 27$ and $\sum fx^2 = 81$. Hence

$$\mu = 2.7, \quad \sigma^2 = \frac{81}{10} - 2.7^2 = 0.81, \quad \sigma = 0.9.$$

81. Each separate group has population standard deviation $\sqrt{8/3} \approx 1.633$. The combined mean is 17 and the combined standard deviation is $\sqrt{166/6} \approx 5.260$. The combined data include the large difference between the two group centres as well as the small variation within each group.
82. Original data: range 4, IQR 3, standard deviation $\sqrt{2} \approx 1.414$. Revised data: range 30, IQR 16 and standard deviation $\sqrt{130.96} \approx 11.444$. All three increase, with the range and standard deviation especially sensitive to the extreme value.
83. Office B is faster on average, but Office A is more consistent. The comparison should also examine skewness, outliers, sample size and whether the two offices handle comparable cases.
84. Variance is measured in hours squared and standard deviation in hours. Conversion to minutes multiplies standard deviation by 60 and variance by $60^2 = 3600$.
85. Equal standard deviations describe equal measured spread in the two samples. They do not show that the distributions have the same centre or shape, nor that the population estimates have equal precision; the sample sizes differ greatly.

Exercises 86–93: grouped data and visual interpretation.

86. (a) Modal class 20–< 30 minutes. (b) Using midpoints, the estimated mean is $980/40 = 24.5$ minutes. The median is $20 + [(20 - 14)/14](10) \approx 24.29$ and the mode is $20 + [4/(4 + 6)](10) = 24$ minutes. (c) The distribution is unimodal and approximately symmetric, with at most slight positive skewness.
87. (a) The difference is 4 percentage points. (b) Relative to Region A, it is $(52/48 - 1)100\% \approx 8.33\%$. (c) Starting the axis at 45% makes the bars' visible heights differ by a large proportion. A scale beginning at zero, a dot plot on a clearly labelled scale, or direct numerical reporting would be less visually exaggerated.
88. (a) Mode and bar chart for nominal party preference. (b) Median and IQR with a box plot or histogram for skewed income. (c) Mean or median with a frequency bar chart or dot plot for the discrete count. (d) Median or mode with an ordered bar chart for ordinal satisfaction. Other well-justified choices are acceptable.
89. With midpoints 5, 15, 25, 35, $n = 20$, $\sum fx = 390$ and $\sum fx^2 = 9100$. Therefore

$$\bar{x} = 19.5, \quad s^2 = \frac{9100 - 390^2/20}{19} \approx 78.684, \quad s \approx 8.870.$$

90. (a) The median class is $40-< 60$. Using the stated boundaries,

$$\tilde{x} = 39.5 + \frac{25 - 17}{20}(20) = 47.5.$$

(b) No midpoint is defined for an open-ended class; a mean estimate would require an assumed endpoint or representative value.

91. Midpoints replace all unknown observations in a class by one value, so within-class variation is lost. Results depend on class widths and boundaries, and open-ended classes require extra assumptions. Any raw data sets with the same counts in each interval produce the same grouped table even if their within-class locations differ.

92. Using midpoints 5, 15, 25,

$$\frac{4(5) + 15x + 6(25)}{10 + x} = 16$$

gives $x = 10$. The total frequency is 20.

93. (a) Successive differences give frequencies 4, 7, 14, 11, 4. (b) The modal class is $10-< 15$. (c) With $N = 40$, the estimated median is

$$10 + \frac{20 - 11}{14}(5) \approx 13.21.$$

Exercises 94–100: responsible statistical interpretation.

94. A higher mean can result from large gains for only some participants while others remain unchanged or lose income. Changes in employment, economic conditions, sample composition, selection into the programme or other factors also prevent a causal conclusion from the before–after mean alone.
95. A convenience sample may systematically exclude or over-represent groups regardless of its size. Increasing sample size can reduce random sampling variability, but it does not correct selection, coverage or non-response bias. A smaller probability sample can therefore represent the target population more defensibly.
96. Check for transcription, unit and measurement errors; verify that the value belongs to the target population and period; examine its influence on conclusions; and assess whether it is a genuine rare case of substantive interest. Exclusion should be documented and a sensitivity analysis reported where appropriate.
97. A numerical mean treats the distances from disagree to neutral and from neutral to agree as equal. Without that assumption, report the median or modal category and the full frequency distribution.
98. Median and IQR resist the influence of long tails and extreme values, making them useful for skewed distributions. They still omit details such as multimodality, gaps, subgroup differences and the exact tail behaviour.

99. Use a scale that does not visually exaggerate differences; label variables, units, groups and sample sizes; use areas and bar lengths proportionally; disclose missing, excluded or revised data; avoid selective time ranges or categories; and provide source and method information.
100. One suitable checklist is: define the population, observational unit and variables; choose a representative collection method; check coding, missingness and measurement quality; select summaries and displays suited to variable type and distribution; verify calculations and examine outliers or subgroups; then communicate results with units, uncertainty, limitations and data-governance safeguards.

C.5 Chapter 5: Probability

Return to the corresponding questions in Chapter 5.

1. $1/2$.
2. $P(A \cup B) = 0.75$; probability of neither = 0.25.
3.

$$1 - \left(\frac{5}{6}\right)^4 = \frac{671}{1296} \approx 0.5177.$$
4. $P(\text{pass}) = 0.80$; $P(\text{online and fail}) = 0.15$; $P(\text{fail} \mid \text{online}) = 0.30$.
5. (a) $5/14$; (b) $15/28$.
6. The events are independent because $P(M)P(S) = 0.6(0.5) = 0.3 = P(M \cap S)$.
7. $P(A \cup B) = 0.70$. They are not independent because $P(A \cap B) = 0$, whereas $P(A)P(B) = 0.12$.
8. $P(D) = 0.019$ and $P(B \mid D) = \frac{0.30(0.04)}{0.019} = \frac{12}{19} \approx 0.6316$.
9. 175 people are in at least one set and 25 are in none.
10. $\frac{1}{2} \cdot \frac{4}{8} = \frac{1}{4}$.
11. (a) The rows of absolute differences are

$$(1, 0, 1, 2, 3), \quad (2, 1, 0, 1, 2), \quad (4, 3, 2, 1, 0),$$

$$(6, 5, 4, 3, 2), \quad (10, 9, 8, 7, 6).$$
 (b) $3/25$; (c) $7/25$.
12. Dog only 25%, both 45%, cat only 12%, neither 18%. (b) 43% do not own a cat; (c) 37% own exactly one type.

13. (a) $0.85^3 = 0.614125$; (b) $0.15^3 = 0.003375$; (c) $1 - 0.15^3 = 0.996625$.
14. (a) $169/463$; (b) $(96 + 214 - 61)/463 = 249/463$.
15. (a) $26/30 = 13/15$. (b) $P(A) = 7/30$, $P(B) = 19/30$ and $P(A \cap B) = 4/30$; since $4/30 \neq (7/30)(19/30)$, the events are not independent.
16. $(5/100)(4/99) = 1/495$.
17. (a) $270/510 = 9/17$; (b) $335/510 = 67/102$; (c) $70/510 = 7/51$; (d) $170/510 = 1/3$; (e) $80/270 = 8/27$; (f) $80/175 = 16/35$.
18. (a) $p = 0.10$; (b) $q = 0.05$, $r = 0.35$.
19. (a) $1/5$; (b) $1/3$; (c) $7/25$.
20. (a) $\frac{\binom{8}{4}\binom{5}{2}}{\binom{13}{6}} = 175/429$; (b) $\frac{\binom{8}{6}}{\binom{13}{6}} = 7/429$; (c) $\frac{\binom{5}{4}\binom{8}{2} + \binom{5}{5}\binom{8}{1}}{\binom{13}{6}} = 37/429$.
21. (a) $\frac{\binom{6}{3}\binom{13}{1}}{\binom{19}{4}} = 65/969$; (b) $\frac{\binom{8}{3}\binom{11}{1} + \binom{8}{4}}{\binom{19}{4}} = 343/1938$; (c) $\frac{\binom{14}{4}}{\binom{19}{4}} = 1001/3876$.
22. (a) $P(Q) = 0.5(0.8) + 0.5(0.3) = 0.55$; (b) $P(\text{cat} | Q) = 0.4/0.55 = 8/11$.
23. The late-path probabilities are $1/10$, $1/5$ and $3/20$. (b) $P(\text{late}) = 9/20$; (c) $P(\text{rain} | \text{late}) = (1/5)/(9/20) = 4/9$.

Exercises 24–33: grouped skills review.

24. (a) 0.82; (b) 0.36; (c) $13/20$; (d) 0.73.
25. (a) $0.45 + 0.38 - 0.20 = 0.63$; (b) $0.62 + 0.50 - 0.30 = 0.82$; (c) $0.40 + 0.45 - 0.70 = 0.15$; (d) $0.28 + 0.35 = 0.63$.
26. (a) $42/120 = 0.35$; (b) $0.18/0.45 = 0.40$; (c) $0.60(0.25) = 0.15$; (d) $30/80 = 0.375$.
27. (a) Independent because $0.4(0.5) = 0.2$. (b) Not independent because $0.3(0.6) = 0.18 \neq 0.15$. (c) Independent because $P(A | B) = P(A)$. (d) The intersection is $0.7 + 0.2 - 0.76 = 0.14 = 0.7(0.2)$, so the events are independent.
28. (a) Intersection 0 and union 0.55. (b) $P(A | B) = 0$. (c) They can be both only when at least one event has probability zero. (d) $A \cap A^c = \emptyset$ and $A \cup A^c = S$.
29. (a) $0.60(0.75) + 0.40(0.35) = 0.59$. (b) $0.50(0.8) + 0.30(0.6) + 0.20(0.4) = 0.66$.
30. (a)

$$P(A | \text{success}) = \frac{0.60(0.75)}{0.59} \approx 0.7627.$$

(b) The non-response probability is 0.34, so

$$P(R_3 | \text{non-response}) = \frac{0.20(0.60)}{0.34} \approx 0.3529.$$

31. (a) $(6/10)(5/9) = 1/3$. (b) $2(6/10)(4/9) = 8/15$. (c) $(9/12)(8/11) = 6/11$. (d) $(9/12)^2 = 9/16$.
32. (a) $1 - 0.8^3 = 0.488$; (b) $1 - 0.65^2 = 0.5775$; (c) $2(0.3)(0.7) = 0.42$; (d) $0.9^4 = 0.6561$.
33. The expressions mean: not A ; A or B or both; A but not B ; and the probability of A given B . A large conditional probability may reflect selection, confounding, common causes or reverse causation rather than a causal effect of B .

Exercises 34–44: events, surveys and probability rules.

34. The probabilities are $280/500 = 0.56$, $190/500 = 0.38$, $130/500 = 0.26$, $(280 + 190 - 130)/500 = 0.68$, and $130/280 = 13/28 \approx 0.4643$, respectively.
35. Rent only = 140 and energy only = 90. At least one = 320, neither = 80, and exactly one = 230 households.
36. Inclusion–exclusion gives

$$520 + 410 + 300 - 180 - 140 - 120 + 60 = 850.$$

Thus 150 use none. Exactly one service is used by 530 residents, while 320 use at least two.

37. The error-free probability is 0.93. The product $500(0.07) = 35$ is the expected or long-run average number of errors over comparable collections; a particular set of 500 need not contain exactly 35 errors.
38. (a) Invalid because probabilities cannot exceed 1. (b) Invalid because an intersection is contained in A . (c) Invalid because a union contains B . (d) Valid because the probabilities sum to 1.
39. $P(A) = 0.40$, $P(B) = 0.45$, $P(A \cup B) = 0.70$, $P((A \cup B)^c) = 0.30$, exactly one = 0.55, and $P(A | B) = 0.15/0.45 = 1/3$.
40. Internet only = 220, smartphone only = 80, exactly one = 300, at least one = 700, and neither = 300 adults.
41. $P(R \cup A) = 0.65 + 0.50 - 0.42 = 0.73$, $P(A | R) = 0.42/0.65 \approx 0.6462$, $P(R | A) = 0.42/0.50 = 0.84$, and neither = 0.27.
42. Rain or cancellation = 0.38; rain only = 0.18; cancellation only = 0.08; neither = 0.62; and $P(\text{cancellation} | \text{rain}) = 0.12/0.30 = 0.40$.
43. (a) The intersection must lie between $0.65 + 0.55 - 1 = 0.20$ and $\min(0.65, 0.55) = 0.55$. (b) It equals $0.65 + 0.55 - 0.85 = 0.35$.
44. Possible explanations include prior academic attainment, family resources, field of study, location, selection into degree study, labour-market conditions and reverse causation in educational choices. The conditional probability is descriptive association.

Exercises 45–55: conditional probability and two-way tables.

45. $P(E) = 270/360 = 0.75$, $P(T) = 0.50$, $P(E | T) = 144/180 = 0.80$, $P(E | T^c) = 126/180 = 0.70$, and $P(T | E) = 144/270 = 8/15 \approx 0.5333$. The events are not independent because $0.80 \neq 0.75$.
46. Within the low-risk group, A has success probability 0.90 and B 0.80; within the high-risk group, A has 0.30 and B 0.20. Overall, A has $39/110 \approx 0.3545$ while B has $82/110 \approx 0.7455$. A contains far more high-risk cases, whereas B contains far more low-risk cases, producing the reversal.
47. $P(\text{Yes}) = 210/400 = 0.525$, $P(\text{Yes} | \text{woman}) = 120/200 = 0.60$, and $P(\text{woman} | \text{Yes}) = 120/210 = 4/7 \approx 0.5714$. The events are not independent because $0.60 \neq 0.525$.
48. Since $P(A | B) = P(A)$, the events are independent. $P(A \cap B) = 0.4(0.5) = 0.20$ and $P(A \cup B) = 0.70$.

49.

$$P(\text{young} | \text{unemployed}) = \frac{0.12(0.25)}{0.06} = 0.50.$$

The two conditionals have different denominators and answer different questions.

50. $P(A | B) = 30/80 = 0.375$, $P(B | A) = 30/70 = 3/7 \approx 0.4286$, and $P(A^c | B) = 50/80 = 0.625$.
51. The probabilities are $3/9 = 1/3$ and $4/9$, respectively. The first selection changes the composition of the remaining files, showing dependence.
52. The four equally likely outcomes are BB, BG, GB, GG . Thus the probabilities are $3/4$ and $(1/4)/(3/4) = 1/3$.
53. $P(\text{error}) = 0.05$, $P(\text{flagged}) = 0.14$, $P(\text{error} | \text{flagged}) = 45/140 = 9/28 \approx 0.3214$, and $P(\text{flagged} | \text{error}) = 45/50 = 0.90$.
54. $P(B) = 0.60(0.40) + 0.40(0.20) = 0.32$, $P(A \cap B) = 0.24$, and $P(A | B) = 0.24/0.32 = 0.75$.
55. $P(E | I)$ is the likelihood of the evidence under innocence; $P(I | E)$ is the posterior probability of innocence after the evidence. Reversal requires prior probabilities of innocence and guilt and the evidence probabilities under both possibilities.

Exercises 56–66: independence, dependence and repeated trials.

56. Both missed $= 0.1^2 = 0.01$; at least one missed $= 1 - 0.9^2 = 0.19$; exactly one missed $= 2(0.1)(0.9) = 0.18$.
57. Both fail $= 0.05(0.08) = 0.004$; at least one works $= 0.996$; at least one fails $= 1 - 0.95(0.92) = 0.126$.

58. No response = $0.7^4 = 0.2401$; at least one = 0.7599 ; exactly one = $4(0.3)(0.7)^3 = 0.4116$; exactly two = $6(0.3)^2(0.7) = 0.2646$.
59. Both incomplete = $(2/10)(1/9) = 1/45$; exactly one = $2(2/10)(8/9) = 16/45$; the stated conditional probability is $1/9$.
60. $P(A) = 0.50$, $P(B) = 0.60$ and $P(A \cap B) = 0.30$, so the product test confirms independence. Also $P(B | A) = 30/50 = 0.60 = P(B)$.
61. $P(A | B) = 0$. Since $P(A \cap B) = 0$ but $P(A)P(B) = 0.14$, the choices are not independent.
62. $P(A^c \cap B^c) = 0.7(0.6) = 0.42$ and $P(A \cup B) = 1 - 0.42 = 0.58$. Complements of independent events retain the product relationship.
63. Exactly two = $3(0.8)^2(0.2) = 0.384$; all three = $0.8^3 = 0.512$; at least two = $0.384 + 0.512 = 0.896$.
64. Without replacement, the probability is $(5/8)(4/7) = 5/14 \approx 0.3571$. With replacement it is $(5/8)^2 = 25/64 \approx 0.3906$.
65. (a) Under independence, both = $0.70(0.60) = 0.42$ and

$$P(\text{written pass} | \text{interview pass}) = 0.60.$$

(b) The joint probability becomes $0.70(0.75) = 0.525$.

66. Household members share income, housing, local labour markets, debts, health exposures and economic shocks. These common causes can make their hardship outcomes positively dependent.

Exercises 67–77: total probability, trees and Bayes' theorem.

67. $P(C) = 0.50(0.80) + 0.30(0.70) + 0.20(0.90) = 0.79$.
68. $P(\text{in person} | C) = 0.18/0.79 \approx 0.2278$ and $P(\text{online} | C) = 0.40/0.79 \approx 0.5063$.
69. The flag probability is $0.08(0.85) + 0.92(0.10) = 0.16$. Hence

$$P(\text{ineligible} | \text{flag}) = \frac{0.08(0.85)}{0.16} = 0.425.$$

Ineligible applications are uncommon, so false flags among the much larger eligible group form a substantial share of all flags.

70. Overall employment is

$$0.35(0.72) + 0.65(0.55) = 0.6095.$$

The posterior is

$$\frac{0.35(0.72)}{0.6095} \approx 0.4135.$$

71. Overall response = $0.60(0.50) + 0.40(0.75) = 0.60$. $P(\text{phone} \mid \text{response}) = 0.30/0.60 = 0.50$.
72. Lateness = $0.50(0.10) + 0.30(0.40) + 0.20(0.80) = 0.33$. $P(\text{storm} \mid \text{late}) = 0.16/0.33 \approx 0.4848$.
73. Overall support = $0.50(0.40) + 0.30(0.55) + 0.20(0.70) = 0.505$. $P(C \mid \text{support}) = 0.14/0.505 \approx 0.2772$.
74. Overall complaint probability = $0.50(0.02) + 0.30(0.04) + 0.20(0.08) = 0.038$. $P(C \mid \text{complaint}) = 0.016/0.038 = 8/19 \approx 0.4211$.
75. At prevalence 0.01,

$$P(\text{condition} \mid +) = \frac{0.01(0.95)}{0.01(0.95) + 0.99(0.10)} \approx 0.0876.$$

At prevalence 0.20, it is $0.20(0.95)/[0.20(0.95) + 0.80(0.10)] \approx 0.7037$. The higher base rate makes true positives a much larger share of all positive results.

76. Let $p = P(I)$. Solving

$$\frac{0.8p}{0.8p + 0.1(1 - p)} = 0.5$$

gives $p = 1/9 \approx 0.1111$.

77. Let $x = P(\text{success} \mid B)$. Then $0.40(0.80) + 0.60x = 0.62$, so $x = 0.50$. Also $P(B \mid \text{success}) = 0.60(0.50)/0.62 \approx 0.4839$.

Exercises 78–88: finite sampling without replacement.

78. (a) $\binom{7}{2}\binom{5}{1}/\binom{12}{3} = 21/44 \approx 0.4773$; (b) $\binom{7}{3}/\binom{12}{3} = 7/44 \approx 0.1591$.
79. No error = $\binom{11}{3}/\binom{15}{3} = 33/91$; exactly one = $\binom{4}{1}\binom{11}{2}/\binom{15}{3} = 44/91$; at least one = $1 - 33/91 = 58/91$.
80. $\binom{6}{2}\binom{14}{2}/\binom{20}{4} = 91/323 \approx 0.2817$.
81. Both support = $(30/50)(29/49) = 87/245 \approx 0.3551$; exactly one = $2(30/50)(20/49) = 24/49 \approx 0.4898$.
82. $\binom{10}{1}\binom{8}{2}/\binom{18}{3} = 35/102 \approx 0.3431$.
83. $1 - \binom{17}{4}/\binom{20}{4} = 493/969 \approx 0.5088$.
- 84.
- $$\frac{\binom{3}{2}\binom{6}{3} + \binom{3}{3}\binom{6}{2}}{\binom{9}{5}} = \frac{25}{42} \approx 0.5952.$$
85. Both senior = $(3/8)(2/7) = 3/28$. Exactly one senior = $2(3/8)(5/7) = 15/28$.
86. $\binom{16}{5}/\binom{20}{5} = 91/323 \approx 0.2817$.

87. Each group has problematic probability 0.20. Thus at least one $= 1 - (0.8)^2 = 0.36$ and both $= (0.2)^2 = 0.04$.
88. Without replacement, $\binom{7}{3}/\binom{10}{3} = 7/24 \approx 0.2917$. With replacement, $(7/10)^3 = 0.343$. The without-replacement probability is smaller because each Type A selection reduces the remaining Type A proportion.

Exercises 89–100: responsible probability interpretation.

89. Private education is associated with prior attainment, family income, location, networks and selection. These variables and reverse causal pathways can produce a large conditional probability without identifying the effect of education itself.
90. The probability of seeing the evidence if a person is innocent is not the probability that the person is innocent after seeing the evidence. The latter requires Bayes' theorem, including base rates and the evidence probability under guilt.
91. When a condition is rare, the large group without the condition can generate many false positives even at a modest false-positive rate. Consequently, true cases may form only a modest fraction of all positive results despite high sensitivity.
92. Check whether the events share causes or units, whether one changes exposure to the other, whether sampling is with replacement, and whether observed conditional probabilities are stable and close to the corresponding marginal probabilities.
93. Mutually exclusive events cannot occur together; exhaustive events cover every possible outcome. For one journey, travelling only by bus and travelling only by train are mutually exclusive, while “uses public transport” and “does not use public transport” are exhaustive complements.
94. The relative increase is $(2/1 - 1)100\% = 100\%$, while the absolute increase is 1 percentage point. Both are needed because the relative statement can sound large while the absolute risk remains small.
95. Complete files may come from applicants with better assistance, simpler cases or greater resources; staff may help promising cases become complete; and eligibility or case complexity can affect both completeness and approval.
96. If missingness is related to the outcome or conditioning group, the observed cases are not representative of the intended denominator. Correct arithmetic then estimates a conditional probability for the observed subset rather than the target population.
97. Across many comparable cases assigned probability 0.70, roughly 70% should occur if the forecasts are calibrated. A single outcome remains uncertain and supplies almost no evidence about long-run calibration by itself.
98. Report group-specific base rates, sensitivity or true-positive rates, false-positive rates, positive predictive probabilities and sample sizes, together with uncertainty and the consequences of each error type.

99. Shared shocks create positive dependence. Independence multiplies marginal probabilities and can therefore assign too little probability to joint adverse outcomes when difficulties tend to cluster within households.
100. One suitable checklist is: define the experiment, population and events; represent the sample space with a table, Venn diagram or tree; choose the complement, addition, multiplication, total probability or Bayes rule; calculate with consistent denominators; check bounds, dependence and base rates; then communicate the result with assumptions, interpretation, uncertainty and limitations.

C.6 Chapter 6: Discrete Random Variables

Return to the corresponding questions in Chapter 6.

1. $a = 4/5$, $E(Y) = 3$, $\text{Var}(Y) = 1$.
2. $k = 0.15$, $E(X) = 1.56$, $\text{Var}(X) = 1.4064$; $E(Y) = -3.88$, $\text{Var}(Y) = 5.6256$.
3. $E(X) = 5$, $\text{Var}(X) = 22/3$; $E(Y) = 13$, $\text{Var}(Y) = 66$.
4. The cumulative probabilities at 0, 1, 2, 3, 4, 5 are 0.08, 0.25, 0.55, 0.80, 0.95, 1; $P(1 < X \leq 4) = 0.70$, $E(X) = 2.37$, and $\sigma_X \approx 1.286$.
5. (a) Valid: the probabilities are non-negative and sum to 1. (b) Invalid: $P(X = -1) = -1/3$. (c) Valid: the probabilities are $15/23, 5/23, 3/23$.
6. The probabilities sum to $4(0.1) + 3(0.2) = 1$. (a) 0.1; (b) 0.5; (c) 0.5.
7. The probabilities at $-1, 0, 1, 3, 4, 5$ are 0.1, 0.1, 0.2, 0.2, 0.3, 0.1; hence $a = 0.2$ and $P(X \geq 3) = 0.6$. The CDF levels are 0, 0.1, 0.2, 0.4, 0.6, 0.9, 1 across the successive cut points $-1, 0, 1, 3, 4, 5$.
8. $k = 0.10$; $P(1 \leq X \leq 3) = 0.85$; $P(X \geq 3) = 0.55$; $P(2 < X < 5) = 0.50$. The CDF levels are 0, 0.20, 0.45, 0.85, 0.95, 1 across 1, 2, 3, 4, 5.
9. $P(X \leq 0.4) = 0.75$; $P(X > 0.3) = 0.40$; $P(X = 0.3) = 0.30$. The PMF at 0.1, 0.2, 0.3, 0.4, 0.5 is 0.05, 0.25, 0.30, 0.15, 0.25.
10. $E(X) = 0.33$, $\text{Var}(X) = 0.0151$; $E(Y) = -6.34$, $\text{Var}(Y) = 0.0604$.
11. The PMF is $p(2) = 8/33$, $p(3) = 2/11$, $p(4) = 32/99$, $p(5) = 25/99$. Thus $P(X < 3 \text{ or } X > 4) = 49/99$, $E(X) = 355/99$ and $\text{Var}(X - 2) = 12080/9801$. The CDF levels are 0, $8/33$, $14/33$, $74/99$, 1 across 2, 3, 4, 5.
12. $\mu = 1.47$; $\sigma = \sqrt{12891}/100 \approx 1.1354$; $P(X > \mu + \sigma) = 0.19$; mode = 1; median = 1.

13. The PMF at 1, 2, 3, 4 is, respectively,

$$\frac{1}{4}, \quad \frac{1}{2}, \quad \frac{15}{64}, \quad \frac{1}{64}.$$

Median = 2; $E(X) = 129/64$; $\text{Var}(3X) = 20151/4096 \approx 4.9197$.

14. $k = 1/10$; $P(X \geq 2) = 0.7$; $E(X) = 2$; $\text{Var}(X) = 1$; mode = 3; median = 2.
15. (a) The probabilities at 0, 1, 2, 3, 4, 5 are 0.10, 0.20, 0.30, 0.22, 0.13, 0.05. (b) $P(1 \leq X \leq 3) = 0.72$ and $P(X \geq 4) = 0.18$. (c) The CDF levels are

$$0, 0.10, 0.30, 0.60, 0.82, 0.95, 1$$

across the successive cut points 0, 1, 2, 3, 4, 5. (d) $E(X) = 2.23$; $\text{Var}(X) = 1.7371$; $\sigma \approx 1.318$; mode = 2; median = 2.

16. (a) The PMF at 0, 1, 2, 3, 4 is 0.12, 0.25, 0.33, 0.20, 0.10. (b) $P(1 < X \leq 3) = 0.53$. (c) $E(X) = 1.91$ and $\text{Var}(X) = 1.3219$. (d) Mode = 2 and median = 2.
17. Expected net outcome = RM 2100; $P(\text{positive outcome}) = 0.60$; expected combined outcome = $20(2100) = \text{RM } 42\,000$. This is a long-run average, not a guaranteed total for a particular set of projects.
18. $E(X) = 2.1$, $E(X^2) = 4.9$ and $E(C) = 100 + 40(2.1) + 10(4.9) = \text{RM } 233$. However, $C(E(X)) = C(2.1) = \text{RM } 228.10$. Expectation cannot generally be moved through a nonlinear function; here $E(X^2) \neq [E(X)]^2$.
19. (a) $E(Y) = 98$, $\text{Var}(Y) = 144$ and $\sigma_Y = 12$. (b) $E(Z) = 0$, $\text{Var}(Z) = 1$ and $\sigma_Z = 1$.
20. Since $a + b = 0.7$ and $a + 2b + 0.3 = 1.4$, $a = 0.3$ and $b = 0.4$. The CDF levels are 0, 0.2, 0.5, 0.9, 1 across 0, 1, 2, 3; $\text{Var}(X) = 0.84$.
21. $E(A) = E(B) = 1$; $\text{Var}(A) = 0.5$ and $\text{Var}(B) = 0.9$. Variable A is more predictable because more probability is concentrated near the common mean, whereas B places more probability at the extremes.
22. $E(G) = -5(0.6) + 3(0.3) + 20(0.1) = -\text{RM } 0.10$; $P(G > 0) = 0.40$; expected combined return over 100 plays = $-\text{RM } 10$. The negative mean represents a small long-run expected loss per play, not the result of every individual play.

23. (a)

x	0	1	2	3
$P(X = x)$	1/30	3/10	1/2	1/6

using $P(X = x) = \binom{6}{x} \binom{4}{3-x} / \binom{10}{3}$. (b) $P(X \geq 2) = 2/3$. (c) $E(X) = 9/5 = 1.8$ and $\text{Var}(X) = 14/25 = 0.56$.

24. $E(X) = 0.42$ and $\text{Var}(X) = 0.5236$. Thus the expected payout is $1200(0.42) = \text{RM } 504$, its standard deviation is $1200\sqrt{0.5236} \approx \text{RM } 868.32$, and the insurer's expected net income is $600 - 504 = \text{RM } 96$ per policy.
25. The CDF levels are 0, 0.10, 0.35, 0.65, 0.90, 1 across 0, 1, 2, 3, 4. Also, $P(1 \leq V \leq 3) = 0.80$, $E(V) = 2$, $\text{Var}(V) = 1.3$, mode = 2 and median = 2.
26. Solving $a + 2(2/5) + 4(3/5 - a) = 14/5$ gives $a = 2/15$, so the probabilities at 1, 2, 4 are $2/15, 2/5, 7/15$. The CDF levels are 0, $2/15, 8/15, 1$ across 1, 2, 4, and $\text{Var}(X) = 34/25 = 1.36$.
27. The estimated PMF at 0, 1, 2, 3, 4 is 0.09, 0.21, 0.325, 0.255, 0.12. Thus $\hat{P}(X \geq 3) = 0.375$, $\bar{x} = 2.105$, estimated variance ≈ 1.29398 and estimated standard deviation ≈ 1.138 . These are based on one sample; another sample may give different relative frequencies, and the sampling method may not represent the population perfectly.

Exercises 28–37: grouped skills review.

28. A Bernoulli PMF has $P(X = 1) = p$ and $P(X = 0) = 1 - p$. The corresponding mean–variance pairs are (0.15, 0.1275), (0.40, 0.24) and (0.65, 0.2275).
29. (a) $k(1 + 2 + 3 + 4) = 1$, so $k = 0.10$. (b) $k = 1 - 0.10 - 0.20 - 0.25 = 0.45$. (c) Invalid: the probabilities sum to $1/2$, not 1. (d) $k(1 + 2 + 4) = 1$, so $k = 1/7$.
30. $P(X \geq 1) = 0.65$, $P(-1 < X < 2) = 0.65$, $P(X \neq 0) = 0.75$ and $P(|X| = 1) = 0.50$.
31. The CDF levels are 0, 0.20, 0.55, 0.85, 1 across the cut points 0, 1, 2, 3. The requested probabilities are 0.55, 0.45 and 0.15.
32. Successive CDF jumps give PMF values 0.10, 0.30, 0.35, 0.25 at $x = 0, 1, 2, 3$. Hence $P(X \geq 2) = 0.60$.
33. (a) 1.4; (b) 6.5; (c) 0.60; (d) 2.1.
34. $E(X) = 1.9$, $E(X^2) = 4.5$, $\text{Var}(X) = 4.5 - 1.9^2 = 0.89$, and $\sigma \approx 0.9434$.
35. For Y , the mean, variance and standard deviation are 13, 36, 6. For Z , they are 5, 4, 2. For W , they are 2.5, 1, 1.
36. Distribution A has mode and median 1. Distribution B is bimodal at 0 and 2, with median 1. The final distribution has mode and median 1.
37. Decision A has expected value $0.4(100) - 0.6(20) = \text{RM } 28$; Decision B has expected value $\text{RM } 25$. A has the larger expectation. $\text{RM } 28$ is a probability-weighted long-run average, not one of A's possible outcomes.

Exercises 38–48: PMFs, CDFs and recorded counts.

38. $P(X \geq 3) = 0.30$, $E(X) = 1.95$, $\text{Var}(X) = 1.2475$ and $\sigma \approx 1.1169$. The mode and median are both 2.
39. The probabilities sum to $0.40 + 3k = 1$, so $k = 0.20$. Across the intervals determined by 0, 1, 2, 3, the CDF levels are
- $$0, \quad 0.20, \quad 0.40, \quad 0.80, \quad 1.$$
- $E(X) = 1.6$ and $\text{Var}(X) = 1.04$.
40. The empirical PMF is 0.15, 0.25, 0.35, 0.20, 0.05. Thus $\hat{P}(X \geq 3) = 0.25$, mean = 1.75, variance = 1.1875 and standard deviation ≈ 1.0897 .
41. The PMF at 0, 1, 2, 3, 4, 5 is 0.08, 0.18, 0.30, 0.24, 0.14, 0.06. $P(1 \leq X \leq 3) = 0.72$ and $P(X \geq 4) = 0.20$. The mean is 2.36, variance 1.7104, standard deviation ≈ 1.3078 , mode 2 and median 2.
42. The CDF jumps give PMF values 0.05, 0.20, 0.35, 0.25, 0.15 at 0, 1, 2, 3, 4. $P(1 < X \leq 3) = 0.60$, $E(X) = 2.25$, $\text{Var}(X) = 1.1875$, mode 2 and median 2.
43. Since $a + b = 0.70$ and $a + 2b + 0.60 = 1.70$, $a = 0.30$ and $b = 0.40$. The CDF levels are 0, 0.10, 0.40, 0.80, 1 across 0, 1, 2, 3. Also $E(X^2) = 3.70$, so $\text{Var}(X) = 3.70 - 1.70^2 = 0.81$.
44. The CDF levels are 0, 0.10, 0.30, 0.65, 0.90, 1 across 0, 1, 2, 3, 4. Hence $P(X \leq 2) = 0.65$, $P(1 < X < 4) = 0.60$, the median is 2, and $P(X > 2) = 0.35$.
45. The PMF at 0, 1, 2, 3 is 0.20, 0.25, 0.35, 0.20. The mode and median are 2, which is also the smallest value with cumulative probability at least 0.75.
46. These variables take distinct recorded categories, but their codes need not represent equal numerical distances. Arithmetic on arbitrary codes can create means and variances without a defensible substantive interpretation.
47. The estimated PMF at 0, 1, 2, 3, 4 is 0.08, 0.22, 0.32, 0.26, 0.12. Thus $\hat{P}(X \geq 3) = 0.38$, mean = 2.12, variance = 1.2656 and standard deviation ≈ 1.1250 . They are sample estimates because the relative frequencies can vary across samples and may reflect sampling bias.
48. Since $k(4 + 3 + 2 + 1) = 1$, $k = 0.10$. The PMF is 0.40, 0.30, 0.20, 0.10, giving CDF levels 0, 0.40, 0.70, 0.90, 1 across 0, 1, 2, 3. The mean is 1 and the variance is 1.

Exercises 49–59: expected value and decisions.

49. $E(X) = 1.55$, while the mode and median are 1. Over many comparable students, the average use approaches about 1.55 sessions per student.

50. $E(X) = 1.1$, $E(X^2) = 1.7$ and $E(C) = 50 + 20(1.1) + 5(1.7) = \text{RM } 80.50$. However, $C(E(X)) = 50 + 20(1.1) + 5(1.1)^2 = \text{RM } 78.05$. A nonlinear function cannot generally be moved through expectation.
51. Expected return $= -10(0.60) + 5(0.30) + 40(0.10) = -\text{RM } 0.50$. Profit probability $= 0.40$ and the expected combined return over 200 plays is $-\text{RM } 100$.
52. Expected outcome $= -2000(0.20) + 5000(0.35) + 9000(0.15) = \text{RM } 2700$. Positive-outcome probability $= 0.50$, and ten comparable projects have expected combined outcome $\text{RM } 27\,000$.
53. $E(X) = 0.50 + 2(0.20) = 0.90$, so the expected total is 450 units. Linearity of expectation gives the total mean even if individual demands are dependent.
54. $E(X) = 0.47$ and $\text{Var}(X) = 0.5291$. Expected payout $= \text{RM } 470$, payout standard deviation $= 1000\sqrt{0.5291} \approx \text{RM } 727.39$, and expected net income $= 650 - 470 = \text{RM } 180$.
55. Both means are $\text{RM } 50$. Decision A has variance 2500, while B has variance 100. Decision B is more predictable.
56. $E(X) = 0.15 + 2(0.05) = 0.25$, so the expected claim payment and break-even premium are $\text{RM } 500$.
57. The expected total is $E(\sum_{i=1}^{300} I_i) = 300(0.40) = 120$. Expectation describes an average over repeated comparable groups, not a guaranteed count.
58. $E[(X - 2)^2] = 0.20(1) + 0.50(0) + 0.30(1) = 0.50$.
59. Expected value $= \text{RM } 100$; median and mode $= \text{RM } 0$. The zero outcome is a better description of the most common individual result.

Exercises 60–70: variance, transformations and predictability.

60. $E(X) = 1.5$, $E(X^2) = 3.3$, $\text{Var}(X) = 1.05$ and $\sigma \approx 1.0247$.
61. Use $\text{Var}(X) = E(X^2) - [E(X)]^2$. Thus the variance is $19 - 16 = 3$ and the standard deviation is $\sqrt{3} \approx 1.732$.
62. For Y , the mean, variance and standard deviation are 29, 144, 12. For Z , they are 0, 1, 1.
63. $E(Z) = 0$ and $\text{Var}(Z) = 1$. Standardisation expresses values in standard-deviation units relative to the original mean.
64. Both means are 2. Distribution A has variance 2, while B has variance 3.2. A is more concentrated around the common mean.
65. For $X + 10$: mean 40, variance 36, standard deviation 6. For $0.5X$: mean 15, variance 9, standard deviation 3. For $2X - 5$: mean 55, variance 144, standard deviation 12.

66. Both distributions have mean 1. Distribution A has variance 0.20, while B has variance 0.90. A is more predictable because most of its probability is concentrated at the mean.
67. The mean is 360 minutes, variance $60^2(4) = 14\,400$ minutes squared, and standard deviation 120 minutes.
68. The variances are 0.09, 0.25, 0.09. Bernoulli variation is greatest when success and failure are equally likely and declines as the distribution concentrates near 0 or 1.
69. The implied variance is $8 - 3^2 = -1$, which is impossible because a variance cannot be negative. At least one reported moment is wrong.
70. Relative frequencies are 0.20, 0.40, 0.30, 0.10. The empirical mean is 1.3, variance 0.81 and standard deviation 0.90.

Exercises 71–81: distributions derived from finite samples.

71. With denominator $\binom{12}{4} = 495$, the PMF at $x = 0, 1, 2, 3, 4$ is

$$\frac{1}{99}, \quad \frac{14}{99}, \quad \frac{14}{33}, \quad \frac{35}{99}, \quad \frac{7}{99}.$$

Thus $P(X \geq 3) = 14/33$, $E(X) = 7/3$ and $\text{Var}(X) = 70/99$.

72. With denominator $\binom{8}{3} = 56$, the PMF at $x = 0, 1, 2, 3$ is $1/56, 15/56, 15/28, 5/28$. The mean is $15/8 = 1.875$.
73. With denominator $\binom{15}{2} = 105$, the PMF at $x = 0, 1, 2$ is $11/21, 44/105, 2/35$. The mean is $8/15$.
74. The CDF levels are 0, $11/21$, $33/35$, 1 across 0, 1, 2. $P(X \geq 1) = 10/21$, and the median is 0 because $F(0) = 11/21 > 1/2$.
75. With denominator $\binom{20}{3} = 1140$, the PMF at 0, 1, 2, 3 is $14/285, 28/95, 44/95, 11/57$. Therefore $P(X \geq 2) = 187/285$ and $E(X) = 9/5 = 1.8$.
76. The PMF at 0, 1, 2 is 0.09, 0.42, 0.49. The mean is 1.4 and variance 0.42.
77. The PMF at 0, 1, 2 is 0.08, 0.44, 0.48. The mean is 1.4 and variance 0.40.
78. The unconditional probabilities at 0, 1, 2 are 0.38, 0.44, 0.18. The mean is 0.80.
79. Sample A has empirical mean 1.15 and variance 0.6275. Sample B has empirical mean 1.50 and variance 0.65.
80. From $a + b = 0.60$ and $0.40 + 2b = 1.10$, obtain $a = 0.25$ and $b = 0.35$. The CDF levels are 0, 0.25, 0.65, 1 across 0, 1, 2, and the variance is 0.59.
81. $k(2 + 3 + 4 + 5) = 1$, so $k = 1/14$. $P(X \geq 2) = (4 + 5)/14 = 9/14$ and $E(X) = 13/7$.

Exercises 82–91: risk, programmes and social decisions.

82. $E(X) = 0.33$ and $\text{Var}(X) = 0.4011$. Expected payout = $1500(0.33) = \text{RM } 495$; payout standard deviation = $1500\sqrt{0.4011} \approx \text{RM } 949.99$; expected net income = $500 - 495 = \text{RM } 5$.
83. Expected cost = $200(0.20) + 500(0.10) = \text{RM } 90$. The variance is 24 900, giving standard deviation $\approx \text{RM } 157.80$. Expected total for 100 cases is $\text{RM } 9000$.
84. Expected net outcome = 0.75 million ringgit. The variance is 2.0625 and standard deviation ≈ 1.436 million ringgit. The mean omits risk, distribution, affordability, timing and non-monetary consequences.
85. Mean payment = $100(0.60) + 500(0.30) + 2000(0.10) = \text{RM } 410$. Median and mode are both $\text{RM } 100$. The high-value tail raises the mean above the payment received by most claimants.
86. $E(X) = 2.25$ and $P(X \geq 3) = 0.40$. Unused slots are 3, 2, 1, 0, 0, giving $E(U) = 3(0.05) + 2(0.20) + 1(0.35) = 0.90$.
87. Expected loss = $5000(0.05) + 20\,000(0.01) = \text{RM } 450$. The standard deviation is $\sqrt{5\,047\,500} \approx \text{RM } 2246.66$. The large spread reflects rare but severe losses.
88. Expected return = $-2(0.70) + 3(0.25) + 18(0.05) = \text{RM } 0.25$. Profit probability = 0.30 and expected total over 100 plays is $\text{RM } 25$.
89. The PMF at 0, 1, 2, 3 is 0.216, 0.432, 0.288, 0.064. The mean is 1.2 and variance 0.72.
90. Both expected benefits are 50. Policy A has standard deviation 50, while B has standard deviation 10. A may appeal when the chance of the high outcome is valued; B may be preferred for predictability and protection against the zero outcome.
91. The fitted distribution describes modelled outcome probabilities. Selection, confounding, external changes and model assumptions can affect those outcomes, so expectation alone does not identify what the programme causes.

Exercises 92–100: responsible interpretation.

92. For example, an outcome of 0 with probability 0.90 and 1000 with probability 0.10 has expected value 100, although the typical and modal outcome is 0 and 100 never occurs.
93. Equal means do not imply equal variance, tail risk, probability of loss or distribution across people. One policy can be predictable while another combines very high and very low outcomes.
94. Sampling variability, an unrepresentative design, non-response, measurement or coding error and changes in the target population can all make the empirical PMF differ from the population PMF.

95. Retain the rounded values and note the rounding discrepancy, or renormalise them by dividing by 0.99. The chosen method and any adjusted probabilities should be disclosed.
96. Expected monetary value can omit rights, fairness, distributional effects, non-market outcomes, risk aversion, irreversibility and uncertainty in the assigned probabilities or valuations.
97. A mixture PMF averages subgroup distributions. Groups may have different means, variances, modes or tail risks that disappear in the aggregate summary.
98. Consider who receives benefits and bears risks, minimum protections, accessibility, disparate errors across groups, transparency, human review, appeal mechanisms and data privacy.
99. A structural break can change the support, probabilities and relationships generating outcomes. Predictions based on the old PMF then describe a regime that no longer applies.
100. One suitable checklist is: define the random variable, population, units and support; obtain or estimate valid non-negative probabilities summing to one; construct the PMF and CDF; calculate probabilities, expectation and spread; check arithmetic, units, sensitivity and subgroup patterns; then communicate assumptions, uncertainty, risk, limitations and conditions for revision.

C.7 Chapter 7: Sequences and Series

Return to the corresponding questions in Chapter 7.

1. $k = 4$ or $k = -3/2$.
2. (a) $a_1 = 9$, $a_n = a_{n-1} + 5$; (b) $a_1 = 2$, $a_n = a_{n-1} - 5$.
3. (a) 5050; (b) 1275.
4. 2920.
5. $a = 52$, $d = -7$, and $S_{21} = -378$.
6. $a = 4$ and $r = 3$.
7. (a) 3, 12, 48, 192; geometric; (b) 7, 12, 17, 22; arithmetic.
8. RM 3 250 000.
9. (a) $a = 50$, $r = 0.8$; (b) $S_{10} \approx 223.16$; (c) $S_{\infty} = 250$.
10. (a) $0.312 + 0.000312 + \dots = 104/333$; (b) $4 + 0.086 + 0.000086 + \dots = 4082/999$.
11. (a) 645; (b) 108297.

12. (a) 720; (b) 6720; (c) $n(n-1)(n-2)$.
13. (a) $x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4$; (b) $x^5 - 10x^4y + 40x^3y^2 - 80x^2y^3 + 80xy^4 - 32y^5$.
14. 103680.
15. $T_{r+1} = \binom{6}{r}(-2)^r y^{6-r} x^{3r}$; coefficient of x^6 : $60y^4$; term independent of x : y^6 ; last two terms: $-192yx^{15} + 64x^{18}$.
16. 67.9373.
17. a_n is the sequence term and S_n is the partial sum. The first four terms are 3, 5, 7, 9, and $S_4 = 24$.
18. $a = -17/2$, $d = 7$.
19. $a_n = S_n - S_{n-1} = 2n + 4$; it is arithmetic with first term 6 and common difference 2.
20. (a) 8, 13, 18, 23, 28; (b) $5/6, 4/3, 11/6, 7/3, 17/6$.
21. $3 + 6 + \cdots + 399 = 26\,733$.
22. (a) $T_n = \frac{1}{4}(\frac{2}{3})^{n-1}$, $T_8 = 32/2187$; (b) $T_n = 4(0.3)^{n-1}$, $T_8 = 0.0008748$; (c) $T_n = (-\sqrt{3})^{n-1}$, $T_8 = -27\sqrt{3}$.
23. $r = -4$, $a = -15/64$.
24. $x = 6$ and $T_5 = 81/4$.
25. The possible first terms are -40 and -8 .
26. $\sqrt[4]{120}$.
27. $n = 10$.
28. 1404.
29. (a) $0.05 + 0.005 + \cdots = 1/18$; (b) $3 + 0.2 + 0.02 + \cdots = 29/9$.
30. (a) 220 computers; (b) 3010 computers.
31. (a) $N = 21$; (b) 27 000 phones.
32. (a) 520 points; (b) 6600 points.
33. (a) RM 138 915; (b) 2034; (c) approximately RM 1 704 814.46.
34. (a) 636; (b) $n(4n^2 + 21n + 35)/6$; (c) $n(n+1)(2n-1)/2$.
35. (a) n^2 ; (b) $2n(n+2)$; (c) 495; (d) 10 671 480.
36. (a) $16 - 96x + 216x^2 - 216x^3 + 81x^4$; (b) $1 - 6x + \frac{27}{2}x^2 - \frac{27}{2}x^3 + \frac{81}{16}x^4$; (c) $-27y^3 + 27xy^2 - 9x^2y + x^3$; (d) $-1024 + 3840x - 5760x^2 + 4320x^3 - 1620x^4 + 243x^5$.

- 37.** $1 + 2x + \frac{7}{4}x^2 + \frac{7}{8}x^3 + \cdots$; $(1.025)^8 \approx 1.2184$ using these terms.
- 38.** General term $T_{r+1} = \binom{10}{r}x^{20-5r}$. (a) $x^{20} + 10x^{15}$; (b) $10x^{-25} + x^{-30}$; (d) 210; (e) $210x^{-10}$; (f) coefficient 120.

Exercises 39–48: grouped skills review.

- 39.** The terms for Cases A–D are 37, -1 , 44 and 195.
- 40.** The corresponding sums are 246, 63, 156 and 945.
- 41.** The terms for Cases A–D are 192, 5, 80 and 1.
- 42.** The corresponding sums are 381, 315, 55 and 121.
- 43.** (a) 16; (b) -20 ; (c) $25/7 \approx 3.5714$; (d) divergent because $|r| = 1.05 > 1$.
- 44.** (a) 590; (b) 650; (c) 2628; (d) 381.
- 45.** (a) 16, 22, 28, 34; (b) $-3, 2, 7$; (c) 12, 36; (d) 6, 18, 54.
- 46.** (a) 336; (b) 120; (c) 66; (d) $n = 10$.
- 47.** (a) $1 + 8x + 24x^2 + 32x^3 + 16x^4$;
 (b) $32 - 80x + 80x^2 - 40x^3 + 10x^4 - x^5$;
 (c) $a^4 - 12a^3b + 54a^2b^2 - 108ab^3 + 81b^4$;
 (d) $1 - 3x + \frac{15}{4}x^2 - \frac{5}{2}x^3 + \cdots$.
- 48.** (a) 1792; (b) 1215; (c) 240; (d)

$$T_{r+1} = \binom{7}{r} 3^{7-r} (-1)^r x^{14-3r}.$$

Exercises 49–59: arithmetic change and public planning.

- 49.** Week 12 has $18 + 11(3) = 51$ cases, and $S_{12} = 12(18 + 51)/2 = 414$ cases.
- 50.** Year 8 has $240 + 7(18) = 366$ homes, and the eight-year total is $8(240 + 366)/2 = 2424$ homes.
- 51.** The year-10 budget is RM 1.92 million and the ten-year total is RM 15.60 million.
- 52.** (a) $B_8 = 600$ cases; (b) $1200 - 75n \leq 300$ gives $n \geq 12$, so month 12 is first; (c) the formula eventually predicts a negative backlog and assumes a constant monthly reduction despite changing caseloads and capacity.
- 53.** $40 + 5(n-1) = 100$ gives cohort 13. The total through that cohort is $13(40+100)/2 = 910$ participants.
- 54.** The year-7 subsidy is RM 870, and the seven-year total is RM 5145.

55. The final row contains $24 + 17(2) = 58$ seats. Total capacity is $18(24 + 58)/2 = 738$ seats.
56. The year-5 salary is RM45 600, and the five-year total is RM204 000. Inflation, promotion, policy changes or interruptions may make the constant increment unrealistic.
57. From $a + 4d = 140$ and $a + 11d = 210$, obtain $d = 10$ and $a = 100$. The first twelve targets sum to 1860 cases.
58. Equality gives $100 + 20(n - 1) = 160 + 10(n - 1)$, so $n = 7$ and each plan supplies 220 units then. Cumulative output through period 7 is 1120 for A and 1330 for B. In period 8, A supplies 240 units and B supplies 230, so A is larger.
59. The fitted sequence has $a = 42$ and $d = -3$, so the tenth value is 15 and $S_{10} = 285$. A changed policy, capacity, demand or measurement process can break the pattern.

Exercises 60–70: geometric growth, decay and accumulation.

60. The year-10 population is $80\,000(1.025)^9 \approx 99\,909$. The sum of the ten annual figures is approximately 896 271 person-years.
61. $N_6 = 15\,000(0.92)^6 \approx 9095$ cases. Since $\ln(0.5)/\ln(0.92) \approx 8.313$, month 9 is the first whole month below half.
62. $A(5) = 175$, while $G(5) \approx 176.23$. Checking whole periods gives $G(t) \leq A(t)$ for $t = 0, 1, 2, 3, 4$ and $G(5) > A(5)$, so $t = 5$ is first. The arithmetic additions remain 15, whereas the geometric additions grow with the current level.
63. Value after five years is $120\,000(0.85)^5 = \text{RM } 53\,244.64$, so the loss is RM66 755.36. Since $\ln(0.5)/\ln(0.85) \approx 4.265$, the value first falls below half after five whole years.
64. The first six waves reach $500(1 - 0.6^6)/(1 - 0.6) = 1191.68$ people in the model. The limiting total is $500/(1 - 0.6) = 1250$ people.
65. The year-5 count is $2000(0.75)^4 \approx 633$. The first five annual counts total approximately 6101.56 participant-years.
66. The year-6 grant is approximately RM2.433 million, and the six-year total is approximately RM13.266 million.

67.

$$0.272727\ldots = 0.27 + 0.0027 + 0.000027 + \cdots = \frac{0.27}{1 - 0.01} = \frac{3}{11}.$$

The ratio is 0.01, whose absolute value is below 1.

68. $12\,000r^2 = 7680$ gives the positive ratio $r = 0.8$. The year-5 number is 4915.2, and the first five annual counts total 40 339.2 subscriber-years.

69. The limiting benefit is $1000/(1 - 0.7) = 10\,000/3 \approx 3333.33$ units. Requiring $1 - 0.7^n \geq 0.95$ gives $n \geq \ln(0.05)/\ln(0.7) \approx 8.399$, so 9 periods are needed. The shrinking positive increments leave an ever-smaller gap to the limit.
70. After five years the arithmetic model gives 600, while the geometric model gives approximately 659.08. After ten years they give 200 and approximately 434.39, respectively. The arithmetic formula becomes negative after sufficiently many years, which is impossible for a non-negative stock.

Exercises 71–80: summation and aggregated quantities.

71. (a) 1275; (b) 2870; (c) 3025.

72. $\sum_{r=1}^{12} (40 + 3r) = 12(40) + 3(78) = 714$ cases.

- 73.

$$\sum_{r=1}^{10} (100 + 5r + 0.2r^2) = 1000 + 275 + 77 = 1352.$$

- 74.

$$\sum_{r=1}^n r(2r + 1) = 2 \sum_{r=1}^n r^2 + \sum_{r=1}^n r = \frac{n(n+1)(4n+5)}{6}.$$

At $n = 20$, the value is 5950.

75. One form is $\sum_{r=0}^9 (25 + 5r)$, whose value is 475.

- 76.

$$a_n = S_n - S_{n-1} = 8n + 6.$$

Thus the outputs are arithmetic with first term 14 and common difference 8. Total output through period 5 is $S_5 = 150$.

77. $\sum_{r=1}^8 (2000 + 300r) = 16\,000 + 10\,800 = \text{RM } 26\,800$.

78. $\sum_{r=1}^{15} r^2 = 15(16)(31)/6 = 1240$ blocks.

79. The complete sum is $30(31)/2 = 465$, so the omitted term is $465 - 438 = 27$.

80. Substituting $r = j + 1$ changes $3r + 2$ to $3j + 5$ and the limits from $1 \leq r \leq n$ to $0 \leq j \leq n - 1$, preserving every term. The common value is

$$\frac{3n(n+1)}{2} + 2n = \frac{n(3n+7)}{2}.$$

Exercises 81–90: binomial coefficients and approximation.

81. The quadratic estimate is

$$1 + 8(0.03) + \binom{8}{2}(0.03)^2 = 1.2652.$$

The calculator value is approximately 1.2667701, so the estimate is low by approximately 0.0015701.

82. The first three terms are

$$100\,000\,000 + 4\,000\,000x + 60\,000x^2.$$

The omitted cubic and quartic terms are relatively small when $|x|/100$ is small.

83. The coefficient is $\binom{8}{3}2^53^3 = 48\,384$.

84.

$$T_{r+1} = \binom{7}{r}(-2)^r x^{14-3r}.$$

For x^5 , $14 - 3r = 5$ gives $r = 3$, so the coefficient is $\binom{7}{3}(-2)^3 = -280$. A constant term would require $14 - 3r = 0$, which has no integer solution.

85. $\binom{n}{2} = 45$ gives $n(n-1) = 90$, hence $n = 10$.

86. Row 8 is

$$1, 8, 28, 56, 70, 56, 28, 8, 1.$$

Choosing r objects to include is equivalent to choosing the remaining $8 - r$ objects to exclude, so opposite entries agree.

87. The two estimates are

$$\begin{aligned} \text{linear: } 1 + 12(0.01) &= 1.12, \\ \text{quadratic: } 1.12 + \binom{12}{2}(0.01)^2 &= 1.1266. \end{aligned}$$

The calculator value is approximately 1.1268250, giving errors about 0.0068250 and 0.0002250, respectively.

88. In $(1+x)^5$, the coefficient of x^r counts the ways to choose exactly r of five components. Exactly two can be selected in $\binom{5}{2} = 10$ ways.

89. The general power of x is $9 - 3r$. A constant requires $r = 3$, giving coefficient $\binom{9}{3}2^6(-1)^3 = -5376$.

90. The coefficient is $\binom{12}{3}(1/2)^3 = 55/2 = 27.5$.

Exercises 91–100: model evaluation and responsible interpretation.

91. A sequence lists each annual cost separately; the corresponding series adds selected annual costs to obtain a cumulative total.

92. Successive ratios are 1.2, so a geometric model with first term 100 and ratio 1.2 is suitable. The next value is 207.36. The differences are not constant, ruling out an arithmetic model.

93. Possible failures include structural policy or economic changes, capacity and resource limits, changing population behaviour, measurement revisions, shocks and parameters that vary over time.
94. An infinite geometric series converges when $|r| < 1$. Even then, the assumed constant ratio, infinite time horizon, independence of periods or meaning of the limiting total may be unrealistic.
95. Check that benefits are measured in common units and periods, are not double-counted, use defensible forecasts or discounting, cover comparable populations and include relevant costs, harms and uncertainty.
96. Equal additions assume a constant absolute change. Constant percentage change assumes each addition depends on the current level, so the increments themselves grow or shrink. Similar early values can therefore diverge substantially later.
97. Truncation omits higher-order terms. Their combined size depends on the input and can alter rankings, thresholds or monetary decisions, so a remainder estimate or comparison with the exact value should accompany the approximation.
98. Check that observations use consistent units and time intervals, that missing or duplicated records are handled, and that signs, category definitions and the summation limits match the intended population and period.
99. Relevant issues include who receives resources, whether rapid growth disadvantages other groups, minimum service guarantees, uncertainty and downside risk, transparency of assumptions, human review, accountability and procedures for revising allocations.
100. One suitable checklist is: define variables, units, population and time index; inspect differences or ratios and choose a model; estimate parameters and calculate terms or sums; check formulas, bounds, units and sensitivity; compare with data and alternatives; then communicate assumptions, uncertainty, distributional effects, limitations and conditions for revision.

C.8 Chapter 8: Linear Equations and Graphs

Return to the corresponding questions in Chapter 8.

1. $x = 4$.
2. $x = \frac{58}{11}$.
3. $x \leq 7$, or $(-\infty, 7]$.
4. (a) $(-4, 6]$; (b) $(-\infty, 3)$; (c) $[-2, \infty)$.
5. $m = -3/2$; x -intercept $(4, 0)$; y -intercept $(0, 6)$.

6. $y = -4x + 13$.
7. $y = -\frac{3}{2}x + \frac{5}{2}$.
8. (a) $F = 1.80d + 4.50$; (b) \$26.10; (c) \$1.80 per kilometre and a fixed charge of \$4.50.
9. (a) \$0.18 per kilowatt-hour and a fixed predicted cost of \$24; (b) \$141; (c) $u = 700$ kilowatt-hours.
10. (a) $x = -7$; (b) $x = 4$; (c) $y = 4$; (d) no solution, since the only algebraic candidate is excluded.
11. Stocks = \$12 000; bonds = \$6 000.
12. Regular hourly wage = \$7.50.
13. (a) $(-\infty, -4]$; (b) $[-1/3, \infty)$; (c) $[12, \infty)$; (d) $(-14/9, \infty)$.
14. (a) $(-4/3, 2]$; (b) $[41, 59)$; (c) $(-11, 5]$.
15. (a) $2x - 3y = 18$; (b) $y = 3x - 9$.
16. Slope = 40 dollars per can; $C(x) = 40x + 250$. The graph is the line segment from $(0, 250)$ to $(160, 6650)$.
17. The scatter plot should place the six supplied observations and the line $p = 5600c - 1100$ on common axes. The slope predicts \$5600 per additional carat. The predicted prices are
- $$p(0.85) = \$3660 \quad \text{and} \quad p(1.20) = \$5620.$$
- A \$4000 diamond is predicted to weigh $5100/5600 \approx 0.911$ carat.
18. The slope is 3.8 feet per additional inch of diameter. For $d = 8$, $h = 49.13 \approx 49$ feet. For $h = 30$, $d \approx 2.97 \approx 3$ inches.
19. (a) $x = 33/4$, one solution. (b) No solution, since the equation reduces to $-5 = 3$. (c) Infinitely many solutions, since both sides are identical.
20. (a) $-3 < x \leq 6$, or $(-3, 6]$. (b) $2/3 < x \leq 14$, or $(2/3, 14]$.
21. (a) $x < -3$ or $x \geq 2$; $(-\infty, -3) \cup [2, \infty)$. (b) -3 does not belong to the set, while 2 does. (c) The complement is $[-3, 2)$.
22. The slope is $-3/2$, so $y = -3x/2 + 2$. The intercepts are $(0, 2)$ and $(4/3, 0)$, and $y(3) = -5/2$.
23. (a) Plan A has fixed cost RM 20 and marginal cost RM 2.50 per gigabyte; Plan B has fixed cost RM 50 and marginal cost RM 1.50 per gigabyte. (b) The plans are equal when $n = 30$, at a cost of RM 95. (c) A is cheaper for $0 \leq n < 30$; B is cheaper for $n > 30$. The sketch consists of the two stated lines meeting at $(30, 95)$.

24. The intercept is the initial RM 240 000 fund and the slope is a decrease of RM 12 000 per month. After eight months, $B = \text{RM } 144\,000$. The fund reaches zero after 20 months; a sensible domain is $0 \leq m \leq 20$.
25. $F(25) = 77^\circ\text{F}$; $C = 35^\circ\text{C}$ when $F = 95^\circ\text{F}$; and $C = (F - 32)/1.8$. Each increase of 1°C corresponds to 1.8°F , while 32°F corresponds to 0°C .
26. The changes give slope 4 throughout, so $y = 4x + 10$. Thus $y(10) = 50$, and $y = 70$ gives $x = 15$. The score rises by four points per additional training hour; the intercept predicts a score of 10 at zero training hours.
27. The calculations are

x	2	4	6	8	10
$\hat{Y}$	48.4	54.8	61.2	67.6	74.0
e	-1.4	1.2	-1.2	2.4	-1.0

The largest absolute residual is 2.4 thousand ringgit at $x = 8$. A positive residual means observed income is above the model prediction; a negative residual means it is below.

28. For 2025, $t = 5$ and $R = 4.9$ million; this is interpolation within the 2020–2026 data period. For 2035, $t = 15$ and $R = 7.1$ million; this is extrapolation and is less reliable because the observed linear trend may not continue that far.
29. (a) $4500 + 18q = 30q$ gives $q = 375$ units and revenue RM 11 250. (b) Profit is predicted when $q > 375$. (c) At $q = 600$, profit is RM 2700. (d) Variable cost is RM 18 per unit, revenue is RM 30 per unit, and fixed cost is RM 4500.
30. Predicted savings are zero at $I = 240/0.18 \approx \text{RM } 1333.33$. At $I = 3000$, $S = \text{RM } 300$, and $S > 0$ when $I > \text{RM } 1333.33$. The intercept predicts savings of $-\text{RM } 240$ at zero income, which may represent borrowing or using previous savings, but it may lie outside the model's useful range.
31. (a) The slope is $(105 - 65)/(30 - 10) = 2$, so $\hat{y} = 2x + 45$. (b) $\hat{y}(20) = 85$. (c) The residual is $82 - 85 = -3$; the observed response is three units below the prediction.
32. The slope predicts 0.35 fewer hours of leisure for each additional paid working hour. The intercept predicts 18 leisure hours when $h = 0$, though this is outside the fitted range. At $h = 40$, $L = 4$ hours. At $h = 70$, the model gives $L = -6.5$ hours, which is impossible and is an extrapolation beyond the observed range.

Exercises 33–42: grouped skills review.

33. (a) $x = 6$; (b) $x = -6$; (c) $x = 7$; (d) $x = 21/5$.
34. (a) Infinitely many solutions; (b) no solution; (c) one solution, $x = -4$.

35. (a) $(-\infty, 5]$; (b) $(-\infty, -4)$; (c) $[11, \infty)$; (d) $[1, 4)$.
36. (a) $(-3, 5]$; (b) $[2, \infty)$; (c) $(-\infty, -4) \cup [1, \infty)$; (d) $-2 \leq x < 6$.
37. The slopes for Cases A–D are 2, $-3/2$, undefined and 0.
38. (a) $y = 3x - 7$; (b) $y = -2x + 3$; (c) $y = -2x + 5$; (d) $y = -\frac{1}{2}x + 6$.
39. The given slope is $3/2$, so parallel lines have slope $3/2$ and perpendicular lines slope $-2/3$. The requested lines are $y = \frac{3}{2}x - 4$ and $y = -\frac{2}{3}x + 3$.
40. (a) $(10, 0)$ and $(0, 4)$; (b) $(4, 0)$ and $(0, 12)$; (c) $(-6, 0)$ and no y -intercept.
41. The costs for Cases A–C are RM 980, RM 2050 and RM 900. In $C = F + vq$, F is fixed cost and v is variable cost per unit; Case C has no fixed cost.
42. Case A: prediction 30, residual 2. Case B: prediction 32, residual -3 . Case C: prediction 12, residual 1.

Exercises 43–53: linear equations and quantitative constraints.

43. Let education receive E . Then $E + 2E + (E + 120\,000) = 720\,000$, so education receives RM 150 000, health RM 300 000 and administration RM 270 000.
44. If the regular rate is w , then $40w + 8(1.5w) = 52w = 780$, so $w = \text{RM } 15$ per hour.
45. With rural count R and urban count U , $U = 2R + 40$ and $U + R = 640$. Thus $R = 200$ and $U = 440$.
46. Break-even satisfies $18\,000 + 45n = 60n$, giving $n = 1200$. Contributions exceed cost for $n > 1200$.
47. $68 = 50 + 10z$ gives $z = 1.8$. When $z = 2.5$, $S = 75$.
48. Let f and c be the two ticket counts. Solving $f + c = 30$ and $15f + 8c = 345$ gives $f = 15$ and $c = 15$.
49. $0.92I = 3680$, so gross income is RM 4000.
50. (a) $63 = 100(378)/N$ gives $N = 600$; (b) $x = 0.725(800) = 580$.
51. $506\,500 = 500\,000 + 8000 - 5000 + M$, so net migration is $M = 3500$ people.
52. $1800 = 1200 + 75t$ gives $t = 8$ years. The intercept is initial rent RM 1200, and the slope is an increase of RM 75 per year.
53. The first equation is an identity and has infinitely many solutions; it represents two equivalent expressions. The second reduces to $10 = 7$ and has no solution; it represents inconsistent requirements.

Exercises 54–64: straight lines, rates and comparisons.

54. The slope is $(78 - 48)/(8 - 2) = 5$, so $y = 5x + 38$ and $y(12) = 98$. The index rises by five units per additional unit of x in the model.
55. The slope is $(1500 - 900)/(50 - 20) = 20$, giving $C = 500 + 20q$. Fixed cost is RM 500 and variable cost RM 20 per unit.
56. $B(10) = \text{RM } 100\,000$. The balance reaches zero at $t = 250/15 \approx 16.67$ months. A suitable domain is $0 \leq t \leq 16.67$.
57. Every ratio $\Delta y/\Delta x$ equals 3, so $y = 12 + 3x$ and $y(12) = 48$.
58. The parallel route is $y = \frac{3}{2}x - 7$. The perpendicular route is $y = -\frac{2}{3}x + 6$.
59. With t measured after 2022, the slope is $(57 - 48)/3 = 3$ percentage points per year, so $A = 48 + 3t$ and the 2028 prediction is 66 percentage points. The slope is not a 3% relative increase per year.
60. $30 + 4x = 60 + 2x$ gives $x = 15$ and common cost 90. Plan A is cheaper for $0 \leq x < 15$, while B is cheaper for $x > 15$.
61. $200 - 5p = 20 + 4p$ gives $p = 20$ and quantity 100. Demand falls by five units per price unit, while supply rises by four units per price unit.
62. The slope is $(3.3 - 2.4)/6 = 0.15$ million per year. The model is $I = 2.4 + 0.15t$, where t is years after 2018. It gives 2.85 million in 2021 and 4.20 million in 2030. The latter is an extrapolation beyond the two observed dates.
63. The intercepts are $(8, 0)$ and $(0, 12)$, and the slope is $-3/2$. The feasible boundary is the closed line segment joining those intercepts in the first quadrant.
64. The annual slope is $12(0.8) = 9.6$ units per year. Changing the time origin changes which date corresponds to input zero and therefore changes the numerical intercept, while leaving the underlying line unchanged.

Exercises 65–75: inequalities and feasible ranges.

65. $0.25I + 300 \leq 1800$ gives $I \leq 6000$. With $I \geq 0$, the feasible interval is $[0, 6000]$ ringgit.
66. $12\,000 + 80n \leq 50\,000$ gives $n \leq 475$, so the greatest whole number is 475 participants.
67. $20 \leq a < 65$, or $[20, 65)$. Age 20 is included, while age 65 is excluded.
68. $6x - 3 > 5x + 7$ gives $x > 10$, or $(10, \infty)$.
69. The constraints give $x \geq 12$ and $x \leq 30$, so the feasible interval is $[12, 30]$.
70. $500 - 18t \geq 200$ gives $0 \leq t \leq 50/3 \approx 16.67$. The greatest whole number of periods is 16.

71. $500 + 35n \leq 2500$ gives $n \leq 2000/35 \approx 57.14$, so the greatest whole-number attendance is 57.
72. $80 - 4t \geq 0$ gives the domain $[0, 20]$.
73. $2500 < I \leq 5000$, or $(2500, 5000]$. The lower threshold is strict, while the upper threshold is included.
74. $40 + 3x < 70 + 1.5x$ gives $x < 20$. For non-negative usage, A is cheaper on $[0, 20)$; the plans are equal at $x = 20$.
75. The algebraic solution is $[-3, 4)$. Intersecting with $x \geq 0$ gives the contextual solution $[0, 4)$.

Exercises 76–86: regression, prediction and residuals.

76. The prediction is $25 + 1.8(12) = 46.6$. The residual is $44 - 46.6 = -2.6$.
77. Predictions are 60, 50, 40, 30 and residuals are 1, -3 , 2, -3 . The largest absolute residual is 3, occurring at $x = 5$ and $x = 15$.
78. The slope is $(43 - 18)/(14 - 4) = 2.5$, so $\hat{y} = 2.5x + 8$ and $\hat{y}(10) = 33$.
79. Each additional year of education is associated with predicted monthly income higher by 0.18 thousand ringgit, or RM 180. The intercept predicts RM 2400 at zero years. Confounding, selection and reverse causation prevent causal interpretations.
80. Predictions are 12, 15, 18, 21, 24, giving residuals 0, 0, 1, -1 , 0. The point at $x = 3$ is above the line and the point at $x = 4$ is below it; the other three lie on the line.
81. For 2023, $t = 4$ and $R = 1.52$; this is interpolation. For 2035, $t = 16$ and $R = 2.48$; this is extrapolation.
82. The residual mean is 0. Positive and negative errors can cancel, so a zero mean does not rule out curvature, unequal spread, outliers or poor prediction outside the sample.
83. Since $x = 1000X$, the model becomes $y = 5 + 2X$. The slope changes from 0.002 per ringgit to 2 per thousand ringgit because the input unit is 1000 times larger.
84. An extreme observation can pull the line towards itself, changing both slope and intercept. The effect is especially strong when its input value is far from the rest of the data.
85. A fitted slope describes association within the model and data. Omitted variables, selection, reverse causation and measurement error can generate the association without a causal effect.
86. At $x = 10$, Group A predicts 40 and Group B predicts 45. A pooled line can conceal different subgroup levels, slopes, sample sizes and error patterns.

Exercises 87–94: intersections, break-even points and decisions.

87. $5000 + 12q = 20q$ gives break-even output $q = 625$ and revenue RM 12 500. Profit occurs for $q > 625$. At $q = 800$, profit is RM 1400.
88. $300 - 4p = 60 + 2p$ gives $p = 40$ and quantity 140. This is the price–quantity pair at which the two modelled quantities agree.
89. $O + T = 120$ and $O = 2T$ give $T = 40$ and $O = 80$ staff-hours.
90. $25 + 2.2n = 55 + 1.2n$ gives $n = 30$ and common cost RM 91. Plan A is cheaper below 30 units and B above 30.
91. (a) No intersection: the distinct lines are parallel. (b) Infinitely many intersections: the equations represent the same line.
92. At $x = 4$, the second line gives $y = 13$. Hence $4a + 5 = 13$, so $a = 2$ and the intersection is $(4, 13)$.
93. $40q \geq 8000 + 25q$ gives $q \geq 1600/3 \approx 533.33$. The smallest whole-number scale is 534, where the surplus is RM 10.
94. Net benefit is $40x - 2000$, which is positive for $x > 50$. At $x = 80$, net benefit is $3200 - 2000 = 1200$ units.

Exercises 95–100: responsible use of linear models.

95. A positive slope can reflect confounding variables, selection, reverse causation or common trends. Causal interpretation requires an appropriate research design and assumptions beyond the line.
96. The intercept is the prediction at input zero. If zero is outside the data range or impossible in context, the intercept is an algebraic feature rather than a meaningful observed baseline.
97. Extrapolation may fail because relationships become nonlinear, capacity constraints bind, populations or definitions change, shocks occur, or the fitted slope is unstable outside the sample.
98. A model can fit historical noise, omit important predictors or face a structural break. Future observations may also come from a different population or range of inputs.
99. Check subgroup representation and residuals, measurement and coding quality, disparate effects or error costs, transparency of the rule, privacy, human review, appeal procedures and ongoing monitoring.
100. One suitable checklist is: define variables, population, units and feasible domain; inspect the data and proposed linear relationship; estimate or derive slope and intercept; calculate predictions, intersections or feasible ranges; check arithmetic, residuals, assumptions, sensitivity and subgroups; then communicate units, uncertainty, limitations, governance safeguards and conditions for revision.

C.9 Chapter 9: Matrices

Return to the corresponding questions in Chapter 9.

1. Row: A ; column: E ; identity: B ; diagonal: B ; square: B, C, D, F ; zero: A ; upper triangular: B, F ; lower triangular: B, D .

2.

$$\begin{aligned} P + Q &= \begin{pmatrix} 5 & -1 & -1 \\ -1 & 1 & 7 \end{pmatrix}, & 2Q - P &= \begin{pmatrix} -2 & -2 & 7 \\ 1 & 8 & 5 \end{pmatrix}, \\ PS &= \begin{pmatrix} 11 & -12 & 0 & -5 \\ -15 & 5 & 8 & 4 \end{pmatrix}, & QR &= \begin{pmatrix} 9 \\ 9 \end{pmatrix}, \\ P^T &= \begin{pmatrix} 4 & -1 \\ 0 & -2 \\ -3 & 3 \end{pmatrix}, & R^T Q^T &= \begin{pmatrix} 9 & 9 \end{pmatrix}. \end{aligned}$$

The products QP , RR , SR and $Q^T S$ are undefined.

3. $M_{11} = -22$, $M_{23} = 12$, $M_{32} = 5$; cofactors $C_{11} = -22$, $C_{23} = -12$, $C_{32} = -5$.

4. (a) $\begin{pmatrix} -12 & -12 & 12 \\ 9 & 21 & -19 \\ 6 & 6 & -10 \end{pmatrix}$; (b) $\begin{pmatrix} 13 & -24 & 15 \\ -27 & -16 & 10 \\ -5 & 1 & -14 \end{pmatrix}$.

5. (a) -14 ; (b) 2 .

6. (a) $\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$; (b) $\frac{1}{5} \begin{pmatrix} 1 & -3 \\ 1 & 2 \end{pmatrix}$; (c) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 2 \\ -2 & -1 & -1 \end{pmatrix}$; (d) $\begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}$.

7. $p = -5$, $q = 4$.

8. (a) EUR 29.21; (b) EUR 14.54. Only MN is defined and its entries are labour costs by unit size (rows) and plant (columns):

$$MN = \begin{pmatrix} 18.68 & 14.54 \\ 29.21 & 22.76 \\ 40.03 & 31.24 \end{pmatrix}.$$

9. $(x, y, z) = (3, -2, 1)$.

$$10. \begin{pmatrix} 5 & 2 & 3 \\ 7 & 3 & 4 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 48 \\ 67 \\ 32 \end{pmatrix}; (x, y, z) = (4, 5, 6).$$

11. Food amounts: $X = 10$ g, $Y = 3$ g, $Z = 12$ g.

12. $\begin{pmatrix} 120 & 80 & 50 \end{pmatrix} A = \begin{pmatrix} 114 & 93 & 43 \end{pmatrix}$. Thus the estimated counts are 114 for policy X, 93 for policy Y and 43 undecided.

13.

$$\mathbf{p}_2 = T\mathbf{p}_1 = \begin{pmatrix} 0.46525 \\ 0.34050 \\ 0.19425 \end{pmatrix}.$$

The projected percentages are approximately 46.53%, 34.05% and 19.43%, respectively.

14. (a) AB has order 2×2 and BA has order 3×3 . (b)

$$AB = \begin{pmatrix} 2 & -3 \\ 18 & 5 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 7 & 4 \\ 2 & -6 & -8 \\ 2 & 19 & 12 \end{pmatrix}.$$

(c) The products have different orders and entries, so they cannot be equal; multiplication order matters.

15. $X = (A + B)/2 = \begin{pmatrix} 2 & 3 \\ 1 & 3 \end{pmatrix}$. Substitution gives $2X - A = B$. Also, $X^T = \begin{pmatrix} 2 & 1 \\ 3 & 3 \end{pmatrix}$ and $\det(X) = 3$.

16. (a) $\det M(k) = k^2 - k - 6 = (k - 3)(k + 2)$. (b) $k = 3$ or $k = -2$. (c) $M(4)^{-1} = \frac{1}{6} \begin{pmatrix} 3 & -2 \\ -3 & 4 \end{pmatrix}$.

17. The first-row cofactors are $C_{11} = -9$, $C_{12} = -7$ and $C_{13} = 5$. Hence $\det(C) = 1(-9) + 2(-7) + 0(5) = -23$.

18. (a)

$$\begin{pmatrix} 1 & 1 \\ 12 & 7 \end{pmatrix} \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 180 \\ 1710 \end{pmatrix}.$$

(b) The inverse is $\begin{pmatrix} -7/5 & 1/5 \\ 12/5 & -1/5 \end{pmatrix}$. (c) $a = 90$ adult tickets and $c = 90$ child tickets.

19. Let $\mathbf{1} = (1, 1, 1)^T$ and $\mathbf{u} = (1, 1)$. Then

$$R\mathbf{1} = \begin{pmatrix} 250 \\ 270 \end{pmatrix}, \quad \mathbf{u}R = \begin{pmatrix} 210 & 190 & 120 \end{pmatrix}.$$

The total sample is 520, so the overall proportions are approximately 0.4038, 0.3654 and 0.2308.

20. (a) A 3×3 matrix times a 3×1 vector gives a 3×1 vector. (b)

$$S\mathbf{w} = \begin{pmatrix} 71.55 \\ 67.75 \\ 75.75 \end{pmatrix}.$$

(c) The ranking is C, A, B. Different weights change the relative influence of the three indicators and may therefore change the ranking.

21. (a) $T = \begin{pmatrix} 0.85 & 0.30 \\ 0.15 & 0.70 \end{pmatrix}$. (b) Both columns sum to 1. (c)

$$\mathbf{p}_1 = \begin{pmatrix} 740 \\ 260 \end{pmatrix}, \quad \mathbf{p}_2 = \begin{pmatrix} 707 \\ 293 \end{pmatrix}.$$

(d) For example, the transition rates are assumed to remain constant and to apply uniformly to the modelled population.

22. A is valid: its entries are non-negative and each column sums to 1. Matrix B is invalid because its first column sums to 0.90. Matrix C is invalid because it contains a negative entry, even though its columns sum to 1.

23. (a) $\det(T) = 0.5$, so $T^{-1} = \begin{pmatrix} 7/5 & -3/5 \\ -2/5 & 8/5 \end{pmatrix}$. (b) $\mathbf{p}_0 = (16/25, 9/25)^T = (0.64, 0.36)^T$.
(c) Multiplication by T returns $(0.62, 0.38)^T$.

24. (a) $R\mathbf{v} = (620, 460)^T$: the plan requires 620 staff hours and 460 material units. (b) With $\mathbf{c} = (25, 8)$,

$$\mathbf{c}R\mathbf{v} = \text{RM } 19\,180.$$

(c) Reversing the order either produces incompatible dimensions or combines quantities whose rows and columns do not share meanings.

25. $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ and

$$X = A^{-1}B = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}.$$

Direct multiplication confirms that $AX = B$.

- 26.** (a) The second row is twice the first, so the rows are dependent and $\det(A) = 0$. (b) Therefore A^{-1} does not exist. (c) With $z = t$, the solutions are $(x, y, z) = (2 - t, 2 - t, t)$ for any real t . (d) The second equation would need to be twice the first, but its right-hand side is 13 rather than 12, so the system is inconsistent.
- 27.** (a) Suitable assumptions include stable transition rates, correctly defined categories, representative data, no major policy shock and similar behaviour across individuals. (b) New observed movements between categories can be converted into updated column proportions and used to replace or re-estimate the matrix entries. (c) Matrix arithmetic only establishes the consequences of the supplied numbers; unreliable assumptions or data can still produce a poor real-world prediction.

Exercises 28–37: grouped skills review.

- 28.** A is a 1×3 row matrix; B is a 3×1 column matrix; C is a 2×2 square, identity and diagonal matrix; D is a 3×3 square and diagonal matrix; E is a 2×2 square, diagonal and zero matrix.

29.

$$A + B = \begin{pmatrix} 3 & 4 \\ 1 & 4 \end{pmatrix}, \quad 2A - B = \begin{pmatrix} 3 & -7 \\ 8 & 8 \end{pmatrix}, \quad 3B = \begin{pmatrix} 3 & 15 \\ -6 & 0 \end{pmatrix}.$$

- 30.** AB is 2×2 and BA is 3×3 . The products are

$$AB = \begin{pmatrix} 2 & -3 \\ 18 & 5 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 7 & 4 \\ 2 & -6 & -8 \\ 2 & 19 & 12 \end{pmatrix}.$$

Their different orders and entries demonstrate that multiplication order matters.

31.

$$A^T = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 4 \end{pmatrix}, \quad (AB)^T = \begin{pmatrix} 2 & 18 \\ -3 & 5 \end{pmatrix}.$$

Direct multiplication gives the same matrix for $B^T A^T$.

- 32.** (a) 10; (b) 11; (c) 0; (d) -6.

- 33.** (a) $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$; (b) $\frac{1}{6} \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$; (c) no inverse because the determinant is zero.

- 34.** (a) $(x, y) = (3, 1)$; (b) $(x, y) = (4, 1)$.

- 35.** The first-row cofactors are -9, -7, 5, so $\det(C) = 1(-9) + 2(-7) + 0(5) = -23$.

36. (a) $X = (A + B)/2 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$; (b) $\mathbf{x} = (3, 1)^\top$.

37. The entries are non-negative and each column sums to 1, so T is valid. Also

$$\mathbf{p}_1 = \begin{pmatrix} 0.50 \\ 0.50 \end{pmatrix}.$$

The model predicts equal shares after one period.

Exercises 38–48: matrices as social-science data tables.

38. Regional totals are $(250, 270, 200)^\top$. Category totals are $(270, 265, 185)$, for overall sample 720. The proportions are 0.3750, approximately 0.3681 and approximately 0.2569.

39.

$$(200, 150, 100)P = (185, 165, 100).$$

40. $S\mathbf{w} = (72.25, 73.75, 72.75, 74.90)^\top$. The ranking is D, B, C, A. Different weights change the influence of each indicator and can change the ranking.

41. $Q\mathbf{p} = (1380, 1520)^\top$ ringgit for Regions 1 and 2.

42.

$$A + B = \begin{pmatrix} 250 & 155 & 105 \\ 185 & 215 & 150 \end{pmatrix}, \quad B - A = \begin{pmatrix} 10 & -5 & 5 \\ 5 & -5 & 10 \end{pmatrix}.$$

Positive entries show increases from quarter A to B and negative entries show decreases.

43. The two successive matrices are

$$\begin{pmatrix} 105 & 126 \\ 84 & 157.5 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 115 & 136 \\ 94 & 167.5 \end{pmatrix}.$$

44. The transpose is 3×2 : its rows become categories and its columns become regions. Each original entry is retained, but its row–column position and reporting perspective are exchanged.

45. $R\mathbf{v}$ is 2×1 and gives total resources; $\mathbf{c}R$ is 1×3 and gives cost per service; $\mathbf{c}R\mathbf{v}$ is a scalar total cost. The product $\mathbf{v}R$ is undefined because the inner dimensions 1 and 2 do not agree.

46. $\mathbf{n}P = (130, 170)$ and $\mathbf{n}P\mathbf{c} = 130(10) + 170(20) = 4700$.

47. Row totals are $(100, 100, 100)^\top$ and column totals are $(65, 75, 85, 75)$.

48. The first-row condition gives $20 + 30 + x = 90$, so $x = 40$. The third-column condition gives $x + 40 = 80$, confirming the result.

Exercises 49–59: determinants, inverses and linear systems.

49. $\det M(k) = k^2 - k - 6 = (k - 3)(k + 2)$, so the singular values are $k = 3$ and $k = -2$.
Also

$$M(4)^{-1} = \frac{1}{6} \begin{pmatrix} 3 & -2 \\ -3 & 4 \end{pmatrix}.$$

50. The inverse is $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, giving $(x, y)^T = (40, 10)^T$.

51.

$$\begin{pmatrix} 1 & 1 \\ 12 & 7 \end{pmatrix} \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 180 \\ 1710 \end{pmatrix},$$

whose solution is $a = 90$ adults and $c = 90$ children.

52.

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 120 \\ 140 \\ 120 \end{pmatrix}.$$

The solution is $(x, y, z) = (40, 60, 20)$.

53. The solution is $(x, y) = (3, 2)$.

54. $A^{-1} = \frac{1}{10} \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix}$, giving $\mathbf{x} = (3, 2)^T$.

55. (a) With $y = t$, the solutions are $(x, y) = (5 - 2t, t)$ for any real t . (b) The second right-hand entry would need to be twice the first, so the system is inconsistent. (c) Both behaviours arise because $\det(A) = 0$ and no unique inverse solution exists.

56. The determinant is -23 , so C^{-1} exists.

57. The inverse is $\begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$, giving $\mathbf{x} = (3, 2)^T$.

58.

$$X = A^{-1}B = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}.$$

Multiplying by A returns B .

59. $\det(A) = 0.01$, while $\det(B) = 1$. For right-hand side $(2, 2.01)^T$, the solution is $(1, 1)^T$; for $(2, 2.02)^T$, it is $(0, 2)^T$. A change of only 0.01 in the data produces a large change in the solution because A is nearly singular.

Exercises 60–70: resource, expenditure and allocation models.

60. $R\mathbf{v} = (620, 460)^\top$: 620 staff hours and 460 material units. Total cost is $\mathbf{c}R\mathbf{v} = 25(620) + 8(460) = \text{RM } 19\,180$.

61. Total expenditure is $(120, 80, 50)(500, 650, 900)^\top = \text{RM } 157\,000$.

62. Resource totals are $R\mathbf{v} = (460, 320, 600)^\top$. The total cost is $\mathbf{c}R\mathbf{v} = \text{RM } 20\,640$.

63.

$$RQ = \begin{pmatrix} 460 & 570 \\ 320 & 330 \\ 600 & 720 \end{pmatrix}, \quad \mathbf{c}RQ = (20\,640, 24\,660).$$

Plan A has the lower modelled resource cost.

64. Multiplying by $(2, 3, 1)^\top$ million gives district allocations $(1.9, 2.5, 1.6)^\top$ million ringgit. These sum to RM 6 million, the full available amount.

65. $\mathbf{p}S = (1340, 1440)$ ringgit for the two regions.

66. $S\mathbf{w} = (71.55, 67.75, 75.75)^\top$, so the ranking is C, A, B.

67. $(300, 200, 100)U = (850, 800, 650)$ total service-use units.

68. $R\Delta\mathbf{v} = (30, 20)^\top$. The original resource vector is $(260, 540)^\top$ and the revised vector is $(290, 560)^\top$, confirming the same change.

69. Matrix multiplication is associative when the dimensions agree, so either grouping gives the same scalar. Reversing factors changes the dimension matching and the represented mapping, and may be undefined or produce a different result.

70. Recalculate $S\mathbf{w}$ for plausible alternative weights, compare scores and rankings, and report which conclusions remain stable. Large changes reveal sensitivity to value judgements in the weights.

Exercises 71–81: transition and state-change models.

71.

$$\mathbf{p}_1 = \begin{pmatrix} 740 \\ 260 \end{pmatrix}, \quad \mathbf{p}_2 = \begin{pmatrix} 707 \\ 293 \end{pmatrix}.$$

72. A is valid. Matrix B is invalid because its first column sums to 0.90. Matrix C is invalid because it contains a negative entry, even though its columns sum to 1.

73. $\mathbf{p}_1 = (0.62, 0.38)^\top$ and $\mathbf{p}_2 = (0.61, 0.39)^\top$.

74. Under the order S, O, U,

$$T = \begin{pmatrix} 0.75 & 0.10 & 0.25 \\ 0.15 & 0.80 & 0.25 \\ 0.10 & 0.10 & 0.50 \end{pmatrix}.$$

Thus $\mathbf{p}_1 = (0.455, 0.365, 0.180)^\top$.

75. Solving $T\mathbf{p} = \mathbf{p}$ with entries summing to 1 gives $\mathbf{p} = (0.6, 0.4)^\top$.

76. $T^{-1} = \begin{pmatrix} 7/5 & -3/5 \\ -2/5 & 8/5 \end{pmatrix}$, giving $\mathbf{p}_0 = (0.64, 0.36)^\top$.

77. $T = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}$.

78. $T_1(0.5, 0.5)^\top = (0.55, 0.45)^\top$, while $T_2(0.5, 0.5)^\top = (0.45, 0.55)^\top$.

79. Completion is absorbing because a completed participant remains completed with probability 1. After two periods, the counts are $(490, 510)^\top$.

80. $T^5\mathbf{p}_0 = (0.5875, 0.4125)^\top$, which is close to the stationary vector $(0.6, 0.4)^\top$.

81. Each multiplication advances the state by one period, so T^n represents n repeated applications. Policy changes, population composition, shocks or behaviour that changes over time can make a constant transition matrix unsuitable.

Exercises 82–91: matrix sensitivity and decision analysis.

82. $X = (A + B)/3 = \begin{pmatrix} 1 & 4/3 \\ 1 & 3 \end{pmatrix}$.

83. $\det A(k) = 4 - 2k$, so $k = 2$ is singular. At this value, the system cannot have a unique inverse solution; depending on $\mathbf{b}$, it has either no solution or infinitely many solutions.

84. With $\mathbf{w}_1$, the scores are $(71.55, 67.75, 75.75)^\top$, ranking C, A, B. With $\mathbf{w}_2$, the scores are $(67.2, 73.0, 61.5)^\top$, ranking B, A, C. The ranking is therefore sensitive to the indicator priorities.

85. The solution is $(x, y) = (60, 40)$.

86.

$$AB = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 2 \\ 0 & 3 \end{pmatrix}.$$

The first transformation changes the input received by the second, so reversing their order changes the outcome.

87. Transposition makes indicators the rows and regions the columns. If the old row labels are retained, values will be attributed to the wrong entities even though the numerical entries are unchanged.
88. $T\mathbf{p} = (0.600, 0.400)^\top$, while $T'\mathbf{p} = (0.608, 0.392)^\top$. The revised assumptions raise the first-state prediction by 0.008.
89. Direct multiplication gives $I_3\mathbf{v} = (4, -2, 7)^\top = \mathbf{v}$. The identity leaves a vector unchanged and represents a no-change transformation.
90. $A\mathbf{v} = (250, 250)^\top$, $B\mathbf{v} = (200, 300)^\top$, and $(A - B)\mathbf{v} = (50, -50)^\top$.
91. The inverse calculation gives $\mathbf{x} \approx (457.14, 542.86)^\top$. These are model-based corrected estimates and depend on the classification-error matrix being accurate and applicable to the current data.

Exercises 92–100: responsible use of matrix models.

92. Matching dimensions is not enough: adjacent rows and columns must represent the same categories in the same order and compatible units. Otherwise a numerical product can be meaningless.
93. Check that transition probabilities are estimated reliably, remain stable over time, apply to the target population, use exhaustive and mutually exclusive states, and are not disrupted by major policy or economic changes.
94. Ask who selected the indicators and weights, whether groups are represented fairly, whether minimum protections are needed, how uncertainty affects ranks, and whether review or appeal is possible.
95. Aggregation combines groups and can conceal different rates, distributions, errors and impacts within them. Subgroup products and residual comparisons should accompany the overall result.
96. A singular matrix means the equations do not identify one unique solution. Consistent dependent equations have infinitely many solutions; inconsistent equations have none.
97. A matrix product propagates assumed numerical relationships. It does not rule out confounding, selection, reverse causation or an incorrect mechanism.
98. Retain the rounded entries and disclose the discrepancy, or renormalise each affected column to sum to 1. The chosen method and all adjusted values should be reported.
99. Aggregate or suppress small cells, apply access controls, remove unnecessary identifiers, assess re-identification risk, use secure release procedures and document any privacy–utility trade-off.

100. One suitable checklist is: define the entities, categories, units and matrix orientation; collect and label compatible data; verify dimensions and model assumptions; perform the matrix calculations; check determinants, totals, probabilities, sensitivity and subgroups; then communicate interpretation, uncertainty, limitations, privacy and governance safeguards, and conditions for revision.

C.10 Chapter 10: Limits and Continuity

Return to the corresponding questions in Chapter 10.

1. (a) $+\infty, -\infty$, does not exist, $-2, -2$. (b) $3, 3, 3, -\infty, -\infty$. (c) $+\infty, 3/2$, does not exist, $1, +\infty$. (d) $2, 2, 2, -2, 4$.
2. (a) 7. (b) The left-hand limit is $2\sqrt{2}$ and the right-hand limit is $-3/7$, so the two-sided limit does not exist.
3. (a) $a = -5/2$; (b) $a = 3, b = -5$.
4. (a) $-\infty$; (b) $+\infty$; (c) $-\infty$; (d) $7/2$.
5. (a) No vertical or horizontal asymptote; a hole at $(2, 4)$. (b) vertical $x = 0$, horizontal $y = 0$. (c) vertical $x = -1$ and $x = 2$, horizontal $y = 1$. (d) No vertical or horizontal asymptote; a hole at $(-1, -8)$.
6. $a = 1$. At $x = -1$, the left-hand limit and function value are 1, while the right-hand limit is 2, so f is not continuous.
7. The limit is 80. The model predicts that participation approaches 80% in the long run. It never equals 80 for finite t because $30/(t+1) > 0$.
8. $N(2) = -8$ and $N(3) = 5$. Since N is continuous, the Intermediate Value Theorem guarantees at least one root in $(2, 3)$. It does not give the exact root or establish uniqueness.
9. (a) 6; (b) $1/6$; (c) $-1/4$; (d) 3.
10. (a) 16; (b) -8 ; (c) -2 ; (d) 3.
11. (a) Removable: the limit is 2, but $f(1) = 5$. (b) Jump: the one-sided limits are -1 and 1. (c) Infinite: $x = 2$ is a vertical asymptote. (d) Removable: after cancellation the limit at -3 is $-1/6$.
12. (a) $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$; (b) $(-\infty, 5/2]$; (c) $(-4, 1) \cup (1, \infty)$; (d) $\mathbb{R}$.
13. (a) $k = 1$; (b) $a = 2$ and $b = 4$.
14. (a) Vertical asymptotes $x = -2, 2$ and horizontal asymptote $y = 3$. (b) Horizontal asymptote $y = 0$; no vertical asymptote. (c) Vertical asymptote $x = 1$ and oblique asymptote $y = 2x + 2$. (d) Vertical asymptote $x = -1$, horizontal asymptote $y = 1$ and a removable hole at $(2, 4/3)$.

15. (a) $5/2$; (b) 0 ; (c) $-\infty$; (d) $+\infty$.
16. (a) $Q(2) = -2$ and $Q(3) = 18$, so a root lies in $(2, 3)$. (b) The successive retained intervals are $[2, 2.5]$, $[2, 2.25]$ and $[2.125, 2.25]$. (c) The midpoint estimate is 2.1875 , with absolute error at most 0.0625 .
17. (a) 85 ; (b) the limit is 60 , representing the model's long-run satisfaction level; (c) from above; (d) $t = \ln 5 / 0.2 = 5 \ln 5 \approx 8.05$ time units.
18. (a) Both one-sided limits are RM 100 . (b) Yes, since $C(2000) = 100$ as well. (c) RM 140 . (d) The marginal charge changes from RM 0.05 to RM 0.08 per additional ringgit of transaction value, although the total charge has no jump at the threshold.
19. (a) $40 - 0.1h$; (b) 40 ; (c) near $q = 100$, one additional unit raises modelled revenue by approximately RM 40 .
20. Since $20e^{-0.3t} > 0$ for every finite t , the model approaches 70 but never reaches it at a finite time. Continuity is a property of the idealised formula; it does not prove that observed approval data are continuous or that the model's assumptions are valid.

Exercises 21–30: grouped skills review.

21. The values are $6, 11, 18, 27, 38, 51, 66$ and 83 .
22. The values are $1, 5/4, 7/5, 3/2, 11/7, 13/8, 5/3$ and $17/10$.
23. Since the quotient simplifies to $x + a$, the values are $4, 6, 8, 10, 12, 14, 16$ and 18 .
24. Rationalisation gives $1/(2m)$, so the values are $1/4, 1/6, 1/8, 1/10, 1/12, 1/14, 1/16$ and $1/18$.
25. None of the two-sided limits exists because the left- and right-hand limits differ by 2 in every case.
26. The values are $2/3, 3/4, 4/5, 5/6, 6/7, 7/8, 8/9$ and $9/10$.
27. For each stated m , the vertical asymptote is $x = m - 1$ and the horizontal asymptote is $y = m$.
28. Continuity requires
- $$c = m(m - 1) + m - 2 = m^2 - 2.$$
- Thus the values are $2, 7, 14, 23, 34, 47, 62$ and 79 .
29. $f_m(m) = -1$ and $f_m(m + 1) = 2m > 0$. Since each polynomial is continuous, a root lies in $(m, m + 1)$.
30. The quotient simplifies to $2a + h$, so the values are $2, 4, 6, 8, 10, 12, 14$ and 16 .

Exercises 31–40: numerical and local behaviour.

31. Both one-sided limits are approximately 70, so the two-sided limit is approximately 70. Since $A(5) = 75$, the recorded point differs from the nearby trend; if the limiting pattern is exact, the discontinuity is removable.
32. The left- and right-hand limits are both 3, so the two-sided limit is 3. Since $W(10) = 5$, the discontinuity is removable.
33. The left-hand limit is RM 1000 and the right-hand limit is RM 0. They are unequal, so the two-sided limit does not exist.
34. $T(u) \rightarrow +\infty$. The model predicts unbounded delay as utilisation approaches full capacity from below.
35. As $t \rightarrow 2$, $1/(t-2)$ becomes unbounded and the sine term continues oscillating between -1 and 1 . Nearby values do not settle, so no limit exists; changing one function value cannot remove the oscillation.
36. The left-hand limit is RM 20 and the right-hand limit is RM 25. The two-sided limit does not exist, although $F(1000) = \text{RM } 20$.
37. (a) 21; (b) 36; (c) 4; (d) 4.
38. $M(100) = 46$, $M(1000) = 40.6$ and $M(10\,000) = 40.06$. The limit is 40, representing the variable cost approached as the fixed amount per case becomes negligible.
39. The left-hand limit is -1 and the right-hand limit is 1 , so the two-sided limit does not exist.
40. A finite table samples only selected inputs and rounding can hide small deviations or oscillation. An algebraic or theoretical argument is required to establish the exact limiting behaviour.

Exercises 41–50: algebraic limit techniques.

41. 10.
42. $1/4$.
43. -1 .
44. 12.
45. $-1/4$.
46. 108.
47. The quotient is $40 + 0.05h$, so the limit is 40 monetary units per additional unit.

48. The quotient is $60 - h$, so the limit is 60 participation units per additional unit of input.
49. The quotient is -5 for every non-zero h , so the limit is -5 . Demand falls by five units per one-unit local price increase.
50. The cancellation is valid only for $x \neq 0$, which is sufficient in a punctured neighbourhood. The original value at 0 is undefined, but nearby values equal $x + 1$, whose limit is 1.

Exercises 51–60: continuity and policy thresholds.

51. $1000k + 200 = 1200$, so $k = 1$.
52. The left-hand limit is $0.02(5000) = 100$, so $d = 100$.
53. The limit is 6, but $R(3) = 10$; the discontinuity is removable.
54. The left-hand limit and value are RM 2.50, while the right-hand limit is RM 4.00. This is a jump discontinuity.
55. The left-hand limit is RM 1200, the right-hand limit and $B(3000)$ are RM 0, and the discontinuity is a jump. Crossing the threshold causes a discrete RM 1200 loss of benefit.
56. Both one-sided limits and $T(20\,000)$ equal RM 1000, so the tax is continuous. The marginal rate changes from 5% to 10%, producing a kink rather than a jump.
57. The discontinuity is infinite because $D(u) \rightarrow +\infty$ as $u \rightarrow 1^-$. The model predicts rapidly worsening delay near full utilisation.
58. $(-5, 2) \cup (2, \infty)$.
59. Define $f(4) = 8$.
60. A genuine jump may be specified by legislation, eligibility rules or administrative practice; an artificial jump may result from incompatible fitted formulas, sparse data or coding error. Check the formal rule, raw observations, metadata and behaviour on both sides of the threshold.

Exercises 61–70: asymptotes and long-run behaviour.

61. The limit is 25. The term $5000/n$ is a fixed amount spread over cases, while 25 is the long-run variable cost per case.
62. $P(0) = 20$ and the limit is 80. Since $3e^{-0.4t} > 0$ for every finite t , the model never reaches 80 at a finite time.
63. The limit is 4, approached from above. The condition $6e^{-0.2t} < 0.5$ gives $t > \ln(12)/0.2 \approx 12.42$ time units.

64. Rewriting gives $E(x) = 5 + 70/(x + 10)$. The limit is 5, approached from above for $x \geq 0$.
65. The vertical asymptote is $x = 20$ and the horizontal asymptote is $y = 50$. As $x \rightarrow 20^-$, $F(x) \rightarrow -\infty$; as $x \rightarrow 20^+$, $F(x) \rightarrow +\infty$.
66. Both models begin at 20 and tend to 80. At $t = 10$, $A(10) \approx 71.88$ and $B(10) \approx 79.60$. The larger rate parameter in B closes the remaining gap more quickly.
67. $I(x) = 3 - 10/(x^2 + 4)$, so the horizontal asymptote is $y = 3$ and the graph approaches it from below.
68. Division gives $C(x) = x - 10 + 200/(x + 10)$. Thus $x = -10$ is a vertical asymptote and $y = x - 10$ is an oblique asymptote. On $x \geq 0$, only the oblique asymptote is relevant to long-run behaviour.
69. The limits are 1 as $x \rightarrow \infty$ and 0 as $x \rightarrow -\infty$. The horizontal asymptotes are $y = 1$ and $y = 0$.
70. Since $|e^{-t} \sin t| \leq e^{-t} \rightarrow 0$, the limit is 70. The sine factor changes sign and the graph equals or crosses the asymptote at multiples of π , so the asymptote is not a bound.

Exercises 71–80: existence and bisection.

71. $N(2) = -7$ and $N(3) = 5$. Since N is continuous, at least one root lies in $(2, 3)$.
72. Excess demand is $G(p) = 80 - 3p$. Since $G(26) = 2$ and $G(27) = -1$, an equilibrium lies in $(26, 27)$. The exact price is $p = 80/3$.
73. $B(2) = -5$ and $B(3) = 9$. Continuity gives at least one balanced point in $(2, 3)$.
74. $S(4) \approx 73.98 < 75$ and $S(5) \approx 75.54 > 75$, so continuity gives a solution in $(4, 5)$. Solving exactly gives $t = \ln 4/0.3 \approx 4.62$.
75. The retained intervals are $[2.5, 3]$, $[2.5, 2.75]$ and $[2.625, 2.75]$. The midpoint is 2.6875 and the error is at most 0.0625.
76. $H(20) = 300$ and $H(25) = 250$, so continuity gives a price in the interval. Solving $500 - 10p = 275$ gives $p = 22.5$.
77. No conclusion follows. For example, $f(x) = x^2 - 1$ is positive at $x = -2$ and $x = 2$ but has roots at $x = -1$ and $x = 1$. A sign change is sufficient for the theorem's root test, not necessary.
78. The function jumps from -1 to 1 and never equals zero. There is no contradiction because J is not continuous on $[-1, 1]$.
79. The values at 2 and 3 have opposite signs. If f is continuous, the theorem guarantees a root in $(2, 3)$.
80. After n steps, midpoint error is at most $2/2^n$. Requiring $2/2^n < 0.001$ gives $n \geq 11$.

Exercises 81–90: instantaneous rates from limits.

81. The quotient is $34 + 0.02h$, so the limit is 34 monetary units per additional unit.
82. The quotient is $60 - 0.1h$, so the limit is 60 monetary units per additional unit.
83. The quotient is $30 - h$, so the limit is 30 benefit units per additional unit of programme scale.
84. The quotient is $60 - 3h$, so the limit is 60 monetary units per year.
85. The quotient is $35 + 0.5h$, so the limit is 35 responses per additional unit of sample input.
86. The quotients are 34.02, 34.002 and 34.0002. They approach the limiting marginal cost 34 as h becomes smaller.
87. The units are people per year. The limit is the instantaneous net population change predicted by the model at time t .
88. The quotient simplifies to 6 for $h \neq 0$, so the limit is 6.
89. For $h < 0$, $|h|/h = -1$; for $h > 0$, it equals 1. The unequal one-sided limits mean that no single instantaneous rate exists at the kink.
90. The left-hand limit is RM 10 per unit and the right-hand limit is RM 15 per unit. The total cost is continuous, but the unequal marginal limits mean that there is no single instantaneous marginal cost at $q = 100$.

Exercises 91–100: model evaluation and responsible interpretation.

91. A limit describes arbitrarily close or long-run behaviour. The difference $A(t) - L$ may remain non-zero for every finite t , as in exponential adjustment models.
92. Continuity is an assumption of the mathematical representation. Observations may be discrete, rounded, noisy or affected by sudden events even when the fitted curve is continuous.
93. Possible reasons include an unstable fitted rate, structural breaks, changing population composition, capacity constraints, policy or economic shocks and measurement error.
94. An asymptote is a feature of a formula's end behaviour. A cap or bound is an external restriction, while a policy target is an intended outcome; none follows automatically from the others.
95. Check the legal or administrative rule, observations immediately before and after the date, coding and measurement changes, and whether separate fitted regimes improve residual behaviour.

96. Rounding can make distinct nearby values appear identical, and a finite table cannot exclude later deviation or oscillation. Exact conclusions require algebraic or theoretical support.
97. The formula may omit queue discipline and behavioural adaptation; parameters may change near capacity; and measurement or rounding error is magnified when the denominator is very small.
98. The theorem does not give the exact location, uniqueness, the number of solutions, a rate of numerical convergence or a causal interpretation.
99. Sampling error can reverse an estimated sign and may remove the apparent bracket. If continuity is doubtful, a sign change need not imply a root. The interval and conclusion should therefore include uncertainty or sensitivity analysis.
100. One suitable checklist is: define variables, units and domain; identify the relevant local or long-run limit; calculate it by a justified numerical, graphical or algebraic method; check one-sided behaviour, continuity, signs and bounds; interpret it within the model; then communicate assumptions, uncertainty, limitations and conditions for revision.

C.11 Chapter 11: Differentiation

Return to the corresponding questions in Chapter 11.

1. (a) -5 ; (b) $3/(2\sqrt{3x})$; (c) $-3/x^2$; (d) $4x - 5$.
2. (a) $-5/(4x^6) + 6x - 1/\sqrt{2x}$; (b) $-7/x^4 + 20/(3x^5)$; (c) $\frac{5}{2}x^{3/2} - \frac{3}{2}x^{-3/2}$; (d) $-5/x^2 - \frac{5}{3}x^{2/3} + \frac{10}{3}x$.
3. (a) $-2/x^3$; (b) $12x^2 + 1/(6x^{3/2})$; (c) $\frac{3}{2}x^{-1/4} - 12x^{-5}$; (d) $12(5x - 2)/[x^2(4 - 5x)^2]$.
4. (a) Revenue rises by \$437.50; average rate = \$8.75 per additional seat. (b) $R'(x) = 60 - 0.05x$. (c) $R(1000) = \$35\,000$ and $R'(1000) = \$10$ per seat; near 1000 units, one additional sale raises revenue by about \$10.
5. (a) $S'(t) = 1/(2\sqrt{t})$. (b) $S(9) = 11$ million dollars and $S'(9) = 1/6$ million dollars per month. (c) $S(10) \approx 11.1667$ million and $S(11) \approx 11.3333$ million by linear approximation.
6. (a) 5 m/s; (b) $3 + h$ m/s; (c) 3 m/s.
7. (a) $v(t) = 3t^2 - 30t + 72 = 3(t - 4)(t - 6)$; (b) $v(2) = 24$ ft/s, $v(5) = -3$ ft/s; (c) $t = 4, 6$ seconds.
8. (a) $k'(x) = 4x^3 - 24x^2 = 4x^2(x - 6)$; (b) $y = -20x + 20$; (c) $x = 0, 6$.
9. (a) $y' = e^x(x + 1)^2$; (b) $y' = (-3x^2 + 2x + 3)/(x^2 + 1)^2$; (c) $y' = 8(2x + 5)^3$.

10. Profit is $\Pi(q) = 60q - 0.2q^2 - 500$. The maximum occurs at $q = 150$, with price $p = 50$ and maximum profit 4000 monetary units.
11. $q(60) = 200$ and $q'(p) = -5$, so $E(60) = -(60/200)(-5) = 1.5$. Demand is elastic: locally, a 1% price increase is associated with an approximate 1.5% decrease in quantity demanded.
12. (a) $y' = e^{2x}(6x^2 + 6x - 2)$; (b) $y' = 4x/(2x^2 + 5)$; (c) $y' = e^{\sqrt{x}}/(2\sqrt{x})$; (d) $y' = (x^2 - 4x - 1)/(x - 2)^2$; (e) $y' = 6(5 - 3x)^{-3}$.
13. (a) $(1, 0)$; (b) $y = x - 1$; (c) $y = -x + 1$.
14. (a) $A'(t) = 24e^{-0.4t}$ and $A''(t) = -9.6e^{-0.4t}$. (b) $A'(0) = 24$ and $A'(5) = 24e^{-2} \approx 3.25$ percentage points per year. (c) The model is increasing for all $t \geq 0$ and is concave downward. (d) The positive first derivative shows that adoption rises, while the negative second derivative shows that its growth rate falls as the saturation level is approached.
15. (a) The stationary points are $(1, 8)$, a local maximum, and $(3, 4)$, a local minimum. (b) Increasing on $(-\infty, 1) \cup (3, \infty)$ and decreasing on $(1, 3)$. (c) The point of inflexion is $(2, 6)$; the curve is concave downward on $(-\infty, 2)$ and upward on $(2, \infty)$.
16. (a) $L(t) = 50 + 20 \ln 5 + 4(t - 4)$. (b) $G(4.1) \approx 50 + 20 \ln 5 + 0.4 \approx 82.59$. (c) Linearisation uses the local tangent; curvature causes the approximation error to grow as the input moves farther from 4.
17. (a) $MC(q) = 25 + 0.1q$ and $MR(q) = 90 - 0.2q$. (b) $\Pi(q) = 65q - 0.15q^2 - 1200$ and $\Pi'(q) = 65 - 0.3q$. (c) $q = 650/3 \approx 216.67$ units; since $\Pi''(q) = -0.3 < 0$, this is a maximum. (d) Maximum profit is approximately RM 5841.67.
18. (a) $\overline{C}(q) = 500/q + 12 + 0.02q$. (b) $\overline{C}'(q) = -500/q^2 + 0.02$, so the minimum occurs at $q = 50\sqrt{10} \approx 158.11$ units. (c) At this output, $500/q = 0.02q$, so $\overline{C}(q) = 12 + 0.04q = C'(q)$.
19. (a) $N(x) = 100x - 2x^2 - 200$. (b) $N'(x) = 100 - 4x$, so the maximum occurs at $x = 25$. (c) $N(25) = 1050$, while $N(0) = N(50) = -200$.
20. The absolute maximum is $W(6) = 41$. The absolute minimum is $W(0) = 5$; the other endpoint gives $W(10) = 25$.
21. (a) $q'(p) = -20e^{-0.02p}$ and $E(p) = 0.02p$. (b) $E(30) = 0.6$, so demand is inelastic; $E(80) = 1.6$, so demand is elastic. (c) The unit-elastic price is $p = 50$. (d) $R'(p) = 1000e^{-0.02p}(1 - 0.02p)$, which changes from positive to negative at $p = 50$; hence revenue is maximised there.
22. (a) $0 \leq p \leq 200$. (b) $E(p) = p/(200 - p)$ for $0 \leq p < 200$. (c) Unit elasticity occurs at $p = 100$. (d) Revenue is maximised at $p = 100$, where $q = 300$ and $R = 30\,000$ monetary units.

23. (a) $dI/dt = 150 - 6t$. (b) $dC/dt = 0.8(dI/dt) = 120 - 4.8t$. (c) At $t = 10$, $dI/dt = 90$ and $dC/dt = 72$. The model predicts that consumption is then increasing at 72 monetary units per unit of time.
24. (a) The signs of f' are positive on $(-\infty, -1)$, negative on $(-1, 2)$ and positive on $(2, \infty)$. (b) f is increasing on $(-\infty, -1) \cup (2, \infty)$ and decreasing on $(-1, 2)$. (c) $x = -1$ is a local maximum and $x = 2$ is a local minimum.
25. (a) The centred estimate is $(63.38 - 62.80)/(5.1 - 4.9) = 2.9$ percentage points per year. (b) The backward estimate is 3.0 and the forward estimate is 2.8 percentage points per year. (c) Near year 5, approval is increasing at about 2.9 percentage points per year. The estimates may differ because the underlying slope changes and the recorded values are rounded or subject to measurement variation.
26. Any three: a positive derivative describes association within the fitted model rather than proving causation; the derivative has only been established on the observed domain, so extrapolation may be unreliable; making x as large as possible ignores constraints, costs and competing objectives; and the fitted model may omit relevant variables or rely on unstable assumptions.

Exercises 27–36: grouped skills review.

27. The derivatives are $2x, 3x^2, 4x^3, 5x^4, 6x^5, 7x^6, 8x^7$ and $9x^8$.
28. In every case $y' = 2ax + b$; substitute the stated values of a and b .
29. In general, $y' = e^x(x^n + nx^{n-1})$ for each stated value of n .
30. For each stated k ,
- $$y' = \frac{k^2 + 1}{(x + k + 1)^2}.$$
31. For each stated k , $y' = k(k + 1)(kx + k - 1)^k$.
32. For each stated k , $y' = k/(kx + k - 1)$.
33. The point is $((k - 1), (k - 1)(2k - 3))$ and the slope is $3k - 4$. Hence
- $$y - (k - 1)(2k - 3) = (3k - 4)[x - (k - 1)].$$
34. The marginal costs are 22, 23.4, 24.8, 26.2, 27.6, 29, 30.4 and 31.8 monetary units per additional unit near output 10.
35. The maximising outputs are 10, 11, 12, 13, 14, 15, 16, 17; the corresponding maximum profits are 50, 61, 74, 89, 106, 125, 146 and 169.
36. The elasticities are approximately 0.667, 0.818, 1, 1.222, 1.5, 1.857, 2.333 and 3. The first two cases are inelastic, the third is unit elastic and the remaining cases are elastic.

Exercises 37–46: rates, tangents and local change.

37. $A'(t) = 6/\sqrt{t}$; $A'(9) = 2$ index points per year.
38. $S'(t) = 6t^2 - 30t + 36$. Thus $S'(2) = 0$ and $S'(5) = 36$ million ringgit per year; spending is locally stationary at $t = 2$ and increasing at $t = 5$.
39. $R(520) - R(500) = 580$, so the average rate is 29 monetary units per unit. Since $R'(q) = 80 - 0.1q$, $R'(500) = 30$.
40. $G(4) = 50 + 10 \ln 5$ and $G'(4) = 2$. The tangent is $y = 50 + 10 \ln 5 + 2(x - 4)$, giving $G(4.2) \approx 50 + 10 \ln 5 + 0.4 \approx 66.49$.
41. The point is $(3, 4)$ and the tangent slope is 2: $y - 4 = 2(x - 3)$. The normal is $y - 4 = -\frac{1}{2}(x - 3)$.
42. $dY/dt = 100 - 4t$ and $dC/dt = 0.75(100 - 4t)$. At $t = 10$ the rates are 60 and 45 monetary units per year, respectively.
43. $P'(t) = 3000e^{0.03t}$, so $P'(5) \approx 3486$ people per year.
44. $N'(t) = -100e^{-0.2t}$, so $N'(4) \approx -44.93$ non-responses per time unit.
45. $B'(x) = 600/(x + 10)^2$, so $B'(20) = 2$ benefit units per additional unit of programme scale.
46. $C'(100) = 21$, while $C(101) - C(100) = 21.03$. The derivative is the close local approximation.

Exercises 47–56: marginal economics and social programmes.

47. $C'(q) = 30 + 0.1q$, so $C'(100) = 40$ monetary units per additional unit.
48. $R'(q) = 120 - 0.4q$, so $R'(200) = 40$ monetary units per additional unit.
49. Profit is $-1000 + 80q - 0.15q^2$. It is maximised at $q = 800/3 \approx 266.67$, with maximum profit ≈ 9666.67 monetary units.
50. $T'(r) = 500 - 40r$, so the maximum occurs at $r = 12.5$, with $T(12.5) = 3125$.
51. Net benefit is $-3x^2 + 120x - 300$. It is maximised at $x = 20$, with maximum net benefit 900.
52. $C'(200) = 20$, while $C(201) - C(200) = 20.02$.
53. $\bar{C}(q) = 1000/q + 10 + 0.04q$. The minimum occurs at $q = 50\sqrt{10} \approx 158.11$, where minimum average cost and marginal cost are both approximately 22.65.
54. $Q'(L) = 50 - L$, so the marginal product at $L = 20$ is 30 units. Total output is maximised at $L = 50$, giving $Q = 1250$.

55. $C'(Y) = 0.8 - 0.00002Y$, so $C'(20\,000) = 0.4$.
56. $A'(n) = 8e^{-0.1n}$ and $A'(10) = 8/e \approx 2.94$. The positive but declining marginal benefit shows diminishing returns.

Exercises 57–66: optimisation and shape.

57. The absolute maximum is 300 at $x = 10$; the absolute minimum is 100 at $x = 0$.
58. Net benefit is $-x^2 + 80x - 500$. Its maximum is 1100 at $x = 40$. Endpoint values are -500 at $x = 0$ and 700 at $x = 60$.
59. The absolute minimum is 500 at $s = 20$. Endpoint comparison gives the absolute maximum 725 at $s = 35$.
60. Stationary points are $(1, 6)$, a local maximum, and $(3, 2)$, a local minimum. The function increases on $(-\infty, 1) \cup (3, \infty)$ and decreases on $(1, 3)$. The point of inflexion is $(2, 4)$.
61. The maximum is 145 at $q = 15$; the minimum is -80 at $q = 0$.
62. Solve $12e^{-0.3t} = 2$. The rate reaches 2 at $t = \ln 6/0.3 \approx 5.97$ years and is below 2 afterwards.
63. Revenue is $R(p) = 800p - 4p^2$. It is maximised at $p = 100$, where $q = 400$ and $R = 40\,000$.
64. Profit is $80q - 0.25q^2 - 1000$. It is maximised at $q = 160$, giving price 60 and profit 5400.
65. The absolute minimum is 40 at $x = 6$; the absolute maximum is 76 at $x = 0$.
66. The continuous optimum is $q = 12.5$. Both $q = 12$ and $q = 13$ give the greatest integer-output profit, 212.

Exercises 67–76: elasticity and response.

67. $E(p) = p/(200 - p)$. Hence $E(50) = 1/3$ is inelastic, $E(150) = 3$ is elastic and unit elasticity occurs at $p = 100$.
68. $E(p) = 0.01p$. Demand is inelastic at $p = 60$, elastic at $p = 140$ and unit elastic at $p = 100$.
69. Since $q'(p) = -750p^{-2.5}$, $E(p) = -pq'(p)/q(p) = 1.5$ at every price. Demand is always elastic.
70. Revenue is $1000pe^{-0.02p}$ and is maximised at $p = 50$. Then $q = 1000/e \approx 367.88$ and $R = 50\,000/e \approx 18\,393.97$.
71. Revenue is $500p - 5p^2$, maximised at $p = 50$. Then $q = 250$ and $R = 12\,500$.

72. $E(p) = p/(p + 10)$, so $E(40) = 0.8$ and demand is inelastic. The elasticity approaches 1 but never reaches it at a finite non-negative price.
73. $N'(f)/N(f) = -0.005$ per ringgit. Locally, a one-ringgit fee increase is associated with an approximate 0.5% decrease in enrolment.
- 74.
- $$E(a) = \frac{40a}{(a + 1)[100 + 40 \ln(a + 1)]},$$
- so $E(9) \approx 0.187$.
75. $\varepsilon_T = (100 - 2r)/(100 - r)$. Thus $\varepsilon_T(20) = 0.75$ and $\varepsilon_T(60) = -0.5$. The negative value means a higher rate locally reduces revenue.
76. The slopes are -2 and -20 , but the quantities are 60 and 600. Both elasticities equal $2/3$, because elasticity measures a proportional rather than absolute response.

Exercises 77–86: approximation and derivative estimates.

77. The linearisation is $L(t) = 50 + 20 \ln 5 + 4(t - 4)$. Thus $G(4.1) \approx 82.5888$, while the model gives approximately 82.5848.
78. $C(100) = 3500$ and $C'(100) = 30$, so $C(102) \approx 3560$. The exact value is 3560.2.
79. Backward estimate = 8, forward estimate = 7 and centred estimate = 7.5 index points per unit time.
80. Backward estimate = 29.95, forward estimate = 30.05 and centred estimate = 30 monetary units per additional unit.
81. $U(5) = 8e^{-0.5}$ and $U'(5) = -0.8e^{-0.5}$. Linearisation gives $U(5.2) \approx 4.7552$; the model value is approximately 4.7562.
82. Since $G''(t) = -20/(t + 1)^2 < 0$, the graph is concave down and its tangent line locally overestimates the function.
83. Backward estimate = 4, forward estimate = -1 and centred estimate = 1.5 units per time. Small observational changes can therefore produce quite different local estimates.
84. $A'(t) = 5 - 0.4t$, so $A'(10) = 1$ approval-index point per year.
85. $P'(5) = 1$ percentage point per year and $P(5) = 37.5\%$. The relative rate is $1/37.5 \approx 2.67\%$ per year.
86. The observed estimate is 0.4 index point per year below the fitted slope. Possible explanations include measurement noise, rounding, model misspecification or a changing local trend.

Exercises 87–100: model evaluation and responsible interpretation.

87. A positive fitted derivative describes conditional association in the model. Confounding, reverse causation, omitted constraints and competing objectives prevent the two stated conclusions.
88. A zero derivative identifies a stationary point only. The derivative may keep the same sign, as for $f(x) = x^3$ at 0, so a sign or second-derivative test is required.
89. The objective may omit distribution, rights, accessibility, externalities, risk, non-monetary outcomes or political constraints.
90. A local maximum exceeds nearby values, while an absolute maximum exceeds every feasible value. Endpoints can exceed all interior stationary values on a closed interval.
91. The domain represents feasibility. An algebraic stationary point outside it cannot describe an attainable decision.
92. Examples: marginal cost has units RM per unit of output, while an approval derivative may have units percentage points per year.
93. The derivative is useful for a smooth function and a small change. Use the exact difference when decisions are discrete, curvature is substantial or the step is not small.
94. Elasticity may change because preferences, income, available substitutes, market composition, policy and the fitted price range differ across periods or groups.
95. A reform can create a structural break in the level, slope or other parameters. Separate equations allow each policy regime to have its own derivative.
96. A jump has no single tangent or finite two-sided derivative. Left- and right-hand changes can describe behaviour on the two sides and the size of the discrete jump should be reported separately.
97. A negative second derivative means the first derivative is falling. The quantity can still increase while $f' > 0$, but at a decreasing rate.
98. Rescaling the input changes its numerical unit and therefore the derivative by the reciprocal scale factor. Converting RM to thousands of RM multiplies a derivative per RM by 1000 when it is expressed per thousand RM.
99. A fitted marginal effect can absorb the influence of omitted variables correlated with the predictor, so it may not represent a controlled or causal change.
100. One suitable checklist is: define variables, units and domain; identify the relevant function and point; differentiate correctly; check signs, units, constraints and numerical reasonableness; interpret the local result within the model; then communicate assumptions, uncertainty, limitations and conditions for revision.

C.12 Chapter 12: Integration

Return to the corresponding questions in Chapter 12.

1. (a) $\ln|4x-5|+C$; (b) $4a^2e^x+C$; (c) $4x+\frac{5}{2}\ln|x|+C$; (d) $\frac{16}{3}e^{3x/4}+C$; (e) $x^{-5}-x^{-3}+C$; (f) $x^5/5-6x^7/7+x^9+C$; (g) $e^x/(2b^2)+C$; (h) $6x^7/7+12x^{5/3}/5+8x+C$.
2. (a) $(x^3-2)^4/4+C$; (b) $(2+\ln\beta)^4/4+C$.
3. (a) $\frac{2}{15}(x^3+1)^{5/2}-\frac{2}{9}(x^3+1)^{3/2}+C$; (b) $\frac{2}{5}(x^2+1)^{5/2}-\frac{2}{3}(x^2+1)^{3/2}+C$; (c) $\frac{3}{2}(x^2-5)^{2/3}+C$; (d) $\frac{7}{18}(t^3+3t^2+10)^{6/7}+C$.
4. (a) 0; (b) -5; (c) 20; (d) 15; (e) 35.
5. (a) $11/6$; (b) $-8\sqrt{3}/5$; (c) 7; (d) $(e^2-3)/2$; (e) $1383/5$; (f) $2-1/e$; (g) $\frac{1}{4}\ln 17$; (h) $1+1/(2e^2)-1/(2e^4)$.
6. $\int 2x/\sqrt{2x-7} \, dx = \frac{1}{3}(2x-7)^{3/2}+7\sqrt{2x-7}+C$; the definite integral is $68/3$.
7. $7/3$ square units.
8. (a) $9/2$ square units; (b) $125/6$ square units.
9. (a) $9/8$ square units; (b) $1/24$ square unit; (c) $16/3$ square units.
10. $\int_{300}^{900} (500-x/3) \, dx = \text{RM } 180\,000$.
11. (a) $T = \sqrt{\ln 55} \approx 2.002$, or about 2 years. (b) Profit = $(27/11 - \ln 55/22)$ thousand RM $\approx \text{RM } 2272.39$.
12. $20(e^{3/2}-e) \approx \text{RM } 35.27$.
13. (a) RM 30 per dictionary; (b) RM 25 000; (c) part (a) divides one total cost by the number of units, whereas part (b) averages the total-cost function over all output levels in the interval.
14. (a) $q_e = 20$, $p_e = 60$; (b) consumer surplus = 600, producer surplus = 400; (c) total surplus = 1000 monetary units.
15. $L(0) = 0$ and $L(1) = 1$. $G = 1 - 2 \int_0^1 x^3 dx = 1 - 1/2 = 1/2$.
16. (a) $C(q) = 900 + 24q + 0.03q^2$; (b) $C(200) = \text{RM } 6900$.
17. (a) $e^{x^3} + C$; (b) $\frac{1}{2}\ln(x^2+4) + C$.
18. $15/4$.
19. (a) $\left(\frac{x}{2} - \frac{1}{4}\right)e^{2x} + C$; (b) 1.

20. Total requests = 504; average rate = 84 requests per hour.
21. Signed area = $5/2$; total geometric area = $13/2$ square units.
22. $32/3$ square units.
23. The trapezoidal estimate is 222 requests.
24. Present value = $12000(1 - e^{-0.12})/0.04 \approx \text{RM } 33\,923.87$.
25. $L(0) = 0$, $L(1) = 1$ and $G = 1 - 2 \int_0^1 x^4 \, dx = 3/5$. This is greater than the value $1/3$ for $L(x) = x^2$, indicating greater inequality for these illustrative Lorenz curves.
26. (a) $N(t) = 4200 - 3000e^{-0.1t}$. (b) $N(5) = 4200 - 3000e^{-0.5} \approx 2380$ participants. (c) The limit is 4200 participants; the model approaches this long-run total as the registration rate tends to zero.
27. (a) $\ln(3x^2 + 4) + C$; (b) $-e^{-t^2/2} + C$; (c) $x \ln x - x + C$; (d) $e^x(x^2 - 2x + 2) + C$.
28. (a) 20; (b) $1/2$; (c) $\ln 9$.
29. (a) $A'(x) = 3x^2 - 2x$; (b) $A(1) = 0$; (c) $A(x) = x^3 - x^2$; (d) $A(2) = 4$, the net accumulation of the stated rate from $x = 1$ to $x = 2$.
30. (a) The signed integral is $4/3$. (b) The total geometric area is 4 square units. (c) The signed integral subtracts the region below the axis, whereas total area adds its magnitude.
31. (a) The average value is $400/3 \approx 133.33$. (b) $c = 5 - 5\sqrt{3}/3 \approx 2.11$ lies in $[0, 5]$ and satisfies $M(c) = 400/3$.
32. The separate trapezoidal areas are 60, 112.5, 108.75 and 105, giving an estimated total of 386.25 arrivals.
33. (a) The trapezoidal estimate is 189. (b) Simpson's estimate is $572/3 \approx 190.67$. (c) The methods use different interpolating shapes between the observed rates; neither need be exact for the unknown underlying curve.
34. (a) Total inflow = 4640 people and total outflow = 2720 people. (b) Net change = 1920 people. (c) The population after eight years is 11 920.
35. (a) Undiscounted income = $500\,000(e^{0.12} - 1) \approx \text{RM } 63\,748.43$. (b) Present value = $500\,000(1 - e^{-0.12}) \approx \text{RM } 56\,539.78$. (c) Future payments receive discount factors below 1, reflecting their lower value at time 0.
36. (a) $q_e = 2\sqrt{10}$ and $p_e = 60$. (b) Consumer surplus and producer surplus are each $160\sqrt{10}/3$. (c) Total surplus is $320\sqrt{10}/3$. It omits considerations such as externalities, distribution, market power or non-price social effects.
37. (a) $G = (\alpha - 1)/(\alpha + 1)$. (b) For $\alpha = 5/2$, $G = 3/7$. (c) Since $3/7 > 1/3$, this illustrative curve represents greater inequality than $L(x) = x^2$.

- 38.** Any three: the fitted rate may not remain valid outside the observed interval; estimated parameters and data contain uncertainty; integration accumulates the modelled association but does not establish causation; omitted variables or structural changes may alter the rate; and an exact integral of an approximate model is still only a model-based estimate.

Exercises 39–48: grouped skills review.

- 39.** The antiderivatives are

$$\frac{x^3}{3} + C, \frac{x^4}{4} + C, \frac{x^5}{5} + C, \frac{x^6}{6} + C, \frac{x^7}{7} + C, \frac{x^8}{8} + C, \frac{x^9}{9} + C.$$

- 40.** In general, $\int (ax^2 + b) dx = ax^3/3 + bx + C$. Substitute the seven stated (a, b) pairs.

- 41.** Each integral is $\ln |ax + b| + C$ for its corresponding pair.

- 42.** For each stated c ,

$$\int 2x(x^2 + c)^c dx = \frac{(x^2 + c)^{c+1}}{c + 1} + C.$$

- 43.** For each non-zero a ,

$$\int xe^{ax} dx = e^{ax} \left(\frac{x}{a} - \frac{1}{a^2} \right) + C.$$

- 44.** The values are 4, 12, 24, 40, 60 and 84, respectively.

- 45.** The areas are 6, 18, 40, 75, 126 and 196 square units.

- 46.** The average values are 12, 16, 21, 27, 34 and 42.

- 47.** The trapezoidal estimates are 28, 31, 34, 37, 40 and 43.

- 48.** The consumer surpluses are 100, 181.5, 288, 422.5, 588 and 787.5 monetary units.

Exercises 49–58: accumulated social quantities.

- 49.** Total calls $= \int_0^8 (120 + 15t) dt = 1440$; average rate $= 1440/8 = 180$ calls per hour.

- 50.** Total inflow $= 11\,000$, total outflow $= 6500$, net change $= 4500$ people and final population $= 89\,500$.

- 51.**

$$N(t) = 20\,000 + 25\,000(1 - e^{-0.2t}).$$

Thus $N(6) \approx 37\,470$ doses and the long-run total is 45 000 doses.

- 52.** Total interviews $= 312$ and average rate $= 52$ per hour. Since $q'(t) = 12 - 4t$, the rate is greatest at $t = 3$ hours, when it equals 58 interviews per hour.

- 53.** $C(t) = 400 + 30t - 2.5t^2$. Its maximum is $C(6) = 490$ complaints and $C(10) = 450$.

54. The average is $260/3 \approx 86.67$. Solving $M(c) = 260/3$ gives $c = 6 - (2/3)\sqrt{21} \approx 2.945$ within the stated interval.

55. The signed accumulation is 0 million ringgit. The total geometric area is 9 million ringgit: losses of 4.5 before $t = 3$ and gains of 4.5 afterwards cancel in the signed integral.

56.

$$\int_0^6 [B(q) - C(q)] \, dq = \int_0^6 (12 + 6q - q^2) \, dq = 108.$$

Equality occurs at $q = 3 + \sqrt{21} \approx 7.58$.

57. Total completions $= \int_0^4 (2000 + 300\sqrt{t}) \, dt = 9600$ homes; average rate $= 2400$ homes per year.

58. $F(t) = 5000 + 800 \ln(t + 1)$ kilograms, so $F(4) \approx 6287.55$ kilograms.

Exercises 59–68: economics and public finance.

59. $C(q) = 1200 + 30q + 0.02q^2$ and $C(500) = \text{RM } 21\,200$.

60.

$$R(q) = 100q - 0.1q^2, \quad C(q) = 1000 + 20q + 0.05q^2.$$

Hence $\Pi(q) = -1000 + 80q - 0.15q^2$, which is maximised at $q = 800/3 \approx 266.67$, with maximum profit $\approx \text{RM } 9666.67$.

61. The two values, in millions of ringgit, are

$$\text{undiscounted benefit} = 100(e^{0.1} - 1) \approx 10.517,$$

$$\text{present value} = \frac{2}{0.03}(1 - e^{-0.15}) \approx 9.286.$$

62. Present value

$$= \int_3^7 1.5e^{-0.04t} \, dt = 37.5(e^{-0.12} - e^{-0.28}) \approx \text{RM } 4.918 \text{ million.}$$

63. $q_e = 100/3 \approx 33.33$ and $p_e = 160/3 \approx 53.33$. Consumer surplus ≈ 1111.11 , producer surplus ≈ 555.56 and total surplus ≈ 1666.67 monetary units.

64. Before tax, $q = 40$, $p = 60$ and total surplus is 1600. After tax, $q = 30$, buyer price 70 and seller price 50. Consumer surplus and producer surplus are each 450, tax revenue is 600 and deadweight loss is 100.

65. Total benefit is 38.4 million ringgit, total cost is 27 million ringgit and accumulated net benefit is RM 11.4 million.

66. Present cost $\approx \text{RM } 9.610$ million and present benefit $\approx \text{RM } 14.773$ million. Hence NPV $\approx \text{RM } 5.162$ million and the benefit–cost ratio is approximately 1.537.

67. Average cost per unit at 500 is RM 55. The average value of C on $[0, 500]$ is approximately RM 17 916.67. The first divides one total cost by output; the second averages total-cost values over all output levels in the interval.
68. The reserve, in millions of ringgit, is $R(t) = 5 + t - 0.15t^2$. It returns to 5 at $t = 20/3 \approx 6.67$ months and $R(10) = 0$.

Exercises 69–78: inequality and distribution.

69. $L(0) = 0$, $L(1) = 1$ and $G = 1 - 2 \int_0^1 x^{5/2} dx = 1 - 4/7 = 3/7 \approx 0.4286$.
70. $G_A = 1/3$ and $G_B = 3/5$. Population B is represented as more unequal because its Lorenz curve is more bowed and its Gini coefficient is larger.
71. The endpoints are correct, but $L(x) - x = x(1 - x) > 0$ for $0 < x < 1$. The curve lies above equality and is concave, contrary to the usual Lorenz-curve conditions.
72. $G_0 = 1/2$. Since $\int_0^1 L_1(x) dx = 7/24$, $G_1 = 5/12$. The coefficient falls by $1/12 \approx 0.0833$.
73. Bottom 40%: $L(0.4) = 0.16$, or 16% of income. Middle 20%: $L(0.6) - L(0.4) = 0.20$, or 20%. Top 40%: $1 - L(0.6) = 0.64$, or 64%.
74. The area is $\int_0^1 (x - x^3) dx = 1/4$. Doubling it gives $G = 1/2$.
75. The pieces agree at $x = 0.8$, where both equal 0.4, and the endpoints are 0 and 1. The area under the curve is $0.16 + 0.14 = 0.30$, so $G = 1 - 2(0.30) = 0.40$.
76. For $L = x^\alpha$, $G = (\alpha - 1)/(\alpha + 1)$. Thus $\alpha = (1 + 0.45)/(1 - 0.45) = 29/11 \approx 2.636$.
77. The bottom half pays $T(0.5) = 0.5^{3/2} \approx 0.3536$, or 35.36%. The top half pays approximately 64.64%.
78. Positive scaling preserves every income share and therefore the Lorenz curve and Gini coefficient. Adding a constant changes incomes proportionally by different amounts and can alter the shares and measured inequality.

Exercises 79–88: numerical integration and recorded data.

79. The trapezoidal estimate is $\frac{1}{2}[50 + 2(70 + 90 + 80) + 60] = 295$ arrivals.
80. Simpson's estimate is $\frac{1}{3}[50 + 4(70) + 2(90) + 4(80) + 60] = 890/3 \approx 296.67$ arrivals, about 1.67 above the trapezoidal estimate.
81. The separate trapezia give $39 + 45 + 44 = 128$ accumulated units.
82. The exact integral is 104. The trapezoidal estimate is 112, an overestimate of 8. Simpson's estimate is 104, exact here because the integrand is quadratic.
83. The successive contributions are 120, 160, 170, 140 and 100, for a total of 690. The cumulative estimate first exceeds 500 at the recorded time $t = 4$.

84. The estimate is $10 + 50 + 25 = 85$ units. Negative endpoint values reduce the signed trapezoidal contribution; they are not converted to positive values.
85. The estimate is $\frac{1}{2}[800 + 2(1000 + 900 + 700) + 600] = 3300$ people over four years.
86. $\frac{1}{2}[10 + 2x + 30] = 50$, so $x = 30$.
87. Secant lines lie above a convex graph, so the trapezoidal rule overestimates. Simpson's rule integrates cubic interpolants exactly, and therefore is exact for polynomials of degree at most three.
88. Recalculate every trapezium or Simpson panel containing the revised observation and update the total. Checks should include time spacing and units, transcription and arithmetic, and comparison with the previous estimate or a sensitivity analysis.

Exercises 89–100: model evaluation and responsible interpretation.

89. The unemployment stock is the number unemployed at a time; entries to and exits from unemployment are flows per unit time. Integrating entries minus exits gives the change in the stock.
90. Antidifferentiation determines a family of totals differing by a constant. An initial total, fixed cost, opening stock or another boundary value identifies that constant.
91. The integral is measured in households. The average value retains the rate unit, households per month.
92. Integration accumulates the fitted rate but does not rule out confounding, selection, reverse causation, measurement error or changes that would have occurred without the programme.
93. The functional form may fail outside the data range; parameters may change; capacity or budget constraints may bind; and policy, demographic or economic shocks may create a structural break.
94. At 3%, present value is approximately RM 86 393.93; at 7%, it is approximately RM 71 916.39. A higher discount rate gives less present weight to later benefits.
95. Surplus may omit externalities, distributional consequences, non-market effects, imperfect information, market power and errors in the estimated demand or supply curves.
96. The Gini coefficient compresses a whole distribution into one number and is scale-invariant. Different Lorenz curves can produce the same area while median income, poverty and the positions of gains or losses differ.
97. A signed integral measures net change, allowing positive and negative effects to cancel. Total geometric area measures the total magnitude of movement or burden regardless of sign.

98. Consider whether a usable formula exists, whether observations are equally spaced and sufficiently numerous, and the required accuracy relative to curvature, data quality and computational effort.
99. Safeguards include transparent data and assumptions, sensitivity and subgroup analysis, independent or human review, appeal and correction procedures, privacy protection and monitoring after allocation.
100. One suitable checklist is: define variables and units; distinguish stocks from flows; specify the rate model and interval; integrate or choose a justified numerical rule; check constants, bounds, units, data and sensitivity; then communicate the result, uncertainty, limitations and conditions for revision.

C.13 Chapter 13: Differential Equations

Return to the corresponding questions in Chapter 13.

1. (a) $\frac{dy}{\sqrt{y+1}} = \frac{x}{x^2+1} dx$; (b) $\frac{dy}{y+4} = \frac{dx}{x-5}$.
2. (a) $2\sqrt{y+1} = \frac{1}{2} \ln(x^2+1) + C$; (b) $\ln|y+4| = \ln|x-5| + C$, equivalently $y+4 = C(x-5)$.
3. (a) $y + 2 \ln|y-2| = x$; (b) $y^3 = x^2 + x + 6$.
4. Dividing by xy gives $4(1 + 3/x) dx + (1 - 3/y) dy = 0$. Hence

$$y + 4x + 12 \ln|x| - 3 \ln|y| = C.$$

5. (a) $P(t) = 200\,000e^{0.03t}$; (b) $P(10) \approx 269\,972$; (c) doubling time $= \ln 2/0.03 \approx 23.10$ years.
6. (a) $A(t) = 100 - 80e^{-0.4t}$; (b) $A(5) \approx 89.17\%$; (c) $t = \ln 4/0.4 \approx 3.47$ years.
7. $U(t) = 4 + 6e^{-0.5t}$. The rate reaches 5% when $t = 2 \ln 6 \approx 3.58$ years.
8. Differentiation gives

$$y' = \frac{24e^{-2t}}{(1 + 3e^{-2t})^2} = 2y \left(1 - \frac{y}{4}\right),$$

$$\text{and } y(0) = 4/(1 + 3) = 1.$$

9. $\tan^{-1} y = x^2/2$, so $y = \tan(x^2/2)$. The largest interval containing 0 is $(-\sqrt{\pi}, \sqrt{\pi})$.
10. The equilibria are $y = 0$ and $y = 6$. For $0 < y < 6$, $y' > 0$, while for $y > 6$, $y' < 0$. Thus, 0 is unstable and 6 is stable on the non-negative state space.
11. (a) $S(t) = 75/(1 + 4e^{-0.4t})$; (b) $S(5) \approx 48.66\%$; (c) $t = \ln 16/0.4 \approx 6.93$ years.
12. (a) $k = \ln(1.4)/4 \approx 0.08412$ per year; (b) $N(10) = 1200e^{10k} \approx 2783$ users; (c) doubling time $= \ln 2/k \approx 8.24$ years.

13. $U(1) \approx 7$, $U(2) \approx 5.5$ and $U(3) \approx 4.75$. The exact value is $4 + 6e^{-1.5} \approx 5.34$, so this Euler approximation is about 0.59 percentage points too low.
14. (a) If P is the number of participants, $dP/dt = 50 - 0.1P$; the equilibrium is $P = 500$. (b) One suitable form is $dN/dt = rN(1 - N/K)$ with $r > 0$; the equilibria are $N = 0$ and $N = K$.
15. (a) First order, autonomous and separable; equilibrium $y = 12$. (b) First order, non-autonomous and separable; $y = \pm 1$ are constant solutions. (c) Second order and autonomous, but not a first-order separable equation. (d) First order, autonomous and separable; equilibria $y = 0, 5$.
16. (a) $P' = 10e^{-0.2t} = 0.2(80 - P)$ and $P(0) = 30$. (b) $P(5) = 80 - 50e^{-1} \approx 61.61$. (c) $t = \ln 5/0.2 \approx 8.05$ time units. (d) The stable equilibrium is 80; 0.2 is the adjustment rate per unit of time, so the remaining gap is multiplied by $e^{-0.2t}$.
17. (a) $y = e^{x^2} - 3$. (b) $y = \sqrt{t^2 + 2t + 4}$. (c) $y = -1 + \sqrt{2e^x + 2}$.
18. (a) $k = \ln(29/40)/3 \approx -0.10719$ per month. (b) $R(t) = 40\,000e^{-0.10719t}$. (c) $R(12) = 40\,000(29/40)^4 \approx 11\,051$ readers. (d) $T_{1/2} = \ln(0.5)/k \approx 6.47$ months. (e) For example, the proportional decline may not remain constant, readership cannot fall indefinitely in the same smooth pattern, or an external event may change interest.
19. (a) $I' = -0.12I$, $I(0) = 100$, so $I(t) = 100e^{-0.12t}$. (b) $I(5) \approx 54.88$. (c) $e^{-0.12} - 1 \approx -0.11308$, or a decrease of about 11.31%. (d) The continuous rate acts throughout the year; the equivalent annual multiplier is $e^{-0.12}$ rather than $1 - 0.12$.
20. (a) $A(t) = 68 - 24e^{-0.25t}$ and $A(4) \approx 59.17$. (b) $24e^{-0.25t} \leq 2$, so $t \geq 4 \ln 12 \approx 9.94$. (c) $A' > 0$ below 68 and $A' < 0$ above 68, so trajectories move towards the stable equilibrium.
21. (a) $C' = 120 - 0.15C$ and the equilibrium is $C = 800$ cases. (b) $C(t) = 800 - 300e^{-0.15t}$. (c) $C(12) \approx 750.41$ cases. (d) $t = \ln 3/0.15 \approx 7.32$ months. (e) At equilibrium, the inflow and outflow rates are equal: $120 = 0.15(800)$; both flows remain positive.
22. (a) The equilibria are $y = 2$ and $y = 6$; 2 is unstable and 6 is stable. (b) $y' < 0$ for $y < 2$, $y' > 0$ for $2 < y < 6$, and $y' < 0$ for $y > 6$. (c) Starting at 4, the solution increases towards 6. (d) Starting between 0 and 2, the model directs the index downwards and eventually outside its meaningful non-negative state space.
23. (a) $M(t) = 90/(1 + 4e^{-0.3t})$. (b) $M(8) \approx 66.04\%$. (c) $t = \ln 4/0.3 \approx 4.62$ years. (d) The maximum rate occurs at $M = 45$ and is $0.3(90)/4 = 6.75$ percentage points per year.
24. (a) $P(t) = 100/(1 + 4e^{-rt})$, and $P(4) = 50$ gives $r = \ln 4/4 \approx 0.34657$ per time unit. (b) $P(8) = 80$. (c) $t = \ln 12/r \approx 7.17$ time units. (d) A different assumed saturation level changes the remaining capacity, the estimated value of r and all later predictions.

25. (a) $E'(0) = 4$ and $L'(0) = 3.6$ users per time unit. (b) $E(t) = 10e^{0.4t}$ and $L(t) = 100/(1 + 9e^{-0.4t})$, so $E(5) \approx 73.89$ and $L(5) \approx 45.09$. (c) $t = \ln 10/0.4 \approx 5.76$. (d) No; the logistic model approaches 100 asymptotically. (e) Logistic growth includes the capacity factor $1 - L/100$, which reduces the effective growth rate as participation rises.
26. (a) $Y(t) = 75 - 40e^{-kt}$ and $Y(3) = 55$ gives $k = \ln 2/3 \approx 0.23105$ per year. (b) The gap from 75 halves every three years. (c) $Y(6) = 65$. (d) For example, the long-run level 75 and adjustment parameter k must remain constant over the forecast period.
27. (a) $M' = 200 - 0.05M$; equilibrium $M = 4000$ and 0.05 has units per month. (b) $I' = 0.30(125 - I)$; equilibrium $I = 125$ and 0.30 has units per year. (c) One suitable form is $A' = rA(1 - A/N)$; equilibria $A = 0, N$ and r has units per unit of time. (d) $T' = 3 - kT$; equilibrium $T = 3/k$, with k measured per year and the constant contribution measured in index points per year.
28. Euler's method with $h = 0.5$ gives $U(0.5) = 8.5$, $U(1) = 7.375$, $U(1.5) = 6.53125$ and $U(2) = 5.89844$. The exact value is $4 + 6e^{-1} \approx 6.20728$, so the error is about -0.309 . With $h = 1$, $U(2) = 5.5$, whose error of about -0.707 is larger.
29. Euler's method gives $P(1) = 26.4$, $P(2) = 34.17216$ and $P(3) \approx 43.17008$. The exact value is $100/(1 + 4e^{-1.2}) \approx 45.35606$, so the Euler estimate is about 2.19 percentage points too low.
30. (a) Any three suitable points: the proportional growth rate is assumed constant; no saturation or resource constraint is included; the first two years may be unrepresentative; parameters may change; and extrapolation over twenty years is highly uncertain. (b) Slowing growth towards a ceiling suggests a logistic model; movement towards a stable target from either side suggests an adjustment model. (c) A fitted trajectory describes association over time and may omit confounding influences; two observations cannot establish a causal mechanism. (d) Re-estimate the parameters, test for a structural break and, if necessary, replace the equation with a piecewise or otherwise revised model reflecting the new policy environment.

Exercises 31–40: grouped skills review.

31. In each case $P(t) = P_0e^{rt}$. The values $P(5)$ are approximately

1105.17, 1465.68, 1889.80, 2386.92, 2967.70, 3644.24, 4430.26.

32. The half-lives are approximately 6.93, 6.30, 5.78, 5.33, 4.95, 4.62 and 4.33 time units.

33. In each case $y(t) = L + (y_0 - L)e^{-kt}$. The values $y(3)$ are approximately

38.05, 42.78, 47.61, 52.54, 57.55, 62.64, 67.79.

34. In each case $P(t) = 100/(1 + 4e^{-rt})$. The values $P(5)$ are approximately
40.46, 42.89, 45.36, 47.84, 50.34, 52.84, 55.32.
35. The equilibria are 0 and K . On the non-negative state space, 0 is unstable and K is stable for every stated value of K .
36. The solutions are $y = e^{nx^2/2}$ for the corresponding values $n = 1, 2, 3, 4, 5, 6, 7$.
37. Differentiation gives $y' = -0.3(y_0 - L)e^{-0.3t} = 0.3(L - y)$, while substitution of $t = 0$ gives $y(0) = y_0$ in every case.
38. The rate parameters are approximately 0.3466, 0.2310, 0.1733, 0.1386, 0.1155, 0.0990 and 0.0866 per year.
39. The one-step estimates are 26.00, 29.27, 32.48, 35.63, 38.72, 41.75 and 44.72.
40. Suitable reasons include structural breaks in behaviour, a changed equilibrium or capacity, altered inflow or outflow mechanisms, demographic or economic shocks, measurement changes and parameters that vary over time.

Exercises 41–50: population and demographic change.

41. $P(t) = 50\,000e^{0.025t}$; $P(10) \approx 64\,201$; doubling time $= \ln 2/0.025 \approx 27.73$ years.
42. (a) $P' = 1200 - 0.04P$ and the equilibrium is 30 000; (b) $P(t) = 30\,000 - 10\,000e^{-0.04t}$; (c) $P(5) \approx 21\,813$.
43. $N(t) = 80\,000e^{-0.06t}$; $N(8) \approx 49\,503$ and the half-life is approximately 11.55 years.
44. Solve $100\,000e^{0.02t} = 130\,000e^{-0.01t}$. Thus $t = \ln(1.3)/0.03 \approx 8.75$ years.
45. $I(t) = 100e^{0.05t}$, so $I(1) \approx 105.13$. The equivalent discrete annual growth rate is $e^{0.05} - 1 \approx 5.127\%$.
46. $k = \ln(1.25)/4 \approx 0.05579$ per year; $N(10) \approx 17\,469$ and doubling time ≈ 12.43 years.
47. The equilibrium is 50 000 and $P(t) = 50\,000 - 30\,000e^{-0.01t}$, giving $P(10) \approx 22\,855$.
48. $N(t) = 12\,000e^{-0.08t}$; $N(5) \approx 8044$ and $t = \ln(12\,000/8000)/0.08 \approx 5.07$ years.
49. $S(t) = 65 - 20e^{-0.2t}$; $S(5) \approx 57.64$ and $S = 60$ at $t = \ln 4/0.2 \approx 6.93$ years.
50. $L(10) = 60\,000$, whereas $E(10) = 50\,000e^{0.2} \approx 61\,070$. The exponential model adds a constant proportion, so its annual additions increase over time.

Exercises 51–60: public services and administrative systems.

51. Equilibrium 2000 cases; $B(t) = 2000 - 1000e^{-0.1t}$ and $B(6) \approx 1451.19$.
52. $A(t) = 500 - 300e^{-0.3t}$ and $A(4) \approx 409.64$. The index tends towards the stable long-run level 500.
53. $U(t) = 5 + 4e^{-0.25t}$; $U = 6$ when $t = \ln 4/0.25 \approx 5.55$ periods.
54. $A(t) = 70 - 30e^{-0.18t}$; $A(6) \approx 59.81$. The remaining gap is at most five when $t \geq \ln 6/0.18 \approx 9.95$.
55. Equilibrium 600; $C(t) = 600 + 200e^{-0.15t}$ and $C(12) \approx 633.06$. The stock falls towards 600.
56. Equilibrium 200; $V(t) = 200 + 100e^{-0.2t}$ and $V(5) \approx 236.79$.
57. Equilibrium 30; $T(t) = 30 - 10e^{-0.1t}$ and $T(10) \approx 26.32$. The constant term adds three points per period, while $0.1T$ represents proportional loss.
58. The equilibrium is 3000 and

$$W(t) = 3000 - 1000e^{-0.05t}.$$

The list reaches 2500 at $t = \ln 2/0.05 \approx 13.86$ months.

59. $B(t) = 120 - 40e^{-0.4t}$; $B(3) \approx 107.95$. The gap half-life is $\ln 2/0.4 \approx 1.73$ periods.
60. After four periods, Model A gives approximately 73.04 and Model B 91.88. The larger rate parameter closes the target gap faster.

Exercises 61–70: political and economic adjustment.

61. $I(t) = 2 + 6e^{-0.3t}$; the model reaches 3% at $t = \ln 6/0.3 \approx 5.97$ periods.
62. $R(t) = 100 - 30e^{-0.15t}$; $R(5) \approx 85.83$ and $R = 90$ at $t = \ln 3/0.15 \approx 7.32$ periods.
63. $D(t) = 100e^{0.04t} - 50$, so $D(10) \approx 99.18$ million. Setting $D' = 0$ gives $D = -50$, outside the non-negative state space.
64. $C(t) = 60 - 15e^{-0.25t}$ and $C(4) \approx 54.48$. The derivative is positive below 60 and negative above 60.
65. Equilibria 20 and 80; 20 is unstable and 80 stable. Movement is downward below 20, upward between 20 and 80, and downward above 80. A start between 0 and 20 is driven towards and eventually below zero, outside the stated context.
66. $G(t) = 40 + 15e^{-0.1t}$ and $G(5) \approx 49.10$; the model predicts a decline towards 40.
67. $Y(3) = 100 - 40e^{-0.6} \approx 78.05$. After the target changes, $Y(6) = 80 + [Y(3) - 80]e^{-0.6} \approx 78.93$.

68. $C(t) = 95 - 30e^{-0.4t}$; $C = 90$ at $t = \ln 6/0.4 \approx 4.48$ periods.
69. Equilibria 40 and 70; 40 is unstable and 70 stable. Starting at 50, the solution increases towards 70.
70. $H = 500$ is stable; $S = 0$ is unstable and $S = 100$ stable on $[0, 100]$; $R = 0$ is stable; and $T = -200$ is the only equilibrium, so non-negative values move away from it and increase.

Exercises 71–80: diffusion and capacity constraints.

71. $A(t) = 100/(1 + 9e^{-0.5t})$; $A(5) \approx 57.51$ and $A = 50$ at $t = \ln 9/0.5 \approx 4.39$.
72. $P(t) = 80/(1 + 3e^{-0.3t})$; $P(6) \approx 53.48$. Maximum growth is $0.3(80)/4 = 6$ units per period at $P = 40$.
73. $p(t) = 1/(1 + 9e^{-0.8t})$; $p(4) \approx 0.7316$, or 73.16%.
74. $P(t) = 100/(1 + 4e^{-rt})$ and $P(3) = 50$ gives $r = \ln 4/3 \approx 0.4621$. Hence $P(6) = 80$.
75. $E(10) = 10e^3 \approx 200.86$, while $L(10) = 100/(1 + 9e^{-3}) \approx 69.06$. Capacity slows logistic growth and keeps it below 100.
76. The time from 20 to 50 is $\ln 4/0.4$, and the time from 50 to 80 is the same. Total time $= 2 \ln 4/0.4 \approx 6.93$ periods.
77. Maximum growth occurs at membership 1000 and equals $0.6(2000)/4 = 300$ members per period.
78. A changed capacity alters the ratio of occupied to remaining capacity, the estimated rate needed to match observations, the inflexion level $K/2$ and the asymptotic prediction K .
79. $S(t) = 70/(1 + 4e^{-0.25t})$ and $S(8) \approx 45.41$. The saturation level 70 is the modelled upper long-run share.
80. Differentiation gives $P' = rK(Ae^{-rt})/(1 + Ae^{-rt})^2 = rP(1 - P/K)$. From $P(0) = K/(1 + A)$, $A = (K - P(0))/P(0)$.

Exercises 81–90: numerical approximation and data checking.

81. Euler gives $y(1) = 22$, $y(2) = 30.4$ and $y(3) = 36.28$. The exact value is $50 - 40e^{-0.9} \approx 33.74$, so the error is about 2.54.
82. With $h = 0.5$, $y(3) \approx 34.91$, whose error of about 1.18 is smaller than the $h = 1$ error.
83. Euler gives $p(1) = 14.5$, $p(2) \approx 20.70$ and $p(3) \approx 28.91$. The exact value is $100/(1 + 9e^{-1.5}) \approx 33.24$.

84. A smaller step follows curvature more closely but requires more iterations. It reduces discretisation error, not error in the assumed mechanism, parameters or data.
85. Euler predicts $C(1) = 500 + [100 - 0.05(500)] = 575$. The residual is $560 - 575 = -15$ cases; the model overpredicts, possibly because inflow was lower or resolution faster than assumed.
86. $k = \ln(1.21)/2 = \ln(1.1) \approx 0.09531$ per period. The prediction is $N(4) = 1464.1$, giving residual $1400 - 1464.1 = -64.1$.
87. $Y(t) = 100 - 60e^{-kt}$ and $Y(2) = 70$ gives $k = \ln 2/2 \approx 0.3466$. The target gap halves every two periods.
88. With $h = 1$, $y(1) = 2$, overshooting the equilibrium 1. With $h = 0.25$, $y(0.25) = 0.5$, remaining within the interval towards the equilibrium.
89. Local truncation error is introduced in one numerical step; accumulated error results from repeated steps; model error comes from the equation or parameters. A smaller step reduces the first two, but not model error.
90. Suitable recursions are

$$P_{n+1} = P_n + hrP_n, \quad Y_{n+1} = Y_n + hk(L - Y_n), \quad S_{n+1} = S_n + hrS_n(1 - S_n/K).$$

Time units must match the rate parameters; values should also be checked for non-negativity and, for logistic models, the bound K .

Exercises 91–100: model evaluation and responsible interpretation.

91. A rising fitted trajectory shows temporal association, not causation. Confounders may include economic conditions, publicity, selection into the programme, demographic change, other policies or reverse causation.
92. For $0 \leq t \leq 4$, $A(t) = 70 - 30e^{-0.2t}$, so $A(4) \approx 56.52$. For $t \geq 4$, $A(t) = 85 + [A(4) - 85]e^{-0.3(t-4)}$, giving $A(7) \approx 73.42$.
93. The equilibrium is $C = 800$, where inflow and resolution are both 120 cases per period. The stock is constant because the flows balance, not because both flows are zero.
94. The prediction is $100\,000e^{0.02(50)} \approx 271\,828$. Long-run growth may be limited by resources, migration, changing fertility or mortality, policy, economic shocks or an unstable rate.
95. At $t = 5$, the $K = 100$ model gives approximately 52.84, while the $K = 120$ model gives approximately 56.72. Capacity affects both the remaining growth space and the predicted trajectory.
96. Two observations can estimate a simple rate but reveal little about curvature, capacity, equilibrium or structural change. Several different models can pass through the same two points.

97. Suitable criteria include observed curvature and bounds, a credible causal mechanism, residual patterns, stability across periods, out-of-sample performance, parameter interpretability and known policy breaks.
98. The parameter 0.2 has units per unit time. Equilibria are 0 and 100; the derivative is positive for $0 < P < 100$ and negative above 100. The meaningful percentage state space is $0 \leq P \leq 100$.
99. Safeguards include transparent assumptions and data, subgroup and fairness checks, sensitivity analysis, human review, appeal or correction mechanisms, privacy protection and monitoring after implementation.
100. One suitable checklist is: define variables and units; state the mechanism and assumptions; formulate the rate equation and initial condition; solve or approximate it; check units, bounds, data and residuals; then communicate predictions, uncertainty, limitations and conditions for revision.

C.14 Chapter 14: Financial Mathematics

Return to the corresponding questions in Chapter 14.

1. Interest = $5000(0.04)(3)$ = RM 600; final amount = RM 5600.
2. $8000(1.05)^6 \approx$ RM 10 720.77.
3. $20\,000/(1.06)^4 \approx$ RM 15 841.87.
4. $(1 + 0.048/12)^{12} - 1 \approx 0.04907$, or approximately 4.907%.
5. Accumulated value $\approx$ RM 11 591.43. Total deposits are RM 10 800, so interest earned is approximately RM 791.43.
6. The monthly payment is approximately RM 1332.25. The total of the 120 payments is

$$120R \approx \text{RM } 159\,869.52.$$
7. Value = $90\,000(0.85)^5 \approx$ RM 39 933.48; loss in value $\approx$ RM 50 066.52.
8. (a) $1.06/1.04 - 1 \approx 0.01923$, or approximately 1.923%; (b) $48\,000(1.06/1.04) \approx$ RM 48 923.08 in the original year's purchasing power.
9. (a) Simple-interest amount = RM 15 240.
 (b) Annual-compound amount $\approx$ RM 15 627.12.
 (c) Continuously compounded amount $\approx$ RM 15 719.57.
 (d) Simple interest is earned only on the original principal; annual and continuous compounding credit interest on accumulated interest, with continuous compounding doing so most frequently.

10. (a) Effective annual rate $= (1 + 0.054/12)^{12} - 1 \approx 5.5357\%$. (b) Equivalent nominal quarterly rate $= 4[(1.0553568)^{1/4} - 1] \approx 5.4243\%$.
11. (a) Value $= 20\,000(1.04)(1.055)(0.98)(1.06) \approx \text{RM } 22\,795.43$. (b) Overall return $\approx 13.977\%$. (c) Equivalent annual effective rate $= [(1.04)(1.055)(0.98)(1.06)]^{1/4} - 1 \approx 3.325\%$.
12. Replacement payment $= 5000(1.05) + 7000(1.05)^{-2} \approx \text{RM } 11\,599.21$.
13. Present value $= 4000a_{\overline{3}|0.06} + 8000(1.06)^{-4} \approx \text{RM } 17\,028.80$.
14. (a) End-of-month deposit $\approx \text{RM } 761.83$.
 (b) Beginning-of-month deposit $\approx \text{RM } 759.55$.
 (c) Each beginning-of-month deposit earns interest for one extra month.
15. Present value $= 6000(1.05)^{-4}a_{\overline{4}|0.05} \approx \text{RM } 17\,503.57$.
16. Present value $= 2400/0.048 = \text{RM } 50\,000$.
17. (a) Monthly payment $\approx \text{RM } 1228.05$.
 (b) Balance after 60 payments $\approx \text{RM } 151\,277.79$.
 (c) Payment 61 contains approximately RM 680.75 interest and RM 547.30 principal.
18. For five years: payment $\approx \text{RM } 1933.28$, total payments $\approx \text{RM } 115\,996.81$ and interest $\approx \text{RM } 15\,996.81$. For ten years: payment $\approx \text{RM } 1110.21$, total payments $\approx \text{RM } 133\,224.60$ and interest $\approx \text{RM } 33\,224.60$.
19. Both the retrospective and prospective methods give $B_{24} \approx \text{RM } 101\,377.55$.
20. Required annual deposit $= 100\,000/s_{\overline{8}|0.05} \approx \text{RM } 10\,472.18$.
21. (a) NPV $\approx \text{RM } 8494.76$. (b) The positive NPV supports accepting the project on the stated financial assumptions. (c) Examples include accessibility, environmental effects, distributional consequences or service quality.
22. (a) Project A has NPV $\approx \text{RM } 95.18$ and Project B has NPV $\approx \text{RM } 997.60$. (b) Project B has the greater financial NPV. (c) Social benefits, distribution, risk and non-monetary effects may differ between the projects.
23. Let $y = 1 + r$. The equation $10\,000 = 6000/y + 6000/y^2$ gives $5y^2 - 3y - 3 = 0$. Hence $r = (3 + \sqrt{69})/10 - 1 \approx 13.07\%$.
24. (a) Exact real increase $= 1.07/1.045 - 1 \approx 2.392\%$.
 (b) Real salary $= 60\,000(1.07/1.045) \approx \text{RM } 61\,435.41$.
25. (a) Straight-line depreciation is RM 12 000 per year and the three-year book value is RM 44 000. (b) $d = 1 - (20\,000/80\,000)^{1/5} \approx 24.21\%$. (c) Reducing-balance value after three years $\approx \text{RM } 34\,822.02$.

26. Present value = $25\,000e^{-0.04(7)} \approx \text{RM } 18\,894.59$.
27. Any three: the total of all repayments; total interest; effective annual rate; fees, insurance and penalties; whether the rate is fixed or variable; and the financial risk of remaining in debt for a longer period.

Exercises 28–37: grouped skills review.

28. The simple interest amounts are

A–D: RM 120, RM 202.50, RM 300, RM 412.50;
E–H: RM 540, RM 682.50, RM 840, RM 1012.50.

29. The accumulated values are approximately

A–D: RM 2433.31, RM 3115.45, RM 3828.84, RM 4574.36;
E–H: RM 5352.90, RM 6165.39, RM 7012.76, RM 7895.96.

30. The present values are approximately

A–D: RM 4113.51, RM 4524.86, RM 4936.21, RM 5347.57;
E–H: RM 5758.92, RM 6170.27, RM 6581.62, RM 6992.97.

31. The effective annual rates are approximately

A–D: 3.660%, 4.282%, 4.907%, 5.536%;
E–G: 6.168%, 6.803%, 7.442%.

32. The accumulated values are approximately RM 1233.56, RM 1480.27, RM 1726.98, RM 1973.69, RM 2220.40, RM 2467.11 and RM 2713.82.

33. The monthly payments are approximately

A–D: RM 448.63, RM 493.50, RM 538.36, RM 583.22;
E–G: RM 628.09, RM 672.95, RM 717.81.

34. The balances after payment 12 are approximately

A–D: RM 5179.39, RM 5697.32, RM 6215.26, RM 6733.20;
E–G: RM 7251.14, RM 7769.08, RM 8287.02.

35. The five-year values are approximately RM 11 809.80, RM 12 990.78, RM 14 171.76, RM 15 352.74, RM 16 533.72, RM 17 714.70 and RM 18 895.68.

36. The exact real growth rates are approximately

A–D: 1.942%, 2.427%, 2.913%, 3.398%;
E–G: 3.883%, 4.369%, 4.854%.

37. The NPVs are approximately

A–D: RM 692.05, RM 1028.55, RM 1365.06, RM 1701.57;
E–G: RM 2038.07, RM 2374.58, RM 2711.08.

Exercises 38–48: household saving and borrowing.

- 38.** (a) RM 5600; (b) RM 5624.32; (c) compounding produces RM 24.32 more.
- 39.** (a) RM 7541.12; (b) deposits RM 7200, interest RM 341.12; (c) actual rates, fees, taxes, deposit timing or missed deposits may differ from the model.
- 40.** The beginning-of-month value is approximately RM 7571.29, about RM 30.16 more than the ordinary annuity because every deposit earns interest for one additional month.
- 41.** (a) RM 513.02 per month; (b) RM 510.72 per month; (c) beginning deposits earn interest for one extra month.
- 42.** (a) RM 579.98; (b) total repayments RM 34 799.04 and interest RM 4799.04; (c) possible omitted costs include fees, insurance or penalties.
- 43.** (a) Balance $\approx$ RM 19 064.66; (b) interest $\approx$ RM 95.32 and principal $\approx$ RM 484.66; (c) interest is calculated on a declining balance.
- 44.** (a) Total paid = $500 + 24(220) =$ RM 5780; finance charge RM 780; (b) the payments occur at different times and the financed balance declines, so a time-value calculation is required.
- 45.** Replacement payment = $3000(1.06) + 5000/1.06 \approx$ RM 7896.98.
- 46.** Final value = $10\,000(1.05)(0.97)(1.08) =$ RM 10 999.80, an overall return of approximately 9.998%. Successive rates compound on changing balances.
- 47.** Annual compounding gives approximately RM 12 166.53; continuous compounding gives RM 12 214.03, a difference of approximately RM 47.50.
- 48.** (a) Future cost $\approx$ RM 68 428.45; (b) lump sum today $\approx$ RM 42 932.86; (c) tuition growth, investment returns, taxes and the study period are uncertain.

Exercises 49–59: housing, education and retirement.**49.**

Payment $\approx$ RM 1705.63,
 total repayments $\approx$ RM 409 350.95,
 total interest $\approx$ RM 159 350.95.

- 50.** For 15 years, payment $\approx$ RM 2029.47 and interest $\approx$ RM 115 303.97. For 25 years, payment $\approx$ RM 1520.32 and interest $\approx$ RM 206 097.44. The longer term lowers the payment but greatly increases total interest and time in debt.
- 51.** Balance after 60 payments $\approx$ RM 210 108.05. Payment 61 contains approximately RM 945.49 interest and RM 760.14 principal.
- 52.** (a) Initial cash RM 35 000 and loan RM 270 000; (b) payment $\approx$ RM 1842.08; (c) taxes, insurance, maintenance, fees, income risk and other debt obligations also affect affordability.
- 53.** Payment $\approx$ RM 491.30 and total interest $\approx$ RM 7164.92.
- 54.** Required quarterly deposit $\approx$ RM 2724.92.
- 55.** (a) RM 286 362.59; (b) RM 300 680.72; (c) advantage $\approx$ RM 14 318.13.
- 56.** Required retirement fund $\approx$ RM 333 551.62.
- 57.** Required fund = $3000/0.05 =$ RM 60 000.
- 58.** Value = $5000 + 5000/0.05 =$ RM 105 000.
- 59.** (a) Exact real return = $1.07/1.04 - 1 \approx 2.885\%$; (b) purchasing-power value after five years $\approx$ RM 115 279.53; (c) subtraction ignores the product of the two rates.

Exercises 60–70: public finance and social programmes.

- 60.** Present value $\approx$ RM 16.222 million.
- 61.** Reserve value $\approx$ RM 3.234 million.
- 62.** NPV $\approx$ RM 1.573 million, so the financial criterion supports acceptance on the stated assumptions.
- 63.** Project A has NPV $\approx$ RM 1.343 million; Project B has NPV $\approx$ RM 2.293 million. Project B has the larger financial NPV.
- 64.** Present value = $20/1.04^5 \approx$ RM 16.439 million. This is the current amount financially equivalent to the future payment at the stated rate.
- 65.** Present value = $1\,000\,000(1.05)^{-3}a_{\overline{5}|0.05} \approx$ RM 3.740 million.
- 66.** Benefit = $1800(1.03)(1.04)(1.05) =$ RM 2024.57; cumulative increase $\approx 12.476\%$.

67. Exact real change = $1.05/1.065 - 1 \approx -1.408\%$, so purchasing power falls.
68. Cost after four years = $2(1.03)^4 \approx \text{RM } 2.251$ million. Actual costs may be affected by policy, wages, technology, demand or shocks not represented by a constant growth rate.
69. Subsidised payment $\approx \text{RM } 1109.20$ and total interest $\approx \text{RM } 66\,206.85$; market payment $\approx \text{RM } 1364.50$ and interest $\approx \text{RM } 127\,480.76$. The modelled interest saving is approximately $\text{RM } 61\,273.92$.
70. Let $y = 1 + r$. Solving $12\,000 = 7000/y + 7000/y^2$ gives $r \approx 10.923\%$. It is the discount rate at which NPV is zero.

Exercises 71–80: organisations and project appraisal.

71. Annual depreciation $\text{RM } 10\,000$; book value after three years $\text{RM } 70\,000$.
72. Value after three years $\text{RM } 51\,200$; accumulated depreciation $\text{RM } 48\,800$.
73. $d = 1 - (25\,000/80\,000)^{1/5} \approx 20.755\%$ per year.
74. Straight-line book value after four years is $\text{RM } 48\,000$. The reducing-balance rate is $1 - (16\,000/80\,000)^{1/8} \approx 18.223\%$, giving value approximately $\text{RM } 35\,777.09$ after four years.
75. NPV $\approx \text{RM } 13\,293.57$.
76. Project A has NPV $\approx \text{RM } 9202.37$; Project B has NPV $\approx \text{RM } 15\,603.16$. Project B has the larger NPV.
77. Present value of benefits $\approx \text{RM } 6.233$ million; present value of costs $\approx \text{RM } 5.890$ million; NPV $\approx \text{RM } 0.342$ million and benefit–cost ratio ≈ 1.058 .
78. (a) Payback period = 2.5 years; (b) NPV $\approx -\text{RM } 105.18$; (c) payback ignores discounting and cash flows after recovery of the initial cost.
79. Present cost of buying $\approx \text{RM } 30\,593.46$; present cost of leasing $\approx \text{RM } 31\,185.95$. Buying is lower by approximately $\text{RM } 592.49$ on the stated assumptions.
80. Sustainable annual grant = $0.045(2\,000\,000) = \text{RM } 90\,000$. Paying $\text{RM } 100\,000$ exceeds the modelled earnings and would reduce capital unless returns or other contributions compensate.

Exercises 81–90: inflation, depreciation and rate sensitivity.

81. Exact real growth = $1.06/1.04 - 1 \approx 1.923\%$; subtraction gives 2%.
82. Exact real return = $1.08/1.05 - 1 \approx 2.857\%$.

83. Overall multiplier = $1.12(0.90) = 1.008$, an increase of 0.8%. The decrease is applied to a different starting value.
84. Base-period purchasing power = $50\,000/1.25 = \text{RM } 40\,000$.
85. Value = $150\,000(0.88)^4 \approx \text{RM } 89\,954.30$; total loss $\approx \text{RM } 60\,045.70$.
86. (a) RM 8000 per year; (b) $d = 1 - (12\,000/60\,000)^{1/6} \approx 23.528\%$; (c) value after three years $\approx \text{RM } 26\,832.82$.
87. Required nominal amount = $10\,000e^{0.03(10)} \approx \text{RM } 13\,498.59$.
88. Exact real discount rate = $1.08/1.03 - 1 \approx 4.854\%$.
89. NPV at 5% is approximately RM 10 612.67; at 8% it is approximately RM 5897.94. A higher discount rate reduces the present value of future returns.
90. At 4%, payment $\approx \text{RM } 1012.45$ and total interest $\approx \text{RM } 21\,494.17$. At 6%, payment $\approx \text{RM } 1110.21$ and interest $\approx \text{RM } 33\,224.60$.

Exercises 91–100: evaluating financial claims and decisions.

91. Effective annual rate = $(1 + 0.12/12)^{12} - 1 \approx 12.683\%$. Monthly interest is added to the balance and earns further interest during the year.
92. (a) Interest RM 1500, total repayment RM 11 500 and monthly payment $\approx \text{RM } 319.44$; (b) the borrower does not retain the full original balance for the entire term, so interest measured against the declining balance is higher than the quoted flat rate suggests.
93. Net multiplier = $1.04(0.99) = 1.0296$, so the return is 2.96%, slightly below the 3% subtraction approximation.
94. Two-week rate = 15%; simple annualised rate = $15\%(26) = 390\%$; effective annual rate = $(1.15)^{26} - 1 \approx 3685.68\%$. Repeated compounding produces an extremely large annual equivalent.
95. Relevant considerations include distribution of benefits and costs, accessibility, environmental effects, rights and fairness, service quality, risk and uncertainty, and strategic or legal constraints. A positive financial NPV measures only monetised consequences under the selected assumptions.

96.

$$\text{NPV}(3\%) \approx \text{RM } 53\,020.28,$$

$$\text{NPV}(8\%) \approx -\text{RM } 128\,991.86.$$

A higher social discount rate places less weight on later benefits and can reverse the decision.

97. Purchasing power after four years = $3000/(1.05)^4 \approx \text{RM } 2468.11$ in today's money. Without indexation, the pension recipient bears the inflation risk.

98. Loan A has payment approximately RM 599.42 and total stated cost RM 22 079.05, including the fee. Loan B has payment approximately RM 606.63 and total cost RM 21 838.61. The lower advertised rate is not the cheaper offer once the fee is included.
99. Possible failures include a variable or reset rate, refinancing, late or extra payments, fees and insurance, changes in loan terms, payment holidays, default or early settlement.
100. A fair comparison should align cash-flow timing; convert rates to comparable effective rates; include fees, taxes and penalties; distinguish nominal from real values; assess uncertainty and sensitivity; and consider affordability, distributional effects and other non-financial consequences.

C.15 Chapter 15: Index Numbers

Return to the corresponding questions in Chapter 15.

1. $I = (3.00/2.40)(100) = 125$; the price is 25% above its base-year value.
2. Base-period salary = $100(2950)/118 = \text{RM } 2500$.
3. $I_{2024/2020} = 115(108)/100 = 124.2$.
4. (a) $(110 + 125 + 95 + 120)/4 = 112.5$; (b) $[5(110) + 2(125) + 2(95) + 120]/10 = 111$.
5. $[3(120) + 2(110) + 5x]/10 = 116$, so $x = 116$.
6. $(132.1/126.4 - 1)(100)\% \approx 4.51\%$.
7. Real income = $54\,000/1.35 = \text{RM } 40\,000$ in base-year purchasing power.
8. $100(1.04)(1.06)(1.03) = 113.5472 \approx 113.55$.
9. $L \approx 123.08$, using base-period quantities; $P \approx 122.96$, using current-period quantities; and $F \approx 123.02$.
10. The composite index is

$$\frac{30(125) + 35(118) + 20(140) + 15(110)}{100} = 123.3.$$

The weighted household cost is 23.3% above its base-period level.

11. Any two suitable reasons, such as different spending weights, regional price variation, substitution between products, or a household purchasing goods not represented closely by the national basket.
12. (a) 25%. (b) Reverse index = $10000/125 = 80$. (c) The percentage changes use different starting values as their denominators. (d) $(1.25)(0.80) = 1$, so the final value equals the original.

13. (a) Employment is approximately 240 000, 250 000, 262 000 and 273 000 in 2021–2024. (b) $(109.2/104.8 - 1)100\% \approx 4.20\%$. (c) The index-point change must be divided by the 2023 index: $4.4/104.8$, not by 100.
14. (a) With 2022 as base, the indices are approximately 78.63, 85.47, 92.31, 100 and 107.69. (b) $126/108 = 107.69/92.31 \approx 1.1667$, an increase of about 16.67%. (c) For example, the old base may be outdated, atypical or based on weights that no longer represent current behaviour.
15. (a) $I_{2024/2020} = 110(1.04)(0.98) = 112.112 \approx 112.11$. (b) Approximately 12.11%. (c) $10000/112.112 \approx 89.20$. (d) Base rent = $2400/1.12112 \approx \text{RM } 2140.72$.
16. (a) Composite index = $[35(108) + 30(115) + 20(125) + 15(102)]/100 = 112.6$. (b) Contributions are food 2.8, housing 4.5, transport 5.0 and other 0.3 index points. (c) Transport contributes most. Contributions depend on both the weight and the component's movement away from 100.
17. (a) Household A: 118.75. (b) Household B: 116.50. (c) Household A has the larger modelled increase. (d) The households assign different expenditure weights to the same price movements, so their experienced inflation differs.
18. (a) $(123.9/118 - 1)100\% = 5.00\%$.
(b) $(129.5/123.9 - 1)100\% \approx 4.52\%$.
(c) $(129.5/118 - 1)100\% \approx 9.75\%$.
(d) The second year's increase applies to the already increased price level, so the annual rates compound.
19. (a) Fully indexed benefit = $1800(132.5/125) = \text{RM } 1908$. (b) 6%. (c) A 5% increase gives RM 1890, equivalent to $1890(125/132.5) \approx \text{RM } 1783.02$ in the earlier period's purchasing power. (d) The recipient's spending pattern may differ from the national basket, or the published index may not represent the recipient's region or household type.
20. (a) Nominal wage index = $3450/3000(100) = 115$.
(b) Real-wage index = $115/112(100) \approx 102.68$.
(c) $3450/1.12 \approx \text{RM } 3080.36$.
(d) Nominal growth is 15%, but purchasing power grows by only about 2.68% after allowing for the higher price level.
21. (a) Purchasing-power indices are $10000/125 = 80$ and $10000/120 \approx 83.33$. (b) $(83.33/80 - 1)100\% \approx 4.17\%$. (c) A lower CPI means a unit of money purchases more of the base-period basket.
22. (a) The fixed-base series is 100, 103, 99.91, 105.9046 and 110.1408. (b) The total increase is approximately 10.14%. (c) The largest annual increase is 6%, occurring in 2025. (d) The 3% rise in 2023 is more than reversed by the following 3% fall because the decrease is applied to the higher level: $1.03(0.97) = 0.9991$.

23. (a) The first two fixed-base indices are 102 and $102(1.05) = 107.1$. (b) The third link is $100(117.81/107.1) = 110$. (c) The index increased by 10% between the second and third linked periods.
24. (a) The totals are $\sum p_0q_0 = 10\,800$, $\sum p_1q_0 = 12\,040$, $\sum p_0q_1 = 10\,700$ and $\sum p_1q_1 = 11\,820$. (b) $L \approx 111.48$, $P \approx 110.47$ and $F \approx 110.97$. (c) The current quantities give less weight to items with the larger price increases, so the current-basket Paasche index is slightly lower. (d) Report the Fisher index, approximately 110.97.
25. (a) Value index $= 11\,820/10\,800(100) \approx 109.44$. (b) Total expenditure rises by about 9.44%, compared with the Fisher price increase of about 10.97%. (c) The value index reflects changes in quantities as well as prices.
26. (a) Region A: $4800/1.20 = \text{RM } 4000$; Region B: $4400/1.05 \approx \text{RM } 4190.48$. (b) Region A has the higher nominal income. (c) Region B has the higher cost-adjusted income by approximately RM 190.48. (d) The cost basket may not represent either individual household, and income alone does not measure public services, wealth, inequality or overall welfare.
27. (a) Solving $110w + 130(40) + 100(60 - w) = 117(100)$ gives $w = 50$; the other-items weight is 10. (b) Contributions are food 5 points, housing 12 points and other items 0 points. (c) Its weight affects how strongly other changing components influence the average, even though it contributes no change in this period.
28. (a) National real wage growth $= (108/106 - 1)100\% \approx 1.89\%$. (b) For the lower-income group, $(108/111 - 1)100\% \approx -2.70\%$. (c) The national average can conceal a loss of purchasing power for groups facing different price changes and expenditure weights.
29. (a) $(200/150 - 1)100\% = 33.33\%$. (b) The report treats a rise of 50 index points as a 50% increase; the percentage must be calculated relative to 150. (c) Any four suitable items: the base period; the two comparison periods; the basket or population covered; the weights and formula; geographical coverage; whether figures are revised or seasonally adjusted; and relevant limitations.

Exercises 30–40: grouped skills review.

30. The indices are 105, 107, 109, 111, 113, 115, 117 and 119.
31. The current values are RM 220, RM 277.50, RM 336, RM 395.50, RM 456, RM 517.50 and RM 580.
32. The reverse indices are approximately 90.91, 89.29, 87.72, 86.21, 84.75, 83.33 and 81.97.
33. (a) 81.82, 90.91, 100;
 (b) 81.25, 89.29, 100;
 (c) 80.70, 87.72, 100;
 (d) 80.17, 86.21, 100;

(e) 79.66, 84.75, 100;

(f) 79.17, 83.33, 100;

(g) 78.69, 81.97, 100.

34. The composite indices are 114, 113.8, 113.6, 113.4, 113.2, 113 and 112.8.

35. The inflation rates are approximately 4.17%, 4.96%, 5.74%, 6.50%, 7.26%, 8.00% and 8.73%.

36. The real-income indices are approximately

104.76, 105.66, 106.54, 107.41, 108.26, 109.09, 109.91.

37. The fixed-base indices are 99.96, 101.97, 104.00, 106.05, 108.12, 110.21 and 112.32.

38. The Laspeyres index is 117.5 in every case. As q_{1A} takes the stated values, the other indices are

q_{1A}	80	85	90	95	100	105	110
P	116.00	116.22	116.43	116.63	116.82	117.00	117.17
F	116.75	116.86	116.96	117.06	117.16	117.25	117.34

39. Any four suitable reasons, such as different expenditure weights, product choices, outlets, regions, household sizes, substitution behaviour or eligibility for subsidised prices. Each changes the relevant basket, weights or prices relative to the national index.

40. (a) The new level is 50% above the original base; (b) $(150/145 - 1)100\% \approx 3.45\%$; (c) $10000/150 \approx 66.67$; (d) the 5 points must be measured relative to the starting index 145, not relative to the base value 100.

Exercises 41–50: public budgets and social protection.

41. (a) $\text{RM } 1200(132.5/125) = \text{RM } 1272$; (b) 6%; (c) $\text{RM } 1248(125/132.5) \approx \text{RM } 1177.36$; (d) a real decrease of approximately 1.89%.

42. (a) $\text{RM } 2832$; (b) actual pension $\text{RM } 2688$, shortfall $\text{RM } 144$; (c) $\text{RM } 2688/1.18 \approx \text{RM } 2277.97$ in base-period money.

43. (a) $\text{RM } 8(1.15)(1.08) = \text{RM } 9.936$ million; (b) $\text{RM } 0.436$ million; (c) for example, costs and population affect the budget multiplicatively and service intensity per person is unchanged.

44. (a) $0.70(112) + 0.20(108) + 0.10(125) = 112.5$; (b) $\text{RM } 4.50$; (c) food contributes 8.4 points, labour 1.6 and transport 2.5.

45. (a) The expenditure indices are 100, 110.24, 119.88 and 123.17; (b) an increase of approximately 23.17%; (c) the rise in average cost per case more than offsets the caseload decline.
46. (a) $109/112(100) \approx 97.32$; (b) purchasing power is approximately 2.68% below the base; (c) nominal allowance RM 1635, with base-period purchasing power approximately RM 1459.82.
47. (a) 118.5; (b) health contributes 9 points, education 3.5 and housing 6; (c) health makes the largest contribution.
48. (a) Region A: RM 2160; Region B: RM 1890; (b) the A income is RM 60 below its threshold, while the B income is RM 60 above its threshold; (c) a regional average basket may not represent the household's needs, prices or composition.
49. (a) $147/140 - 1 = 5\%$; (b) $145/140 - 1 \approx 3.57\%$; (c) $145/147 - 1 \approx -1.36\%$.
50. The statement is correct only when 115 is compared directly with the base index 100. From 110 to 115, the increase is $(115/110 - 1)100\% \approx 4.55\%$.

Exercises 51–60: wages and living standards.

51. (a) Nominal wage growth = 10% and inflation = 6%; (b) exact real growth = $1.10/1.06 - 1 \approx 3.77\%$; (c) approximately RM 3320.75 in the earlier period's money.
52. (a) Household A: 117.3; Household B: 113.55; (b) Household A; (c) it assigns more weight to food and transport, whose indices are relatively high.
53. (a) National-CPI real-wage index ≈ 102.86 ; lower-income-CPI index ≈ 97.30 ; (b) these imply a 2.86% gain and a 2.70% loss, respectively; (c) groups face different baskets and price changes.
54. (a) The real-salary series is approximately 100, 101.96, 100.93, 101.75; (b) 2021; (c) nominal growth must be compared with the different price-level increase in each year.
55. (a) Group M: RM 4480; Group W: RM 3776; (b) the gap falls from 20% to approximately 15.71% of Group M's wage; (c) an index measures proportional change, not the underlying wage level.
56. (a) Real-wage index = $108/104(100) \approx 103.85$;
(b) unit-labour-cost index = $108/115(100) \approx 93.91$;
(c) real wages rose modestly while nominal labour cost per unit of output fell, subject to the quality and comparability of both indices.
57. (a) $121/110(100) = 110$; (b) $110/105(100) \approx 104.76$; (c) averages conceal variation in benefit levels and recipient characteristics.
58. (a) Inflation is 7% and the wage adjustment is 4.2%; (b) $1.042/1.07 - 1 \approx -2.62\%$; (c) the agreement produces a real-wage decline.

59. (a) Region A: RM 4000; Region B: approximately RM 4190.48; (b) A ranks higher nominally, but B ranks higher after the stated cost adjustment; (c) the figures omit household needs, public services, wealth and other dimensions of welfare.
60. (a) $125/118(100) \approx 105.93$; (b) real median income is approximately 5.93% above the base; (c) a median does not describe every household or the distribution around the middle.

Exercises 61–70: housing, regions and transport.

61. (a) $112(105)(98)/100^2 = 115.248$; (b) approximately RM 1728.72; (c) the final link reduces the preceding level by 2%, but the index remains above the 2022 base.
62. (a) 123.5; (b) RM 2223; (c) renters and owners purchase different housing services and face different expenditure shares.
63. (a) Region A: RM 420 000; Region B: RM 500 000; (b) Region A; (c) Region B; (d) the index omits the different base price levels.
64. The rebased series is approximately 83.33, 92.59, 100, 111.11. The second-to-fourth ratio is $120/100 = 111.11/92.59 \approx 1.20$ in either series.
65. (a) 125.5; (b) approximately RM 3.14; (c) fuel contributes $0.40(35) = 14$ index points.
66. (a) $92/108(100) \approx 85.19$; (b) ridership per person is approximately 14.81% below the base; (c) possible explanations include service changes, fares, remote work or alternative transport, but the index does not identify the cause.
67. (a) $110/118(100) \approx 93.22$; (b) current budget RM 55 million, full adjustment RM 59 million, giving a RM 4 million shortfall; (c) examples include deterioration, traffic volume, standards or the condition of the road network.
68. Thresholds are RM 2000 in A and RM 1520 in B. The A income is RM 100 below its threshold; the B income is RM 80 above its threshold.
69. (a) $165/150 - 1 = 10\%$; (b) $150/165(100) \approx 90.91$; (c) the point change must be divided by the starting index 150.
70. (a) $0.40(110) + 0.35(125) + 0.25(105) = 114$; (b) expenditure weights reflect how much is spent rather than how many people reside in each city; (c) residents have different baskets, incomes and local prices.

Exercises 71–80: education, health and public services.

71. (a) $118/130(100) \approx 90.77$; (b) approximately $60\%(118/130) = 54.46\%$; (c) the scholarship covers a smaller share of tuition than in the base period.
72. (a) 126.5; (b) RM 303.60; (c) medicine contributes $0.50(40) = 20$ index points.

73. (a) $108/105(100) \approx 102.86$; (b) measured output per staff member is about 2.86% higher; (c) the index may omit quality, case complexity, outcomes or staff working conditions.
74. (a) $119/112(100) = 106.25$; (b) $106.25/107(100) \approx 99.30$; (c) real budget per student is about 0.70% below the base.
75. (a) $75\%(1.044) = 78.3\%$; (b) 3.3 percentage points; (c) the relative increase is 4.4%, measured against the original 75% rate.
76. (a) $120(115)/100 = 138$; (b) RM 2.76 million; (c) utilisation increased as well as prices.
77. (a) 109.7; (b) health contributes 3.2 points, education 5.25 and income 1.25; (c) the indicators have different meanings, measurement quality and normative importance.
78. The old-weight index is 112; the new-weight index is 108.5. The lower new result follows from placing less weight on the component with index 120 and more on the component with index 95.
79. (a) $112(125)/100 = 140$; (b) RM 14 million; (c) part of the increase finances 25% more doses rather than merely compensating for cost per dose.
80. The simple average is 97.5. It is misleading because a lower waiting-time index is favourable while a higher satisfaction index is favourable; their directions and units must first be made comparable.

Exercises 81–90: production, trade and the environment.

81. (a) $90/120(100) = 75$; (b) emissions per unit of output are 25% below the base; (c) other pollutants, resource use or total impacts may follow different patterns.
82. (a) $130(92)/100 = 119.6$; (b) higher prices outweigh the reduction in use, so expenditure rises by 19.6%; (c) multiplying assumes compatible definitions, coverage and weighting.
83. (a) $L = 580/500(100) = 116$; $P = 588/510(100) \approx 115.29$; $F \approx 115.65$; (b) $V = 588/500(100) = 117.6$; (c) the value index also reflects the changed quantities.
84. The fixed-base series is 100, 95, 104.5, 108.68. The overall increase is 8.68%. Annual changes compound, so $-5 + 10 + 4 = 9\%$ is only an additive approximation.
85. (a) 110.9; (b) manufacturing contributes 7.5 points, services 3.2 and agriculture 0.2; (c) sector shares change as the economy and data coverage evolve.
86. The estimated volume index is $125/110(100) \approx 113.64$, suggesting export volume about 13.64% above the base under the index decomposition.
87. (a) $118(105)/100 = 123.9$; (b) $123.9/110(100) \approx 112.64$; (c) consumption, tariffs and household circumstances vary across the population.

88. (a) RM 64; (b) $(60/50)/(128/100)(100) = 93.75$; (c) the real charge is 6.25% below its base-period strength.
89. The final index is $100(1.10)(0.90) = 99$. The 10% fall is applied to the larger post-increase value, so the final level is 1% below the starting level.
90. The rebased series is approximately 111.11, 133.33, 100, 150. Division of every value by the same positive constant preserves ordering and all ratios.

Exercises 91–100: evaluating and communicating indices.

91. An index of 132 means the measured basket is 32% above its base-period level, not 132% higher. The report should state the base period, basket, population, geographical coverage and date.
92. (a) -0.6 index points; (b) $(124.8/125.4 - 1)100\% \approx -0.48\%$; (c) provisional observations may change after additional data, corrections or revised methods become available.
93. (a) $100/80(100) = 125$; (b) $100/95(100) \approx 105.26$; (c) an unusually low base exaggerates the apparent subsequent increase.
94. $30(110) + w(125) + (70 - w)(95) = 100(111.5)$ gives $w = 40$; the third component also has weight 30.
95. $40(120) + 35x + 25(108) = 100(114.9)$ gives $x = 114$. Its contribution is $0.35(14) = 4.9$ index points.
96. (a) $F = \sqrt{124(118)} \approx 120.96$; (b) 6 index points; (c) consumers may substitute towards goods whose relative prices rose less, so current-quantity and base-quantity weights differ.
97. (a) Older-household index = 123.5; national index = 117.25; (b) the older-household basket has greater weight on healthcare and housing, whose indices are high; (c) subgroup indices reveal distributional effects concealed by a national average.
98. Rebased Series A is approximately 90.91, 100, 110. Series B is 91, 100, 111; both show similar growth, but may differ because of formula choice, rounding, data coverage, weights or revisions.
99. (a) 110; (b) $(1100/1.20)/1000(100) \approx 91.67$; (c) quality adjustment attempts to separate pure price change from changes in product characteristics.
100. Suitable questions concern:
- the base period and periods compared;
 - the represented population, region and basket;
 - the weights;
 - the formula and linking method;

- data quality, uncertainty and revisions; and
- whether the change is in index points or percentage terms.

Subgroup differences and the decision being informed also matter. An index describes measured relative change; causal explanation requires additional evidence.

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Editorial note

Unless otherwise stated, the surrounding exposition, derivations, notation, worked calculations, exercises, and TikZ diagrams were independently written and typeset for this book. Reference materials were reorganised, corrected where necessary, and adapted to maintain consistent notation throughout the document.

This book was developed and typeset in Prism with assistance from OpenAI language models. AI-assisted work included editorial restructuring, copy-editing, LaTeX drafting, and the implementation of selected diagrams. The author remained responsible for selecting the scope, reviewing the mathematics, adapting source material, and approving the final presentation.

About the Author

Asfa Izzat was born in Ipoh, Perak. Although mathematics was not initially his strongest subject at school, this did not prevent him from developing a lasting interest in it. His experience as a student shaped the central aim of this book: to present foundation mathematics as a connected body of ideas that can be learned progressively through clear

explanations, worked examples, and carefully arranged practice. He began developing the book while studying at the Centre for Foundation Studies in Science, Universiti Malaya.

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